References with abstracts for QWIM project: Retirement portfolios in quantitative wealth and investment management QWIM

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Abstract

This document contains a curated list of references (and their abstracts) to be considered as literature review for this QWIM project. Two recordings are also made available: 1) overview of QWIM (Quantitative Wealth and Investment Management) and 2) my recommendations on timeline and steps for the QWIM project.

The recommendation is to analyze these references in the following order

1) Section 1.2: Top recommendations within main references
2) (Optional) Section 1.1: Main references
3) (Optional) Section 1.3: Comprehensive list of references

For the the first 3-4 weeks of this project it is recommended to combine A) literature review (performed in multiple stages, described below) with B) identification and retrieval of the relevant datasets.

The staged approach for the literature review is meant to primarily focus on the the most relevant information, such that at the end of each stage you can decide which references to retain for next stage(s) and, respectively, which references to eliminate from analysis.

1) First stage: decide (only based on reading their abstracts) which references to retain for next stage.
2) Second stage: for the retained references, also read the conclusion and the section containing numerical results, to decide which references to retain for next stage.
3) Third stage: for the retained references, identify whether
   - the method/model described in reference is applicable to the use case(s) for this QWIM project
   - there is an existing coding implementation; if not mentioned in the reference, you may need to reach out to the author(s).
4) Final stage(s): compare the retained references to make the final selection of the reference(s) you want to use for your contribution to this QWIM project.

This QWIM project is meant to be done as a team, with each team member selecting from the references at least 1 (one) quantitative model/method associated to the main topic(s) of this QWIM project, and with implementations done through a coding framework designed by your team (in collaboration with your advisers).

The coding framework incorporates both individual components (selected models/methods, at least one per team member), and common components (for analysis, comparison, visualization, etc.).

Individual components of your team's coding framework

- model/method selected by first team member
- model/method selected by second team member
- ...
- model/method selected by n-th team member

Common components of your team's coding framework

- Data retrieval
- Data analysis
- Tests
- Comparison with benchmarks
- Comparison among selected models/methods
- Analysis of results
- Interactive dashboard
- Report generation

Recommendation is to use Python for the coding framework that your team will design and use for this QWIM Project, incorporating carefully selected Python and R libraries.

This QWIM project places strong emphasis on the combination [model selection/implementation PLUS convincing your audience on advantages and usefulness of the selected models], rather than just the model selection/implementation. That is why this QWIM project requires an interactive dashboard. I will also provide guidance how to collaborate as a team in updating the interactive dashboard to include, show and analyze each and every model considered by your team.

Regarding which type of interactive dashboard to select, my recommendation is to use **Shiny** (in Python or in R), with **streamlit** a secondary rcommendation.

Each team will create a private Github repository for their QWIM project. I will place there Python codes for a Shiny QWIM interactive dashboard, such that your team can update it as needed for your project.

A recording of my presentation on Shiny dashboard (in Python) for QWIM is made available. The list of webpages presented in the recording include Shiny Python guide and how it compares with other interactive dashboards, gallery of Shiny dashboards, 4-hour masterclass on Shiny for Python.

It is strongly recommended to leverage existing coding implementations and Python/R packages selected based on careful analysis.

To give a few examples, it is recommended to use **polars** for tabular data structures, **kedro** for an automated coding framework for a data science project (such as this QWIM project), and **uv** for project and package management.

To assist you in the selection of coding implementations and Python/R packages, a curated list will be separately provided to your team.

The list of your team's deliverables for this QWIM project includes

- Written report (executive summary, literature review, description of selected models, numerical results, analysis and comparison, conclusion)
- interactive dashboard

Contents

1 Relevant references

1.1 Main references

List of references:

Ackerley et al. (*To spend or not to spend?* 2022)

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1.2 Top recommendations within main references

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1.3.25 Software implementations and frameworks

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Arnold et al. (“Social Security Decisions: Should Recipients Opt for Early Payments?” 2023)
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Milevsky (“Modeling Human Longevity and Life Tables,” 2020)
Milevsky (“Intelligent Drawdown Rates,” 2020)
Milevsky (“Modeling the Risk of Sequence-of-Returns,” 2020)
Rook (“Multivariate Density Modeling for Retirement Finance,” 2017)
Wang et al. (“reslife: Residual Lifetime Analysis Tool in R,” 2023)

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Abeliansky, A. and Strulik, H. (2021). "Health and Aging Before and after Retirement." In: *SSRN e-Print*.

We investigate health and aging before and after retirement for specific occupational groups. We use five waves of the Survey of Health, Aging, and Retirement in Europe (SHARE) and construct a frailty index for elderly men and women from 10 European countries. Occupational groups are classified according to low vs. high education, blue vs. white collar color, and high vs. low physical or psychosocial job burden. Controlling for individual fixed effects, we find that, regardless of the used classification, workers from the first (low status) group display more health deficits at any age and accumulate health deficits faster than workers from the second (high status) group. We instrument retirement by statutory retirement ages ("normal" and "early") and find that the health of workers in low status occupations benefits greatly from retirement, whereas retirement effects for workers in high status occupations are small and frequently insignificant. We also find that workers from low status occupations always have higher health deficits, i.e. we find evidence for an occupational health gradient that widens with increasing age, before and after retirement.

Abeysekera, S. P. and Rosenbloom, E. S. (2002). "Optimal Rebalancing for Taxable Portfolios." In: *The Journal of Wealth Management* 5(3), pp. 42–49.

The authors focus on the issue of rebalancing taxable portfolios and overcoming the capital gains lock-in effect. They propose an integer linear program, developed to maximize the expected after-tax return over a desired planning horizon, given an initial portfolio with unrealized gains and losses. They then offer an integer quadratic program, developed for minimizing the variance of the return given an initial portfolio with unrealized gains and losses. They conclude by suggesting that these models allow investors to explicitly take into consideration their personal tax situation when rebalancing their portfolios.

Ackerley, A., Nefouse, N., and Nikles, D. (2022). *To spend or not to spend?* Tech. rep. BlackRock.

This was not what we expected to find: on average across all wealth levels, most current retirees still had 80% of their pre-retirement savings after almost two decades of retirement according to research conducted jointly with the Employee Benefit Research Institute (EBRI). One-third even grew their assets over the course of retirement. While that sounds like good news, the reasons why some retirees did not spend their assets may be complicated. Certainly some may have benefited from greater access to defined benefit pensions, more income replacement from Social Security, strong real estate appreciation and an investment market that generally delivered strong returns and high interest rates. On the other hand, some retirees may not have spent their savings due to uncertainty, perhaps even hoarding assets at the expense of fully supporting the retirement lifestyle they might have enjoyed. In order to better understand why retirees either spent down assets or held on to them, BlackRock engaged Greenwald and Associates to conduct 19 in-depth interviews and survey 1,510 retirees.

Interviews for the survey portion of this study were conducted online in February 2018. To qualify for this study participants had to be fully retired for at least five years are between the ages of 60 and 80, and have investable assets between $200,000 and $2,500,000. Survey results have been weighted to reflect the overall make up of Americans retired for five years or longer. In a similarly sized random sample survey of general population respondents, the margin of error (at the 95% confidence level) for the total population in this study would be plus or minus approximately 2.5 percentage points.

We identified six key themes and drivers of retiree spending behavior that emerged from the findings and offer "lessons learned" advice for participants and retirees, employers and advisors.

1) Retirees prefer to keep their assets untouched
2) Retirees more often plan to spend consistently - increasing with age
3) Retirees retain their accumulation mindset
4) Retirees with pension income least likely to spend down. Because they don't need to.
5) Men and women approach finances in retirement differently, and mostly for good reason
6) More recent retirees are less optimistic

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Adcock, C., Areal, N., Armada, M. R., Cortez, M. C., Oliveira, B., and Silva, f. (2017). "Portfolio Performance Measurement: Monotonicity with Respect to the Sharpe Ratio and Multivariate Tests of Correlation." In: *SSRN e-Print*.

This paper reports an investigation into methods of portfolio performance measurement. The work is motivated first by equivocal empirical evidence reported by several authors about the correlation of performance measures with the Sharpe ratio. Secondly it is motivated by recent work which specifies that performance measures will be monotone functions of the Sharpe ratio if portfolio returns follow the same location-scale distribution. The paper demonstrates that the class of location-scale distributions is broader than previously reported. It presents conditions under which monotonicity with respect to the Sharpe ratio will fail. The paper shows that for large sample sizes the correlation between pairs of performance measures that are functions of the Sharpe ratio is unity. The correct null hypothesis for tests of correlation is therefore rho=1. Two multivariate tests of this null hypothesis are presented. The new tests are used to carry out of a comprehensive study of performance measurement for a set over ninety UK investment trusts.

Adra, S. and Menassa, E. (2025). "Waiting for Wealth Accumulation: The Investment Experience in Time." In: *The Journal of Investing*, joi.2025.1.350.

The authors investigate how the passive equity investment experience unfolds in time. They introduce the concept of a "wealth plateau"-the minimum time interval during which accumulated wealth remains above a preset level. They estimate the waiting times needed for diverse plateau objectives (POs) to be reached and analyze the influence of such factors as the cyclically adjusted price-to-earnings (CAPE) ratio, unemployment, inflation, and the yield curve on achieving these objectives. Their results highlight how these variables' effects vary based on plateau goals and investment styles. They also emphasize the trade-offs investors encounter when making intertemporal choices to shorten the time required to achieve their POs. By acknowledging the significance of time-based framing on the analytics and practice of goal-based investing, their findings provide insights for academics and retail investors.

Akyildirim, E., Gambara, M., Teichmann, J., and Zhou, S. (2025). "Randomized signature methods in optimal portfolio selection." In: *Quantitative Finance* 25(2), pp. 197–216.

We present convincing empirical results on the application of Randomized Signature Methods for non-linear, non-parametric drift estimation for a multi-variate financial market. Even though drift estimation is notoriously inaccurate due to small signal to noise ratio, one can still try to learn optimal non-linear maps from past data to conditional expectations of future returns for the purposes of portfolio optimization. Randomized Signatures, in contrast to classical signatures, allow for high dimensional markets and provide features on the same scale. We do not contribute to the theory of Randomized Signatures here, but rather present our empirical findings on portfolio selection in real world settings including real market data and transaction costs.

Alankar, A., DePalma, M., and Scholes, M. (2013). "An Introduction to Tail Risk Parity: Balancing Risk to Achieve Downside Protection." In: *SSRN e-Print*.

Tail Risk Parity (TRP) adapts the risk balancing techniques of Risk Parity in an attempt to protect the portfolio at times of economic crisis and reduce the cost of the protection in the absence of a crisis. In measuring expected tail loss we use a proprietary implied expected tail loss (ETL) measure distilled from options market information.

Whereas Risk Parity focuses on volatility, Tail Risk Parity defines risk as expected tail loss, something that hurts investors more than volatility. Risk Parity is a subset of Tail Risk Parity when asset returns are normally distributed and or volatility adequately captures tail loss risk. Hence, when the risk of tail events is negligible, Tail Risk Parity allocations will resemble Risk Parity allocations.

Tail Risk Parity seeks to reduce tail losses significantly while retaining more upside than Risk Parity or other mean variance optimization techniques. It is very difficult to construct portfolios under symmetric risk measures (such as volatility as used in Risk Parity) that do not penalize both large losses and, unfortunately, large gains. Tail Risk Parity aims to protect investments against large losses when investors can least afford them, when systemic crises unfold and correlations spike unexpectedly. This is exactly when the marginal utility of an extra dollar is highest.

Our research suggests that a Tail Risk Parity approach hedges the risk of large losses more cheaply than using the options market (historically we estimate savings of about 50.

We believe that Tail Risk Parity offers an attractive solution for investors seeking balanced investment portfolios that can cost effectively reduce exposures to tail losses.

Alderson, M. J. and Betker, B. L. (2017). "Does the Benefit of Deferring Social Security Offset the Opportunity Cost to Do So?" In: *Journal of Financial Planning*.

This study examined the extent to which deferring the claiming date for Social Security benefits preserved an individual's tax-deferred retirement assets, thus enhancing the ability to provide for constant real consumption. The analysis incorporated the relevant reductions and credits for early and late claiming, mortality risk, stochastic investment returns, and the availability of longevity insurance. Results provide guidance for financial planners to help clients make the optimal claiming decision. For single and married individuals with substantial savings, the deferral of benefits reduced the risk of retirement asset exhaustion. The results demonstrate the unique nature of the Social Security asset in the retirement portfolio as well as the value of self-insuring against longevity risk.

Alekseev, G., Giglio, S., Maingi, Q., Selgrad, J., and Stroebel, J. (2022). "A Quantity-Based Approach to Constructing Climate Risk Hedge Portfolios." In: *SSRN e-Print*.

We propose a new methodology to build portfolios that hedge the economic and financial risks from climate change. Our quantity-based approach exploits information on how mutual fund managers trade in response to idiosyncratic changes in their climate risk beliefs. We exploit two types of idiosyncratic belief shocks: (i) instances when fund advisers experience local extreme heat events that are known to shift climate change beliefs, and (ii) instances when fund managers change the language in shareholder disclosures to express concerns about climate risks. We use the funds' observed portfolio changes around such idiosyncratic belief shocks to predict how investors will reallocate their capital in response to aggregate climate news shocks that shift the beliefs and asset demands of many investors and thus move equilibrium prices. We show that a portfolio that is long stocks that investors tend to buy after experiencing negative idiosyncratic climate belief shocks, and short stocks that investors tend to sell, appreciates in value in periods with negative aggregate climate news shocks. Our quantity-based portfolios have superior out-of-sample hedge performance compared to portfolios constructed using existing alternative methods. The key advantage of the quantity-based approach is that it learns from rich cross-sectional trading responses rather than time-series price information, which is particularly limited in the case of newly emerging risks such as those from climate change. We also demonstrate the versatility of the quantity-based approach by constructing successful hedge portfolios for aggregate unemployment and house price risk.

Alexandrova, M. and Gatzert, N. (2019). "What Do We Know About Annuitization Decisions?" In: *Risk Management and Insurance Review* 22(1), pp. 57–100.

Against the background of aging societies and increasing life expectancies, the protection of individuals from outliving their savings has become increasingly relevant. Annuities represent insurance against longevity risk and can prevent old-age poverty. The aim of this article is to present the current state of theoretical, empirical and experimental evidence with regard to annuitization decisions. Toward this end, we conduct a systematic literature review that includes 89 articles. Based on this, we study welfare effects of mandatory annuitization, annuitization rates and the optimal fraction of wealth to be annuitized, as well as determinants of retirees' choice to annuitize and their impact on annuity demand. Finally, we present possible solutions for overcoming the low uptake of annuities based on its causes. One main result is that behavioral biases in annuitization decisions particularly require considerably more theoretical research and empirical evidence, and that theoretical models already appear to well explain empirically observed annuitization rates.

Ali, Y., Fang, M., Sota, P. A. A., Taylor, S., and Wang, X. (2019). "Social Security Benefit Valuation, Risk, and Optimal Retirement." In: *Risks* 7(4), p. 124.

We develop valuation and risk techniques for the future benefits of a retiree who participates in the American Social Security program based on their chosen date of retirement, the term structure of interest rates, and forecasted life expectancy. These valuation methods are then used to determine the optimal retirement time of a beneficiary given a specific wage history and health profile in the sense of maximizing the present value of cash flows received during retirement years. We then examine how a number of risk factors including interest rates, disease diagnosis, and mortality risks impact benefit value. Specifically, we utilize principal component analysis in order to assess both interest rate and mortality risk. We then conduct numerical studies to examine how such risks range over distinct income and demographic groups and finally summarize future research directions.

Aliaga-Diaz, R., Ahluwalia, H., Zhu, V., Donaldson, S., Daga, A., and Pakula, D. (2021). *Vanguard's Life-Cycle Investing Model (VLCM): A general portfolio framework for goals-based investing*. Tech. rep. Vanguard.

Investors have multiple goals throughout their lifetime, each requiring them to make complex, interconnected decisions about saving, spending, and asset allocation. We present a framework for making asset allocation decisions based on an investor's goals, preferences, and personal circumstances and factoring in the uncertainty of asset returns. The Vanguard Life-Cycle Investing Model (VLCM) is a proprietary model for glide-path construction that can assist in the creation of custom investment portfolios for retirement as well as nonretirement goals, such as saving for college. The VLCM embodies key principles of life-cycle investing theory, including a utility-based framework encompassing risk aversion and time preference. It also incorporates important behavioral finance considerations such as loss aversion and income shortfall aversion. The use of the VLCM enables cost-benefit analysis of glide-path customization, evaluation of risk-return trade-offs of various asset and sub-asset allocation choices, and multiple portfolio analytics of the probability of success and odds of income sufficiency. Based on VLCM's analytical framework, we find that risk-aversion levels are the dominant factor behind the broad stock-bond split in the glide path, affecting both glide-path slope and ending allocation.

Allain, S. (2022). "Long Term Care Planning For Mature Clients: Insuring Against Longevity and Healthcare Risks with Modern Solutions." In: *Investments and Wealth Monitor* 2022(September/October).
Because we are living longer, longevity and healthcare risks now intersect. The magnitude of these two unsystematic risks is so large that, as part of any retirement plan, it makes sense to both assess, and put a plan in place, to mitigate these threats to retirement security.

Alleva, B. (2016). "Discount Rate Specification and the Social Security Claiming Decision." In: *SSRN e-Print*.
Choosing the claiming age that maximizes the expected present value of lifetime Social Security retirement benefits requires a survival function to account for an individual's prospective longevity along with the specification of a rate by which to discount the future benefit payments for each claiming age. This article evaluates optimal claiming ages for prospective beneficiaries across a range of 81 real discount rate options (specified in increments of one-tenth of 1 percent) from 0 percent to 8 percent, considering the survival functions for men and women born in 1952. It examines the implications of choosing a given rate as well as the sensitivity of the optimal claiming age to a specific rate choice.

Alleva, B. (2017). "Social Security Retirement Benefit Claiming-Age Combinations Available to Married Couples." In: *SSRN e-Print*.
The rules for claiming Social Security retired-worker benefits are complex in large part because they offer a potential claimant flexibility in choosing a claiming age most to his or her advantage. The complexity of those rules is multiplied for a married couple, as the potential eligibility for spousal benefits, the couple's age difference, and other factors must also be considered. The number of claiming-age combinations available to a married couple varies widely for couples with different circumstances. This note explores the claiming rules, contingent situations, claiming-age combinations, and benefit amounts available to married couples across a range of respective birth years and benefit levels based on respective earnings histories.

Alleva, B. J. (2015). "Minimizing the Risk of Opportunity Loss in the Social Security Claiming Decision." In: *The Journal of Retirement* 3(1), pp. 67–86.
The optimal age to claim Social Security benefits is often defined as that which maximizes expected lifetime benefits. Expectations are not always met however, and this optimal claim age is unlikely to be that which would have maximized lifetime benefits received by the eventual age at death. The optimal claim age then carries the potential for an opportunity loss, or the risk of receiving less than the maximum possible, which varies with age at death. This study evaluates the claiming decision by first computing and presenting the maximizing claim age for each possible death age over the retirement horizon, and then determining the actual opportunity loss for each death age for claiming at the otherwise optimal claim age. The potential for opportunity loss over the retirement horizon is then computed for the optimal claim age. If a participant can target a period within the retirement horizon for which the values total lifetime benefits higher than the remainder, the optimal claim age for that target period will entail less risk, or less potential for opportunity loss, than that for optimizing over the entire retirement horizon. Such a strategy could effectively address gaps in a retirement income portfolio.

Almadi, H., Rapach, D. E., and Suri, A. (2014). "Return Predictability and Dynamic Asset Allocation: How Often Should Investors Rebalance?" In: *The Journal of Portfolio Management* 40(4), pp. 16–27.
To exploit return predictability via dynamic asset allocation, investors face the important practical issue of how often to rebalance their portfolios. More frequent rebalancing uses statistically and economically significant short-horizon return predictability to aggressively pursue the dynamic investment opportunities afforded by changes in expected returns. However, the degree of return predictability typically appears stronger at longer horizons, which, along with lower transaction costs, favors less frequent rebalancing. The authors analyze the performance effects of rebalancing frequency in the context of dynamic portfolios constructed from monthly, quarterly, semi-annual, and annual return forecasts for U.S. stocks, bonds, and bills, where the dynamic portfolios rebalance at the same frequency as the forecast horizon. Along the transaction-cost/rebalancing frontier, monthly (annual) rebalancing provides the greatest outperformance when unit transaction costs are below (above) approximately 50 basis points, and dynamic portfolios based on annual rebalancing typically outperform the benchmarks for unit transaction costs well in excess of 400 basis points.

Alsabah, H., Capponi, A., Ruiz Lacedelli, O., and Stern, M. (2021). "Robo-Advising: Learning Investors Risk Preferences via Portfolio Choices." In: *Journal of Financial Econometrics* 19(2), pp. 369–292.
We introduce a reinforcement learning framework for retail robo-advising. The robo-advisor does not know the investor risk preference but learns it over time by observing her portfolio choices in different market environments. We develop an exploration-exploitation algorithm that trades off costly solicitations of portfolio choices by the investor with autonomous trading decisions based on stale estimates of investor risk aversion. We show that the approximate value function constructed by the algorithm converges to the value function of an omniscient robo-advisor over a number of periods that is polynomial in the state and action space. By correcting for the investor mistakes, the robo-advisor may outperform a stand-alone investor, regardless of the investor opportunity cost for making portfolio decisions.

Ameriks, J., Briggs, J., Caplin, A., Shapiro, M. D., and Tonetti, C. (2024). "The Long-Term-Care Insurance Puzzle: Modeling and Measurement." In: *SSRN e-Print*.
Individuals face significant late-in-life risks, prominently including the need for long-term care (LTC). Yet, they hold little long-term care insurance (LTCI). In this paper we use a structural model and a purpose-designed dataset to understand the determinants of insurance demand. We distinguish between a fundamental lack of desire to insure, crowd out from existing insurance, and unmet demand due to poor products available in the market. The model features individual-specific non-homothetic health-state-dependent preferences over normal consumption, consumption when in need of long-term care, and bequests, which are estimated using strategic survey questions. To account for differences between the modeled and measured insurance products, we study not only individuals' holdings of LTCI, but also their stated demand for an idealized product that mirrors that in the model. We find that many individuals would purchase LTCI and receive a large consumer surplus if it were a better product, while many others do not want to purchase even high-quality actuarially fair LTCI due to the values of their heterogeneous state-dependent preferences, their demographics, and their financial situation.

An, B.-J. and Sachdeva, K. (2021). "Missing the Target? Retirement Expectations and Target Date Funds." In: *SSRN e-Print*.
This paper measures the cost of biased retirement expectations for investors in target-date funds. Using survey data, we show that respondents systematically underestimate their long-run labor participation on average by 4.8 years, with errors having meaningful cross-sectional relationships. We use these insights to build a life-cycle model of target-date funds to measure the costs of biased expectations. Calibrations suggest that errors in expectations compound over time, costing the median respondent 4% of total wealth, equivalent to 0.2% a year. These estimates suggest that the choice architecture of target-date funds should be modified to improve the financial adequacy of future retirees.

Anantharaman, D. and Henderson, D. (2016). "Understanding Pension Liabilities: A Closer Examination of Discount Rates." In: *SSRN e-Print*.
The discount rate represents a critical choice in accounting for corporate defined-benefit pension plans due to the long-term nature of pension liabilities. U.S. GAAP and IFRS mandate the AA corporate bond rate. Their requirement is subject to much debate, with the risk-free rate and the expected return on pension assets (EROA) often proposed as alternatives. We examine which of these rates best fits pension values as perceived by equity- and debt-market participants. For equity values, the EROA rate dominates, while for credit ratings, the AA rate dominates. For financially healthy firms, however, discounting at the EROA produces the best fit for both equity values and credit ratings. In contrast, for firms nearing financial distress, discounting at the AA rate provides the best fit. We also find that the accumulated benefit obligation measure consistently dominates the projected benefit obligation in explanatory power for credit ratings. Overall, we find that market participants adjust GAAP-recognized pension obligations in both amount and discount rate, in a manner that appears consistent with firm-specific economic realities.

Anarkulova, A., Cederburg, S., O'Doherty, M. S., and Sias, R. W. (2022). "The Safe Withdrawal Rate: Evidence from a Broad Sample of Developed Markets." In: *SSRN e-Print*.
We use a comprehensive new dataset of asset-class returns in 38 developed countries to examine a popular class of retirement spending rules that prescribe annual withdrawals as a constant percentage of the retirement account balance. A 65-year-old couple willing to bear a 5% chance of financial ruin can withdraw just 2.26% per year, a rate materially lower than conventional advice (e.g., the 4% rule). Our estimates of failure rates under conventional withdrawal policies have important implications for individuals (e.g., savings rates, retirement timing, and retirement consumption), public policy (e.g., participation rates in means-tested programs), and society (e.g., elderly poverty rates).

Ando, M. and Fukasawa, M. (2025). "When to efficiently rebalance a portfolio." In: *Quantitative Finance*, pp. 1–11.
A constant weight asset allocation is a popular investment strategy and is optimal under a suitable continuous model. We study the tracking error for the target continuous rebalancing strategy by a feasible discrete-in-time rebalancing under a general multi-dimensional Brownian semimartingale model of asset prices. In a high-frequency asymptotic framework, we derive an asymptotically efficient sequence of simple predictable strategies.

Andrews, E. S., Turner, J. A., Schmitt, R., and Sun, W. (2024). "Delayed Social Security Claiming: Why Do So Few Participants Delay Claiming Past Age 65?" In: *The Journal of Retirement* 12(1), pp. 7–21.
The authors investigate why only a minority of retirees follow the recommendations of economists and financial journalists and delay claiming Social Security old-age benefits until at least their Full Retirement Age. The authors argue that low levels of savings and a reluctance to draw down financial assets contribute to a reluctance to delay Social Security claiming after retirement. The authors apply the numerical optimization simulation model of Sun and Webb (2010), using their code, to show the effects on optimal claiming age of longer life expectancy, lower real interest rates, and differing levels of liquidity constraints. The authors find that lower real interest rates and increased life expectancy increase the optimal claiming past age 65, but only for those not liquidity constrained (men) and not already at the highest claiming age (women). The optimal claiming age for people with a high level of liquidity ranges from 68 to 70, while for people with a low level of liquidity, it ranges from 65 to 66. The authors' empirical work supports these findings.

Ansari, S. N. (2024). "Quantitative Models for US Equity Trading Strategies Using Complex Derivatives in Wealth Management." In: *SSRN e-Print*.
The integration of quantitative models in wealth management has transformed the landscape of private capital management, especially within the US equity markets. These models leverage advanced mathematical techniques and computational tools to devise sophisticated trading strategies that involve complex derivatives. By systematically applying these models, wealth managers can optimise portfolio performance, manage risks, and align investment strategies with client goals. This paper explores the role of quantitative models in developing US equity trading strategies using complex derivatives, emphasising their utility in systematic wealth management.

El-Ansary, A. (2008). "Alpha insurance: A computational framework to measure hedging demands for active investors." In: *Journal of Asset Management* 9(5), pp. 310–320.
The purpose of the paper is to develop the concept of portfolio insurance against active managers' stock selection risks. The insurance premium is estimated through the use of exotic options and the impact on investors' utility is analysed within a multi-moment efficient frontier framework. For illustration, the suggested methodology is applied to the Swiss Market Index and employed to estimate the hedging demands faced by investors when portfolio choice problem is considered in a multi-period framework.

Anson, M. (2024). "Thinking Outside the Benchmark: Part II." In: *The Journal of Portfolio Management* 51(1), pp. 123–131.
Thinking outside the benchmark requires institutional investors to break away from the traditional asset class lines that apply to most portfolio construction processes. Most versions of strategic asset allocation focus on just that-on asset classes. This builds into the complementary concept of a total portfolio approach (TPA) that advocates that institutional investors step away from traditional asset class buckets and consider economic factors that drive the risk and return profile for a pool of capital. Every investment is broken down into its main economic drivers and then added to the total portfolio to assess its overall risk and return profile. Asset class lines become descriptions, but not restrictions, on how the portfolio is constructed. An example of TPA is the investment focus on environmental, social and governance (ESG) investments. ESG is more a theme and less an asset class, and it has the potential to allow investors to think outside of benchmarks and away from traditional asset class descriptions.

Antell, J. and Vaihekoski, M. (2025). "Long-Term Equity Investing and Withdrawal Rules: How Not to Die Penniless." In: *SSRN e-Print*.
This paper provides evidence on the outcomes of several different withdrawal policies for a long-term equity investor with a motive to preserve the real value of their assets and maximize the withdrawals. Using data for the US and Finnish stock markets from 1913 to 2023, we find that historically the maximum endowment-preserving withdrawal rates would have been 1.83 and 10.95 percent of the initial investment at the end of 1912 for Finland and the US, respectively. The maximum withdrawal rate varies with the timing of the initial investment. The average maximum withdrawal rates for the first one hundred years are 6.63 percent for Finland and 7.79 percent for the USA. A rearview look shows that following a rule where the withdrawal is a given fixed rate of the nominal value of the portfolio all the while either disallowing reductions in nominal withdrawals

(Finland) or keeping them at least 90% level of the previous nominal withdrawal (USA), would have historically provided the highest average withdrawals in real terms. Looking forward, both countries offer withdrawal rates of four or even five percent with reasonable risk if one allows the withdrawals to be adjusted for stock market development.

Apicella, G., Molent, A., and Gaudenzi, M. (2024). "The Life Care Annuity: enhancing product features and refining pricing methods." In: *arXiv e-Print*.

The state-of-the-art proposes Life Care Annuities, that have been recently designed as variable annuity contracts with Long-Term Care payouts and Guaranteed Lifelong Withdrawal Benefits. In this paper, we propose more general features for these insurance products and refine their pricing methods. We name our proposed product "GLWB-LTC". In particular, as to the product features, we allow dynamic withdrawal strategies, including the surrender option. Furthermore, we consider stochastic interest rates, described by a Cox-Ingersoll-Ross process. As to the numerical methods, we solve the stochastic control problem involved by the selection of the optimal withdrawal strategy through a robust tree method, which outperforms the Monte Carlo approach. We name this method "Tree-LTC", and we use it to estimate the fair price of the product, as some relevant parameters vary, such as, for instance, the entry age of the policyholder. Furthermore, our numerical results show how the optimal withdrawal strategy varies over time with the health status of the policyholder. Our findings stress the important advantage of flexible withdrawal strategies in relation to insurance policies offering protection from health risks. Indeed, the policyholder is given more choice about how much to save for protection from the possible disability states at future times.

Armour, P. and Hung, A. (2017). "Drawing down Retirement Wealth: Interactions between Social Security Wealth and Private Retirement Savings." In: *SSRN e-Print*.

Individual financial planning for retirement in the US is increasingly important, given the trend away from employer-provided defined benefit (DB) plans, the rising Social Security (SS) Full Retirement Age (FRA), and retiring baby boomers. A key financial decision that Americans make is how to draw on their retirement wealth across various sources, including both privately saved retirement funds and SS benefits. For SS retirement benefits, the main decision is at what age to claim, with claiming before the FRA resulting in lower monthly benefits, and claiming later leading to higher benefits. The terms of this tradeoff have changed in recent years: since 2003, the FRA has risen from 65 and will gradually increase to 67 by 2027, representing a drop in the present value of SS benefits. Meanwhile, defined contribution (DC) plans have gained in popularity, presenting retirees with more control over their private retirement wealth. The changing dynamics of both SS wealth and the private retirement decision space underscore the need for examining how individuals make decisions across their entire portfolio of retirement wealth. We use HRS survey data matched to SS administrative data to study how households integrate SS benefits into their general retirement income plans. We find starkly different non-SS retirement asset decumulation patterns across individuals who claim at different ages, with those claiming before the FRA drawing down pension and IRA wealth faster than those who claim at or after the FRA. However, the earliest claimants, those who claim SS retirement benefits exactly at age 62, are a highly heterogeneous group, consisting of both low-income, high expected mortality individuals as well as individuals with high pension holdings. We further find that this earliest claimant group is more likely to have retired and begun decumulating non-SS retirement assets even before age 62; however, this group's median and average assets are not substantially lower than later claimants. An analysis of claiming behavior by non-SS retirement wealth holdings shows that individuals with more retirement savings were overwhelmingly likely to claim between the ages of 62 and the FRA. On the other hand, those with the least retirement savings are more likely to either claim SS benefits as early as possible, either in the form of disability benefits or retirement benefits, or they would delay claiming SS retirement benefits until after the FRA. Moreover, birth cohorts facing higher FRAs tend to delay claiming SS retirement benefits on average; however, those most affected by this reduction in SS wealth - those with few other retirement assets - are the least reactive. Finally, households that do delay claiming as the FRA rises also tend to delay retirement and drawing down their non-SS retirement assets, indicating complementarity between SS and non-SS decumulation decisions and strong spillovers from changes in both SS and private retirement policy.

Armour, P. and Knapp, D. (2021a). *The Changing Picture of Who Claims Social Security Early*. Tech. rep. AARP Public Policy Institute.

From the early 1990s to 2010s, the fraction of individuals claiming Social Security retirement benefits at the early eligibility age (EEA) of 62 declined from over half of the eligible population to less than a third. Although claiming at 62 can make financial sense for some, Social Security benefit rules result in reduced monthly benefits of up to 30 percent relative to waiting to claim benefits until full retirement age. Such a considerable benefit reduction persists for life- and threatens financial security at later ages. This report highlights differences over time between individuals who claim Social Security retirement benefits at age 62 and those who claim after age 62. It documents how these populations have changed in terms of education, labor force participation, pension plan participation, and wealth-and how demographic, employment, and economic characteristics relate to claiming age. This report is the first in a two-part series; the second report focuses on the later-in-life financial and well-being consequences of early claiming.

The authors identify the demographic, employment, and economic characteristics associated with those who claim at the EEA and relate these characteristics to people who claim after it. For both men and women, individuals who claim Social Security benefits at 62, as compared with those who claim later:

1) Have less education, on average;
2) Are more likely to live in rural areas;
3) Are more likely to already have a worklimiting health condition;
4) Have a lower reported likelihood of living to age 75;
5) Are less likely to have a job before turning age 62;
6) If employed before turning age 62, earn less, on average;
7) If employed before turning age 62, are more likely to have a physically demanding job.

With respect to wealth and pension comparisons of age-62 claimers to later claimers, Armour and Knapp find individuals who claim Social Security benefits at 62:

1) Are substantially more likely to already be receiving pension income;
2) Are less likely to have defined contribution plan assets.

The authors caution that even after accounting for the interrelationship of observable demographic, employment, and economic factors using regression analysis, these characteristics account for only a small fraction of the shift toward later claiming. Other reasons-such as longer life expectancy, a greater ability to work longer, better understanding of the financial benefits of waiting to claim Social Security benefits, and stronger incentives to delay claiming due to changes in Social Security benefit rules-likely influence people to wait to claim their benefits.

Finally, although the novel coronavirus pandemic has not yet been associated with increases in claiming Social Security retirement benefits (Goss and Glenn 2020), as short-term furloughs turn into long-term unemployment, the findings in this report suggest that employment losses may lead to earlier claiming-in particular among those with less education and those living in more rural areas. As the second report in this series makes clear, the financial consequences of early claiming can be long lasting.

Armour, P. and Knapp, D. (2021b). *The Consequences of Claiming Social Security Benefits at Age 62*. Tech. rep. AARP Public Policy Institute.

Deciding when to claim Social Security benefits comes with an important tradeoff: claiming benefits at the early eligibility age (EEA) of 62 allows beneficiaries earlier access to their monthly Social Security payments, but these payments will be lower for life than if the beneficiary had waited. Despite the reduction in monthly benefits, almost one-third of people still claim Social Security retirement benefits at the EEA. Delayed claiming is often promoted by financial advisers and retirement experts for its higher monthly benefits (there is an almost 8 percent boost in monthly benefits for each year someone delays claiming). Nevertheless, little is known about whether claiming Social Security benefits at the EEA leads to substantively different short- and long-term outcomes than does claiming later. This report, the second in a two-part series, focuses on the consequences for individuals who claim at the EEA; it estimates both short- and long-term differences between otherwise similar individuals who claim Social Security retirement benefits at age 62 and those who claim later.

KEY FINDINGS In comparing matched age-62 claimants against later claimants with similar characteristics, Armour and Knapp find:

1) Age-62 claimants stop working earlier and cash out defined contribution plans sooner.
2) Age-62 claimants have lower household income throughout their 60s and 70s (on average, $10,000-$20,000 less in annual household income).

3) Age-62 claimants in their 70s have substantially less liquid wealth than do later claimants at similar ages (on average, 27 percent lower at age 74 and 43 percent lower at age 80).
4) Measures of financial hardship and mortality rates are not statistically or substantively different between the two claiming groups into their 70s.

The findings of this report are limited in a couple of significant ways that may underestimate the effects of claiming early. First, given the time frame, HRS data do not allow the authors to track households past age 80, which likely understates the relative advantage of claiming later since the benefits of delayed claiming grow substantially in claimants' 80s. Second, individuals observed in their late 70s claimed in the 1990s or early 2000s-and thus faced a lower penalty for early claiming than more recent cohorts, who were subject to different Social Security benefit rules. In addition, more recent cohorts are living longer; therefore, someone turning 62 today has a stronger incentive to delay claiming, all else equal.

Armour, P. and Knapp, D. (2022). "The consequences of claiming Social Security benefits at age 62." In: *Journal of Pension Economics and Finance*, pp. 1–27.

Delaying claiming of Social Security old-age benefits past the earliest eligibility age, age 62, raises the monthly benefit for a person's life. Despite arguments from both proponents and opponents of delayed claiming in academia and public discourse, little is known about whether claiming decisions lead to substantively different outcomes. We compare differences in outcomes between age-62 claimants and otherwise similar later claimants that are matched on health, employment, and financial characteristics at age 60. We find that age-62 claimers are substantially less likely to work after 62 and have persistently lower income into their 70s. Differences in assets emerged in the 70s, with early claimants having lower wealth, but we find no differences in mortality or self-reported financial hardship. The difference in wealth is driven primarily by a growth in wealth among later claimants rather than substantial decumulation by age-62 claimants.

Armstrong, J., Buescu, C., and Dalby, J. (2024). "Optimal post-retirement investment under longevity risk in collective funds." In: *arXiv e-Print*.

We study the optimal investment problem for a homogeneous collective of n individuals investing in a Black-Scholes model subject to longevity risk with Epstein–Zin preferences. and with preferences given by power utility. We compute analytic formulae for the optimal investment strategy, consumption is in discrete-time and there is no systematic longevity risk. We develop a stylised model of systematic longevity risk in continuous time which allows us to also obtain an analytic solution to the optimal investment problem in this case. We numerically solve the same problem using a continuous-time version of the Cairns–Blake–Dowd model. We apply our results to estimate the potential benefits of pooling longevity risk over purchasing an insurance product such as an annuity, and to estimate the benefits of optimal longevity risk pooling in a small heterogeneous fund.

Arnold, T., Earl, J. H., and Marshall, C. D. (2024). "Required Minimum Distribution (RMD) Spreadsheet Calculators Based on the SECURE Act of 2022." In: *The Journal of Wealth Management* 27(2), pp. 99–106.

The Setting Every Community Up for Retirement Enhancement Act (SECURE Act) of 2022 made a second round of changes (relative to the SECURE Act of 2019) to the required minimum distribution (RMD) schedule for individual retirement accounts (IRAs) and defined contribution retirement plans. Excel spreadsheet calculators are developed to calculate the new annual RMD cash flows throughout retirement for those who are retired and for those who are planning to retire. The spreadsheet calculators also allow savings to accrue with interest if the RMD is in excess of expected annual costs.

Arnold, T., Earl, J. H., Marshall, C. D., and Schwartz, A. (2017). "Excel Calculators for Determining Retirement Accumulation and Disbursement Information." In: *The Journal of Wealth Management* 20(2), pp. 94–101.

This article provides two different retirement calculators to help investors plan for retirement savings and retirement disbursements. The calculators are created in Excel and require minimal programming, which makes them not only practical but also easy to implement. Furthermore, the calculators can address multiple investor scenarios and can be integrated to produce a reasonably thorough retirement plan.

Arnold, T., Earl, J. H., Marshall, C. D., and Schwartz, A. (2018). "Using "Equivalent Tax Rates" to Determine Tax-Efficient Retirement Investment and Withdrawal." In: *The Journal of Wealth Management* 21(2), pp. 55–69.

The authors propose an equivalent tax rate framework to compare the tax-efficient accumulation and withdrawal of retirement funds across multiple investment vehicles. The method allows for the consideration of a given investment vehicle expected return, time horizon, and tax treatment relative to a tax-deferred retirement account receiving pre-tax contributions and possible employer matching of contributions. The metric is comparable across alternative investment accounts irrespective of the specific marginal tax rate in place during the accumulation or distribution stages.

Arnold, T., Earl, J. H., and Nixon, T. D. (2023). "Social Security Decisions: Should Recipients Opt for Early Payments?" In: *The Journal of Wealth Management* 25(4), pp. 8–16.
Two Excel-based templates are developed to help determine when it is optimal for starting to receive monthly social security benefits. The decision information accounts for uncertain life expectancy by implementing a rate of return that should be set, at a minimum, to the individual's expected return on investments or based on a metric provided in the article that considers potential life expectancy.

Arnott, R., Li, F., and Linnainmaa, J. (2024). "Smart Rebalancing." In: *Financial Analysts Journal* 80(2), pp. 26–51.
The sometimes vast gap between live results and paper portfolio performance is caused in part by trading costs, discontinuous trading, and missed trades or other frictions, along with asset management fees. Smart beta and factor strategies are not exempt from this sort of 'implementation shortfall." This paper provides new evidence on the efficacy of prioritizing transactions so as to focus portfolio turnover on the trades that offer the strongest signals and hence the highest potential performance impact. Rebalancing filters of this sort can capture much of the factor premia for a long-only paper portfolio while cutting turnover and trading costs relative to a fully rebalanced portfolio.

Arnott, R. D., Harvey, C. R., and Markowitz, H. (2019). "A backtesting protocol in the era of machine learning." In: *The Journal of Financial Data Science* 1(1), pp. 64–74.
Machine learning offers a set of powerful tools that holds considerable promise for investment management. As with most quantitative applications in finance, the danger of misapplying these techniques can lead to disappointment. One crucial limitation involves data availability. Many of machine learning early successes originated in the physical and biological sciences, in which truly vast amounts of data are available. Machine learning applications often require far more data than are available in finance, which is of particular concern in longer-horizon investing. Hence, choosing the right applications before applying the tools is important. In addition, capital markets reflect the actions of people, which may be influenced by others actions and by the findings of past research. In many ways, the challenges that affect machine learning are merely a continuation of the long-standing issues researchers have always faced in quantitative finance. While investors need to be cautious, more cautious than in past applications of quantitative methods new tools offer many potential applications in finance. In this article, the authors develop a research protocol that pertains both to the application of machine learning techniques and to quantitative finance in general.

Arnott, R. D., Sherrerd, K. F., and Wu, L. (2013). "The Glidepath Illusion ... and Potential Solutions." In: *The Journal of Retirement* 1(2), pp. 13–28.
Target-date investment strategies purport to meet the two primary objectives of any retirement savings program: maximizing the real value of investors' nest eggs while minimizing uncertainty around prospective income in retirement. The authors demonstrate that the classic glidepath approach to retirement investing moving from equity-centric to bond-centric investing as we age does not meet these objectives. The authors summarize the flaws in traditional glidepath implementation and explore illustrative changes to the rules-based, mechanistic solution for retirement planning that can improve the expected outcome for investors, using simulations to test alternatives. Their findings show that, even with simple rules-based approaches, there are better ways to achieve our financial objectives for retirement.

Arshanapalli, B. and Nelson, W. (2012). "Asset Allocation Options for Wealth Accumulation." In: *The Journal of Wealth Management* 14(4), pp. 22–27.
What is the best asset allocation scheme for retirement investing and wealth accumulation? The authors examine seven schemes, including the 110 minus age rule of thumb and strategies employed by popular retirement mutual funds. Historically, the 100 percent equity portfolio has offered the highest returns with little extra risk. However, if the authors lower equity premiums or bootstrap historical results, then retirement mutual funds and the 110 minus age rule of thumb perform somewhat better than 100 percent equity portfolios in adverse circumstances. But overall, a 100 percent equity portfolio provides the highest wealth accumulation.

Assabil, S. E. and Mcleish, D. L. (2021). "Assessing the Impact of Longevity Risk for Countries with Limited Data." In: *The Journal of Retirement* 8(3), pp. 62–75.
The impact of longevity risk has not been well studied in most developing countries due to the lack of suitable mortality data. As a result, most pension companies in these countries (especially those on the African continent) do not account for longevity risk in their annual valuation. This can even lead to their collapse if steps are not taken to address it. In this work, we develop a method of assessing longevity risk where there is a severe lack of mortality data. The method is based on the assumption that there is a nearly linear relationship between annuitant's hazard function and their mortality at higher ages (post-retirement age), which permits

approximating with the Gompertz model. We tried the method on mortality data from Ghana and the results are consistent with those in the standard literature. That is, longevity risk is a treat to pension companies in Ghana even though, in the case of Ghana, this treat has partially been mitigated by the high-interest rate in the country. With this method, pension and life companies that are not able to account for longevity risk as a result of lack of data or newly formed pension companies with even two-year mortality data will be able to assess the longevity risk they face without relying on data or models from other countries.

Atkins, A. B. and Caliendo, F. N. (2009). "Strategies for maximizing social security benefits." In: *The Journal of Wealth Management* 12(1), pp. 25–31.

People nearing retirement face a well-known decision: When should Social Security benefits be initiated?

The answer may not seem obvious since there is a key trade-off involved. The retiree can choose low benefits for a longer period of time, or high benefits for a shorter period of time, or something in between. The optimal initiation date that maximizes Social Security wealth is a quantitative question and it depends on the life expectancy of the person and on the real rate of return they expect to earn on their investments, among other things.

The authors attempt to provide some answers to this practical and important question. Their focus throughout is on relatively high-wealth individuals who will receive the maximum Social Security benefits. They show how the government Social Security benefit calculator makes recommendations on the optimal age to initiate benefits be inappropriate for high-wealth individuals.

They conclude with the following rough rules of thumb

Take benefits early (i.e., at age 62): If an individual expects to make a good return on their investments. This is true regardless of their life expectancy.

Take benefits later (i.e., at age 70): If an individual expects to make modest investment returns and they expect to live a long time.

Atra, R. J. and Pae, Y. (2023). "Should Glidepaths Be Gender-Specific?" In: *The Journal of Wealth Management* 26(2), pp. 52–66.

Predetermined glidepaths are useful tools for asset allocation and for limiting behavioral biases, but do not account for all the characteristics of an investor, leading to highly varied failure rates between men and women. Our bootstrapping analysis indicates that income levels and earnings patterns, along with life expectancy, have the largest contributions to the differences in failure rates with other issues, such as Social Security, having more modest impacts. The results also suggest that aggressive allocations on the part of men may be a rational attempt to achieve retirement failure rates comparable to women.

Augustyniak, M., Badescu, A., and Bégin, J.-F. (2023). "A discrete-time hedging framework with multiple factors and fat tails: On what matters." In: *Journal of Econometrics* 232(2), pp. 416–444.

This article presents a quadratic hedging framework for a general class of discrete-time affine multi-factor models and investigates the extent to which multi-component volatility factors, fat tails, and a non-monotonic pricing kernel can improve the hedging performance. A semi-explicit hedging formula is derived for our general framework which applies to a myriad of the option pricing models proposed in the discrete-time literature. We conduct an extensive empirical study of the impact of modelling features on the hedging effectiveness of S&P 500 options. Overall, we find that fat tails can be credited for half of the hedging improvement observed, while a second volatility factor and a non-monotonic pricing kernel each contribute to a quarter of this improvement. Moreover, our study indicates that the added value of these features for hedging is different than for pricing. A robustness analysis shows that a similar conclusion can be reached when considering the Dow Jones Industrial Average. Finally, the use of a hedging-based loss function in the estimation process is investigated in an additional robustness test, and this choice has a rather marginal impact on hedging performance.

Avanzi, B. and De Felice, L. (2024). "Optimal strategies for the decumulation of retirement savings under differing appetites for liquidity and investment risks." In: *Decisions in Economics and Finance*.

A retiree's appetite for risk is a common input into the lifetime utility models that are traditionally used to find optimal strategies for the decumulation of retirement savings. In this work, we consider a retiree with potentially differing appetites for the key financial risks of decumulation: liquidity risk and investment risk. We set out to determine whether these differing risk appetites have a significant impact on the retiree's optimal choice of decumulation strategy. To do so, we design and implement a framework which selects the optimal decumulation strategy from a general set of admissible strategies in line with a retiree's goals, and under differing appetites for the key risks of decumulation. Overall, we find significant evidence to suggest that a retiree's differing appetites for different decumulation risks will impact their optimal choice of strategy at retirement. Through an illustrative example calibrated to the Australian context, we find results which are consistent with actual behaviours in this jurisdiction (in particular, a shallow market for annuities), which lends support to our framework and may provide some new insight into the so-called annuity puzzle. The Python script that was used to obtain the results in Sect. 7.1 is available on https://github.com/agi-lab/decumulation-risk. The run-time is around 80 h using a 2.6 GHz 6-Core Intel i7 processor.

Avanzi, B. and Felice, L. de (2023). "Optimal Strategies for the Decumulation of Retirement Savings under Differing Appetites for Liquidity and Investment Risks." In: *arXiv e-Print*.

A retiree's appetite for risk is a common input into the lifetime utility models that are traditionally used to find optimal strategies for the decumulation of retirement savings. In this work, we consider a retiree with potentially differing appetites for the key financial risks of decumulation. We set out to determine whether these differing risk appetites have a significant impact on the retiree's optimal choice of strategy. To do so, we design and implement a framework which selects the optimal decumulation strategy from a general set of admissible strategies in line with a retiree's goals, and under differing appetites for the key risks of decumulation. Overall, we find significant evidence to suggest that a retiree's differing appetites for different decumulation risks will impact their optimal choice of strategy at retirement. Through an illustrative example calibrated to the Australian context, we find results which are consistent with actual behaviours in this jurisdiction (in particular, a shallow market for annuities), which lends support to our framework and may provide some new insight into the so-called annuity puzzle.

Aw, E. N. W., Jiang, J. Q., Sivin, G. Y., and Xie, H. (2024). "Direct Indexing Myths and Facts." In: *The Journal of Beta Investment Strategies* 15(3), pp. 10–19.

Although many investors are mindful of mutual funds and ETFs as investment instruments for indexing strategy, few are familiar with direct indexing strategies designed to replicate any stock market index using a separately managed account (SMA) format. Historically, the SMA format required significant computations as each client account required individual management. Recent advancements in computational power, however, allow for the growth of direct indexing, providing clients with customizability, not only to take the benefit of tax-loss harvesting but also to implement other tailored investment objectives. The results provided in this article suggest that there exist benefits associated with tax-loss harvesting even though tax-loss harvesting defers but does not discharge tax obligations. Furthermore, the article also reveals that maintaining a reasonable ex-post tracking error to achieve the indexing goal while obtaining a positive tax-loss benefit is feasible.

Azman, S. and Pathmanathan, D. (2022). "The GLM framework of the Lee-Carter model: a multi-country study." In: *Journal of Applied Statistics* 49(3), pp. 1752-763–12.

The Lee-Carter model is a well-known model in modeling mortality. We aim to compare three probability models (Poisson, negative binomial and binomial) based on the Generalized Linear Model (GLM) framework of the Lee-Carter model. These models are applied to mortality data for 10 selected countries (Japan, United States, United Kingdom, Australia, Sweden, Spain, Belgium, Canada, Netherlands and Bulgaria) and the fit of these models is assessed using the deviance statistics and standardized residuals against fitted value plot. Among these three models, the negative binomial Lee-Carter model gave the best fit based on the deviance statistics and estimates of the log of deaths.

Bachmann, K., Hens, T., and Stossel, R. (2014). "Designing A Risk Profiler: Which Measures Predict Risk Taking?" In: *SSRN e-Print*.

In this paper we assess the suitability of different risk profiling measures when individuals are involved in a process of discovering their willingness to take risks over different decision modes. The latter involve decisions under ambiguity, decisions after gaining experience and receiving outcome information on previous decisions. We find that risk taking is associated with individuals' risk preferences in all decision modes but not with their investment experience. Although simulated experience improves the risk awareness and supports a higher risk taking, it cannot substitute the assessment of risk preferences and in particular the assessment of individual's loss aversion. In contrast, self-assessed risk tolerance measures are not suitable for predicting risk taking in any decision mode. If risk preferences cannot be assessed, only the gender can be used as a predictor of risk taking.

Bacinello, A. R., Maggistro, R., and Zoccolan, I. (2024). "Risk-neutral valuation of GLWB riders in variable annuities." In: *Insurance: Mathematics and Economics* 114, pp. 1–14.

In this paper we propose a model for pricing GLWB variable annuities under a stochastic mortality framework. Our set-up is very general and only requires the Markovian property for the mortality intensity and the asset price processes. The contract value is defined through an optimization problem which is solved by using dynamic programming. We prove, by backward induction, the validity of the bang-bang condition for the set of discrete withdrawal strategies of the model. This result is particularly remarkable as in the insurance literature either the existence of optimal bang-bang controls is assumed or it requires suitable conditions. We assume constant interest rates, although our results still hold in the case of a Markovian interest rate process. We present extensive numerical examples, modelling the mortality intensity as a non mean reverting square root process and the asset price as an exponential Levy process, and compare the results obtained for different parameters and policyholder behaviours.

Bae, S., Lee, Y., Kim, W. C., Kim, J. H., and Fabozzi, F. J. (2024). "Goal-based investing with goal postponement: multistage stochastic mixed-integer programming approach." In: *Annals of Operations Research*.
This paper introduces a multistage stochastic mixed-integer programming model designed for a goal-based investing (GBI) problem, incorporating the option of goal postponement. Our model allows individuals to defer the fulfillment of their goals within a predefined timeframe. We emphasize the advantages of incorporating goal postponement into the GBI framework, including its ability to accommodate stage-preference ambiguity, address mistiming issues, and enhance utility for individuals. Theoretical results of a GBI problem with goal postponement are presented, and to tackle large-scale multistage GBI problems, we employ a decomposition algorithm known as stochastic dual dynamic integer programming (SDDiP). Numerical results demonstrate that the option to postpone a goal proves especially advantageous when goals are exposed to high inflation rates, and SDDiP emerges as a computationally efficient approach for handling large-scale GBI problems.

Bai, Z. and Wallbaum, K. (2020). "Optimizing Pension Outcomes Using Target-Driven Investment Strategies: Evidence from Three Asian Countries with the Highest Old-Age Dependency Ratio." In: *Asia-Pacific Journal of Financial Studies* 49(4), pp. 652–682.
As a response to unforeseeable market turbulence – such as the 2008 financial crisis and the most recent market drawdown triggered by the COVID-19 pandemic – we propose a new pension investment strategy that could better protect a long-term pension plan in volatile market conditions. Over a hypothetical 20-year pension scheme and various target volatility scenarios, we show that our newly proposed strategy, which attaches a target volatility mechanism to a lifecycle strategy, could provide more effective capital protection and risk control for pension investment vehicles. Our results are robust with a consideration of transaction costs.

Bailey, D. H., Borwein, J. M., and Lopez de Prado, M. (2017). "Stock Portfolio Design and Backtest Overfitting." In: *Journal of Investment Management* 15(1), pp. 75–87.
In mathematical finance, backtest overfitting connotes the usage of historical market data to develop an investment strategy, where too many variations of the strategy are tried, relative to the amount of data available. Backtest overfitting is now thought to be a primary reason why investment models and strategies that look good on paper often disappoint in practice. Models and strategies suffering from overfitting typically target the specific idiosyncrasies of a limited dataset, rather than any general behavior, and, as a result, often perform erratically when presented with new data. In this study, we address overfitting in the context of designing a mutual fund or investment portfolio as a weighted collection of stocks. Very often a newly minted equity-based fund of this type has been designed by an exhaustive computer-based search of some sort to obtain an optimal weighting that exhibits excellent performance based, say, on the past 10 or 20 years' historical market data, and the fund often highlights this backtest performance.

Bailey, R., DeShetler, A., Leming, J., Weber, S. M., Youssef, J., and Young, J. A. (2021). *Planning for health care costs in retirement*. Tech. rep. Vanguard.
Vanguard engaged Mercer Health & Benefits to develop a model to forecast health care costs for U.S. retirees. Vanguard believes retirement planning frameworks should be adapted as follows: 1) Planning for annual health care insurance premiums and out-of-pocket expenses at retirement should be distinct from planning for long-term care expenses. 2) Some research estimates health care costs in retirement as a lifetime lump sum. We believe a better planning framework considers these costs as annual expenses personalized to an individual's health status, coverage choices, retirement age, and loss of any employer subsidies. For a typical 65-year-old woman, the Mercer-Vanguard model predicted an annual health care expense of $5,100 for 2020. 3) During their working years, some people should save at higher rates to account for potential future incremental health care spending. This group includes those with generous employer benefits that may not be offered in retirement and those with higher risk of chronic conditions because of family history or current health status 4) Long-term care costs represent a separate and difficult planning challenge because of the wide distribution of potential outcomes. Half of the population will never incur them-but everyone faces a small but meaningful risk of requiring costly care for multiple years.

Bairoliya, N. and McKiernan, K. (2021). "Revisiting Retirement and Social Security Claiming Decisions." In: *SSRN e-Print*.
Why do individuals retire and claim their Social Security benefits at the age they do? Understanding the key drivers of these decisions has been an important topic of research as it can help guide policy discussions on the impact of potential reforms to the Social Security program. We revisit this crucial question by exploring new sources of heterogeneity in these decisions as well as novel mechanisms governing these trade-offs. Using data from the Health and Retirement Study and the Understanding America Survey, we first document (1) important heterogeneities in social security claiming behavior of men by their education and marital status, (2) strong correlations between health, labor supply and benefit claiming decisions and (3) significant misinformation related to Social Security program knowledge and survival chances at older ages. We then build a life-cycle model of consumption, savings, labor supply, and Social Security application decisions as well as heterogeneity in education, marital status and SS program knowledge. The model includes uncertainty in health, subjective survival, wages, and job separation as well as rich details of the U.S. Social Security program to understand why a majority of individuals claim Social Security benefits prior to their normal retirement age, despite large penalties associated with these early benefit claims. We show that the estimated model can closely match the claiming behavior as seen in the data and also produce differences in SS claims along the dimensions of heterogeneity considered. Counterfactual experiments indicate that precautionary motives, misinformation, and preferences governing future discounting as well as altruism, together, go a long way in explaining overall claiming behavior. Together, these forces can explain a third of the overall early benefit claims and two-thirds of age 62 claims-with varying intensities across education and marital groups.

Banerjee, S. (2019). *Asset Decumulation or Asset Preservation? What Guides Retirement Spending?* Tech. rep. Employee Benefit Research Institute EBRI.
The Employee Benefit Research Institute (EBRI) undertook a study examining the extent to which the non-housing assets of certain retirees changed during their first 20 years of retirement (or until death, if earlier).
The study relied on income and asset data from the Health and Retirement Study (HRS), the most comprehensive survey of older Americans in the country, and on spending data from the Consumption and Activities Mail Survey (CAMS), a supplement to HRS.
Highlights:

1) The study shows that retirees generally exhibit very slow decumulation of assets.
2) More specifically, within the first 18 years of retirement, individuals with less than $200,000 in non-housing assets immediately before retirement had spent down (at the median) about one-quarter of their assets; those with between $200,000 and $500,000 immediately before retirement had spent down 27.2 percent. Retirees with at least $500,000 immediately before retirement had spent down only 11.8 percent within the first 20 years of retirement at the median.
3) While some retirees do spend down most of their assets in the first eighteen years following retirement, about one-third of all sampled retirees had increased their assets over that period.
4) Pensioners were much less likely to have spent down their assets than non-pensioners. During the first 18 years of retirement, the median non-housing assets of pensioners (who started retirement with much higher levels of assets) had gone down only 4 percent, compared to 34 percent for non-pensioners.
5) The median ratio of household spending to household income for retirees of all ages hovered around one, inching slowly upward with age. This suggests that majority of retirees had limited their spending to their regular flow of income and had avoided drawing down assets, which explains why pensioners, who had higher levels of regular income, were able to avoid asset drawdowns better than others

All numbers are measured in 2015 dollars.

Banks, J. and Crawford, R. (2022). "Managing Retirement Incomes." In: *Annual Review of Economics* 14(1), pp. 181–204.
In this article we discuss the state of the literature relating to the decumulation of retirement wealth and the management of retirement incomes. On the one hand, life-cycle models that allow for strong bequest motives and for the effects of medical expense risks have been shown to be able to rationalize retirees' wealth, income, and consumption trajectories. On the other, studies of individual asset choices and portfolio decisions seem to suggest low levels of financial literacy and engagement as well as non-negligible consequences of age-related cognitive decline on financial decision making. We argue that future work should try to reconcile these two sets of conflicting findings into a coherent and holistic evidence base to inform policy, because issues concerning

the management of retirement incomes, and insurance against different risks in retirement more generally, will become increasingly important for future cohorts of retirees.

Banos, D. R., Ortiz-Latorre, S., and Font, O. Z. (2024). "A functional variational approach to pricing path dependent insurance policies." In: *arXiv e-Print*.

The main purpose of this work is the derivation of a functional partial differential equation (FPDE) for the calculations of equity-linked insurance policies, where the payment stream may depend on the whole past history of the financial asset. To this end, we employ variational techniques from the theory of functional Itô calculus.

Banyai, A., Tatay, T., Thalmeiner, G., and Pataki, L. (2024). "The Impact of Rebalancing Strategies on ETF Portfolio Performance." In: *Journal of Risk and Financial Management* 17(12), p. 533.

This research explores the efficacy of rebalancing strategies in a diversified portfolio constructed exclusively with exchange-traded funds (ETFs). We selected five ETF types: short-term U.S. Treasury bonds, U.S. equities, global commodities, U.S. real estate investment trusts (REITs), and a multi-strategy hedge fund. Using a 10-year historical period, we applied a unique simulation model to generate random portfolios with varying asset weights and rebalancing tolerance bands, assessing the impact of rebalancing premiums on portfolio performance. Our study reveals a significant positive correlation ($r = 0.6492$, $p < 0.001$) between rebalancing-weighted returns and the Sharpe ratio, indicating that effective rebalancing enhances risk-adjusted returns. Support vector regression (SVR) analysis shows that rebalancing premiums have diverse effects. Specifically, equities and commodities benefit from rebalancing with improved risk-adjusted returns, while bonds and REITs demonstrate a negative relationship, suggesting that rebalancing might be less effective or even detrimental for these assets. Our findings also indicate that negative portfolio rebalancing returns combined with positive rebalancing-weighted returns yield the highest average Sharpe ratio of 0.4328, highlighting a distinct and reciprocal relationship between rebalancing effects at the asset and portfolio levels. This research highlights that while rebalancing can enhance portfolio performance, its effectiveness varies by asset class and market conditions.

Bar, M. and Gatzert, N. (2022). "Products and Strategies for the Decumulation of Wealth during Retirement: Insights from the Literature." In: *North American Actuarial Journal* 27(2), pp. 322–340.

The question how individuals should (optimally) annuitize their wealth remains of high relevance in light of longevity risk and volatile capital markets. In this article, we first present traditional and innovative products and strategies for the decumulation of wealth during retirement, based on a review of 72 selected academic articles in peer-reviewed journals. We further identify relevant factors that generally influence the conception of these products from the retirees' perspectives, and derive implications for product developers, before concluding with avenues of future research. Our results indicate that innovative suggestions often comprise tontine-like structures, exploit actuarial and accounting smoothing in various ways, defer annuitization to higher ages, or combine it with long-term care options, for instance. Key areas of future research in this field include the consideration of both insurer and retiree perspectives in the analysis of products, using behavioral considerations when evaluating the retirees' perspective, and taking into account the impact of costs or expenses. While recent articles increasingly consider these aspects, manifold opportunities for future research remain.

Barro, D., Canestrelli, E., and Consigli, G. (2019). "Volatility versus downside risk: performance protection in dynamic portfolio strategies." In: *Computational Management Science* 16(3), pp. 433–479.

Volatility-based and volatility targeting approaches have become popular among equity fund managers after the introduction in 1993 of the VIX, the implied volatility index on the S&P500 at the Chicago Board of Exchange (CBOE), followed, in 2004, by futures and option contracts on the VIX: since then we have assisted to an increasing interest in risk control strategies based on market signals. In January 2000 also the FTSE implied volatility index (FTSEIVI) was introduced at the London Stock Exchange. As a result, specifically in the US, portfolio strategies based on combinations of market indices and derivatives have been proposed by Stock Exchanges and investment banks: one such example is the S&P500 protective put index (PPUT). Early in 2016, relevant to the definition of optimal bond-equity strategies, CBOE launched an Index called TYVIX/VIX featuring an investment rotation strategy based jointly on signals coming from the VIX and the 10-year Treasury Yield implied volatility (TYVIX). All these are rule-based portfolio strategies in which no optimization methods are involved. While rather effective in reducing the downside risk, those index-based portfolio approaches do not allow an optimal risk-reward trade-off and may not be sufficient to control financial risk originated by extreme market drops. To overcome these limits we propose an optimization-based approach to portfolio management jointly focusing on volatility and tail risk controls and able to accommodate effectively the return payoffs associated with option strategies, whose cost as market volatility increases may become excessive. The model is based on a mean absolute deviation formulation and tested in the US equity market over the 2000-2016 period and with a focus on three periods of high volatility, in 2000, 2001 and 2008. The results confirm that optimal volatility controls produce better risk-adjusted returns if compared with rule-based approaches. Moreover the portfolio return distribution is dynamically shaped depending on the adopted risk management approach.

Basu, A. K., Byrne, A., and Drew, M. E. (2011). "Dynamic Lifecycle Strategies for Target Date Retirement Funds." In: *The Journal of Portfolio Management* 37(2), pp. 83–96.
Lifecycle funds offered to retirement plan participants gradually reduce exposure to stocks as the funds approach the target date of the participants' retirement.The authors show that such deterministic switching rules produce inferior wealth outcomes for the investor compared to strategies that dynamically alter the allocation between growth and conservative assets based on cumulative portfolio performance relative to a set target.The dynamic allocation strategies proposed in this article exhibit almost stochastic dominance over strategies that unidirectionally switch assets without consideration of portfolio performance.

Basu, A. K. and Drew, M. E. (2014). "The value of tail risk hedging in defined contribution plans: what does history tell us." In: *Journal of Pension Economics and Finance* 14(03), pp. 240–265.
Hedging against tail events in equity markets has been forcefully advocated in the aftermath of recent global financial crisis. Whether this is beneficial to long horizon investors like employees enrolled in defined contribution (DC) plans, however, has been subject to criticism. We conduct historical simulation since 1928 to examine the effectiveness of active and passive tail risk hedging using out of money put options for hypothetical equity portfolios of DC plan participants with 20 years to retirement. Our findings show that the cost of tail hedging exceeds the benefits for a majority of the plan participants during the sample period. However, for a significant number of simulations, hedging result in superior outcomes relative to an unhedged position. Active tail hedging is more effective when employees confront several panic-driven periods characterized by short and sharp market swings in the equity markets over the investment horizon. Passive hedging, on the other hand, proves beneficial when they encounter an extremely rare event like the Great Depression as equity markets go into deep and prolonged decline.

Basu, A. K. and Wiafe, O. K. (2017). "Impact of persistent bad returns and volatility on retirement outcomes." In: *Finance Research Letters* 21, pp. 201–205.
We examine wealth outcomes and risk of ruin faced by retirees due to persistent bad returns and high volatility in equity markets occurring at different stages of their retirement. Our results show poor equity returns persisting over long periods can put retirement security to serious risk but volatile market conditions actually have the opposite impact. The timing of such persistent bad returns and volatility (early or late stages of retirement) is critical and has differing effects on retirement outcomes. The results are robust to varying portfolio allocations to equities although the precise impacts are different.

Bateman, H., Stevens, R., and Lai, A. (2015). "Risk Information and Retirement Investment Choice Mistakes Under Prospect Theory." In: *Journal of Behavioral Finance* 16(4), pp. 279–296.
We assess alternative presentations of investment risk using a discrete choice experiment which asked subjects to rank three investment portfolios for retirement savings across nine risk presentation formats and four underlying risk levels. Using Prospect Theory utility specifications, we estimate individual-specific parameters for risk preferences in gains and losses, loss aversion, and error propensity variability. Our results support presentations that describe investment risk using probability tails. Risk preferences and error propensity were found to vary significantly across sociodemographic groups and levels of financial literacy. Our findings should assist regulatory efforts to disclose risk information to the mass market.

Bauman, T., Gasperov, B., Begusic, S., and Kostanjcar, Z. (2023). "Deep Reinforcement Learning for Robust Goal-Based Wealth Management." In: *arXiv e-Print*, pp. 69–80.
Goal-based investing is an approach to wealth management that prioritizes achieving specific financial goals. It is naturally formulated as a sequential decision-making problem as it requires choosing the appropriate investment until a goal is achieved. Consequently, reinforcement learning, a machine learning technique appropriate for sequential decision-making, offers a promising path for optimizing these investment strategies. In this paper, a novel approach for robust goal-based wealth management based on deep reinforcement learning is proposed. The experimental results indicate its superiority over several goal-based wealth management benchmarks on both simulated and historical market data.

Baynes, L. and Renzi-Ricci, G. (2022). "Exploring the equity-bond relationship in a low-rate environment with unsupervised learning." In: *Journal of Investment Strategies* 11(2), pp. 1–10.
Some investors have become concerned about the low-interest-rate environment and its impact on the role that bonds play in a multi-asset portfolio. In order to analyze the equity hedging property of government bonds, we

apply a simple but powerful machine learning technique called k-means clustering to periods with low interest rates. Our findings show that government bonds have historically acted as intended in an equity-bond portfolio, with typically positive bond performance in equity-down scenarios. Although there are some periods in which both equities and bonds fall, these can be viewed as ordinary parts of market volatility and distinct from the typical outcomes that can be considered as recurrent market states. The results of this alternative and complementary approach - which has not, to the best of the authors' knowledge, been used before to study equity-bond diversification benefits - supplement the existing literature by providing further evidence of the added value that bonds bring to a strategic multi-asset portfolio.

Beattie, K. (2022). "Longevity Risk: Plan for the Unknown." In: *Investments and Wealth Monitor*.

Life expectancy is underestimated. Forty-three percent of retirees underestimate their own life expectancy by at least five years.Underestimating life expectancy, combined with having too short a planning horizon, can result in inadequate income later in retirement. Pensions, Social Security, and annuities are uniquely suited to addressing and combating longevity risk. It is critical we help investors understand a more complete view of their life expectancy probabili- ties and the implications for retirement income planning.

Bell, D. N. F., Comerford, D. A., and Douglas, E. (2020). "How do subjective life expectancies compare with mortality tables? Similarities and differences in three national samples." In: *The Journal of the Economics of Ageing* 16 (100241).

Estimates of personal longevity play a vital role in decisions relating to asset accumulation and decumulation. Subjective life expectancy (SLE) is a measure of individuals' expectation of remaining years of life. Either explicitly or implicitly, it is a key determinant of consumption and savings behaviour, and may be guided by a person's own health and health behaviours. The Gateway to Global Aging, a platform for the Health and Retirement Study's (HRS) family of population surveys, provides harmonised longitudinal datasets for many countries, each based on individual survey responses from respondents aged 50 and above. In this paper, we analyse SLE three of these datasets: the English Longitudinal Study of Ageing (ELSA), The Irish Longitudinal Study of Ageing (TILDA) and Healthy Ageing in Scotland (HAGIS). First, we focus on measurement of SLE, followed by the SLE differential - the discrepancy between SLE and mortality risk indicated by population life tables. One novel finding from our analysis is that the SLE differential is positive for Ireland and is negative for Scotland and England. This difference does not appear to be explained by differences of survey design or population characteristics.

Belzile, L. R., Davison, A. C., Gampe, J., Rootzén, H., and Zholud, D. (2022). "Is There a Cap on Longevity? A Statistical Review." In: *Annual Review of Statistics and Its Application* 9(1).

There is sustained and widespread interest in understanding the limit, if there is any, to the human life span. Apart from its intrinsic and biological interest, changes in survival in old age have implications for the sustainability of social security systems. A central question is whether the endpoint of the underlying lifetime distribution is finite. Recent analyses of data on the oldest human lifetimes have led to competing claims about survival and to some controversy, due in part to incorrect statistical analysis. This article discusses the particularities of such data, outlines correct ways of handling them, and presents suitable models and methods for their analysis. We provide a critical assessment of some earlier work and illustrate the ideas through reanalysis of semisupercentenarian lifetime data. Our analysis suggests that remaining life length after age 109 is exponentially distributed and that any upper limit lies well beyond the highest lifetime yet reliably recorded. Lower limits to 95% confidence intervals for the human life span are about 130 years, and point estimates typically indicate no upper limit at all.

Bengen, W. P. (2021). "The Planner's Toolkit for Managing Retirement Withdrawal Plans." In: *Journal of Financial Planning* 34(4), pp. 74–80.

Since 1993, I've been researching sustainable withdrawals from retirement portfolios and searching for something: a complete, soup-to-nuts approach to managing the withdrawal process-from construction of the original withdrawal plan, through monitoring the plan and identifying deviations, to applying corrections, as needed. I present in this article my version of the financial planner's complete toolkit for managing withdrawal plans. As with any area of financial planning, you can't launch the initial plan and expect it to be appropriate for the client forever. A withdrawal plan needs to be actively managed to determine if the plan no longer suits a client's needs, and to make appropriate changes to bring the plan back into focus.

In a way, this article is more of an overview rather than a complete delineation of the process. My approach relies heavily on graphical elements: dozens of charts are required as part of the toolkit. Furthermore, owing to the large number of variables involved, withdrawal planning is imbued with an inherent complexity. Accordingly,

there simply isn't room in an article of this length to provide you with the level of detail demanded for a full exposition of my methods. So, consider this a test drive and please wait for the production models to be released in the near future.

I'll proceed in three steps:

1) Establishing the withdrawal plan (determining, among other variables, the withdrawal scheme and initial withdrawal rate).
2) Monitoring the plan for deviations.
3) Making corrections to the plan, as required (or as desired by the client).

These three steps may be repeated several times during the life of the client's plan.

Benhamou, E., Ohana, J.-J., and Guez, B. (2024). "GRIP: Graphical Models Revealing Insights for Portfolio Replication - A Learning Approach." In: *SSRN e-Print*.

This paper presents a new and effective methodology for decoding strategies in the context of investment portfolios. The proposed approach relies on Dynamic Bayesian Graphical Models, which are powerful tools for capturing complex relationships and dependencies in data over time. Using these models, we can accurately decode the hidden strategy within the investment universe. By leveraging Dynamic Bayesian Graphical Models, we calculate dynamic weights that exhibit the most stable allocation rules. Through extensive experimentation on various investment scenarios, we demonstrate that our approach achieves high accuracy in decoding strategies. The method's reliance on Dynamic Bayesian Graphical Models enables it to effectively uncover hidden patterns and relationships within the investment data, leading to improved portfolio allocation decisions and robust generalization across different market conditions.

Benz, C. (2021). *When Is the 'Right' Time to File for Social Security Benefits?* Tech. rep. Morningstar.

Author Mike Piper discusses when it makes sense to claim Social Security early and when delaying is advisable. Many consumers have heard the advice to delay claiming Social Security until later in life–past age 62 and perhaps up until age 70–with the goal of claiming a higher lifetime benefit. But is delaying Social Security always advisable? Not necessarily. In a recent episode of The Long View podcast, author Mike Piper shared instances when claiming earlier is advisable, as well as when claiming benefits later makes sense. He also discussed the free tool he created to help with Social Security claiming decisions, Open Social Security. Piper writes the popular Oblivious Investor blog, and he is also the author of several books about Social Security, taxes, and retirement, including Can I Retire?, Social Security Made Simple, Investing Made Simple, and Taxes Made Simple.

Beracha, E., Downs, D. H., and MacKinnon, G. (2017). "The 4% rule: Does real estate make a difference?" In: *Journal of Property Research* 34(3), pp. 181–210.

This paper examines the wealth maximisation and preservation effects of including commercial real estate in retirement-phase portfolio management. Prior research addresses the role of real estate during the wealth-accumulation phase of the investor lifecycle; however, little is known about the contribution of real estate during the invest-and-spend, or decumulation, phase. To address this issue, we estimate short-fall risk based on the widely known 4% Rule. We use pricing data for multiple asset classes and simulation techniques, combined with a robust correlation structure, to examine: short-fall risk sensitivity to alternative spending rules; the impact of public vs. private real estate allocations; wealth preservation as an investment objective; and the effect of real estate on upside, or wealth maximisation, potential. We find short-fall risk in a decumulation portfolio decreases with substantial allocations to real estate. This result holds for a portfolio including either public or private real estate. Additionally, and under most conditions, the best performing decumulation-phase portfolios include a real estate allocation with both public and private real estate exposure. These results have significant implications for investors, whether they be retirees, plan administrators or endowments, as well as financial economists studying the lifecycle of investment decisions.

Bernard, C., De Vecchi, C., and Vanduffel, S. (2024). "Robust assessment of life insurance products." In: *Annals of Operations Research*.

In this paper we propose a robust assessment for the premium of a standard life insurance contract with respect to the uncertainty on the estimated residual lifetime distribution function. Specifically, we provide a method to derive the range of values that the premium of a given contract can attain when considering all residual lifetime distribution functions that satisfy an L2 distance constraint to a reference distribution function. Furthermore, we show that the L2 distance constraint can be used as flexible starting point to include further information regarding future mortality.

Bernhardt, T. and Donnelly, C. (2018). *Pension Decumulation Strategies: A State-of-the-Art Report*. Tech. rep. Risk Insight Lab, Heriot-Watt University.
This report is focused on income withdrawal and investment strategies that are proposed for decumulation in retirement. Popular strategies which are used in practice and specifications of how to decide the income and investment strategy are reviewed.
One of the most popular rules, the SWR or "4% rule", is very simple and easy to implement. The retiree invests in a constant-mix strategy and withdraws an inflation-indexed income for 30 years. There is intended to be a 90% chance of the strategy being successful.
However the constant-mix investment strategy is criticized by Scott et al. (2009) as being too expensive. In many possible future scenarios, there is an excess of assets at the end of 30 years. That means that the retiree has followed a wasteful strategy. They have paid for surpluses in excess of their stated goal. Moreover, there is a chance that the retiree could exhaust their funds before 30 years are up. Scott et al. (2009) show that options or a different dynamic strategy could be employed to give either better outcomes for the same cost or the same outcomes for a lower cost.
As shown repeatedly in the literature, optimal investment strategies are dynamic strategies. They are usually not constant-mix strategies, although their exact dynamics depend on the optimization problem. In contrast, many drawdown products in the market are constant-mix strategies. Optimal withdrawal rates, when allowed to be a decision variable, also vary over time. How much they vary will depend on the problem specification. However, if the problem set-up allows for risk aversion, then the withdrawal rate will vary less as the investor becomes more risk averse.
Annuities or modern tontines should be part of any decumulation strategy, particular at older ages. They are a cost-efficient strategy which either eliminate or significantly reduce the chance of running out of money. In general, the annuity or modern tontine should be incorporated into the retiree's assets when their mortality is the main driver of the decumulation strategy. This is the case when buying these products has a higher value than trading solely in the financial market.
There are many approaches to specifying the decumulation problem. Utility theory-based methods, such as Expected Utility Theory and Cumulative Prospect Theory, allow for investor risk aversion. This latter property can give intuitively appealing results. However, the problem specification is not easy to communicate.
Minimizing the difference between the actual consumption and a desired consumption via the minimization of a quadratic function has also been studied. Closed-form solutions are difficult to obtain.
Probabilistic methods, such as minimizing the probability of ruin, appear more objective. However, as they don't allow for investor risk aversion, the results can potentially be unattractive. The problems have to be carefully specified in order to obtain an agreeable solution.
Habit formation tries to match actual consumption behaviour closely. The investor seeks to keep their consumption above a weighted average of their past consumption. It may be difficult to justify such a formulation to the investor. Similarly, a problem specification (confusingly called drawdown) where the probability of the investor's wealth falling below some fraction of their running maximum wealth is minimized, may also be hard to explain as a decumulation goal.

Bessembinder, H., Chen, T.-F., Choi, G., and Wei, K. C. J. (2025). "How Should Investors' Long-Term Returns Be Measured?" In: *Financial Analysts Journal*, pp. 1–30.
We assess measures of long-horizon investment outcomes and clarify underlying trading strategy interpretations. We focus attention on a measure we call the "sustainable return," defined as the rate of periodic withdrawal for consumption consistent with the preservation of real capital. We use this notion to highlight the role of return sequence risk, which is distinct from risk in the overall level of returns. We illustrate this and several other long-horizon measures in a global stock sample, emphasizing limitations of the arithmetic and geometric means of short-interval returns and the necessity in many contexts to consider the reinvestment of interim cash flows.

Bessler, W., Opfer, H., and Wolff, D. (2017). "Multi-asset portfolio optimization and out-of-sample performance: an evaluation of Black Litterman, mean-variance, and naive diversification approaches." In: *The European Journal of Finance* 23(1), pp. 1–30.
The Black Litterman model aims to enhance asset allocation decisions by overcoming the problems of mean-variance portfolio optimization. We propose a sample-based version of the Black Litterman model and implement it on a multi-asset portfolio consisting of global stocks, bonds, and commodity indices, covering the period from January 1993 to December 2011. We test its out-of-sample performance relative to other asset allocation models and find that Black Litterman optimized portfolios significantly outperform na ive-diversified portfolios (1/N

rule and strategic weights), and consistently perform better than mean-variance, Bayes Stein, and minimum-variance strategies in terms of out-of-sample Sharpe ratios, even after controlling for different levels of risk aversion, investment constraints, and transaction costs. The BL model generates portfolios with lower risk, less extreme asset allocations, and higher diversification across asset classes. Sensitivity analyses indicate that these advantages are due to more stable mixed return estimates that incorporate the reliability of return predictions, smaller estimation errors, and lower turnover.

Bessler, W. and Wolff, D. (2017). "Portfolio Optimization with Industry Return Prediction Models." In: *SSRN e-Print*.

We postulate that utilizing return prediction models with fundamental, macroeconomic, and technical indicators instead of using historical averages should result in superior asset allocation decisions. We investigate the predictive power of individual variables for forecasting industry returns in-sample and out-of-sample and then analyze multivariate predictive regression models including OLS, a regularization technique, principal components, a target-relevant latent factor approach, and forecast combinations. The gains from using industry return predictions are evaluated in an out-of-sample Black-Litterman portfolio optimization framework. We provide empirical evidence that portfolio optimization utilizing industry return prediction models significantly outperform portfolios using historical averages and those being passively managed.

Betzuen Alvarez, A. J. and Betzuen Zalbidegoitia, A. (2024). "The Role of Sex in the Assessment of Return and Downside Risk in Decumulation Financial Planning." In: *Risks* 12(9), p. 142.

This paper aims to assess the return and downside risk of a decumulation portfolio established at the retirement age of a senior, with a determined lifetime horizon differentiated by the sex of the citizen. To measure the portfolio's return and downside risk, two ratios conditioned by seniors' risk attitude towards portfolio failure are employed: the downside Sortino ratio and the downside risk-return ratio. Unlike other research in the field, this manuscript provides three portfolio compositions catering to different senior investment profiles: aggressive, moderate, and conservative. Additionally, it offers a decumulation horizon conditioned by the sex-specific life expectancy of the individual, instead of offering different scenarios for conducting a sensitivity analysis. Lastly, this study was conducted across three socioeconomically distinct countries: the US, Spain, and Japan. The results clearly demonstrate that both sex and nationality significantly influence the selection of the optimal decumulation portfolio composition aimed at exhausting funds by the senior's demise.

Bevilacqua, M. and Tunaru, R. (2021). "The SKEW index: Extracting what has been left." In: *Journal of Financial Stability* 53, p. 100816.

This study disentangles a measure of implied skewness that is related to downward movements in the U.S. equity index from the corresponding implied skewness that is associated with upward movements. A positive SKEW index is constructed from S&P 500 call options, whereas a negative SKEW index is constructed from the S&P 500 put options. We show that the positive SKEW is linked to market sentiment, whereas the negative SKEW is related to existing tail risk measures. The negative SKEW is proposed as a more objective prudent tail risk measure, and it is found to be able to predict recessions, market downturns, and uncertainty indicators up to one year in advance. The predictive power of the negative SKEW is also confirmed when we control for other tail risk measures and also out-of-sample.

Bhansali, V. (2014). *Tail risk hedging: Creating Robust Portfolios for Volatile Markets*. McGraw-Hill. 272 pp.

Tail Risk Hedging is built on the author's practical experience applying macroeconomic forecasting and quantitative modeling techniques across asset markets. Using empirical data and charts, he explains the consequences of diversification failure in tail events and how to manage portfolios when this happens. He provides an easy-to-use, yet rigorous framework for protecting investment portfolios against tail risk and using tail hedging to play offense. Tail Risk Hedging explores how to: Generate profits from volatility and illiquidity during tail-risk events in equity and credit markets; Buy attractively priced tail hedges that add value to a portfolio and quantify basis risk; Interpret the psychology of investors in option pricing and portfolio construction; Customize explicit hedges for retirement investments; Hedge risk factors such as duration risk and inflation risk; Managing tail risk is today's most significant development in risk management, and this thorough guide helps you access every aspect of it. With the time-tested and mathematically rigorous strategies described here, including pieces of computer code, you get access to insights to help mitigate portfolio losses in significant downturns, create explosive liquidity while unhedged participants are forced to sell, and create more aggressive yet tail-risk-focused portfolios. The book also gives you a unique, higher level view of how tail risk is related to investing in alternatives, and of derivatives such as zerocost collars and variance swaps. Volatility and tail risks are here to stay, and so should your clients' wealth when you use Tail Risk Hedging for managing portfolios.

Bhansali, V. (2008). "Tail Risk Management." In: *The Journal of Portfolio Management* 34(4), pp. 68–75.
In the current deleveraging episode, the severity and simultaneous realization of low-probability events across a number of strategies has brought portfolio tail-risk hedging to the center of investors' attention. In this article, the author discusses the basic principles and implementation considerations behind portfolio tail-risk management, emphasizing the importance of viewing tail risk as the consequence of a systemic shock during which normally uncorrelated markets become correlated due to a reduction in liquidity. This approach suggests that tail risk should be measured by performing macro scenario analysis and that hedges should be picked based on their cost and joint performance in periods of stress relative to the portfolio being hedged. Since liquidity reduction is a typical consequence of systemic shocks, the author also discusses specific operational issues that come to the fore in such events.

Bhansali, V. (2015). "Tail-Risk Management for Retirement Investments." In: *The Journal of Retirement* 2(3), pp. 78–86.
One of the insights of behavioral finance is the tendency for small investors to overreact to market swings. Even a well-structured portfolio may be vulnerable to panic selling in a downturn.
The article explains how a policy of tail-risk hedging could deter such behavior by putting a floor on drops in the portfolio. Instead of maintaining an overly conservative stance and an unnecessarily low share of equities, investors could hedge their portfolios by relying on options, which have become a much more feasible tool for the small investor in recent years.
The article discusses the relationship between the time to retirement and the need to hedge. Investors nearing retirement have to adopt a more defensive position or hedge explicitly, but younger investors can be more aggressive and buy options with lower strike prices.

Bhansali, V. (2018). "Right Tail Hedging: Managing Risk When Markets Melt Up." In: *The Journal of Portfolio Management*.
Popular academic and practitioner lore claims that buying options, whether puts or calls, is a negative expected return investment and hence should not be undertaken by rational, risk-neutral investors. However, real-world considerations such as the possibility of large jumps and imitative behavior of traders in both bull and bear markets can reverse this conclusion. In particular, the potential for positive economic shocks such as those observed since the 2016 U.S. election may make call options-based strategies superior to buy and hold strategies. In this article, the author extends his work published in The Journal of Portfolio Management in 2007 on left tail or downside tail risk hedging to address upside hedging, which has become increasingly relevant in an environment of low yields, elevated asset prices, low credit spreads, indexation, and multidecadal lows in call option prices.

Bhansali, V., Chang, L., Holdom, J., and Johnson, M. (2024a). "Option Total Return and Active Option Portfolio Management." In: *The Journal of Portfolio Management* 50(4), pp. 75–87.
The authors introduce the concept of total return for options and option portfolios. By taking local derivatives of the option price with respect to the underlying drivers in an option-pricing model (i.e., delta, gamma, vega, theta), one can decompose daily option price changes into the sum of the changes in these underlying drivers. By accumulating the contribution from the underlying drivers, one can compute each driver's contribution to the total performance of a portfolio of options. The authors discuss the relevance of this framework to applications such as tail-risk hedging and discuss how the approach can be useful to the active management of option portfolios. Their approach provides a natural extension of the concept of total return that is widely used for the evaluation of traditional investment portfolios.

Bhansali, V., Chang, L., Holdom, J., and Rappaport, M. (2020). "Monetization Matters: Active Tail Risk Management and the Great Virus Crisis." In: *The Journal of Portfolio Management* 47(1), pp. 16–28.
The authors discuss monetization strategies for both left- and right-tail risk hedging to illustrate the potential benefits of active management of hedges. In particular, by including actual data from the sharp COVID-19 pandemic-related market correction and subsequent rebound of 2020, the authors quantify how monetization strategies have the ability to improve the performance of portfolio hedges. This extends previous work on active tail risk hedging published in this journal. The authors conclude that active management of tail hedging can result in significant increases in the efficacy of tail hedging.

Bhansali, V., Fabozzi, F. J., Harlow, R., Kobor, A., Niehaus, J., Small, C., and Weisman, A. (2024b). "Applications of derivatives for portfolio risk management." In: *Journal of Asset Management* 25(6), pp. 552–578.
In this article, five in-depth illustrations of practical applications of various derivatives for risk control for asset management are provided. The illustrations are presented using stock index futures, interest-rate derivatives

(Treasury futures and interest rate swaps), options, and equity swaps. The cases presented bridge the gap between theoretical finance and practical application, making it invaluable for those involved in risk management for portfolio managers.

Bianchi, M. and Briere, M. (2021a). "Robo-Advising for Small Investors." In: *SSRN e-Print*.

We study the effects of robo-advising on investors' attention, trading, and performance on a large set of Employees Saving Plans covering a representative sample of French employees. We find that relative to self-managing, accessing the robo services is associated to an increase in the time investors spend to follow their portfolios and to an increase in trading activities. After having taken up the robo, investors are willing to increase their investment, bear more risk, and to rebalance their portfolio in a way to keep their allocation closer to the target. They also experience higher risk-adjusted returns. These effects tend to be stronger for investors with smaller portfolios, who are less likely to have access to traditional advice. Our results shed light on the dynamics of investors' trust towards the robo service and suggest that automated advice can promote financial inclusion.

Bianchi, M. and Briere, M. (2021b). "Robo-Advising: Less AI and More XAI?" In: *SSRN e-Print*.

We start by considering some of the key reasons behind the academic and industry interest in robo-advisors. We discuss how robo-advice could potentially address some fundamental problems in investors' decision making as well as in traditional financial advice. We then move on to some of the ongoing issues regarding the future of robo-advice. Firstly, the role Artificial Intelligence (AI) plays, and should play, in robo-advice. Secondly, how far should the personalisation of robo-advice recommendations go. Third, how trust in automated financial advice can be generated and maintained. Fourth, whether robots are perceived as complements or substitutes to human decision-making. Our conclusion outlines some thoughts on what the next generation of robo-advisors might look like. We highlight the importance of recent insights in Explainable AI and how new forms of AI applied to financial services would benefit from importing insights from economics and psychology to design effective human/robot interaction.

Biggs, A. G., Chen, A., and Munnell, A. (2021). "The Consequences of Current Benefit Adjustments for Early and Delayed Claiming." In: *SSRN e-Print*.

Workers have the option of claiming Social Security retirement benefits at any age between 62 and 70, with later claiming resulting in higher monthly benefits. These higher monthly benefits reflect an actuarial adjustment designed to keep lifetime benefits equal, for an individual with average life expectancy, regardless of when benefits are claimed. The actuarial adjustments, however, are decades old. Since then, interest rates have declined; life expectancy has increased; and longevity improvements have been much greater for high earners than low earners. This paper explores how changes in longevity and interest rates have affected the fairness of the actuarial adjustment over time and how the disparity in life expectancy affects the equity across the income distribution. It also looks at the impact of these developments on the costs of the program and the progressivity of benefits. The paper found that:

1) The increases in life expectancy and the decline in interest rates argue for smaller reductions for early claiming and a smaller delayed retirement credit for later claiming.
2) Specifically, the benefit at 62 should equal 77.5 percent, as opposed to 70.0 percent, of the full age-67 benefit, and the benefit at 70 should equal 119.9 percent, instead of 124.0 percent, of the full benefit.
3) The outdated actuarial adjustments are a modest moneymaker for the program - about $1.9 billion in 2018, with most of the gains coming from those claiming at 62, who are typically lower earners.
4) Surprisingly, the correlations between earnings and life expectancy and between earnings and claiming behavior have only modest implications for both the cost and progressivity of Social Security benefits.
5) Finally, the cost and distributional effects of earnings-related life expectancy and claiming cannot be addressed through the actuarial adjustments for early and late claiming. They reflect the fact that high earners get their large benefits for a long time and low earners get their more modest benefits for a shorter time.

The policy implications of the findings are:

- Increases in life expectancy and the decline in interest rates suggest smaller reductions for early claiming and a smaller delayed retirement credit for later claiming.
- Accounting for differential mortality would involve changing benefits, and is not a problem that can be solved by tinkering with the actuarial adjustments.

This study considered Consequences of Current Benefit Adjustments for Early and Delayed Claiming.

Biggs, A. G. (2017). "The life cycle model, replacement rates, and retirement income adequacy." In: *The Journal of Retirement* 4(3), pp. 96–110.
The key insight of the life cycle model in economics is that a household consumption at any given time is determined not so much by its current income as by the total income available to the household over its lifetime. A replacement rate can be a useful tool in approximating the life cycle model predictions for how households prepare for retirement. The Social Security Administration Office of the Chief Actuary (OACT) publishes two different calculations of retirement income replacement rates, each of which finds that Social Security benefits replace about 40% of a typical retiree pre-retirement earnings. Some interpret these figures as indicating that Social Security benefits are insufficiently generous and that U.S. households total retirement saving is inadequate. But OACT two methods for calculating replacement rates both violate the life cycle model in a meaningful way. OACT career-average earnings replacement rates, in which lifetime earnings are first indexed upward by the rate of economywide wage growth, exaggerates by roughly one-fifth the real value of earnings available to a household for consumption over its lifetime. This overstatement lowers a household measured ability to replace their pre-retirement earnings. OACT final-earnings replacement rates effectively compare Social Security retirement benefits to pre-retirement earnings only in the years in which the individual worked, ignoring the life cycle model prediction that household consumption is a function of long-term average earnings, including years in which a household member was not employed. A replacement-rate calculation more consistent with the life cycle model would compare retirement income to an average of real earnings calculated over a significant number of years. Such an approach would find substantially higher replacement rates for the typical retiree. It is important both for Social Security policy and the analysis of overall retirement savings adequacy that replacement-rate calculations build on the insights of the life cycle model that guides most economic analysis of retirement saving.
Bjerring, T., Ross, O., and Weissensteiner, A. (2017). "Feature selection for portfolio optimization." In: *Annals of Operations Research* 256, pp. 21–40.
Most portfolio selection rules based on the sample mean and covariance matrix perform poorly out-of-sample. Moreover, there is a growing body of evidence that such optimization rules are not able to beat simple rules of thumb, such as 1/N. Parameter uncertainty has been identified as one major reason for these findings. A strand of literature addresses this problem by improving the parameter estimation and/or by relying on more robust portfolio selection methods. Independent of the chosen portfolio selection rule, we propose using feature selection first in order to reduce the asset menu. While most of the diversification benefits are preserved, the parameter estimation problem is alleviated. We conduct out-of-sample back-tests to show that in most cases different well-established portfolio selection rules applied on the reduced asset universe are able to improve alpha relative to different prominent factor models.
Blake, D. and Cairns, A. J. G. (2020). "Longevity risk and capital markets: the 2018-19 update." In: *Annals of Actuarial Science* 14(2), pp. 219–261.
This Special Issue of the Annals of Actuarial Science contains 12 contributions to the academic literature all dealing with longevity risk and capital markets. Draft versions of the papers were presented at Longevity 14: The Fourteenth International Longevity Risk and Capital Markets Solutions Conference that was held in Amsterdam on 20-21 September 2018. It was hosted by the Pensions Institute at City, University of London and the Netspar Network for Studies on Pensions, Ageing and Retirement.
Blanchett, D. (2014). "Addressing Key Retirement Risks." In: *The Journal of Retirement* 2(2), pp. 67–80.
This article introduces a new model to jointly determine the optimal allocation to equities, real assets, and annuities for a retiree with a given set of preferences. This model improves on a number of existing retirement income models by directly incorporating a cost associated with deviating from the investor's target risk preference, jointly testing the effects of four preference variables aversion to volatility, strength of a bequest motive, preference for sustainable expenditure, and sensitivity to the risk of future inflation using a dynamic withdrawal strategy and taking a probabilistic approach to mortality.
Blanchett, D. (2015). "The Value of Goals-Based Financial Planning." In: *Journal of Financial Planning*.
The financial planning profession is built on helping people accomplish goals. While investing appropriately is generally an important part of accomplishing a goal, achieving a goal often requires advice beyond selecting investments based on alpha and beta. Past research on the topic of goals-based financial planning has focused primarily on determining optimal portfolios to fund different types of goals. In contrast, the focus of this research is how to determine which goals should be funded, as well as how to optimally save toward goals over time. A utility model based on prospect theory was used to determine the optimal funding strategy for a household. The results suggest that using a goals-based framework to determine which goals to fund and how to fund them

can lead to an increase in utility-adjusted wealth of 15.09 percent for a hypothetical household versus a na ive strategy focused only on funding retirement. This is equivalent to generating an annual alpha of 1.65 percent for the lifetime of the base scenario household.

Blanchett, D. (2018). "Health shocks and subsequent retiree spending." In: *The Journal of Retirement* 6(1), pp. 55–69.

Healthcare expenses are receiving increasing attention from retirees (and the media) today and many financial planners are starting to think about how to potentially incorporate these expenses into a financial plan. This article explores some of the key considerations associated with modeling healthcare-related expenses for retirees with a focus on the impact of these expenses on future retiree spending (and consumption) using data from the Health and Retirement Study. We find that households that experience unexpected out-of-pocket healthcare expenses tend to decrease spending by more than those who do not. For example, if healthcare expenses in a given year exceed 10% of total spending (excluding health insurance premiums, which we define as a health shock), then future total spending, nondurable spending, and consumption drop by approximately 5% more than households that do not experience the shock. Financial advisors (and households) interested in including healthcare expenses as part of a financial plan should be aware of this effect since it can potentially reduce the actual financial implications associated with a retiree health shock (or other retiree healthcare expenditures).

Blanchett, D. (2020a). "The Benefit of Diversified Guaranteed Income for Retirees: Combining Immediate Fixed and Variable Annuities." In: *Retirement Management Journal* 2(3).

This paper explores the potential benefits of developing a retirement income that considers both immediate fixed annuities (IFA) and immediate variable annuities (IVA) using a stochastic utility model combined with a scenario framework. Optimal annuity allocations vary considerably across household type, but certainty equivalent retirement income increases by 20 percent, on average, when incorporating annuities. Total annuity allocations increase when both IFAs and IVAs are considered, and retirees realize only approximately two-thirds of the benefits of annuitization when just one annuity type is considered. IVA allocations were typically higher than IFA allocations because most households already have a base level of fixed guaranteed income (through Social Security); therefore, IVAs can be a unique diversifier from a retirement-income perspective. Overall, this analysis strongly suggests retirees (and financial advisors) should consider annuities as part of a retirement-income strategy, and that they should consider different types of annuities to create the best possible plan.

Blanchett, D. (2020b). "The value of allocating to annuities." In: *The Journal of Retirement* 8(1), pp. 40–52.

This article explores the costs and benefits associated with different approaches a financial advisor could take toward using (or not using) annuities across his or her entire book of clients (i.e., for an entire cohort of retirees). Fees are used to proxy the implied total expenses (or costs) for the two approaches, and not surprisingly, the relative fees between the portfolio and the annuity has a significant impact on the respective benefit of each. For example, a financial advisor would be better off not recommending annuities at all (on average), if the investment-only approach is considered to be low cost and the annuity approach is assumed to be high cost. If the fees for both approaches are assumed to be moderate (i.e., the implied total expenses of the strategies are equal), which is the most realistic scenario for financial advisors providing high-quality advice to clients, the average annuity allocation across scenarios is approximately 30% and helping retirees allocate to annuities optimally generates a total of 73 basis points of alpha-equivalent benefit compared with a financial advisor that does not consider annuities as part of the retirement plan. The marginal alpha-equivalent benefit associated with optimal annuitization strategies, which is estimated by focusing on the total additional benefit generated from allocating to annuities based on the percentage of assets actually used to purchase the annuity, is closer to 150 basis points-a level of increased risk-adjusted performance that would be almost impossible for a financial advisor to overcome using traditional investments (i.e., ignoring annuities).

Blanchett, D. (2021a). "Do Advisors Improve 401(k) Plans?" In: *The Journal of Retirement* 8(4), pp. 26–42.

Research on the impact of 401(k) plan advisors has focused predominately on the investment menu or participant-level allocation decisions. This research takes a more holistic perspective and reviews the differences between advised plans across a variety of domains with the use of data from 5500 Employee Benefits Security Administration filings from 2016. Advised plans with assets from \$1 million to \$50 million were clearly performing "better" than those without, and the advisor impact generally declined as plan assets increased. Although this research suggests advisors are adding value, correlation is not causation, and additional research on the actual impact of 401(k) advisors is warranted.

Blanchett, D. (2021b). "Minding the Gap in Subjective Mortality Estimates." In: *The Journal of Retirement* 9(2), pp. 8–20.

This article explores the accuracy of subjective life expectancy estimates using data primarily from the from the Health and Retirement Study (HRS). Although individuals appear to some sense about their likelihood of survival (i.e., their subjective mortality), there are notable gaps in these estimates, consistent with past research. Evidence suggests that although subjective estimates may be relatively accurate, on average, and that households appear to do a relatively good job considering various objective factors (for example, health status), there are often significant errors in individual forecasts, and households do not appear to correctly consider all the relevant objective factors (such as income and smoking). Therefore, financial planners need to educate themselves on how to better model and personalize mortality assumptions into financial plans versus relying on purely subjective estimates to ensure that planning assumptions are as accurate as possible.

Blanchett, D. (2021c). "The Benefit of Diversified Guaranteed Income for Retirees: Combining Immediate Fixed and Variable Annuities." In: *SSRN e-Print* 2(3).

This paper explores the potential benefits of developing a retirement income that considers both immediate fixed annuities (IFA) and immediate variable annuities (IVA) using a stochastic utility model combined with a scenario framework. Optimal annuity allocations vary considerably across household type, but certainty equivalent retirement income increases by 20 percent, on average, when incorporating annuities. Total annuity allocations increase when both IFAs and IVAs are considered, and retirees realize only approximately two-thirds of the benefits of annuitization when just one annuity type is considered. IVA allocations were typically higher than IFA allocations because most households already have a base level of fixed guaranteed income (through Social Security); therefore, IVAs can be a unique diversifier from a retirement-income perspective. Overall, this analysis strongly suggests retirees (and financial advisors) should consider annuities as part of a retirement-income strategy, and that they should consider different types of annuities to create the best possible plan.

Blanchett, D. (2022). "Protected Lifetime Income Benefits: The Future of Longevity Protection." In: *Investments and Wealth Monitor* 2022(November/December).

More solutions are being created to help retirees address longevity risk. For example, one strategy provides lifetime income with a benefit amount that evolves throughout retirement based entirely on account performance. We refer to this strategy as a "protected lifetime income benefit" or PLIB. This article contrasts the efficacy of a PLIB against three other common types of annuities: A a single premium immediate annuity or SPIA, A a deferred income annuity or DIA, A and a variable annuity (VA) with a guaranteed lifetime withdrawal benefit or GLWB focused on economic value. Overall, we find that th.

Blanchett, D. (2023). "Redefining the Optimal Retirement Income Strategy." In: *Financial Analysts Journal*.

This paper introduces a cohesive series of models designed to improve retirement income projections. First, the retirement income goal (i.e., liability) is decomposed based on assumed spending elasticity (e.g., "needs" and "wants"). Second, spending is assumed to evolve throughout retirement using a dynamic withdrawal strategy leveraging the funded ratio concept. Third, optimal strategies are determined using an expected utility model based on prospect theory, which also yields a client-friendly outcomes metric. Overall, this framework can result in advice and guidance that is notably different than models using more basic (and common) assumptions, especially approaches relying on probability of success-related metrics.

Blanchett, D. and Cormier, W. (2021). "Right-Sizing Retirement: Exploring the Retirement Consumption Gap in Early Retirement." In: *Journal of Financial Planning* 34(2), pp. 68–81.

Research suggests some households underspend in retirement, resulting in a "retirement consumption gap." This paper explores this effect, specifically during the first 10 years of retirement, and incorporates both household assets and pre-retirement spending using data from the Health and Retirement Study. Only 18 percent of households have enough wealth to cover pre-retirement consumption when they retire, which suggests most households will not be able to maintain their pre-retirement lifestyle in retirement-a finding consistent with other general estimates of the retirement readiness of Americans. Real financial assets decline for 65 percent of households during the first 10 years of retirement, with a median real decline of 35 percent. Real retiree spending declines for 75 percent of households during the first 10 years of retirement, with a median annual decline of approximately 2 percent per year. This suggests financial planners should consider changes in retirement spending that are less than inflation as part of a retirement plan. The percentage of households that can fund their retirement consumption increases dramatically during the first 10 years of retirement, from 18 percent to 48 percent, primarily due to reductions in spending. This suggests households "right-size" their spending early in retirement to better align with available resources. It is not clear, though, to what extent this behavior persists further into retirement (due to data limitations). Many well-funded households could increase consumption, but appear not to do so (i.e., exhibit a retirement consumption gap). Potential reasons include the desire to leave a bequest, uncertain medical expenses (especially late in retirement), uncertain life expectancy, etc. While this group is a minority of retiree households, understanding what drives this behavior is especially important to financial planners since this group tends to have the most accumulated wealth and are, therefore, more likely to seek the services of, or use, a financial planner.

Blanchett, D., Finke, M., and Pfau, W. (2018). "Low Returns and Optimal Retirement Savings." In: *How Persistent Low Returns Will Shape Saving and Retirement*. Oxford University Press.

Lifetime financial outcomes relate closely to the sequence of investment returns earned over the life cycle. Higher return assumptions allow individuals to save at a lower rate, withdraw at a higher rate, retire with a lower wealth accumulation, and enjoy a higher standard of living. While analysis of this topic is often based on historical investment performance, present bond yields are historically low and equity prices are quite high, suggesting that individuals will likely experience lower returns in the future. This implies the need for higher savings rates, lower withdrawal rates, a larger nest egg at retirement, and a lower lifetime standard of living. We show that lower-income workers will need to save about 50 percent more if low rates of return persist in the future, and higher-income workers will need to save nearly twice as much in a low return environment compared to the optimal savings using historical returns.

Blanchett, D. and Finke, M. S. (2019). "Should Annuities be Purchased from Tax-Sheltered Assets?" In: *SSRN e-Print*.

Retirees who purchase an annuity may assume that retirement savings accounts are ideal for funding retirement income. Annuities, however, are a tax-favored investment. We investigate the relative benefits of purchasing an annuity from tax-deferred and taxable accounts for various payout levels, tax rates, asset tax efficiency, and assumed portfolio rates of return. We find considerable evidence that investors are better off using non-qualified accounts to purchase annuities, although the benefits vary significantly by investor characteristics and the tax efficiency of investments held in non-qualified savings. In some cases, selecting the right account increases the after-tax income by over 10% which is equivalent to approximately 50 basis points of added portfolio return. A 15-year deferred annuity purchased from non-qualified vs. qualified bonds earning 4% provides over 30% more after-tax income for a 65-year old with a 40% marginal tax rate.

Blanchett, D. and Finke, M. S. (2021). "Guaranteed Income: A License to Spend." In: *SSRN e-Print*.

Prior research finds that retirees don't spend nearly as much as they could from their investments. Economic theory provides both rational and behavioral explanations for under-spending among retirees with high non-annuitized wealth. Longevity risk will result in lower spending among rational, risk-averse retirees. Retirees may also exhibit behavioral preferences that make them far more comfortable spending from income than assets. We explore how the composition of retirement assets is related to retirement spending and find that retirees who hold a higher percentage of their wealth in guaranteed income spend more than retirees whose wealth consists primarily of non-annuitized assets. Marginal estimates suggest that investment assets generate about half of the amount of additional spending as an equal amount of wealth held in guaranteed income. In other words, retirees will spend twice as much each year in retirement if they shift investment assets into guaranteed income wealth. The size of the effect suggests that the explanation for under-spending non-annuitized savings is likely both a behavioral and a rational response to longevity risk.

Blanchett, D. and Finke, M. S. (2023). "Fixed Rate Annuities: Are High Returns An Anomaly?" In: *Journal of Investment Consulting* 22(1), pp. 63–69.

Fixed-rate annuities, also called multi-year guaranteed annuities, or MYGAs, offer expected returns protected by state guaranty associations similar to other safe investments such as certificates of deposit, money market accounts, or Treasury bonds. Yields on MYGAs have risen considerably, in relative terms, in recent years compared to historical levels. MYGAs offered by highly rated insurers had a 100-basis-point (bps) yield premium above Treasury bonds in 2020 and 2021, with premiums exceeding 200 bps from lower-rated insurers. Expected MYGA returns were approximately 100 bps higher than yields on similarly rated corporate bonds and have significantly lower default risk. We explore this apparent anomaly and discuss possible explanations including rent extraction from less sophisticated or inattentive consumers, and an increase in portfolio risk among insurers owned by private equity.

Blanchett, D. and Finke, M. S. (2024). "Retirees Spend Lifetime Income, Not Savings." In: *SSRN e-Print*.

The shift to defined contribution savings plans means that more retirees must fund spending from savings. Prior studies find that there appears to be a behavioral resistance to spending down savings after retirement in a manner that is consistent with life cycle models. We explore how lifetime income, wage income, capital income, qualified savings, and nonqualified savings are used to fund retirement spending. Retirees spend far more from lifetime income than other categories of wealth. Approximately 80% of lifetime income is consumed, on average. In contrast, only about half of available savings and other income sources are consumed. Withdrawal rates from savings are well below typical guidance and the analysis suggests that converting savings into lifetime income could increase retirement consumption significantly, especially for married households.

Blanchett, D., Finke, M. S., and Guillemette, M. A. (2016a). "Who Exhibits Time Varying Risk Aversion?" In: *SSRN e-Print*.

There is growing empirical evidence that risk preferences change based on financial market conditions. This paper explores individual predictors of time varying risk aversion among participants in U.S. defined contribution plans using a unique dataset with daily responses to a risk tolerance questionnaire. We find that older investors (ages 51-65) are more susceptible to time varying risk aversion. Among older investors, variable risk preference was greatest for participants with the smallest account balances and the lowest incomes, but was unrelated to equity allocation within their retirement portfolio. Much of the variation in the aggregate risk tolerance score can be attributed to variation in future long-term equity performance expectations among older investors. We discuss how target-date funds have the potential to reduce losses from poor market timing that may result from time varying risk aversion.

Blanchett, D., Finke, M. S., and Pfau, W. D. (2016b). "Required Retirement Savings Rates Today." In: *SSRN e-Print*.

Recent asset pricing studies suggest that demand for stocks since 1980 has driven expected returns below their historical average. The current yield of risk-free assets in the U.S. is also well below historical bond yields. This decrease in bond yields, coupled with increases in longevity, has doubled the cost of funding a real dollar of income in retirement since 1980 for a 65-year-old retiree. Many common financial planning practices are surprisingly sensitive to asset returns, and advisors need to understand the challenges clients will face if high asset prices persist. Results from a life cycle planning model show that savings rates would need to rise sharply for households hoping to maintain the same standard of living in retirement if real asset returns are low. Low expected returns also have a surprisingly strong impact on the amount of savings needs to fund legacy goals and a negative impact on client spending throughout their life cycle. Advisors may need to modify expected returns in planning software to provide clients with more realistic projections on meeting long-term spending goals.

Blanchett, D. and Kaplan, P. D. (2018). "Beyond the Glide Path:The Drivers of Target-Date Fund Returns." In: *The Journal of Retirement* 5(4), pp. 25–39.

The authors explore the relative importance of the three primary drivers of target-date fund (TDF) performance: equity (or market) exposures (which, across a series vintages, combine to form a glide path), style exposures (intrastock and intrabond allocations), and other investment selection decisions (e.g., manager selection and the active/passive decision, as well as any other residual risk exposures). They find that overall equity exposure drives only about 25% of the variation in returns across TDFs versus approximately 30% for style and 45% for selection, on average. Consequently, the analysis of the riskiness of a given TDF must be based on more than the overall weight given to equites.

Blanchett, D. and Li, Z. (2023). "ESG Fund Allocations among New, Do-It-Yourself Defined Contribution Plan Participants." In: *The Journal of Retirement* 11(1), pp. 56–73.

Investment strategies focused on environmental, social, and governance (ESG) issues have been receiving increased interest among defined contribution (DC) plan sponsors, consultants, and regulators. This article explores the allocation decisions of 9,324 newly enrolled DC participants who are self-directing their accounts in a DC plan that offers at least one ESG fund. The analysis suggests that demand for ESG funds is relatively low, with ESG fund allocations and holding levels being lower than random chance would suggest. While there are some clear demographic preferences for ESG funds (e.g., among younger participants with higher incomes), ESG allocations appear to be primarily a function of weak preferences, driven by naive diversification, although ESG allocations are significantly higher in plans where general ESG usage is more elevated. ESG funds have the potential to drive participants away from professionally managed investment options, such as target-date funds, resulting in lower risk-adjusted returns for participants if they are simply added to core menus. Overall, this analysis suggests that plan sponsors should take a thoughtful and cautious approach when considering adding ESG funds to an existing core menu.

Blanchett, D., Lutter, S., and Huston, S. J. (2023). "Using Confidence and Risk Tolerance to Build Financial Wellness." In: *SSRN e-Print*.

Risk tolerance is an important investor preference that is commonly used to determine portfolio risk levels with financial advisors, although it also has more general implications with how investors decide to invest in risky assets on their own. While research on risk tolerance has primarily focused investor demographic variables, such as gender, income, financial wealth, even recent market performance; financial confidence also has the potential impact risk attitudes (yet relatively little research exists on this topic). This research uses the responses to an online financial wellness assessment offered from June 2020 to December 2021 with 62,819 respondents. We have detailed demographic data available on respondents, as well as perceptions information on both subjective (self-assessed) financial wellness and objective financial wellness scores (based on responses to five questions) in addition to specific noted financial stressors (e.g., student loans, credit card debt, saving for retirement, etc.). Less than a quarter (23%) of respondents in our dataset report being "Confident" regarding their finances. Individuals who report higher financial confidence have significantly higher levels of risk tolerance, an effect that persists when controlling for a myriad of demographic variables. Increasing allocations to risky assets (e.g., equities) due to improving financial confidence has the potential to significantly increase retirement savings. For example, using a hypothetical model we estimate balances could be 11% higher at retirement among investors who are financially confident, controlling for other factors. The results of the analysis suggest improving financial wellbeing has the potential to improve the long-term wealth of households through not only improving overall financial behaviors but also increasing expected returns on savings to the extent it better enables investors to invest more aggressively (assuming a positive equity risk premium).

Blanchett, D. M. (2017). "The Impact of Guaranteed Income and Dynamic Withdrawals on Safe Initial Withdrawal Rates." In: *Journal of Financial Planning* 30(4), pp. 42–52.

Optimal safe initial withdrawal rates can differ significantly across relatively similar retiree household scenarios and to a much larger extent than has been commonly noted in retirement income research. Important variables to consider are the amount of guaranteed income, the extent to which the household can adjust consumption, the risk of the investment portfolio, the return assumptions used for projections, and the degree of income stability desired by the retiree. Guaranteed income had the largest impact on the estimated safe initial withdrawal rates among the variables considered. Those withdrawal rates varied by more than 4 percentage points for different levels of guaranteed income. Although recent research using forward-looking returns suggested that 3 percent is a more appropriate safe initial withdrawal rate for retirees today, this analysis suggested that 4 percent is likely still a much better starting place for many retirees, although the optimal target varies considerably. For financial planners using probability of success-based metrics, targeting a lower success level (e.g., a 75 percent success rate versus a 95 percent success rate) is likely more appropriate, although the optimal target is highly dependent on retiree characteristics.

Blanchett, D. M., Finke, M., and Pfau, W. D. (2013). "Low Bond Yields and Safe Portfolio Withdrawal Rates." In: *The Journal of Wealth Management* 16(2), pp. 55–62.

The majority of research on sustainable withdrawal strategies has used either historical rolling time periods or a stochastic (Monte Carlo) simulation process based on long-term averages, where the expected return of an asset class is the same for each year of the simulation. While these approaches may be reasonable to describe long-term averages, we believe they are less useful when there is a significant and sustained deviation, such as the current low bond yield market. This article introduces a model that takes into account current bond yields and allows them to drift toward a higher value during retirement, using an autoregressive model based primarily on historical relationships between asset classes. This approach can better replicate the actual bond returns a current or near retiree can expect during retirement both now and in the future. Using this model, we find a significant reduction in safe initial withdrawal rates, with a 4 percent initial real withdrawal rate having approximately a 50 percent probability of success over a 30-year period. It finds that a retiree who wants a 90 percent probability of achieving a retirement income goal with a 30-year time horizon and a 40 percent equity portfolio would only have an initial withdrawal rate of 2.8 percent. Such a low withdrawal rate would require 42.9 percent more savings if the retiree wanted to pull the same dollar value out of the portfolio annually as he or she would get with a 4 percent withdrawal rate from a smaller portfolio.

Bloch, D. A. (2024). "Futuretesting Quantitative Strategies." In: *SSRN e-Print*.

Building quantitative strategies with a Backtesting engine leads to overfitting, which produces results that cannot be replicated in the future. To assess the robustness of a strategy and avoid overfitting, it is necessary to perform multiple backtests on different datasets. However, there is no market dataset spanning the whole time-space domain as we only observe a single realisation per security. We must therefore rely on generating synthetic data that is statistically close to the market data. We present the Futuretesting engine as a framework for generating realistic synthetic market data used for testing and optimising trading strategies, performing risk management, and analysing risk models. We utilise it to compute probabilistic Risk/Return profiles for a set of trading strategies everywhere in the time-space domain. We perform multiple futuretests on different datasets and under different scenarios to obtain a distribution of performance metrics. Given some accuracy measures, such as the mean-variance criterion, we can then find the optimum strategy maximising returns and minimising risk. Finally, we test this optimum strategy on real market data on different securities and over different periods of time. First, we introduce quantitative principles and give an overview of quantitative trading strategies, focusing on trading with moving averages as well as trading with options. Next, we formulate the optimum trading problems, present backtesting strategies, discuss the calibration of trading rules, and introduce the modelling of trailing stops. Then, we present our probabilistic framework, detail the creation of market scenarios, present the tools used for measuring the performance of the strategies, and provide quantitative trading algorithms. To conclude, we consider a stochastic model to generate market returns, perform a Futuretesting on a few quantitative strategies, and display results.

Bodily, S. E. and Furman, B. (2016). "Long-Term Care Insurance Decisions." In: *Decision Analysis* 13(3), pp. 173–191.

The purchase of long-term care (LTC) insurance is a difficult lifetime choice made in the face of highly uncertain risks, including mortality, morbidity, timing and length of LTC, and portfolio investment risk. Many individuals do not know how to think about this decision properly and, in the face of too much anecdotal and too little objective information, will not proactively decide. We used Monte Carlo simulation modeling with detailed, experience-based distributions for LTC uncertainties and their correlations to project investment growth to death given alternative levels of LTC insurance. Using constant risk aversion, we calculate certainty equivalents for the resulting distributions of final holdings at death. Decisions were separated for male and female individuals and group and individual market insurance opportunities. Sensitivity analysis was conducted varying age, cost of coverage, starting investment amount, risk tolerance, return on portfolio investment, inflation, and length of LTC coverage. Optimality results suggest low levels of coverage or no insurance, with higher use of insurance only for individuals who are young, have low risk tolerance, low starting portfolio amounts, or combinations of these characteristics. While the contribution of this work is to assist individual decision making, it will also be informative to policy makers and insurance companies.

Bogan, V. L., Geczy, C. C., and Grable, J. E. (2020). "Financial planning: A research agenda for the next decade." In: *Financial Planning Review* 3(2).

We provide an informed discussion about challenges, opportunities and the future of research and practice in the field of financial planning over the next 10 years. As editors of Financial Planning Review, using a mix-methods approach and a survey of subject-matter expert views, we outline what we believe are some of the future key themes of financial planning. We also present an overview of the challenges and opportunities facing researchers who are working to build, inform, and expand the financial planning body of literature. Further, we discuss the financial planning-related topics that would benefit most from increased research and study.

Bollen, N. P. B. and Posavac, S. (2018). "Gender, risk tolerance, and false consensus in asset allocation recommendations." In: *Journal of Banking & Finance* 87, pp. 304–317.

We study the impact of gender on asset allocation recommendations. Graduate business students and professional wealth managers are randomly assigned a male or female client. Participants recommend an allocation and choose an allocation for themselves. Male students choose a riskier allocation than female students, consistent with existing evidence of a gender difference in risk tolerance, and recommend a riskier allocation. In contrast, male and female wealth managers choose and recommend the same allocation, indicating that male and female finance professionals feature similar risk preferences. In both samples, a subject's allocation choice is the strongest predictor of the recommendation provided.

Boon, L.-N., Briere, M., and Werker, B. J. M. (2020). "Systematic longevity risk: to bear or to insure?" In: *Journal of Pension Economics and Finance* 19(3), pp. 409–441.

We compare two contracts for managing systematic longevity risk in retirement: a collective arrangement that distributes the risk among participants, and a market-provided annuity contract. We evaluate the contracts' appeal with respect to the retiree's welfare, and the viability of the market solution through the financial reward to the annuity provider's equityholders. We find that individuals prefer to bear the risk under a collective arrangement than to insure it with a life insurers' annuity contract subject to insolvency risk (albeit small). Under realistic capital provision hypotheses, the annuity provider is incapable of adequately compensating its equityholders for bearing systematic longevity risk.

Boreiko, D. and Massarotti, F. (2020). "How Risk Profiles of Investors Affect Robo-Advised Portfolios." In: *Frontiers in Artificial Intelligence* 3.

Automated financial advising (robo-advising) has become an established practice in wealth management, yet very few studies have looked at the cross-section of the robo-advisors and the factors explaining the persistent variability in their portfolio allocation recommendations. Using a sample of 53 advising platforms from the US and Germany, we show that the underlying algorithms manage to identify different risk profiles, although substantial variability is evident even within the same investor types' groups. The robo-advisor expertise in a particular asset class seems to play a significant role, as does the geographical location, while the breadth of the offered investment choice (number of portfolios) across the robo-advisors under study does not seem to have an effect.

Boscaljon, B., Filbeck, G., and Ho, C.-C. (2008). "Rebalancing strategies for creating efficient portfolios." In: *The Journal of Investing* 17(2), pp. 93–103.

This article applies an annual rebalancing strategy to create portfolios of industry leaders and compares the efficiency of these portfolios with the S&P 500 Index and the CRSP market index portfolios. For the last four decades, value-weighted portfolios consisting of as few as eight or nine securities formed from industry leaders are more efficient with annual rebalancing than the S&P 500 and are indistinguishable from the CRSP market index portfolio. The findings suggest important implications for choosing appropriate benchmarks for measuring tracking error.

Botteron, S., Courbage, C., and Wagner, J. (2024). "On the Motivations for Purchasing Long-Term Care Insurance: Protecting Bequest and Unreliability of Family Care." In: *Risks* 12(8), p. 124.

Family considerations are known to influence the decision to buy long-term care (LTC) insurance. This paper uses a Swiss survey to identify the characteristics of individuals willing to purchase LTC insurance, either to protect their children's bequest or because they cannot rely on family for care. First, it shows that the presence or absence of children plays an important role in the two motivations for buying LTC insurance. Second, it shows that individuals from the French-speaking part of Switzerland and those with lower self-perceived health are more likely to buy LTC insurance because of the unreliability of family care. On the other hand, individuals with higher self-perceived health and those with a right and center political orientation are more likely to buy LTC insurance for reasons of bequest protection. The results provide insights into designing more targeted strategies to promote LTC insurance.

Bouchey, P. (2024). "The Direct Index Playbook for Wealth Managers." In: *The Journal of Beta Investment Strategies* 15(3), pp. 118–131.

Direct indexing is more than just a separate account that provides passive exposure to an index. It's a way for investors to track an index, customize their exposure, and generate realized capital losses. These losses create a tax benefit and also the opportunity to integrate the direct index into a number of wealth management strategies. This article is a playbook that describes the various applications of direct indexing in wealth management.

Bouchey, P., Edwards, S., and Cavallo, S. (2023). "Using a Direct Index in a Core–Satellite Portfolio." In: *The Journal of Beta Investment Strategies*.

The value of direct indexing in enhancing overall after-tax returns has been well established. But where does direct indexing fit in an investor's portfolio? In this article, we demonstrate the potential benefits of building a portfolio using a core-satellite approach. We consider multiple investment vehicles and approaches-passive ETFs, direct indexing, and active investment strategies-and create several possible implementation paths for investors to consider.

Brauer, K. (2021). "Nudged into Better Portfolios and Lower Risk: Robo-Advice and Savings Decisions." In: *SSRN e-Print*.

Mutual fund and ETF savings plans (SPs) are becoming an increasingly important part of retail investor portfolios and many investors now adopt robo-advisors to obtain guidance on SP choices. Using data from a large online bank that introduced a robo-advising tool, I explore how robo-advice changes investors' SP choices and document three main results. First, default options improve robo-advice users' fund choices towards lower-cost and more diversified funds. Second, many investors - also those who previously held all-equity portfolios - adhere to the default asset allocation that is associated with a 50% equity exposure, although they could construct riskier SP portfolios through the robo-advisor. Third, I document considerable heterogeneity in longer-term adherence to robo-advisor recommendations. First-time SP users are more inert and stick to the robo-advisor's proposed asset allocation while experienced SP users quickly readjust their equity exposure away from the robo-advisor's recommendation. My results emphasize the power of defaults in all-digital robo-advisory services and highlight how they can improve fund choices while at the same time push investors into unsuitable asset allocations.

Bravo, J. M., Ayuso, M., Holzmann, R., and Palmer, E. (2023). "Intergenerational actuarial fairness when longevity increases: Amending the retirement age." In: *Insurance: Mathematics and Economics* 113, pp. 161–184.
Continuous longevity improvements and population ageing have led countries to modify national public pension schemes by increasing standard and early retirement ages in a discretionary, scheduled, or automatic way, and making it harder for people to retire prematurely. To this end, countries have adopted alternative retirement age strategies, but our analyses show that the measures taken are often poorly designed and consequently misaligned with the pension scheme's ultimate goals. This paper discusses how to implement automatic indexation of the retirement age to life expectancy developments while respecting the principles of intergenerational actuarial fairness and neutrality among generations of the respective policy scheme design. With stable demographic conditions, we show in policy designs in which extended working lives translate into additional pension entitlements, the pension age must be automatically updated to keep the period in retirement constant. Alternatively, policy designs that pursue a fixed replacement rate are consistent with retirement age policies targeting a constant balance between active years in the workforce and years in retirement. Under conditions of population ageing, the statutory pension age will have to increase at a faster rate to meet the intergenerational equity criteria. The empirical strategy employed a Bayesian Model Ensemble approach to stochastic mortality modelling to address model risk and generate forecasts of intergenerationally and actuarially fair pension ages for 23 countries from 2000 to 2050. The findings show that the pension age increases needed to accommodate the effect of longevity developments on pay-as-you-go equilibrium and to reinstate equity between generations are sizeable and well beyond those employed and/or legislated in most countries. A new wave of pension reforms may be at the doorsteps.

Bravo, J. M. (2019). "Funding for longer lives. Retirement wallet and risk-sharing annuities." In: *Ekonomiaz: Revista vasca de economia*.
Longevity increases and population ageing create challenges for all societal institutions, particularly those providing retirement income, health care, and long-term care services. At the individual level, an obvious question is how to ensure all retirees have an adequate, secure, stable and predictable lifelong income stream that will allow them to maintain a target standard of living for however long the individual lives. In this paper we introduce and discuss the concept of retirement wallet representing the multiple income and service sources individuals and their families will have to fund for longer lives. We then address the main decumulation risks and options, including the adoption of a given longevity insurance strategy, of a programmed withdrawal strategy and of an investment strategy. The main payout options available for allocating assets accumulated in pension plans are discussed, particularly the role of traditional and innovative investment and longevity risk-sharing structures. We provide illustrative results for the price of innovative participating longevity-linked life annuities (PLLAs) that link benefits to the dynamics of both a longevity index and an interest rate adjustment factor using Spanish mortality and financial market data.

Bravo, J. M. V. (2020). "Addressing the Pension Decumulation Phase of Employee Retirement Planning." In: *Who Wants to Retire and Who Can Afford to Retire?* IntechOpen.
Longevity increases and population ageing create challenges for all societal institutions, particularly those providing retirement income, healthcare, and long-term care services. At the individual level, an obvious question is how to ensure all retirees have an adequate, secure, stable, and predictable lifelong income stream that will allow them to maintain a target standard of living for, however, long the individual lives. In this chapter, we review and discuss the main pension decumulation options by explicitly modelling consumers' behaviour and objectives though an objective function based on utility theory accounting for consumption and bequest motives and different risk preferences. Using a Monte-Carlo simulation approach calibrated to US financial market and mortality data, our results suggest that purchasing a capped participating longevity-linked life annuity at retirement including embedded longevity and financial options that allow the annuity provider to periodically revise annuity payments if observed survivorship and portfolio outcomes deviate from expected (or guaranteed) values at contract initiation deliver superior welfare results when compared with classical annuitization and non-annuitization decumulation strategies.

Breach, T., D'Amico, S., and Orphanides, A. (2020). "The term structure and inflation uncertainty." In: *Journal of Financial Economics* 138(2), pp. 388–414.
To assess the importance of inflation risk for nominal Treasury yields, a novel quadratic term structure model with time-varying inflation risk is estimated using survey-based inflation uncertainty. The resulting yield decomposition captures very diverse macroeconomic dynamics of inflation and real risk premiums (large and positive during the 1980s but small and negative post-2008) and generates sensible high-frequency estimates of expected

inflation and real short rates over a long sample. The explicit link between the model-implied factors and macro fundamentals reveals that short- but not long-run fluctuations are unspanned by yields, consistent with an interest rate policy unresponsive to transient inflation shocks.

Bronshtein, G., Scott, J., Shoven, J. B., and Slavov, S. N. (2020). "Leaving big money on the table: Arbitrage opportunities in delaying social security." In: *The Quarterly Review of Economics and Finance* 78, pp. 261–272.
Even though delaying Social Security is equivalent to purchasing a very favorably priced annuity, almost everyone takes Social Security at or before their full retirement age. Many who take Social Security early simultaneously report additional annuity income. This combination can create an arbitrage opportunity where an individual could explicitly or implicitly sell their relatively high priced annuity, use the proceeds to delay Social Security and secure higher income for life. Several hundred thousand, perhaps millions, of households fail to take advantage of this arbitrage opportunity, with a resulting loss that ranges up to $250,000.

Brown, R. (2024). "Ex-Post Cherry Picking." In: *The Journal of Wealth Management* 27(3), pp. 28–47.
Financial planners and institutional investment consultants are making a mistake-one that is problematic and meaningful. They are basing their forward-looking capital market assumptions (CMAs) unduly on US stock/bond returns during the post-industrial era (the last 75 years). This is a problem because US returns during this period were exceptional and non-repeatable, as the result of relative American exceptionalism during the post-WWII era. But this is nothing new, exceptional, rare, or unexpected. Quite the opposite: it is fully expected, normal, and even vanilla. Throughout history (the last 1,000 years), there has always been a top-dog nation or economy. The determination of which nation would hold this role was determined by relative comparative advantage, defined within the context of current-day technologies and economic structures. India, China, Portugal, Spain, the Netherlands, France, the UK, and-most recently-the US all held this enviable position. But for good reasons, no nation has ever been able to maintain permanent dominance. The US is no different, and its rise and inevitable fall are consistent with the last 1,000 years of economic development. As a consequence, setting forward-looking CMAs based on a rare and exceptional period drawn from history (essentially, ex-post cherry picking both the geography and time period) is offensive, and a clear violation of due diligence and fiduciary duty. We can do better as advisors/consultants to our clients. This article attempts to lay out the data that supports the argument that American exceptionalism was real, genuine, and vital during the post-industrial era, and that this exceptionalism was driven by clear and unambiguous relative advantages, which are no more. As a consequence, forecasting that future US investment market returns will continue to be 'higher return' and 'lower risk' than the rest of the world is unwarranted and lacks foundation.

Brown, R. A. (2018). "Intelligent Rebalancing." In: *The Journal of Investing* 27(1), pp. 31–42.
The rebalancing rule applied to a multiasset class portfolio will have significant impact on both risk and return. All portfolios managed to a fixed-weight policy asset allocation using non-market-cap weights require periodic rebalancing because failure to rebalance drives realized asset weights further and further away from policy with the passage of time. The most common approach to rebalancing adopted by the registered investment advisor, broker/dealer, and bank trust department segments of the investment industry is fixed time (e.g., monthly or quarterly). This article demonstrates that frequent fixed-time rebalancing rules are the most harmful to both risk and return, serving to increase risk and to reduce return. Improvements can be made by rebalancing less frequently. Additional enhancements to both risk and return can be obtained by following rebalancing rules that reflect recent market behaviors (i.e., are path dependent).

Brown, T. (2022). "Women in Retirement: Living Longer, Less Money Saved." In: *The Journal of Retirement* 9(4).
We frequently read or hear about women saving less money for retirement while at the same time living longer. Women need to fund more years in retirement than men. Does this apply to all groups of women, regardless of race/ethnicity? It's often stated this is not a retirement issue but rather an issue of pay inequity prior to retirement. Data suggests it is a issue of unequal pay, but that doesn't mean the retirement industry can't act to improve the situation for women employees. Let's look more closely, think more deeply, focus more acutely. What can be done to help women of all races/ethnicities? A good start is to design a stronger defined contribution plan, implement auto features at greater levels, understand the employee base, and create effective, customized communication. These actions will drive women of all races/ethnicities forward to a more equal retirement footing with men.

Browning, C., Guo, T., Cheng, Y., and Finke, M. S. (2020). "Spending in Retirement." In: *SSRN e-Print*.
Retirement savings adequacy estimates are often based on the assumption that individuals spend the same amount every year in retirement, and that the withdrawal rate to fund spending is based on spending down a percentage of retirement savings. We simulate safe consumption rates and compare the amount that wealthy

retirees are spending to the amount they could safely spend given different asset return assumptions and investment portfolio allocations. Retirees in the top quintile of financial wealth are spending nowhere near an amount that would place them in danger of running out of money. In fact, the average financial assets of wealthy retirees increased during this time period and most retirees spent less than their income. Setting aside 40% of financial assets to cover uncertain longevity, medical costs, and bequests still results in a consumption gap as high as 47.3.% among higher-wealth retirees.

Bruder, B., Caudamine, L., Do, H., Gisimundo, V., Granjon, C., and Xu, J. (2023). "Optimal Decumulation Strategies for Retirement Solutions." In: *SSRN e-Print*.
Traditional pension systems based on pay-as-you-go (PAYG) have become more difficult to sustain in developed countries as a result of continued population ageing, fewer children in the family and weak economic growth. Over the past decades, pension systems have gradually shifted from defined-benefit (DB) plan to defined-contribution (DC) plan. In DC pension schemes, there is no guarantee of pension payments after retirement, so it is important to develop appropriate decumulation strategies to efficiently convert wealth into income. The first part of this working paper systematically reviews two types of decumulation strategies: safe withdrawal rate methods and optimal control theory-based methods. The second part presents simulations of these strategies. In addition, we investigate the robustness of these strategies by simulating non-normally distributed returns of risky assets, such as Student's t distribution and skew normal distribution. Finally, we test these strategies with forward-looking financial data simulated by Amundi Cascade Asset Simulation Model.

Brunel, J. L. P. (2015). *Goals-Based Wealth Management*. Hoboken, NJ, USA: Wiley. 212 pp.
Goals-Based Wealth Management is a manual for protecting and growing client wealth in a way that changes both the services and profitability of the firm. Written by a 35-year veteran of international wealth education and analysis, this informative guide explains a new approach to wealth management that allows individuals to take on a more active role in the allocation of their assets. Coverage includes a detailed examination of the goals-based approach, including what works and what needs to be revisited, and a clear, understandable model that allows advisors to help individuals to navigate complex processes. The companion website offers ancillary readings, practice management checklists, and assessments that help readers secure a deep understanding of the key ideas that make goals-based wealth management work. The goals-based wealth management approach was pioneered in 2002, but has seen a slow evolution and only modest refinements largely due to a lack of wide-scale adoption. This book takes the first steps toward finalizing the approach, by delineating the effective and ineffective aspects of traditional approaches, and proposing changes that could bring better value to practitioners and their clients. (1) Understand the challenges faced by the affluent and wealthy (2) Examine strategic asset allocation and investment policy formulation (3) Learn a model for dealing with the asset allocation process (4) Learn why the structure of the typical advisory firm needs to change High-net-worth individuals face very specific challenges. Goals-Based Wealth Management focuses on how those challenges can be overcome while adhering to their goals, incorporating constraints, and working within the individual's frame of reference to drive strategic allocation of their financial assets.

Brunel, J. L. P. (2020). "Extending the Goals-Based Framework to Comprise Both Investment and Financial Planning." In: *The Journal of Wealth Management* 22(44), pp. 21–26.
This article seeks to fill a void in the literature on goals-based planning. Most of the current work covers cases where clients already possess significant financial assets that they plan on totally or partially spending down, through expenses as well as various dynastic or philanthropic transfers. Yet, planners–be they focused on income/savings/expense management issues or conduct their work in the asset management sphere–have to deal with at least two other potential cases. Our analysis suggests that the goals-based planning approach has the potential to inject more texture into conversations with all clients. It also shows that there are significant, and at times even dramatic, differences in the ways one might deal with 1) an individual who is already wealthy and spending down his or her wealth, 2) another who may be wealthy, but has unrealized non-financial wealth and ongoing current savings inflows, and 3) yet another who has a significant income and savings power but not enough accumulated financial wealth to live on. This is nothing more than the classical case of the difference between human and financial capital. Individuals in an asset decumulation mode have more financial than human capital, while those who are in an accumulation mode have more human than financial capital. As one might expect, this describes a full spectrum, and our "accumulation/decumulation case" falls somewhere between these extremes.

Brunel, J. L. P. (2023). "Looking Back at the Past 25 Years and Forward to the Next 25." In: *The Journal of Wealth Managemen* 26(1), pp. 14–19.

Earlier in the second half of the 20th century, asset management evolved dramatically as it became common practice to revisit certain principles which had underpinned the institutional world: the introduction of quantitative practices in general and the "discovery" of the efficient frontier in particular forced a rethink of risk-taking and thus portfolio construction. Index investing probably is the single most visible development still visible today. The private wealth world waited longer to embark on the same path of reformation, but made up for its initial lateness by innovating at a fast pace since the early 1990s. Three crucial changes accompanied a gradual shift from the inherited wealth, as the main client, to the advent of new wealth, which was arguably more prepared to ask for new approaches. The recognition that taxes matter and that tax day is everyday was, chronologically, the first, though it capitalized in part on work that had been carried out in the institutional world with certain insurance companies. The advent of open architecture simply reflected a reality which already existed in the institutional world, but was not necessarily appreciated by managers who were "product-driven." This allowed a broadening of the strategies offered to individuals, including the use of alternative and illiquid strategies. Finally, the recognition of the fact that individuals have multiple goals, multiple time horizons and different risk tolerances for different goals brought Goals-Based Wealth Management, with a key catalyst being the work of Das, Markowitz, Scheid and Statman in 2010. Looking ahead, one can see that a lot still has to be done, with an important element probably involving the integration of insurance into the wealth management mix.

Bruni, R., Cesarone, F., Scozzari, A., and Tardella, F. (2016). "Real-world datasets for portfolio selection and solutions of some stochastic dominance portfolio models." In: *Data in Brief* 8, pp. 858–862.

A large number of portfolio selection models have appeared in the literature since the pioneering work of Markowitz. However, even when computational and empirical results are described, they are often hard to replicate and compare due to the unavailability of the datasets used in the experiments. We provide here several datasets for portfolio selection generated using real-world price values from several major stock markets. The datasets contain weekly return values, adjusted for dividends and for stock splits, which are cleaned from errors as much as possible. The datasets are available in different formats, and can be used as benchmarks for testing the performances of portfolio selection models and for comparing the efficiency of the algorithms used to solve them. We also provide, for these datasets, the portfolios obtained by several selection strategies based on Stochastic Dominance models (see "On Exact and Approximate Stochastic Dominance Strategies for Portfolio Selection" (Bruni et al. [2])). We believe that testing portfolio models on publicly available datasets greatly simplifies the comparison of the different portfolio selection strategies.

Bruni, R., Cesarone, F., Scozzari, A., and Tardella, F. (2017). "On exact and approximate stochastic dominance strategies for portfolio selection." In: *European Journal of Operational Research* 259(1), pp. 322–329.

New type of approximate stochastic dominance designed for portfolio selection. Equivalent to minimizing the expected shortfall of the portfolio below the benchmark. An easily solvable LP model for the practical implementation of our approach. Extensive empirical comparison of stochastic dominance models for portfolio selection. One recent and promising strategy for Enhanced Indexation is the selection of portfolios that stochastically dominate the benchmark. We propose here a new type of approximate stochastic dominance rule which implies other existing approximate stochastic dominance rules. We then use it to find the portfolio that approximately stochastically dominates a given benchmark with the best possible approximation. Our model is initially formulated as a Linear Program with exponentially many constraints, and then reformulated in a more compact manner so that it can be very efficiently solved in practice. This reformulation also reveals an interesting financial interpretation. We compare our approach with several exact and approximate stochastic dominance models for portfolio selection. An extensive empirical analysis on real and publicly available datasets shows very good out-of-sample performances of our model.

Bryzgalova, S., Huang, J., and Julliard, C. (2021). "Bayesian solutions for the factor zoo: we just ran two quadrillion models." In: *SSRN e-Print*.

We propose a novel, and simple, Bayesian estimation and model selection procedure for cross-sectional asset pricing. Our approach, that allows for both tradable and non-tradable factors, and is applicable to high dimensional cases, has several desirable properties. First, weak and spurious factors lead to diffuse, and centered at zero, posteriors for their market price of risk, making such factors easily detectable. Second, posterior inference is robust to the presence of such factors. Third, we show that flat priors for risk premia lead to improper marginal likelihoods, rendering model selection invalid. Therefore, we provide a novel prior, that is diffuse for strong factors but shrinks away useless ones, under which posterior probabilities are well behaved, and can be used for factor and (non necessarily nested) model selection, as well as model averaging, in large scale problems. We apply our method to a very large set of factors proposed in the literature, and analyse 2.25 quadrillion possible models, gaining novel insights on the empirical drivers of asset returns.

Buehler, H., Phillip, M., and Wood, B. (2022). "Deep Bellman Hedging." In: *SSRN e-Print*.
We present an actor-critic-type reinforcement learning algorithm for solving the problem of hedging a portfolio of financial instruments such as securities and over-the-counter derivatives using purely historic data. The key characteristics of our approach are: he ability to hedge with derivatives such as forwards, swaps, futures, options; incorporation of trading frictions such as trading cost and liquidity constraints; applicability for any reasonable portfolio of financial instruments; realistic, continuous state and action spaces; and formal risk-adjusted return objectives. Most importantly, the trained model provides an optimal hedge for arbitrary initial portfolios and market states without the need for re-training. We also prove existence of finite solutions to our Bellman equation, and show the relation to our vanilla Deep Hedging approach.

Buetow, G. W. and Hanke, B. (2020). "How Long is Long Enough?" In: *The Journal of Retirement* 88(22), pp. 39–48.
Defined contribution (DC) plan (DCP) fiduciaries are often faced with conflicting perspectives when executing their responsibilities and terminating underperforming active managers are involved. Consultants, investment managers, and capital market intermediaries generally argue that more time is needed to prove that an investment strategy is suboptimal. These parties often have inherent conflicts of interest to argue for active management over passive management. Most conflicts are economic in nature, but some are steeped in intellectual hubris. In this article the authors enter the fray from the plan participants' (PPs) side. Ultimately, the very essence of a DCP is to offer a menu of investment options that enable PPs to optimize wealth aggregation in a diversified manner using a multiasset class solution. We show that PPs are better served when fiduciaries monitor active investment managers and replace them with passive alternatives in a timely manner if they underperform. Too often plan fiduciaries churn a DCP by replacing underperforming active funds with other active funds.

Buetow, G. W., Hanke, B., and Zagonov, M. (2020). "Active management in defined contribution plans." In: *The Journal of Retirement* 7(4), pp. 61–79.
We analyze the problem that fiduciaries face when monitoring and selecting from a universe of active mutual funds within a defined contribution (DC) plan. In a DC plan, a fiduciary must recognize that there are two levels of decision makers, namely the fiduciary, who decides which funds will comprise the DC plan and the individual plan participants, who must decide which funds to invest in and the timing of their investment. Moreover, plan participants, and to some degree the fiduciary, need to be able to make investment decisions without being investment professionals. We find that due to the general lack of consistency in performance of mutual funds, fiduciaries and plan participants would be better served by selecting passive rather than active funds across the US equity mutual fund space. Moreover, the most consistently outperforming funds tend to have meaningfully higher tracking errors relative to their stated benchmarks, which makes effective asset allocation in a DC plan more difficult.

Butt, A., Khemka, G., Lim, W., and Warren, G. J. (2023). "Primer on Retirement Income Strategy Design and Evaluation." In: *Society of Actuaries*.
This 'handbook' covers a wide range of aspects around retirement income strategies including: member attributes that matter; types of income objectives; nature of income risk; approaches to investing and drawing down on available assets; stochastic modelling; and the use of utility and metrics to characterise outcomes and identify a suitable retirement strategy. The analysis is supported excel models that are available to download. The Primer is based around the idea of three types of income objectives - an income floor, income target and income optimization - and shows how this framing flows into both strategy selection and evaluation with each objective calling for different strategies and a different set of metrics. Access may be gained through the SOA website at https://www.soa.org/resources/research-reports/2023/ret-income-strat-de/.

Butt, A., Khemka, G., and Warren, G. J. (2021). "Principles and Rules for Translating Retirement Objectives into Strategies." In: *SSRN e-Print*.
This article sets out principles and decision rules for setting appropriate drawdown and investment strategies during retirement given an individual's objectives and risk tolerance. In particular, we highlight how the suitable drawdown strategy can relate to the objective, and how annuities might be combined with other investments to help manage risk. Our aim is to offer research-based guidance to product providers, financial advisers and individuals in formulating retirement strategies.

Cairns, A. J. G., Blake, D., Dowd, K., and Coughlan, G. (2020). "A Two-Factor Model for Stochastic Mortality with Parameter Uncertainty: Theory and Calibration." In: *SSRN e-Print* (s).

We use a case study of a pension plan wishing to hedge the longevity risk in its pension liabilities at a future date. The plan has the choice of using either a customised hedge or an index hedge, with the degree of hedge effectiveness being closely related to the correlation between the value of the hedge and the value of the pension liability. The key contribution of this paper is to show how correlation and, therefore, hedge effectiveness can be broken down into contributions from a number of distinct types of risk factors. Our decomposition of the correlation indicates that population basis risk has a significant influence on the correlation. But recalibration risk as well as the length of the recalibration window are also important, as is cohort effect uncertainty. Having accounted for recalibration risk, additional parameter uncertainty has only a marginal impact on hedge effectiveness. Finally, the inclusion of Poisson risk only starts to become significant when the smaller population falls below about 10,000 members over age 50. Our case study shows that, at least for medium and large pension plans, longevity risk can be substantially hedged using index hedges as an alternative to customised longevity hedges. As a consequence, when the hedger's population involves more than about 10,000 members over age 50, index longevity hedges (in conjunction with the other components of an ALM strategy) can provide an effective and lower cost alternative to both a full buy-out of pension liabilities or even to a strategy using customised longevity hedges.

Camp, J., Kuselias, S., and Manley, S. (2025). "To Roth or Not to Roth? When Making After-Tax Contributions to a 401(k) Makes Sense." In: *The Journal of Wealth Management* 27(4), pp. 68–79.

Popular media regularly exhort readers and investors to strongly consider after-tax contributions to retirement accounts. Although the advice is sometimes caveated, the implication is that sophisticated investors use Roth accounts. However, this appears to shortchange the extraordinary value of delaying a tax payment for decades when making pre-tax contributions via Traditional retirement accounts. In this article, we look at the present value of tax payments associated with two retirees who save exactly the same amounts over time but choose different retirement vehicles, a Roth 401(k) and a Traditional 401(k). Assuming no major shifts in tax policy, we find that for the majority of taxpayers, Traditional retirement contributions ultimately save taxpayers a substantial amount of money on total tax payments across their lifetime. Our results should strongly caution media outlets and financial advisors recommending Roth retirement accounts for a broad swath of taxpayers. Unless there are significantly unfavorable changes to the tax code in the future, it is likely that only a minority of taxpayers will benefit from lower taxes when saving only through a Roth.

Campasano, J. (2024). "Mitigating Risk with Conditional Option Strategies." In: *The Journal of Alternative Investments* 27(2), pp. 73–89.

Although hedging strategies protect equity portfolios during downturns they impair performance in the long run. This study exams a risk mitigation strategy that conditions on the level of the VIX Index. When the VIX Index is above its historical average, a call is written on an index portfolio. When it is below its historical average, a protective put is purchased. Over the sample period 2005 to 2022 the strategy outperforms an index portfolio, traditional call writing, and put protection strategies on both an absolute and relative basis.

Campbell, J. Y. and Sigalov, R. (2022). "Portfolio choice with sustainable spending: A model of reaching for yield." In: *Journal of Financial Economics* 143(1), pp. 188–206.

We show that reaching for yield-a tendency to take more risk when the real interest rate declines while the risk premium remains constant-results from imposing a sustainable spending constraint on an otherwise standard infinitely lived investor with power utility. When the interest rate is initially low, reaching for yield intensifies. The sustainable spending constraint also affects the response of risk-taking to a change in the risk premium, which can even change sign. In a variant of the model where the sustainable spending constraint is formulated in nominal terms, low inflation also encourages risk-taking.

Canarella, G., Gil-Alana, L. A., Gupta, R., and Miller, S. M. (2020). "Modeling US historical time-series prices and inflation using alternative long-memory approaches." In: *Empirical economics* 58, pp. 1491–1511.

We consider two important features of the historical US price data (1774-2015), namely the data persistence and cyclical structure. We first consider the persistence of the series and focus on standard long-memory models that incorporate a peak at the zero frequency. We examine different models with respect to the deterministic terms, including nonlinear deterministic trends of the Chebyshev form. Then, we investigate a more general model that includes both persistence and cyclicality of the series and, thus, includes two fractional integration parameters, one at the zero (long-run) frequency and the other at the nonzero (cyclical) frequency. We model the cyclical structure as a Gegenbauer process. This specification outperforms the standard long-memory specifications. We find that the order of integration at the zero frequency is about 0.5, and the one at the cyclical frequency is about 0.2 with cycles repeating approximately every 6 years, producing mean-reverting long-memory effects at both the zero and cyclical frequencies. Fitting the values to this model, however, we discover the presence of a break that, according to the methods employed, takes place at around 1940-1941. The results indicate the prevalence of the long-run or zero component with a much higher degree of persistence during the second post-1940-1941 subsample, suggesting important implications for monetary policy.

Caner, M., Fan, Q., and Li, Y. (2024). "Navigating Complexity: Constrained Portfolio Analysis in High Dimensions with Tracking Error and Weight Constraints." In: *arXiv e-Print*.

This paper analyzes the statistical properties of constrained portfolio formation in a high dimensional portfolio with a large number of assets. Namely, we consider portfolios with tracking error constraints, portfolios with tracking error jointly with weight (equality or inequality) restrictions, and portfolios with only weight restrictions. Tracking error is the portfolio's performance measured against a benchmark (an index usually), and weight constraints refers to specific allocation of assets within the portfolio, which often come in the form of regulatory requirement or fund prospectus. We show how these portfolios can be estimated consistently in large dimensions, even when the number of assets is larger than the time span of the portfolio. We also provide rate of convergence results for weights of the constrained portfolio, risk of the constrained portfolio and the Sharpe Ratio of the constrained portfolio. To achieve those results we use a new machine learning technique that merges factor models with nodewise regression in statistics. Simulation results and empirics show very good performance of our method.

Capolongo, A. and Pacella, C. (2021). "Forecasting inflation in the euro area: countries matter!" In: *Empirical Economics* 61, pp. 2477–2499.

We construct a Bayesian vector autoregressive model with three layers of information: the key drivers of inflation, cross-country dynamic interactions, and country-specific variables. The model provides good forecasting accuracy with respect to the popular benchmarks used in the literature. We perform a step-by-step analysis to shed light on which layer of information is more crucial for accurately forecasting medium-run euro area inflation. Our empirical analysis reveals the importance of including the key drivers of inflation and taking into account the multi-country dimension of the euro area. The results show that the complete model performs better overall in forecasting inflation excluding energy and unprocessed food over the medium term. We use the model to establish stylized facts on the euro area and cross-country heterogeneity over the business cycle.

Capponi, A. (2023). "A Continuous Time Framework for Sequential Goals Investment Management." In: *Fields CFI Workshop on Quantitative Methods for Wealth Management*.

We develop a continuous time framework for sequential goals-based investing, where the objective is to maximize expected weighted utility from goal fundedness. A stochastic factor process drives price dynamics, goal amounts, and cash infusion into the client's portfolio. Using a generalization of the mean-variance efficient frontier, we show that the computational complexity of the procedure to recover the optimal control can be significantly reduced by rewriting the HJB equation in terms of standard deviation and mean of minimum-variance portfolio returns, and covariance of such portfolio returns with the state process. Our analysis highlights the fundamental tradeoff between immediate goal consumption versus saving towards future goal liabilities. We find that it is optimal to fund an expiring goal up to the level where the marginal benefit of additional fundedness is exceeded by the marginal opportunity cost of subtracting wealth from future goals. An investor with all-or-nothing utility is more risk averse towards an approaching goal deadline if she is well funded, but also takes excessive risk if she is not on track with upcoming goals, compared to an investor who maximizes the weighted expected goal fundedness.

Capponi, A., Olafsson, S., and Zariphopoulou, T. (2020). "Personalized Robo-Advising: Enhancing Investment through Client Interaction." In: *arXiv e-Print*.

Automated investment managers, or robo-advisors, have emerged as an alternative to traditional financial advisors. The viability of robo-advisors crucially depends on their ability to offer personalized financial advice. We introduce a novel framework, in which a robo-advisor interacts with a client to solve an adaptive mean-variance portfolio optimization problem. The risk-return tradeoff adapts to the client's risk profile, which depends on idiosyncratic characteristics, market returns, and economic conditions. We show that the optimal investment strategy includes both myopic and intertemporal hedging terms which are impacted by the dynamics of the client's risk profile. We characterize the optimal portfolio personalization via a tradeoff faced by the robo-advisor between receiving client information in a timely manner and mitigating behavioral biases in the risk profile communicated by the client. We argue that the optimal portfolio's Sharpe ratio and return distribution improve if the robo-advisor counters the client's tendency to reduce market exposure during economic contractions when the market risk-return tradeoff is more favorable.

Capponi, A. and Zhang, Y. (2022). "Goal Based Investment Management." In: *SSRN e-Print*.
We develop a continuous time framework for goals-based investing, where the objective is to maximize the priority-weighted fundedness of the clients' goals. We show that the value function of the control problem can be recovered as the unique viscosity solution to a nonlinear Hamilton-Jacobi Bellman equation. Our analysis highlights the fundamental tradeoff between immediate goal consumption versus savings toward future goal liabilities. We show that it is optimal to fund an expiring goal up to the level where the marginal benefit of consuming wealth is exceeded by the marginal opportunity cost of funding future goals. This tradeoff depends crucially on the interplay between goal amounts, goal deadlines and guaranteed income stream. Our numerical comparative statics analysis reveals that the risk aversion towards an upcoming goal increases if its deadline is approaching or if its priority relative to future goals increases.

Capponi, A. and Zhang, Y. (2024). "A Continuous Time Framework for Sequential Goal-Based Wealth Management." In: *Management Science*.
We develop a continuous time framework for sequential goals-based wealth management. A stochastic factor process drives asset price dynamics and the client's goal amount and income. We prove the weak dynamic programming principle for the value function of our control problem, which we show to be the unique viscosity solution of the corresponding Hamilton-Jacobi-Bellman equation. We develop an equivalent and computationally efficient representation of the Hamiltonian, which yields the optimal portfolio within a factor-dependent opportunity set defined by the maximum and minimum variance hypersurfaces. Our analysis shows that it is optimal to fund an expiring goal up to the level where the marginal benefit of additional fundedness is exceeded by the opportunity cost of diverting wealth from future goals. An all-or-nothing investor is more risk averse toward an approaching goal deadline if well funded, but more risk seeking if not on track with upcoming goals, compared with an investor with flexible goals.

Capponi, A. and Zhang, Z. (2020). "Risk Preferences and Efficiency of Household Portfolios." In: *arXiv e-Print*.
We propose a novel approach to infer investors' risk preferences from their portfolio choices, and then use the implied risk preferences to measure the efficiency of investment portfolios. We analyze a dataset spanning a period of six years, consisting of end of month stock trading records, along with investors' demographic information and self-assessed financial knowledge. Unlike estimates of risk aversion based on the share of risky assets, our statistical analysis suggests that the implied risk aversion coefficient of an investor increases with her wealth and financial literacy. Portfolio diversification, Sharpe ratio, and expected portfolio returns correlate positively with the efficiency of the portfolio, whereas a higher standard deviation reduces the efficiency of the portfolio. We find that affluent and financially educated investors as well as those holding retirement related accounts hold more efficient portfolios.

Carbonneau, A. (2020). "Deep Hedging of Long-Term Financial Derivatives." In: *arXiv e-Print*.
This study presents a deep reinforcement learning approach for global hedging of long-term financial derivatives. A similar setup as in Coleman et al. (2007) is considered with the risk management of lookback options embedded in guarantees of variable annuities with ratchet features. The deep hedging algorithm of Buehler et al. (2019a) is applied to optimize neural networks representing global hedging policies with both quadratic and non-quadratic penalties. To the best of the author's knowledge, this is the first paper that presents an extensive benchmarking of global policies for long-term contingent claims with the use of various hedging instruments (e.g. underlying and standard options) and with the presence of jump risk for equity. Monte Carlo experiments demonstrate the vast superiority of non-quadratic global hedging as it results simultaneously in downside risk metrics two to three times smaller than best benchmarks and in significant hedging gains. Analyses show that the neural networks are able to effectively adapt their hedging decisions to different penalties and stylized facts of risky asset dynamics only by experiencing simulations of the financial market exhibiting these features. Numerical results also indicate that non-quadratic global policies are significantly more geared towards being long equity risk which entails earning the equity risk premium.

Carl, U. (2017). "Understanding Rebalancing and Portfolio Reconstitution." In: *SSRN e-Print*.
This paper analyses the impact of rebalancing and portfolio reconstitution on portfolio returns and factor exposures. Varying the rebalancing frequency and the portfolio reconstitution frequency leads to distinct patterns in relative factor exposures. These patterns are symmetric for rebalancing and portfolio reconstitution. Short term reversal drives the returns at high frequencies, momentum at intermediate frequencies, while value and long term reversal stand out at low frequencies. The variation in returns at different frequencies can be linked to macro-economic variables, in particular the cross-sectional volatility.

Cassidy, D. P., Peskin, M. W., Siegel, L. B., and Sexauer, S. (2013). "Be Kind to Your Retirement Plan - Give It a Benchmark." In: *The Journal of Retirement* 1(1), pp. 81–90.
The performance of any retirement plan needs to be gauged against an appropriately chosen benchmark. We explain the importance of making transparent the risks taken in retirement portfolios, especially QDIA portfolios, and indicate how benchmarks help with this task. Benchmarks should be chosen or constructed for both the accumulation and decumulation phases of a plan, and should have a risk level appropriate to the age and situation of investors. For example, the S&P 500 is a poor benchmark to use when investors are five years from retirement. Given the importance of income to retirees, a benchmark for decumulation should be based on matching assets to liabilities or planned spending, and it should minimize longevity, investment, counterparty, and liquidity risks. We give an example of a decumulation benchmark, which is based on a strategy of holding TIPS and deferred life annuities. This is a call to action: Only by adopting the best practice of choosing and using an appropriate benchmark for retirement portfolios-accumulation and decumulation-can fiduciaries execute the responsibility they have agreed to take on.

Cavaglia, S., Scott, L., Blay, K., and Gupta, T. (2022). "Equity factors for multi-asset class portfolios: a strategic asset allocation perspective." In: *Journal of Asset Management* 23, pp. 100–113.
This paper highlights the long run, strategic benefits of factor premia as a complement (overlay) to an underlying exposure to equities and bonds. We provide a utility-based framework for evaluating alternative strategies and in particular account for the impact of extreme and undesirable events to long run wealth accumulation. We present evidence suggesting that an overlay of equity premia to a reference portfolio can enhance the likelihood of achieving wealth accumulation goals and can smooth the transition path to achieving those goals. These results can be attributed to both long positions and short positions in contrast to recent findings suggesting shorts fail to add value. The benefits of the factor premia overlay additionally extend to the decumulation or retirement stage as reflected in an enhancement to the coverage ratio. Taken together, these findings suggest that the equity factor premia strategies we present can be utilized to support welfare enhancing gains.

Cavalcante Filho, E., Tassinari Moraes, F., and De-Losso, R. (2021). "Unskilled Fund Managers: Replicating Active Fund Performance With Few ETFs." In: *SSRN e-Print*.
This paper uses Exchange Traded Funds (ETFs) instead of risk factors as benchmarks to examine active mutual fund performance distribution. While transaction costs are included in the ETF returns, that is not true regarding risk factors, making it more challenging to characterize extraordinary performances via alphas. Assessments are based on the estimation of the skilled funds proportion defined by Barras et al. (2010). After evaluating several ETF combinations, we conclude that sets of 3 to 5 ETFs replicate most levels of active fund performance. Finally, we propose specific ETF selection algorithms, whereby we estimate that 95% of active management funds fail to generate value for their investors. Alphas calculated with ETFs are higher than those using risk factors, and the difference is similar to the transaction costs required for investing in risk factor portfolios (Frazzini et al. (2012)).

Cazalet, W., Curtil, D., Fabozzi, F. J., Hixon, S., Rudin, A., Sathyajit, R., Stavena, J., and Upadhyay, S. (2024). "Derivative applications to asset allocation and multi-asset management." In: *Journal of Asset Management* 25(6), pp. 531–551.
This article provides applications of derivatives to asset allocation and multi-asset management. The four applications include using futures for top-down asset allocation, deploying portable alpha strategies using derivatives to achieve desired convexity in payoff profiles, developing effective hedging strategies, and using derivatives for active speculative views by proprietary traders.

Cesarone, F., Di Paolo, A., Bufalo, M., and Orlando, G. (2024). "A Benchmark-Asset Principal Component Factorization For Index Tracking on Large Investment Universes." In: *SSRN e-Print*.
We develop an innovative methodology based on a benchmark-asset principal component factorization to determine a tracking portfolio that replicates the performance of a benchmark by investing in a subset of assets of a large investment universe.We compare three variants of our index tracking procedure with a standard optimization-based approach and with three variants of a naive strategy based on selecting the assets most correlated with the benchmark.The empirical analysis highlights that our approach shows index tracking abilities similar to the optimization-based portfolio selection model but with lower turnover and faster running times of about four orders of magnitude.

Cesarone, F., Moretti, J., and Tardella, F. (2018). "Why Small Portfolios Are Preferable and How to Choose Them." In: *SSRN e-Print*.

One of the fundamental principles in portfolio selection models is minimization of risk through diversification of the investment. However, this principle does not necessarily translate into a request for investing in all the assets of the investment universe. Indeed, following a line of research started by Evans and Archer almost 50 years ago, we provide here further evidence that small portfolios are sufficient to achieve almost optimal in-sample risk reduction with respect to variance and to some other popular risk measures, and very good out-of-sample performances. While leading to similar results, our approach is significantly different from the classical one pioneered by Evans and Archer. Indeed, we describe models for choosing the portfolio of a prescribed size with the smallest possible risk, as opposed to the random portfolio choice investigated in most of the previous works. We find that the smallest risk portfolios generally require no more than 15 assets. Furthermore, it is almost always possible to find portfolios that are just 1% more risky than the smallest risk portfolios and contain no more than 10 assets. The preference for small optimal portfolios is also justified by recent theoretical results on the estimation errors for the parameters required by portfolio selection models. Our empirical analysis is based on some new and on some publicly available benchmark data sets often used in the literature.

Cesarone, F., Mottura, C., Ricci, J. M., and Tardella, F. (2019). "On the stability of portfolio selection models." In: *SSRN e-Print*.

One of the main issues in portfolio selection models consists in assessing the effect of the estimation errors of the parameters required by the models on the quality of the selected portfolios. Several studies have been devoted to this topic for the minimum variance and for several other minimum risk models. However, no sensitivity analysis seems to have been reported for the recent popular Risk Parity diversification approach, nor for other portfolio selection models requiring maximum gain-risk ratios.Based on artificial and real-world data, we provide here empirical evidence showing that the Risk Parity model is always the most stable one in all the cases analyzed. Furthermore, the minimum risk models are typically more stable than the maximum gain-risk models, with the minimum variance model often being the preferable one.

Chalmers, J. and Reuter, J. (2020). "Is conflicted investment advice better than no advice?" In: *Journal of Financial Economics* 138(2), pp. 366–387.

The benefit of investment advice depends on the quality of advice and the investor's counterfactual portfolio. We use changes in the Oregon University System Optional Retirement Plan to highlight the impact of plan design on the counterfactual portfolios of advice seekers. When brokers are available and target date funds (TDFs) are not, brokers help participants with high predicted demand for advice bear market risk, but they recommend higher-commission options. When brokers are removed and TDFs are added, new high-predicted-demand participants primarily invest in TDFs, which offer similar market risk but higher Sharpe ratios than the broker-advised portfolios within our sample.

Chang, C. E., Krueger, T. M., and Witte, H. D. (2024). "Mutual Fund and ETF Selection for Retirement Portfolios Using an ESG Mandate Criterion." In: *The Journal of Retirement*.

Impact investing is a growing trend, but the appropriate role of environmental, social, and governance (ESG) considerations in investment decisions is becoming more controversial. Conflicts between the Securities and Exchange Commission and states, along with the increase of "ESG" funds with only a nominal commitment to ESG ideals, have left individuals who prioritize ESG principles without clear guidance on retirement planning. To provide direction to ESG-minded investors, this research examines thousands of mutual funds and ETFs, comparing funds with ESG mandates to non-mandated funds. The study found that funds with ESG mandates have higher levels of short-term ESG-based sustainability and long-term economic sustainability, according to Morningstar's Globe Ratings and Moat Ratings, respectively. Mandated funds tend to have earned higher historical risk-adjusted returns and have higher anticipated returns. Although there are differences between mutual funds and ETFs, investors who use ESG factors to select funds are likely to have already done well and can expect continued outperformance.

Chang, H.-Y., Sheu, D.-F., and Chen, S.-Y. (2010). "A Framework for Assessing Individual Retirement Planning Investment Policy Performance." In: *The Journal of Wealth Management* 13(3), pp. 38–49.

Because there are more and more older people in Taiwan, how retirees can maintain their past consumption level is an important issue. This article assesses the performance of individual investment planning policies for retirement. Because factors such as investment performance, risk, and taxation must be considered in making investment policy choices, the authors apply the analytic hierarchy process (Satty [1980]) to design a framework for evaluating individual investment policy performance. The research findings show that the most important criterion is investment performance. These results provide some suggestions for retirement planning.

Chang, L. and Shi, Y. (2023). "Forecasting mortality rates with a coherent ensemble averaging approach." In: *ASTIN Bulletin* 53(1), pp. 2–28.
Modeling and forecasting of mortality rates are closely related to a wide range of actuarial practices, such as the designing of pension schemes. To improve the forecasting accuracy, age coherence is incorporated in many recent mortality models, which suggests that the long-term forecasts will not diverge infinitely among age groups. Despite their usefulness, misspecification is likely to occur for individual mortality models when applied in empirical studies. The reliableness and accuracy of forecast rates are therefore negatively affected. In this study, an ensemble averaging or model averaging (MA) approach is proposed, which adopts age-specific weights and asymptotically achieves age coherence in mortality forecasting. The ensemble space contains both newly developed age-coherent and classic age-incoherent models to achieve the diversity. To realize the asymptotic age coherence, consider parameter errors, and avoid overfitting, the proposed method minimizes the variance of out-of-sample forecasting errors, with a uniquely designed coherent penalty and smoothness penalty. Our empirical data set include ten European countries with mortality rates of 0-100 age groups and spanning 1950-2016. The outstanding performance of MA is presented using the empirical sample for mortality forecasting. This finding robustly holds in a range of sensitivity analyses. A case study based on the Italian population is finally conducted to demonstrate the improved forecasting efficiency of MA and the validity of the proposed estimation of weights, as well as its usefulness in actuarial applications such as the annuity pricing.
Changwony, F. K., Campbell, K., and Tabner, I. T. (2021). "Savings goals and wealth allocation in household financial portfolios." In: *Journal of Banking and Finance* 124 (106028).
We investigate how savings goals relate to wealth allocation and how this relationship is moderated by financial advice and numerical ability. Using panel data from a large household survey we find that as the number and the time horizon of savings goals increases, portfolios shift from safe assets to both fairly safe assets and risky assets. We also find that households with access to multiple sources of financial advice and independent financial advice hold more fairly safe and risky assets and that independent financial advice enhances the influence of savings goals on wealth allocation to fairly safe and risky assets. Overall we find that the possession of savings goals is associated with long term saving activity, and this is particularly evident for those with low levels of numerical ability. By enabling the formation of savings goals, the financial planning process can facilitate long-term investment in risky assets.
Charteris, A. and McCullough, K. (2020). "Tracking error vs tracking difference: Does it matter?" In: *Investment Analysts Journal* 49(3), pp. 269–287.
Fund fact sheets are intended to provide investors with information necessary to make investment decisions. For passive funds, the inclusion of cumulative returns for the fund and benchmark enable investors to measure the fund's tracking performance using tracking difference. However, fund managers rely on tracking error to measure tracking performance, which is rarely presented. We evaluate the differences between these two metrics to ascertain whether the use of one or the other measure by investors could impact their investment decision. Results reveal that tracking error and tracking difference capture different elements of tracking performance, with varying rankings across the two measures for a sample of United States (US) funds. The empirical findings are robust to an adjustment for serial correlation, periods of extreme market volatility and varying measurement horizons. Recommendations for industry practice are made in light of these findings.
Chaudhuri, S. E. and Lo, A. W. (2019). "Dynamic Alpha: A Spectral Decomposition of Investment Performance Across Time Horizons." In: *Management Science* 65(9), pp. 4440–4450.
The value added by an active investor is traditionally measured using alpha, tracking error, and the information ratio. However, these measures do not characterize the dynamic component of investor activity, nor do they consider the time horizons over which weights are changed. In this paper, we propose a technique to measure the value of active investment that captures both the static and dynamic contributions of an investment process. This dynamic alpha is based on the decomposition of a portfolio's expected return into its frequency components using spectral analysis. The result is a static component that measures the portion of a portfolio's expected return resulting from passive investments and security selection and a dynamic component that captures the manager's timing ability across a range of time horizons. Our framework can be universally applied to any portfolio and is a useful method for comparing the forecast power of different investment processes. Several analytical and empirical examples are provided to illustrate the practical relevance of this decomposition.
Chen, A., Ferrari, G., and Zhu, S. (2023a). "Striking the Balance: Life Insurance Timing and Asset Allocation in Financial Planning." In: *arXiv e-Print*.

This paper investigates the consumption and investment decisions of an individual facing uncertain lifespan and stochastic labor income within a Black-Scholes market framework. A key aspect of our study involves the agent's option to choose when to acquire life insurance for bequest purposes. We examine two scenarios: one with a fixed bequest amount and another with a controlled bequest amount. Applying duality theory and addressing free-boundary problems, we analytically solve both cases, and provide explicit expressions for value functions and optimal strategies in both cases. In the first scenario, where the bequest amount is fixed, distinct outcomes emerge based on different levels of risk aversion parameter γ: (i) the optimal time for life insurance purchase occurs when the agent's wealth surpasses a critical threshold if $\gamma \in (0, 1)$, or (ii) life insurance should be acquired immediately if $\gamma > 1$. In contrast, in the second scenario with a controlled bequest amount, regardless of γ values, immediate life insurance purchase proves to be optimal.

Chen, A., Ferrari, G., and Zhu, S. (2023b). "Striking the Balance: Life Insurance Timing and Asset Allocation in Financial Planning." In: *SSRN e-Print*.

This paper investigates the consumption and investment decisions of an individual facing uncertain lifespan and stochastic labor income within a Black-Scholes market framework. A key aspect of our study involves the agent's option to choose when to acquire life insurance for bequest purposes. We examine two scenarios: one with a fixed bequest amount and another with a controlled bequest amount. Applying duality theory and addressing free-boundary problems, we analytically solve both cases, and provide explicit expressions for value functions and optimal strategies in both cases. In the first scenario, where the bequest amount is fixed, distinct outcomes emerge based on different levels of risk aversion parameter γ: (i) the optimal time for life insurance purchase occurs when the agent's wealth surpasses a critical threshold if $\gamma \in (0, 1)$, or (ii) life insurance should be acquired immediately if $\gamma > 1$. In contrast, in the second scenario with a controlled bequest amount, regardless of γ values, immediate life insurance purchase proves to be optimal.

Chen, A., Fuino, M., Sehner, T., and Wagner, J. (2022a). "Valuation of long-term care options embedded in life annuities." In: *Annals of Actuarial Science*, pp. 1–27.

In most industrialised countries, one of the major societal challenges is the demographic change coming along with the ageing of the population. The increasing life expectancy observed over the last decades underlines the importance to find ways to appropriately cover the financial needs of the elderly. A particular issue arises in the area of health, where sufficient care must be provided to a growing number of dependent elderly in need of long-term care (LTC) services. In many markets, the offering of life insurance products incorporating care options and LTC insurance products is generally scarce. In our research, we therefore examine a life annuity product with an embedded care option potentially providing additional financial support to dependent persons. To evaluate the care option, we determine the minimum price that the annuity provider requires and the policyholder's willingness to pay for the care option. For the latter, we employ individual utility functions taking account of the policyholder's condition. We base our numerical study on recently developed transition probability data from Switzerland. Our findings give new and realistic insights into the nature and the utility of life annuity products proposing an embedded care option for tackling the financing of LTC needs.

Chen, A., Hieber, P., and Rach, M. (2021a). "Optimal retirement products under subjective mortality beliefs." In: *Insurance: Mathematics and Economics* 101(A), pp. 55–69.

Many empirical studies confirm that policyholder's subjective mortality beliefs deviate from the information given by publicly available mortality tables. In this study, we look at the effect of subjective mortality beliefs on the perceived attractiveness of retirement products, focusing on two extreme products, conventional annuities (where the insurance company takes the longevity risk) and tontines (where a pool of policyholders shares the longevity risk). If risk loadings and charges are neglected, a standard expected utility framework, without subjective mortality beliefs, leads to the conclusion that annuities are always preferred to tontines (Yaari (1965), Milevsky and Salisbury (2015)). In the same setting, we show that this result is easily reversed if an individual perceives her peer's life expectancies to be lower than the ones used by the insurance company. We prove that, assuming such subjective beliefs, there exists a critical tontine pool size from which the tontine is always preferred over the annuity. This suggests that tontines might be perceived as much more attractive than suggested by standard expected utility theory without subjective mortality beliefs.

Chen, A., Li, H., and Schultze, M. B. (2022b). "Tail index-linked annuity: A longevity risk sharing retirement plan." In: *Scandinavian Actuarial Journal*, pp. 1–26.

This paper proposes an innovative retirement product focusing on longevity risk sharing, a contract we refer to as tail index-linked annuity (TILA). Specifically, the proposed TILA pays out variable annual payments, which will be equal to a regular nominal amount when a reference survival index is lower than a predetermined threshold (i.e. normal evolution of longevity risk), and a reduced, index-dependent payment when the threshold is passed (i.e. highly unfavorable evolution of longevity risk). The proposed TILA aims at not only improving the benefits of the policyholders, which has been the focus in recent literature on innovative retirement products, but also reducing the longevity risk exposure of the insurer, particularly for advanced retirement ages. Using real-world mortality data and a stochastic multi-population mortality model, we find that the proposed TILA leads to higher expected lifetime utility than regular annuities for policyholders with different degrees of risk aversions. Meanwhile, numerical analysis shows that the proposed TILA could greatly mitigate the solvency risk of the insurer, leading to a substantially lower loss probability and expected (tail-) loss than regular annuities in the presence of a longevity shock, and therefore could reduce the insurer's required solvency capital under the latest solvency regulations.

Chen, A., Nguyen, T., Qian, L., and Yang, Z. (2025a). "The role of health in consumption and portfolio decision-making: Insights from state-dependent models." In: *Journal of Computational and Applied Mathematics* 457, p. 116290.

In this paper, we study an optimal consumption and asset allocation problem accounting for the fact that individuals' utility differs across various health states. Our study is based on the assumption that an individual's health status evolves through a semi-Markov process, where the transition probabilities are contingent on both the present state and the duration spent in that state. The optimal form of stock investment and of consumption is determined analytically. The optimal consumption level is significantly shaped by the integration of health-related factors and can be represented as the inverse of the marginal utility function with respect to time and state price density. Additionally, we introduce a Lagrangian multiplier that can be derived by solving a fixed point problem. While our optimal solutions are applicable to a wide range of utility functions, we provide numerical illustrations specifically using the power utility function in a model encompassing three distinct health states. Transition probabilities are estimated from real data collected by the China Insurance Regulatory Commission (CIRC), using a high-order polynomial Perks formula. We find that the investor's consumption is higher when health is good, but lower when care is needed.

Chen, A., Rach, M., and Sehner, T. (2019). "On the Optimal Combination of Annuities and Tontines." In: *SSRN e-Print*.

Tontines, retirement products constructed in such a way that the longevity risk is shared in a pool of policyholders, have recently gained vast attention from researchers and practitioners. Typically, these products are cheaper than annuities, but do not provide stable payments to policyholders. This raises the question whether, from the policyholders' viewpoint, the advantages of annuities and tontines can be combined to form a retirement plan which is cheaper than an annuity and carries less risk than a tontine. In this article, we analyze and compare three approaches of combining annuities and tontines in an expected utility framework: The introduced in Chen et al. (2019), a product very similar to the tonuity which we call and a portfolio consisting of an annuity and a tontine. We show that the payoffs of a tonuity or an antine can be replicated by a portfolio consisting of an annuity and a tontine. Consequently, policyholders achieve higher expected utility levels when choosing the portfolio over the novel retirement products tonuity and antine.

Chen, A. and Munnell, A. H. (2021a). *How Much Taxes Will Retirees Owe on Their Retirement Income?* Tech. rep. Center for Retirement Research at Boston College.

To evaluate their retirement resources, households approaching retirement will examine their Social Security statements, defined benefit pensions, defined contribution balances, and other financial assets. However, many households may forget that not all of these resources belong to them; they will need to pay some portion to federal and state government in taxes. It is unclear, however, just how large the tax burden is for the typical retired household and for households with different income levels. This project aims to shed light on the tax burdens that retirees face by estimating lifetime taxes for a group of recently retired households. The project uses data from the Health and Retirement Study (HRS) linked to administrative earnings to determine Social Security benefits and administrative records on state of residence to estimate state tax liabilities. Income is then projected over the expected retirement of each household. Federal and state taxes, are estimated with TAXSIM, for each household on its reported and projected income.

The paper found that:

- These estimates show that households in the aggregate will have to pay about 6 percent of their income in federal and state income taxes.
- But this liability rests primarily with the top quintile of the income distribution.

- For the lowest four quintiles, taxes are negligible - ranging from 0 percent to 1.9 percent.
- In contrast, the average liability is 3 percent for the top quintile, 16.4 percent for the top 5 percent, and 22.7 percent for the top 1 percent.

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Chen, A. and Munnell, A. H. (2021b). "How Much Taxes Will Retirees Owe on Their Retirement Income?" In: *SSRN e-Print*.

To evaluate their retirement resources, households approaching retirement will examine their Social Security statements, defined benefit pensions, defined contribution balances, and other financial assets. However, many households may forget that not all of these resources belong to them; they will need to pay some portion to federal and state government in taxes. It is unclear, however, just how large the tax burden is for the typical retired household and for households with different income levels. This project aims to shed light on the tax burdens that retirees face by estimating lifetime taxes for a group of recently retired households. The project uses data from the Health and Retirement Study (HRS) linked to administrative earnings to determine Social Security benefits and administrative records on state of residence to estimate state tax liabilities. Income is then projected over the expected retirement of each household. Federal and state taxes, are estimated with TAXSIM, for each household on its reported and projected income.

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The policy implications of the findings are that taxes are meaningful for the top quintile and an important consideration for many households reliant on 401(k)/IRA or financial assets for security in retirement.

Chen, A., Haberman, S., and Thomas, S. (2020). "The implication of the hyperbolic discount model for the annuitisation decisions." In: *Journal of Pension Economics and Finance* 19(3), pp. 372–391.

The low demand for immediate annuities at retirement has been a long-standing puzzle. We show that a hyperbolic discount model can explain this behaviour and results in the attractiveness of long-term deferred annuities. With a set of benchmark assumptions, we find that retirees would be willing to pay a much higher price than the actuarial fair price for annuities with longer deferred periods. Moreover, if governments were to introduce a pre-commitment device which requires pensioners to make annuitisation decisions around 10 years before retirement, the take up rate of annuities could become higher.

Chen, A., Haberman, S., and Thomas, S. (2023c). "Adaptive Retirement Planning, Sustainable Withdrawals, and Deferred Annuities." In: *The Journal of Retirement* 10(3), pp. 96–119.

In this article, we integrate investment decisions in the post-retirement period with the inclusion of a deferred annuity (DA) to provide a lifetime decumulation solution. We use the perfect withdrawal rate (PWR) as a tool to make recommendations on withdrawal rates and asset allocations. We illustrate how cheap it is to use a DA to deal with longevity risk. Moreover, if an individual wants to maximize median PWR, they should allocate almost 100% in stocks. If an individual wants to maximize minimum PWR, they should allocate 40% - 60% in stocks. A substantial stocks component should therefore be maintained throughout the retirement period as a new "normal" asset allocation.

Chen, A., Haberman, S., and Thomas, S. (2025b). "The Role of the Annuity Price in Decumulation Strategies with Deferred Annuities." In: *The Journal of Retirement* 12(3), pp. 44–67.

In this article, we consider post-retirement decumulation strategies after the purchase of a deferred annuity (DA) and explore the sensitivity of the retirement incomes and asset allocation decisions to the magnitude of the deferred annuity prices. We use the Perfect Withdrawal Amount (PWA) as a tool to understand the role of the different annuity price determinant factors and identify optimal asset allocations. We find that, when the selected deferred period is very long, the maximum PWA during the decumulation stage is insensitive to the annuity price loading. Moreover, the selection of the deferred period is a very sensitive input that affects annual withdrawals.

Chen, A., Haberman, S., and Thomas, S. H. (2021b). "Combining flexible Asset Allocation, Sustainable Withdrawals, and Deferred Annuities to provide an Adaptive Lifelong Investing Solution." In: *SSRN e-Print*.

In this paper, we integrate investment decisions in the post-retirement decumulation period with that of the deferred annuity purchase to provide a lifetime decumulation solution. Based on Monte Carlo simulation and historical experience, we use the Perfect Withdrawal Rate (PWR) as a tool to make recommendations on withdrawal rates and asset allocations for different levels of risk preferences. We have a few potentially important findings. First, we illustrate how cheap it is to use a deferred annuity (especially with a deferred period of more than 15 years) as a solution to deal with longevity risk and maintain control of retirement wealth with the investor. Second, we find that if an individual wants to maximise median PWR, he/she should allocate almost 100% in stocks regardless of the length of chosen decumulation period. If an individual wants to maximise minimum PWR, he/she should allocate around 40% - 60% in stocks; therefore, a substantial stocks component should be maintained even if the individual is very risk averse. This then links to our final conclusion on a re-defined Glidepath: if an individual can accept a lower than 50% risk of failure, he/she should move from stocks to bonds as he/she becomes older; however a certain percentage in stocks should be maintained through the decumulation phase.

Chen, A., Haberman, S., and Thomas, S. (2017). "Optimal Decumulation Strategies During Retirement with Deferred Annuities." In: *SSRN e-Print*.

Since greater flexibility in accessing pension savings has been given to defined contribution pensioners, retirees are in need of advice on how to spend down their savings to make retirement income last throughout their lifetime. Deferred annuities have been discussed extensively in recent years as a retirement solution and have been recommended in the OECD Roadmap for the Good Design of Defined Contribution Pension Plans (OECD2016). Assuming a world where deferred annuities are available, we propose two utility maximising decumulation strategies comprising a deferred annuity purchased at retirement and optimal consumption and savings before the commencement of the annuity. A retiree who is concerned about longevity risk and wants to retain a certain level of liquidity is advised to spend 21.6% on a 15-year deferred annuity or 9.13% on a 20-year deferred annuity. A retiree who simply wants to use annuities to maximise overall satisfaction from retirement consumption is advised to spend 61.83% on a 6-year deferred annuity. We compare our strategies with other available decumulation strategies in the market, hence verifying the merits of the design. Moreover, the stability of our results are examined after allowing for consumption smoothness, social income benefits, a target replacement ratio and a bequest motive.

Chen, M., Shirazi, M., Forsyth, P. A., and Li, Y. (2023d). "Machine Learning and Hamilton-Jacobi-Bellman Equation for Optimal Decumulation: a Comparison Study." In: *arXiv e-Print*.

We propose a novel data-driven neural network (NN) optimization framework for solving an optimal stochastic control problem under stochastic constraints. Customized activation functions for the output layers of the NN are applied, which permits training via standard unconstrained optimization. The optimal solution yields a multi-period asset allocation and decumulation strategy for a holder of a defined contribution (DC) pension plan. The objective function of the optimal control problem is based on expected wealth withdrawn (EW) and expected shortfall (ES) that directly targets left-tail risk. The stochastic bound constraints enforce a guaranteed minimum withdrawal each year. We demonstrate that the data-driven approach is capable of learning a near-optimal solution by benchmarking it against the numerical results from a Hamilton-Jacobi-Bellman (HJB) Partial Differential Equation (PDE) computational framework.

Chen, W. and Langrene, N. (2020). "Deep neural network for optimal retirement consumption in defined contribution pension system." In: *arXiv e-Print*.

In this paper, we develop a deep neural network approach to solve a lifetime expected mortality-weighted utility-based model for optimal consumption in the decumulation phase of a defined contribution pension system. We formulate this problem as a multi-period finite-horizon stochastic control problem and train a deep neural network policy representing consumption decisions. The optimal consumption policy is determined by personal information about the retiree such as age, wealth, risk aversion and bequest motive, as well as a series of economic and financial variables including inflation rates and asset returns jointly simulated from a proposed seven-factor economic scenario generator calibrated from market data. We use the Australian pension system as an example, with consideration of the government-funded means-tested Age Pension and other practical aspects such as fund management fees. The key findings from our numerical tests are as follows. First, our deep neural network optimal consumption policy, which adapts to changes in market conditions, outperforms deterministic drawdown rules proposed in the literature. Moreover, the out-of-sample outperformance ratios increase as the number of training iterations increases, eventually reaching outperformance on all testing scenarios after less than 10 minutes of training. Second, a sensitivity analysis is performed to reveal how risk aversion and bequest motives change the consumption over a retiree's lifetime under this utility framework. Third, we provide the optimal consumption rate with different starting wealth balances. We observe that optimal consumption rates are not proportional to initial wealth due to the Age Pension payment. Forth, with the same initial wealth balance and utility parameter settings, the optimal consumption level is different between males and females due to gender differences in mortality.

Chen, W., Minney, A., Toscas, P., Koo, B., Zhu, Z., and Pantelous, A. A. (2021c). "Personalised drawdown strategies and partial annuitisation to mitigate longevity risk." In: *Finance Research Letters* 39 (101644).
Despite the importance of drawdown strategies under a defined contribution system with increased longevity risk, little guidance to retired and retiring members has been forthcoming from superannuation funds. This paper provides a do-it-yourself drawdown design for members of superannuation funds along with comparison studies on a range of retirement income strategies under an array of realistic scenarios. A stochastic economic scenario generator is used to simulate the uncertain outcomes of different drawdown strategies during retirement. The impact of annuitisation for mitigating longevity risk under government pension rules and the selection of personalised drawdown and annuitisation strategies for retirement are examined.

Chen, Y. and Israelov, R. (2024). "Exclude with Impunity: Personalized Indexing and Stock Restrictions." In: *Financial Analysts Journal* 80(2), pp. 7–25.
Using simulated historical backtests, we study the impact of stock exclusions on the performance of passive and active portfolios. We find that at low to moderate numbers, stock exclusions have very little influence on passive portfolios. Their effects on active portfolios vary by the factor in consideration and the portfolio construction method, but the magnitudes are much smaller than suggested by the percentage of stocks being excluded. We find similar patterns with industry-concentrated exclusions. Overall, our results suggest that investors should feel comfortable excluding a fairly large number of stocks before experiencing any significant deterioration in their investment performance.

Chen, Z., Feng, R., Li, H., and Yang, T. (2024). "Coping with longevity via hedging: Fair dynamic valuation of variable annuities." In: *Insurance: Mathematics and Economics* 117, pp. 154–169.
This paper introduces a fair valuation framework for pricing variable annuity liabilities and their embedded guarantee riders within a dynamic multi-period context. We focus on variable annuities featuring the Guaranteed Lifetime Withdrawal Benefit (GLWB) rider, which exposes policyholders to both financial and longevity risks. We employ a fair dynamic valuation method that is market-consistent, actuarially-consistent, and time-consistent. Our findings demonstrate that this approach effectively establishes fair management fee rates, aligning with prior research and industry surveys. Furthermore, we highlight the potential for significant reductions in liability valuation, and consequently, GLWB rider pricing, through effective management of longevity risk within the insurer's net liability.

Cheng, Y., Gibson, P., and Guo, T. (2017). "Life Quality and Health Costs in Late Retirement." In: *SSRN e-Print*.
Individuals are living longer due to the advancement of medical technology and nutrition quality. Are the elderly enjoying retirement in those extended years with good quality of life, or, are they simply alive? Using data from the Health and Retirement Study (HRS) and the Consumption and Activities Mail Survey (CAMS), this study contributes to the literature by presenting empirical evidence on how individuals spend time in retirement. The results show that retirees on average do not spend their time significantly different throughout retirement. Most life tasks such as reading the paper or magazines, listening to music, playing sports or exercising, visiting others, and house cleaning are similar among retirees in different age groups. We also present evidence that retirees on average experience a spike in medical expenses late in retirement. We compare systematic withdrawal strategies with and without health costs risk quantifying the impact on portfolio sustainability.

Chhabra, A. B. (2015). *The Aspirational Investor: Taming the Markets to Achieve Your Life's Goals*. HarperBusiness. 245 pp.
The Chief Investment Officer of Merrill Lynch Wealth Management explains why goals, not markets, should be the primary focus of your investment strategy-and offers a practical, innovative framework for making smarter choices about aligning your goals to your investment strategy. The Aspirational Investor is a practical, innovative approach to managing wealth based on key goals and the careful allocation of risks rather than responding to the whims of the financial markets. Chhabra introduces his "Wealth Allocation Framework," which accommodates the three seemingly incompatible objectives that must underpin every sound wealth management plan: the need for financial security in the face of known and unknowable risks; the need to maintain current living standards over time despite inflation; and the need to pursue aspirational goals for wealth creation.

Chien, C.-L. (2019). *Enhancing Retirement Success Rates in the United States*. Springer International Publishing. 113 pp.

Dr. Chien's book aims to help middle class retirees get a better handle on their retirement finances by ensuring that all of their key household assets are considered in the planning process. It is essential reading for retirees and their advisors who are seeking how to best position assets for a successful retirement.

Chiu, Y.-F., Hsieh, M.-H., and Tsai, C. (2019). "Valuation and analysis on complex equity indexed annuities." In: *Pacific-Basin Finance Journal* 57 (101175).

Equity-indexed annuities (EIAs) are popular products that eliminate the downside risk while still providing upside potential. Among the three major categories of EIAs, ratchet EIAs are the most popular. Ratchet EIAs with quanto features emerge due to differences in asset returns across countries. The literature covers the pricing of the EIAs that are not quantos, and this paper fills the hole. In deriving pricing formulas, we add an exchange rate model as well as a foreign risk-free rate model to the pricing framework of Black and Scholes. Our formulas cover quanto ratchet EIAs for both compound and simple versions that may have a return cap and employ two types of geometric return averaging. The numerical analyses illustrate how contract features and market parameters affect contract values. The results also highlight the significance of quantos in contract pricing.

Chong, W. F., Cui, H., and Li, Y. (2022). "Pseudo-Model-Free Hedging for Variable Annuities via Deep Reinforcement Learning." In: *arXiv e-Print*.

This paper proposes a two-phase deep reinforcement learning approach, for hedging variable annuity contracts with both GMMB and GMDB riders, which can address model miscalibration in Black-Scholes financial and constant force of mortality actuarial market environments. In the training phase, an infant reinforcement learning agent interacts with a pre-designed training environment, collects sequential anchor-hedging reward signals, and gradually learns how to hedge the contracts. As expected, after a sufficient number of training steps, the trained reinforcement learning agent hedges, in the training environment, equally well as the correct Delta while outperforms misspecified Deltas. In the online learning phase, the trained reinforcement learning agent interacts with the market environment in real time, collects single terminal reward signals, and self-revises its hedging strategy. The hedging performance of the further trained reinforcement learning agent is demonstrated via an illustrative example on a rolling basis to reveal the self-revision capability on the hedging strategy by online learning.

Chow, V., Gu, J., and Wang, Z. (2021). "Implied Equity Premium and Market Beta." In: *SSRN e-Print*.

We extend the ex-ante mean-variance (SVIX) models of Martin (2017) and Martin-Wagner (2019) to a mean-variance-asymmetry (AVIX) framework for incorporating higher-moment and co-moment risk in asset pricing. AVIX is a risk-neutral measure of the left-tail asymmetries in return that corrects the SVIX approach's downside bias. The options implied market beta of equity is a weighted sum of the beta of SVIX and that of AVIX. Empirically, the implied betas possess significant predictability of risk/return relationship and the hedging ability against bear/crashing markets. We develop an investible portfolio MKT that mimics realized outcomes on the implied market index adjusted for volatility-asymmetry.

Clare, A., Glover, S., Seaton, J., Smith, P. N., and Thomas, S. (2020). "Measuring sequence of returns risk." In: *The Journal of Retirement* 8(1), pp. 65–79.

The authors discuss the nature and importance of the concept of sequence risk, the risk that a bad return occurs at a particularly unfortunate time, such as around the point of maximum accumulation or the start of decumulation. This concept is especially relevant in the context of retirement savings, where the implications for withdrawal rates of a bad return can be particularly severe. They show how the popular glidepath or target-date savings products are very exposed to such risk. Three different measures of sequence risk are proposed, each of which is intended to inform investors of the probability that a chosen investment strategy may not deliver desired withdrawal rates, and hence these measures are intended to aid investment choices; conventional performance measures such as Sharpe or Sortino ratios are only indirectly related to this ability to achieve a given withdrawal experience. Finally, the authors note that, using US data, very simple portfolios comprising equities and bonds can achieve very low probabilities of failure to achieve popular desired withdrawal rates, such as 5% a year, as long as the equity component is smoothed by switching in and out of cash using a simple trend-following rule.

Clare, A., Seaton, J., Smith, P. N., and Thomas, S. (2017). "Reducing Sequence Risk Using Trend Following and the CAPE Ratio." In: *Financial Analysts Journal* 73(4), pp. 91–103.

The risk of experiencing bad investment outcomes at the wrong time, or sequence risk, is a poorly understood but crucial aspect of the risk investors face -particularly those in the decumulation phase of their savings journey, typically over the period of retirement financed by a defined contribution pension scheme. Using US equity return data for 1872-2014, we show how this risk can be significantly reduced by applying trend-following investment strategies. We also show that knowing a valuation ratio, such as the cyclically adjusted price-to-earnings (CAPE) ratio, at the beginning of a decumulation period is useful for enhancing sustainable investment income.

Clare, A., Seaton, J., Smith, P. N., and Thomas, S. (2021a). "Perfect Withdrawal in a Noisy World: Investing Lessons with and without Annuities while in Drawdown between 2000 and 2019." In: *The Journal of Retirement* 9(1), pp. 9–39.
This article shows how the relatively new concept of Perfect Withdrawal Rate can be used in assessing the appropriate sustainable withdrawal amounts from a pot of wealth. This concept can be applied equally to private retirement funds, endowments, and charities - and indeed in any context requiring regular withdrawals from an initial source of funds. The subject of estimating sustainable withdrawal rates usually falls back on describing the likely minimum safe withdrawal possibilities for various portfolio constructions over different decumulation periods. This analysis employs either a long period of historical data or a recombination of data in the form of Monte Carlo simulations. To illustrate the power of the Perfect Withdrawal concept, the article considers the case of someone who initiated retirement on January 1, 2000, at age 65 and, with the benefit of actual investment returns, assesses investment and withdrawal rate options and lessons to be learned from this experience. The article also introduces the concept and a methodology for purchasing a delayed annuity so that at age 85 (on December 31, 2019), the hypothetical retiree is fully transitioned from investment income to annuity income for the rest of their life, no matter how long that may be.

Clare, A., Seaton, J., Smith, P. N., and Thomas, S. (2023a). "Using the Minimum Acceptable Annual Withdrawal with the Perfect Withdrawal Rate Rule." In: *The Journal of Retirement* 11(1), pp. 34–55.
This study shows how to use the concept of Perfect Withdrawal Rates to quantify appropriate retirement withdrawal amounts year by year based on changing returns and (probably) in conversation with an adviser, using both historical strings of returns and Monte Carlo simulations. The authors show how the 'strings' approach is superior (i.e., more 'realistic') in practice to Monte Carlo. The preferred withdrawal strategy is 'adaptive' as information is updated, and their suggested measure of success is how well a withdrawal strategy performs against a targeted withdrawal financial amount, most likely chosen in discussion between a retiree and an advisor, and which is labelled as the minimum acceptable annual withdrawal (MAAW) rate. The authors suggest that this offers a very simple, practical, and intuitively appealing measure of success for the performance of portfolios in the decumulation context. They also discuss the use of delayed and deferred annuities in this context, as well as the associated target residual financial sum.

Clare, A., Seaton, J., Smith, P. N., and Thomas, S. (2024a). "The Science of flexible Retirement Choices: Switching Retirement Savings into an Annuity." In: *The Journal of Retirement* 12(2), pp. 59–75.
In this article we consider the choice that retirees might make between drawing down from their pension pot or the purchase of an annuity. A key finding is that in a world of loss aversion, across a very wide range of assumptions, there is almost always a crossover point during retirement at which moving out of drawdown and into an annuity can be the optimal strategy. This suggests that the pension industry should investigate the construction of a hybrid, 'flex-first, fix-later' pension product. We show that a hybrid approach can produce much higher levels of happiness, especially at older ages, than staying wholly in drawdown or buying an annuity at the point of retirement.

Clare, A., Seaton, J., Smith, P. N., and Thomas, S. (2025). "Saving to Decumulate: A Lifetime Journey Based on the Perfect Contribution Rate." In: *The Journal of Retirement*, jor.2025.1.180.
This paper examines the relationship between the accumulation (contribution) and decumulation (withdrawal) phases of pension saving across a person's economic life. These are concepts that are usually considered "independent," but for which we find a surprisingly strong empirical relationship, generated most likely by the long-run mean reversion of returns. Using evidence from a long run of US data since 1870, we introduce the idea of a perfect contribution rate (PCR), which is the annual contribution required over a person's savings' life that will allow the individual to achieve a real wealth target prior to the start of decumulation, assuming perfect foresight of asset returns. In particular, there is an interesting positive relation between PCR and subsequent perfect withdrawal rate (PWR), which we believe reflects mean reversion in asset returns over long time periods. In other words, if an agent has to save larger amounts to achieve their target wealth when investment returns are poor, then typically the subsequent (possible) withdrawal rates may also be large. We also introduce the idea of the perfect retirement ratio (PRR)-the ratio of the PWR to the previous PCR-and find this to be highly time-dependent, with significant policy implications.

Clare, A. D., Seaton, J., Smith, P. N., and Thomas, S. H. (2019). "Absolute Momentum, Sustainable Withdrawal Rates and Glidepath Investing in US Retirement Portfolios from 1925." In: *SSRN e-Print*.
A significant part of the development in pension provision in many countries is the emergence of Date Funds or TDFs. In this paper we examine the proposition of de-risking through life and the guidance offered by TDFs in the decumulation phase following retirement. We investigate the withdrawal experience associated with Glidepath Investing in the US since 1925 for conventional bond-equity portfolios. We find one very powerful conclusion: that smoothing the returns on individual assets by simple absolute momentum or trend following techniques is a potent tool to enhance withdrawal rates, often by as much as 50% per annum! And, perhaps of even greater social relevance is that it removes the -tail of unfortunate withdrawal rate experiences, i.e. the bad luck of a poor sequence of returns early in decumulation. We show that diversifying assets over time by switching between an asset and cash in a systematic way is potentially more important for the retirement income experience than diversifying one portfolio across asset classes. We also show that Glidepath investing is only sensible within a few years of the target date. This finding provides succour to enthusiasts for target date investing in the face of the growing hostility in the literature.

Clare, A. D., Seaton, J., Smith, P. N., and Thomas, S. H. (2021b). "Can sustainable withdrawal rates be enhanced by trend following?" In: *International Journal of Finance & Economics* 26(1), pp. 27–41.
We examine the consequences of alternative popular investment strategies for the decumulation of fundsinvested for retirement through a defined contribution pension scheme. We examine in detail the viability of specific 'safe' withdrawal rates including the '4%-rule' of Bengen. We find two powerful conclusions. First that smoothing the returns on individual assets by simple trend following techniques is a potent tool to enhance withdrawal rates.Second, we show that while diversification across asset classes does lead to higher withdrawal rates than simple equity/bond portfolios, "smoothing" returns in itself is far more powerful a tool for raising withdrawal rates. In fact,smoothing the popular equity/bond portfolios (such as the 60/40 portfolio) is in itself an excellent and simple solution to constructing a retirement portfolio. Alternatively, trend following enables portfolios to contain more risky assets, and the greater upside they offer, for the same level of overall risk compared to standard portfolios. To anticipate our empirical findings, we find two powerful conclusions: Smoothing the returns on individual assets by simple trend following techniques (or similar) is a potent tool to enhance withdrawal rates. Although diversification across asset classes does lead to higher withdrawal rates than simple equity/bond portfolios, "smoothing" returns is a far more powerful tool for raising withdrawal rates; in fact, smoothing the popular equity/bond portfolios (such as the 60/40 portfolio) is an excellent and simple solution to constructing a retirement portfolio.

Clare, A. D., Seaton, J., Smith, P. N., and Thomas, S. H. (2022). "Measuring Success in Decumulation: The Minimum Acceptable Annual Withdrawal Rate (MAAW)." In: *SSRN e-Print*.
This paper proposes a new method of assessing the sustainability of withdrawals from a given pot of wealth in retirement or indeed in any similar context such as intergenerational family wealth or charity/endowment spending. We use the recent concept of Perfect Withdrawal Rates to assess appropriate withdrawal amounts year by year based on changing returns and (probably) in conversation with an adviser. The preferred withdrawal strategy is 'adaptive' and the focus here is on devising a new measure of success of a portfolio retirement strategy which reflects the aspirations of both retiree and adviser in their practical communications. We introduce a new measure of success to gauge how well a withdrawal strategy performs against a target withdrawal amount, the Minimum Acceptable Annual Withdrawal Rate (MAAW). We suggest that this offers a very practical and intuitively appealing measure of success for the performance of portfolios in the decumulation context.

Clare, A. D., Seaton, J., Smith, P. N., and Thomas, S. H. (2023b). "The Science of flexible Retirement Choices: Switching Retirement Savings into an Annuity." In: *SSRN e-Print*.
In this paper we consider the choice that retirees might make between drawing down from their pension pot and the purchase of an annuity. A key finding of our research that in a world of 'loss aversion', across a very wide range of assumptions, there is almost always a 'crossover point' during retirement at which moving out of drawdown into an annuity can be the optimal strategy. This suggests that the pensions industry should investigate the construction of a hybrid, 'flex-first, fix-later' pension product. We show that a 'hybrid' approach can produce much higher levels of happiness, especially at older ages, than staying wholly in drawdown or from buying an annuity at the point of retirement.

Clare, A. D., Seaton, J., Smith, P. N., and Thomas, S. H. (2024b). "Saving to Decumulate: a Lifetime Journey, Introducing the Perfect Contribution Rate and its Relation to the Perfect Withdrawal RatE." In: This paper examines the relationship between the accumulation (the contribution) and decumulation (withdrawal) phases of pension saving across one's economic life. These are concepts which are usually considered 'independent' but for which we find a surprisingly strong empirical relationship generated most likely by the long-run mean reversion of returns. Using evidence from a long run of US data since 1870 we introduce the idea of a Perfect Contribution Rate which is the annual contribution required over one's savings' life which will allow the individual to achieve a real wealth target prior to the start of decumulation assuming perfect foresight of asset returns. In particular, there is an interesting positive relation between PCR and subsequent PWR which we believe reflects mean reversion in asset returns over long time periods. In other words, if an agent has to save larger amounts to achieve their target wealth since investment returns are poor, then typically the subsequent (possible) withdrawal rates may also be large.

Clark, R., Maurer, R., and Mitchell, O. S. (2018a). "How Persistent Low Returns Will Shape Saving and Retirement." In: *How Persistent Low Returns Will Shape Saving and Retirement*. Oxford University Press.
Financial market developments over the past decade have undermined what was once thought to be conventional wisdom about saving, investment, and retirement spending. Foremost among these is the depressingly persistent and extended period of low capital market returns, driving concerns about how to rethink saving and investments in what can be called the "new normal." This chapter introduces the themes of the book: how we arrived at our current state of affairs and what changes need to be made to achieve adequate retirement incomes for future retirees, and exploring new designs for pension plan sponsors. With increasing life expectancy adding to the problem of low market returns, the chapter urges policymakers to start reforming now to ensure retirement financial security.

Clark, R. L., Hammond, R. G., Morrill, M. S., and Vanderweide, D. (2018b). "Annuity options in public pension plans: the curious case of social security leveling." In: *The Journal of Retirement* 6(1), pp. 33–44.
Social Security Leveling is an annuity option that allows defined benefit pension participants to receive a level income throughout retirement by taking a larger pension benefit before they reach Social Security eligibility. This option can enable retirees to access a greater proportion of their employer pension wealth at younger ages and have a smoother income stream in retirement, at the cost of a lower pension payment during later retirement years. To evaluate the desirability of this option, a retiree must consider pricing, anticipated life expectancy, and income needs throughout retirement. We show that plan design features of this annuity option favor those who expect a shorter-than-average period of retirement and who have higher personal discount rates. This study uses detailed administrative records on all North Carolina state and local government retirees from 2009 through 2014. Among individuals claiming retirement benefits before age 62, over 20% chose the Social Security Leveling annuity option. Using multivariate regression analysis, this study finds evidence that North Carolina public sector retirees choose the Social Security Leveling option in a manner that is consistent with predictions.

Collins, P. J. (2016). "Annuities and retirement income planning." In: *CFA Foundation Research Foundation Briefs*.
Annuitization is one asset management strategy for retirees seeking to secure lifetime income. The US annuity marketplace offers a variety of annuity contracts, including single premium annuities, advanced life deferred annuities, variable annuities with lifetime income guarantee riders, and ruin contingent deferred annuities. Advisers seeking to provide guidance to clients in or near retirement can benefit by understanding (1) the arguments both for and against annuitization and (2) how a client's interests might be best represented in the marketplace. Important annuity contract provisions are highlighted and briefly discussed so the adviser can become more familiar with retirement-planning options.

Collins, P. J., Lam, H. D., and Stampfli, J. (2015). "How Risky is Your Retirement Income Risk Model?" In: *Financial Services Review* 24(3).
Adequately sustaining lifetime income is a critical portfolio objective for retired investors. This article provides a brief review of various retirement income modeling approaches including historical back testing, Monte Carlo simulations, and other risk modeling techniques. Implausible assumptions underlying risk models may mislead investors concerning the risk and return expectations of their investment strategies. We compare risk models, evaluate their credibility, and demonstrate how overly-simplified models may distort the risks retired investors face. Failure rate differences are stark: 4 percent at the low end versus 49 percent at the high end. The article ends with general comments regarding model risk and practitioner investment advice.

Collins, P. J. and Stampfli, J. (2019). "Sequence of Returns Risk Reconsidered." In: *Retirement Management Journal* 8(1).
In the process of evaluating various retirement income strategies, it appears that, for investors not endowed with substantial wealth relative to consumption demands, sequence of returns risk is operative throughout retirement. We explore, in an asset-liability modeling context, the reasons why sequence risk exists throughout the planning

horizon and why it can be particularly acute at the end of an investor's life span. Given the nature of this risk, prudent asset management benefits from developing appropriate risk metrics, and from implementing credible monitoring, evaluation, and communication procedures. Two case studies focus on sequence of returns risk. They present risk metrics designed to answer the following questions: (1) is the investor's retirement income strategy feasible; (2) if yes, is it sustainable; and (3) does it allow sufficient flexibility to provide security in the face of financial shocks? The risk metrics employ information derived from both investment simulation models and actuarial calculations.

Consigli, G. and Di Tria, M. (2018). "Asset-liability management and goal-based investing for retail business." In: *International Journal of Financial Engineering and Risk Management* 2(4), p. 308.

The industry of online personal financial services is expected over the next years to absorb an increasing share of households and individuals' savings and investment decisions with a parallel expansion of tailored decision tools and underlying methodological developments. In this article we present a dynamic stochastic optimisation model formulated to tackle a long-term optimal wealth management problem-based explicitly on the introduction of consumption and investment goals with a terminal inflation-adjusted retirement target. By embedding a goal-based investing philosophy in a dynamic framework we provide a reference modelling approach for increasingly popular households asset-liability management services. In a discrete setting we show that a dynamic stochastic programming formulation will lead to a highly realistic representation and solution of an otherwise hardly manageable optimisation problem and it is consistent with computer-aided decision support tools' operational requirements.

Consigli, G., Iaquinta, G., Moriggia, V., di Tria, M., and Musitelli, D. (2012). "Retirement planning in individual asset liability management." In: *IMA Journal of Management Mathematics* 23(4), pp. 365–396.

Increasing financial pressure on State-controlled pension systems has caused, over the last two decades or so, an unprecedented effort by private pension funds (PFs) and insurance companies to issue new types of retirement vehicles. This article investigates the effects of such widespread phenomenon from the perspective of individual asset liability management. A multistage stochastic programming problem has been formulated with investment opportunities including PFs, unit-linked contracts and variable life annuities. The introduction of a specific risk measure with respect to a desirable retirement income stream and a planning horizon spanning the entire individuals' working life helps to analyse the implications of observed market dynamics on retirement strategies. We present comparative results focusing on the retirement planning problem for three representative individuals carrying different time horizons but common retirement goals. The results show the benefits over traditional pension accumulation plans of dynamic strategies based on mixed portfolios of retirement products.

Coppola, M., Russolillo, M., and Simone, R. (2020). "On the management of retirement age indexed to life expectancy: a scenario analysis of the Italian longevity experience." In: *The Journal of Risk Finance* 21(3), pp. 217–231.

Purpose This paper aims to measure the financial impact on social security system of a recently proposed indexation mechanism for retirement age by considering the Italian longevity experience. The analysis is motivated by the progressive increase in life expectancy at advanced age, which is rapidly bringing to the fore noticeable socio-economic consequences in most industrialized countries. Among those, the impact on National Social Security systems is particularly relevant if people live longer than expected; this will lead to greater financial exposure for pension providers. Design/methodology/approach Referring to the Italian population for illustrative purposes, the authors contemplate different scenarios for mortality projection methods and for the implementation of pension age shift while accounting for gender and cohort gaps and model risk. Synthetic indicators to measure the impact of the indexation mechanism on social security system are introduced on the basis of pension cash flows. Findings An indexation policy that manages gender gap while adjusting retirement age for varying life expectancy is proposed. As a result, sustainability of public retirement expenditure is improved. Originality/value The paper is a concise scenario analysis of the reduction of costs and risks that pension providers would have if the system resorted to link retirement age to life expectancy. The ideas fostered by the paper follow a recent proposal of the Authors on a flexible retirement scheme that deals with model risk for mortality projection and accounts for gender gap in mortality rates.

Correia, C. L., Sayre, R., and Allen, J. C. (2017). "Avoiding the Medicaid Trap: A Step-by-Step Guide to Estate Planning." In: *The Journal of Retirement* 5(2), pp. 96–103.

Until an elderly relative suddenly needs long-term care and cannot afford to pay for it, most Americans do not appreciate how difficult the Medicaid legal system is to navigate. With long-term services ranging from a median monthly average of USD1,473 for limited services to USD7,698 for comprehensive services, providing long-term care for a loved one forces the majority of Americans to apply for Medicaid benefits. Many individuals find themselves trapped to transfer any of their estate prior to applying for Medicaid because of the five-year lookback penalty. This article discusses seven estate planning steps that will help individuals protect their net worth while also securing eligibility for Medicaid benefits.

Cosares, S. (2013). "Generating a Family of Optimal Glide Paths for Investment Planning with Target Dates." In: *The Journal of Wealth Management* 16(3), pp. 43–53.
The article presents an optimization model for an approach to financial planning whereby investment decisions over time follow a (decreasing) glide path. With this strategy, an investment advisor would recommend riskier assets for the immediate future in order to take advantage of their larger expected returns. The author presumes that, over time, the proportion of more conservative investments in the portfolio will increase. Thus, it is likely to achieve expected investment-gain targets while reducing the risk of severe losses in wealth at the time when the money is actually needed (the target dates). There are a number of benefits to front-loading investment risk in this way, even though it comes with an increase in overall risk, as measured by the variance in the random variable representing total accumulated wealth. To accommodate varying objectives among investors regarding this trade-off, the model includes a parameter for glide-path intensity that allows the investor to select the level of deviation from the risk-minimizing solution, which would tend to recommend investments having roughly the same risk return level for every year in the planning horizon (i.e., a straight-line strategy). The model then focuses on the traditional multi-goal and multitime-horizon problem and uses simulations to demonstrate the benefits associated with a flexible implementation of the glide-path strategy that further minimizes the risk of missing a goal.

Costa, P., Pakula, D., and Clarke, A. S. (2021). *Fuel for the F.I.R.E.: Updating the 4% rule for early retirees*. Tech. rep. Vanguard.
Followers of the F.I.R.E. movement-Financial Independence Retire Early-have long relied on the "4% rule" to determine what they can withdraw from their portfolios over perhaps 40 or 50 years in retirement. The 4% rule, however, was originally designed for investors with a 30-year retirement horizon. And the simplifying assumptions it embeds about future returns, diversification, and fees can limit a retirement plan's viability over both shorter and longer horizons. Using Vanguard's principles for investing success, this research paper illustrates how F.I.R.E. investors can improve their chances of financing an early retirement.

Crook, M. and Baredes, M. (2015). "Total Wealth Allocation: Liquidity, Longevity, and Legacy." In: *The Journal of Wealth Management* 18(3), pp. 18–26.
In general, households use old technology for asset allocation decisions. Target-risk portfolio models are pervasive across the industry, and risk-profile questionnaires continue to form the foundation for strategic asset allocation decision-making. Better technology would utilize liability relative optimization and a holistic balance sheet and incorporate behavioral finance considerations. However, these concepts have not been integrated into most practitioners' businesses, even though the core work on each was completed 20-30 years ago. The authors propose a scalable, total wealth model for integrating liability relative optimization into households' portfolios. The basis for the model is an asset segmentation approach that refl ects the nuances of a households' balance sheet and future objectives and enables liability relative optimization and total wealth considerations to be applied appropriately.

Crook, M. and Pickens, N. (2025). "Portable Alpha for the (Taxable) Masses: Can Capital Efficient Funds Live Up to the Hype?" In: *The Journal of Beta Investment Strategies* 16(1), pp. 101–115.
Capital efficient retail products offer the promise of institutional-style portable alpha in an exchange traded fund (ETF) or mutual fund wrapper. Previous articles indicate that one specific type of capital efficient product, a 90/60 equity/bond strategy, has two potential efficiency-enhancing uses within investment portfolios. First, some have shown that 90/60 strategies can be used as an efficient replacement for long-only equity positions. Second, others suggest that 90/60 strategies can be used as replacements for traditional stock and bond allocations to implement a portable alpha strategy. We investigate both of these claims for investors that would otherwise invest in a portfolio of globally diversified equities and high-quality US bonds.

Crook, M. and Sutedja, R. (2017). "Will Long-Term Care Ruin Retirement Plans?" In: *The Journal of Retirement* 4(3), pp. 42–50.
Long-term care expenses represent a known unknown in retirement planning. A large majority of households will need some sort of long-term care support as they age, and costs for certain types of long-term care, like nursing homes, can run over USD100,000 per year. However, most families will not incur expenses of this order. Who is at risk? We use a simulation-based framework to analyze how often long-term care expenses are likely

to cause ruin in an otherwise prudently constructed financial plan. Our framework also enables us to estimate long-term care usage by type of service and calculate a distribution of total care costs for a particular household. We find that roughly 85% of older couples will utilize long-term care. Long-term care expenses impact financial plan sustainability at a declining rate as wealth increases from USD1 million to USD10 million. At a portfolio value of USD1 million, adding long-term care expenses to the simulation results in ruin (defined as depleting the portfolio entirely prior to the death of both members of the couple) about 30% of the time. With certain caveats discussed in the article, we estimate that failure rates increase by 9 percentage points for USD5 million households, and increase by about 5 percentage points for USD10 million households. We also find that female same-sex households are particularly at risk due to greater longevity and higher incidence of long-term care usage. In summary, financial plans that do not incorporate long-term care expenses can significantly overestimate the long-term sustainability of the plan.

Crook, M. W. (2019). "Liabilities Matter: Improving Target Date Glidepath Construction through Liability Adaptive Asset Allocation." In: *The Journal of Retirement* 7(1), pp. 44–57.

Target date glidepath funds have become extraordinarily popular in defined contribution plans during the past decade. Most glidepath strategies are based, fundamentally, on the idea that investors have bond-like human capital that slowly depletes during their career, and therefore investors should complement their human capital depletion with a declining equity glidepath as they progress toward retirement. Despite the theoretical attractiveness, the human capital approach unfortunately falls flat from an empirical and practical standpoint. In this article the author proposes a new liability-adaptive approach for target date glidepath construction. By moving away from a human capital approach and instead focusing on future retirement liabilities, the author shows that plan sponsors can improve outcomes for participants. The author also believes this framework enables sponsors to better communicate the underlying assumptions in their glidepath offering to participants who are planning for retirement. Finally, the author shows how financial advisors can use the same framework with individual investors and families to improve outcomes on a customized basis.

Cvitanic, J., Kou, S., Wan, X., and Williams, K. L. (2020). "Pi portfolio management: reaching goals while avoiding drawdowns." In: *SSRN e-Print*.

We propose an approach to portfolio selection that explicitly takes into account investors simultaneous investment objectives, such as achieving target return levels and avoiding specific drawdowns. Our approach is consistent with both standard and non-standard risk preferences, such as those of prospect theory. Instead of asking the investor to choose between lotteries, transforming this into an estimate of the risk preferences, and then selecting the portfolio accordingly, we propose to directly offer investors a choice between lotteries" with varying probabilities of experiencing target levels of profits and losses. Our approach enables investors to flexibly assess the effectiveness of portfolio choices under various conditions. We discuss implementation considerations and compare our approach to traditional mean-variance portfolio selection.

Czasonis, M., Kritzman, M., and Turkington, D. (2024). "Full-Scale Currency Hedging." In: *Journal of Investment Management* 22(2).

After years of spirited debate, most investors agree that to minimize the risk currencies add to a portfolio they should hedge its currency exposures based on its betas relative to the currencies to which it is exposed. However, this notion of hedging makes sense only if the betas reliably reflect the co-occurrences of the cumulative returns of the portfolio and currencies over the investor's horizon. And this will be true only if the correlations of currencies with the portfolio and with each other are constant across the return intervals used to estimate them and stationary through time. Neither of these conditions holds empirically. The authors propose a new currency hedging technique called full-scale hedging, which explicitly considers the full distribution of horizon co-occurrences.

D'Acunto, F. and Rossi, A. G. (2021). "New Frontiers of Robo-Advising: Consumption, Saving, Debt Management, and Taxes." In: *SSRN e-Print*.

Traditional forms of robo-advice were targeted to help individuals make portfolio allocation decisions. Based on the balance-sheet view of households, the scope for robo-advising has been expanding to many other personal-finance choices, such as households' saving and consumption decisions, debt management, mortgage uptake, tax management, and lending. This chapter reviews existing research on these new functions of robo-advising with a special emphasis on the questions that are still open for researchers across several disciplines. We also discuss the attempts to optimize jointly all personal-finance decisions, which we label "holistic robo-advisors". We conclude by assessing fruitful avenues for research and practice in finance, computer science, marketing, decision science, information systems, law, and sociology.

D'Amico, G., Singh, S., and Selvamuthu, D. (2024). "Optimal investment-disinvestment choices in health-dependent variable annuity." In: *Insurance: Mathematics and Economics* 117, pp. 1–15.
This paper exploits the influence of the policyholder's health status on the optimal time at which the policyholder decides to stop paying health-dependent premiums and starts withdrawing health-dependent benefits from a variable annuity (VA) contract accompanied by a guaranteed lifelong withdrawal benefit (GLWB). A mixed continuous-discrete time model is developed to find the optimal time for withdrawal regime initiation. The model determines the investment and disinvestment triggers according to the market conditions for both dynamic and static cases. In the static case, the optimal time is computed at the policy's inception time. In contrast, in the dynamic case, the optimal initiation time is achieved by recursive calculation of the exercise frontier of a real deferral option. Another finding is the sensitivity analysis of the contract concerning the insurance fee and the age of the policyholder.
Da Fonseca, J. (2024). "Pricing guaranteed annuity options in a linear-rational Wishart mortality model." In: *Insurance: Mathematics and Economics* 115, pp. 122–131.
This paper proposes a new model, the linear-rational Wishart model, which allows the joint modelling of mortality and interest rate risks. Within this framework, we obtain closed-form solutions for the survival bond and the survival floating rate bond. We also derive a closed-form solution for the guaranteed annuity option, i.e., an option on a sum of survival (floating rate) bonds, which can be computed explicitly up to a one-dimensional numerical integration, independent of the model dimension. Using realistic parameter values, we provide a model implementation for these complex derivatives that illustrates the flexibility and efficiency of the linear-rational Wishart model.
Dadashi, H. (2020a). "Optimal investment strategy post retirement without ruin possibility: A numerical algorithm." In: *Journal of Computational and Applied Mathematics* 363, pp. 325–336.
We give a numerical algorithm based on the policy iteration method to find approximations of solution of an HJB equation arising from the portfolio optimization problem post retirement for a member of a defined contribution plan. In our numerical algorithm, we apply a fully implicit finite difference scheme to the HJB equation on a bounded domain. In formulating the optimal control problem, the loss function is written using a target function during the decumulation phase and at the terminal time. Following Di Giacinto et al. (2014), a minimum guarantee for the final annuity is considered. Finally, we give the simulation results for the final annuity, optimal investment strategy and optimal wealth process, for different levels of risk aversion and different loss functions.
Dadashi, H. (2020b). "Optimal investment–consumption problem: Post-retirement with minimum guarantee." In: *Insurance: Mathematics and Economics* 94, pp. 160–181.
We study the optimal investment-consumption problem for a member of defined contribution plan during the decumulation phase. For a fixed annuitization time, to achieve higher final annuity, we consider a variable consumption rate. Moreover, to have a minimum guarantee for the final annuity, a safety level for the wealth process is considered. To solve the stochastic optimal control problem via dynamic programming, we obtain a Hamilton-Jacobi-Bellman (HJB) equation on a bounded domain. The existence and uniqueness of classical solutions are proved through the dual transformation. We apply the finite difference method to find numerical approximations of the solution of the HJB equation. Finally, the simulation results for the optimal investment-consumption strategies, optimal wealth process and the final annuity for different admissible ranges of consumption are given. Furthermore, by taking into account the market present value of the cash flows before and after the annuitization, we compare the outcomes of different scenarios.
Daher, M., Dahling, D., Pritchard, R., and Tseng, B. (2020). *From Accumulation to Decumulation – Why It Matters and What Plan Sponsors Should Know*. Tech. rep. Fidelity Investments.
Through conversations with Fidelity clients – and consistent with industry surveys – we have noted increasing interest from DC plan sponsors in allowing retirees to stay in plan.
Retirees and pre-retirees need help balancing risks – including investment, longevity, liquidity, and utilization risk – to translate accumulated savings into sustainable income.
Fidelity's recordkeeping data shows that older DC participants may not fully appreciate the importance of appropriate asset allocation and diversification as they age.
Sponsors wanting to allow participants to stay in plan should consider offering post-retirement tools and education along with strategies and products that can help retirees to allocate their assets appropriately, generate income, and safely draw down savings during retirement.
Dai, M., Lei, Y., Liu, H., and Yang, C. (2024). "Optimal Tax-Timing with Transaction Costs." In: *SSRN e-Print*.

We develop a dynamic portfolio model incorporating capital gains tax (CGT), year-end taxation, and transaction costs. We find that transaction costs affect loss deferrals much more than gain deferrals, and such effects are asymmetric in the presence of accumulated realized gains and losses. Our model can help explain the puzzle that even when investors face equal long-term/short-term CGT rates or almost zero interest rates, they may still defer realizing large capital losses. In addition, it provides several unique, empirically testable predictions.

Dai, W. and Medhat, M. (2021). "US Inflation and Global Asset Returns." In: *SSRN e-Print*.
We study the relation between US inflation and the performance of global asset classes (including bonds, stocks, industry portfolios, factor premiums, commodities, and REITs), both over a long sample period (1927-2020) and over the most recent 30 years (1991-2020). We find that most assets had positive average real returns in both low- and high-inflation years. While average real returns were lower in years with higher inflation for most assets, many of the differences are not statistically reliable, especially among non-bond assets and in more recent times. We also find mostly weak correlations over time between nominal returns and inflation, including contemporaneous, lagged, expected, and unexpected inflation. The notable exceptions are energy stocks and commodities, where there are reliably positive correlations with both expected and unexpected inflation, but our results also suggest both assets are too volatile to be an effective inflation hedge. Our results confirm the potential of most asset classes to outpace inflation over the long term and suggest that, for investors prioritizing the preservation of purchasing power, inflation-indexed securities may be a more appropriate inflation hedge than commonly suggested alternatives.

Dai, Y., Guo, D., Leggio, K. B., and Lien, D. (2025). "Optimal Decumulation: The Case for Quarterly Withdrawals of Retirement Savings." In: *The Journal of Investing* 34(2), pp. 113–127.
There is no shortage of recommendations for ways retirees can decumulate their lifetime savings during retirement. The topic is not one retirees consider lightly; most retirees fear outliving their funds more than they fear dying. In this article we consider three alternatives to the much-maligned 4% rule and find retirees are best served by withdrawing funds quarterly using a strategy based upon a multiple of the IRS' Required Minimum Distribution (RMD) tables. Our results eliminate the likelihood of retirees exhausting funds during the conservative retirement timespan of 35 years while also maximizing funds available during their retirement years.

Dang, D.-M., Forsyth, P. A., and Vetzal, K. R. (2017). "The 4 percent strategy revisited: a pre-commitment mean-variance optimal approach to wealth management." In: *Quantitative Finance* 17(3), pp. 335–351.
In contrast to single-period mean-variance (MV) portfolio allocation, multi-period MV optimal portfolio allocation can be modified slightly to be effectively a down-side risk measure. With this in mind, we consider multi-period MV optimal portfolio allocation in the presence of periodic withdrawals. The investment portfolio can be allocated between a risk-free investment and a risky asset, the price of which is assumed to follow a jump diffusion process. We consider two wealth management applications: optimal de-accumulation rates for a defined contribution pension plan and sustainable withdrawal rates for an endowment. Several numerical illustrations are provided, with some interesting implications. In the pension de-accumulation context, Bengen (1994)?s [J. Financial Planning, 1994, 7, 171?180], historical analysis indicated that a retiree could safely withdraw 4 percent of her initial retirement savings annually (in real terms), provided that her portfolio maintained an even balance between diversified equities and U.S. Treasury bonds. Our analysis does support 4 percent as a sustainable withdrawal rate in the pension de-accumulation context (and a somewhat lower rate for an endowment), but only if the investor follows an MV optimal portfolio allocation, not a fixed proportion strategy. Compared with a constant proportion strategy, the MV optimal policy achieves the same expected wealth at the end of the investment horizon, while significantly reducing the standard deviation of wealth and the probability of shortfall. We also explore the effects of suppressing jumps so as to have a pure diffusion process, but assuming a correspondingly larger volatility for the latter process. Surprisingly, it turns out that the MV optimal strategy is more effective when there are large downward jumps compared to having a high volatility diffusion process. Finally, tests based on historical data demonstrate that the MV optimal policy is quite robust to uncertainty about parameter estimates. In contrast to single-period mean-variance (MV) portfolio allocation, multi-period MV optimal portfolio allocation can be modified slightly to be effectively a down-side risk measure. With this in mind, we consider multi-period MV optimal portfolio allocation in the presence of periodic withdrawals. The investment portfolio can be allocated between a risk-free investment and a risky asset, the price of which is assumed to follow a jump diffusion process. We consider two wealth management applications: optimal de-accumulation rates for a defined contribution pension plan and sustainable withdrawal rates for an endowment. Several numerical illustrations are provided, with some interesting implications. In the pension de-accumulation context, Bengen (1994)?s [J. Financial Planning, 1994, 7, 171?180], historical analysis indicated that a retiree

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Daraei, D. and Sendova, K. (2024). "Determining Safe Withdrawal Rates for Post-Retirement via a Ruin-Theory Approach." In: *Risks* 12(4), p. 70.
To ensure a comfortable post-retirement life and the ability to cover living expenses, it is of utmost importance for individuals to have a clear understanding of how long their pre-retirement savings will last. In this research, we employ a ruin-theory approach to model the inflows and the outflows of retirees' portfolios. We track all transactions within the portfolios of retired clients sourced by a registered investment provider to Canada's Financial Wellness Lab at Western University. By utilizing an advanced ruin model, we calculate the mean and the median time it takes for savings to be exhausted, the probabilities of exhaustion of funds within the retirees' expected remaining lifetime while accounting for the observed withdrawal rates, and the deficit at ruin if a retiree has used up all of their savings. We also account for gender as well as for the risk tolerance of retired clients using a K-Means clustering algorithm. This allows us to compare the financial outcomes for female and male retirees and to enhance some findings in the literature. In the final phase of our study, we compare the results obtained by our methodology to the 4% rule which is a widely used approach for post-retirement spending. Our results show that most retirees can withdraw safely more than they currently do (around 2.5%). A withdrawal rate of about 4.5% is proved to be safe, but it might not provide sufficient income for most retirees since it yields approximately CAD 20,000 per year for male retirees in the highest risk tolerance group who withdraw about 4.5% annually.

Das, S. and Varma, S. (2020). "Dynamic Goals-Based Wealth Management using Reinforcement Learning." In: *Journal of Investment Management* 18 (2), pp. 1–20.
We present a reinforcement learning (RL) algorithm to solve for a dynamically optimal goal-based portfolio. The solution converges to that obtained from dynamic programming. Our approach is model-free and generates a solution that is based on forward simulation, whereas dynamic programming depends on backward recursion. This paper presents a brief overview of the various types of RL. Our example application illustrates how RL may be applied to problems with path-dependency and very large state spaces, which are often encountered in finance.

Das, S. R., Ostrov, D., Radhakrishnan, A., and Srivastav, D. (2020a). "Dynamic portfolio allocation in goals-based wealth management." In: *Computational Management Science* 17, pp. 613–640.
We report a dynamic programming algorithm which, given a set of efficient (or even inefficient) portfolios, constructs an optimal portfolio trading strategy that maximizes the probability of attaining an investor's specified target wealth at the end of a designated time horizon. Our algorithm also accommodates periodic infusions or withdrawals of cash with no degradation to the dynamic portfolio's performance or runtime. We explore the sensitivity of the terminal wealth distribution to restricting the segment of the efficient frontier available to the investor. Since our algorithm's optimal strategy can be on the efficient frontier and is driven by an investor's wealth and goals, it soundly beats the performance of target date funds in attaining investors' goals. These optimal goals-based wealth management strategies are useful for independent financial advisors to implement behavioral-based FinTech offerings and for robo-advisors.

Das, S. R., Ostrov, D., Radhakrishnan, A., and Srivastav, D. (2022a). "Dynamic optimization for multi-goals wealth management." In: *Journal of Banking and Finance* 140 (106192).
We develop a dynamic programming methodology that seeks to maximize investor outcomes over multiple, potentially competing goals (such as upgrading a home, paying college tuition, or maintaining an income stream in retirement), even when financial resources are limited. Unlike Monte Carlo approaches currently in wide use in the wealth management industry, our approach uses investor preferences to dynamically make the optimal determination for fulfilling or not fulfilling each goal and for selecting the investor's investment portfolio. This can

be computed quickly, even for numerous investor goals spread over different or concurrent time periods, where each goal may be all-or-nothing or may allow for partial fulfillment. The probabilities of attaining each (full or partial) goal under the optimal scenario are also computed, so the investor can ensure the algorithm accurately reflects their preference for the relative importance of each of their goals. This approach vastly outperforms static portfolio strategies and target-date funds, widely used in the wealth management industry.

Das, S. R., Ostrov, D., Radhakrishnan, A., and Srivastav, D. (2023a). "Lifestyle, Longevity, and Legacy Risks with Annuities in Retirement Portfolio Decumulation." In: *The Journal of Wealth Management*.
Investors planning for retirement balance three Ls: (1) lifestyle risk, hoping to maintain a consumption stream that provides a chosen standard of living; (2) longevity risk, hoping to remain solvent throughout their lifetime; and (3) legacy risk, hoping to leave a bequest to their heirs. We solve this multiple objective problem for a wide range of consumption and annuitization scenarios. For each scenario, we apply dynamic programming to optimally evolve the investments in the non-annuitized portion of the portfolio so as to minimize longevity risk. Our dynamic programming approach has the advantages of (1) generating results that are far superior to what standard Monte Carlo methods, static portfolios, and target date fund glide paths can provide and (2) not requiring utility functions, which are hard to specify for individuals. We show that investors who want to minimize their longevity and legacy risk and who are unable to annuitize their full consumption stream are best off avoiding even partial annuitization of their portfolio. For investors who are able to annuitize their full consumption stream, we quantify their longevity versus legacy risk trade-offs, enabling them to select the best annuity for their needs.

Das, S. R., Ostrov, D. N., Casanova, A., Radhakrishnan, A., and Srivastav, D. (2020b). "Combining Investment and Tax Strategies for Optimizing Lifetime Solvency under Uncertain Returns and Mortality." In: *SSRN e-Print*.
We consider an investor who is looking to maximize their probability of remaining solvent throughout their lifetime by using an algorithm that aims to optimize their investment allocation strategy and optimize their tax strategy for withdrawal allocations between tax deferred accounts (TDAs), Roth accounts, and taxable stock and bond accounts. Our optimization works with stochastic investment returns and stochastic mortality. We find that optimizing the investment strategy (via dynamic programming) has a much larger impact on the investor remaining solvent than optimizing the tax strategy (via Monte Carlo and numerical optimization). This result is key to effectively optimizing both strategies simultaneously. We show that our optimized investment strategy soundly beats a standard target date fund strategy, while our novel optimized tax strategy displays the optimal desired properties suggested by non-stochastic tax optimization research.

Das, S. R., Ostrov, D. N., Casanova, A., Radhakrishnan, A., and Srivastav, D. (2022b). "Optimal Goals-Based Investment Strategies For Switching Between Bull and Bear Markets." In: *The Journal of Wealth Management* 24(4), pp. 8–36.
We solve a dynamic, long-horizon, goals-based wealth management problem, given different investment regimes. In a world with a good regime (bull market) and a bad regime (bear market), an investor who is cognizant that regime switching occurs has the potential to do better than an investor who assumes only one regime. However, models with more than one regime incur the additional risk of regime uncertainty. Investors must be able to predict which regime is governing the market with reasonable levels of confidence, or they can be worse off than investors who assume just one regime. Using data from recent history, we develop a framework that determines how accurate regime prediction needs to be to achieve gains from a regime-cognizant goals-based investing approach.

Das, S. R., Ostrov, D. N., Radhakrishnan, A., and Srivastav, D. (2023b). "Efficient Goal Probabilities: A New Frontier." In: *Journal of Investment Management* 21(3).
In goals-based wealth management (GBWM), an investor looks to maximize the probabilities of attaining each of n goals over time. Because the goals are in competition for potentially limited financial resources, their relative importance must be specified, which we do by assigning utility weights to each goal. Given these weights, dynamic programming can determine both the optimal investment strategy and the optimal strategy for when to fulfill versus forgo each goal. This yields the optimal goal probabilities for fulfilling each goal. By altering the utility weights, we show how to generate the efficient goal probability frontier (EGPF), an (n - 1)-dimensional hypersurface of the optimized goal probability combinations. Just as the classic efficient frontier in mean-variance portfolio optimization allows investors to understand the trade-offs under the best circumstances between their portfolio's mean and variance, the EGPF allows the investor to under- stand the trade-offs under the best circumstances between the probabilities of attaining each of their goals - without needing to see or understand

the goals' underlying utility weights. We extend our EGPF framework to determine the minimal initial wealth needed to attain specified probabilities for each completely or partially fulfilled goal.

Das, S. R. and Ross, G. (2021). "The Role of Options in Goals-Based Wealth Management." In: *SSRN e-Print*.
We develop a facile methodology using dynamic programming for goals-based wealth management over long horizons where rebalancing uses the standard securities and also derivative securities. A kernel density estimation approach is developed to accommodate any number of derivative assets, solving a high dimensional problem with fast computation. The approach easily accommodates skewed and fat-tailed distributions. Portfolio performance is much better with the use of options, especially for investors with aggressive goals.

Das, S. R. and Ross, G. (2023). "The Role of Options in Goals-Based Wealth Management." In: *Journal of Investment Management* 21(1).
We develop a methodology using dynamic programming for goals-based wealth management over long horizons where portfolio rebalancing uses the standard securities and also derivative securities. A kernel density estimation approach is developed to accommodate derivative assets, solving a high-dimensional problem with fast computation. The approach accommodates skewed and fat-tailed distributions. Portfolio performance is better with the use of options, especially for investors with aggressive goals. The improved performance arises because options unlock additional leverage, which is useful for reaching upside goals. Calls are preferred to puts unless upside goals are modest. The framework is extensible with periodic withdrawals and multiple goals, while being cognizant of downside risk.

Davidow, A. B. (2022). "Goals-Based Investing: Moving Beyond the 60/40 Portfolio." In: *Investments and Wealth Monitor*.
Wealth advisors should use goals-based investing to solve the needs of HNW families by adjusting their relationships and value proposition to better align with client goals. HNW families have complex needs, and goals-based invest- ing is designed to deal with them indi- vidually. Goals-based investing is preferable to the naive 60/40 portfolio for both wealth advisors and HNW families.

Davidow, A. B. (2023). "Building Better Portfolios with Alternative Investments: A Goals-Based Framework." In: *Investment and Wealth Monitor* (July/August), pp. 19–29.
This article considers asset allocation and portfolio construction, and the impact of adding alternative investments to a diversified portfolio. We will use case studies to illustrate how to allocate alternative investments and show the impact they have on a traditional portfolio.

Davies, S. (2023). "A Review: The Financial Impact for Socially Responsible Investors." In: *SSRN e-Print*.
The purpose of this document is to provide an overview of the financial impact of socially responsible investing (SRI) practices for investors, in particular, the impact of excluding socially undesirable stocks and putting more weight on socially desirable stocks. The review covers only the financial performance of SRI investing and is agnostic to the morality or externalities of such practices. On the theory side, the review discusses the expected financial trade-offs associated with SRI investing. On the empirical side, the review covers the historical return performance of various SRI investment strategies and discusses the expected returns going forward. The review also considers the management fees associated with delegated SRI investing and the impact on portfolio quality. Finally, the review provides a representative example to quantify the overall net financial impact, which shows that the at-retirement wealth losses are not insignificant.

Davis, J. and Klein, S. (2022). "Beyond Asset Allocation: Three Pillars of a Robust Retirement." In: *Investments and Wealth Monitor*.
Without assuming prophetic skill, it is difficult to evaluate an asset allocation on a stand-alone basis. Instead, we suggest that retirees focus on what we call "the three pillars of a robust retire- ment," which comprise strategies that make retirement cash flows more predictable: a realistic and beneficial spending plan, increasing tax efficiency, and maximized Social Security benefits.

De Angelis, T., Milazzo, A., and Stabile, G. (2024). "On variable annuities with surrender charges." In: *arXiv e-Print*.
In this paper we provide a theoretical analysis of Variable Annuities with a focus on the holder's right to an early termination of the contract. We obtain a rigorous pricing formula and the optimal exercise boundary for the surrender option. We also illustrate our theoretical results with extensive numerical experiments. The pricing problem is formulated as an optimal stopping problem with a time-dependent payoff which is discontinuous at the maturity of the contract and non-smooth. This structure leads to non-monotonic optimal stopping boundaries which we prove nevertheless to be continuous and regular in the sense of diffusions for the stopping set. The lack of monotonicity of the boundary makes it impossible to use classical methods from optimal stopping. Also

more recent results about Lipschitz continuous boundaries are not applicable in our setup. Thus, we contribute a new methodology for non-monotone stopping boundaries.

De La Pena, J. I., Fernandez-Ramos, M. C., and Garayeta, A. (2021). "Cost-Free LTC Model Incorporated into Private Pension Schemes." In: *International Journal of Environmental Research and Public Health* 18(5) (2268).
Long-term care coverage is not integrated into an individual's retirement strategy. It is an additional public health service that is not considered into private pension funds. Nevertheless, this coverage is not sufficient due to the problems of financial sustainability of the public pension systems. However, there are large sums in pension plans dedicated to paying retirement pensions that can be transformed into support for long-term care coverage. This paper develops a mechanism of pension transformation through the different mortality of the beneficiary when becoming a dependent beneficiary. This mechanism allows the beneficiary to convert their pension to LTC support at their own choice, without increasing the cost of the private pension scheme. The proposed model provides consistency in the pension that a retiree receives and adapts it to a retiree's life expectancy: the retiree receives a higher pension when he/she needs it most.

de Mendonça, H. F., Garcia, P. M., and Vicente, J. V. M. (2021). "Rationality and anchoring of inflation expectations: An assessment from survey-based and market-based measures." In: *Journal of Forecasting* 40(6), pp. 1027–1053.
The aim of this paper is twofold. Firstly, we test the rationality of survey-based and market-based inflation expectations. Secondly, we investigate whether they indicate a different performance of the central bank in anchoring inflation expectations. Briefly, this paper verifies if inflation expectations' proxies display the same features regarding rationality and anchoring. Using data from the Brazilian market, we present robust evidence that both survey-based and market-based inflation expectations have useful content to explain realized inflation. Moreover, we find that these proxies of inflation expectations provide different assessments of the central bank's ability to anchor inflation expectations. The findings point out that central banks must monitor both survey-based and market-based inflation expectations to improve their monetary policy conduct.

De Nard, G., Ledoit, O., and Wolf, M. (2025). "Improved Tracking-Error Management for Active and Passive Investing." In: *The Journal of Portfolio Management* 51(4), pp. 40–62.
Tracking-error management is largely absent from the academic literature but ubiquitous in real life: Most portfolio managers are tied to a benchmark. Some of them aim to track a benchmark (such as the S&P 500), which is not necessarily a trivial task because the benchmark often contains assets that are difficult or expensive to trade. In this case, the objective is to minimize tracking error. Other managers aim to take on an active tilt without deviating too much from a benchmark. In this case, the objective is to control tracking error. In both cases, managers need an estimator of the covariance matrix of many (excess) returns for their objective. This article demonstrates the benefit of sophisticated shrinkage estimators (in conjunction with multivariate GARCH models) to this end, relative to the commonly used sample covariance matrix.

de Villiers, J. U. and Roux, E.-M. (2019). "Reframing the Retirement Saving Challenge: Getting to a Sustainable Lifestyle Level." In: *Journal of Financial Counseling and Planning* 30(2), pp. 277–288.
An increasing number of individuals will be unable to retire comfortably amidst an international retirement savings crisis. Research suggests that behavioral factors contribute to inadequate retirement savings. We present a procedure that reframes the retirement savings decision, aimed at alleviating some of the negative effects of the behavioral factors. This procedure shifts the focus from the required wealth at retirement (the future) to the lifestyle an individual can afford to maintain now (the present). A sustainable lifestyle level (SLL) approach is expressed mathematically and illustrated with practical examples. The SLL approach offers a practical tool for retirement planning professionals to present recommendations that are simple and easy to understand for individuals faced with complex retirement planning decisions.

Deelstra, G., Devolder, P., and Melis, R. (2021). "Optimal annuitisation in a deterministic financial environment." In: *Decisions in Economics and Finance* 44(1), pp. 161–175.
The global reforms to public pension schemes over the last thirty years have progressively reduced individuals' post-retirement social security income. In order to compensate for this, individuals join pension funds and individual plans to increase their wealth at retirement. These types of fully funded plans generally give individuals the opportunity to withdraw the capital accumulated into their scheme or to convert it into an annuity. In this paper, we analyse individuals' post-retirement choices to allocate the wealth at retirement between consumption, risk-free investments and a life annuity. We develop a discrete time optimisation model, in a deterministic framework, with a constant relative risk aversion (CRRA) utility function. We study the effect of a bequest

motive and the annuity rate used by the insurer on the optimal choice. Several numerical applications are presented to illustrate the optimal annuitisation decision results and the optimal consumption paths.

Deelstra, G., Devolder, P., and Roelants du Vivier, B. (2024). "Impact of correlation between interest rates and mortality rates on the valuation of various life insurance products." In: *ASTIN Bulletin* 54(3), pp. 569–599.
In this paper, we question the traditional independence assumption between mortality risk and financial risk and model the correlation between these two risks, estimating its impact on the price of different life insurance products. The interest rate and the mortality intensity are modelled as two correlated Hull and White models in an affine set-up. We introduce two building blocks, namely the zero-coupon survival bond and the mortality density, calculate them in closed form and perform an investigation about their dependence on the correlation between mortality and financial risk, both with theoretical results and numerical analysis. We study the impact of correlation also for more structured insurance products, such as pure endowment, annuity, term insurance, whole life insurance and mixed endowment. We show that in some cases, the inclusion of correlation can lead to a severe underestimation or overestimation of the best estimate. Finally, we illustrate that the results obtained using a traditional affine diffusive set-up can be generalised to affine jump diffusion by computing the price of the zero-coupon survival bond in the presence of jumps.

Deguest, R., Martellini, L., Milhau, V., Suri, A., and Wang, H. (2015). *Introducing a Comprehensive Investment Framework for Goals-Based Wealth Management*. Tech. rep. EDHEC Risk Institute.
Any investment process should start with a thorough understanding of the investor problem. Individual investors do not need investment products with alleged superior performance; they need investment solutions that can help them meet their goals subject to prevailing dollar and risk budget constraints. In this new publication, EDHEC-Risk Institute develops a general operational framework that can be used by financial advisors to allow individual investors to optimally allocate to categories of risks they face across all life stages and wealth segments so as to achieve personally meaningful financial goals.

Dehm, C. (2020). "Stochastic mortality : modelling and optimal investment." PhD thesis. University of Ulm.
Nothing is more certain than death, nothing more uncertain than its hour. Even today the timepoint of death is unknown, but death will occur with certainty. In this work, the uncertainty of death time is included in our considerations of mortality modelling, optimal investment as insured under uncertain death time and optimal investment as insurance company under mortality risk. This work provides three contributions to scientific literature: In the first part, we suggest causality-based modifications of the Gompertz law and the Lee Carter model with focus on main death causes. In an application on German health data, we identify three main death causes, namely 'diseases of circulatory system', 'neoplasms' and 'diseases of the respiratory system', completed by modelling 'accumulated remaining diseases'. Using VAR(p)-time series models, we forecast life expectancy and life annuity for the total population as well as for the genders separately. The causal Lee Carter model for men matches the holistic Lee Carter model best, with nearly same life expectancy and life annuity. Moreover, we find that the predominant marginal impact on life expectancy are 'neoplasms' with an average loss of about 5 years. In the second contribution, we consider an optimal investment problem inspired by a life insurance contract. These contracts usually have an option-like form with the additional difficulty that the timepoint of benefit payment (e.g. death of the insured) is uncertain. We generalize the known Merton problem in two directions: we consider non-concave optimization with uncertain investment maturity. In fact, we consider general non-concave utility functions and prove an existence and optimality result. To illustrate our findings, we do a numerical study, which suggests that the occurrence of a possible premature stop leads to a reduced performance of the optimal solution in terms of certainty equivalent. Due to the long term of an insurance contract, a typical life insurance company is subject to interest and investment risks as well as mortality risk. In our third contribution, we consider these risks as jump-diffusion processes. Through (interest) zero-coupon bond and longevity bond, interest rate risk and mortality risk can be (partially) traded on the capital market. In this Levy-market, we derive an extended HJB equation system to be able to find closed-form solutions of the optimal trading strategy and the expected terminal value. We prove necessity and sufficiency of the HJB equation approach. Moreover, we show a general L2-asymptotic of truncated Levy processes.

DeJong, J. C. and Robinson, J. H. (2022). "A Case Study in Sequence Risk: A 20-Year Retrospective on the Impact of the 2000-2002 and 2007-2009 Bear Markets on Retirement Nest Egg Sustainability." In: *The Journal of Wealth Management* 24(4), pp. 37–68.
The first decade of the 2000s saw two prolonged bear markets, both of which produced declines of over 50% in value in the S&P 500 Index from peak to trough. In total, the ten-year period from March 1999 through February 2009 saw the stock market lose more than 30 percent of its value. This period represented the worst

decade for the U.S. stock market since the Great Depression. For consumers who had the misfortune of retiring in 1999, this "lost decade" embodied the definition of sequence of returns risk. Researchers at the time could only speculate as to the impact such sharply negative returns might have on the long-term sustainability of retiree spending portfolios.

del Carmen Valls Martínez, M., Santos-Jaén, J. M., Amin, F.-u., and Martín-Cervantes, P. A. (2021). "Pensions, Ageing and Social Security Research: Literature Review and Global Trends." In: *Mathematics* 9(24), p. 3258.
Pension systems are one of the fundamental pillars of the welfare state. The ageing of the population caused by longer life expectancy and low birth rates has led to a crisis in the public pension system in developed countries. Changes for the system's sustainability are necessary, and the scientific literature on the subject is abundant, especially in recent years. This article aims to carry out a bibliometric analysis of the research carried out to date, highlighting, in turn, future lines of research. The study was carried out on a total of 1287 articles published from 1936 to 2021 and found in the Scopus database. The SciMAT, VOSviewer, and Datawrapper tools were used to analyse the most important articles, authors, countries, and institutions by volume of production and citations, as well as the relationships between them. Likewise, the most important keywords and their evolution over time were highlighted, obtaining the main focus of the research. In addition to the general analysis, a specific study was carried out in the area of Mathematics. The results show that the leading countries are the United Kingdom, the USA, and the Netherlands. On the other hand, the lead subject area in which these articles have been published is Economics, Econometrics, and Finance. The research trends are sustainability, pension reform related to ageing, and pension insurance.

DeLibero, R. and Pfau, W. D. (2017). "Life Insurance as a Retirement Income Tool." In: *SSRN e-Print*.
Given its tax-preferential treatment, careful study is warranted to determine whether life insurance can play an important role in an overall retirement portfolio. This study develops hypothetical scenarios for different types of individuals with varying ages and distribution periods, while using a historical outlook to determine the proper structure of a variable universal life insurance policy. We compare a variable universal life policy to different investment vehicles (both in qualified and non-qualified accounts) on an after-tax basis in order to better understand the potential tradeoff for tax-deferral and insurance fees within the life insurance.

Delong, L. and Chen, A. (2016). "Asset allocation, sustainable withdrawal, longevity risk and non-exponential discounting." In: *Insurance: Mathematics and Economics* 71, pp. 342–352.
The present paper studies an optimal withdrawal and investment problem for a retiree who is interested in sustaining her retirement consumption above a pre-specified minimum consumption level. Apparently, the withdrawal and investment policy depends substantially on the retiree's health condition and her time preferences (subjective discount factor). We assume that the health of the retiree can worsen or improve in an unpredictable way over her lifetime and model the retiree's mortality intensity by a stochastic process. In order to make the decision about the consumption and investment policy more realistic, we assume that the retiree applies a non-exponential discount factor (an exponential discount factor with a small amount of hyperbolic discounting) to value her future income. In other words, we consider an optimization problem by combining four important aspects: asset allocation, sustainable withdrawal, longevity risk and non-exponential discounting. Due to the non-exponential discount factor, we have to solve a time-inconsistent optimization problem. We derive a non-local HJB equation which characterizes the equilibrium optimal investment and consumption strategy. We establish the first-order expansions of the equilibrium value function and the equilibrium strategies by applying expansion techniques. The expansion is performed on the parameter controlling the degree of discounting in the hyperbolic discounting that is added to the exponential discount factors. The first-order equilibrium investment and consumption strategies can be calculated in a feasible way by solving PDEs.

Dempster, M. A. H., Kloppers, D., Medova, E., Osmolovsky, I., and Ustinov, P. (2016). "Lifecycle Goal Achievement or Portfolio Volatility Reduction?" In: *The Journal of Portfolio Management* 42(2), pp. 99–117.
This article is concerned with the use of currently available technology to offer individuals, financial advisors, and pension fund financial planners detailed prospective financial plans tailored to an individual's financial goals and obligations. By taking account of all an individual's prospective cash flows, including servicing current liabilities, and simultaneously optimizing prospective spending, saving, asset allocation, tax, and insurance, etc. using dynamic stochastic optimization, the authors compare the results of their goal-based fully dynamic strategy with the financial advisory industry's representative current best practices. These include piecemeal fixed-allocation portfolios for specific goals, target-date retirement funds, and fixed real-income post-retirement financial products, all using Markowitz mean-variance optimization based on the very general goal of minimizing portfolio volatility for a specific portfolio expected return over a finite horizon. Making use of the same data and

marketcalibrated Monte Carlo stochastic simulation for all the alternative portfolio strategies, the authors find that flexibility is of key importance for both individual portfolio and spending decisions. The authors measure superiority by the certainty-equivalent increase in expected utility of individual lifetime consumption (gamma) and the extra initial capital required by an individual to put the dominated strategy on the same expected-utility footing as the optimal dynamic strategy (initial capital gap). They find that the adaptive dynamic goal-based portfolio strategy's performance is far superior to all the industry's Markowitz-based approaches. These empirical results should put paid to the commonly held view that the extra complexity of holistic dynamic stochastic models is not worth the marginal extra value obtained from their use.

Dempster, M. A. H. and Medova, E. A. (2011). "Asset Liability Management for Individual Households." In: *British Actuarial Journal* 16(2).
Personal finance is a challenging topic which can benefit from a scientific approach to individual financial planning. This paper presents an individual asset liability management (iALM) model for life cycle planning which uses the methodology of dynamic stochastic optimisation and incorporates ideas from both classical and behavioural finance. Its implementation is in the form of a decision support tool for use by financial advisers or wealth managers. The investment universe is given by a set of indices for major asset classes and their returns are simulated forward over the lifetime of a household. On the liability side the foreseen cash flows of incomes and outgoings are simulated and punctuated by life events such as illness and death. The household's utility function is constructed for each time period over a range of monetary values in terms of household financial goals and preferences. Taxes and pension savings are treated using the tax shielded saving accounts specific to a national jurisdiction in terms of constraints in the optimisation sub-models. The paper goes on to present an analysis of iALM model recommendations for a representative UK household, together with an evaluation of the sensitivity of the financial plan generated to changes in market environments such as the 2007-9 crisis. The promise of this new technology is to bring modern decision support tools to individual investors in order to facilitate custom designed consumption, savings and investment policies.

Denault, M. and Simonato, J.-G. (2022). "A note on a dynamic goal-based wealth management problem." In: *Finance Research Letters* 46 (Part B) (102404).
This short note suggests two improvements for solving the goal-based wealth management problem proposed by Das, Ostrov, Radhakrishnan and Srivastav (2020). The first suggestion smoothes and improves the convergence of the approximate solutions towards the underlying, continuous solution either by using analytic solutions at the penultimate time point or by adjusting the wealth grid. The second suggestion pertains to fast matrix products and is purely computational but has a large impact on the time required to solve the problem. We also propose a more consistent approximation for the calculation of the return parameters.

Dew-Becker, I. and Giglio, S. (2023). "Cross-Sectional Uncertainty and the Business Cycle: Evidence from 40 Years of Options Data." In: *American Economic Journal: Macroeconomics* 15(2), pp. 65–96.
This paper presents a novel and unique measure of cross-sectional uncertainty constructed from stock options on individual firms. Cross-sectional uncertainty varied little between 1980 and 1995 and subsequently had three distinct peaks-during the tech boom, the financial crisis, and the coronavirus epidemic. Cross-sectional uncertainty has had a mixed relationship with overall economic activity, and aggregate uncertainty is much more powerful for forecasting aggregate growth. The data and moments can be used to calibrate and test structural models of the effects of uncertainty shocks. In international data, we find similar dynamics and a strong common factor in cross-sectional uncertainty.

Dew-Becker, I., Giglio, S., and Kelly, B. (2021). "Hedging macroeconomic and financial uncertainty and volatility." In: *Journal of Financial Economics* 142(1), pp. 23–45.
We study the pricing of shocks to uncertainty and volatility using a wide-ranging set of options contracts covering a variety of different markets. If uncertainty shocks are viewed as bad by investors, they should carry negative risk premiums. Empirically, however, uncertainty risk premiums are positive in most markets. Instead, it is the realization of large shocks to fundamentals that has historically carried a negative premium. In other words, we find that the return premium for gamma is negative, while that for vega is positive. These results imply that it is jumps, for which exposure is measured by gamma, not forward-looking uncertainty shocks, measured by vega, that drive investors' marginal utility. In further support of the jump interpretation, the return patterns are more extreme for deeper out-of-the-money options.

Diamond, S., Boyd, S., Greenberg, D., Kochenderfer, M., and Ang, A. (2023). "Optimal Claiming of Social Security Benefits." In: *The Journal of Retirement* 10(3), pp. 33–46.

Using a life-cycle framework, we compute the optimal age to claim Social Security benefits. If the ratio of a worker's wealth to their primary insurance amount (PIA) – the monthly benefit payable at their full retirement age – exceeds a certain threshold, they should defer Social Security for at least a year or claim immediately. The optimal threshold depends on mortality assumptions and individual preferences, but is less sensitive to capital market assumptions. The threshold wealth to PIA ratio increases from 5.5 for men and 5.2 for women at age 62, to 11.1 for men and 10.4 for women at age 69. For example, a man with a PIA of $2,000 should delay past age 69 unless his wealth is less than $22,200, illustrating that all except those with low levels of wealth should delay claiming. Those with wealth below these levels should claim, because maintaining their consumption requires immediate Social Security payments.

Dichtl, H., Drobetz, W., and Wambach, M. (2014). "Where is the value added of rebalancing? A systematic comparison of alternative rebalancing strategies." In: *Financial Markets and Portfolio Management*.
This study compares the performance of different rebalancing strategies under realistic market conditions by reporting statistical significance levels. Our analysis is based on historical data from the United States, the United Kingdom, and Germany and comprises three different classes of rebalancing (periodic, threshold, and range rebalancing). Despite cross-country differences, our history-based simulation results show that all rebalancing strategies outperform a buy-and-hold strategy in terms of Sharpe ratios, Sortino ratios, and Omega measures. The differences in risk-adjusted performance are not only statistically significant, but also economically relevant. However, the choice of a particular rebalancing strategy is of only minor economic importance.

Dickson, J. M., Bruno, M. A., and Wong, B. C. (2018). *A "BETR" approach to Roth conversions*. Tech. rep. Vanguard.
Investors typically decide whether to convert to a Roth IRA from a traditional IRA by comparing their current and expected future marginal tax rates. The rule of thumb has been that higher future tax rates make a conversion more desirable, while lower ones make it less so. (Given that future tax rates are uncertain for many reasons, many investors may want to diversify this tax risk through partial conversions.) We introduce a break-even tax rate (BETR) that yields a more accurate view of what future tax rate would make an investor indifferent to a conversion.

DiJoseph, M. A. (2020). *Spending guidelines to help ease retirees' market worries*. Tech. rep. Vanguard.
Staying the course does not have to mean standing still. In fact, Vanguard research explains how you can help clients build adaptable, resilient, and sustainable retirement spending plans. Our research paper From assets to income: A goals-based approach to retirement spending shows you how to help clients implement a personalized spending strategy. The resulting plan is a customized, responsive formula that can help reduce clients' anxiety and stress about their ability to meet retirement income goals, regardless of the market environment.

DiLellio, J. and Ostrov, D. N. (2019). "Constructing Tax Efficient Withdrawal Strategies for Retirees with Traditional 401(k)/IRAs, Roth 401(k)/IRAs, and Taxable Accounts." In: *SSRN e-Print*.
We construct an algorithm for United States retirees that computes individualized tax-efficient annual withdrawals from IRAs/401(k)s, Roth IRAs/Roth 401(k)s, and taxable accounts. Our algorithm applies a new approach that generates an individualized strategy that results in consistent improvements over non-individualized withdrawal strategies currently advocated by financial institutions and academics. Among other results, we quantifiably demonstrate why retirees should avoid, not seek, dividend producing stocks in their taxable accounts. Our model, which can work to optimize either portfolio longevity or the bequest to an heir, accommodates many salient tax code features, including dividends, different taxable lots, conversions, and required minimum distributions.

Dillschneider, Y., Maurer, R., and Schober, P. (2020). "Dynamic Portfolio Choice with Annuities When the Interest Rate Is Stochastic." In: *SSRN e-Print*.
This paper studies the optimal life cycle consumption and portfolio choice problem taking into account annuity risk due to stochastic interest rates. When the purchase of annuities is restricted to the retirement date, the annuitant is exposed to the risk of meeting low interest rates at the purchase date. This annuity risk can be diversified by spreading annuity purchases over the whole pre-retirement period. The numerical results of our life cycle model show that such temporal diversification enhances welfare for retirees up to 9% of certainty equivalent consumption and that welfare gains increase with higher interest rate risk.

Dillschneider, Y., Maurer, R., and Schober, P. (2024). "Optimal annuitization under stochastic interest rates." In: *ASTIN Bulletin* 54(3), pp. 626–651.
The decision about when and how much to annuitize is an important element of the retirement planning of most individuals. Optimal annuitization strategies depend on the individual's exposure to annuity risk, meaning the

possibility of meeting unfavorable personal and market conditions at the time of the annuitization decision. This article studies optimal annuitization strategies within a life-cycle consumption and portfolio choice model, focusing on stochastic interest rates as an important source of annuity risk. Closing a gap in the existing literature, our numerical results across different model variants reveal several typical structural effects of interest rate risk on the annuitization decision, which may however vary depending on preference specifications and alternative investment opportunities: When allowing for gradual annuitization, annuity risk is temporally diversified by spreading annuity purchases over the whole pre-retirement period, with annuity market participation starting earlier in the life cycle and becoming more extensive with increasing interest rate risk. Ruling out this temporal diversification possibility, as embedded in many institutional settings, incurs significant welfare losses, which are increasing with higher interest rate risk, together with larger overall demand for annuitization.

Ding, G. and Marazzina, D. (2021). "Sensitivity of Optimal Retirement Problem to Liquidity Constraints." In: *arXiv e-Print*.
In this work we analytically solve an optimal retirement problem, in which the agent optimally allocates the risky investment, consumption and leisure rate to maximise a gain function characterised by a power utility function of consumption and leisure, through the duality method. We impose different liquidity constraints over different time spans and conduct a sensitivity analysis to discover the effect of this kind of constraint.

Ding, Y., Li, Y., and Song, R. (2022). "Statistical Learning for Individualized Asset Allocation." In: *arXiv e-Print*.
We establish a high-dimensional statistical learning framework for individualized asset allocation. Our proposed methodology addresses continuous-action decision-making with a large number of characteristics. We develop a discretization approach to model the effect from continuous actions and allow the discretization level to be large and diverge with the number of observations. The value function of continuous-action is estimated using penalized regression with generalized penalties that are imposed on linear transformations of the model coefficients. We show that our estimators using generalized folded concave penalties enjoy desirable theoretical properties and allow for statistical inference of the optimal value associated with optimal decision-making. Empirically, the proposed framework is exercised with the Health and Retirement Study data in finding individualized optimal asset allocation. The results show that our individualized optimal strategy improves individual financial well-being and surpasses benchmark strategies.

Diris, B., Palm, F., and Schotman, P. (2015). "Long-Term Strategic Asset Allocation: An Out-of-Sample Evaluation." In: *Management Science* 61(9), pp. 2185–2202.
We evaluate the out-of-sample performance of a long-term investor who follows an optimized dynamic trading strategy. Although the dynamic strategy is able to benefit from predictability out-of-sample, a short-term investor using a single-period market timing strategy would have realized an almost identical performance. The value of intertemporal hedge demands in strategic asset allocation appears negligible. The result is caused by the estimation error in predicting the predictors. A myopic investor only needs to predict one-period-ahead expected returns, but hedge demands also require accurate predictions of the predictor variables. To reduce the problem of errors in optimized portfolio weights, we consider Bayesian procedures. Myopic and dynamic portfolios are similarly affected by such modifications, and differences in performance become even smaller.

Dixon, M. F., Halperin, I., and Bilokon, P. (2020). *Machine Learning in Finance: from theory to practice*. Springer International Publishing. 548 pp.
This book introduces machine learning methods in finance. It presents a unified treatment of machine learning and various statistical and computational disciplines in quantitative finance, such as financial econometrics and discrete time stochastic control, with an emphasis on how theory and hypothesis tests inform the choice of algorithm for financial data modeling and decision making. With the trend towards increasing computational resources and larger datasets, machine learning has grown into an important skillset for the finance industry. This book is written for advanced graduate students and academics in financial econometrics, mathematical finance and applied statistics, in addition to quants and data scientists in the field of quantitative finance. Machine Learning in Finance: From Theory to Practice is divided into three parts, each part covering theory and applications. The first presents supervised learning for cross-sectional data from both a Bayesian and frequentist perspective. The more advanced material places a firm emphasis on neural networks, including deep learning, as well as Gaussian processes, with examples in investment management and derivative modeling. The second part presents supervised learning for time series data, arguably the most common data type used in finance with examples in trading, stochastic volatility and fixed income modeling. Finally, the third part presents reinforcement learning and its applications in trading, investment and wealth management. Python code examples are provided to support the readers' understanding of the methodologies and applications. The

book also includes more than 80 mathematical and programming exercises, with worked solutions available to instructors. As a bridge to research in this emergent field, the final chapter presents the frontiers of machine learning in finance from a researcher's perspective, highlighting how many well-known concepts in statistical physics are likely to emerge as important methodologies for machine learning in finance.

Dixon, M. F. and Halperin, I. (2020). "G-Learner and GIRL: Goal Based Wealth Management with Reinforcement Learning." In: *SSRN e-Print*.
We present a reinforcement learning approach to goal based wealth management problems such as optimization of retirement plans or target dated funds. In such problems, an investor seeks to achieve a financial goal by making periodic investments in the portfolio while being employed, and periodically draws from the account when in retirement, in addition to the ability to re-balance the portfolio by selling and buying different assets (e.g. stocks). Instead of relying on a utility of consumption, we present G-Learner: a reinforcement learning algorithm that operates with explicitly defined one-step rewards, does not assume a data generation process, and is suitable for noisy data. Our approach is based on G-learning (Fox et al., 2015) - a probabilistic extension of the Q-learning method of reinforcement learning. In this paper, we demonstrate how G-learning, when applied to a quadratic reward and Gaussian reference policy, gives an entropy-regulated Linear Quadratic Regulator (LQR). This critical insight provides a novel and computationally tractable tool for wealth management tasks which scales to high dimensional portfolios. In addition to the solution of the direct problem of G-learning, we also present a new algorithm, GIRL, that extends our goal-based G-learning approach to the setting of Inverse Reinforcement Learning (IRL) where rewards collected by the agent are not observed, and should instead be inferred. We demonstrate that GIRL can successfully learn the reward parameters of a G-Learner agent and thus imitate its behavior. Finally, we discuss potential applications of the G-Learner and GIRL algorithms for wealth management and robo-advising.

Dixon, M. F. and Halperin, I. (2021). "Goal-based wealth management with generative reinforcement learning." In: *Risk (Cutting edge)*.
A combination of machine learning techniques provides multi-period portfolio optimisation. Matthew Dixon and Igor Halperin develop a reinforcement learning (RL) approach to goal-based wealth management problems such as optimisation of retirement plans or target date funds. They present G-Learner: a reinforcement learning algorithm that does not assume a data generation process and is suitable for noisy data. Their approach is based on G-learning, a probabilistic extension of the Q-learning method of reinforcement learning. In addition to G-Learners, which solves the direct RL problem, they develop GIRL, a G-learning inverse RL algorithm to infer the investor reward function from the observed trading actions.

Dom, M. S., Howard, C., Jansen, M., and Lohre, H. (2024). "Beyond GMV: Raising the Bar for Evaluating Covariance Matrix Estimators." In: *SSRN e-Print*.
When validating variance-covariance (VCV) estimators based on the ex-post volatility of the global minimum variance (GMV) portfolio, we confirm the academic backing for considering shrinkage and covariance dynamics in modeling the VCV for equity portfolio construction. Yet, the underlying test portfolios are often impractical due to their high leverage, concentration, turnover, and transaction costs. Resorting to more practical GMV and risk-parity portfolios, we reveal a considerably reduced opportunity set for alternative VCV estimators. Although the implicit shrinkage in asset weight constraints makes further shrinkage redundant, accounting for the dynamics in the time-series dependence of asset returns remains statistically relevant.

Dom, M. S., Howard, C., Jansen, M., and Lohre, H. (2025). "Beyond GMV: the relevance of covariance matrix estimation for risk-based portfolio construction." In: *Quantitative Finance* 25(3), pp. 403–419.
We empirically analyze the relevance of variance-covariance (VCV) estimators in equity portfolio construction. While traditional analyses of unconstrained global minimum-variance (GMV) portfolios support using shrinkage and modeling covariance dynamics, the resulting portfolios are often impractical due to high leverage, concentration, and costs. By examining constrained GMV and risk parity portfolios, we find a significantly reduced opportunity for alternative VCV estimators to outperform the sample estimator. Specifically, we show that a long-only portfolio with asset-level constraints and a transaction cost penalty produces similar results to shrinkage-based methods, even when using the sample VCV estimator. However, accounting for time-series dynamics in asset returns remains statistically relevant for volatility reduction. Our findings emphasize how the interaction between VCV estimators and portfolio construction choices shapes both the statistical and practical outcomes in portfolio management.

Dong, B., Xu, W., Sevic, A., and Sevic, Z. (2020). "Efficient willow tree method for variable annuities valuation and risk management." In: *International Review of Financial Analysis* 68, p. 101429.

Variable annuities (VAs) with various guarantees are popular retirement products in the past decades. However, due to the sophistication of the embedded guarantees, most existing methods only focus on the one of embedded guarantees underlying one specified stochastic model. The method to evaluate VAs with all guarantees and manage its risk is very limited, except for the Monte Carlo method. In this paper, we propose an efficient willow tree method to evaluate VAs embedded with all popular guarantees on the market underlying various stochastic models. Moreover, our tree structure is also applicable to compute dollar delta, value at risk (VaR) and conditional tail expectation (CTE) in hedging and risk-based capital calculation. Numerical experiments demonstrate the accuracy and efficiency of our method in pricing and managing the risk of VAs.

Dong, Z.-L., Zhu, M.-X., and Xu, F.-M. (2022). "Robo-advisor using closed-form solutions for investors' risk preferences." In: *Applied Economics Letters*.

In this article, we design a robo-advisor which has a bi-level framework. The framework enables it to handle a large amount of assets using fast algorithms in the lower level. The proposed robo-advisor can utilize the closed-form solutions for investors' risk preferences based on corresponding portfolio choices. A dynamic weight is applied to update investors' risk preferences. Numerical results based on real data in Chinese stock market show that our proposed robo-advisor can accurately estimate the risk preferences of investors and outperform the benchmark formed by market indexes.

Donnelly, C. and Bernhardt, T. (2018). *Pension decumulation strategies: a state-of-the-art report*. Tech. rep. Heriot Watt University.

The authors are part of the ARC research project "Minimising Longevity and Investment Risk while Optimising Future Pension Plans". The goal is to develop new pension product designs that keep the customers' needs at the forefront. As a first step, this report was written to familiarize the project team with the existing knowledge on decumulation strategies. There is a UK-focus in the report. The report is aimed at the reader who is familiar with the actuarial world, but are not necessarily familiar with stochastic control techniques. For this reason, technical descriptions of the mathematical methods used to solve many of the problems studied in the cited literature are not included. Questions relating to property, taxes, regulations and solvency have been ignored in order to keep the focus on what is the state-of-the-art on the fundamental question of "how should I decumulate in retirement"? The literature on decumulation strategies is vast and only a fraction of it is included. Omission of relevant material is unintentional.

Donnelly, C., Khemka, G., and Lim, W. (2022). "Investing for retirement: Terminal wealth constraints or a desired wealth target?" In: *European Financial Management*.

We investigate how well different investment strategies can give pre-retirees more certainty about their income in retirement, whilst allowing them to benefit from taking investment risk. Under an expected utility-maximizing framework, we find that a loss aversion utility function gives a high degree of certainty about its desired wealth target and is robust to different market models. Imposing terminal wealth constraints does not improve the certainty of achieving the desired target enough to counter-balance the increased chance of obtaining a lower income. The power utility function is not robust to different market models and becomes too risk-averse with wealth constraints.

Drew, M. E., Walk, A. N., West, J., and Cameron, J. (2014). "Improving retirement adequacy through asset class prioritization." In: *Journal of Financial Services Marketing* 19(4), pp. 291–303.

Highly risk-averse retirees are generally advised to adopt a fixed spending strategy such as the 4% withdrawal rule. To prevent the premature depletion of a retirement portfolio, the rule attempts to proxy as the "safe withdrawal rate". But a constant withdrawal rate means that retirees accumulate unspent surpluses when markets outperform and face spending shortfalls when markets underperform. While a safe withdrawal rate can prevent spending shortfalls, the opportunity cost of unspent surpluses associated with this strategy can be extreme. We apply a range of basic investment decision rules to a retirement portfolio applying various withdrawal rates and examine the probability of shortfalls over a retirement horizon. Using a block bootstrap simulation technique, we examine decision rules relating to stock and bond investments. Our results show that retirement portfolios with a bias towards stocks coupled with a decision rule that sources withdrawals from bonds and cash before stocks significantly outperforms alternative withdrawal strategies, despite the inherent increase in volatility. This finding is in direct contrast to the safe withdrawal rate conventions used in contemporary financial advice models.

Drew, M. E., Walk, A. N., and West, J. (2015). "Conditional Allocations to Real Estate: An Antidote to Sequencing Risk in Defined Contribution Retirement Plans." In: *The Journal of Portfolio Management* 41(6), pp. 82–95.

In this article, the authors investigate the potential for real estate as an asset class to be exploited to protect against sequencing risk (or path dependency) in defined contribution retirement funds. Their results suggest that allocating both listed and unlisted real estate assets to retirement portfolios, even if very minor, can enhance the risk-return profile and probability of successfully achieving retirement outcomes. Using a bootstrap simulation approach, the authors test for a range of asset allocations that include real estate. In addition, they examine the sensitivity of real estate performance to changes in monetary policy to optimize portfolio outcomes for fund managers who actively seek exposure to real estate assets. Their findings indicate that the performance of real estate is highly dependent on monetary policy settings that, when used in a dynamic asset allocation process, have the potential to enhance portfolio returns while limiting the impact of downside risk.

Drew, M. E., Walk, A. N., and West, J. M. (2016). "Withdrawal Capacity in the Face of Expected and Unexpected Health and Aged-Care Expenses During Retirement." In: *The Journal of Retirement* 3(3), pp. 77–94.
We examine the consequences of taking account of costs associated with age-related health treatment and agedcare services during the retirement phase. Simulating asset return data using historical bootstrap simulation, we derive an optimal withdrawal income during retirement using dynamic optimization techniques. The greatest risk to income sustainability occurs when unexpected health costs combine with above-average longevity for conservative investors. High costs of health treatment without a commensurate adjustment in asset allocation toward assets with a less conservative risk-return profile risk premature wealth depletion. The risk of ruin can be mitigated through a dynamic life-cycle strategy during the retirement phase.

Drew, M. E. and West, J. M. (2021). "Retirement Income Sufficiency through Personalised Glidepaths." In: *Financial Analysts Journal* 77(2), pp. 5–20.
Portfolio "glidepaths" accommodate growth in a worker's early working life and transition to lower risk settings as the worker nears retirement. The success of this design hinges on its objective of amassing wealth at the date of retirement, but the design offers little in the way of a solution for the provision of income during retirement. The relevant risk for workers is retirement income uncertainty. Allocating investments through time to satisfy an income goal is not equivalent to the maximisation of wealth at retirement. We demonstrate that glidepaths can be personalised for individuals to maximise expected retirement income sufficiency under a range of assumptions, including longevity risk.

Dreyer, A. and Pogorelova, L. (2018). "The Effect of Income-Dependent Mortality on Withdrawal Strategies." In: *The Journal of Retirement* 6(1), pp. 70–81.
Recent literature documented significant dependency between individual lifetime earnings and life expectancy. In light of this phenomenon, we examine the effect of using salary-fitted mortality tables on optimized withdrawal strategies. We find that although earnings rank meaningfully affects expected retirement length, the impact of modeling salary-fitted mortality is more nuanced and depends on the investor risk-aversion parameter. In the base case, a risk-neutral investor is willing to adjust her optimal withdrawal strategy roughly in proportion to the change in expected retirement length and can therefore experience a substantially larger misallocation of withdrawals. In contrast, for a risk-averse investor, a 10% income-related difference in expected retirement length does not necessitate a sizable adjustment of an optimized withdrawal strategy; the unconditional mortality assumption causes annual withdrawal misallocation of only about 1% to 3% of pre-retirement consumption. Given this difference in results, we stress the need for a comprehensive and cohesive framework when analyzing the full implications of a retirement horizon assumption.

Driessen, J. and Kuiper, I. (2019). "Rebalancing for Long Term Investors: Why It Pays to Do Less." In: *SSRN e-Print*.
In this study we show that the rebalance frequency of a multi-asset portfolio has only limited impact on the utility of a long-term passive investor. Although continuous rebalancing is optimal, the loss of a suboptimal strategy corresponds to up to only 30 basis points of the initial wealth of the investor, assuming market returns are unpredictable and transaction costs can be ignored. Our results suggest that reducing transaction costs clearly outweighs the benefit of frequent rebalancing. When we study a setting where asset returns are predictable, we find that a long-term investor that ignores this predictability underestimates the benefit of less frequent rebalancing. In this setting, limiting the frequency to at least once every quarter results in significant higher utility, even without transaction costs.

Du, Y. (2021). "Essays on Portfolio Choice and Health over the Life Cycle." PhD thesis. Temple University.
This dissertation examines the effect of health and its associated variables on households' consumption and portfolio choices over life cycle. The first two essays constitute my job market paper, which explains why the risky portfolio share rises in wealth from two health mechanisms: endogenous health investment and medical

expenditure risk. The third chapter explores the effect of health and health risk on households' optimal consumption and portfolio decisions over life cycle.
Chapter 1 titled "PORTFOLIO CHOICE AND HEALTH ACROSS WEALTH: EMPIRICAL EVIDENCE" illustrates the empirical relationship between the portfolio puzzle and the heterogeneity of health variables across wealth. Classic financial theory suggests that under the assumption of no borrowing constraints and no mean-reverting stock returns, households should hold a constant risky portfolio in spite of their wealth, ages and life horizons (Samuelson (1969) and Merton (1969, 1971)). Yet data from the Survey of Consumers Finances (SCF) show that the risky portfolio share of financial assets increases in wealth. In the literature, this is called the "portfolio puzzle". Meanwhile, various sources of data indicate that, compared with the non-wealthy households, the wealthy people have better health, longer life horizon, higher out of pocket medical spending with lower uncertainty, and more health care time. All these facts suggest a novel correlation between the portfolio puzzle and the heterogeneity of health variables across wealth and provide a robust empirical foundation to explain the portfolio puzzle from a health perspective.
In Chapter 2 titled "A LIFE CYCLE MODEL OF PORTFOLIO CHOICE AND HEALTH", a life cycle model with endogenous health investment and medical expenditure risk is proposed to capture the key empirical features in the first chapter. This calibrated model remarkably matches the U.S. data. I find that endogenous health investment is essential to explain the portfolio puzzle: if health is exogenous without investment, the model can only could deliver 7.2% of the risky share gap across wealth. Medical expenditure risk is less important and has a larger effect on the non-wealthy group. If I abstract from medical expenditure risk, the risky shares increase for both groups: 24% for the low wealth group and 5% for the wealthy group. This life cycle model provides new insights into how health affects households' financial behavior.
Chapter 3 titled "OPTIMAL CONSUMPTION AND PORTFOLIO CHOICE WITH HEALTH RISK" investigates the effect of health and health risk on households' optimal consumption and portfolio allocations over the life cycle. The simulation results show that consumption, savings in bonds, and savings in stocks all increase with health. The risky portfolio share, which is the ratio of savings in stocks to the total financial assets, demonstrates the same tendency for both health states over the life cycle: at the very young age, the risky portfolio share is relatively high. Starting from the middle age, this share falls significantly and keeps steady until the end of life. For most of the lifetime, the risky portfolio share is positively related with health. These results emphasize the importance of health and its associated risk in consumption and portfolio decisions.

Duffy, S., Finke, M. S., and Blanchett, D. (2021). "The Value of Delayed Social Security Claiming for Higher-Earning Women." In: *SSRN e-Print*.
This study estimates the expected benefit from delayed Social Security claiming for higher-earning, healthier women who can expect to receive more future income payments than other Americans. The expected net present value of Social Security payments from delayed claiming for healthy women is $179,999, or more than twice the value of delayed claiming for a male of average health. The benefit from delayed claiming ranges from a low of $16,548 per year to $29,395 per year for healthy women. Even average health women gain at least $132,202 in Social Security wealth by waiting until age 70 to claim Social Security income benefits. Significant gains to women who delay claiming are robust to a 2% increase in real discount rates and to a 21% reduction in income that could occur if the Social Security trust fund depletes in 2026. The added benefits from reduced longevity and inflation risk suggest that the failure to delay claiming results in a significant loss in both retirement wealth and expected welfare.

Dunham, L. M. and Washer, K. M. (2020). "Using Life Tables for Retirement Planning." In: *The Journal of Wealth Management* 23(3), pp. 28–36.
This study examines two methods for determining retirement accumulations. The first method assumes that the client will live to a very old age. This method may be appropriate for clients who are highly risk averse and/or want to self-insure the risk of outliving their savings. The second method employs life expectancy tables to weight retirement cash flows by survival probabilities. The target accumulation under this method is much lower and more easily attainable for clients than the first method. The primary drawback of the second method is that, from a statistical point of view, about 55% of clients will outlive their savings. Because of this, risk-sharing, perhaps through a deferred life annuity, would be advisable unless clients have other viable options available (e.g., family support) in the event that their savings are used up. We believe there is value in both methods and that some clients will prefer a higher, more ambitious target, whereas others will prefer a lower, more attainable target.

Dutta, S. and Jain, S. (2023). "Precision versus Shrinkage: A Comparative Analysis of Covariance Estimation Methods for Portfolio Allocation." In: *arXiv e-Print*.

In this paper, we perform a comprehensive study of different covariance and precision matrix estimation methods in the context of minimum variance portfolio allocation. The set of models studied by us can be broadly categorized as: Gaussian Graphical Model (GGM) based methods, Shrinkage Methods, Thresholding and Random Matrix Theory (RMT) based methods. Among these, GGM methods estimate the precision matrix directly while the other approaches estimate the covariance matrix. We perform a synthetic experiment to study the network learning and sample complexity performance of GGM methods. Thereafter, we compare all the covariance and precision matrix estimation methods in terms of their predictive ability for daily, weekly and monthly horizons. We consider portfolio risk as an indicator of estimation error and employ it as a loss function for comparison of the methods under consideration. We find that GGM methods outperform shrinkage and other approaches. Our observations for the performance of GGM methods are consistent with the synthetic experiment. We also propose a new criterion for the hyperparameter tuning of GGM methods. Our tuning approach outperforms the existing methodology in the synthetic setup. We further perform an empirical experiment where we study the properties of the estimated precision matrix. The properties of the estimated precision matrices calculated using our tuning approach are in agreement with the algorithm performances observed in the synthetic experiment and the empirical experiment for predictive ability performance comparison. Apart from this, we perform another synthetic experiment which demonstrates the direct relation between estimation error of the precision matrix and portfolio risk.

Dziwisch, A., Krahnhof, P., and Zureck, A. (2021). "Empirical determination of sustainable withdrawal rates considering historical yields and inflation rates in Germany." In: *Zeitschrift fur die gesamte Versicherungswissenschaft* 110, pp. 117–132.

On account of the current low interest rate phase, which is most likely to continue in the coming years, the average yields to be achieved in the bond, time deposit and savings product sectors are declining, so that risk-averse investors in particular have few opportunities to generate return-oriented retirement provisions.

This scientific article analyzes the level of a possible safe withdrawal rate for diversified pension portfolios, considering historical returns and inflation rates. Consequently, this article provides immediate practical added value for a possible retirement provision.

The evaluation is based on the consideration of historical returns of the stock and bond market in Germany. To determine a safe withdrawal rate, the development of portfolios with different compositions and inflation-adjusted withdrawal rates are simulated over periods of 15 to 35 years. In this simulation, the risky part of the portfolio is represented by German equities, the low risk part by German government bonds.

To sum up, the empirical results show a maximum safe withdrawal rate of 4%. The underlying portfolio is composed of 50% equities and 50% government bonds. Particularly due to the outlined demographic change in Germany as well as the ongoing low-interest phase, the empirical study can provide significant theoretical and practical insights.

Edesess, M. (2017). "Rebalancing: A Comprehensive Reassessment." In: *SSRN e-Print*.

Investment advisors, investment books, and investment columns almost never fail to advise investors to rebalance their portfolios. Numerous academic articles have been written about it, most of them arguing that it adds value. Yet the existence and precise nature of that value have remained a subject of intense debate. Most past articles on rebalancing have reduced the comparison of rebalancing with alternatives such as buy-and-hold to a single summary statistic, either expected return or median return. Due to the different skews in the probability distributions, however, comparing a single summary statistic does not support the overly broad conclusions that have often been reached. Comparisons that take the entire distributions into account provide better insight. Results of those comparisons are mixed as to whether rebalancing adds value. This article analyzes the arguments that have been advanced for rebalancing, identifying their points of validity and their flaws. It presents the results of original empirical research, simulations, and mathematical derivations that together identify how often and to what extent rebalancing enhances or degrades the performance of an investment portfolio.

Ehsani, S., Harvey, C. R., and Li, F. (2023). "Is Sector Neutrality in Factor Investing a Mistake?" In: *Financial Analysts Journal* 79(3), pp. 95–117.

Stock characteristics have two sources of predictive power. First, a characteristic might be valuable in identifying high or low expected returns across industries. Second, a characteristic might be useful in identifying individual stock expected returns within an industry. Past studies generally find that the firm-specific component is the strongest predictor, leading many to sector neutralize their factor exposures. We show both analytically and empirically that the average long-short investor is more likely to benefit from hedging out sector bets, whereas the long-only investor should, on average, avoid sector neutralization.

El Bernoussi, R. and Rockinger, G. M. (2019). "Rebalancing with transaction costs: theory, simulations, and actual data." In: *SSRN e-Print*.

In the absence of transaction costs and the presence of independent returns, a buy-and-hold strategy can be shown to generate higher expected returns than a fixed-weight strategy, where the portfolio weights are regularly readjusted/rebalanced to some initial level. This higher expected return comes, however, at a cost: higher volatility. The resulting trade-off leads to different rankings of the Sharpe ratio depending on the statistical moments of the assets. We theoretically discuss causes affecting the ranking of the Sharpe ratio, and we introduce an easy-to-implement methodology to deal with proportional transaction costs. Under transaction costs, the buy-and-hold strategy as the cheaper approach should be the winner. In various simulation experiments, we investigate the relevance of transaction costs on rebalancing strategies. Eventually, we consider a realistic portfolio with a risk-free asset, bonds, and various stock indices that allows us to demonstrate that in practice, as long as transaction costs remain small, rebalancing has value for realistic portfolios.

Elkamhi, R., Fabozzi, F. J., Lee, J. S. H., Salerno, M., Vatanen, K., and Vohra, S. (2024a). "Applications of FX derivatives to portfolio management." In: *Journal of Asset Management* 25(6), pp. 600–616.

This article demonstrates the use of foreign-exchange (FX) derivatives in portfolio management, highlighting the strategic applications of FX forwards, futures, swaps, and options. It begins by detailing how these derivatives help institutional investors mitigate the adverse effects of currency fluctuations on internationally diversified portfolios. A significant focus is placed on currency hedging with derivatives overlays, which consolidate currency exposures across asset classes into a centralized management function, thereby enhancing overall risk management. The article also delves into the strategic uses of FX options, which offer flexible, tailored risk management strategies crucial for handling the complex dynamics of global financial markets. Through real-world examples and theoretical insights, the article illustrates the critical role of FX derivatives in stabilizing portfolio returns and managing exposure to currency risks.

Elkamhi, R., Lee, J. S. H., and Salerno, M. (2024b). "(Re)Balancing Act: The Interplay of Private and Public Assets in Dialing the Asset Allocation." In: *The Journal of Portfolio Management* 50(7), pp. 162–182.

The growing trend of sovereign wealth and pension funds to allocate more toward private investments has made the management of asset allocation more complex. Traditional rebalancing methods, such as fixed-weights rebalancing, encounter problems when applied to private assets, as their illiquidity and lags in appraisal valuations pose challenges. During financial crises, the delayed and smoothed valuations of private assets lead them to be overweight in portfolios as public assets decline in values. Rebalancing the underweight public assets can increase leverage usage and, more importantly, deteriorate the fund's liquidity position. To address these challenges, this article proposes a holistic rebalancing strategy: rebalance a portfolio to the desired factor allocation by complementing the factor exposures of existing private assets with an allocation to public assets that overall delivers the required factor allocation. This approach safeguards the liquidity position of a fund during market downturns by maintaining a stable risk and leverage profile. It presents a dynamic and risk-aware approach for rebalancing portfolios with private assets.

Elkins, K. (2019). *Stanford analyzed 292 retirement strategies to determine the best one – here's how it works*. Tech. rep. CNBC.

In 2017, the Stanford Center on Longevity analyzed 292 different retirement income strategies and determined the best way for most people to withdraw their savings. It's called the "spend safely in retirement strategy" (SSiRS) and involves two basic components: delaying Social Security benefits and creating an "automatic retirement paycheck." In a new 2019 report conducted by the Stanford Center on Longevity and the Society of Actuaries, "Viability of the Spend Safely in Retirement Strategy," the research team took a deeper dive into the SSiRS and explored different ways to implement it. The strategy is designed to be used by middle-income workers and retirees, which the report defines as those having USD1 million or less in retirement savings. Here's a closer look at how the two key components of the SSiRS work.

Ellis, C., Moenig, T., and Volkman Wise, J. (2023). "Registered Index-Linked Annuities in Qualified Retirement Plans." In: *SSRN e-Print*.

Many Americans continue to be financially underprepared for their retirement. Automatic enrollment in employer-sponsored qualified retirement plans (QRPs) has helped, but the target date funds commonly used as default investment choices have their own problems. We propose a target-date version of the recently developed Registered Index-Linked Annuities and we suggest that these "TD-RILAs" provide a more cost-effective and more transparent way to attain a diversified equity exposure, the level of which decreases over time. A theoretical analysis explores the optimal structure of TD-RILAs and their comparison to target date funds from an investor's perspective. A large-scale lab experiment sheds further light on investors' preferences, focusing on the importance of product transparency, the employer's default investment choice, and the role of information in improving financial literacy. We conclude that TD-RILAs would be a suitable addition to QRPs and may even rival target date funds as Qualified Default Investment Alternatives.

Elnekave, R. (2007). "The mathematics of savings and retirement planning." In: *The Journal of Wealth Management* 10(3), pp. 87–92.
Saving and retirement planning currently rely on techniques that have proven to be failure prone at best, and completely inadequate under the worst circumstances. Further, existing methods do not provide an understanding of the drivers behind the success or failure of savings plans. This lack of insight, combined with a deficient framework, contributes to the adverse outcomes experienced by many pension plans and individuals. This article introduces a new perspective on modeling savings that borrows extensively from the field of hydrology. To accomplish this, the author applies a modified version of mathematical techniques used by hydrologists in dam building to savings and retirement planning. The goal is to present an intuitive decision framework, based on a novel concept. In the process, the author hopes to open a new path in the dialogue to improve the state of the art in savings and retirement planning.

Envestnet Research (2020). *Capital Sigma: The Advisor Advantage*. Tech. rep. Envestnet.
We recently updated our quantitative research study, "Capital Sigma: The Advisor Advantage," which explores how select sources of advisor-created value can produce meaningful levels of excess return. For advisors to deliver the most alpha to client portfolios, we recommend a focus on five sources: financial planning, asset selection and allocation, investment selection, systematic rebalancing and tax management. The sum total of these five sources creates what we call Capital Sigma. In fact, the combination of successfully implementing these five sources has produced around 3% of value add annually –results that are consistent with PMC research first published in 2014.

Erdemlioglu, D. and Joliet, R. (2019). "Long-term asset allocation, risk tolerance and market sentiment." In: *Journal of International Financial Markets, Institutions and Money* 62, pp. 1–19.
This paper studies optimal equity portfolios with long-term horizon under heterogeneous risk aversion levels. We focus on European stocks and empirically show that contemporaneous excess returns of semi-active strategies are negatively associated with market conditions and sentiment. Consistent with our long-horizon perspective, we find that the effects of sentiment measures on semi-active portfolio returns are sizeable and economically relevant, particularly in bull (post-crisis) periods, even after controlling for the five Fama-French factors, momentum, macro indicators and political uncertainty shocks either globally or country-wise. By contrast, the effects of sentiment measures on the passive (benchmark) portfolio appear to be negligible. The results further indicate that realized portfolio returns generated from our long-term strategies are considerably resilient to the episodes of flight-to-safety (risk-off) regimes.

Eschenbach, T. G. and Lewis, N. A. (2019). "Risk, standard deviation, and expected value: when should an individual start social security?" In: *The Engineering Economist* 64(1), pp. 24–39.
In choosing when to start collecting Social Security, the differences in expected net present values (NPVs) are small-but the corresponding standard deviations are not. Starting earlier is less risky. The case analyzed is single individuals in the U.S. system, but the methodology can be applied to couples and to the systems of other nations. Considering risk and return together places Social Security in the same risk/return framework as other capital investments. Behavioral, situational, and qualitative factors that often dominate decisions on when to start are linked with quantitative approaches to longevity risk and mortality risk.

Estrada, J. (2014). "The Glidepath Illusion: An International Perspective." In: *The Journal of Portfolio Management* 40(2), pp. 52–64.
Target-date funds have become a core product for investors who are saving for retirement. These funds periodically reduce their allocations to stocks and increase their allocations to bonds and cash, becoming more conservative as retirement approaches. This lifecycle strategy implies that investors are aggressive with little capital and conservative with much more capital, which may be suboptimal in terms of wealth accumulation. This article evaluates three alternative types of strategies, including contrarian strategies that follow a glidepath opposite that of target-date funds; that is, they become more aggressive as retirement approaches. The results from a comprehensive sample that spans more than 19 countries, two regions, and 110 years suggest that, relative to lifecycle strategies, the alternative strategies considered here provide investors with higher upside potential,

more limited downside potential, and higher uncertainty although that uncertainty is largely limited to how much higher their terminal wealth is expected to grow with these strategies.

Estrada, J. (2016). "The Retirement Glidepath: An International Perspective." In: *The Journal of Investing* 25(2), pp. 28–54.

All individuals need to decide how much to save during their working years, how much to spend during retirement, and the asset allocation of their portfolio in both periods. A portfolio's exposure to stocks and bonds affects key variables, such as the probability of portfolio failure, degree of downside protection, and expected bequest. How this exposure should evolve during retirement is the ultimate issue explored in this article. After considering declining-equity, rising-equity, and static glidepaths, the comprehensive international evidence from 19 countries and the world market over 110 years ultimately suggests that both an all-equity portfolio and a 60/40 stock/bond allocation are simple and very effective strategies for retirees to implement.

Estrada, J. (2017a). "Maximum Withdrawal Rates: A Novel and Useful Tool." In: *Journal of Applied Corporate Finance* 29(4), pp. 134–137.

The failure rate is arguably the tool most widely used to evaluate retirement strategies. This tool has several shortcomings, one of which is that it does not account for the very different bequests that different strategies may leave behind. The maximum withdrawal rate is a novel tool that, besides dealing with this problem, is useful for 1) providing a comprehensive evaluation of retirement strategies, and 2) evaluating the likelihood of sustaining a target sequence of inflation adjusted withdrawals.

Estrada, J. (2017b). "Refining the Failure Rate." In: *The Journal of Retirement* 4(3), pp. 63–76.

The importance of determining how often a retirement strategy failed in the past, and how likely it is to fail in the future, can hardly be overstated. For this reason, the failure rate has become an essential tool when evaluating these strategies. However, this variable is silent about how long into the retirement period a strategy failed; two strategies that sustained withdrawals for 10 and 25 years of a 30-year retirement period have both failed, but a retiree would be far from indifferent between them. The two variables proposed here, which seek to refine and complement, not to replace, the failure rate aim to correct this shortcoming.

Estrada, J. (2017c). "Replacing the Failure Rate: A Downside Risk Perspective." In: *SSRN e-Print*.

The evaluation of retirement strategies is typically based on the failure rate, although some alternatives have been proposed recently, such as risk-adjusted success (RAS), which aims to capture not just the failure but also the success of the strategies evaluated. The main shortcoming of RAS is that it measures risk with the standard deviation, which penalizes strategies that leave large bequests. The variable proposed here, downside risk-adjusted success (D-RAS), addresses this shortcoming by taking a downside risk perspective, measuring risk with the semideviation, and therefore with volatility below a chosen benchmark. This modification leads to the selection of more plausible strategies than those selected by both the failure rate and RAS.

Estrada, J. (2018a). "From Failure to Success: Replacing the Failure Rate." In: *The Journal of Wealth Management* 20(4), pp. 9–21.

The failure rate is arguably the variable most widely used in the evaluation of retirement strategies. Its main shortcoming, evaluating how often a strategy fails but not by how much it does, is overcome by shortfall years, which considers precisely this information. The joint use of the failure rate and shortfall years is an improvement over using just the failure rate but implies the use of two variables, rather than just one, which do not always point in the same direction. This article introduces a new variable, years sustained, that focuses on success rather than on failure. The ratio between its mean and standard deviation, risk-adjusted success, is the single variable proposed here to be used in a comprehensive evaluation of retirement strategies.

Estrada, J. (2018b). "Maximum withdrawal rates: an empirical and global perspective." In: *The Journal of Retirement* 5(3), pp. 57–71.

Standard analysis of retirement strategies involves evaluating their failure rate. One of the shortcomings of this approach is that a strategy may have a low failure rate and at the same time leave a large unintended bequest. Maximum withdrawal rates, by definition, exhaust a portfolio by the end of the retirement period, thus leaving no bequest; they can be used both to assess the likelihood of sustaining any chosen level of inflation-adjusted withdrawals, and more generally to evaluate retirement strategies. This article provides a comprehensive historical perspective on maximum withdrawal rates in 21 countries over 115 years with 11 asset allocations ranging from 100% stocks to 100% bonds.

Estrada, J. (2019). "Managing to target: dynamic adjustments for accumulation strategies." In: *SSRN e-Print*.

Planning for retirement, particularly during the accumulation period, largely consists of setting a target value for the retirement portfolio and implementing a policy aimed at hitting that target. Financial plans are inevitably based on expected returns, which are likely to be different from those an individual experiences during the accumulation period. Thus, when the portfolio deviates from the path outlined in the plan, the individual can choose between a static policy of sticking to his plan and simply hope to hit the target, or dynamic policies designed to keep the portfolio close to its path. This article evaluates three types of such dynamic policies, broadly referred to as to target (M2T), that adjust the periodic contributions or the portfolio asset allocation. The results reported show that some of the dynamic policies outlined outperform a static policy, and adjusting contributions is far superior to adjusting the asset allocation.

Estrada, J. (2020a). "Retirement Planning: From Z to A." In: *The Journal of Retirement* 88(22), pp. 8–22.
Retirement planning is an issue that must be tackled early and solved backward. It must be tackled early because little can be done if an individual is not on the right path and has only a few years left to work. It must be solved backward because it makes little sense to aim for a portfolio that may not be able to sustain the desired lifestyle in retirement. This article introduces an approach that integrates the working period and the retirement period; leads individuals to consider all relevant variables at the beginning of their journey; and enables them to start saving early to build a target portfolio designed specifically to sustain a desired retirement. The analytical framework introduced yields closed-form solutions for the target retirement portfolio and the contributions that need to be made during the working years to hit that target. The framework proposed is illustrated with an empirical base case, sensitivity analysis, and Monte Carlo simulations.

Estrada, J. (2020b). "Sequence Risk: Is It Really a Big Deal?" In: *SSRN e-Print*.
Financial planners are keenly aware of, and routinely warn clients about, sequence risk; that is, the possibility of facing a sequence of low returns early in retirement that may force retirees to scale down the plans they had made. This really is a scary scenario, but one that the evidence here shows that retirees are not very likely to encounter. A new and refined definition of sequence risk is advanced in this article, linking this type of risk to the sustainability of a withdrawal strategy. Furthermore, three ways of assessing sequence risk are proposed, among them one that enables a retiree to monitor the sustainability of his withdrawal strategy periodically and to introduce adjustments when necessary. The ultimate message of this article is that retirees should be informed, but not obsess, about sequence risk.

Estrada, J. (2020c). "Target-Date Funds, Glidepaths, and Risk Aversion." In: *SSRN e-Print*.
Target-date funds feature asset allocations that become increasingly conservative as investors approach retirement. An important shortcoming of this strategy is that it is suboptimal in terms of capital accumulation, which begs the question of why these funds are so popular. A possible answer is that investors become more risk averse as they age, gradually favoring more downside protection as they approach retirement. The main issue explored in this article is how much more risk averse would investors need to become during their working years to select asset allocations similar to those in target-date funds; the evidence here shows that investors would have to roughly double their risk aversion during the last 25 years of their working period. An intuitive interpretation of this result, based on how much an individual would pay to avoid a gamble, is also discussed.

Estrada, J. (2020d). "Target-Date Funds, Glidepaths, and Risk Aversion." In: *The Journal of Wealth Management* 23(3), pp. 50–60.
Target-date funds feature asset allocations that become increasingly conservative as investors approach retirement. An important shortcoming of this strategy is that it is suboptimal in terms of capital accumulation, which begs the question of why these funds are so popular. A possible answer is that investors become more risk averse as they age, gradually favoring more downside protection as they approach retirement. The main issue explored in this article is how much more risk averse would investors need to become during their working years to select asset allocations similar to those in target-date funds; the evidence here shows that investors would have to roughly double their risk aversion during the last 25 years of their working period. An intuitive interpretation of this result, based on how much an individual would pay to avoid a gamble, is also discussed.

Estrada, J. (2021a). "Sequence Risk: Is It Really a Big Deal?" In: *The Journal of Investing* 30(6), pp. 47–69.
Financial planners are keenly aware of, and routinely warn clients about, sequence risk; that is, the possibility of facing a sequence of low returns early in retirement that may force retirees to scale down significantly the plans they had made. This really is a scary scenario, but one that the evidence here shows that retirees are not very likely to encounter. A new and refined definition of sequence risk is advanced in this article, linking this type of risk to the sustainability of a withdrawal strategy. Furthermore, three ways of assessing sequence risk are proposed, among them one that enables a retiree to monitor the sustainability of his withdrawal strategy periodically and to introduce adjustments when necessary. The ultimate message of this article is that retirees should be informed, but not obsess, about sequence risk.

Estrada, J. (2021b). "The Gain-Pain Index: Asset Allocation for Individual (And Other?) Investors." In: *SSRN e-Print*.

Individual investors typically determine their asset allocation using investor questionnaires, which can be viewed as black boxes that generate a result without highlighting the benefits and costs of the portfolios considered. This article introduces an asset allocation tool, the gain-pain index (GPI), that overcomes this shortcoming. The tool proposed incorporates two critical variables found in investor questionnaires, the portfolio holding period and the investor's risk tolerance, and broadens the definition of risk beyond volatility by also considering the probability of suffering a loss and the magnitude of the loss. The model is used to determine optimal asset allocations for 21 countries and the world market, for different holding periods and levels of risk aversion.

Estrada, J. (2021c). "The Sustainability of (Global) Withdrawal Strategies." In: *Journal of Financial Planning* 34(11), pp. 82–98.

The most important financial issue retirees have to deal with is whether their strategy will be able to sustain all the withdrawals they expect to make and a bequest they aim to leave. It is critical to periodically monitor the evolution of a financial plan in order to detect early signs of trouble, which may lead a retiree to introduce dynamic adjustments to a strategy.

To that purpose, this article features two tools, a sustainability test and the sustainable withdrawal, and shows how to apply them. This article also discusses the empirical evidence on both tools based on a comprehensive sample of 22 markets over a 120-year period.

Estrada, J. (2022). "The Gain-Pain Index: Asset Allocation for Individual (and Other?) Investors." In: *The Journal of Wealth Management* 25(2), pp. 34–48.

Individual investors typically determine their asset allocation using investor questionnaires, which can be viewed as black boxes that generate a result without highlighting the benefits and costs of the portfolios considered. This article introduces an asset allocation tool-the gain-pain index-that overcomes this shortcoming. The proposed tool incorporates two critical variables found in investor questionnaires-the portfolio holding period and the investor's risk tolerance-and broadens the definition of risk beyond volatility by considering the probability of suffering a loss and the magnitude of the loss. The model is used to determine optimal asset allocations for 21 countries and the world market for different holding periods and levels of risk aversion.

Estrada, J. (2023). "Retirement Planning: Is One Number Enough?" In: *SSRN e-Print*.

Academics, practitioners, and investors at large typically have a preference for making decisions based on a single variable that is supposed to be maximized or minimized. This approach is usually appropriate but it may also be misleading. When selecting an optimal retirement strategy a retiree may aim to maximize the coverage ratio, a novel metric superior to the failure rate. This article suggests to focus on the whole distribution of coverage ratios instead, or at least on some percentiles that may be of particular interest to a retiree. Although such an approach may not be as neat as making decisions based on optimizing a single variable, it does enable the consideration of the relevant trade-offs a retiree needs to evaluate in order to find an ideal retirement strategy.

Estrada, J. (2024). "Retirement Planning: The Volatility-Adjusted Coverage Ratio." In: *The Journal of Retirement* 12(1), pp. 40–60.

The important decisions that retirees have to make to try to achieve their financial goals during retirement often stem from models used by financial planners. Despite the important role it plays in many of those models, the failure rate has several limitations and many alternatives have been proposed. This article introduces a new metric, the volatility-adjusted coverage ratio, which incorporates the benefit (the coverage ratio) and the cost (the volatility of the portfolio) of the strategies considered. Application of this new metric, which improves upon both the failure rate and the coverage ratio, is illustrated by determining the optimal asset allocation, for several initial withdrawal rates, for twenty-two global markets. The overall results show that maximizing the volatility-adjusted coverage ratio typically yields optimal asset allocations that are, first, more conservative than those stemming from maximizing the coverage ratio or the expected utility of the coverage ratio; and second, more consistent with those featured by target-date funds.

Estrada, J. and Kritzman, M. (2018). "Evaluating Retirement Strategies: A Utility-Based Approach." In: *SSRN e-Print*.

Retirees need to make two critical financial decisions, namely, the withdrawal rate and the asset allocation of their portfolios. We propose a methodology that retirees, and particularly advisors, could use to make these decisions in an optimal way. We introduce a new variable, the coverage ratio, and a theoretical approach, based on utility. Our approach can be used to make optimal decisions during both the accumulation and the retirement period, but we illustrate it by focusing on the latter, and particularly on the choice of an optimal asset allocation.

We find that the strategies selected by our utility-based approach are in general somewhat more aggressive than those selected by the failure rate and other existing approaches.

Estrada, J. and Kritzman, M. (2019). "Toward Determining the Optimal Investment Strategy for Retirement." In: *The Journal of Retirement* 7(1), pp. 35–42.

Investors who are about to retire are first and foremost concerned with supporting their spending needs throughout retirement. But they also derive satisfaction from growing their wealth beyond what is needed to support consumption in order to leave a bequest to their heirs or chosen charities. The predominant metric for evaluating retirement investment strategies is the failure rate. However, it doesn't distinguish between strategies that fail early in retirement from those that fail near the end of retirement, and it doesn't account for potential bequests. To overcome these shortcomings the authors propose a new metric, the coverage ratio, which is more comprehensive and informative than the failure rate. In addition, they propose a utility function to evaluate the coverage ratio, which penalizes shortfalls more than it rewards surpluses. Finally, the authors use their proposed framework to determine the optimal allocation to stocks and bonds using historical and simulated returns.

Fabozzi, F. J. and Fabozzi, F. A. (2022). "A Primer on Hedging with Stock Index Futures." In: *The Journal of Derivatives*.

In this article, the authors discuss the various approaches and issues associated with hedging with stock index futures. The most common approach used in practice is based on minimizing the variance of the hedge within the mean-variance framework to obtain the optimal hedge ratio. In determining the optimal hedge ratio, consideration must be given to the basis risk to which a fund is exposed when using stock index futures. An optimal hedge ratio based on variance minimization is the slope coefficient estimated from an ordinary least squares (OLS) regression of the returns of the portfolio to be hedged on the returns of the stock index futures contract. The estimated slope coefficient is referred to as beta. The optimal hedge ratio can be further refined by adjusting for the beta estimated from an OLS regression of the return on the underlying stock index on the return on the stock index futures. A criticism of the OLS model is twofold. The first is that there are statistical issues in estimating beta using the basic OLS regression model. Several models that employ advanced econometric techniques have been proposed for estimating hedge ratios. The second criticism is that the OLS model assumes a constant hedge ratio, despite the theoretical and empirical evidence showing the hedge ratio should be time varying. Evidence suggests that employing advanced econometric models to estimate the slope coefficient offers little improvement in hedging effectiveness-and even if there is some improvement, the modeling cost may not justify the extra effort. As for the second criticism, the well-known autoregressive conditional heteroscedasticity (ARCH) and the generalized ARCH (GARCH) have been used to allow for time-varying hedge ratios. Although some studies have reported that ARCH and GARCH can improve hedge effectiveness, the effort may not be warranted due to the additional modeling as well as the time-varying hedge ratios involving rebalancing the portfolio periodically, which can add significantly to the cost of hedging.

Fabozzi, F. J. and Lopez de Prado, M. (2018). "Being Honest in Backtest Reporting: A Template for Disclosing Multiple Tests." In: *The Journal of Portfolio Management* 45(1), pp. 141–147.

Selection bias under multiple testing is a serious problem. From a practitioner perspective, failure to disclose the impact of multiple tests of a proposed investment strategy to clients and senior management can lead to the adoption of a false discovery. Clients will lose money, senior management will misallocate resources, and the firm may be exposed to reputational, legal, and regulatory risks. From the perspective of academic journals that publish evidence supporting an investment strategy, the failure to address selection bias under multiple testing threatens to invalidate large portions of the literature in empirical finance. In this article, the authors propose a template that practitioners should use to fairly disclose multiple tests involved in an alleged discovery when pitching strategies to clients and senior management. The same template could be used by contributors to academic journals so that referees, and ultimately readers, can assess the strategy. By disclosing this information, those who are charged with making the final decision about a discovery can evaluate the probability that the purported discovery is false.

Fan, J., Weng, H., and Zhou, Y. (2021). "Optimal estimation of functionals of high-dimensional mean and covariance matrix." In: *arXiv e-Print*.

Motivated by portfolio allocation and linear discriminant analysis, we consider estimating a functional $mu^T\Sigma^{-1}\mu$ involving both the mean vector μ and covariance matrix Σ. We study the minimax estimation of the functional in the high-dimensional setting where $\Sigma^{-1}\mu$ is sparse. Akin to past works on functional estimation, we show that the optimal rate for estimating the functional undergoes a phase transition between regular parametric rate and some form of high-dimensional estimation rate. We further show that the optimal rate is attained by a carefully designed plug-in estimator based on de-biasing, while a family of naive plug-in estimators are proved to fall short. We further generalize the estimation problem and techniques that allow robust inputs of mean and covariance matrix estimators. Extensive numerical experiments lend further supports to our theoretical results.

Fan, Y.-A., Murray, S., and Pittman, S. (2013). "Optimizing retirement income: an adaptive approach based on assets and liabilities." In: *The Journal of Retirement* 1(1), pp. 124–135.

Many retirees financial resources are insufficient to sustain their desired expenditures, which makes clear the need for more efficient retirement asset allocation strategies. We propose an approach to retirement investing that directly incorporates spending targets within a multiperiod framework that improves performance with respect to spending sustainability and bequest motives. We propose an model, which is dynamic and allows for recourse decisions and permits changes in asset allocation in response to market outcomes as the retiree ages. Practitioners can use the adaptive model to increase the sustainability of their clients spending and the funding of bequests, thereby enhancing their clients welfare in their later years.

Fan, Z., Londono, J. M., and Xiao, X. (2022). "Equity tail risk and currency risk premiums." In: *Journal of Financial Economics* 143(1), pp. 484–503.

We find that an option-based equity tail risk factor is priced in the cross section of currency returns; more exposed currencies offer a low risk premium because they hedge against equity tail risk. A portfolio that buys currencies with high equity tail beta and shorts those with low beta extracts the global component in the tail factor. The estimated price of risk of this novel global factor is consistently negative in currency carry and momentum portfolios, and in portfolios of other asset classes, suggesting that excess returns of these strategies can be partially understood as compensations for global tail risk.

Farahani, A. and Yan, M. (2014). *Constructing a Liability Hedging Portfolio: A Guide to Best Practices for US Pension Plans*. Tech. rep. Cambridge Associates.

To construct an effective liability hedging portfolio, a key first step is to evaluate the variety of ways liabilities can be calculated and discounted and to identify the most relevant liability metric for a plan sponsor's circumstances. Plan sponsors should also define the acceptable level of surplus risk and carefully consider the appropriate duration of the liability hedging assets-shorter duration assets often have a meaningful mismatch to most defined benefit pension liabilities. Plan sponsors should not construct a liability hedge in isolation, as the size and composition of both the growth portfolio and liability hedge will have important implications for each other. Since the end goal is to maximize portfolio return at a controlled level of risk, the liability hedging portfolio should not attempt to perfectly immunize a cash flow stream-an extremely difficult task when considering real world constraints of corporate bond issuance, transaction costs, and liquidity-but rather optimize duration, curve, and credit spread exposures within the broader context of the total plan's risks related to both its assets and liability. We recommend plan sponsors consider the use of more complex fixed income structures to manage their liability hedge programs. Most notably, we believe prudent use of derivatives (and, implicitly, some degree of portfolio leverage) should be considered, as bond futures, interest rate swaps, swaptions, and other techniques provide far more flexibility to optimize the portfolio when compared to a plain vanilla physical bond portfolio. Derivatives may also allow for a more capital-efficient liability hedging allocation, which we believe is paramount when looking at the low level of return provided by fixed income in the current environment. Still, even with these derivatives-based strategies, a "perfect hedge"is not attainable. Plan sponsors should think carefully about the active management implications of constructing the liability hedge, particularly for those plans that target low surplus volatility and maintain heavy allocations to fixed income. In normal environments, a plan with the majority of its liability hedging assets devoted to bond managers (typically with moderate tracking error) will not be able to generate the same level of value add as a plan with a more diversified portfolio invested in higher active risk strategies such as equities, hedge funds, and private investments. Best practices when developing a liability hedging framework include: Understanding and defining the liability, 1) Quantifying acceptable surplus risk, 2) Acknowledging the basis risks in replicating the liability hedging benchmark, 3) Monitoring the interaction of credit and equities, 4) Emphasizing capital efficiency given the lower returns of liability hedging assets, and 5) Considering derivatives as part of the toolkit.

Faria, M. C. (2021). "An Examination of Target Date Fund Glidepath Construction." In: *SSRN e-Print*.

The paper examines Target Date Fund glidepath construction as well as asset allocation strategies. We consider characteristics, features, and assumptions in solving the multi-period portfolio selection problem. This paper discusses the theoretical underpinnings of deterministic, adaptive, and stochastic models as well as some interesting alternative strategies. Furthermore, we also propose research topics for future consideration that the author believes may improve upon the existent model frameworks.

Farrington, R. (2021). *The Best Order of Operations For Saving For Retirement*. Tech. rep. The College Investor. So, what is the best order of operations for saving for retirement? Let me break it down for you, and show you the exact strategy I'm using as well.

1) Save in Your 401k (Up To The Match)
2) Save The Max In Your IRA
3) Continue To Max Your 401k Contributions
4) Max Your HSA
5) Side Hustle And Do A SEP IRA
6) Save in a Standard Brokerage Account
7) Be Smart About Social Security

We have presented The Order Of Operations For Saving.

Fellowes, M., Fichtner, J. J., Plews, L., and Whitman, K. (2019). *The Retirement Solution Hiding in Plain Sight: How Much Retirees Would Gain by Improving Social Security Decisions*. Tech. rep. Capital One Investing. Social Security now accounts for about one-third of all income annually received by U.S. retirees, amounting to $1 trillion in annual benefits. While impactful, research consistently finds that the financial effect of Social Security could be even greater if more people waited to enroll, since monthly benefits can increase in value if retirees delay claiming. But, we don't know how much is annually lost from households making the sub-optimal decision about when to claim Social Security, how many are making mistakes, or who is making those wrong decisions. To explore these questions, we utilize new technology invented by United Income and data sponsored by the Social Security Administration, finding:

- Retirees will collectively lose $3.4 trillion in potential income that they could spend during their retirement because they claimed Social Security at a financially sub-optimal time, or an average of $111,000 per household. The average Social Security recipient would receive 9 percent more income in retirement if they made the financially optimal decision about when to claim this retirement benefit.
- Current retirees will collectively lose an estimated $2.1 trillion in wealth because they made the sub-optimal decision about when to claim Social Security, or an average of $68,000 per household. Most retirees will lose wealth in their 60s and early 70s if they choose to optimize Social Security, but will be wealthier in their late 70s through the rest of their lives.
- Only 4 percent of retirees make the financially optimal decision about when to claim Social Security. About 57 percent of retirees would build more wealth through their life if they waited to claim until they were 70 years old (when only 4 percent of retirees currently claim), while only 6.5 percent of retirees would have more wealth if they claimed prior to turning 64 (when over 70 percent of retirees currently claim benefits).
- About 21 percent of those at risk of not affording retirement (or having enough income to cover their expected cost of living) would see an improvement in their chances if they claimed Social Security at the optimal time. Among those retirees at risk that start with a greater than 10 percent chance of affording retirement, 95 percent see their chances of affording retirement improve by an average of 28 percent.
- Elderly poverty could be cut by nearly 50 percent if all retirees claimed Social Security at the financially optimal time. In particular, about 13 percent of people over the age of 70 are expected to live in poverty at some point, which is estimated to fall to 7 percent if retirees had claimed Social Security at the optimal time -a rate that could potentially fall even further if they earned additional income while they waited to claim Social Security.

This report finds that nearly no retirees are making the financially optimal decision about Social Security, and that the costs of those mistakes are high for retiring households, particularly those at risk of not being able to afford retirement. In addition, since making the optimal decision means sacrificing wealth in the near-term, we think it is unlikely more people will make the right decision without a policy intervention. There are numerous difficulties associated with solving this problem, though, which will require a thorough and diverse process for addressing. Among the topics for consideration should be the eligibility age range rules, which were last materially modified in 1983. Since 92 percent of retirees are expected to be better off waiting to claim until at least their 65th birthday, claiming before should ideally be an exception for those who demonstrably need to claim benefits before the full retirement age. Means-testing rules may be one way to address this, though an easier place to start would be to change how the Social Security Administration frames claiming age options to the public. Instead of portraying age 62 as the "early eligibility age," for instance, claiming at age 62 could instead be labeled as the "minimum benefit age" while age 70 could be labeled as the "maximum benefit age." The Social Security Administration could also be provided with resources to improve utilization of the policy it administers, perhaps in partnership with third-party fiduciaries. With the potential to put $2.1 trillion wealth and $3.4 trillion in income in the pockets of retirees, policymakers should be focused on improving this program.

Feng, R. and Jing, X. (2017). "Analytical valuation and hedging of variable annuity guaranteed lifetime withdrawal benefits." In: *Insurance: Mathematics and Economics* 72, pp. 36–48.

Variable annuity is a retirement planning product that allows policyholders to invest their premiums in equity funds. In addition to the participation in equity investments, the majority of variable annuity products in today's market offer various types of investment guarantees, protecting policyholders from the downside risk of their investments. One of the most popular investment guarantees is known as the guaranteed lifetime withdrawal benefit (GLWB). In current market practice, the development of hedging portfolios for such a product relies heavily on Monte Carlo simulations, as there were no known closed-form formulas available in the existing actuarial literature. In this paper, we show that such analytical solutions can in fact be determined for the risk-neutral valuation and delta-hedging of the plain-vanilla GLWB. As we demonstrate by numerical examples, this approach drastically reduces run time as compared to Monte Carlo simulations. The paper also presents a novel technique of fitting exponential sums to a mortality density function, which is numerically more efficient and accurate than the existing methods in the literature.

Ferrari, G. and Zhu, S. (2023). "Optimal Retirement Choice under Age-dependent Force of Mortality." In: *arXiv e-Print*.

This paper examines the retirement decision, optimal investment, and consumption strategies under an age-dependent force of mortality. We formulate the optimization problem as a combined stochastic control and optimal stopping problem with a random time horizon, featuring three state variables: wealth, labor income, and force of mortality. To address this problem, we transform it into its dual form, which is a finite time horizon, three-dimensional degenerate optimal stopping problem with interconnected dynamics. We establish the existence of an optimal retirement boundary that splits the state space into continuation and stopping regions. Regularity of the optimal stopping value function is derived and the boundary is proved to be Lipschitz continuous, and it is characterized as the unique solution to a nonlinear integral equation, which we compute numerically. In the original coordinates, the agent thus retires whenever her wealth exceeds an age-, labor income- and mortality-dependent transformed version of the optimal stopping boundary. We also provide numerical illustrations of the optimal strategies, including the sensitivities of the optimal retirement boundary concerning the relevant model's parameters.

Fichtner, J. J. and Seligman, J. S. (2018). "Retirement Saving and Decumulation in a Persistent Low-Return Environment." In: *How Persistent Low Returns Will Shape Saving and Retirement*. Oxford University Press.

The current retirement environment presents challenges, not only over the period for which interest rates remain low, but also once interest rates appreciably increase. This chapter addresses two related questions: first, how have households responded to the current low interest rate environment, and second, are there alternative responses or investments which households might do well to consider? We employ the Health and Retirement Study to first investigate impacts of the low interest rate on savings, wealth, and asset allocation. We also report on a subset of households who were relatively successful at building and preserving wealth over this period. Following this, we consider alternative portfolio and wealth management strategies targeting increases in equities and delayed claiming of Social Security in terms of their potential to add value in persistent low return environments.

Fidelity Research (2019). *Why work with a financial advisor*. Tech. rep. Fidelity.

Industry studies estimate that professional financial advice can add between 1.5% and 4% to portfolio returns over the long term, depending on the time period and how returns are calculated. A one-on-one relationship with an advisor is not just about money management. A financial advisor can help provide ongoing financial planning so you can have peace of mind while pursuing your life goals. The financial planning process includes defining your goals, understanding your current situation, and identifying the key steps to move forward. Beyond long-term goals like retirement, and shorter-term ones like buying a house, education, or travel, holistic financial planning can also include legacy planning, family support, health care, insurance, and charitable giving.

Fieberg, C., Hornuf, L., and Streich, D. (2024). "Using Large Language Models for Financial Advice." In: *SSRN e-Print*.

We study whether large language models (LLMs) are able to generate suitable financial advice and which model features are associated with higher-quality advice. To this end, we elicit portfolio recommendations from 32 LLMs for 64 investor profiles differing with respect to their risk tolerance and capacity, home country, sustainability preferences, gender, and investment experience. We find that LLMs are generally capable of generating financial advice as the recommendations can in fact be implemented, take into account important investor characteristics when determining market and risk exposure, and display historical performance in line with assumed risks. We further find that foundation models and larger models generate portfolios that are easier to implement and more sensitive to an investor's circumstances than fine-tuned models and smaller models. We find some evidence consistent with LLMs inheriting human biases as domestic exposures substantially exceed benchmark weights (home bias). On the other hand, we find no evidence of gender-based discrimination found in human financial advice. Based on these results, we discuss the potential application of LLMs in the financial advice context.

Finefrock, C. J., Gradisher, S. M., and Nitz, C. M. (2015). "Long-Term Care Insurance: Comparisons for Determining the Best Options for Clients." In: *Journal of Financial Planning* 38(2), pp. 36–43.
An estimated 8.36 million people need some type of long-term care (LTC) service annually. The American Association for Long-Term Care Insurance (AALTCI) recommends that individuals purchase LTC policies in their mid-50s. The cost for LTC is increasing and currently averages $43,472 annually for adult day service center, to $87,600 for a private nursing home room. Studies show that most Americans are not financially prepared for LTC needs. This study evaluates various LTC solutions, including traditional LTC, variable annuity combination products, fixed annuity combination products, variable life insurance combinations, universal life combinations, and "hybrid" LTC. Results suggest the areas of differentiation include: cost per unit of coverage, case design flexibility, certainty of coverage, liquidity, investment upside, death benefit to heirs, and degree of medical underwriting required.

Finke, M. (2019). *Bill Sharpe Seeks a Better Retirement Income Solution*. Tech. rep. ThinkAdvisor.
Sharpe still comes up with simple truths that can rock even a veteran's understanding of retirement planning. To Sharpe, the role of a financial advisor is to understand the range of options all clients face in order to build an income that efficiently balances risk and return. The advisor needs to understand these tradeoffs, and he or she needs to be able to explain them to a client. He's even written a book that's available on his Stanford website to help advisors "focus more on communicating possible outcomes and helping the clients understand the options." Sharpe's critique of the 4% rule is that the methodology is supposed to safely provide a straight line of after-inflation income using a portfolio of random returns over an unknown lifetime. But when you don't know what asset returns will look like and how long you'll live, a retiree has two lifestyle choices: (1) live well and risk running out of money, or (2) be conservative and risk giving too much to your heirs.

Finke, M. and Pfau, W. (2015). "Reduce Retirement Costs with Deferred Income Annuities Purchased before Retirement." In: *Journal of Financial Planning* 28(7), pp. 40–49.
Most deferred income annuities (DIAs) have a relatively short deferral period, making them particularly appealing to clients nearing retirement who value the ability to plan on a fixed, nominal income stream after retirement. Unlike long-term deferral period annuities (longevity insurance) that are primarily meant to protect against longevity risk, a short-term deferral period annuity can provide a steady income to pre-fund retirement spending over the entire retirement life cycle. The purchase of a DIA before retirement is particularly valuable for clients who would have maintained a stock allocation lower than 70 percent. Retirees would have been better off without a DIA in simulations where portfolio returns are very high or when retirees die early. Findings suggest that short deferral period annuities can reduce the cost of funding retirement, provide longevity protection, and offer behavioral benefits to clients concerned about near-retirement market performance.

Finke, M. S. and Blanchett, D. (2016). "An Overview of Retirement Income Strategies." In: *Journal of Investment Consulting* 17(1), pp. 22–30.
This paper provides an introduction to strategies that can be used to turn a retirement portfolio into income. Defined contribution savings present a retirement planning challenge when asset returns, longevity, and spending needs are uncertain. We present the fundamental trade-offs of managing an investment portfolio and decumulating investments in retirement, and we discuss the potential benefit of financial products that address portfolio and spending risks. Simulations illustrate the trade-offs among legacy, liquidity, and shortfall risk. We dis-cuss a comprehensive strategy that combines longevity protection, portfolio allocation, and dynamic spending based on realized portfolio returns to provide a guideline for professionals building retirement income plans for clients. We then provide an overview of the benefits of adding products such as annuities to a traditional investment

portfolio to reduce longevity risk and increase expected spending. The primary purpose of this paper is to provide a fundamental overview of current research and theory on retirement income planning.

Fisch, J. E. and Turner, J. A. (2018). "Making a Complex Investment Problem Less Difficult:Robo Target-Date Funds." In: *The Journal of Retirement* 5(4), pp. 40–45.

Investing is a complicated affair, particularly for people with low financial literacy. Target-date funds are designed to make investing easy for pension participants. To simplify the employee decision, many defined contribution plans offer employees a target-date fund based on only one piece of data employee expected retirement date. The employee may be placed in a target-date fund as the default plan if the employee does not make an active choice. Target-date funds one-size-fits-all approach generally does not provide the appropriate level of risk for all employees who plan to retire in a given year. The authors address that issue in this article. Their proposal has three parts. First, they propose that target-date funds should allow greater personalization of investments by offering participants a conservative, moderate, and risky fund for each target-date. While pension participants currently have the option of choosing a more or less risky target-date fund by choosing a later or earlier target-date than their actual retirement date, many pension participants lack the sophistication to take advantage of that option. Second, they suggest that pension plans incorporate robo advisers to help participants identify the appropriate level of risk and appropriate target-date fund based on their personal circumstances. Third, they propose on-the-spot financial education, to be provided when a participant is selecting a target-date fund, to help participants understand the implications of risk level and target-date fund choice for both pension growth and the range of possible outcomes.

Fischer, M., Jensen, B. A., and Koch, M. (2023a). "Optimal Retirement Savings over the Life Cycle: A Deterministic Analysis in Closed Form." In: *SSRN e-Print*.

In this paper, we explore the life cycle consumption-savings problem in a stylized model with a risk-free investment opportunity, a tax-deferred retirement account, and deterministic labor income. Our closed form solutions show that liquidity constraints can be severely binding; in particular in situations with a high growth rate of labor income, in which retirement saving is optimally postponed. With a tax-deferred account, it is always optimal to save in this (illiquid) account first before saving in the (liquid) taxable account in order to satisfy the needs for consumption smoothing. The optimal retirement savings pattern is far from the widespread practice of contributing a fixed fraction of current labor income over the working life to a tax-deferred environment.

Fischer, M., Jensen, B. A., and Koch, M. (2023b). "Optimal retirement savings over the life cycle: A deterministic analysis in closed form." In: *Insurance: Mathematics and Economics* 112, pp. 48–58.

In this paper, we explore the life cycle consumption-savings problem in a stylized model with a risk-free investment opportunity, a tax-deferred retirement account, and deterministic labor income. Our closed form solutions show that liquidity constraints can be severely binding; in particular in situations with a high growth rate of labor income, in which retirement saving is optimally postponed. With a tax-deferred account, it is always optimal to save in this (illiquid) account first before saving in the (liquid) taxable account in order to satisfy the needs for consumption smoothing. The optimal retirement savings pattern is far from the widespread practice of contributing a fixed fraction of current labor income over the working life to a tax-deferred environment.

Flint, E. J., Seymour, A. J., and Chikurunhe, f. (2016). "In Search of the Perfect Hedge Underlying." In: *SSRN e-Print*.

This report attempts to answer the question: What underlying portfolio should one use to hedge an active fund? We introduce a framework which allows us to conduct analysis on simulated realistic active portfolios in order to build intuition as to how hedge mismatch error affects the level of protection from a given hedge. We show that for typical market conditions, hedge effectiveness improves dramatically when using a hedge portfolio that more accurately reflects the underlying portfolio. This has clear consequences for using generic index options to hedge highly active portfolios. We also showcase several active return and tracking error decompositions that allow one to precisely quantify and thus manage the sources of risk and rewards within a given portfolio. We then discuss a mixed integer quadratic programming formulation that enables us to search across a large investment universe in order to find the subset of stocks that most closely replicates a given portfolio's performance, whilst simultaneously complying with realistic market constraints. Motivated by these three elements, we introduce several alternative hedging methods for the fund manager to implement a better hedge for their active portfolio. In this report, we focus specifically on the use of long-only and long/short custom basket options as a means of creating an appropriate portfolio hedge.

Fonseca, R., Michaud, P.-C., Galama, T., and Kapteyn, A. (2021). "Accounting for the Rise of Health Spending and Longevity." In: *Journal of the European Economic Association* 19(1), pp. 536–579.

We estimate a stochastic life-cycle model of endogenous health spending, asset accumulation, and retirement to investigate the causes behind the increase in health spending and longevity in the United States over the period 1965-2005. Accounting for changes over time in taxes, transfers, Social Security, income, health insurance, smoking and obesity, and technological progress, we estimate that technological progress is responsible for half of the increase in life expectancy over the period. Substantial growth in health spending over the period is largely the result of growth in economic resources and the generosity of health insurance, with a modest role for medical technological progress. The growth in spending does not come from changes in a single source, but sources jointly interacted to increase spending: complementarity effects explain up to 26.3% of the increase in health spending. Overall, for those born in 1940, the combined changes in resources and health insurance that occurred over the period are valued at 35.7% of lifetime consumption.

Forsyth, P. (2021). "Two stage decumulation strategies for DC plan investors." In: *International Journal of Theoretical and Applied Finance* 24(01) (2150007).

Optimal stochastic control methods are used to examine decumulation strategies for a defined contribution (DC) plan retiree. An initial investment horizon of 15 years is considered, since the retiree will attain this age with high probability. The objective function reward measure is the expected sum of the withdrawals. The objective function tail risk measure is the expected linear shortfall with respect to a desired lower bound for wealth at 15 years. The lower bound wealth level is the amount which is required to fund a lifelong annuity 15 years after retirement, which generates the required minimum cash flows. This ameliorates longevity risk. The controls are the withdrawal amount each year, and the asset allocation strategy. Maximum and minimum withdrawal amounts are specified. Specifying a short initial decumulation horizon, results in the optimal strategy achieving: (i) median withdrawals at the maximum rate within 2-3 years of retirement (ii) terminal wealth larger than the desired lower bound at 15 years, with greater than 90% probability and (iii) median terminal wealth at 15 years considerably larger than the desired lower bound. The controls are computed using a parametric model of historical stock and bond returns, and then tested in bootstrap resampled simulations using historical data. At the 15 year investment horizon, the retiree has the option of (i) continuing to self-manage the decumulation policy or (ii) purchasing an annuity.

Forsyth, P. (2023). "A Stochastic Control Approach to Defined Contribution Plan Decumulation: "The Nastiest, Hardest Problem in Finance"." In: *Fields CFI Workshop on Quantitative Methods for Wealth Management*.

We pose the decumulation strategy for a Defined Contribution (DC) pension plan as a problem in optimal stochastic control. The controls are the withdrawal amounts and the asset allocation strategy. We impose maximum and minimum constraints on the withdrawal amounts, and impose no-shorting no-leverage constraints on the asset allocation strategy. Our objective function measures reward as the expected total withdrawals over the decumulation horizon, and risk is measured by Expected Shortfall (ES) at the end of the decumulation period. We solve the stochastic control problem numerically, based on a parametric model of market stochastic processes. We find that, compared to a fixed constant withdrawal strategy, with minimum withdrawal set to the constant withdrawal amount, the optimal strategy has a significantly higher expected average withdrawal, at the cost of a very small increase in ES risk. Tests on bootstrapped resampled historical market data indicate that this strategy is robust to parametric model misspecification.

Forsyth, P., Vetzal, K. R., and Westmacott, G. (2017a). "Target Wealth: The Evolution of Target Date Funds." In: *SSRN e-Print*.

Target Date Funds have become very popular with investors saving for retirement. The main feature of these funds is that investors are automatically switched from high risk to low risk assets as retirement approaches. However, our analysis brings into question the rationale behind these funds. Based on a model with parameters fitted to historical returns, and also on model independent bootstrap resampling, we find that constant proportion strategies give virtually the same results for terminal wealth at the retirement date as target date strategies. This suggests that the vast majority of Target Date Funds are serving investors poorly. However, if we allow the asset allocation strategy to adapt to the current level of the total portfolio value, significantly lower risk of terminal wealth can be achieved, at no cost to its expected value.

Forsyth, P., Vetzal, K. R., and Westmacott, G. (2018). "Management of Withdrawal Risk Through Optimal Life Cycle Asset Allocation." In: *SSRN e-Print*.

Retirees who do not have defined benefit pension plans typically must fund spending from accumulated savings. This leads to the risk of depleting these savings, i.e. withdrawal risk. We analyze this risk through full life cycle optimal dynamic asset allocation, including the accumulation and decumulation phases. We pose the asset allocation strategy as a problem in optimal stochastic control. Various possible objective functions are tested and compared using metrics such as the probability of portfolio depletion, the median of the remaining portfolio value, and conditional value at risk (CVAR). The control problem is solved using a Hamilton-Jacobi-Bellman formulation, based on a parametric model of the underlying stochastic processes and a variety of objective functions. Monte Carlo simulations which use the optimal controls are presented to evaluate the performance metrics. These simulations are based on both the parametric model and bootstrap resampling of 91 years of historical data. Based primarily on the resampling tests, we conclude that target-based approaches which seek to establish a safety buffer of wealth at the end of the decumulation period appear to be superior to strategies which directly attempt to minimize risk measures such as the probability of portfolio depletion.

Forsyth, P. A. (2022a). "A Stochastic Control Approach to Defined Contribution Plan Decumulation: "The Nastiest, Hardest Problem in Finance"." In: *North American Actuarial Journal* 26(2), pp. 227–251.
We pose the decumulation strategy for a defined contribution (DC) pension plan as a problem in optimal stochastic control. The controls are the withdrawal amounts and the asset allocation strategy. We impose maximum and minimum constraints on the withdrawal amounts, and impose no-shorting no-leverage constraints on the asset allocation strategy. Our objective function measures reward as the expected total withdrawals over the decumulation horizon, and risk is measured by expected shortfall (ES) at the end of the decumulation period. We solve the stochastic control problem numerically, based on a parametric model of market stochastic processes. We find that, compared to a fixed constant withdrawal strategy, with minimum withdrawal set to the constant withdrawal amount the optimal strategy has a significantly higher expected average withdrawal, at the cost of a very small increase in ES risk. Tests on bootstrapped resampled historical market data indicate that this strategy is robust to parametric model misspecification.

Forsyth, P. A. (2022b). "Short term decumulation strategies for underspending retirees." In: *Insurance: Mathematics and Economics* 102, pp. 56–74.
There is growing empirical evidence that many retirees are decumulating their assets very slowly, if at all. This fact is in stark contrast to the usual lifecycle models of spending. It appears that these underspending retirees adjust their withdrawals to avoid reducing their assets. In order to appeal to this class of retirees, we use optimal stochastic control techniques which maximize a multi-objective risk-reward problem. The reward is the total of withdrawals (over a five year period), while risk is based on a left tail measure. Our controls for this problem are the withdrawal amount per quarter, and the stock-bond asset allocation. We allow flexible withdrawals (even zero). This added flexibility results in a high probability of (i) retaining 90% of real wealth at the end of five years, and (ii) significant total spending over the five years. We suggest that these types of strategies will be appealing this underspending group of retirees.

Forsyth, P. A., Li, Y., and Vetzal, K. R. (2017b). "Are target date funds dinosaurs? Failure to adapt can lead to extinction." In: *arXiv e-Print*.
Investors in Target Date Funds are automatically switched from high risk to low risk assets as their retirements approach. Such funds have become very popular, but our analysis brings into question the rationale for them. Based on both a model with parameters fitted to historical returns and on bootstrap resampling, we find that adaptive investment strategies significantly outperform typical Target Date Fund strategies. This suggests that the vast majority of Target Date Funds are serving investors poorly.

Forsyth, P. A. and Vetzal, K. R. (2019). "Optimal asset allocation for retirement saving: deterministic vs. time consistent adaptive strategies." In: *Applied Mathematical Finance* 26(1), pp. 1–37.
We consider optimal asset allocation for an investor saving for retirement. The portfolio contains a bond index and a stock index. We use multi-period criteria and explore two types of strategies: deterministic strategies are based only on the time remaining until the anticipated retirement date, while adaptive strategies also consider the investor accumulated wealth. The vast majority of financial products designed for retirement saving use deterministic strategies (e.g., target date funds). In the deterministic case, we determine an optimal open loop control using mean-variance criteria. In the adaptive case, we use time consistent mean-variance and quadratic shortfall objectives. Tests based on both a synthetic market where the stock index is modelled by a jump-diffusion process and also on bootstrap resampling of long-term historical data show that the optimal adaptive strategies significantly outperform the optimal deterministic strategy. This suggests that investors are not being well served by the strategies currently dominating the marketplace.

Forsyth, P. A., Vetzal, K. R., and Westmacott, G. (2019). "Management of Portfolio Depletion Risk through Optimal Life Cycle Asset Allocation." In: *North American Actuarial Journal* 23(3), pp. 447–468.
Members of defined contribution (DC) pension plans must take on additional responsibilities for their investments, compared to participants in defined benefit (DB) pension plans. The transition from DB to DC plans

means that more employees are faced with these responsibilities. We explore the extent to which DC plan members can follow financial strategies that have a high chance of resulting in a retirement scenario that is fairly close to that provided by DB plans. Retirees in DC plans typically must fund spending from accumulated savings. This leads to the risk of depleting these savings, that is, portfolio depletion risk. We analyze the management of this risk through life cycle optimal dynamic asset allocation, including the accumulation and decumulation phases. We pose the asset allocation strategy as an optimal stochastic control problem. Several objective functions are tested and compared. We focus on the risk of portfolio depletion at the terminal date, using such measures as conditional value at risk (CVAR) and probability of ruin. A secondary consideration is the median terminal portfolio value. The control problem is solved using a Hamilton-Jacobi-Bellman formulation, based on a parametric model of the financial market. Monte Carlo simulations that use the optimal controls are presented to evaluate the performance metrics. These simulations are based on both the parametric model and bootstrap resampling of 91 years of historical data. The resampling tests suggest that target-based approaches that seek to establish a safety margin of wealth at the end of the decumulation period appear to be superior to strategies that directly attempt to minimize risk measures such as the probability of portfolio depletion or CVAR. The target-based approaches result in a reasonably close approximation to the retirement spending available in a DB plan. There is a small risk of depleting the retiree funds, but there is also a good chance of accumulating a buffer that can be used to manage unplanned longevity risk or left as a bequest.

Forsyth, P. A., Vetzal, K. R., and Westmacott, G. (2020). "Optimal Asset Allocation for DC Pension Decumulation with a Variable Spending Rule." In: *ASTIN Bulletin* 50(2), pp. 419–447.

We determine the optimal asset allocation to bonds and stocks using an annually recalculated virtual annuity (ARVA) spending rule for DC pension plan decumulation. Our objective function minimizes downside withdrawal variability for a given fixed value of total expected withdrawals. The optimal asset allocation is found using optimal stochastic control methods. We formulate the strategy as a solution to a Hamilton-Jacobi-Bellman (HJB) Partial Integro Differential Equation (PIDE). We impose realistic constraints on the controls (no-shorting, no-leverage, discrete rebalancing) and solve the HJB PIDEs numerically. Compared to a fixed-weight strategy which has the same expected total withdrawals, the optimal strategy has a much smaller average allocation to stocks and tends to de-risk rapidly over time. This conclusion holds in the case of a parametric model based on historical data and also in a bootstrapped market based on the historical data.

Forsyth, P. A., Vetzal, K. R., and Westmacott, G. (2021). "Optimal control of the decumulation of a retirement portfolio with variable spending and dynamic asset allocation." In: *ASTIN Bulletin* 51(3), pp. 905–938.

We extend the Annually Recalculated Virtual Annuity (ARVA) spending rule for retirement savings decumulation (Waring and Siegel (2015) Financial Analysts Journal, 71(1), 91-107) to include a cap and a floor on withdrawals. With a minimum withdrawal constraint, the ARVA strategy runs the risk of depleting the investment portfolio. We determine the dynamic asset allocation strategy which maximizes a weighted combination of expected total withdrawals (EW) and expected shortfall (ES), defined as the average of the worst 5% of the outcomes of real terminal wealth. We compare the performance of our dynamic strategy to simpler alternatives which maintain constant asset allocation weights over time accompanied by either our same modified ARVA spending rule or withdrawals that are constant over time in real terms. Tests are carried out using both a parametric model of historical asset returns as well as bootstrap resampling of historical data. Consistent with previous literature that has used different measures of reward and risk than EW and ES, we find that allowing some variability in withdrawals leads to large improvements in efficiency. However, unlike the prior literature, we also demonstrate that further significant enhancements are possible through incorporating a dynamic asset allocation strategy rather than simply keeping asset allocation weights constant throughout retirement.

Fox, S. (2020). "Linking metrics to objectives: retirement saving, spending, and active management." In: *The Journal of Retirement* 7(3), pp. 6–29.

Although financial advisors and institutions are increasingly urged to adopt objectives or goals-based investment strategies, there is a need to more clearly define what is meant by investment objectives and their effects on outcomes. In a controlled modeling experiment, I formalize two investor personas (Accumulator and Decumulator) and two investment strategies (Target Cash flows and Target Wealth) to provide a deeper understanding of the effects on investor "success" or "failure." In addition, applying characteristics that are potentially available in actively managed funds, such as differences among investments in terms of beta, alpha, and sequence of returns, enables the demonstration of several surprising effects, including changes in the distribution of investment returns and, of importance, the trajectory of investor outcomes in an objectives-based framework.

Frank, L. R. and Brayman, S. (2016). "Combining Stochastic Simulations and Actuarial Withdrawals into One Model." In: *Journal of Financial Planning* 29(11), pp. 44–53.

This paper explored a method that combined actuarial approaches for calculating withdrawals in retirement with Monte Carlo simulations. The model recalculated withdrawals for each scenario within each simulation, with a new simulation beginning for each year based on individual capital remaining and adjusted time horizons using mortality tables. The model approach made a direct connection between the various elements making up spending decisions. There were important differences between a simulation that used decision rules to shape income and altered initial withdrawal assumptions, and a model that recast each subsequent year in isolation. The combination of stochastic simulations and period-adjusting actuarial withdrawal techniques proposed here has been termed Continually Adjusting Stochastic Actuarial Model (CASAM).

Frank, L. R., Mitchell, J. B., and Blanchett, D. M. (2012). "An Age-Based, Three-Dimensional Distribution Model Incorporating Sequence and Longevity Risks." In: *Journal of Financial Planning* 25(3).

The authors develop an age-based, three-dimensional distribution model that illustrates a retiree's yearly transition through retirement, including superannuated years (very old age), to demonstrate the impact of longevity risk

The model builds on "Probability-of-Failure-Based Decision Rules to Manage Sequence Risk in Retirement" by Frank, Mitchell, and Blanchett (2011) to simultaneously:

1) Establish an age-based distribution model
2) Incorporate current-age life expectancy
3) Address survivorship into superannuated ages
4) Address market sequence risk
5) Develop a method to rationally incorporate retiree goals of either consumption or inheritance, and switch between the two as desired

Past research has primarily focused on fixed, non-age-specific distribution periods. The model developed in this paper demonstrates:

- How longevity probability can be developed into dynamic, yearly adjustable distribution periods
- These dynamic distribution periods can then be combined with stochastic (Monte Carlo) real returns
- The distribution period (DP) can be dynamically managed, as the retiree ages, by varying the percentiles of longevity
- The withdrawal rate is very sensitive to DP, and the DP is very sensitive to life expectancy; thus more attention should be paid to longevity risk than has been in past research
- The probability of failure (POF) remains a useful metric to evaluate exposure of a retiree's portfolio to market sequence risk, and a POF-based decision rule applies to both the asset allocation effect and the age effect on withdrawal rates

A working paper, which includes appendices with data that demonstrate how these figures are developed, and additional data, is available at SSRN: `www.ssrn.com/abstract=1849983`.

Franklin, M. B. (2021). *Maximizing Social Security Retirement Benefits*. Tech. rep. InvestmentNews.

This is the guide for consumers and financial advisers about how this critical source of guaranteed income fits into an overall retirement plan.

Topics include:

- The ABC's of claiming Social Security
- How Social Security benefits are taxed
- What survivors need to know to maximize their benefits
- What claiming strategies are disappearing for most new retirees
- How recent changes to Social Security claiming rules affect married couples and divorced spouses

Plus, much more including new rules & strategies to help maximize benefits.

Fraser, S. P. and Payne, B. C. (2018). "Bond Tents: Reshaping the Equity Glide Slope at the End of Wealth Accumulation." In: *The Journal of Wealth Management* 21(2), pp. 27–38.

Asset allocation is widely accepted as a primary driver of portfolio returns over time. Conventional investment practice recommends that investors reduce their risk and thus increase their allocation to fixed-income investments as they approach retirement. This research extends the literature by examining the shape of the fixed-income glide slope prior to retirement. The results suggest that portfolios with high fixed-income allocations, or bond tents, can serve investors well, and nonlinear glide slopes can produce favorable portfolio characteristics for investors.

French, E. and Jones, J. B. (2017). "Health, Health Insurance, and Retirement: A Survey." In: *Annual Review of Economics* 9(1), pp. 383–409.

The degree to which retirement decisions are driven by health is a key concern for both academics and policy makers. In this review, we survey the economic literature on the health-retirement link in developed countries. We describe the mechanisms through which health affects labor supply and discuss how these mechanisms interact with public pensions and public health insurance. The historical evidence suggests that health is not the primary source of variation in retirement across countries and over time. Furthermore, the decline of health with age can only explain a small share of the decline in employment near retirement age. Health considerations nonetheless play an important role, especially in explaining cross-sectional variation in employment and other outcomes within countries. We review the mechanisms through which health affects retirement and discuss recent empirical analyses.

Friedenthal, M. (2020). "The Third Generation of Financial Planning." In: *Advisor Perspectives*.

By introducing additional variables to the simulation process within a financial plan, a more realistic and reliable representation of probable outcomes will be achieved. The third generation of financial planning incorporates dynamic correlations between stocks, bonds, and inflation as well as dynamic volatilities of each data series. It incorporates dynamic risk directive expectations that evolve as clients migrate towards and through their cash-flow series. Layering into the simulation process the customizable, last-survivor probabilities per annum facilitates an encompassing probability of outliving one's money, currently the most robust metric in planning technology.

Fuino, M. and Wagner, J. (2018). "Long-term care models and dependence probability tables by acuity level: New empirical evidence from Switzerland." In: *Insurance: Mathematics and Economics* 81, pp. 51–70.

Abstract Due to the demographic changes and population aging occurring in many countries, the financing of long-term care (LTC) poses a systemic threat. The scarcity of knowledge about the probability of an elderly person needing help with activities of daily living has hindered the development of insurance solutions that complement existing social systems. In this paper, we consider two models: a frailty level model that studies the evolution of a dependent person through mild, moderate and severe dependency states to death and a type of care model that distinguishes between care received at home and care received in an institution. We develop and interpret the expressions for the state- and time-dependent transition probabilities in a semi-Markov framework. Then, we empirically assess these probabilities using a novel longitudinal dataset covering all LTC needs in Switzerland over a 20-year period. As a key result, we are the first to derive dependence probability tables by acuity level, gender and age for the Swiss population. We find that the transition probabilities differ significantly by gender, age and time spent in the frailty level and type of care states.

Fullmer, R. K. (2009). "A Framework for Portfolio Decumulation." In: *Journal of Investment Consulting* 10(1), pp. 63–71.

Before the bear market of 2008-09, the conventional wisdom was that conservative asset allocations might not provide sufficient returns for the long horizons that many retirees face. Therefore, new retirees needed significant allocations to equity. The advice often came with a call to invest more aggressively in the early years of retirement, then taper the equity exposure later in retirement. Arguments for this approach typically present results of Monte Carlo simulation showing that higher equity allocations, particularly early in retirement, can increase the probability of sustaining higher withdrawal rates from the portfolio. But these arguments fall victim to at least two fallacies.

The first fallacy is tragically ironic. Even though the advice is based on probability theory, it overlooks a primary tenet emphasized by Blaise Pascal, the founder of probability theory. Pascal realized that while probability matters, so do consequences. In fact, Pascal emphatically warned against playing the odds without considering the consequences. While it may be true that higher equity exposures can increase the likelihood of sustaining higher withdrawal rates, they also increase the magnitude of failure in unsuccessful cases. Students of investment theory may recognize this oversight as a form of "the fallacy of time-diversification of risk." By making the portfolio more aggressive, the risk of achieving poor returns puts the entire financial plan in peril. The retiree may go broke early, face a lower standard of living, or abandon retirement altogether. So everyone who advises retirees must seriously ponder the following question: Even though higher equity allocations may raise the probability of success for a retirement income plan, does this necessarily mean that higher equity allocations are less risky?

The second fallacy involves an assumption underlying most simulation methodologies. The assumption is that the investment strategy is predetermined at the start of the investment horizon. Examples of predetermined strategies include allocations that remain static over the entire horizon or that use glide paths similar to that of target date funds, in which the equity exposure tapers over time. The only way such an assumption makes sense, however, is if the investor's risk exposure is static over time. But for retirees who depend on nest eggs for living expenses, this assumption is never correct. Holding more-aggressive portfolios early in retirement and less-aggressive allocations later in retirement leads to the retiree taking the most investment risk precisely when it is most risky to do so - when the risk of outliving the portfolio (longevity risk) is greatest.

Fullmer, R. K. (2012). "Modern Portfolio Decumulation A New Strategy for Managing Retirement Income." In: *Someday Rich: Planning for Sustainable Tomorrows Today*. Wiley, pp. 249–271.

This chapter presents Richard K. Fullmer's paper, titled "Modern Portfolio Decumulation." It is a logical extension of his mis-measurement of risk work. It describes a new framework for efficient portfolio construction in the decumulation phase of the investment lifecycle, in which an investor who has accumulated assets over time wishes to use those assets to fund ongoing living expenses. It combines elements of both investment theory and actuarial science, introducing an effective way to manage longevity risk in the portfolio. The chapter discusses that the conventional standard for accumulation of wealth is not optimal for decumulation. It outlines an alternative approach for investing in the decumulation phase, bringing focus upon dynamic asset allocation models.

Fullmer, R. K. and Tzitzouris, J. A. (2014). "Evaluation of Target-Date Glide Paths withinDefined Contribution Plans." In: *The Journal of Retirement* 1(4), pp. 75–94.

In this article, the authors present a methodology to support sponsors of defined-contribution (DC) plans when evaluating the allocation to equity over a target-date fund's glide path. When evaluating glide paths, sponsors must pay heed to two primary goals: The first is to generate lifetime income consistently over the course of retirement; the second is to limit the risk of capital loss near and during retirement, which is particularly important for participants who withdraw balances over short horizons. The challenge of glide path selection is striking a compromise between these competing goals. This compromise cannot be achieved via objective analysis alone and must also be informed by the subjective horizon and risk preferences of the sponsor acting as agent for the plan participants. The authors find that higher-equity glide paths offer greater efficacy for lifetime income replacement, while lower-equity glide paths offer greater efficacy for stable account balances with lower risk of capital losses. These conclusions hold over a wide range of plan characteristics and assumptions, although the relative magnitude of the trade-off depends on characteristics unique to each plan.

Fulton, C. and Hubrich, K. (2021). "Forecasting US Inflation in Real Time." In: *Econometrics* 9(4) (36).

We analyze real-time forecasts of US inflation over 1999Q3-2019Q4 and subsamples, investigating whether and how forecast accuracy and robustness can be improved with additional information such as expert judgment, additional macroeconomic variables, and forecast combination. The forecasts include those from the Federal Reserve Board's Tealbook, the Survey of Professional Forecasters, dynamic models, and combinations thereof. While simple models remain hard to beat, additional information does improve forecasts, especially after 2009. Notably, forecast combination improves forecast accuracy over simpler models and robustifies against bad forecasts; aggregating forecasts of inflation's components can improve performance compared to forecasting the aggregate directly; and judgmental forecasts, which may incorporate larger and more timely datasets in conjunction with model-based forecasts, improve forecasts at short horizons.

Fusato, D. (2022). "Private Risk Pooling: A Fresh Approach to Longevity Risk." In: *Investments and Wealth Monitor* (2022), November/December.

New products that work in innovative ways to pool longevity risk are now becoming available to help investors deal with some of the uncertainties of living longer in retirement.

Gabudean, R., Gomes, F., Michaelides, A., and Zhang, Y. (2021). "On Optimal Allocations of Target-Date Funds." In: *The Journal of Retirement* 9(2), pp. 58–79.

We study optimal life-cycle portfolio allocation and its application to target-date fund (TDF) design. We show that optimal TDF allocation must be explicitly linked to a savings rate; for example, a higher savings rate generating higher financial wealth accumulation should necessarily come with a more conservative TDF. Further, we quantify the extent to which accumulated wealth, the investing environment, and participant characteristics affect TDF allocations and compare the resulting optimal allocations against the observed universe of US TDF products. Plan sponsors and asset managers can use these findings to improve TDF selection and design.

Gabudean, R. C. and Weiss, R. A. (2019). "How to Evaluate Target-Date Funds: A Practical Guide." In: *The Journal of Retirement* 6(4), pp. 68–81.

The authors describe and showcase a framework to analyze life-cycle, or target-date, investment strategies with intuitive, practical metrics. Anchored in the complex theory of life-cycle investing, we propose a comprehensive collection, or , of measures to capture various aspects of the problem: wealth-focused versions of return and risk concepts; the link between retirement contributions and portfolio returns; the strategy ability to support income in retirement; and behavior around retirement, a period when risk to invested wealth is greatest. These metrics are synthesized into one holistic view based on investor-specific preferences, highlighting the connection between investor characteristics and the life-cycle investment problem.

Gan, G. (2013). "Application of data clustering and machine learning in variable annuity valuation." In: *Insurance: Mathematics and Economics* 53(3), pp. 795–801.

We study the pricing of a large portfolio of VA policies. A clustering method is used to select representative policies. A machine learning method is used to estimate the guarantee value. The proposed method performs well in terms of accuracy and speed. The valuation of variable annuity guarantees has been studied extensively in the past four decades. However, almost all the studies focus on the valuation of guarantees embedded in a single variable annuity contract. How to efficiently price the guarantees for a large portfolio of variable annuity contracts has not received enough attention. This paper fills the gap by introducing a novel method based on data clustering and machine learning to price the guarantees for a large portfolio of variable annuity contracts. Our test results show that this method performs very well in terms of accuracy and speed.

Gao, X. and Sun, L. (2021). "Modeling retirees' investment behaviors in the presence of health expenditure risk and financial crisis risk." In: *Economic Modelling* 94, pp. 442–454.

This paper examines the impact of both health risks and financial market turmoil on individual investment decisions during retirement. Using a panel data set compiled from the U.S. Health and Retirement Survey, we find that health shocks increase retirees' disposable income uncertainty and compromise their risk-taking capacity. This is especially the case for those with moderate levels of wealth. However, we find that health risk alone cannot explain the empirically observed lower level of risky asset investment behaviors. It is the presence of extreme events in the financial markets, in combination with health risks, that significantly reduce retirees' incentives to invest in risky assets. Compared with younger investors, retirees' shorter investment horizon and higher health risks render them more vulnerable to the detrimental consequences of wealth evaporation caused by financial crises. Our results indicate that effective portfolio management for retirees must take into account both their health risks and the occurrence of financial market tail risks.

Gard, R. and Gremm, M. (2018). "Two Measures of Financial Risk Tolerance from Questionnaire Data." In: *SSRN e-Print*.

Set-based principal component analysis extracts two independent measures of financial risk tolerance from questionnaire responses. These measures are less noisy than most scoring methods. They are also more robust because they provide a redundant assessment of financial risk tolerance. This analysis reveals that typical risk questionnaires measure financial risk tolerance and income-related risk tolerance. These two factors are unrelated, which suggests that income-related questions should be dropped from financial risk questionnaires. We also examine how financial risk tolerance depends on financial literacy and gender.

Garivaltis, A. (2019). "Exact replication of the best rebalancing rule in hindsight." In: *Journal of Derivatives* 26(4), pp. 35–53.

This article prices and replicates the financial derivative whose payoff at T is the wealth that would have accrued to a 1USD deposit into the best continuously-rebalanced portfolio (or fixed-fraction betting scheme) determined in hindsight.

Ge, W. (2017). "Stress-Testing Volatility Risk Premium Harvesting Strategies Based on S&P 500 Index Options." In: *The Journal of Index Investing* 8(1), pp. 37–46.

The volatility risk premium (VRP), commonly accessed by writing equity index options, can help investors enhance portfolio returns. Some investors, however, have an aversion to options selling, especially put options, because they fear significant losses in a financial crisis. This article examines this aversion by analyzing the performance of two VRP-harvesting strategies based on writing out-of-the-money SandP 500 Index options (the dedicated VRP and the overlay VRP constructs) in market distress and comparing them with the SandP 500

Index. All series are stress-tested with five historical crisis episodes in the last three decades and three additional extreme scenarios modeled after the worst stock market crashes in the past century. The analysis reveals that the VRP strategies can achieve their objective of outperforming the market and mitigating investors' losses during financial distress. The performance of the VRP constructs in a crisis depends on two main factors: the beta of the VRP strategy and the speed of the market crash. This article concludes that investors' aversion to option selling may be unjustified and they may benefit significantly from the VRP at all times, especially in today's market, when expected returns from traditional assets are subdued.

Ge, W. (2018). "Alternatives to alternative assets: assessing S&P 500 index option strategies as hedge fund replacements." In: *The Journal of Alternative Investments* 20(4), pp. 69–80.
Hedge funds should deliver good risk-adjusted returns and provide diversification for investors portfolios. However, the performance of hedge funds has deteriorated in recent years and many investors may not find investment in hedge funds suitable, due to the high costs, illiquidity, and lack of transparency. The focus of this study is to examine whether equity index option-based strategies may deliver the same coveted ideal hedge fund return characteristics, that is, good risk-adjusted returns, mitigated risk measures, and added diversification. A total of seven S&P 500 Index-based generic option-writing strategies, named CP100 to CP0, are analyzed and compared with a comprehensive set of fourteen Credit Suisse hedge fund indexes using volatility or beta as matching metrics. The three series of CP100, CP80, and CP50, can be viewed as a solid suite of liquid alternatives to hedge funds. They match the fourteen Credit Suisse hedge fund indexes on beta levels, but deliver better risk-adjusted returns. Their annual returns and risk measures are better than or on par with the hedge fund indexes, except for the Credit Suisse Global Macro Index, which may be matched by a levered version of the CP100 option strategy. This suite of three option strategies are transparent, liquid, inexpensive, and easy to implement. They can achieve the same goals of hedge funds: attractive risk-adjusted returns, subdued risks, and added diversification. They may play multiple and important roles in institutional investors portfolios.

Ge, W. (2019a). "Optimal Glide Path Selection for Low-Volatility Assets." In: *The Journal of Index Investing* 10(3), pp. 70–84.
Target-date funds (TDFs) have become a popular choice for retirement plans, and the concept of a glide path is essential for TDFs. One of the key assumption behind glide paths is that the main component of a retirement plan should be growth assets invested in the overall equity market example, the S&P 500 Index. Researchers have recently challenged this assumption and argued for using smart beta factors for retirement purposes. Among them, the factor of low volatility may be uniquely suitable for retirement investing. This article studies the use of low-volatility assets for the purpose of retirement planning and the choice of ideal glide paths. This study is agnostic about the means by which the low-volatility risk/return profile is achieved and analyzes four series with diminished volatility constructed with different methodologies. The article concludes that low-volatility assets may indeed help retirement investors achieve their objectives, though such investors must choose ideal glide paths carefully to suit the selected low-volatility series. When equipped with proper low-volatility assets and carefully chosen glide paths, retirement plan managers may both improve the odds that their plans succeed and increase the expected final wealth levels.

Ge, W. (2019b). "Using the volatility risk premium to mitigate the next financial crisis." In: *The Journal of Wealth Management* 22(3), pp. 85–97.
Ten years have passed since the trough of the Global Financial Crisis, and the US equity market has experienced one of the longest stretches of ascent in history. Some investors have started questioning whether the US stock market is overvalued and if a recession is on the way. It is usually impossible to predict the next financial crisis. An investor best course of action may be to adjust the investment portfolio to be resilient against potential market headwinds. This paper argues for utilizing the Volatility Risk Premium (VRP), specifically option-selling VRP strategies, to mitigate the losses the portfolio may suffer from a future financial crisis. Such a VRP strategy, if implemented with out-of-the-money equity index options, can help investors cushion the losses from an equity market crash and recover more quickly than the broad equity market. Investors can utilize the VRP by itself or combine it with traditional equity to construct the most suitable investment strategies. This paper further examines implementation choices of such strategies and stress test their performance with four representative crises from the past three decades.

Ge, W. (2019c). "Utilizing Low-Volatility Assets To Mitigate Sequence Risk In Retirement Investing." In: *The Journal of Investing* 28(5), pp. 85–100.
Sequence risk is a subtle risk factor retirement investment managers face that can hinder their plans for success. This article illustrates the damaging effects that sequence risk can have on retirement plans and how asset

volatility can exacerbate this risk. The study analyzes an asset-centric approach that utilizes low-volatility assets to mitigate sequence risk, as their diminished volatility implies more certainty for retirement plans. It further examines the return-risk profiles of four candidate low-volatility series, compares their performance either as stand-alone series or in the context of 60/40 portfolios with the traditional equity represented by the S&P 500 Index, and finally employs two sets of Monte Carlo simulations to assess the effectiveness of such assets for the task of sequence risk mitigation. The study demonstrates that 60/40 portfolios with low-volatility components can significantly reduce the uncertainty of retirement investment outcomes in terms of both narrower final wealth ranges and reduced failure rates.

Geisler, G., Harden, B., and Hulse, D. (2021). "A Comparison of the Tax Efficiency of Decumulation Strategies." In: *Journal of Financial Planning* 34(3), pp. 72–89.

This article examines several decumulation strategies for a client approaching retirement with a mix of tax-favored retirement accounts and taxable accounts, each holding appreciated stocks and taxable bonds.

The analysis applies these strategies to three "nest egg" scenarios to determine their resulting portfolio lives, while examining tax consequences throughout those lives.

In these scenarios, the more tax-efficient decumulation strategies last from 1 to over 11 percent longer than both the Conventional Wisdom and Schwab's recently marketed decumulation strategies.

The results provide insights that financial planning professionals can use to tailor tax-efficient decumulation recommendations that better fit a client's particular situation, increasing how long their wealth lasts during retirement.

Geisler, G. and Hulse, D. (2016). "The Taxation of Social Security Benefits and Planning Implications." In: *Journal of Financial Planning* 29(5), pp. 52–63.

Up to 85 percent of Social Security benefits could be taxable, with the percentage increasing as income increases. Additional income can cause additional Social Security benefits to be taxable, a so-called "tax torpedo." The tax effect of this tax torpedo depends on the tax bracket in which the benefits are taxed. This effect can be as high as 21.25 percent of the additional income, but it is less than 10 percent in many circumstances. This tax effect is an important factor to consider when deciding whether to start Social Security benefits at age 62 or age 70.

Gelmini, M. and Uberti, P. (2024). "The equally weighted portfolio still remains a challenging benchmark." In: *International Economics* 179, p. 100525.

This research replicates the paper "Optimal Versus Naive Diversification: How Inefficient is the 1/N Portfolio Strategy?", DeMiguel et al. (2009b). Similar to the referring paper, working in the mean-variance context, we compare the out-of-sample performance of the same investment strategies on the basis of standard metrics (Sharpe ratio, certainty equivalent and turnover). We consider proportional transaction costs and estimation rolling windows of limited length. Our study updates the original paper for many interesting aspects. First, to exclude that the empirical evidence of DeMiguel et al. (2009b), whose data stopped in 2004, could depend on very specific market behavior, we use an updated version of the original databases that contains the returns of the last 20 years. Recent data are characterized by a few severe systemic events, the 2008 global financial crisis and the shock related to the pandemic, and a generally higher level of price volatility than the previous periods. In our opinion, this variation in the market's conditions makes the replication very interesting. Second, we introduce the Equally Risk Contribution (ERC) portfolio within the allocation strategies under comparison. This allocation rule is strictly related to the mean-variance approach when the variance is used as the referring risk measure and it constitutes a very interesting alternative investment benchmark. Moreover, using real data, we study whether a variation of the holding period or the length of the estimation window can modify the performance of all the strategies under comparison. Our findings confirm the results of DeMiguel et al. (2009b), i.e. that the equally weighted portfolio still remains a challenging benchmark to beat. Nevertheless, we find a few significant differences: the number of strategies that outperform naive diversification is larger due to the increased market volatility; limiting the impact of transaction costs by investing in a portfolio with a stable allocation as the ERC, or modifying the lengths of the estimation window and the holding period, is not sufficient to beat naive diversification systematically.

Gemmo, I., Rogalla, R., and Weinert, J.-H. (2020). "Optimal portfolio choice with tontines under systematic longevity risk." In: *Annals of Actuarial Science* 14(2), pp. 302–315.

We derive optimal portfolio choice patterns in retirement (ages 66-105) for a constant relative risk aversion utility maximising investor facing risky capital market returns, stochastic mortality risk, and income-reducing health shocks. Beyond the usual stocks and bonds, the individual can invest his assets in tontines. Tontines are cost-efficient financial contracts providing age-increasing, but volatile cash flows, generated through the pooling of mortality without guarantees, which can help to match increasing financing needs at old ages. We find that a tontine invested in the risk-free asset dominates stock investments for older investors without a bequest motive. However, with a bequest motive, it is optimal to replace the tontine investment over time with traditional financial assets. Our results indicate that early in retirement, a tontine is only an attractive investment option, if the tontine funds are invested in a risky asset. In this case, they crowd out stocks and risk-free bonds in the optimal portfolios of younger investors. Over time, the average optimal portfolio weight of tontines decreases. Introducing systematic mortality risks noticeably reduces the peak allocation to tontines.

Gerrard, R., Kyriakou, I., Nielsen, J. P., and Vodička, P. (2023). "On optimal constrained investment strategies for long-term savers in stochastic environments and probability hedging." In: *European Journal of Operational Research* 307(2), pp. 948–962.

In this paper, we derive constrained optimal investment strategies for long-term savers who are interested in investing their funds in stocks, but are afraid of potentially losing, for example, their retirement income. We call this probability hedging as it is determined by the probability of landing up within bounds that are agreed from interaction with the investor. We show that our strategies can be derived under different utility functions and multifactor model assumptions. We prove that the probability measure varies with the utility function choice and that the logarithmic utility, in particular, results in an intuitive probability hedge under the physical measure. This makes it easier to communicate, without putting at risk the financial advice conducted by potentially misrepresenting the realism of the theoretical results. Our strategy is also shown to yield a better distribution of the terminal wealth than traditional hedging approaches.

Ghilarducci, T., Papadopoulos, M., and Webb, A. (2022). "The Illusory Benefit of Working Longer on Retirement Financial Preparedness: Rethinking Advice That Working Longer Increases Retirement Income." In: *The Journal of Retirement* 9(4).

This article demonstrates that although working longer can, in theory, substantially improve financial preparedness for retirement, it yields much smaller improvements in practice. Multivariate analysis using data from the Health and Retirement Study reveals that working longer improves financial preparedness only for workers who also delay claiming Social Security. But most older workers combine work with Social Security benefits, do not increase their financial wealth, and miss out on the delayed retirement credit. For many, early claiming is a rational choice because their low and often part-time earnings fall short of projected post-retirement income.

Ghilarducci, T., Radpour, S., and Webb, A. (2019). "New evidence on the effect of economic shocks on retirement plan withdrawals." In: *The Journal of Retirement* 6(4), pp. 7–19.

Using data from the Survey for Income and Program Participation (SIPP), this study investigates the relationship between withdrawals from 401(k) and IRA accounts and household level economic shocks such as job loss, job change, divorce, and the onset of poor health. Workers in low-wage households are more likely to withdraw from their accounts than those in middle and high income households, in part because they are more likely to withdraw when they experience a shock and also experience more shocks. Shocks are associated with about 20% of all retirement account withdrawals and exacerbate pre-existing inequalities in financial preparation for retirement.

Glickman, M. M. and Hermes, S. (2015). "Why Retirees Claim Social Security at 62 and How It Affects Their Retirement Income: Evidence from the Health and Retirement Study." In: *The Journal of Retirement* 2(3), pp. 25–39.

Despite higher monthly benefits for those who delay, many people still claim Social Security retirement benefits at age 62, the earliest age of eligibility. This study uses data from the Health and Retirement Study to examine the demographic and occupational characteristics associated with early claiming, as well as the retirement income of early claimers compared with those who delay. The authors find that several work-related factors may cause people to claim Social Security benefits early and suggest they may face challenges in continuing to work at older ages. For example, those who worked in physically demanding blue-collar jobs were 55% more likely to claim benefits prior to their full retirement age, after controlling for other factors, compared with those in all other occupations. Other factors, such as having lower expectations of living to age 75 significantly increase the likelihood of claiming early. The median income for those who claim at full retirement age or later was 45% higher after claiming benefits than for those who claimed early and 33% higher at age 72. Even when comparing early and delayed claimers with similar total income after claiming, average household income for delayed claimers was higher at age 72 than for early claimers.

Godin, F., Hamel, E., Gaillardetz, P., and Hon-Man Ng, E. (2023). "Risk allocation through shapley decompositions, with applications to variable annuities." In: *ASTIN Bulletin* 53(2), pp. 311–331.

This paper introduces a flexible risk decomposition method for life insurance contracts embedding several risk factors. Hedging can be naturally embedded in the framework. Although the method is applied to variable annuities in this work, it is also applicable in general to other insurance or financial contracts. The approach relies on applying an allocation principle to components of a Shapley decomposition of the gain and loss. The implementation of the allocation method requires the use of a stochastic on stochastic algorithm involving nested simulations. Numerical examples studying the relative impact of equity, interest rate and mortality risk for guaranteed minimal maturity benefit (GMMB) policies conclude our analysis.

Goldberg, L. R., Cai, T., and Hand, P. (2021). "Tax-Rate Arbitrage: Realization of Long-Term Gains to Enable Short-Term Loss Harvesting." In: *SSRN e-Print*.

We look at an enhanced loss-harvesting strategy, tax-rate arbitrage, which exploits the differential between short- and long-term tax rates. Our study relies on ATBAT, an After-Tax Back-Testing Analysis Tool that lets us examine tax-managed strategies over numerous historical periods. For the ideal tax-rate arbitrage investor, one who is subject to federal only tax rates, who has a long horizon and a planned liquidation date, and who launches the strategy from all cash, tax-rate arbitrage generated an average of 0.78% in excess after-tax active return at a 10-year horizon relative to a standard loss-harvesting strategy. Other investors with different profiles may benefit from tax-rate arbitrage but typically to a lesser extent.

Goldman Sachs Asset Management Research (2021). *Retirement Realities: Time For Change: Retirement Survey & Insights Report 2021*. Tech. rep. Goldman Sachs Asset Management.

Goldman Sachs Asset Management released the findings of its Retirement Survey & Insights Report, which seeks to learn directly from plan participants about their experience preparing for, transitioning to, and managing their finances in retirement.

The survey revealed that financial hardships have affected retirement savings for 89% of workers, compared to 17% of retirees. More than half of the retirees polled retired earlier than expected, and health was the top concern among retirees. The youngest respondents are expecting to leave the workforce much earlier, however, with 25% of Generation Z respondents planning to retire before the age of 55.

Overall, survey responses highlight the importance of employer retirement programs for both retired and currently working individuals, with 56% of retired respondents (and 38% of working respondents) surveyed stating that they derive their source of retirement education from employer retirement programs.

Responses were analyzed from two distinct populations: those currently in retirement and those currently working, with the latter cohort spanning a range of age groups, including Generation X, Millennials and Generation Z individuals, in geographies across the US.

Notable highlights from the report include:

1) More than half of retired respondents retired earlier than expected and health was the top reason listed for retired individuals. Health reasons topped the list of reasons for retirement, with 24% of respondents forced into retirement due to health concerns, and with only 3% of respondents having worked longer than expected. Notably, only 9% of participants indicated that having sufficient retirement savings is a key trigger for retirement.

2) Interrupted savings may be causing shortfalls in retirement plans. Eighty-nine percent of working individuals say financial hardships impacted their retirement savings, compared to 17% of retired individuals. Eighty-three percent of workers say paying down existing loans impacted their retirement savings, compared to 15% of retirees.

3) Younger workers may need to plan with more realistic assumptions. Twenty-five percent of Generation Z respondents plan to retire before age 55, while roughly 30% of those under age 40 believe they need 60% or less of their pre-retirement income in retirement, compared to Generation X respondents, who feel they need 80% of their pre-retirement income in retirement.

4) Employer-sponsored retirement programs are the most used source of retirement education. Survey responses highlight the importance of employer retirement programs for both retired and currently working individuals, with 56% of retired respondents surveyed stating that they derive their source of retirement education from employer retirement programs. Additionally, offering new programs or continuing to expand on existing ones is an important value driver for any organization.

The survey was conducted between July and August 2021 in the United States, with responses from 1,237 participants (613 working Generation X, Millennials and Generation Z individuals and 624 retired individuals (aged 60-75)).

Goldstein, B. (2018). "The glide path less traveled: A critical examination of assumptions, outcomes, and glide path specification." In: *SSRN e-Print*.

This paper defines key assumptions, derived from empirical data, to deliver highly accurate and representative simulations of participant outcomes. The paper constructs an alternative glide path design that we believe more efficiently balances major retirement savings risks - longevity/shortfall, return sequence, and inflation - than the industry average glide path. The paper demonstrates that an alternative glide path - one that begins with higher equity allocations than the industry average, follows a steeper slope of asset shifts, and arrives at a risk-parity solution at the point of retirement– has less variability in outcomes than the industry average.

Golts, M. and Jones, G. C. (2023). "4×4 Goal Parity." In: *SSRN e-Print*.

In the 4x4 Asset Allocation framework all assets and liabilities in any portfolio should be thought of as means contributing to the following four ends: 1) Liquidity maintenance: nominally safe and quickly accessible "cash-like" pool of assets, 2) Income generation: relatively regular, certain and near-term cash payments, 3) Preservation of (real) capital: assets expected to retain their value over time, 4) Growth: more volatile assets and strategies expected to generate long-run returns. We illustrate the investor-specific quantification of these four goals with decades of empirical data for a wide variety of liquid and private assets, fund indices, and alternative premia strategies. Further, using 1970-2022 data, we simulate strategic Goal Parity Balanced portfolios seeking to have all the four goals equally "powered," as well as strategic Goal Tilted portfolios emphasizing one of the goals. Over the 50+ years of our simulation, a Goal Parity portfolio achieves highly competitive risk-adjusted returns relative to traditional benchmarks, in particular delivering shallower drawdowns in various crisis and inflationary periods. Our Growth-tilted portfolio diversifies and outperforms equities with shallower drawdowns, and our Preservation- tilted portfolio has a very distinct performance pattern: it lags equities during bull markets but delivers robust performance in bear markets. Overall, we have empirically demonstrated that the 4x4 approach provides a powerful goal-based, investor-specific asset management framework.

Golts, M. and Jones, G. C. (2025). "Goal Parity." In: *The Journal of Investing* 34(2), pp. 14–42.

Following "4x4" Asset Allocation" by Cron and Golts (2022), all assets and liabilities in any portfolio should be thought of as means contributing to the following four ends: 1) Liquidity maintenance: nominally safe and quickly accessible "cash-like" pool of assets 2) Income generation: relatively regular, certain and near-term cash payments 3) Preservation of (real) capital: assets expected to retain their value over time 4) Growth: more volatile assets and strategies expected to generate long-run returns. We illustrate the investor-specific quantification of these four goals with decades of empirical data for a wide variety of liquid and private assets, fund indexes, and alternative premium strategies. Further, using 1970-2022 data, we simulate strategic Goal Parity Balanced portfolios, seeking to have all four goals equally "powered," as well as strategic Goal Tilted portfolios emphasizing one of the goals. Over the 50+ years of our simulation, Goal Parity portfolios achieve highly competitive risk-adjusted returns relative to traditional benchmarks, in particular delivering shallower drawdowns in various crisis and inflationary periods. Our Growth-tilted portfolio diversifies and outperforms equities with shallower drawdowns, and our Preservation-tilted portfolio has a very distinct performance pattern: it lags equities during bull markets but delivers robust performance in bear markets. Overall, we empirically illustrate a powerful goal-based, investor-specific asset management framework.

Gongola, J. and Sharon, A. (2020). "The Well-Tempered Retiree: Fine-Tuning Investor Behaviors in Decumulation." In: *Investments and Wealth Monitor*.

We conducted a study to better under- stand the interplay between investor psychology and retirement-oriented decisions and goals. It was designed by PIMCO and conducted online within the United States by The Harris Poll in December 2020, with 758 U.S. adults 55 years and older, including 255 with $500,000 to $999,999 in liquid net wealth and 503 with more than $1 million in liquid net wealth. We asked investors about important retirement decisions, such as timing Social Security benefit claims and their approach to withdraw- ing from savings and investments, and we took measures of their individual human differences, including their sense of their own health and longevity, finan- cial literacy, risk priorities, and desire for bequest or legacy. Then we explored and mapped how these factors inter- mix, probing the psychological mechanisms that drive these relationships.

Goodman, B. and Richardson, D. P. (2019). "Achieving Retirement Income Security: A Comparison of Guaranteed Lifetime Withdrawal Benefit, Systematic Withdrawal and Partial Variable Annuity Strategies." In: *SSRN e-Print*.

Many retirement income products attempt to satisfy multiple and sometimes conflicting objectives because retirees desire products that provide retirement security, inflation protection, liquidity, asset growth and the potential for an estate. In this paper, we used historical data over the past 90 years to conduct simulations and analyze how a Guaranteed Lifetime Withdrawal Benefit (GLWB), systematic withdrawal, and partial Variable Immediate Annuity (VIA) strategies performed in meeting these multiple objectives. With the exception of retirement income starting dates at the beginning of the Great Depression, all three strategies performed well in providing income throughout retired life. The partial VIA and GLWB strategies provided better "peace of mind" retirement income products, while the systematic withdrawal strategy offered the greatest flexibility in managing retirement assets. Overall, we conclude that a partial VIA income strategy comprised of a VIA and supplemental liquid asset account would have provided the best mixture of income generation, risk management, and estate potential for the majority of cohorts.

Goodman, L. and Meiselman, B. S. (2024). "Retirement Tax Shields: A Cohort Study of Traditional and Roth Accounts." In: *SSRN e-Print*.

Previous research has shown the theoretical circumstances under which Roth accounts outperform traditional accounts. In this paper, we inform the traditional-or-Roth choice for current workers by examining the retrospective tax benefits of each account type for the 2003 cohort of retirees. We use the actual history of marginal tax rates at the individual level to estimate that the average tax shields of traditional and Roth retirement accounts—the excess value of retirement consumption financed by those accounts relative to brokerage accounts—were 73% and 46% respectively. Traditional accounts were better for this cohort largely due to tax cuts in 2001 and 2003. Under a counterfactual with inflation-adjusted 2019 law, the average tax shields for traditional and Roth accounts would have been much closer (37% and 29% respectively), and around one third of savers could have consumed more during retirement by saving in Roth accounts rather than in traditional accounts. This is the first paper to apply the history of marginal tax rates—informed by administrative income records—to compare Roth and traditional accounts for a particular cohort of retirees.

Goossens, J. T., Knoef, M., and Ponds, E. (2022). "Can Estimated Risk and Time Preferences Explain Real-Life Financial Choices?" In: *SSRN e-Print*.

We combine experimentally elicited preferences with administrative micro data to study actual financial decision-making. Firstly, we simultaneously elicit and estimate risk and time preferences in a real-life context, with horizons up to 10 years, for more than 1000 pension fund participants. We estimate a present-bias factor of 0.88, an annual discount rate of 3.91% and a CRRA utility curvature of 0.97. Secondly, using an expected utility framework, we show that the individually estimated preferences explain actual retirement decisions up to 82% of our sample for reasonable indifference intervals. Freedom of choice by means of a lumpsum-like annuity creates annual potential welfare gains up to 4.41%, but realized welfare gains are lower or even negative.

Goulding, C. L. and Harvey, C. R. (2025). "Investment Base Pairs." In: *SSRN e-Print*.

Modern finance remains constrained by legacy portfolio construction techniques. While the conventional quantile approaches (e.g., long top 30%/short bottom 30%) and linear weighting schemes dominate cross-sectional strategies, we show that these methods discard crucial cross-asset information. We offer a new approach by decomposing common signals such as value, momentum, and carry, into "base pair" portfolios. Each signal-driven long-short position is shaped by five key drivers: own-asset predictability, cross-asset predictability, signal correlation, and signal mean and variance imbalances. Using 1,710 futures pair portfolios spanning equities, bonds, currencies, and commodities formed from common signal types, we show that targeting top pairs can triple average returns at fixed leverage over 20 years: the aggregate "All" portfolio rises from 3.4% to 10.4% annualized. Equity Value climbs from 3.6% to 14.3% and Currency Momentum reverses a -3.0% loss to a 10.3% gain. By harvesting cross-asset information and eliminating "junk" pairs, this approach offers a robust improvement over the status quo across diverse asset classes and time periods.

Govorun, M., Latouche, G., and Loisel, S. (2015). "Phase-type aging modeling for health dependent costs." In: *Insurance: Mathematics and Economics* 62, pp. 173–183.

In the present paper we develop recursive algorithms to evaluate the distribution of the net present value (abbreviated as "NPV") of a health care contract. The duration of the program is a random variable representing the lifetime of an individual. We suggest a discrete time phase-type approach to model individual health care costs. In this approach, annual health care costs depend naturally on the health state of the individual. We also derive the distribution of the NPV assuming that annual health care costs are iid random variables. We demonstrate analytically that, under special parametrization, the model with iid costs gives a similar expectation of the NPV to the one of the model with health dependent costs. We propose techniques to evaluate the impact of health related events and demonstrate it on numerical examples. Based on Canadian government data on health expenditures, we study the impact on the NPV of the health cost structure by age.

Grable, J. E., Lyons, A. C., and Heo, W. (2019). "A test of traditional and psychometric relative risk tolerance measures on household financial risk taking." In: *Finance Research Letters* 30, pp. 8–13.
This article provides a comparative test of two ways in which individual relative risk tolerance is commonly measured in research and practice. Results suggest that the popular method of measuring gambles across lifetime income provides less insight into household investment behavior than the psychometric approach that is often used to derive a risk profile measure.

Grealish, A. and Kolm, P. N. (2021). "Robo-Advisors Today and Tomorrow: Investment Advice Is Just an App Away." In: *The Journal of Wealth Management* 24(3), pp. 144–155.
Over the past decade robo-advisors have gone from offerings from a handful of start-ups to an established and fast-growing segment of the wealth management industry. Robo-advisors use technology to translate core retail investing principles-establishing an investment plan, seeking broad diversification, weighting cost and value, and accounting for taxes-into automated platforms with easy-to-use interfaces. Intuitive user experience backed by well-established investment processes allow individual investors to create and execute investment strategies with little or no interaction with a financial professional, resulting in large economies of scale for robo-advisors and low costs for clients. In this article, we discuss how robo-advisors assess client risk tolerance, build, and recommend portfolios, manage risk, and optimize taxes. We discuss the implementation of goal-based investing, socially responsible (ESG) investing, and smart beta strategies on robo-advisory platforms. Additionally, we examine robo-advisor performance during the market downturn in March 2020, the first significant market drawdown since their introduction, and find portfolio performance in agreement with broadly diversified stock and bond holdings. Consistent with expectations, robo-advisors also reported increased tax-loss harvesting activity during this downturn.

Grealish, A. and Kolm, P. N. (2023). "Robo-advisory: From investing principles and algorithms to future developments." In: *Machine Learning and Data Sciences for Financial Markets*. Cambridge University Press, pp. 60–85.
Advances in financial technology have led to the development of easy-to-use online platforms referred to as robo-advisors or digital-advisors, offering automated investment and portfolio management services to retail investors. By leveraging algorithms embodying well-established investment principles and the availability of exchange traded funds (ETFs) and liquid securities in different asset classes, robo-advisors automatically manage client portfolios that deliver similar or better investment performance at a lower cost as compared to traditional financial retail services. In this chapter we explore how robo-advisors translate core investing principles and best practices into algorithms. We discuss client onboarding and algorithmic approaches to client risk assessment and financial planning. We review portfolio strategies available on robo-advisor platforms and algorithmic implementations of ongoing portfolio management and risk monitoring. Since robo-advisors serve individual retail investors, tax management is a focal point on most platforms. We devote substantial attention to automated implementations of a number of tax optimization strategies, including tax-loss harvesting and asset location. Finally, we explore future developments in the robo-advisory space related to goal-based investing, portfolio personalization, and cash management.

Greiner, S. P. and Stoyanov, S. V. (2019). "Portfolio scoring by expected risk premium." In: *The Journal of Portfolio Management* 45(4), pp. 83–90.
In this article, the authors discuss a general method for ranking portfolios that places few limitations on the portfolio constituents other than using publicly traded assets. The ranking scores reflect the expected reward investors would require for accepting the risks of the portfolio in the context of an asset pricing framework. The scores are computed through a factor model that acknowledges the factor return correlations. The authors illustrate the approach with a large universe of exchange-traded funds assuming a linear model with Fama-French-Carhart factors wherein factor premiums (i.e., expected returns) are proportional to factor volatilities. The empirical analysis implies that the most significant factors from the Fama-French-Carhart factor set driving the premiums are the market and the momentum factors.

Grennon, T. (2015). "A Dynamic Asset Allocation Approach for Selecting a 401K QDIA." In: *SSRN e-Print*.
The selection of the asset allocation and target date mutual funds that are used as qualified default investment alternative (QDIA) options for 401k participants are driven today by strategic asset allocation, risk and target retirement start dates. This paper details a dynamic asset allocation approach to be used for selecting a QDIA which uses forward looking return assumptions and the Sharpe Ratio for selection.

Grennon, T. (2016). "A Dynamic Approach to Retirement Income." In: *SSRN e-Print*.

We are all well aware of the pitfalls of a tradition systematic withdrawal approach to retirement income especially in light of what we know about sequence return risk. What if instead of simply assuming we earn 10% on stocks, 5% on bonds, and 2.5% on cash equivalents and withdrawing 4% we utilize forward looking return assumptions based on fundamentals (stocks) and yields (bonds, cash)? The following paragraphs explore the idea and test the concept of a safe relative withdrawal rate.

Guan, G., Liang, Z., and Ma, X. (2024). "Optimal annuitization and asset allocation under linear habit formation." In: *Insurance: Mathematics and Economics* 114, pp. 176–191.

This paper studies the optimal consumption-investment-annuitization problem for a retiree with linear consumption habits. We explore the effect of consumption habits on the decision to annuitize when annuities are purchased as a lump sum. The problem is formulated as a combined stopping-control problem. We derive optimal annuitization time, investment, and consumption strategies by a generalized dual method and habit reduction method. We investigate the influence of various factors on the annuitization time and derive optimal annuitization time, which is a barrier strategy. The numerical simulations reveal several interesting results. Our results demonstrate that risk aversion, subjective hazard rate, and consumption habits all play a role in shaping annuitization decisions. Furthermore, we offer a new explanation for the rarity of voluntary annuitization among retirees.

Gudkov, N., Ignatieva, K., and Ziveyi, J. (2019). "Pricing of guaranteed minimum withdrawal benefits in variable annuities under stochastic volatility, stochastic interest rates and stochastic mortality via the componentwise splitting method." In: *Quantitative Finance* 19(3), pp. 501–518.

This paper values guaranteed minimum withdrawal benefit (GMWB) riders embedded in variable annuities assuming that the underlying fund dynamics evolve under the influence of stochastic interest rates, stochastic volatility, stochastic mortality and equity risk. The valuation problem is formulated as a partial differential equation (PDE) which is solved numerically by employing the operator splitting method. Sensitivity analysis of the fair guarantee fee is performed with respect to various model parameters. We find that (i) the fair insurance fee charged by the product provider is an increasing function of the withdrawal rate; (ii) the GMWB price is higher when stochastic interest rates and volatility are incorporated in the model, compared to the case of static interest rates and volatility; (iii) the GMWB price behaves non-monotonically with changing volatility of variance parameter; (iv) the fair fee increases with increasing volatility of interest rates parameter, and increasing correlation between the underlying fund and the interest rates; (v) the fair fee increases when the speed of mean-reversion of stochastic volatility or the average long-term volatility increases; (vi) the GMWB fee decreases when the speed of mean-reversion of stochastic interest rates or the average long-term interest rates increase. We investigate both static and dynamic (optimal) policyholder's withdrawal behaviours; we present the optimal withdrawal schedule as a function of the withdrawal account and the investment account for varying volatility and interest rates. When incorporating stochastic mortality, we find that its impact on the fair guarantee fee is rather small. Our results demonstrate the importance of correct quantification of risks embedded in GMWBs and provide guidance to product providers on optimal hedging of various risks associated with the contract.

Guidolin, M., Hansen, E., and Lozano-Banda, M. (2018). "Portfolio performance of linear SDF models: an out-of-sample assessment." In: *Quantitative Finance* 18(8), pp. 1425–1436.

We evaluate linear stochastic discount factor models using an ex-post portfolio metric: the realized out-of-sample Sharpe ratio of mean-variance portfolios backed by alternative linear factor models. Using a sample of monthly US portfolio returns spanning the period 1968-2016, we find evidence that multifactor linear models have better empirical properties than the CAPM, not only when the cross-section of expected returns is evaluated in-sample, but also when they are used to inform one-month ahead portfolio selection. When we compare portfolios associated to multifactor models with mean-variance decisions implied by the single-factor CAPM, we document statistically significant differences in Sharpe ratios of up to 10 percent. Linear multifactor models that provide the best in-sample fit also yield the highest realized Sharpe ratios.

Gunther, S. and Hieber, P. (2024). "Analyzing the interest rate risk of equity-indexed annuities via scenario matrices." In: *Insurance: Mathematics and Economics* 114, pp. 15–28.

The financial return of equity-indexed annuities depends on an underlying fund or investment portfolio complemented by an investment guarantee. We discuss a so-called cliquet-style or ratchet-type guarantee granting a minimum annual return. Its path-dependent payoff complicates valuation and risk management, especially if interest rates are modelled stochastically. We develop a novel scenario-matrix (SM) method. In the example of a Vasicek-Black-Scholes model, we derive closed-form expressions for the value and moment-generating function of the final payoff in terms of the scenario matrix. This allows efficient evaluation of values and various risk measures, avoiding Monte-Carlo simulation or numerical Fourier inversion. In numerical tests, this procedure proves to converge quickly and outperforms the existing approaches in the literature in terms of computation time and accuracy.

Guo, D. (2019). "A Statistical Response to Challenges in Vast Portfolio Selection." PhD thesis. University of Waterloo.
The thesis is written in response to emerging issues brought about by an increasing number of assets allocated in a portfolio and seeks answers to puzzling empirical findings in the portfolio management area. Over the years, researchers and practitioners working in the portfolio optimization area have been concerned with estimation errors in the first two moments of asset returns. The thesis comprises several related chapters on our statistical inquiry into this subject. Chapter 1 of the thesis contains an introduction to what will be reported in the remaining chapters. A few well-known covariance matrix estimation methods in the literature involve adjustment of sample eigenvalues. Chapter 2 of the thesis examines the effects of sample eigenvalue adjustment on the out-of-sample performance of a portfolio constructed from the sample covariance matrix.

Guo, D., Boyle, P. P., Weng, C., and Wirjanto, T. S. (2019). "When Does The 1/N Rule Work?" In: *SSRN e-Print*.
We propose a "1/N favorability index" to measure how favorable a market is to holding a 1/N portfolio. This index reflects the extent of difficulty for an optimized portfolio to outperform the 1/N portfolio in a specific market. A single-factor model predicts that bull markets are accompanied by a high 1/N favorability index and vice versa. We validate the model implication that the 1/N portfolio is more difficult to beat in bull markets using stock return datasets from a number of countries as well as the classic datasets used by DeMiguel et al. (2009). Our results imply that the reported good performance of the 1/N portfolio in the US equity market can be partially attributed to the long-run bullish trend in the market which gives rise to the high favorability of the market to the 1/N portfolio.

Guo, I., Langrené, N., Loeper, G., and Ning, W. (2022). "Portfolio optimization with a prescribed terminal wealth distribution." In: *Quantitative Finance*, pp. 1–15.
This paper studies a portfolio allocation problem, where the goal is to reach a prescribed wealth distribution at a final time. We study this problem with the tools of optimal mass transport. We provide a dual formulation which is solved with a gradient descent algorithm. This involves solving an associated Hamilton-Jacobi-Bellman and Fokker-Planck equations with a finite difference method. Numerical examples for various prescribed terminal distributions are given, showing that we can successfully reach attainable targets. We then consider adding consumption during the investment process, to take into account distributions that are either not attainable, or sub-optimal.

Gupta, A. and Tayal, V. K. (2023). "Using Monte Carlo Methods for Retirement Simulations." In: *arXiv e-Print*.
Retirement prediction helps individuals and institutions make informed finan-cial, lifestyle, and workforce decisions based on estimated retirement portfolios. This paper attempts to predict retirement using Monte Carlo simulations, allow-ing one to probabilistically account for a range of possibilities. The authors propose a model to predict the values of the investment accounts IRA and 401(k) through the simulation of inflation rates, interest rates, and other perti-nent factors. They provide a user case study to discuss the implications of the proposed model.

Gupta, A. and Li, L. (2004). "A Modeling Framework for Optimal Long-Term Care Insurance Purchase Decisions in Retirement Planning." In: *Health Care Management Science* 7(2), pp. 105–117.
The level of need and costs of obtaining long-term care (LTC) during retired life require that planning for it is an integral part of retirement planning. In this paper, we divide retirement planning into two phases, pre-retirement and post-retirement. On the basis of four interrelated models for health evolution, wealth evolution, LTC insurance premium and coverage, and LTC cost structure, a framework for optimal LTC insurance purchase decisions in the pre-retirement phase is developed. Optimal decisions are obtained by developing a trade-off between post-retirement LTC costs and LTC insurance premiums and coverage. Two-way branching models are used to model stochastic health events and asset returns. The resulting optimization problem is formulated as a dynamic programming problem. We compare the optimal decision under two insurance purchase scenarios: one assumes that insurance is purchased for good and other assumes it may be purchased, relinquished and re-purchased. Sensitivity analysis is performed for the retirement age.

Gupta, A. and Li, L. (2007). "Integrating long-term care insurance purchase decisions with saving and investment for retirement." In: *Insurance: Mathematics and Economics* 41(3), pp. 362–381.

Risk related to long-term care (LTC) is high for the elderly. Planning for LTC is now regarded as the 'third leg' of retirement planning. In this paper, planning for LTC is integrated with saving and investment decisions for an integrated approach to retirement planning. Optimal LTC insurance purchase decisions are obtained by developing a trade-off between post-retirement LTC costs and LTC insurance premiums paid and coverage received. Integrating insurance purchase with wealth evolution, consisting of saving and investment decisions, allows addressing affordability issues. Two-way branching models are used for the stochastic health events and asset returns. The problem, formulated as a nonlinearly constrained mixed-integer optimization problem, is solved using a heuristic. Sensitivity analyses are performed for initial health and wealth status. Some important aspects of an individual's behavioral preferences are also addressed in this framework to provide more robust decision support.

Gupta, K. (2020). "Optimal investment and consumption strategy for a retiree under stochastic force of mortality." MA thesis. Simon Fraser University.

With an increase in the self-driven retirement plans during past few decades, more and more retirees are managing their retirement portfolio on their own. Therefore, they need to know the optimal amount of consumption they can afford each year, and the optimal proportion of wealth they should invest in the financial market. In this project, we study the optimization strategy proposed by Delong and Chen (2016). Their model determines the optimal consumption and investment strategy for a retiree facing (1) a minimum lifetime consumption, (2) a stochastic force of mortality following a geometric Brownian motion process, (3) an annuity income, and (4) non-exponential discounting of future income. We use a modified version of the Cox, Ingersoll, and Ross (1985) model to capture the stochastic mortality intensity of the retiree and, subsequently, determine a new optimal consumption and investment strategy using their framework. We use an expansion method to solve the classic Hamilton-Jacobi-Bellman equation by perturbing the non-exponential discounting parameter using partial differential equations.

Gupta, S. and Sharma, A. K. (2023). "Inflation Hedging Potential of Infrastructure Sector and Sub-Sector Returns - Evidence from Emerging Markets." In: *The Journal of Alternative Investments*.

This article tests the inflation hedging ability of infrastructure investments in both the short and long run in four of the major emerging markets. Based on the Generalized Fisher Hypothesis (GFH), we use Fama and Schwert's framework and Granger causality tests for short-term insights, and Engle and Granger co-integration for a long run analysis, to examine the inflation hedging effectiveness of various listed indexes (composite infrastructure, energy, communication, utilities, common stock, and property). The findings indicate that infrastructure and its sub-sectors constitute a more effective hedge against inflation in the long run than in the short run. The empirical results also stress the presence of an inherent heterogeneity within the infrastructure sector, and suggest that real estate and infrastructure should be considered separately in investors' portfolios. Our results have implications for investors, including institutional and retail, provided they have longer investment horizons, and for other market players to see infrastructure in the light of a unique asset class.

Ha, S. and Fabozzi, F. J. (2022). "A lifetime allocation with human capital: implications for target date fund." In: *Journal of Asset Management* 23(5), pp. 365–375.

In this paper, we propose a new target date fund that incorporates human capital. The proposed glide path for the target date fund is based on the capital asset pricing model in a way such that the asset allocation of the target date fund with human capital is matched with that of the global market portfolio with human capital. This new target date fund provides a customized retirement solution satisfying different human capital risk profiles.

Habib, F., Huaxiong, H., Mauskopf, A., Nikolic, B., and Salisbury, T. S. (2018). "Optimal allocation to deferred income annuities." In: *SSRN e-Print*.

In this paper we employ a lifecycle model that uses utility of consumption and bequest to determine an optimal Deferred Income Annuity (DIA) purchase policy. We lay out a mathematical framework to formalize the optimization process. The method and implementation of the optimization is explained, and the results are then analyzed. We extend our model to control for asset allocation and show how the purchase policy changes when one is allowed to vary asset allocation. Our results indicate that (i.) refundable DIAs are less appealing than non-refundable DIAs because of the loss of mortality credits; (ii.) the DIA allocation region is larger under the fixed asset allocation strategy due to it becoming a proxy for fixed-income allocation; and (iii.) when the investor is allowed to change asset-allocation, DIA allocation becomes less appealing. However, a case for higher DIA allocation can be made for those individuals who perceive their longevity to be higher than the population.

Habib, F., Huaxiong, H., and Milevsky, M. A. (2017). "Approximate solutions to retirement spending problems and the optimality of ruin." In: *SSRN e-Print*.
Milevsky and Huang (2011) investigated the optimal retirement spending policy for a utility-maximizing retiree facing a stochastic lifetime but assuming deterministic investment returns. They solved the problem using techniques from the calculus of variations and derived analytic expressions for the optimal spending rate and wealth depletion time under the Gompertz law of mortality. Of course, in the real world financial returns are stochastic as well as lifetimes, raising the question of whether their qualitative insights and approximations are generalizable or practical.

Hagelstein, P., Lackner, I., Otto, J., Perona, A., and Piziak, R. (2019a). "Fixed and Dynamic Asset Allocation in the Accumulation Phase." In: *Journal of Finance and Investment Analysis* 8(1), pp. 1–12.
In this paper, we consider the historical real returns of fixed and dynamic allocation portfolios consisting of equities and short term bonds over thirty year time horizons, where fixed real contributions are made to the portfolios annually. In particular, we consider both the scenario where the investor annually rebalances a portfolio to a fixed ratio as well as the scenario where the investor's annual contribution has a fixed ratio but the portfolio is never subsequently rebalanced. These results provide investors in the accumulation phase historical data that may provide a useful guide to asset allocation decisions. Of particular interest is that, over the 88 thirty-year time intervals considered, dynamic allocation portfolios had a better overall performance than fixed allocation portfolios, and that both fixed and dynamic allocation portfolios strongly benefited from a heavy equity exposure.

Hagelstein, P., Lackner, I., Otto, J., Perona, A., and Piziak, R. (2019b). "Markowitz Portfolios with Graham Bands in the Accumulation Phase." In: *The Journal of Wealth Management* 22(3), pp. 41–48.
In this article, the authors consider historical real returns of tax-exempt portfolios consisting of equities and short-term bonds over 90 different 30-year time periods from 1900 to 2018, in which fixed real contributions were made to the portfolios annually. Two main types of portfolios are considered. The first is a portfolio in which the investor annually contributed evenly between equities and bonds and never rebalanced the portfolio. The second is a portfolio with Graham bands, in which the investor annually contributed evenly between equities and bonds but rebalanced the portfolio to an overall 50/50 allocation whenever the equity or the bond portion of the portfolio exceeded 75% of the overall portfolio value. The authors also consider analogous results when short-term bonds were replaced by less liquid guaranteed income instruments, such as TIAA Traditional, providing higher yields. These results are intended to provide useful historical data to assist investors in the accumulation phase and their advisors in asset allocation decisions.

Haghani, V. and White, J. (2025). "Out of the Frying Pan and Into the Fire: Selling A Highly Appreciated Stock Without Paying Taxes?" In: *SSRN e-Print*.
This paper will critically examine a tax-efficient strategy for selling highly appreciated assets without incurring capital gains taxes. The strategy, known as leveraged direct index tax loss harvesting, aims to transition a concentrated single-stock position into a diversified portfolio over approximately ten years while minimizing tax liabilities. However, the strategy may not be as beneficial as it appears. Over time, the costs of fees, financing frictions, and tracking risks can outweigh the potential tax savings, and potentially leave investors worse off than if they had simply sold the asset and reinvested in a diversified portfolio.

Halen, N., Faust, K., and Taylor, T. (2020). "Understanding the True Cost of Health Care in Retirement." In: *Retirement Management Journal*.
After years of working hard and diligently saving to prepare for retirement, many individuals expect they'll finally be able to slow down, pursue hobbies, and enjoy the next phase of their lives. However, the truth facing retirees today is that numerous financial risks and uncertainties threaten their ability to spend their hard-earned money and maintain the quality of life they desire. Perhaps the greatest worry for those in, or near, retirement is whether they will be able to afford rising healthcare costs, particularly unplanned out-of-pocket expenses. Retirees' concerns around health care are not without merit, yet a closer look reveals a different picture. Our research concludes that healthcare expenses for many retirees are a small percentage of total spending and are far less variable than most people think, making them easier to plan for properly than conventional wisdom suggests. Two healthcare-related risks increase spending variability in retirement dramatically: long-term care events and longevity. An appropriate strategy for managing healthcare expenses in retirement is to plan for known or diversifiable risks and insure the unknown or undiversifiable risks. This study has three primary goals. First, to assess and quantify the variability of out-of-pocket healthcare costs to determine whether retiree concerns around funding these expenses are warranted. Second, to identify what factors drive variability in

spending and quantify their potential financial impact. Third, to identify solutions financial professionals can utilize with clients to reduce spending variability and achieve better outcomes in retirement.

Haley, M. R. (2017). "K-fold cross validation performance comparisons of six naive portfolio selection rules: how naive can you be and still have successful out-of-sample portfolio performance?" In: *Annals of Finance* 13, pp. 341–353.

Recent research reports that optimal portfolio selection models often perform worse than equal-weight naive diversification in out-of-sample testing. This paper extends this line of inquiry by comparing the out-of-sample performance of the equal-weight naive strategy to the out-of-sample performance of five alternative naive strategies, each of which derives from a simple heuristic that does not require any optimization. Out-of-sample portfolio performance is assessed by mean, standard deviation, skewness, and Sharpe ratio; k-fold cross validation is used as the out-of-sample testing mechanism. The results indicate that the proposed naive heuristic rules exhibit strong out-of-sample performance, in most cases superior to the equal-weight naive strategy. These findings are consequential for at least two reasons: first, if these simple heuristic-based rules outperform the equal-weight naive strategy, then by transitivity they can outperform the mean-variance- and shortfall-optimal portfolio rules that have been shown in the literature to be inferior to the equal-weight naive rule, which further emphasizes the out-of-sample fragility of "optimal"methods; and second, among naive diversification strategies, some appear more robust in out-of-sample testing than others, hence the proposed methods may be useful when forming mixed portfolio selection models wherein a naive strategy is combined with an optimal strategy to improve performance.

Hallerbach, W. G. (2014). "Disentangling rebalancing return." In: *Journal of Asset Management* 15(5), pp. 301–316.

The use of portfolio rebalancing as a profitable strategy (or volatility harvesting) is a hot topic. Indeed, it is interesting to know what the impact of periodic rebalancing is on the growth rate of a portfolio. Unfortunately, the terminology used in the literature is confusing. Terms such as diversification return and rebalancing return are used interchangeably to indicate the growth rate that a rebalanced portfolio can earn in excess of a buy-and-hold portfolio. The literature is also confused in specifying this excess growth rate from rebalancing. In this paper, we investigate the full return from rebalancing and decompose it into the volatility return and the dispersion discount. We prove some general results regarding these components and present some simple approximations which provide direct insight into the driving forces behind these building blocks. We consider a pro forma US asset portfolio over the period 1974-2013, which allows us to investigate the relative magnitude of the discussed effects and their time variation.

Hallstein, S., Liebler, D., and Maurer, R. (2024). "Rethinking the Annuity Puzzle: The Role of Loss Aversion and Money-Back Guarantees." In: *SSRN e-Print*.

We contribute to the resolution of the annuity puzzle by incorporating loss aversion into the preference structures of individuals. Specifically, individuals are averse to the case where death occurs before the sum of payouts exceeds the sum of annuity premiums paid. In realistically calibrated life cycle models with optimal consumption and portfolio choices of risk-averse investors with a bequest motive, we demonstrate that this loss aversion can explain the low demand for life-only annuities. Furthermore, the model predictions align with the evidence that individuals predominantly purchase life annuities with money-back guarantees.

Halperin, I. (2023). "SCOP: Schrodinger Control Optimal Planning for Goal-Based Wealth Management." In: *SSRN e-Print*.

We consider the problem of optimization of contributions of a financial planner such as a working individual towards a financial goal such as retirement. The objective of the planner is to find an optimal and feasible schedule of periodic installments to an investment portfolio set up towards the goal. Because portfolio returns are random, the practical version of the problem amounts to finding an optimal contribution scheme such that the goal is satisfied at a given confidence level. This paper suggests a semi-analytical approach to a continuous-time version of this problem based on a controlled backward Kolmogorov equation (BKE) which describes the tail probability of the terminal wealth given a contribution policy. The controlled BKE is solved semi-analytically by reducing it to a controlled Schrodinger equation and solving the latter using an algebraic method. Numerically, our approach amounts to finding semi-analytical solutions simultaneously for all values of control parameters on a small grid, and then using the standard two-dimensional spline interpolation to simultaneously represent all satisficing solutions of the original plan optimization problem. Rather than being a point in the space of control variables, satisficing solutions form continuous contour lines (efficient frontiers) in this space.

Han, N.-w. and Hung, M.-w. (2012). "Optimal asset allocation for DC pension plans under inflation." In: *Insurance: Mathematics and Economics* 51(1), pp. 172–181.

In this paper, the stochastic dynamic programming approach is used to investigate the optimal asset allocation for a defined-contribution pension plan with downside protection under stochastic inflation. The plan participant invests the fund wealth and the stochastic interim contribution flows into the financial market. The nominal interest rate model is described by the Cox Ingersoll Ross (Cox et al., 1985) dynamics. To cope with the inflation risk, the inflation indexed bond is included in the asset menu. The retired individuals receive an annuity that is indexed by inflation and a downside protection on the amount of this annuity is considered. The closed-form solution is derived under the CRRA utility function. Finally, a numerical application is presented to characterize the dynamic behavior of the optimal investment strategy. We model the optimal asset allocation for DC pension plan under inflation. A minimum guarantee on the amount of retirement annuity is considered. We emphasize the importance of the inflation-indexed bond in the pension portfolio. The investments in indexed bonds would increase with the risk-aversion. The investments in indexed bonds would increase with the guarantee provided.

Hanna, S. and Kim, K. T. (2017). "Treatment of Inflation in Retirement Planning Calculations: An Improved Method." In: *Journal of Financial Planning* 30(1), pp. 44–53.

The typical treatment of inflation in retirement planning textbooks is complex and often not reasonable in terms of the amount to contribute, the first year being dependent on the inflation rate assumption. The typical financial planning approach requires an assumption about the future inflation rate, which can be difficult to justify based on the economic history in the United States and other countries. Economists typically put all amounts and interest rates in inflation-adjusted terms, which is simpler and can be a more rational approach to long-term planning. It can be difficult to comprehend the meaning of future amounts in nominal terms; therefore, doing projections in inflation-adjusted terms may help clients better understand projections. In calculating the capital needs for retirement, inflation-adjusted rates of return for each step should be chosen based on portfolio choices appropriate for before and after retirement.

Hao, H. (2019). "A Regime-Aware Agent-Based Framework for Financial Planning." PhD thesis. Princeton University.

The vulnerability of individuals planning for retirement has been growing due to the conversion from defined-benefit plans to defined-contribution plans, the steady increase in life longevity, and the uncertainty of asset returns under an ever-changing global environment. A serious problem is the lack of appropriate planning for retirement. How much should an individual save beyond the Social Security tax in order to maintain a reasonable lifestyle after retirement? This paper designs a framework to facilitate the process of setting realistic goals for financial planning, featuring the concept of agent-based simulations. The framework also provides policy-rule guidelines for the agent to search for an optimal strategy. Additionally, a micro-macro analysis enables us to analyze a cohort of representative agents and aggregate the individual results on the macro-level. The simulation module employs a regime-based Monte Carlo simulation of multiple asset categories, a factor-based diversifying asset allocation approach, and a collection of dynamic policy-rule-based investment strategies. Empirical results, consisting of a downside risk simulation for university endowments, a sustainability assessment for the Social Security fund, and a personal goal-based retirement planning, demonstrate stylized applications of the planning framework.

Harbron, G. L., Bloore, W., and Zorn, J. (2019). *Withdrawal order: making the most of retirement assets*. Tech. rep. Vanguard.

Retirees in the UK face more challenges today than ever before when it comes to retirement planning. The decline of defined benefit pensions in favour of less expensive (for the employer) defined contribution plans, pension freedoms, increased longevity and other factors have made retirement planning considerably more complex. Arguably foremost among these challenges is how to convert the retiree's accumulated retirement savings into a sustainable income that may need to last 30 years or more.While reducing spending and working longer are the most effective ways of extending the life of a retirement portfolio, increasing the portfolio's net after-tax return can also have apositive impact. One way to accomplish this is to select the proper withdrawal order when deciding which accounts to spend from. This paper looks at three withdrawal orders across three crystallisation strategies. Using our Vanguard Capital Markets Model (VCMM), we simulate the impact of withdrawal order and crystallisation strategy on a number of success metrics over a 30-year time horizon. Our analysis shows that, for most investors, withdrawing from taxable accounts first provides the best results.

Harcourt, D., Daglish, T., and Ulm, E. R. (2024). "Analytic valuation of guaranteed lifetime withdrawal benefits with a modified ratchet." In: *Insurance: Mathematics and Economics* 118, pp. 59–71.

Guaranteed Lifetime Withdrawal Benefits (GLWBs) are an increasingly popular add-on to Variable Annuities, offering a guaranteed stream of payments for the remainder of the policyholder's life. GLWBs have typically been priced using numerical methods such as finite difference schemes or Monte Carlo simulations; obtaining accurate and precise solutions using these methods can be very computationally expensive. In this paper, we extend an existing method for analytic pricing of these policies to a more general fee structure. We introduce a novel variation on the commonly offered ratchet rider that more directly addresses policyholder motivation around lapse-and-reentry behaviour. We then modify our pricing method to accommodate this new rider and compare it to the existing annual ratchet with respect to a policyholder's incentive to lapse such a policy.

Harris, M. (2019). "Should a Retiree File for Social Security at 62 or 70?" In: *The Journal of Retirement* 7(2), pp. 51–59.

Retirees can file for Social Security benefits as soon as they are 62 years old, but payments will become larger if they delay up to a maximum of 70 years of age. Many people believe the best strategy is to wait until later to file in order to maximize benefit payments. In this article, we reexamine that advice and add two considerations. First, we discuss how the claiming decision is made and recognize that in many cases the individual's goal is to support new lifestyle options, not to maximize Social Security withdrawals. Next, we consider alternative sources of income and how using those resources affects the claiming decision. The article concludes with example scenarios that demonstrate how a financial advisor might advise clients on the best strategy for their particular situation.

Harvey, C. R., Hoyle, E., Rattray, S., Sargaison, M., Taylor, D., and Van Hemert, O. (2019). "The best of strategies for the worst of times: can portfolios be crisis proofed?" In: *The Journal of Portfolio Management* 45(5), pp. 7–28.

In the late stages of long bull markets, a popular question arises: What steps can an investor take to mitigate the impact of the inevitable large equity correction? Hedging equity portfolios is notoriously difficult and expensive. In this article, the authors analyze the performance of different tools that investors could deploy. For example, continuously holding short-dated S&P 500 put options is the most reliable defensive method but also the most costly strategy. Holding safe-haven US Treasury bonds produces a positive carry but may be an unreliable crisis-hedge strategy because the post-2000 negative bond-equity correlation is a historical rarity. Long gold and long credit protection portfolios sit between puts and bonds in terms of both cost and reliability. Dynamic strategies that performed well during past drawdowns include futures time-series momentum (which benefits from extended equity sell-offs) and a quality strategy that takes long (short) positions in the highest (lowest) quality company stocks (which benefits from a flight-to-quality effect during crises). The authors examine both large equity drawdowns and recessions. They also provide some out-of-sample evidence of the defensive performance of these strategies relative to an earlier, related article.

Harvey, C. R., Hoyle, E., Rattray, S., and van Hemert, O. (2020a). "Strategic Risk Management: Out-of-Sample Evidence from the COVID-19 Equity Selloff." In: *SSRN e-Print*.

Over the 2016-2019 period, we released a series of research papers on the topic of "strategic risk management", or the embedding of risk management into investment strategy design. We show that key risk controls that we introduced materially helped during the sharp equity market selloff in February-March 2020, when the COVID-19 pandemic accelerated. First, faster trend following and long-short profitability stock strategies performed well during the equity market selloff. Second, responsive volatility targeting reduced positions dramatically ahead of the most volatile period in March 2020, and so improved both the return and risk profile at that time. Third, strategic rebalancing rules helpfully called for keeping an underweight in equities (not rebalancing back to target) at the end of February 2020, regardless of using 1-, 3-, or 12-month trend systems to base the rebalancing rule on.

Harvey, C. R., Liu, Y., and Saretto, A. (2020b). "An Evaluation of Alternative Multiple Testing Methods for Finance Applications." In: *The Review of Asset Pricing Studies* 10(2), pp. 199–248.

In almost every area of empirical finance, researchers confront multiple tests. One high-profile example is the identification of outperforming investment managers, many of whom beat their benchmarks purely by luck. Multiple testing methods are designed to control for luck. Factor selection is another glaring case in which multiple tests are performed, but numerous other applications do not receive as much attention. One important example is a simple regression model testing five variables. In this case, because five variables are tried, a t-statistic of 2.0 is not enough to establish significance. Our paper provides a guide to various multiple testing methods and details a number of applications. We provide simulation evidence on the relative performance of different methods across a variety of testing environments. The goal of our paper is to provide a menu that researchers can choose from to improve inference in financial economics.

Hawkins, H. and Gambhir, S. (2025). "Quantifying Patience: Comprehensive Analysis of Minimum Holding Periods Using Multiple Resampling and Parametric Methods." In: *SSRN e-Print*.

Determining how long investors must remain committed to a strategy before confidently expecting outperformance over a benchmark is a central question in portfolio management. Similarly, understanding how long it takes for the portfolio itself to generate positive returns, even without considering a benchmark is equally important. Relying on short-run metrics or a single simulation method can yield misleading conclusions due to market volatility, structural shifts, & non-normal return distributions [1, 2]. This paper develops a comprehensive statistical framework that integrates multiple resampling & simulation approaches to estimate both underperformance (negative excess returns) & negative absolute return probabilities over holding periods from one to ten years. We focus on traditional methods-IID, block bootstrap, stationary bootstrap [3, 4]-& also consider a non-Gaussian parametric simulation (Student's t-distribution) [1, 5] to capture heavier tails & the possibility of extreme events. Using 19.5 years of monthly portfolio & benchmark data, we first estimate how quickly outperformance becomes likely. We then apply the same techniques to the portfolio's own returns (without subtracting a benchmark) to determine when positive returns become probable. Finally, we employ survival analysis (Kaplan-Meier) [6] to understand the distribution of time-to-first-positive-return & time-to-first-outperformance after initial holding periods By comparing these diverse methodologies & scenarios, we provide a practical, multifaceted guide that helps long-term investors & advisors set realistic expectations, maintain confidence in their chosen strategies, & understand the patience required for both relative & absolute performance goals.

He, Y. (2021). "Optimization-based Tail Risk Hedging." MA thesis. University of Toronto.

This thesis presents a mixed risk-return optimization framework for selecting long put option positions to hedge the tail risk of investments. We formulate tractable optimization models by utilizing hypothetical portfolios that roll put options on a constant basis. Variance and sample Conditional Value-at-Risk (CVaR) are used as risk measures. Firstly, the tail risk hedging for a single asset such as the S 500 ETF is considered. Our proposed models are tested against the out-of-sample historical S 500 index values as well as the values of the index paired with long put options of varying strike prices. The optimized hedged portfolios could provide sufficient protection in market downturns while not losing significant returns in a longer investment horizon without explicitly predicting future market behavior. This is achieved by dynamically adjusting the positions of put options with different protection levels according to the market trends. Allocations to different put options are analyzed under various market trends and investor risk aversion levels. Then our framework is extended to multiple underlying assets and put options associated with them by constructing a hypothetical portfolio consisting of a combination of each put option and the associated underlying asset. The optimized strategies overcome traditional drawbacks associated with protective put strategies, as well as outperform the strategies of investing directly in the underlying assets and holding constant long positions in put options.

Hejazi, S. A., Jackson, K. R., and Gan, G. (2017). "A Spatial Interpolation Framework for Efficient Valuation of Large Portfolios of Variable Annuities." In: *arXiv e-Print*.

Variable Annuity (VA) products expose insurance companies to considerable risk because of the guarantees they provide to buyers of these products. Managing and hedging these risks requires insurers to find the value of key risk metrics for a large portfolio of VA products. In practice, many companies rely on nested Monte Carlo (MC) simulations to find key risk metrics. MC simulations are computationally demanding, forcing insurance companies to invest hundreds of thousands of dollars in computational infrastructure per year. Moreover, existing academic methodologies are focused on fair valuation of a single VA contract, exploiting ideas in option theory and regression. In most cases, the computational complexity of these methods surpasses the computational requirements of MC simulations. Therefore, academic methodologies cannot scale well to large portfolios of VA contracts. In this paper, we present a framework for valuing such portfolios based on spatial interpolation. We provide a comprehensive study of this framework and compare existing interpolation schemes. Our numerical results show superior performance, in terms of both computational efficiency and accuracy, for these methods compared to nested MC simulations. We also present insights into the challenge of finding an effective interpolation scheme in this framework, and suggest guidelines that help us build a fully automated scheme that is efficient and accurate.

Hens, T. and Nordlie, T. (2024). "How good are LLMs in risk profiling?" In: *SSRN e-Print*.

This study compares OpenAI's ChatGPT-4 and Google's Bard with bank experts in determining investors' risk profiles. We find that for half of the client cases used, there are no statistically significant differences in the risk profiles. Moreover, the economic relevance of the differences is small. However, the LLMs are not good in explaining the risk profiles.

Hens, T., Schenk-Hoppe, K. R., and Woesthoff, M.-H. (2020). "Escaping the backtesting illusion." In: *The Journal of Portfolio Management* 46(4), pp. 81–93.
Two tests can help asset managers to develop more robust investment strategies: an impact test and a survival test. Both tests complement the backtest, in which one checks how a proposed investment strategy would have performed in the past. The impact test considers the performance of the strategy when assets under management grow (crowdedness), and it checks the impact that growth in assets under management in competing strategies has on the proposed strategy (cross impact). The survival test considers the effect of the long-term evolution of assets under management in competition for market capital. Using Shiller S&P 500 index and bond market data, we show that time-series momentum (relative strength) performs best in the backtest and the impact test but that an expected relative cash-flow rule (relative dividend yield) has the best long-term survival properties.
Henshaw, K., Koffi, C. H. A., Pamen, O. M., and Zeineddine, R. (2024). "On the valuation of life insurance policies for dependent coupled lives." In: *arXiv e-Print*.
In this paper, we investigate a complex variation of the standard joint life annuity policy by introducing three distinct contingent benefits for the surviving member(s) of a couple, along with a contingent benefit for their beneficiaries if both members pass away. Our objective is to price this innovative insurance policy and analyse its sensitivity to key model parameters, particularly those related to the joint mortality framework. We employ the QP-rule (described in Section 2: The General Setup), which combines the real-world probability measure P for mortality risk with risk-neutral valuation under Q for financial market risks. The model enables explicit pricing expressions, computed using efficient numerical methods. Our results highlight the interdependent risks faced by couples, such as broken-heart syndrome, providing valuable insights for insurers and policyholders regarding the pricing influences of these factors.
Hicks, W. (2019). *How Much Is An Adviser Worth?* Tech. rep. Sapling Wealth Management.
Often clients try to quantify what good financial advice is worth. After all, how do you determine if the value of the service is greater than its cost? However, this is a tricky proposition, as it depends on how you define "good". Nevertheless, several reports have broached this topic and I have attempted to give a brief overview of the strengths and weaknesses of a few that I believe are quite substantive. Clients should aim to understand these reports in the context of their own personal situation. Identify your own personal areas that require the most support and think about if you are are getting all of the focus that you need.
Hight, G. N. and Haley, J. D. (2021). "Low-Risk Benchmarking Transcends Rebalancing Methods." In: *The Journal of Investing* 30(2), pp. 7–30.
The dismal performance of managed investment and the success of equal allocation and minimum variance models prompt these questions: Can rebalancing driven by minimum variance and maximum diversification rebalancing outperform naive models? Can a minimum variance model produce higher risk-adjusted returns than a maximum diversification model when security selection favors low-correlated assets? This study uses expected shortfall to measure risk, bootstrapping to transform fat-tailed distributions so they are suitable for t-tests, and factor analysis to help explain the relative performance of the models. The minimum variance and maximum diversification models outperformed naive models, and the minimum variance models produced higher risk-adjusted returns than the maximum diversification model. Market factors adequately explained differences in returns between rebalancing models, but adding a factor for the effect of rebalancing procedures improved all models. All models constrained by the lower risk benchmark produced higher risk-adjusted returns than corresponding models constrained by the higher risk benchmark. This outcome suggests many rebalancing models – perhaps even return-based models – could produce superior risk-adjusted returns if lower risk benchmarks constrain risk.
Hilliard, J. E. and Hilliard, J. (2015). "Rebalancing versus Buy and Hold: Theory, Simulation and Empirical Analysis." In: *SSRN e-Print*.
We consider returns from rebalanced and buy and hold portfolios consisting of the same stocks. Theoretical properties are derived using Jensen's inequality and the H older's Defect Formula. Simulations are used to confirm theory and to investigate ambiguous cases where theory is silent. Rebalancing generally increases the Sharpe Ratios and decreases total return volatility. Buy and hold produces greater expected return. Results are more opaque with respect to expected geometric means. Our empirical tests are based on portfolios composed of the risk-free asset and market valued weighted CRSP total returns. While rebalancing reduces volatility, these tests largely favor the buy and hold strategy due to the high relative returns of the index vis-a-vis the risk-free asset.

Hoechle, D., Ruenzi, S., Schaub, N., and Schmid, M. (2023). "Financial Advice and Retirement Savings." In: *SSRN e-Print*.

We use a unique dataset from a large retail bank to examine the impact of financial advice on personal retirement savings. We document that retirement-related financial advice is associated with an increase in tax-exempt retirement accounts and equity investments, both at the extensive as well as the intensive margin. Our analysis suggests a causal link. We find no evidence that advisors particularly help typically disadvantaged clients (female, poorer, less-educated). Additional investments into retirement accounts and equities primarily come from external sources and checking accounts. The bank also benefits from the provision of retirement-related financial advice.

Hoffstein, C., Faber, N., and Braun, S. (2020). "Rebalance Timing Luck: The (Dumb) Luck of Smart Beta." In: *SSRN e-Print*.

Prior research and empirical investment results have shown that portfolio construction choices related to rebalance schedules may have non-trivial impacts on realized performance. We construct long-only indices that provide exposures to popular U.S. equity factors (value, size, momentum, quality, and low volatility) and vary their rebalance schedules to isolate the effects of "rebalance timing luck." Our constructed indices exhibit high levels of rebalance timing luck, often exceeding 100 basis points annualized, with total impact dependent upon the frequency of rebalancing, portfolio concentration, and the nature of the underlying strategy. As a case study, we replicate popular factor-based index funds and similarly find meaningful performance impacts due to rebalance timing luck. For example, a strategy replicating the S&P Enhanced Value index saw calendar year return differentials above 40% strictly due to the rebalance schedule implemented. Our results suggest substantial problems for analyzing any investment when the strategy, its peer group, or its benchmark is susceptible to performance impacts driven by the choice of rebalance schedule.

Hofmann, P. (2021a). "Quantifying the Tax Benefit of Retirement Accounts for Better Client Decisions (Part 1)." In: *Advisor Perspectives*.

How should clients use tax-advantaged accounts? By quantifying the tax benefits of those accounts, we can refine recommendations on asset location and the traditional-versus-Roth decision. Specifically, low-return assets, even if otherwise taxed at high rates, may not benefit much from being placed in tax-advantaged accounts. Also, the choice between traditional and Roth contributions is asymmetric: Traditional accounts provide an opportunity for a higher tax benefit, but risk incurring a tax cost. Roth contributions provide greater certainty of a positive tax benefit.

Hofmann, P. (2021b). "Quantifying the Tax Benefit of Retirement Accounts for Better Client Decisions (Part 2)." In: *Advisor Perspectives*.

In my previous article, I showed how the value of tax-advantaged accounts depends on the tax-free investment income earned in those accounts. This article digs deeper into this dynamic to further demonstrate why the conventional wisdom of "stocks in taxable, bonds in tax advantaged" is not reliable. Compounding of tax-free investment income, protecting against tax policy changes, and avoiding taxes on inflation are all potential reasons for holding stocks in tax-advantaged accounts.

Hollstein, F. and Prokopczuk, M. (2023). "Managing the Market Portfolio." In: *Management Science* 69(6), pp. 3675–3696.

We analyze the relation between time-series predictability and factor investing. We use a large set of financial, macroeconomic, and technical variables to time-series-manage the market portfolio. A combination of the out-of-sample market excess return forecasts of all variables yields a managed market portfolio that generates alphas relative to cross-sectional factor models that exceed 5% per annum. More broadly, the relation between time-series evaluation measures and (multifactor) alphas is weakly positive but complex. The variables' predictability for future returns is more important than that for volatility. Finally, we document that managed market portfolios based on lagged factor realizations also perform well. The replication files for this article are available.

Hopkins, J. (2022). "Retirement Planning for Life Stages." In: *Journal of Financial Planning* 35(11).

Age is just a number, and different clients will hit different financial milestones at differ- ent times. For example, your clients who are physicians or lawyers might hit certain financial milestones faster than your clients in lower-paying careers. But there are different planning elements that you want to make sure clients are covering at different stages of their lives. While some of this information might seem basic to you, I've organized it by decade to help you keep your clients on track in each of these life stages.

Hopkins, J. P., Ragatz, J. A., and Galli, C. (2016). "Ethical issues in retirement income planning: an advisor perspective." In: *The Journal of Retirement* 4(1), pp. 112–130.

The Ethical Issues in Retirement Income Planning study gathered the perceptions of expert retirement income planners. The good news is that the retirement income planners surveyed expressed a high level of satisfaction with the retirement planning profession ethical climate, as 64% of respondents believed that the overall ethical culture in the retirement income industry was solid as compared to 36% that expressed some concern. However, advisors were not without concerns. When asked to rank those concerns, a dominant theme emerged through the survey quantitative and qualitative findings: Respondents believed that ethical violations were the result of a lack of education and not purposeful malfeasance. The number one ethical concern respondents identified was financial elder abuse. Respondents were not overly concerned that financial advisors were the primary perpetrators but were worried about the industry ability to identify cases of financial elder abuse and exploitation, noting that most retirement advisors had not been adequately trained to recognize the signs of abuse, which are often subtle and ambiguous. Respondents reported that they saw very few acts of outright fraud or deceit. Respondents generally believed that retirement advisors, while well-intentioned, often lacked sufficient knowledge in three main areas: Social Security, Medicare, and tax planning. They feared that these knowledge limitations might lead practitioners to make recommendations that were not in the best interest of their clients. An additional and related concern was that consumers would not be able to act as a check since in many cases they lacked the financial literacy to understand the complex products or plans offered to them. In some cases, clients simply trusted their advisor and remained disengaged from the planning process. Respondents also perceived that this level of client dependence reflected a general lack of understanding of the financial advisor pivotal role in the process of planning for retirement.

Horan, S. (2005). *Tax-Advantaged Savings Accounts and Tax-Efficient Wealth Accumulation*. Tech. rep. CFA Institute Research Foundation.

Until recently, the issue of tax-efficient investing has been largely overlooked by the mainstream literature. And simple heuristics to guide investors and their advisors are not always as obvious as they might initially seem. This monograph explores central issues surrounding the use of tax-deferred investment accounts as a means of accumulating wealth and presents a useful framework, grounded in basic time-value-of-money concepts, that can be readily implemented by investment professionals (U.S. as well as non-U.S. based) in various tax environments (current as well as those resulting from changes in the tax code).

Horn, M. and Oehler, A. (2020). "Automated portfolio rebalancing: Automatic erosion of investment performance?" In: *Journal of Asset Management* 21(6), pp. 489–505.

Robo-advisers enable investors to establish an automated rebalancing strategy for a portfolio usually consisting of stocks and bonds. Since households' portfolios additionally include further frequently tradable assets like real estate funds, articles of great value and cash(-equivalents), we analyze whether households would benefit from a service that automatically rebalances a portfolio which additionally includes the latter assets. In contrast to previous studies, this paper relies on real-world household portfolios, which are derived from the German central bank's (Deutsche Bundesbank) Panel on Household Finances (PHF)-Survey. We compute the portfolio performance increase/decrease that households would have achieved by employing rebalancing strategies instead of a buy-and-hold strategy in the period from September 2010 to July 2015 and analyze whether subsamples of households with certain sociodemographic and socioeconomic characteristics would have benefited more from portfolio rebalancing than other household subsamples. The empirical analysis shows that the analyzed German households would not have benefited from an automated rebalancing service and that no subgroup of households would have significantly outperformed another subgroup in the presence of rebalancing strategies.

Horneff, V., Love, D. A., and Maurer, R. (2024). "Rules of Thumb and Retirement Accounts." In: *SSRN e-Print*.

We examine the welfare costs of applying common rules of thumb for saving, investment, 401k contributions, and withdrawals in an environment that includes a realistic treatment of taxation, Social Security benefits, 401k-plan details, and uncertainty in income, longevity, and asset returns. We test the performance of commonly recommended rules, such as investing 100-minus-age percent of assets in stocks, contributing 6-10% of income to a 401k account, or withdrawing the required minimum distribution (RMD) during retirement. We find that target-date rules for asset allocation generate moderate welfare losses, with one-time compensating variation amounts in the \$1,500-\$5,000 range for 401k allocation and \$600-\$1,300 for outside savings. Rules of thumb for 401k contributions generate even smaller welfare losses, in the \$300-\$500 range, reflecting the fact that the matching limit on 401ks provides a strong incentive to save at the threshold for many years of the working life. Withdrawing exactly the RMD proves to be a less successful strategy, though a compound strategy of withdrawing the maximum of a fixed percentage and the RMD is more effective. More generally, we find that rules of thumb may be reasonable alternatives to optimal decisions if the costs of optimizing are sufficiently high.

Horneff, V., Maurer, R., and Mitchell, O. (2017). "How Persistent Low Expected Returns Alter Optimal Life Cycle Saving, Investment, and Retirement Behavior." In: *SSRN e-Print*.

This Chapter explores how an environment of persistent low returns influences saving, investing, and retirement behaviors, as compared to what in the past had been thought of as more "normal" financial conditions. Our calibrated lifecycle dynamic model with realistic tax, minimum distribution, and Social Security benefit rules produces results that agree with observed saving, work, and claiming age behavior of U.S. households. In particular, our model generates a large peak at the earliest claiming age at 62, as in the data. Also in line with the evidence, our baseline results show a smaller second peak at the (system-defined) Full Retirement Age of 66. In the context of a zero-return environment, we show that workers will optimally devote more of their savings to non-retirement accounts and less to 401(k) accounts, since the relative appeal of investing in taxable versus tax-qualified retirement accounts is lower in a low return setting. Finally, we show that people claim Social Security benefits later in a low interest rate environment.

Horneff, V., Maurer, R., and Mitchell, O. S. (2018a). "How Persistent Low Expected Returns Alter Optimal Life Cycle Saving, Investment, and Retirement Behavior." In: *How Persistent Low Returns Will Shape Saving and Retirement*. Oxford University Press.

This chapter explores how an environment of persistent low returns influences saving, investing, and retirement behaviors, compared to what in the past had been conceived of as "normal" financial conditions. Using a calibrated life cycle dynamic model with realistic tax, minimum distribution, and social security benefit rules, we can mimic the large peak at the earliest claiming age at 62 that is seen in the data. Also in line with the evidence, our baseline results show a smaller second peak at the (system-defined) Full Retirement Age of 66. In the context of a zero-return environment, we show that workers will optimally devote more of their savings to non-retirement accounts and less to 401(k) accounts, since the relative appeal of investing in taxable versus tax-qualified retirement accounts is lower in a low return setting. Finally, we show that people claim social security benefits later in a low interest rate environment.

Horneff, V., Maurer, R., and Mitchell, O. S. (2018b). "Putting the Pension Back in 401(k) Retirement Plans: Optimal versus Default Longevity Income Annuities." In: *SSRN e-Print*.

A recent US Treasury regulation allowed deferred longevity income annuities to be included in pension plan menus as a default payout solution, yet little research has investigated whether more people should convert some of the 15 trillion they hold in employer-based defined contribution plans into lifelong income streams. We investigate this innovation using a calibrated lifecycle consumption and portfolio choice model embodying realistic institutional considerations. Our welfare analysis shows that defaulting a small portion of retirees 401(k) assets (over a threshold) is an attractive way to enhance retirement security, enhancing welfare by up to 20% of retiree plan accruals.

Horneff, V., Maurer, R., and Mitchell, O. S. (2021). "Do Required Minimum Distribution 401(K) Rules Matter, and for Whom? Insights from a Lifecycle Model." In: *SSRN e-Print*.

Tax-qualified vehicles helped U.S. private-sector workers accumulate $25Tr in retirement assets. An often-overlooked important institutional feature shaping decumulations from these retirement plans is the "Required Minimum Distribution" (RMD) regulation, requiring retirees to withdraw a minimum fraction from their retirement accounts or pay excise taxes on withdrawal shortfalls. Our calibrated lifecycle model measures the impact of RMD rules on financial behavior of heterogeneous households during their worklives and retirement. We show that proposed reforms to delay or eliminate the RMD rules should have little effects on consumption profiles but more impact on withdrawals and tax payments for households with bequest motives.

Horneff, V., Maurer, R., and Mitchell, O. S. (2023a). "Do required minimum distribution 401(k) rules matter, and for whom? Insights from a lifecycle model." In: *Journal of Banking and Finance* 154 (106941).

Tax-qualified vehicles have helped U.S. private-sector workers accumulate $33Tr in retirement plans. An often-overlooked important institutional feature shaping decumulations from these plans is the "Required Minimum Distribution" (RMD) regulation requiring retirees to withdraw a minimum fraction from their retirement accounts or pay excise taxes on withdrawal shortfalls. Our calibrated lifecycle model measures the impact of RMD rules on heterogeneous households' financial behavior during their work lives and in retirement. The model shows that reforms delaying or eliminating the RMD rules have little effect on consumption profiles, but they would influence withdrawals and tax payments for households with bequest motives.

Horneff, V., Maurer, R., and Mitchell, O. S. (2023b). "Fixed and Variable Longevity Annuities in Defined Contribution Plans: Optimal Retirement Portfolios Taking Social Security into Account." In: *SSRN e-Print*.

This paper investigates retirees' optimal purchases of fixed and variable longevity income annuities using their defined contribution (DC) plan assets and given their expected Social Security benefits. As an alternative, we also evaluate using plan assets to boost Social Security benefits through delayed claiming. We determine that including deferred income annuities in DC accounts is welfare enhancing for all sex/education groups examined. We also show that providing access to well-designed variable deferred annuities with some equity exposure further enhances retiree wellbeing, compared to having access only to fixed annuities. Nevertheless, for the least educated, delaying claiming Social Security is preferred, whereas the most educated benefit more from using accumulated DC plan assets to purchase deferred annuities.

Horneff, V., Maurer, R., and Mitchell, O. S. (2023c). "Optimal Retirement Portfolios with Fixed and Variable Longevity Annuities in Defined Contribution Plans taking Social Security into Account." In: *SSRN e-Print*.

This paper examines how two instruments-annuities with lifelong benefits purchased using defined contribution (DC) plan assets, and social security annuities-should be considered jointly to optimize household lifetime well-being. Understanding how these interact is of key importance in order to generate efficient retirement portfolios. Additionally, there is likely to be substantial heterogeneity in the demand for longevity annuities across the retiree population, depending on their assets inside and outside tax-qualified retirement plans, their mortality assumptions, and their accrued Social Security benefits. Therefore, as an alternative, we also evaluate using plan assets to boost social security benefits through delayed claiming. We determine that including deferred income annuities (DIAs) in DC accounts is welfare enhancing for all sex/education groups examined. We also show that providing access to variable deferred income annuities with some equity exposure (similar to participating annuities) further enhances retiree wellbeing, compared to having access only to fixed annuities. Nevertheless, for the least educated, delaying claiming social security benefits is preferred, whereas the most educated benefit more from using accumulated DC plan assets to purchase deferred annuities.

Horneff, W. J., Maurer, R. H., Mitchell, O. S., and Dus, I. (2008). "Following the rules: Integrating asset allocation and annuitization in retirement portfolios." In: *Insurance: Mathematics and Economics* 42(1), pp. 396–408.

Financial advisers have developed standardized payout strategies to help Baby Boomers manage their money in their golden years. Prominent among these are phased withdrawal plans offered by mutual funds including the self-annuitization or default rules encouraged under US tax law, and fixed payout annuities offered by insurers. Using a utility-based framework, and taking account of stochastic capital markets and uncertain lifetimes, we first evaluate these rules on a stand-alone basis for a wide range of risk aversion. Next, we permit the consumer to integrate these standardized payout strategies at retirement and compare the results. We show that integrated strategies can enhance retirees' well-being by 25 percent 50 percent for low/moderate levels of risk aversion when compared to full annuitization at retirement. Finally, we examine how welfare changes if the consumer is permitted to switch to a fixed annuity at an optimal point after retirement. This affords the retiree the chance to benefit from the equity premium when younger, and exploit the mortality credit in later life. For moderately risk-averse retirees, the optimal switching age lies between 80 and 85.

Hosseini, R., Kopecky, K. A., and Zhao, K. (2022). "The evolution of health over the life cycle." In: *Review of Economic Dynamics* 45, pp. 237–253.

We construct a unified objective measure of health status: the frailty index, defined as the cumulative sum of all adverse health indicators observed for an individual. Using this index, we document four stylized facts on health dynamics over the life cycle and show that they are robust to other ways of constructing the index. We also compare the frailty index with self-reported health status and find significant differences in their dynamics and ability to predict health-related outcomes. Finally, we propose and estimate a stochastic process for frailty dynamics over the life cycle accounting for mortality bias. Our frailty measure and dynamic process can be used to study the evolution of health over the life cycle and its economic implications.

Hou, W. (2020). *How Accurate Are Retirees' Assessments of Their Retirement Risk?* Tech. rep. Center for Retirement Research at Boston College.

Retirees with limited financial resources face numerous risks, including out-living their money (longevity risk), investment losses (market risk), unexpected health expenses (health risk), the unforeseen needs of family members (family risk), and even retirement benefit cuts (policy risk). This study systematically values and ranks the financial impacts of these risks from both the objective and subjective perspectives and then compares them to show the gaps between retirees' actual risks and their perceptions of the risks in a unified framework. It finds that 1) under the empirical analysis, the greatest risk is longevity risk, followed by health risk; 2) under the subjective analysis, retirees perceive market risk as the highest-ranking risk due to their exaggeration of market volatility; and 3) the longevity risk and health risk are valued less in the subjective ranking than in the objective ranking, because retirees underestimate their life spans and their health costs in late life.

Hoyle, E. (2024). "The Glidepath Confusion." In: *SSRN e-Print*.
Using long histories of financial and economic data for the UK and US, we quantify the risk associated with the asset-allocation decisions of pension savers and pensioners. When saving, our focus is on glidepaths: programmes that dictate asset allocations based on the age of the saver, or the number of working years they have left before retirement. We confirm our intuition that savers who allocate more to stocks than bonds have better investment outcomes on average, since stocks have historically generated superior real returns. We also examine the extent to which we can use our choice of glidepath to control tail risks and the variation in outcomes. This turns out to be rather limited, with the real returns of assets being the principal driver of risk. We find that a saver who is unlucky enough to experience a left-tail investment outcome is unlikely to have fared much better had they followed a different glidepath. Using bootstrapping techniques, we discover that over-allocating to stocks may be more damaging in the left tail than historical sample paths suggest. We take a novel look at the impact of fees on pension accumulation, quantifying the effect of investment performance and fee compounding on retirement wealth. Finally, we look at decumulation. We consider how both the level of income required in retirement and the choice of asset allocation can affect the probability of ruin.

Hsu, Y.-C., Lin, H.-W., and Vincent, K. (2017). *Do Cross-Sectional Stock Return Predictors Pass the Test without Data-Snooping Bias?* Tech. rep. Institute of Economics Academia Sinica.
This study examines the possible data-snooping bias as a competing explanation for the anomalies in the cross-section of stock returns. We posit that the exhaustive standalone searches for profitable strategies could lead to recommending spuriously predictive variables. In order to explore the severity of this problem, we use a multiple testing method to evaluate the profitability of portfolios constructed by these predictors. Our empirical analyses suggest that over half of the findings based on individual testing method are no longer statistically significant after we adjust for data-snooping bias. Excluding the micro-cap stocks before portfolios construction and applying the notion of economic significance in this study further weaken the evidence for predictability.

Hsu, P.-H., Han, Q., Wu, W., and Cao, Z. (2018). "Asset allocation strategies, data snooping, and the 1 / N rule." In: *Journal of Banking & Finance* 97, pp. 257–269.
Using a series of advanced tests from White's (2000) Check to correct for data-snooping bias, we assess the out-of-sample performance of various portfolio strategies relative to the naive 1/N rule. When we analyze 16 basic portfolio strategies, 126 learning strategies, and nearly 2,000 extended strategies, we find that some strategies outperform the 1/N rule in conventional tests that do not account for data-snooping bias. However, after we use the new tests that control for such bias, we find that none or very few of these strategies outperform the 1/N rule. Thus, our finding underscores the necessity to control for data-snooping bias when making asset allocation decisions.

Hu, G., Tzucker, R., and Sooben, K. (2024). "When Inflation-Protection Assets Fail to Protect." In: *The Journal of Portfolio Management* 50(9), pp. 142–150.
Inflation, the terrifying beast, was gradually put to sleep over the past 40 years. For a while, people had forgotten its presence. In fact, since 2008, most monetary and fiscal policies have been fighting deflation, not inflation. It is only natural that when it was woken up by a combination of COVID-19-induced supply chain destruction and monstrous fiscal stimulus, investors were not prepared. They soon found that most inflation-protection assets, such as Treasury inflation-protected securities (TIPS), real estate, commodities, gold, and even bitcoins lost money in that inflationary environment, mainly as a result of global central banks tightening monetary policies to fight inflation. In this article, the authors first study the rationale behind the ineffectiveness of traditional inflation-protection assets, then provide a solution based on a carefully designed structure using centrally cleared inflation swaps. With a small twist, the authors also propose a simple one-stop shop structure to match real liabilities for institutions such as pensions, endowments, and foundations. Lastly, for allocators who are concerned about the intrinsic leverage in a derivative-based strategy, the authors suggest a cheap tail risk hedging mechanism using oversupplied inflation options to mathematically contain the downside risk of our structure.

Hu, W. (2021). "A Comprehensive Study of Guaranteed Minimum Maturity Benefit and Guaranteed Minimum Death Benefit under Regime-switching Models." PhD thesis. North Carolina State University.
In this dissertation, we investigate the pricing and hedging of Guaranteed Minimum Benefits (GMBs) in life annuity products, and we extend the existing framework by assuming the underlying asset dynamics evolve under a regime-switching jump-diffusion environment. First, two most popular GMB contracts, Guaranteed

Minimum Maturity Benefit (GMMB) and Guaranteed Minimum Death Benefit (GMDB), are characterized by comprehensive quantitative models that assess both the market risk on the liability side and the revenue risk on the asset side from the viewpoint of insurers. Then, five stochastic mortality models (Cox-Ingersoll-Ross process model, Vasicek process model, Ornstein-Uhlenbeck process model, Feller process model and a two-factor mortality model) are utilized to describe the mortality risk embedded in GMBs, and their parameters are estimated by calibrating to the mortality data of the United States male population. After comparing the Akaike information criterion (AIC) and Bayesian information criterion (BIC) criteria, we find that the Feller process model fits the real data best among the five models. In addition, numeric solutions for net liabilities, fair rate of fees and Greeks (sensitivities of net liabilities with respect to model parameters) of GMMB and GMDB are derived by an accurate and fast Fourier Space Timestepping (FST) algorithm. Numerical results based on Monte Carlo simulations are also provided for comparative purpose. Moreover, an unhedged and three statically hedged portfolios which are constructed by different methods (linear algebra method, optimization method and regularization method) are introduced, and their performances are assessed by comparing the short-term and the long-term volatility, Value-at-risk (VaR) and expected shortfall (CVaR). The results indicate that, in the short-term, the portfolio constructed by optimization method performs the best and all three hedged portfolios perform better than unhedged portfolio in the long-term. Finally, these hedging results provide insurers some guidelines to hedge according to their risk tolerances. For example, if an insurer is more concerned about the short-term volatility, a linear algebra method is preferred although it requires frequent rebalancing. In addition, if one is more concerned about the transaction cost, a optimization method is more preferable and it does not need to be rebalanced every year. The regularization method is suitable for an incomplete market which only have few instruments for hedging.

Huang, H. and Milevsky, M. A. (2016). "Longevity risk and retirement income tax efficiency: A location spending rate puzzle." In: *Insurance: Mathematics and Economics* 71, pp. 50–62.

In this paper we model and solve a retirement consumption problem with differentially taxed accounts, parameterized by longevity risk aversion. The work is motivated by some observations on how Canadians de-accumulate financial wealth during retirement which seem rather puzzling. While the Modigliani lifecycle model can justify a variety of (pre-tax) de-accumulation or draw down rates depending on risk preferences, the existence of asymmetric taxes implies that certain financial accounts should be depleted faster than others. Our analysis of data from the Survey of Financial Security indicates that Canadian retirees maintain approximately two-thirds of their financial wealth in tax-sheltered accounts and a third in taxable accounts regardless of age. The ratio of taxable to tax-sheltered wealth increases slightly or remains relatively constant depending on household income which is not what one would expect from the lifecycle model. Indeed, using our model we cannot locate a plausible tax function that justifies a constant - account ratio - regardless of age. For example under flat rates taxable accounts should be depleted well before tax-sheltered accounts are ever touched. The account ratio should go to zero quite rapidly in the absence of government mandated withdrawals. We also demonstrate that under progressive income taxes withdrawals are made from both accounts but at different rates depending on account size, pension income and longevity risk preferences. Again, the - account ratio - should eventually decline. We postulate that this sort of behavior is likely due to irrational considerations linked to mental accounting, etc. It remains to be seen whether this will persist over time and under a more careful analysis of Canadian cohorts or if retirees in other countries exhibit the same behavior.

Huang, M. and Yu, S. (2020). "A new procedure for resampled portfolio with shrinkaged covariance matrix." In: *Journal of Applied Statistics* 47(44), pp. 642–652.

Dealing with estimation error is an important issue when we implement the mean-variance paradigm for portfolio construction. To tackle the problem, two approaches are proposed in literature, the portfolio resampling technique introduced by Michuad and the well-known shrinkaged covariance matrix method. There are certain evidences on the advantages of shrinkaged covariance over portfolio resampling, however, it is unclear whether a combination of the two approaches could produce a better performance compared with using shrinkaged covariance alone. In this paper, we propose a new algorithm to integrated linear or nonlinear shrinkage estimation with resampled portfolio to achieve a further improvement. Our method are demonstrated via extensive simulation and application in active portfolio management process.

Huang, N., Li, J., and Ross, A. (2020). *Housing Wealth Shocks, Home Equity Withdrawal, and the Claiming of Social Security Retirement Benefits*. Tech. rep. Centre For Research On Successful Ageing, Singapore Management University.

This paper examines the impact of changes in house prices on when eligible individuals start receiving Social Security benefits. If house prices increase, financially constrained households may draw upon the additional home equity to finance expenses and delay receipt of Social Security in order to have increased lifetime monthly benefits. To address concerns that house price changes are correlated with unobserved local demand shocks, we use a control function approach and employ two different instrumental variables. We find that individuals delay Social Security claiming when house prices increase during the housing boom. The probability of claiming within two years after becoming eligible decreases by 8.67-8.81 percent for every 10 percent increase in house prices. We also find that the total home loan amount increases in response to the price appreciation, indicating households are drawing upon their home equity to finance consumption and delay receiving Social Security.

Huaxiong, H., Milevsky, M. A., and Salisbury, T. S. (2017). "Retirement Spending and Biological Age." In: *SSRN e-Print*.

We solve a retirement lifecycle model in which the consumer's age does not move in lockstep with calendar time. Instead, biological age increases at a stochastic non-linear rate in chronological age, which one can think of as working with a clock that occasionally moves backwards in time. Our paper is inspired by the growing body of medical literature that has identified biomarkers of aging which – practically speaking – offer better estimates of expected remaining lifetime and future mortality rates. It isn't farfetched to argue that in the not-too-distant future of wearable technology, personal age will be more closely associated with biological time vs. calendar age or time. Thus, after introducing our stochastic mortality model we derive optimal consumption rates in a classic Yaari (1965) framework adjusted to our proper clock and time. In addition to the normative implications of having access to biological age, our positive objective is to partially explain the cross-sectional heterogeneity in retirement spending rates at any given chronological age. In sum, we argue that biological age is not a sufficient statistic for making economic decisions and you need information about both your ages to behave rationally.

Hubble, A. and Grable, J. E. (2019). "The efficient frontuzzle: what investment risk profiling still fails to solve." In: *The Journal of Investing* 28(6), pp. 55–72.

A global study of over 200 professional financial advisors is undertaken to test how risk profile factors are used to make investment portfolio allocation recommendations. When presented with identical risk profile information, similar to what a competent financial advisor would normally collect from a prospective client, respondents are asked to recommend a portfolio allocation among equity, fixed income, and cash for five hypothetical client scenarios. The results find that financial advisors, using their professional judgment, inconsistently puzzled together the presented risk profile factors into portfolio recommendations, on average doing little more than applying the heuristic 100-minus-age rule to recommendations. These troubling results highlight the regulatory need for uniform risk profile evaluation guidance in fiduciary contexts.

Hugonnier, J., Pelgrin, f., and St-Amour, P. (2017). "Closing down the shop: optimal health and wealth dynamics near the end of life." In: *SSRN e-Print*.

Near the end of life, health declines, mortality risk increases and curative is replaced by uninsured long-term care, accelerating the fall in wealth. Whereas standard explanations emphasize inevitable aging processes, we propose a com- plementary closing down the shop justification where agents decisions affect their health and the timing of death. Despite preferring to live, individuals optimally deplete their health and wealth towards levels associated with high death risk and indifference between life and death. Reinstating exogenous aging processes reinforces the relevance of closing down. Using HRS data for elders, a structural estimation of the closed-form decisions identifies and tests conditions for these strategies to be optimal and confirm their economic relevance. We also discuss why policy intervention to reduce the incidence of closing down, although feasible, is not warranted.

Hunt, A. and Blake, D. P. (2020). "Basis Risk and Pensions Schemes: A Relative Modelling Approach." In: *SSRN e-Print*.

For many pension schemes, a shortage of data limits their ability to use sophisticated stochastic mortality models to assess and manage their longevity risk. In this study, we develop a relative model for mortality, which compares the evolution of mortality rates in a sub-population with that observed in a larger reference population. We apply this relative approach to data from the CMI Self-Administered Pension Scheme study, using UK population data as a reference. We then use the relative approach to investigate the potential differences in the evolution of mortality rates between these two populations and find that, in many practical situations, basis risk is much less of a problem than is commonly believed.

Hwang, I., Xu, S., and In, F. (2018). "Naive versus optimal diversification: Tail risk and performance." In: *European Journal of Operational Research* 265(1), pp. 372–388.

It is well documented in portfolio optimization that naive diversification outperforms optimal mean-variance diversification because the latter is subject to severe estimation error. Our study provides an alternative explanation for the outperformance of naive diversification by examining the tail risk of naive diversification relative to optimal mean-variance diversification. We utilize a rolling-sample approach and compare the out-of-sample performance and tail risk of various optimal strategies to that of the naive diversification strategy. Using portfolios consisting of individual stocks, we show that for portfolios containing relatively small number of stocks, naive diversification outperforms optimal mean-variance diversification and is less exposed to tail risk. However, for relatively large number of stocks in the portfolio, naive diversification maintains its superior performance but increases tail risk and results in more concave portfolio returns. These results imply that the outperformance of naive diversification acts as compensation for the increase in tail risk and concavity.

Hyams, S. D., Smith, A. E., Squirrell, C. M., Warren, G. J., Warren, O. H., and Willetts, P. J. (2020). "Saving for retirement: rules of thumb." In: *British Actuarial Journal* 25 (e7).

Rules of thumb (RoTs) are proposed as a means of promoting higher levels of Defined Contribution (DC) pension saving and to help stimulate debate about the high and uncertain cost of pension provision, leading to the development of solutions. The Lifetime Pension Contribution (LPC) tells young people what pension contribution is required over a full working life to achieve a decent retirement income, calculated as 23% of average UK earnings. Another RoT is that each 1% of earnings provides a pension of 1.5% of earnings. Other RoTs show how costs vary by retirement age and if the saver's retirement planning is on track. The current high cost of pensions is partly due to low interest rates and the inefficiencies of the DC market, with inadequate bulk purchasing power and risk sharing. RoTs might help encourage higher employer contributions, either through automatic enrolment or on a voluntary basis.

Hyndman, R. J. (2023). "Lee-Carter models: The wider context." In: *International Journal of Forecasting* 39(3), pp. 1053–1054.

The Lee-Carter model can be considered a special case of some larger model classes that have been well-studied by statisticians and econometricians. For example, it is a special case of a dynamic factor model. When x is a continuous variable (such as age), the model can be considered within the framework of functional data model Brouhns, Denuit, and Vermunt (2002) showed that it is helpful to embed a Lee-Carter type model within a Generalized Linear Modelling (GLM) framework to account for the count nature of deaths. Once we realise that a Lee-Carter model is a special case of a functional time series factor model, it becomes clear that it will also be useful to consider these within a more general GLM-like framework.

Iannarone, N. G. (2018). "Rethinking automated investment adviser disclosure." In: *SSRN e-Print*.

Today, investment advisers, including robo-advisers, push dense disclosures to consumer investors who may not be able to fully comprehend their import. Tomorrow, robo-advisers and their innovative technology, if embraced by regulators and consumers, can facilitate and cultivate a trusted collaborative relationship between investor and adviser where the adviser learns from the investor and provides education tailored to the investor, helping to solve vexing consumer protection issues and lessen the impending retirement crisis. More and more of us are comfortable making decisions without human involvement and entirely through a digital interface. Despite consumers increasing comfort relying upon machines to direct and guide their decisions, most Americans nevertheless lack basic foundational knowledge in finance and investing. Even those Americans who are able to save for retirement lack financial sophistication and may not be able to understanding the complex products in which they are invested. Robo advice may not be the solution for these problems, as robo-advisers present a potential black box problem where the digital investment advisers may not be capable of explanation beyond the information input and advice output. In such a world, it is not only the new technology that should be questioned. The existing standards through which the new technology is regulated should similarly be explored. An understanding of potential benefits and potential harms may provide lessons that better allow us to protect consumer investors and better prepare for the so-called retirement crisis. This essay explores the parameters of the pre-existing regulatory regime as applied to a new landscape of investing with a focus on the key regulatory tool of disclosure. This short essay examines the challenges of delivering appropriate disclosures in a new world of robo-advisory advice. It suggests a reconceptualization of the aim of disclosure to capitalize on robo-advisers ability to truly know their clients and shift the burden of investor education from the customer to a fiduciary level adviser capable of tailoring advice and education to each investor at a level she is capable of understanding. The essay concludes with a recommendation: robo-advisers and investor collaboration via an active and iterative disclosure regime that encourages investor comprehension, understanding, and learning.

Iborra, R. and Chabane, I. (2020). "Strategic Currency Hedging in Multi-Asset Portfolios." In: *The Journal of Investing*.
The authors provide a strategic currency hedging framework for multi-asset portfolios, helping global investors better manage foreign currency exposures by balancing hedging costs and risk. The authors focus primarily on G4 currencies-US dollar (USD), Euro (EUR), Pound sterling (GBP), and Japanese yen (JPY)-but also provide some insights for other G7 currencies. Starting from a single-asset-class perspective, the framework shows that developed market fixed-income instruments should generally be fully hedged back to the investor currency. US investors should fully hedge foreign exchange risk when buying European equities. Japanese equities should be left unhedged for all investors due to the Japanese yen value as a safe haven during market panics. Conversely, Japanese investors buying international equities should fully hedge currency risk but may find it optimal to leave some of their US bond exposures unhedged to minimize hedging costs while retaining some diversification benefits. The picture is also nuanced for European investors where the cost-benefit trade-off points to partial hedging as the optimal approach. When it comes to multi-asset investing, hedge ratio optimality also becomes a function of cross-asset correlations, which leads to interesting nuances of approach across domiciles. The authors believe their research provides global investors with useful information to calibrate strategic hedge ratios.

Idzorek, T., Stempien, J., and Voris, N. (2013). "Bait and Switch: Glide Path Instability." In: *The Journal of Investing* 22(1), pp. 74–82.
With target date funds widely expected to account for more than 50 percent of all defined contribution assets in the near future, it is difficult to overstate their importance on the retirement security of America. Yet there are very few tools and statistics for monitoring and evaluating target date funds. This article highlights a previously unrecognized characteristic of target date fund families: Glide paths aren't necessarily stable over time, and in some cases change dramatically. We document the stability, or lack of stability, of the glide paths for the largest target date fund families and introduce quantitative measures of glide path stability, including our preferred measure, the Glide Path Stability Score.

Idzorek, T. M. (2023). "Personalized Multiple Account Portfolio Optimization." In: *Financial Analysts Journal* 79(3), pp. 155–170.
I develop a multi-account alpha-tracking error framework that simultaneously optimizes across an investor's multiple accounts with different tax treatments, existing holdings, tax lots, and opportunity sets while considering taxes and trade costs in a single optimization. The objective function includes an optional term for an investor's nonpecuniary preferences, such as various environmental, social, and governance (ESG) characteristics. By running the multi-account optimizer regularly, it also serves as a personalized asset location optimizer, tax-loss harvester, portfolio rebalancer, roll-over optimizer, and new client onboarding transition optimizer that simultaneously considers the numerous interconnected tradeoffs to produce ongoing personalized portfolio management.

Ielpo, f., Merhy, C., and Simon, G. (2017). *Engineering Investment Process: Making Value Creation Repeatable*. Elsevier. 430 pp.
The book explores the quantitative steps of a financial investment process. The authors study how these steps are articulated in order to make any value creation, whatever the asset class, consistent and robust. The discussion includes factors, portfolio allocation, statistical and economic backtesting, but also the influence of negative rates, dynamical trading, state-space models, stylized facts, liquidity issues, or data biases. Besides the quantitative concepts detailed here, the reader will find useful references to other works to develop an in-depth understanding of an investment process.

Ilmanen, A. and Maloney, T. (2015). "Portfolio Rebalancing, Part 1: Strategic Asset Allocation." In: *SSRN e-Print*.
This paper shows that many reasonable rebalancing processes may achieve similar benefits of maintaining a portfolio's risk characteristics, but also that price momentum effects can benefit some processes more than others. The authors seek to confirm that rebalancing to target allocations helps to maintain more stable portfolio risk characteristics and improved diversification, and to show that many different methods confer similar long-term risk benefits. Rebalancing can be considered an active contrarian strategy when compared to a buy-and-hold benchmark. The authors discuss the main considerations when designing a rebalancing process, and use a simple empirical analysis to demonstrate the drivers of relative performance over four decades. Asset prices have tended to exhibit trending behavior, and this has favored less-frequent rebalancing schedules or wider tolerance bands, which allow trends to play out between rebalances . When designing a rebalancing process, investors should balance their tolerance for short-term variations in portfolio weights against their expectation or belief that

multi-month price momentum effects will continue in the future. For liquid portfolios, cost considerations may be secondary.

Ilmanen, A. and Rauseo, M. (2018). "Intelligent Risk Taking." In: *How Persistent Low Returns Will Shape Saving and Retirement*. Oxford University Press.

Peoples' ability to consume in retirement is a function of how much they save, how they invest, and what those investments return over the lifecycle. This chapter explores what rate of return is needed to deliver a comfortable retirement based on current savings rates, as well as intelligent ways to construct portfolios to achieve this rate of return. Based on reasonable long-term return assumptions, defined contribution portfolios as frequently constructed today are unlikely to achieve this required rate of return. By relaxing existing constraints and taking advantage of well-known and broadly accepted investment themes, this required rate of return can be achieved with a well-diversified portfolio, which may lead to a more consistent portfolio across different economic environments.

Irlam, G. (2020a). *AACalc Asset Allocation Calculator*. URL: `https://www.aacalc.com/`.

There are many calculators on the web. Unfortunately they all produce different results. We take a scientific approach, explain how our results were derived, and admit what is uncertain.

Irlam, G. (2020b). *AI Planner*. URL: `https://www.aiplanner.com/`.

AIPlanner uses deep reinforcement learning to compute a near optimal spending and investment strategy to maximize lifetime well-being. AIPlanner is optimized for the U.S. tax code.

Irlam, G. (2020c). "Lifetime Portfolio Selection: Using Machine Learning." In: *SSRN e-Print*.

The first half of this paper provides a primer on machine learning and relates it to problems in financial planning. Machine learning proves capable of handling real world complications and details which classical financial planning approaches cannot handle; e.g. taxes, reversion to the mean, and time varying yield curves. The second half of this paper provides a case study of the use of reinforcement learning for making asset allocation, annuitization, and consumption decisions in retirement planning. Machine learning typically delivers within a few percent of the theoretical optimal solution on highly abstract problems. Machine learning is found to outperform other approaches for more complex financial planning problems whose optimal solution is not known. The value of SPIAs and inflationindexed bonds is highlighted by the machine learning approach.

Irlam, G. (2020d). "Machine learning for retirement planning." In: *The Journal of Retirement* 8(1), pp. 32–29.

Machine learning provides a new approach to solving problems in many fields. This article explores the use of machine learning to solve the retirement portfolio problem: deciding how much wealth to consume and how to allocate the remainder. After first reviewing existing approaches to the portfolio problem, this article looks in detail at the use of reinforcement learning. For simple financial scenarios where the optimal solution is known, reinforcement learning is found to deliver to within a few percent of the optimal solution. For more complicated financial scenarios, with no known optimal solution, reinforcement learning outperforms other common approaches. Reinforcement learning proves capable of optimizing highly complex financial models, including the effects of income taxes, mean-reverting asset classes, and time-varying bond yield curves, all of which other approaches cannot handle. Reinforcement learning appears to be the first fundamentally new approach to the portfolio problem in over 50 years.

Irlam, G. (2020e). "Multi Scenario Financial Planning via Deep Reinforcement Learning AI." In: *SSRN e-Print*.

Financial planning via deep reinforcement learning holds much promise. One implementation, AIPlanner, delivered near optimal financial results, but had a major shortcoming. It required training a separate neural network model for each financial scenario. This paper describes extending AIPlanner so that a small family of trained neural network models are capable of rapidly producing financial plans for a wide variety of financial scenarios. Additionally AIPlanner is extended to produce results over the lifecycle, both pre and post retirement, and for couples, as well as individuals. A reasonably realistic income tax model is incorporated. And finally, a more realistic stock model is used. Over the lifecycle, compared to the best discovered alternative strategy, reinforcement learning was found to effectively deliver 14% more retirement consumption.

Irlam, G. and Tomlinson, J. (2014). "Retirement Income Research: What Can We Learn from Economics?" In: *The Journal of Retirement* 1(4), pp. 118–128.

Research on retirement income planning has, for many years, followed two separate tracks. Financial planning practitioners have developed guidelines for withdrawals from savings and asset allocation by testing and fine-tuning various rules of thumb. Economists have applied a different approach, based on life-cycle finance, aimed at maximizing the utility of lifetime consumption, using dynamic programming techniques to optimize retirement withdrawals and asset allocations. This article seeks to use a non-mathematical explanation to help financial planners and others who are not familiar with the economics approach to develop a conceptual understanding of how life-cycle finance and dynamic programming can be applied to retirement income planning. It also compares the performance of optimized recommendations produced by dynamic programming with various rule-of-thumb strategies that have been popular in the planning literature. It attempts to demonstrate the power that the economics approach can bring to improving retirement income planning and argues that more communication between economists and practitioners can help open up new approaches to research and improved practical applications.

Iskhakov, F., Thorp, S., and Bateman, H. (2015). "Optimal Annuity Purchases for Australian Retirees." In: *Economic Record* 91(293), pp. 139–154.

We develop and simulate a stochastic lifecycle model to investigate optimal annuity purchases at retirement. Retirees can invest in risky assets, purchase fairly priced immediate or deferred lifetime annuities, and are eligible for a targeted safety net pension. We match baseline parameters to current Australian settings and conduct scenario analyses over a wide range of individual preferences and financial market outcomes. Except where individuals need to insure a consumption floor, both immediate and deferred annuity purchases are largely crowded out by the means-tested public pension. Welfare losses caused by zero annuitisation are small compared with the losses caused by completely annuitising all savings, particularly if wealth at retirement is low. Decumulation policy should ensure individuals are well informed of the insurance value of annuities and accommodate diverse choices.

Israelov, R. (2019). "Pathetic protection: the elusive benefits of protective puts." In: *The Journal of Alternative Investments*.

Conventional wisdom is that put options are effective drawdown protection tools. Unfortunately, in the typical use case, put options are quite ineffective at reducing drawdowns versus the simple alternative of statically reducing exposure to the underlying asset. This article investigates drawdown characteristics of protected portfolios via simulation and a study of the CBOE SandP 500 5% Put Protection Index. Unless option purchases and their maturities are timed just right around equity drawdowns, they may offer little downside protection. In fact, they could make things worse by increasing rather than decreasing drawdowns and volatility per unit of expected return.

Israelov, R. and Ndong, D. N. (2024). "Equity Tail Protection Strategies Before, During, and After COVID-19." In: *The Journal of Portfolio Management* 50(4), pp. 53–74.

Tail risk is usually an important consideration for investors, and a desire to limit catastrophic loss has led to significant interest in protection strategies. Across four distinct market periods surrounding the COVID-19 pandemic, the authors explore four common tail-risk mitigation strategies: (1) a put protection strategy, (2) a long volatility protection strategy using options, (3) a long VIX futures protection strategy, and (4) a long VIX call overlay. The authors' analysis found that three main themes emerge. First, hedging is expensive. Second, the variable equity exposure embedded in option strategies is a source of risk and path dependence. Finally, a hedger's decision on whether to delta-hedge their option exposure to isolate the option convexity, or to maintain an unhedged position, materially impacts performance in nonforecastable ways. The authors also acknowledge the number of well-reasoned arguments both in favor and against implementing tail-risk hedging strategies. They find that the dispersion of outcomes across only the few strategies explored in this article is notable and that there is likely no easy solution for tail-risk hedging. Those who implement protection strategies should plan for the possibility that their hedges make things worse in times of stress.

Israelov, R. and Nielsen, L. N. (2015). "Still not cheap: portfolio protection in calm markets." In: *SSRN e-Print*.

Recent equity volatility is near all-time lows. Option prices are also low. Many analysts suggest this represents a good opportunity to purchase put options for portfolio insurance. It is well-known that portfolio insurance is expensive on average, but what about in calm markets? History suggests it still is. We investigate the relationship between option richness and volatility across ten global equity indices. Option prices may be low, but their expected values tend to be even lower.

Israelov, R. and Tummala, H. (2018). "An Alternative Option to Portfolio Rebalancing." In: *The Journal of Derivatives* 25(3), pp. 7–32.

Portfolio managers, whether running a passive or an active strategy, typically select target weights on the various asset classes and securities they hold. The prime example is choosing the mix between stocks and bonds. But as market prices evolve, actual weights drift away from the targets. This leads to tracking error and transaction costs in order to rebalance positions to the desired levels. Rebalancing entails selling assets that have gone up in value and increasing exposure to those whose prices have dropped. As Israelov and Tummala point out, this is how a short position in a call option behaves, and they show how to design a hedging strategy based on writing calls that offsets much of the basis risk in trying to run a portfolio with fixed asset weights. An empirical illustration shows that their proposed overlay strategy would have performed well historically in reducing unintended timing risk in a 60/40 stock/bond portfolio. An additional benefit of the plan is that by selling options, it harvests the volatility risk premium embedded in their prices, which enhances returns at the same time risk is being reduced.

Israelsen, C. L. (2017). "Retirement Portfolio Realities: The Mathematics of Survival." In: *International Journal of Trade, Economics and Finance* 8(4), pp. 194–197.

This paper deals with several important retirement questions: "How much money do I need in my investment portfolio at the start of retirement?" and "How much can I safely withdraw from my investment portfolio during the retirement years?" This paper introduces the novel concept of "RAM" or Retirement Account Multiple. The mathematics of income replacement in retirement are demonstrated and the historical survival of two different retirement portfolios are reviewed over a 90-year period from 1926-2015.

Ivanov, K. and Tian, W. (2023). "Optimal Early Retirement with Target Wealth." In: *SSRN e-Print*.

Conventional wisdom often recommends that retirees accumulate 10 to 20 times their annual income for retirement. We examine the validity of this dominating financial advice from a portfolio choice perspective while characterizing the optimal retirement time with a target wealth level. Utilizing a life-cycle model, we find that the early retirement effect reduces consumption, and the retiree invests an increasing percentage of wealth in stocks with more wealth. This conventional wisdom is only optimal for a short expected lifetime, a sharp decline of post-retirement retirement, or an aggressive agent. Moreover, the model predictions explain varying target wealth for retirement across different generations.

Jaconetti, C. M., DiJoseph, M. A., Jr., F. M. K., Pakula, D., and Lobel, H. (2020). *From assets to income: A goals-based approach to retirement spending*. Tech. rep. Vanguard.

Although the population and life expectancies of U.S. retirees are increasing, portfolio yields remain at historically low levels. As defined benefit income becomes less commonly available, the need for informed retirement portfolio spending strategies is more critical, and yet more complex, than ever. The stakes are high, and the impact of subpar decisions can be severe. Because every investor's financial situation is unique, there is no one-size-fits-all solution. But developing and implementing a personal spending strategy can reduce anxiety and stress about the ability to meet retirement income goals. Retirees who hold the majority of their assets in tax-deferred accounts can turn thoseassets into income by setting up an automatic withdrawal plan. They can also purchase an investment specifically designed to provide regular distributions. Those whose portfolios contain a significant portion of taxable assets can add value by working with an advisor to develop a goals-based strategy. Whatever spending strategy you choose, the complexity and consequences of the process underscore the need for and value of skillful guidance.

Jaeger, M., Krugel, S., Marinelli, D., Papenbrock, J., and Schwendner, P. (2020). "Understanding machine learning for diversified portfolio construction by explainable AI." In: *SSRN e-Print*.

In this paper, we construct a pipeline to investigate heuristic diversification strategies in asset allocation. We use machine learning concepts ("explainable AI") to compare the robustness of different strategies and back out implicit rules for decision making.In a first step, we augment the asset universe (the empirical dataset) with a range of scenarios generated with a block bootstrap from the empirical dataset.Second, we backtest the candidate strategies over a long period of time, checking their performance variability. Third, we use XGBoost as a regression model to connect the difference between the measured performances between two strategies to a pool of statistical features of the portfolio universe tailored to the investigated strategy. Finally, we employ the concept of Shapley values to extract the relationships that the model could identify between the portfolio characteristics and the statistical properties of the asset universe. We test this pipeline for studying risk-parity strategies with a volatility target, and in particular, comparing the machine learning-driven Hierarchical Risk Parity (HRP) to the classical Equal Risk Contribution (ERC) strategy.In the augmented dataset built from a multi-asset investment universe of commodities, equities and fixed income futures, we find that HRP better matches the volatility target, and shows better risk-adjusted performances. Finally, we train XGBoost to learn the difference between the realized Calmar ratios of HRP and ERC and extract explanations.The explanations provide fruitful ex-post indications of the connection between the statistical properties of the universe and the strategy performance in the training set. For example, the model confirms that features addressing the hierarchical properties of the universe are connected to the relative performance of HRP respect to ERC.

Jang, C., Clare, A., and Owadally, I. (2022). "Glide paths for a retirement plan with deferred annuities." In: *Journal of Pension Economics and Finance*, pp. 1–17.

We construct investment glide paths for a retirement plan using both traditional asset classes and deferred annuities (DAs). The glide paths are approximated by averaging the asset proportions of stochastic optimal investment solutions. The objective function consists of power utility in terms of secured retirement income from purchased DAs, as well as a bequest that can be withdrawn before retirement. Compared with conventional glide paths and investment strategies, our DA-enhanced glide paths provide the investor with higher welfare gains, more efficient investment portfolios and more responsive retirement income patterns and bequest levels to different fee structures and personal preferences.

Jang, C., Owadally, I., and Clare, A. D. (2024). "Life-Cycle Investment and Housing Decisions with Longevity Annuities and Reverse Mortgage." In: *SSRN e-Print*.

We investigate optimal life-cycle consumption, investment, and housing decisions with longevity annuities and reverse mortgage. A risk-averse investor can hold financial assets, such as cash, bonds, and stocks, and can invest in housing, through renting and purchasing. Variable-rate mortgages are available. At retirement, the investor can release his housing equity through a reverse mortgage product. Longevity annuities are also available to support his income at advanced ages. We use multi-stage stochastic programming to solve the optimization problem numerically. Our numerical results show that longevity annuities can enhance non-housing consumption in retirement and the proceeds from reverse mortgage may raise not only housing and non-housing consumption in retirement but also non-housing consumption prior to retirement. Reverse mortgage also changes housing preference from renting to owning with a lower regular mortgage debt. Longevity annuities can help different risk-averse investors choose a suitable consumption stream in retirement. When social security cannot guarantee a high level of the replacement ratio, our model finds that longevity annuity and reverse mortgage can prevent housing and non-housing consumption drops in retirement. The housing ownership preference over rental however becomes more intense.

Janssen, R., Kramer, B., and Boender, G. (2013). "Life Cycle Investing: From Target-Date to Goal-Based Investing." In: *The Journal of Wealth Management* 16(1), pp. 23–32.

The authors propose the use of goal based investing- or private ALM, as they prefer to call it- to tailor a dynamic investment strategy to the needs of individual clients. They argue that this approach is superior to the - one-size-fits-all,- target-date-oriented static allocation path used in most current life cycle funds. They present the two pillars of their approach: the methodology for obtaining financial and economic scenarios, and the methodology of the goal-oriented dynamic allocation strategy. This approach reduces the risks and improves the feasibility of meeting the clients' goals.

Jennings, W. W. (2023). "Toward a Literature of Wealth Management." In: *The Journal of Wealth Management* 26(1), pp. 39–48.

We characterize the wealth management literature that The Journal of Wealth Management helped define over the last quarter century and then assess its impact. Eight key areas of emphasis include goals-based asset allocation, tax barriers, the extended portfolio, asset location, after-tax asset allocation, tax-loss harvesting, concentrated portfolios, and qualitative matters. The Journal of Wealth Management has contributed valuably in each of these areas and is increasingly well regarded among academic researchers. Some gaps in the literature, however, remain.

Jennings, W. W. and Payne, B. C. (2020). "Corrections Should Trigger Rebalancing." In: *The Journal of Wealth Management* 23(33), pp. 37–49.

We start from the provocative contention that if an investor did not rebalance earlier this year (2020), then that investor does not have an asset allocation policy. More specifically, that investor does not have a rebalancing policy to preserve their asset allocation. We then work backward from a presumption that a correction needs to induce a rebalancing trade and develop an algebraic solution for the width of the rebalancing range to ensure it does so. Across a range of reasonable assumptions, rebalancing ranges need to be tighter than traditional norms.

Jennings, W. W. and Reichenstein, W. R. (2001). "Estimating the Value of Social Security Retirement Benefits." In: *The Journal of Wealth Management* 4(3), pp. 14–29.

The authors start with a simple question: what is the value of an individual's assets that can be used to satisfy retirement income needs, focusing more specifically on Social Security benefits? Consistent with an approach introduced in earlier works published in The Journal of Wealth Management, they then ask how these benefits affect the individual's current asset mix. In particular, they delve further into earlier conclusions that the present value of projected Social Security payments be included as a 'bond' in personal portfolios. They note that individuals' portfolios are usually substantially different when the value of Social Security is included than when it is excluded, and demonstrate that the profession has been miscalculating individuals' "true"portfolios by excluding Social Security. If individuals optimize their traditional portfolios, which exclude Social Security, then they have excessively conservative, sub-optimal true -portfolios.

Jensen, B. A. (2023). "A Note on a (Not so) Well-Known Annuity Formula." In: *SSRN e-Print*.
This note gives a simple derivation of the Taylor series approximation to annuity payments without using differential calculus. Additionally we give the similar approximation to the yield/APR in an annuity contract. These approximations are demonstrated both numerically and graphically.

Jeon, J., Kwak, M., and Park, K. (2021). "Horizon effect on optimal retirement decision." In: *SSRN e-Print*.
We study an optimal consumption, investment, life insurance, and retirement decision of an economic agent who has an option to retire early any time before the mandatory retirement date. We conduct a thorough theoretical analysis for the optimal retirement problem with general utility function in the presence of mandatory retirement date, which leads to the optimal stopping problem in finite horizon. Furthermore, different marginal utility of consumption before and after retirement is considered, which can provide an explanation for the retirement-consumption puzzle, while it makes the problem technically more challenging. Based on the theory of partial differential equation, we analyze the variational inequality arising from the dual problem and establish the duality theorem. We show that the optimal retirement decision is determined by the time-varying optimal retirement wealth boundary, and we provide an integral equation representation for the optimal retirement wealth boundary, which can be solved accurately and efficiently by using recursive integration method. As an extension, the case with stochastic labor income is also considered. The properties of the optimal strategies are provided with emphasis on the role of the mandatory retirement date and the impact of having a retirement option on the optimal financial decisions.

Jeon, J., Kwak, M., and Park, K. (2023). "Horizon effect on optimal retirement decision." In: *Quantitative Finance* 23(1), pp. 123–148.
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Jiang, S., Zuo, W., Guo, Z., and Tuljapurkar, S. (2022). "When to Claim a Pension: The Effect of Uncertainty in Ages at Death." In: *SSRN e-Print*.
When should you claim a pension (such as Social Security) in the US? Much current advice focuses on the increase in the annual pension benefit as individuals delay claiming a pension, an increase due to the actuarial fairness of the system. However, individuals face two risks: whether they live to the planned claiming age, and how long they live after they claim. Here we show that for both individuals and couples, the coefficient of variation of the (combined) lifetime benefit, calculated as the ratio of the standard deviation to the average, increases as individuals/couples delay claiming, indicating a rising risk relative to benefit. We use a utility analysis to find an optimal portfolio under different risk aversion levels. We conclude that individuals and couples should consider both average benefit and risk when deciding on a claiming age, as with other investment decisions.

Johnston, K., Hatem, J., Carnes, T., and Kosedag, A. (2019). "An empirical evaluation of dynamic vs static withdrawal strategies." In: *Managerial Finance* 45(12), pp. 1509–1525.
Purpose The purpose of this paper is to compare simple dynamic withdrawal strategies with the static withdrawal method, examining not only failure rates and ending wealth but also spending. All withdrawal strategies are adjusted for the Internal Revenue Service's (IRS) required minimum distribution (RMD). In addition, this study investigates the use of small company stocks (SCS) in place of large company stocks (LCS). Results indicate SCS portfolios are superior to large. When returns are poor, some dynamic strategies will not ensure income for life. This study demonstrates that the simplest dynamic strategy is superior to two popular dynamic strategies. Design/methodology/approach Using historical overlapping periods, different withdrawal strategies

are examined. Previous studies focused on failure rates and ending wealth. As discussed in Milevsky (2016) different statistical distributions can have similar tail properties (prob of failure) but dissimilar risk and return profile. The detailed examination of both spending and use of small stocks advances the literature in this area. Findings Results indicate that use of small stocks is superior to using large stocks in the portfolios. When US historical stock returns are adjusted downward, there is the potential that some dynamic strategies will not ensure income for life. This study demonstrates that the simplest dynamic strategy is superior to two popular dynamic strategies. Originality/value This paper is the first to examine, in detail, annual spending results for the retiree. Second, it is shown that, overall, SCS are superior to LCS for all stock/bond allocations. Even though absolute downside risk increases slightly, this increase in downside risk is dominated by the upside potential. In other words, the positive skewness of small stock returns along with the cumulative effects of compounding at a higher rate increases both the available wealth for spending and ending wealth. Third, IRS's RMDs are taken into account for every withdrawal strategy examined. Lastly, it demonstrates that the simplest dynamic strategy is superior to two popular dynamic strategies.

Jones, M. W. (2014). "Seeking diversification through efficient portfolio construction using cash-based and derivative instruments." In: *British Actuarial Journal* 19, pp. 468–498.

Diversification across asset classes has declined markedly in the last decade and concerns abound about the ability (or lack) of the financial system to weather another 2008-like event. There is a growing volume of academic literature seeking to derive measurement variables for detecting systemic risk. There is clearly a role for the actuarial profession within this discussion and we hope that this paper can engender further discussions and papers on this specific issue. Just as diversification across conventional asset classes has decreased there has been greater attention paid to the broader array of potential investment strategies that can be accessed via derivatives. The paper explores the prudent management of derivatives in pension funds and general asset portfolios to improve portfolio efficiency both in terms of implementing investment strategies and in broadening the range of investment opportunities for building an efficient investment portfolio from a risk-based perspective.

JP Morgan Asset Management Research (2021). *Guide to Retirement*. Tech. rep. JP Morgan Asset Management.

Updated annually, the Guide to Retirement provides an effective framework for supporting retirement planning conversations with clients. It includes charts and graphs to help you explain complex topics in a clear and concise manner. A description and audio commentary are available for every slide.

Jung, B. (2017). *Your advisor...worth more than 1%?* Tech. rep. Russell Investments.

Value of an advisor is still more than 1% As a follow up to my previous blog post on the value of an advisor, I wanted to provide you with even more evidence that suggests your advisory fee is a very good deal for the value you deliver to your clients.

Jung, J. (2020). *Estimating Markov Transition Probabilities Between Health States Using U.S. Longitudinal Survey Data*. Tech. rep. Towson University.

We use data from two representative U.S. household surveys, the Medical Expenditure Panel Survey (MEPS) and the Health and Retirement Study (Rand-HRS) to estimate Markov transition probability matrices between health states over the lifecycle from age 20-95. We use non-parametric and parametric methods and control for individual characteristics such as age, gender, race, education, income as well as cohort effects. We align two year transition probabilities from HRS with one year transition probabilities in MEPS using a stochastic root method. We find that the non-parametric counting method and the regression specifications based on ordered logit models produce similar results over the lifecycle. However, the counting method overestimates the probabilities of transitioning into bad health states. In addition, we find that young women have worse health prospects than their male counterparts but once individuals get older, being female is associated with transitioning into better health states with higher probabilities than men. We do not find significant differences of the conditional health transition probabilities between African Americans and the rest of the population. We also find that the lifecycle patterns are stable over time. Finally, we discuss issues with controlling for time effects, sample attrition, and other modeling issues that can arise with categorical outcome variables.

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Kabuche, D., Sherris, M., Villegas, A. M., and Ziveyi, J. (2024). "Pooling functional disability and mortality in long-term care insurance and care annuities: A matrix approach for multi-state pools." In: *Insurance: Mathematics and Economics* 116, pp. 165–188.

Mortality risk sharing pools including group self-annuitisation, pooled annuity funds and tontines have been developed as an effective solution for managing longevity risk. Although they have been widely studied in the literature, these mortality risk sharing pools do not consider individual health or functional disability status nor the need for long-term care (LTC) insurance at older ages. We extend these pools to include functional disability and chronic illness and present a matrix-based methodology for pooling mortality risk across heterogeneous individuals classified by functional disability states and chronic illness statuses. We demonstrate how individuals with different health risks can more equitably share mortality risk in a pooled annuity design. A multi-state pool is formed by pooling annuitants considering both longevity and LTC risks and determining the actuarially fair benefits based on individuals' health states. Our methodology provides a general structure for a pooled annuity product that can be applied for general multi-state models. We present an extensive analysis with numerical examples using the US Health and Retirement Study (HRS) data. Our results compare expected annuity benefits for individuals in poor health to those in good health, show the effects of incorporating systematic trends and uncertainty, assess how the valuation of the expected annuity payments interacts with the assumptions used for the multi-state model and assess the impact of pool size.

Kahler, J. R., Clarke, A., and Bruno, M. A. (2020). *HSAs: An off-label prescription for retirement saving*. Tech. rep. Vanguard.

Choosing a health insurance plan involves a complex analysis of premiums, deductibles, out-of-pocket maximums, and tax costs. The right choice depends on an individual's policy options, budget, and expected health care needs.The rapid growth of high-deductible health plans (HDHPs) and health savings accounts (HSAs) adds more complexity to the decision but also creates a unique opportunity to save for retirement and other long-term goals. We explore these underappreciated savings opportunities by reviewing the what, why, and how of HSAs. 1) The what HSAs are tax-sheltered savings accounts available to those enrolled in a high-deductible health plan (HDHP). An HDHP charges lower premiums than traditional insurance plans but comes with higher deductibles and out-of-pocket maximums. An HSA provides tax benefits to help defray these higher costs. 2) The why In tax-advantaged accounts such as IRAs, 401(k) plans, and 529 college savings plans, you pay taxes now or later. With an HSA, the "now" or "later" can become "never." These tax savings can compound to produce higher returns than those available from other accounts 3) The how The how depends on how much you can afford to save. For those with enough savings capacity, the best strategy is to treat the HSA as a long-term investment account by paying for current medical expenses out of pocket. If you have less capacity to save, the decision is more complex, involving both investment and behavioral considerations.

Kan, R. and Wang, X. (2024). "Optimal Portfolio Choice with Unknown Benchmark Efficiency." In: *Management Science* 70(9), pp. 5627–6482.

When a benchmark model is inefficient, including test assets in addition to the benchmark portfolios can improve the performance of the optimal portfolio. In reality, the efficiency of a benchmark model relative to the test assets is ex ante unknown; moreover, the optimal portfolio is constructed based on estimated parameters. Therefore, whether and how to include the test assets becomes a critical question faced by real world investors. For such a setting, we propose a combining portfolio strategy, optimally balancing the value of including test assets and the effect of estimation errors. The proposed combining strategy can work together with some existing estimation risk reduction strategies. In both empirical data sets and simulations, we show that our proposed combining strategy performs well. The replication files for this article are available.

Kaplan, P. D. (2006). "Asset Allocation with Annuities for Retirement Income Management." In: *The Journal of Wealth Management* 8(4), pp. 27–40.

The author starts with a statement of the problem: at, in, or near retirement, investors need to make specific decisions about how they are going to use their savings to generate income during retirement and meet their estate goals when they die. If they withdraw too much each year for spending needs, they face the risk of running out of assets before they die. This is often called risk. If they withdraw too little, they may not be able to meet their immediate expenses and support a certain lifestyle. Investors also need to decide how to invest their savings among asset classes and annuities. The author explores retirement income solutions in a simple setting to illustrate the trade-offs that retired investors face regarding how much income they can generate, how much short-term risk they are exposed to, how large an estate they can expect to leave, and how likely they are not to run out of assets before dying (the probability).

Kaplan, P. D. and Idzorek, T. M. (2024). "The Importance of Joining Lifecycle Models with Mean-Variance Optimization." In: *Financial Analysts Journal*, pp. 1–7.

For nearly three-quarters of a century, there has been a large separation between lifecycle finance models stemming from numerous Nobel laureates and the single-period mean-variance optimization-oriented models starting with Markowitz. Recent advances allow for a new class of models that combine both lifecycle models and mean-variance models. This new class of models uses lifecycle models to answer key financial planning questions and then mean-variance optimization models to answer investment questions. Goals-based models are often silent on many financial planning questions addressed by lifecycle finance and should thus be joined with lifecycle models.

Kazak, E. and Pohlmeier, W. (2019). "Testing out-of-sample portfolio performance." In: *International Journal of Forecasting* 35(2), pp. 540–554.

This paper studies the quality of portfolio performance tests based on out-of-sample returns. By disentangling the components of the out-of-sample performance, we show that the observed differences are driven largely by the differences in estimation risk. Our Monte Carlo study reveals that the puzzling empirical findings of inferior performances of theoretically superior strategies result mainly from the low power of these tests. Thus, our results provide an explanation as to why the null hypothesis of equal performance of the simple equally-weighted portfolio compared to many theoretically-superior alternative strategies cannot be rejected in many out-of-sample horse races. Our findings turn out to be robust with respect to different designs and the implementation strategies of the tests. For the applied researcher, we provide some guidance as to how to cope with the problem of low power. In particular, we make use of a novel pretest-based portfolio strategy to show how the information regarding performance tests can be used optimally.

Kazak, E. and Pohlmeier, W. (2020). *Portfolio Pretesting with Machine Learning*. Tech. rep. University of Lancaster.

This paper exploits the idea of pretesting to choose between competing portfolio strategies. We propose an estimator for a portfolio weight vector, which optimally trades off between Type I and Type II errors when choosing the best investment strategy. Furthermore we accommodate the idea of bagging in the portfolio testing problems, which helps to avoid sharp thresholding and reduces the amount of portfolio turnover. Our approach borrows from both shrinkage and forecast combination literature. The portfolio weights of our strategy are weighted averages of the portfolio weights from a set of stand-alone strategies. More specifically, the weights are generated from a pseudo out-of-sample portfolio pretesting, such that they reflect the probability that a given strategy will be overall best performing. Contrary to previous approaches, the shrinkage intensity is continuously updated to incorporate the most recent information in the rolling window forecasting set-up. We show that the bagged pretest estimator performs exceptionally well, especially when combined with adaptive smoothing. The resulting strategy allows for a flexible and smooth switch between the underlying strategies and is shown to outperform the corresponding stand-alone strategies.

Kelly, D. P. and Roy, K. (2021). *Annuities: An essential slice of the retirement pie*. Tech. rep. J.P. Morgan Asset Management.

When it comes to retirement planning, the greatest investment tools investors can exploit are diversification and annuitization. Annuities have become an essential slice of the retirement pie for Americans approaching and living in retirement. As part of a retirement income solution, annuities also can be an ideal complement to other portfolios. We believe they are very well positioned in today's uncertain market environment. Unlike other financial instruments, these contracts issued by insurance companies combine the features of investing and insurance in a single solution.

They offer a range of flexible benefits, including:

1) Lifetime income: Annuities offer a variety of guaranteed income options that can start immediately or at a future date.

2) Growth potential: The earlier your clients invest, the more opportunity they'll have for long-term growth

potential from a broad range of investment choices. Growth will vary depending on the performance of the investment options you choose.

3) Tax-deferral: Taxes can have a big impact on long-term investment returns – making the benefit of a tax-deferred variable annuity especially attractive for high net worth and affluent investors

4) Legacy options: Many variable annuity contracts offer special riders (for a fee) to provide death benefit protection for loved ones.

We believe annuities are very well positioned in today's uncertain market environment.

Kenigsberg, M. B., Mazumdar, P. D., and Feinschreiber, S. (2014). "Return Sequence and Volatility: Their Impact on Sustainable Withdrawal Rates." In: *The Journal of Retirement* 2(2), pp. 81–98.

This article explores estimates for a sustainable withdrawal rate (SWR) for a typical retiree and examines two important factors that challenge sustainability: sequence of returns and volatility of returns. Using historical analysis, the article explores the implications of sequence of returns risk and suggests strategies for countering it, including annuitization and adaptive withdrawal strategies. The article also explores volatility risk, separated as much as possible from sequence risk, and finds that the relationship between volatility, return, and SWR is not linear. In fact, for a given horizon and degree of confidence, the relationship between the Sharp ratio and the rate of return displays an inflection point, where the level of return required per unit of risk reaches a minimum.

Kessy, S. (2023). "Mortality forecasting with ensembles and combinations." PhD thesis. UNSW.

Many alternative approaches for selecting mortality models and calibration periods have been proposed. The usual practice is to base forecasts on a single mortality model se- lected using in-sample goodness-of-fit measures and an arbitrarily chosen calibration period. However, cross-validation measures are increasingly being used in calibration period selection, model selection, and model combination methods are becoming a common alternative to using a single mortality model and calibration period. First, we propose a stacked regression ensemble that optimally combines different mortal- ity models to reduce out-of-sample mean squared errors and mitigate model uncertainty. Stacked regression uses a meta-learner to approximate horizon-specific weights by minimizing a cross-validation criterion for each forecasting horizon. The horizon-specific weights determine a mortality model combination customized to each horizon. We use 44 popula- tions from the Human Mortality Database (HMD) to compare the stacked regression ensemble with alternative methods. We show that, using one-year-ahead to 15-year-ahead out-of-sample mean squared errors, the stacked regression ensemble improves mortality forecast accuracy by 13% - 49% for males and 19% - 90% for females over individual mortality models. Second, we propose an automated procedure to select diverse multiple starting points to fit a mortality model and weights to combine the out-of-sample forecasts from the selected periods using methods like lasso regression. Using 19 male mortality data from HMD, combining forecasts from multiple fitting periods produces lower mean squared error of mortality rate forecasts than fitting the mortality models to the longest calibration period. For example, we show that the gain in the forecast accuracy of mortality rate forecasts combined based on the Age-Period-Cohort model relative to the longest fitting period is between 11.7% and 31.5% across forecast horizons for 19 male populations. Lastly, we propose a new interpretable stacked regression ensemble (ISRE), which expresses a standard stacked regression ensemble in terms of diverse mortality features from individual mortality models, which directly allows for interpreting the contribution of each feature to the out-of-sample mortality forecasts. Our empirical experiments based on US male data from HMD show that ISRE can attain similar out-of-sample forecast error as the standard stacked regression ensemble while allowing for model interpretability.

Khang, K. and Clarke, A. S. (2020). *Safeguarding retirement in a bear market*. Tech. rep. Vanguard.

What is the effect of "sequence-of-return" risk-the risk of receiving a concentrated series of particularly poor returns-on retirees who depend on a financial portfolio to generate income? We provide a quantitative answer to this question by examining the cohorts that would have retired during or near six major U.S. bear markets since 1926. Compared to otherwise similar investors retiring during the same periods, and assuming constant real-dollar withdrawals, the unlucky ones with a poor sequence of returns were 31% more likely to outlive their wealth, had 11% lower retirement income streams,and left 37% smaller bequests. These adverse effects can be mitigated with an adaptive withdrawal strategy. By countering a decline in portfolio value with an incremental decrease in planned withdrawal amounts, even those bearing the worst sequence-of-return risk could have eliminated the possibility of premature portfolio depletion and increased their bequests by 20%. These improvements would have required a manageable reduction in retirement income on the order of a 5% decrease in the first five years and effectively no change over the whole 35 years of retirement.

Khang, K. and Picca, A. (2021). "Waiting for the Next Factor Wave: Daily Rebalancing around Market Cycle Transitions." In: *The Journal of Portfolio Management* 47(2), pp. 127–144.
To deliver historically observed factor premiums, long-only factor investing relied heavily on a small number of periods, when factors realized outsized returns in the midst of changing market leadership. This article shows that by rebalancing factor funds more frequently during these periods – rebalancing on a daily basis instead of monthly or biannually – investors would have achieved significantly higher factor premiums, effectively doubling the historically observed premiums of many factors. These findings indicate that to harvest factor premiums to their maximal potential, skill is needed on the part of the fund manager-an ability to tell the right moment to aggressively rebalance.

Khanna, K. and Chauhan, V. (2017). "A study of risk profiling and investment choices of retail investor." In: *SSRN e-Print*.
This research is an attempt to explore the association between various demographical factors, risk profile and investment decision of retail investors. All measures were tested for reliability through computation of Cronbach Alpha. The Alpha coefficient value was found 0.741 for variables like wealth, risk profile asset class, fixed return, mutual fund return, equity, real estate and gold commodity. Further, the chi-Square and crammer-v statistics has been used to interpreting the association between these factors. And it revealed that the investors do not always behave rationally and their choice of investments is decided by their risk profile and other demographical factors such as age, gender, income, wealth etc. This research is also useful for portfolio managers to construct the right portfolio for the investors according to their needs and preferences.

Khemka, G., Butt, A., and Mehry, S. (2024). "A Building Block Approach to Retirement Income Design." In: *SSRN e-Print*.
This paper addresses the retirement income planning problem from the perspective of the four main building blocks of retirement income: state pension, mortality credits, investment strategies and drawdown schedules. We detail how these building blocks interact to form a retiree's overall retirement income portfolio, and what trade-offs and interactions must be considered. We find that while access to each building block increases the retiree's certainty equivalent consumption, the most substantial contributor to this increase is from utilization of the mortality credit building block (i.e., annuities).

Khemka, G., Lim, W., and Donnelly, C. (2023). "Aligning Retirement Saving Goals with Inflation." In: *SSRN e-Print*.
Pension investors are exposed to the risk of inflation eroding the purchasing power of their savings. Their risk is particularly high due to saving for retirement taking place over many decades. However, few pre-retirement investment strategies incorporate explicit inflation-proofing. It is shown that ignoring inflation is costly in terms of a retiree's welfare, with reductions of up to 25% possible for the average retiree. More risk averse investors face even larger reductions.In a second study, constraints on the amount of pension savings at retirement are imposed. Constraints are used to give the retiree more certainty about the level of pension savings at the date of their retirement. Such constraints may be expressed in either real or nominal terms. However, ignoring inflation by using nominal constraints gives a potential reduction in welfare of up to 36% for the average retiree. The results illustrate that nominal constraints are ineffective at reducing the risk of inflation.The conclusion is that pre-retirees ignore inflation at their peril. It must be included explicitly in retirement savings targets to improve retirement outcomes. Consequently, there should be greater investment in an index-linked bond or a similar asset.

Kieren, P. and Weber, M. (2022). "When saving is not enough – wealth decumulation in retirement." In: *Journal of Pension Economics and Finance* 21(3), pp. 446–473.
We field a large online survey to study preferences and hypothetical product choices for phased withdrawal accounts and compare their demand to the demand of annuities. We find that most individuals prefer phased withdrawal accounts with dynamic withdrawal rates and equity-based asset allocation. Additionally, when offered the opportunity to exchange the phased withdrawal account with an annuity, most individuals decline to annuitize. Our results suggest that policymakers should consider offering combined solutions of phased withdrawals and annuities. Retirees who are averse to full annuitization could preserve some of their accumulated wealth while also acquiring protection against longevity risk.

Kim, W. C., Kwon, D.-G., Lee, Y., Kim, J. H., and Lin, C. (2020). "Personalized goal-based investing via multi-stage stochastic goal programming." In: *Quantitative Finance* 20(3) (3), pp. 515–526.
In this paper, we propose a goal-based investment model that is suitable for personalized wealth management. The model only requires a few intuitive inputs such as size of wealth, investment amount, and consumption goals

from individual investors. In particular, a priority level can be assigned to each consumption goal and the model provides a holistic solution based on a sequential approach starting with the highest priority. This allows strict prioritization by maximizing the probability of achieving higher priority goals that are not affected by goals with lower priorities. Furthermore, the proposed model is formulated as a linear program that efficiently finds the optimal financial plan. With its simplicity, flexibility, and computational efficiency, the proposed goal-based investment model provides a new framework for automated investment management services.

Kinniry Jr., F. M., Jaconetti, C. M., DiJoseph, M. A., Zilbering, Y., and Bennyhoff, D. G. (2016). *Putting a value on your value: quantifying Vanguard Advisor's Alpha*. Tech. rep. Vanguard.
The value proposition of advice is changing. The nature of what investors expect from advisors is changing. And fortunately, the resources available to advisors are evolving as well In creating the Vanguard Advisor's Alpha concept in 2001, we outlined how advisors could add value, or alpha, through relationship-oriented services such as providing cogent wealth management via financial planning, discipline, and guidance, rather than by trying to outperform the market. Since then, our work in support of the concept has continued. This paper takes the Advisor's Alpha framework further by attempting to quantify the benefits that advisors can add relative to others who are not using such strategies. Each of these can be used individually or in combination, depending on the strategy. We believe implementing the Vanguard Advisor's Alpha framework can add about 3% in net returns for your clients and also allow you to differentiate your skills and practice. Like any approximation, the actual amount of value added may vary significantly, depending on clients' circumstances.

Kintzel, D. (2020). "Income Sustainability in Retirement: A Case Study of the Life-Cycle Account." In: *The Journal of Retirement* 7(3), pp. 31–45.
This article uses Monte Carlo analysis to simulate returns for a popular retirement portfolio strategy and the income derived from it in retirement in order to calculate the chance of it being depleted while the retiree is still alive. I analyze and contrast several withdrawal strategies from the account to illustrate the important trade-offs between income needs and account sustainability. Furthermore, the simulations are compared with payouts from an annuity to show the important differences between each vehicle ability to provide income. The primary findings are that portfolios have a higher chance of account instability with advanced age and income needs, whereas annuities may provide less income earlier in retirement, but the income is sustainable for life.

Kintzel, D. and Turner, J. A. (2020). "Provision of Longevity Insurance Annuities." In: *Financial Analysts Journal* 76(4), pp. 119–133.
Longevity insurance annuities are deferred annuities that begin payment at advanced ages, such as age 82. They provide insurance against running out of money in old age. They simplify the retirement spend-down calculations in that they can be used to convert a problem with an unknown end date (date of death) to one with a fixed end date, which is the start of the longevity insurance annuity. They may allow retirees to have riskier portfolios because they are a steady source of income. In this study, we compared the provision of longevity insurance annuities in the private and public sectors. We analyzed the need for longevity insurance annuities using Monte Carlo simulations that demonstrate the risk of running out of money in a 401(k) account and modeled how longevity annuities might work in practice.

Kitces, M. (2015). "Sequence of return risk and safe withdrawal rates." In: *Retirement Income Conference*.
Watching a portfolio experience market volatility in the first few years of retirement can be terrifying to a new retiree, raising legitimate questions of whether there's a danger that early declines plus ongoing withdrawals could lead to a retirement spending shortfall. And as the safe withdrawal rate research has shown, that danger is real - in fact, it's been dubbed the "sequence of return" risk to retirement spending, a recognition of the reality that even if returns average out in the long run, it doesn't matter if ongoing withdrawals deplete the portfolio before the "good" returns finally show up. Yet the caveat is that while sequence of return risk is real, it's not necessarily just about the danger of getting a severe bear market on the eve of retirement. In fact, a deeper look at the data reveals that there is remarkably little relationship between returns in the first year or two of retirement, and the safe withdrawal rate that can be sustained in the portfolio, even if retirement starts out with a market crash. Instead, it turns out that the true driver of sequence of return risk and safe withdrawal rates are the returns that the retiree earns over the first decade - and specifically, the real returns over the first decade, that provide an indication of whether the retirement portfolio will have produced enough real growth to keep up with inflation-adjusted spending for the rest of retirement. Fortunately, though, bad decades of returns are not entirely random, and instead can be reasonably predicted by long-term market valuation trends, providing retirees with at least a few tools to manage the dangers of sequence of return risk through adjusting asset

allocation in retirement and setting a reasonable initial withdrawal rate in light of the market conditions that exist - and the potential for a bad decade of returns - when their retirement begins.

Kitces, M. and Richards, C. (2019). *Kitces & Carl Ep 01: How To Value The Value Of Financial Planning*. URL: https://www.kitces.com/blog/kitces-carl-richards-talk-value-of-financial-planning-intangible/.

One of the biggest challenges comprehensive financial planners face today is that "everyone" says they do financial planning. Yet in practice, what they actually do, whether it's really comprehensive financial planning, or even what constitutes "financial planning" as opposed to just ad hoc financial advice, has no clear and consistent definition within the industry. And if we can't figure out what the differences are within the industry, then it's almost impossible for the public to understand the differences between the services offered (including any potential conflicts) by a whole bunch of people who do different stuff (frequently playing by different sets of rules) yet say they all do the same thing! Not to mention trying to help the public understand the value (both financially tangible and intangible) of the financial planning choices available and being offered to them. Fortunately, both Morningstar, Vanguard, and Envestnet have done some research to try and quantify an answer to this very question. Their studies suggest that, even when we ignore any potential excess investment return, the advice that advisors provide around the dollars that they are managing can amount to an additional 1.5-3.0% over what a client would have likely gotten had they "done it themselves" But, outside of "helping clients avoid big mistakes" by re-allocating their portfolios or talking them off the proverbial "ledge" when volatility ticks up, there's arguably another aspect to helping clients that transcends the "quantifiable" aspect of the profession. Because most clients don't have anywhere else they can go and feel safe talking about this topic of money, that is so important yet remains a taboo topic of discussion, and for many carries an enormous amount of baggage. And those hurdles only expand geometrically when it's a couple that's sitting across the desk. In fact, perhaps the greatest problem of trying to explain the value of financial advice, especially while the industry is in the midst of such rapid change, is that we are still tying all our value conversations back to the portfolio-based model... to the extent that, even when trying to quantify the behavioral aspects of planning, industry studies still scale them to a percentage of assets. Even though an increasingly amount of the value of financial planning advice is specifically about the advice that occurs outside of the portfolio! Alternatively, perhaps instead of trying to put a dollar value on the advice we provide as advisors at all, we might just consider framing it in terms of the real-world outcomes for clients. For instance, what if we as advisors simply said to a couple having stressful arguments about money that, "We help couples have better conversations about money, so they're not as awkward"? And then let them figure out if it's valuable enough to them to pay for it? Because, if an advisor can help reduce marital stress and reduce the number of marriages that end out in divorce because of money issues, then clients might consider that a whole lot more valuable that we might have ever imagined anyway! Ultimately, then, the key point is to recognize that as financial planning continues to expand beyond just building and managing diversified asset allocated portfolios, the key task as an industry is to figure out how we can frame that value beyond the context of "investable assets". Because if we want to better communicate our value to prospects and clients, especially beyond the portfolio, we must first be willing to fully embrace and acknowledge that value ourselves.

Kitces, M. E. and Pfau, W. D. (2014). "Retirement Risk, Rising Equity Glidepaths, and Valuation-Based Asset Allocation." In: *SSRN e-Print*.

This research investigates two types of dynamic asset allocation strategies (predetermined equity glidepaths and valuation-based asset allocation) for retirees using U.S. historical data. We analyze fixed asset allocations, traditional declining equity glidepaths, rising equity glidepaths, accelerated traditional and rising glidepaths, valuation-based allocations tethered around a fixed allocation, and glidepaths with valuation-based overlays. With U.S. historical data, it is difficult to beat a strategy which maintains a consistently high allocation to stocks (especially as measured by terminal median wealth), to the extent that a retiree's risk tolerance allows for this, and subject to the caveat that high stock allocations cannot always be expected to do as well in the future. However, when we consider retirements beginning in varying valuation environments (as defined by the level of Robert Shiller's cyclically-adjusted price-earnings ratio relative to its then-current historical median), we find the potential for different dynamic allocation strategies to help retirees sustain higher spending levels with lower average stock allocations in certain situations. When retirements begin in overvalued market environments (which reflects the situation for new retirees today), an accelerated rising equity glidepath has shown much potential to provide downside risk protection for retirees by minimizing equity exposure when an adverse market event would have the greatest impact. In other valuation environments, historical worst-case scenario sustainable withdrawal rates were highest with valuation-based asset allocation strategies, which maintain a midrange average stock allocation but adjust higher or lower when markets are deemed undervalued or overvalued, respectively.

Klement, J. (2015). "Investor Risk Profiling: An Overview." In: *SSRN e-Print*.
In this discussion of investor risk profiling, current risk-profiling practice is reviewed and contrasted with regulatory demands and recent research findings.

Klement, J. (2018). "Risk Profiling and Tolerance: Insights for the Private Wealth Manager." In: *Research Foundation Publications of CFA Institute*.
If risk aversion and willingness to take on risk are driven by emotions and we as humans are bad at correctly identifying them, the finance profession has a serious challenge at hand – how to reliably identify the individual risk profile of a retail investor or high-net-worth individual. In this series of CFA Institute Research Foundation briefs, we have asked academics and practitioners to summarize the current state of knowledge about risk profiling in different key areas.

Ko, H., Lee, S., and Lee, J. (2023). "Dynamic Investment Strategy to Safeguard Retirement Portfolio and Insights into Retiree Behavior." In: *SSRN e-Print*.
In this study, we investigate the implications of dynamic investment strategies, particularly portfolio insurance methods, in post-retirement portfolio management amid bearish market conditions. Leveraging an extensive dataset from the US stock market spanning 1928 to 2018, we evaluate performance based on pension-level and downside risk metrics and examine retiree portfolio choices using Expected Utility and Prospect Theory. Our findings empirically confirm the pronounced evidence of the financial merits of the Synthetic Put and Constant Proportion Portfolio Insurance strategies over traditional static methods. Evaluations based on utility theories illuminate the economic benefits of dynamic investment strategies for retirees, factoring in varied risk profiles and economic propensities. Our research advances retirement portfolio literature by providing critical economic insights and reshaping perspectives on risk management and sustainability of retirement funds in volatile markets.

Ko, H., Lee, S., and Lee, J. (2024). "Sequence and longevity risks of South Korean retirees: Insights and potential remedies." In: *Pacific-Basin Finance Journal* 83, p. 102263.
This study investigates the retirement landscape in South Korea, highlighting major challenges like sequence and longevity risks. Our analysis reveals that over half of South Koreans are not well-prepared for retirement. For example, retirees risk depleting 56.3% of their portfolio in 25 years, increasing to 97.6% in poor market conditions. We propose a dynamic withdrawal strategy to mitigate these risks, backed by empirical evidence. Additionally, we consider policy reforms like delaying the retirement age to enhance portfolio sustainability. This research is a significant contribution to discussions on effective retirement strategies, benefiting pension experts, regulators, policymakers, and retirees in South Korea.

Kobor, A. and Muralidhar, A. (2020). "Targeting Retirement Security with a Dynamic Asset Allocation Strategy." In: *Financial Analysts Journal* 76(3), pp. 38–55.
The goal of investing for retirement is to secure a target level of income that maintains the individual's preretirement lifestyle. Current "safe harbor" glide-path products shift investments from stocks to bonds on the basis of the individual's age. This approach is unlikely to secure a target retirement income because the glide path is focused on the wrong goal. We tested a dynamic asset allocation strategy that takes no view of future market performance and is based on a retirement income goal. We show how this dynamic strategy could dominate standard portfolio choices. The article introduces a new way to think about intermediate retirement targets and explores the implications of the dynamic asset allocation strategy for the level of savings required to achieve a retirement goal.

Kohlscheen, E. (2021). "What does machine learning say about the drivers of inflation?" In: *SSRN e-Print*.
This paper examines the drivers of CPI inflation through the lens of a simple, but computationally intensive machine learning technique. More specifically, it predicts inflation across 20 advanced countries between 2000 and 2021, relying on 1,000 regression trees that are constructed based on six key macroeconomic variables. This agnostic, purely data driven method delivers (relatively) good outcome prediction performance. It's out of sample root mean square errors (RMSE) systematically beat even the in-sample benchmark econometric models, with a 28% RMSE reduction relative to a naive AR(1) model and a 8% RMSE reduction relative to OLS. Overall, the results highlight the role of expectations for inflation outcomes in advanced economies, even though their importance appears to have declined somewhat during the last 10 years.

Koijen, R. S. J., Nijman, T. E., and Werker, B. J. M. (2011). "Optimal Annuity Risk Management." In: *Review of Finance* 15(4), pp. 799–833.

This paper studies the life-cycle consumption and portfolio choice problem taking account of annuity risk at retirement. The study allows for government-provided annuity income. Optimally, households allocate retirement wealth to nominal, inflation-linked and variable annuities, and condition this choice on the state of the economy. The case in which there are limitations in the types of annuities that are available is also considered and the costs of annuity market incompleteness are quantified. Subsequently, the paper determines how investors optimally anticipate annuitization before retirement. The conclusion is that ignoring annuity risk before and at retirement can be economically costly.

Konicz, A. K., Pisinger, D., and Weissensteiner, A. (2016). "Optimal retirement planning with a focus on single and joint life annuities." In: *Quantitative Finance* 16(2), pp. 275–295.

We optimize the asset allocation, consumption and bequest decisions of a couple with an uncertain lifetime. The asset menu consists of zero coupon bonds and pure endowments with different maturities, whole life annuities and stocks. The pure endowments pay either fixed or variable benefits, and, similarly to the whole life annuities, are contingent on either a single or a joint lifetime. We model the stock returns and the parameters for the term structure with a vector autoregressive model, thus considering time-varying investment opportunities. To find the optimal solution we use a multi-stage stochastic programming approach, which allows for including complex surrender charges on pure endowments and annuities, as well as transaction costs on stocks and bonds. Our findings indicate that despite high surrender charges, households should invest in a wide combination of life contingent products with different maturities and underlying financial risk.

Konicz, A. K., Pisinger, D., and Weissensteiner, A. (2015). "Optimal annuity portfolio under inflation risk." In: *Computational Management Science* 12(3), pp. 461–488.

The paper investigates the importance of inflation-linked annuities to individuals facing inflation risk. Given the investment opportunities in nominal, real, and variable annuities, as well as cash and stocks, we investigate the consumption and investment decisions under two different objective functions: (1) maximization of the expected CRRA utility function (2) minimization of squared deviations from an inflation-adjusted target. To find the optimal decisions we apply a multi-stage stochastic programming approach. Our findings indicate that independently of the considered ob- jective function and risk aversion, real annuities are a crucial asset in every portfolio. In addition, without investing in real annuities, the retiree has to rebalance the portfolio more frequently, and still obtains the lower and more volatile real consumption.

Konstantinov, G. S. and Fabozzi, F. J. (2024). "Improvements in Global Bond Portfolio Risk Management and Performance by Hedging the Components of Total Risk with Derivatives." In: *The Journal of Fixed Income* 34(3), pp. 26–42.

In this article, we provide an in-depth illustration of how to use derivatives for hedging foreign exchange (FX) risks in global bond portfolios. It focuses on a yield-curve-based approach, using factor models to effectively decompose and manage both currency and interest-rate exposures. The central methodology illustrated is the analysis of yield curves to comprehensively assess and mitigate the FX risks embedded within global bond portfolios. Employing a seven-factor model, which incorporates FX carry, value, and momentum, among other factors, we illustrate how to explain portfolio returns and manage the associated currency risks. Considerable emphasis is placed on understanding the interplay between bond pricing, currency volatility, and the strategic use of FX options to mitigate risk. This approach described is crucial for portfolio managers seeking to optimize their management of multicurrency exposures, by aligning hedging strategies with the portfolio's base currency to improve both performance and risk control. The illustration clearly demonstrates the complexity of FX hedging and the critical role of integrated yield curve analysis in global investment strategies.

Kontoghiorghes, A. P. (2024). "Do personal taxes affect investment decisions and stock returns?" In: *Journal of Financial Economics* 162, p. 103927.

This paper studies the causal effects of personal investment taxes on stock returns and the financial decisions of companies. I exploit a change in legislation in 2013 which allowed stocks listed on the Alternative Investment Market, a sub-market of the London Stock Exchange, to be held in capital gains and dividend tax-exempt investment accounts for the first time. Using a difference-in-differences approach, I find that excess stock returns decreased by their pre-legislation change effective tax rate, and that firms adjusted their capital structure and increased their spending on dividends, capital, and labour, in-line with the "traditional view" of corporate investment.

Koo, B., Pantelous, A. A., and Wang, Y. (2020). "Novel Utility-Based Life Cycle Models to Optimize Income in Retirement in the Presence of Heterogeneous Preferences." In: *SSRN e-Print*.

The global shift towards defined-contribution pension schemes has been accompanied by asymmetric risks and new responsibilities for households to plan and fund effectively their own retirement over the years. In this study, expressing and combining preferences for consumption, investment, bequest, public pension entitlement and the choice of reverse mortgage products, we develop several utility-based life cycle models to facilitate the complex decision-making process that retired households are required to follow to optimize their retirement income. This optimal policy is given in the form of either an analytical or a numerical solution using stochastic dynamic programming. The timing of this paper coincides with the launch of a reverse mortgage style loan, offered by the Australian federal government and allowing retired households to receive an income stream by taking out a loan against the equity in their home. Calibration is performed using real Australian household data.

Kopa, M., Moriggia, V., and Vitali, S. (2018). "Individual optimal pension allocation under stochastic dominance constraints." In: *Annals of Operations Research* 260(1-2), pp. 255–291.

An individual investor has to decide how to allocate his/her savings from a retirement perspective. This problem covers a long-term horizon. In this paper we consider a 40-year horizon formulating a multi-criteria multistage program with stochastic dominance constraints in an intermediate stage and in the final stage. As we are dealing with a real problem and we have formulated the model in cooperation with a commercial Italian bank, the intermediate stage corresponds to a possible withdrawal allowed by the Italian pension system. The sources of uncertainty considered are: the financial returns, the interest rate evolution, the investor's salary process and a considerable withdrawal event. We include a set of portfolio constraints according to the pension plan regulation. The objective of the model is to minimize the Average Value at Risk Deviation measure and to satisfy wealth goals. Three different wealth target formulations are considered: a deterministic wealth target (i.e. a comparison between the accumulated average wealth and a fixed threshold) and two stochastic dominance relations-the first order and the second order-introducing a benchmark portfolio and then requiring the optimal portfolio to dominate the benchmark. In particular, we prove that solutions obtained under stochastic dominance constraints ensure a safer allocation while still guaranteeing good returns. Moreover, we show how the withdrawal event affects the solution in terms of allocation in each of the three frameworks. Finally, the sensitivity and convergence of the stochastic solutions and computational issues are investigated.

Kostakis, A., Mu, L., and Otsubo, Y. (2023). "Detecting political event risk in the option market." In: *Journal of Banking and Finance* 146 (106624).

This study shows that the option market can ex ante detect and quantify the effects of political event risk. Focussing on the 2016 UK referendum on EU membership, we find that the Risk-Neutral Distribution extracted from GBPUSD futures options whose expiry spans the referendum date becomes bimodal and the Implied Volatility curve exhibits an unusual W-shape. To the contrary, the corresponding effects for FTSE100 are found to be very limited. The large swings in expectations regarding the event outcome during the referendum night allow us to observe the counterfactual and validate the ex ante information revealed in the option market.

Kourtidis, I. and Anderson, S. W. (2024). "Enhancing Tax-Loss Harvesting Returns in Direct Indexing by Increasing Stock Dispersion." In: *The Journal of Beta Investment Strategies* 15(3), pp. 108–117.

Direct indexing (DI) combined with tax-loss harvesting (TLH) is a popular approach to generate capital losses to offset a client's capital gains while attempting to track a specified index. The goal is to increase after-tax excess returns by postponing or reducing tax payments. More TLH opportunities exist when the index constituents have a higher dispersion of returns and are therefore more likely to be at a loss. To create more TLH opportunities within DI, the authors propose a method to modify the target index to increase dispersion based on the realized volatility of stocks. They show that as the average constituent volatility increases, the effectiveness of DI TLH also increases against plain DI TLH that does not utilize this method. The authors further show that this method can significantly increase the after-tax excess returns versus plain DI TLH at the same tracking error per unit of additional returns. Further increases of dispersion can increase after-tax excess returns even more, although at a gradually higher cost of tracking error.

Krasner, S., Liberman, J., Sosner, N., and Brenner, S. (2024a). "Levering Up to Do Good: Direct Long-Short Investing and Charitable Giving." In: *The Journal of Beta Investment Strategies* 15(3), pp. 34–65.

The authors use historical strategy simulations to evaluate the advantages of donating appreciated stock in the context of tax-aware long-short factor strategies. Their main findings are as follows. First, long-short strategies have a higher donation capacity than long-only investments, and their donation capacity increases with leverage. Second, long-short strategies have a higher donation efficiency than long-only investments. Similar to donation capacity, donation efficiency increases with leverage. Third, long-short strategies receive a larger loss-realization boost from the donation of appreciated stocks than long-only investments, which also increases with leverage.

Fourth, by removing appreciated positions from the strategy portfolio, higher donation targets reduce the tax costs of modifying the long-short strategy-for example, transitioning it to a long-only portfolio. Finally, when stock donations are done without replenishment, long-short portfolios offer pretax and after-tax performance that is far superior to that of long-only investment. Here, too, pretax and after-tax values achieved with long-short investments increase with leverage.

Krasner, S., Liberman, J., Sosner, N., and, s. filler sydney (2024b). "Levering Up to Do Good: Direct Long-Short Investing and Charitable Giving." In: *SSRN e-Print*.

We use historical strategy simulations to evaluate the advantages of donating appreciated stock in the context of tax-aware long-short factor strategies. Our main findings are as follows. First, long-short strategies have a higher donation capacity than long-only investments, and their donation capacity increases with leverage. Second, long-short strategies have a higher donation efficiency than long-only investments. Similar to donation capacity, donation efficiency increases with leverage. Third, long-short strategies receive a larger loss-realization boost from donation of appreciated stocks than long-only investments, which also increases with leverage. Fourth, by removing appreciated positions from the strategy portfolio, higher donation targets reduce the tax costs of modifying the long-short strategy, for example, transitioning it to a long-only portfolio. Finally, when stock donations are done without replenishment, long-short portfolios offer pre-tax and after-tax performance that is far superior to that of long-only investment. Here too, pre-tax and after-tax value achieved with long-short investments increases with leverage.

Krasnopolsky, M. and Ashton, M. (2018). "Why Pairing LDI with De-Risking Glide Paths Produces Inferior Pension Fund Outcomes." In: *The Journal of Investing* 27(supplement), pp. 58–64.

Combining traditional Liability Driven Investment (LDI) with funded status responsive de-risking strategies involves inconsistent treatment of risks in these two elements of what has become a popular pension strategy. This inconsistency causes irreconcilable conflicts in their execution and imperils the positive pension fund outcome. This article provides a critique of the combined LDI/De-risking Glide Path strategy as currently implemented by many pension plan managers and also provides an example of an alternative solution that improves pension plan outcomes. Our prescription for the pension de-risking glide path approach differs from conventional wisdom, resulting in faster de-risking, without undesirable market betas that are unrelated to the liability. It also avoids illiquid assets that pension funds often gravitate toward in their quest for returns, takes fewer credit risks, and seeks more alpha risks.

Kritzman, M. (2017). "Target-Date Funds: A Regime-Based Approach." In: *The Journal of Retirement* 5(1), pp. 96–105.

Target-date funds automatically shift the asset mix at regular intervals toward safer assets as the fund's target date, which may be several decades away, draws near. A target-date fund is designed to reduce exposure to loss as the investor moves closer to retirement and has fewer years to offset potential losses. The implicit assumption of a target-date fund is that exposure to loss from risky assets increases as time decreases; hence the shift to safer assets as the investment horizon shrinks. However, another implicit assumption of target-date funds is that the standard deviation of assets is relatively stationary. The typical approach for determining the glide path assumes implicitly that the asset's annualized standard deviation will not change much during any subperiod within the longer horizon. This second assumption is contrary to experience. Subperiod volatility is highly nonstationary, although it is not entirely unpredictable. Investors would be better served by the strategy this author describes, of not only managing the effect of time on exposure to loss but also accounting for the nonstationarity of risk.

Kuntz, L.-C. (2018). "Portfolio Strategies with Classical and Alternative Benchmarks." PhD thesis. Georg August University of Gottingen.

This dissertation addresses different key elements in portfolio management. It intends to improve and analyze influences on portfolio strategies and their performance. Likewise, it aims at the systematization and extension of benchmark specifications as well as their effect on portfolio strategies. Each chapter focuses on a different aspect of developing and implementing portfolio strategies. The dissertation seeks to contribute to the advancement of portfolio strategies by making the performance generating process and influences on it more comprehensible and transparent. In doing so, it attempts to strengthen the awareness of the impact of the exact design of portfolio strategies and benchmarks on the resulting portfolio and its performance. The key findings of this dissertation can be summarized as follows: The benchmark specification, especially in terms of the investible universe and the inherent risk conception, has substantial influence on the explicit design and performance of portfolio strategies. In general, the specification of the benchmark and design of portfolio strategies should be carefully considered and the implementation should be well thought out. Alternative risk conceptions, such as regret risk, can be applied to portfolio selection and lead to clearly different portfolio compositions. Moreover, timing strategies can be improved by choosing a careful investment approach on the basis of distributional regressions. All empirical work 3 of this thesis has in common that it pursues different ideas to set up portfolio strategies while explicitly addressing the benchmark specification used for the implementation and evaluation of said strategies.

Kuselias, S., Perreault, S. J., and Shafer, M. (2021). "The Financial and Tax Considerations of Social Security and Early Retirement." In: *The Journal of Wealth Management* 24(3), pp. 90–98.

While individuals who delay the receipt of social security benefits are entitled to larger monthly payments, 57% of recipients elect to receive benefits early. This article compares the present value of social security benefits when taken early to the present value of benefits taken at the full or maximum retirement ages, taking into account factors such as average life expectancy and the time value of money. Furthermore, we expand on prior research by investigating how tax rates may affect these comparisons. Overall, our analyses suggest that taking social security early may be advantageous for many individuals.

Lacombe, D. and Diavatopoulos, D. (2023). "Bayesian Estimation as an Alternative to Monte Carlo Analysis in Financial Planning: An Application Using S&P 500 Returns." In: *SSRN e-Print*.

Monte Carlo techniques are ubiquitous in financial planning and related fields. However, the assumptions underlying Monte Carlo techniques can be unrealistic in many modeling contexts. One criticism of Monte Carlo techniques is the distributional assumption underlying the technique, which usually translates into assuming normally distributed data. Bayesian statistical techniques are an alternative estimation strategy which allows one to estimate the underlying distribution as well as make model comparisons between potential candidate distributions. In addition, Bayesian techniques allow for an intuitive interpretation of results, which makes Bayesian techniques more useful than standard Monte Carlo simulation. We illustrate these ideas using returns from the S&P 500 index over a period of 50 years. The assumption of normality for these data leads to economically significant incorrect inferences and interpretation.

Lalive, R., Magesan, A., and Staubli, S. (2020). *The Impact of Social Security on Pension Claiming and Retirement: Active vs. Passive Decisions*. Tech. rep. NBER.

We exploit a unique Swiss reform to identify the importance of passivity, claiming social security benefits at the Full Retirement Age (FRA). Sharp discontinuities generated by the reform reveal that raising the FRA while imposing small early claiming penalties significantly delays pension claiming and retirement, but imposing large penalties and holding the FRA fixed does not. The nature of the reform allows us to identify that between 47 and 69% of individuals are passive, while imposing additional structure point identifies the fraction at 67%. An original survey of Swiss pensioners reveals that reference-dependent preferences is the main source of passivity.

Lambert, T. E. and Tobe, C. B. (2024). ""Safe' Annuity Retirement Products and a Possible US Retirement Crisis." In: *SSRN e-Print*.

This paper examines a looming possible crisis in many Americans' retirement plans due to the proliferation of annuity products in their retirement investment portfolios. As defined benefit pension plans have almost completely disappeared as a means of retirement savings and have been replaced by defined contribution retirement plans over the last 40 to 50 years, a great number of private and public sector defined contribution retirement plans have become laden with insurance contracts called annuities. Of the remaining solid defined benefit plans many, through a process called Pension Risk Transfer are being converted to high-risk single entity annuities. Such products have been sold to employers and employees as 'safe' and 'guaranteed' financial instruments that that are just as good as a defined retirement benefit plan backed by Federal PBGC (Pension Benefit Guarantee Corporation) insurance. The results of the analysis in this paper calls this into question, and with so many of these annuities having ties to investments and loans related to risky assets, the authors find that many annuity products are exposed to systemic risk that could lead to a bust in the pensions of many retirees and soon-to-be retirees. The 'Emperor has no Clothes' as the life insurance industry has poured billions of dollars into advertising, lobbying, commissions & trade articles with misinformation on annuities with everyone afraid to call out the obvious fiduciary problems. To invest in annuities one must look the other way at one of most basic investment principals -diversification, i.e., 'do not put your eggs in one basket." Excessive monopolistic profits through secret spread fees have remained hidden with no US Federal regulation or oversight. This paper shows the drawbacks, weaknesses, and pitfalls of annuities as investments for retirement plans as well as the injustices of such plans toward lower income workers.

Lambregts, T. R. and Schut, F. T. (2020). "Displaced, disliked and misunderstood: A systematic review of the reasons for low uptake of long-term care insurance and life annuities." In: *The Journal of the Economics of Ageing* 17, pp. 100236+.

With aging populations, the role of private insurance in financing late-in-life risks is likely to grow. Yet, demand for long-term care insurance (LTCI) and life annuities (hereafter annuities) is very limited and lags behind economic projections. This systematic literature review surveys the large number of theoretical and empirical studies analyzing this contradiction. We examine the LTCI and annuity puzzles separately and show which factors limit demand for insurance against both late-in-life risks. Our systematic search rendered 3,945 unique hits and findings of 187 studies were integrated in our analyses. Results hereof suggest that holding of both insurance products is systematically impeded by substitution by social security, adverse selection, nonstandard preferences and limited rationality due to low financial literacy and risk unawareness. Furthermore, insurance holding is concentrated among wealthier and subjectively healthier individuals. A comprehensive approach addressing all four reasons for low uptake may increase insurance holding most effectively and may particularly empower people with lower socio-economic status to make well-informed decisions.

Larson, S. (2022). "Required Minimum Distributions as a Retirement Strategy: The Tradeoff Between RMD Volatility and the Expected Number of Dollars Paid Out." In: *Journal of Financial Planning* 35(1), pp. 60–67.

The intent of this paper is to help financial advisers prepare for a client discussion about the required minimum distributions (RMD) withdrawal strategy, which many Americans follow.

Clients who plan to follow this strategy need to be advised that the asset allocation they choose affects the volatility in the RMDs.

It also affects the expected total number of dollars paid out over a client's retirement life expectancy.

Based on historic returns, the tradeoff between the volatility in RMDs and the expected number of dollars paid out is examined.

Larson, S. J. (2018). "Strategy: Assessing the Impact of Required Minimum Distributions on the 4 Percent Rule." In: *Journal of Financial Service Professionals* 72(1), pp. 55–63.

This paper assesses the impact of required minimum distributions on the 4 percent rule. The main finding pertains to ending retirement account balances. They are likely to be overstated when required minimum distributions are ignored in an analysis, and this would create a false sense of security with regard to longevity risk and legacy goals.

Lassance, N., Kan, R., and Wang, X. (2024). "The distribution of out-of-sample returns of estimated optimal portfolios." In: *SSRN e-Print*.

We derive a stochastic representation for the joint distribution of the out-of-sample mean and variance of a large class of portfolio rules that combines the sample optimal mean-variance portfolio with the sample global minimum-variance portfolio. Our results allow the combining coefficients to be either constant or estimated from historical data. Such a representation enables us to obtain the distributions and moments, asymptotically and in finite samples, of three out-of-sample portfolio performance measures, i.e., return, utility, and Sharpe ratio. These results are useful for a variety of applications, and we detail two of them as illustrations. Our paper provides a comprehensive toolkit that researchers can use to evaluate the out-of-sample performance of existing portfolio rules and develop better portfolio rules in the future.

Laster, D., Suri, A., and Vrdoljak, N. (2012). "Systematic Withdrawal Strategies for Retirees." In: *The Journal of Wealth Management* 15(3), pp. 36–49.

The high-profile debate on spending in retirement often misses a crucial point: No single spending rate suits all retirees. This article develops guidance on spending rates for retirees. Taking into account uncertain asset returns, inflation, and longevity, the authors identify spending rates and asset allocations that minimize the expected lifetime shortfall for a systematic withdrawal plan. The analysis shows that sustainable spending rates depend critically on a client's age, gender, and risk tolerance. It also demonstrates that retirees can reduce the risk of outliving their wealth and increase their expected future bequest by allocating at least some of their portfolios to equities. Moreover, the younger the retiree and the greater her spending needs, the more she should allocate to equities. Finally, because their wealth must last longer, couples should allocate more to equities and spend at a more moderate rate than those who are single.

Laster, D., Vrdoljak, N., and Suri, A. (2014). "Pitfalls in Retirement." In: *The Journal of Retirement* 1(1), pp. 91–99.

The market turmoil of recent years has left many who are in or near retirement uncertain about how best to protect their nest eggs. This article offers insights into the key financial pitfalls to which retirees are prone. Investors can achieve their retirement goals by spending sustainably, maintaining equity exposure, addressing longevity and inflation risks, and devising a sound plan with which they feel comfortable.

Laster, D., Vrdoljak, N., and Suri, A. (2016). "Strategies for Managing Retirement Risks." In: *The Journal of Retirement* 4(1), pp. 11–18.

Retirees commonly face four risks that can threaten to derail their retirement: longevity, health care, sequence of returns, and inflation. These risks pose a multifaceted challenge that requires a combination of strategies, such as carefully choosing when to claim Social Security, allocating assets to lifetime income annuities, adopting a systematic withdrawal strategy, and planning for long-term care. This article offers a range of ideas and insights to help retirees achieve a more secure retirement.

Latham, L. (2021). *Bridging the Gap Between Accumulation and Decumulation for Participants*. Tech. rep. T. Rowe Price.

Defined contribution plan participants are increasingly keeping retirement balances in plan, and a growing number of plan sponsors are interested in retaining these balances.

Information gleaned from focus groups suggests that participants have misperceptions about the value of staying in plan. Some participants do not even know that staying in plan is an option after retirement.

Participants approaching retirement age are inundated with information and even "advice" from sources outside their plan, but the quality of this information and advice can be uneven, misguided, or ill-informed. Plan sponsors can position themselves as a reputable resource and guide.

If plan sponsors want to maintain retirees in plan, they should not keep it a secret. It's crucial that they engage with participants early and often.

Lau, S.-H. P. and Ying, Y. (2024). "Deferred Annuities with Gender-Neutral Pricing: Benefitting Most Women Without Adversely Affecting Too Many Men." In: *SSRN e-Print*.

Many countries emphasize gender equality and ban gender-based annuity pricing, leading to more heterogeneous health characteristics and market inefficiency. Governments may respond by offering deferred annuities when annuitants' health characteristics are more similar at an earlier age. The two combined policies benefit most female annuitants without adversely affecting too many male annuitants, contrasting to the well-known result that banning gender-based pricing benefits all women but adversely affects all men, when only immediate annuities are available. Within each gender group, these two policy interventions benefit annuitants with average health, but adversely affect those on either end.

Le Guenedal, T. and Roncalli, T. (2022). *Portfolio Construction with Climate Risk Measures*. Tech. rep. Amundi Asset Management.

Because of the 2015 Paris Agreement, the development of ESG investing and the emergence of net zero emission policies, climate risk is certainly the most important topic and challenge for asset owners and managers now and will remain so over the next five years. It considerably changes portfolio allocation and the investment framework of both passive and active investors. The goal of this paper is to conduct a survey of the various climate risk measures that are available in the asset management industry and the practices of portfolio construction that use these metrics.

Therefore, the first part of this paper lists the different climate risk metrics - e.g., carbon footprint, carbon transition pathway, carbon transition and physical risks. The second part is dedicated to portfolio optimization, in particular portfolio decarbonization and portfolio alignment (Paris-based benchmarks and net zero carbon objective). Among the different findings, two are of great importance for investors. First, portfolio decarbonization is more difficult when we include scope 3 carbon emissions. Indeed, optimizing using the sum of scopes 1, 2 and 3 emissions leads to a portfolio with more tracking error risk than using direct plus first tier indirect carbon emissions. Second, portfolio alignment is more complex than portfolio decarbonization. Since aligning portfolios with scope 3 is becoming the standard approach to climate portfolio construction, the impact on portfolio management may be substantial, and the divergence between carbon investing and traditional investing will increase.

Lee, W. and Liu, P. (2021). "Work Harder: Diligent Rebalancing and Investment Horizon." In: *The Journal of Portfolio Management*.

In an age of constant information flow, investors have to strike the right balance between incorporating incremental information into a portfolio and managing turnover before the end of the investment horizon. Using a tractable yet realistic model of asset and signaldynamics, the authors provide analytical rigor to justify the optimality of the common practice of updating investment views and portfolio positions as new information arrives. In analyzing different approaches to rebalancing, their results indicate that regularly updating the conditional information set achieves a higher information ratio than either waiting to rebalance until after the investment horizon is reached or using regular rebalancing based on a moving average of the past information set. The authors provide guidance on turnover management in conjunction with the information horizon of the signal.

Lee, Y., Kim, Y., Sanz-Cruzado, J., McCreadie, R., and Lee, Y. (2024a). "Stock Recommendations for Individual Investors: A Temporal Graph Network Approach with Mean-Variance Efficient Sampling." In: *arXiv e-Print*.
Recommender systems can be helpful for individuals to make well-informed decisions in complex financial markets. While many studies have focused on predicting stock prices, even advanced models fall short of accurately forecasting them. Additionally, previous studies indicate that individual investors often disregard established investment theories, favoring their personal preferences instead. This presents a challenge for stock recommendation systems, which must not only provide strong investment performance but also respect these individual preferences. To create effective stock recommender systems, three critical elements must be incorporated: 1) individual preferences, 2) portfolio diversification, and 3) the temporal dynamics of the first two. In response, we propose a new model, Portfolio Temporal Graph Network Recommender, PfoTGNRec, which can handle time-varying collaborative signals and incorporates diversification-enhancing sampling. On real-world individual trading data, our approach demonstrates superior performance compared to state-of-the-art baselines, including cutting-edge dynamic embedding models and existing stock recommendation models. Indeed, we show that PfoTGNRec is an effective solution that can balance customer preferences with the need to suggest portfolios with high Return-on-Investment. The source code and data are available at `https://anonymous.4open.science/r/ICAIF2024-E23E`.

Lee, Y., Kim, Y., Sanz-Cruzado, J., Mccreadie, R., and Lee, Y. (2024b). "Stock Recommendations for Individual Investors: A Temporal Graph Network Approach with Mean-Variance Efficient Sampling." In: *Proceedings of the 5th ACM International Conference on AI in Finance*. ICAIF '24. ACM, pp. 795–803.
Recommender systems can be helpful for individuals to make well-informed decisions in complex financial markets. While many studies have focused on predicting stock prices, even advanced models fall short of accurately forecasting them. Additionally, previous studies indicate that individual investors often disregard established investment theories, favoring their personal preferences instead. This presents a challenge for stock recommendation systems, which must not only provide strong investment performance but also respect these individual preferences. To create effective stock recommender systems, three critical elements must be incorporated: 1) individual preferences, 2) portfolio diversification, and 3) the temporal dynamics of the first two. In response, we propose a new model, Portfolio Temporal Graph Network Recommender PfoTGNRec, which can handle time-varying collaborative signals and incorporates diversification-enhancing sampling. On real-world individual trading data, our approach demonstrates superior performance compared to state-of-the-art baselines, including cutting-edge dynamic embedding models and existing stock recommendation models. Indeed, we show that PfoTGNRec is an effective solution that can balance customer preferences with the need to suggest portfolios with high Return-on-Investment. The source code and data are available at `https://github.com/youngandbin/PfoTGNRec`.

Lefrancois, R., Mamidipudi, P., and Li, J. (2020). "Expectation Risk: A Novel Short-Term Risk Measure for Long-Term Financial Projections." In: *SSRN e-Print*.
Presenting the results of long-term financial projections, particularly those of common interest in retirement planning such as portfolio value and retirement income, can be difficult using traditional risk measures . Long-term projected distributions tend to be very wide and offer limited insight to practitioners into what kind of adjustments might be required along the way to avoid undesirable outcomes. Short-term projections, where available, lack information about final, long-term outcomes. As a complement to the pure long-term and short-term risk measures associated with these distributions, we propose a third measure, expectation risk, which reflects the short-term risk in long-term outcomes. This measure is derived from the short-term distribution of long-term expectations. In the following, we demonstrate how to calculate these distributions for simple cases of portfolio value and retirement income. Further, we show how expectation risk can be used to estimate the magnitude of short-term adjustments that may be required to obtain a desired long-term income.

Leganza, J. M. (2023). "The Effect of Required Minimum Distributions on Intergenerational Transfers." In: *SSRN e-Print*.
How do households use retirement savings accounts in retirement? The answer to this question is important for tax policy pertaining to retirement savings. I shed light on this question by studying how households respond to Required Minimum Distribution (RMD) regulations, which mandate withdrawals from retirement accounts upon reaching a specified age. Using data from the Health and Retirement Study and a regression discontinuity design, I estimate the causal effects of aging into RMD regulations. First, I establish the direct effects of RMDs in my setting and show a sharp increase in withdrawals from Individual Retirement Accounts (IRAs). Next, I provide new evidence on the indirect effects of RMDs and show a concurrent, discontinuous increase in inter vivos transfers. The results indicate that some households ultimately use IRAs to facilitate intergenerational gifts,

holding wealth in the tax-advantaged accounts until required to take distributions and then passing resources to children.

Leggio, K. B., Lien, D., and Dai, Y. (2024). "The Death of the 4% Rule: New Market Conditions and Retiree Needs Require Rethinking Retirement Spending." In: *The Journal of Retirement* 11(4), pp. 40–60.

The dilemma of how to effectively decumulate retirement savings poses a critical question: Can retirees withdraw 4% of their retirement wealth each year? This study delves into this inquiry, leveraging Monte Carlo simulation to evaluate six decumulation strategies. Unlike conventional studies that assume normal distributions for stock and bond returns, the authors utilize a machine learning algorithm to identify the most suitable distributions. Their results reveal that a static 4% withdrawal rate adjusting for inflation from the initial portfolio may heighten the risk of retirees depleting their savings prematurely. Instead, the authors advocate a sustainable approach: retirees should align annual withdrawals with the remaining portfolio value, accommodating market changes. Considering the delicate balance of securing long-term financial well-being while preventing early fund depletion, the authors identify two superior strategies: adhering to either age-based withdrawal percentages from the Internal Revenue Service's Required Minimum Distribution tables or to the fixed 6% rate, adjusted for inflation, both tied to the remaining portfolio value.

Lehner, S. (2025). "Why Do Robo-Advisors Systematically Fail? A Call for Individualized Solutions." In: *SSRN e-Print*.

Robo-advisors promise low-cost, diversified portfolios but often overlook key householdspecific risks and Merton's hedging demands. This paper develops a dynamic portfolio choice model incorporating correlated labor income, housing wealth, tax considerations, and lifecycle needs-factors standard robo-advice commonly ignores. By explicitly modeling hedging demands, alpha-seeking behavior, liquidity needs, and multiple asset classes, we show how oversimplified robo allocations can lead to significant utility losses. Our results highlight the value of a richer, multi-period advisory framework that accounts for household-specific exposures. Finally, we suggest practical enhancements for robo platforms, including broader asset menus, dynamic rebalancing, and better client-data integration.

Leung, T. and Ward, B. (2018). "Dynamic index tracking and risk exposure control using derivatives." In: *Applied Mathematical Finance* 25(2), pp. 180–212.

We develop a methodology for index tracking and risk exposure control using financial derivatives. Under a continuous-time diffusion framework for price evolution, we present a pathwise approach to construct dynamic portfolios of derivatives in order to gain exposure to an index and/or market factors that may be not directly tradable. Among our results, we establish a general tracking condition that relates the portfolio drift to the desired exposure coefficients under any given model. We also derive a slippage process that reveals how the portfolio return deviates from the targeted return. In our multi-factor setting, the portfolio realized slippage depends not only on the realized variance of the index but also the realized covariance among the index and factors. We implement our trading strategies under a number of models, and compare the tracking strategies and performances when using different derivatives, such as futures and options.

Levantesi, S. and Menzietti, M. (2012). "Managing longevity and disability risks in life annuities with long term care." In: *Insurance: Mathematics and Economics* 50(3), pp. 391–401.

The aim of the paper is twofold. Firstly, it develops a model for risk assessment in a portfolio of life annuities with long term care benefits. These products are usually represented by a Markovian Multi-State model and are affected by both longevity and disability risks. Here, a stochastic projection model is proposed in order to represent the future evolution of mortality and disability transition intensities. Data from the Italian National Institute of Social Security (INPS) and from Human Mortality Database (HMD) are used to estimate the model parameters. Secondly, it investigates the solvency in a portfolio of enhanced pensions. To this aim a risk model based on the portfolio risk reserve is proposed and different rules to calculate solvency capital requirements for life underwriting risk are examined. Such rules are then compared with the standard formula proposed by the Solvency II project.

Levine, J. (2021). "No Investment Fee Is Small, Long Term." In: *arXiv e-Print*.

This note presents a simple "back of the envelope" formula estimating that an annual investment fee expense ϵ%, compounding for N years, consumes almost $N\epsilon$% of an investment's value. For example, an investment with 1% annual fee compounding for 30 years pays almost 30% of its final value to fees over this period. This approximation captures how annual fees can compound over a decade or more to cumulative costs an order of magnitude higher. The formula is simple enough for rapid hand calculation.

Lewis, N. D. (2009). "Is There a Role for Commodities in Long-Term Wealth Accumulation?" In: *The Journal of Wealth Management* 12(2), pp. 130–137.
The first three years of the 21st century brought one of the worst bear markets in U.S. history, with equity markets around the world falling around 40% in real terms. This was followed in 2008 and 2009 by a global slump in economic activity and stock market price declines not seen since the great depression. The poor performance of traditional asset classes and concern over their future prospects have raised interest in the inclusion of alternative assets as strategic sources of long-term wealth creation. In sharp contrast to the recommendations of modern portfolio theory, a vast majority of high-net-worth investors are not well diversified. The article outlines the role of commodities in long-term wealth maximization. It demonstrates an alternative approach to measuring risk and diversification by addressing the question: How much can I realistically expect to gain by strategically investing in commodities over my working life? The method is easy to understand and can be easily implemented. Even individual investors with small portfolios can use the approach to gauge their own diversification benefits and risk exposure.

Lewis, W. C. and Caliendo, F. N. (2006). "Tax-Deferred Retirement Saving." In: *The Journal of Wealth Management* 8(4), pp. 12–16.
The authors first observe that both the widespread popularity of saving in tax-deferred retirement vehicles such as IRAs, Keogh plans, and 401(k) programs and the magnitude of the dollar flows into such saving are almost prima facie evidence that at the margin they provide significant gains over saving in traditional taxable accounts, at least for those wanting to defer some part of lifetime consumption to their retirement years. They focus on the gains arising solely from shifting taxable income from the working (i.e., accumulation) years to the retirement (i.e., distribution) years. They turn to the advantages of tax-deferred investing and to the motivation for focusing on the intertemporal shifting of taxable income. Finally, they propose an informal model of the gains from such income shifting and illustrate the magnitude of the gains for levels of annual wage income ranging from 50,000 to 200,000. They show that the gain (as measured by the difference in after-tax retirement income from tax-deferred and taxable accounts) increases with wage income and ranges from 1.7 percent to 11.4 percent. A formal theoretical model is provided in the appendix to the article.

Li, H. and Hyndman, R. J. (2021). "Assessing mortality inequality in the U.S.: What can be said about the future?" In: *Insurance: Mathematics and Economics* 99, pp. 152–162.
This paper investigates mortality inequality across U.S. states by modeling and forecasting mortality rates via a forecast reconciliation approach. Understanding the heterogeneity in state-level mortality experience is of fundamental importance, as it can assist decision making for policymakers, health authorities, as well as local communities who are seeking to reduce inequalities and disparities in life expectancy. A key challenge of multi-population mortality modeling is high dimensionality, and the resulting complex dependence structures across sub-populations. Moreover, when projecting future mortality rates, it is important to ensure that the state-level forecasts are coherent with the national-level forecasts. We address these issues by first obtaining independent state-level forecasts based on classical stochastic mortality models, and then incorporating the dependence structure in the forecast reconciliation process. Both traditional bottom-up reconciliation and the cutting-edge trace minimization reconciliation methods are considered. Based on the U.S. total mortality data for the period 1969-2017, we project the 10-year-ahead mortality rates at both national-level and state-level up to 2027. We find that the geographical inequality in the longevity levels is likely to continue in the future, and the mortality improvement rates will tend to slow down in the coming decades.

Li, H. and Shi, Y. (2021). "Mortality Forecasting with an Age-Coherent Sparse VAR Model." In: *Risks* 9(2), p. 35.
This paper proposes an age-coherent sparse Vector Autoregression mortality model, which combines the appealing features of existing VAR-based mortality models, to forecast future mortality rates. In particular, the proposed model utilizes a data-driven method to determine the autoregressive coefficient matrix, and then employs a rotation algorithm in the projection phase to generate age-coherent mortality forecasts. In the estimation phase, the age-specific mortality improvement rates are fitted to a VAR model with dimension reduction algorithms such as the elastic net. In the projection phase, the projected mortality improvement rates are assumed to follow a short-term fluctuation component and a long-term force of decay, and will eventually converge to an age-invariant mean in expectation. The age-invariance of the long-term mean guarantees age-coherent mortality projections. The proposed model is generalized to multi-population context in a computationally efficient manner. Using single-age, uni-sex mortality data of the UK and France, we show that the proposed model is able to generate more reasonable long-term projections, as well as more accurate short-term out-of-sample forecasts

than popular existing mortality models under various settings. Therefore, the proposed model is expected to be an appealing alternative to existing mortality models in insurance and demographic analyses.

Li, H., Zhou, R., and Ji, M. (2023a). "Nonlinear Modeling of Mortality Data and Its Implications for Longevity Bond Pricing." In: *Risks* 11(12), p. 207.

Human mortality has been improving faster than expected over the past few decades. This unprecedented improvement has caused significant financial stress to pension plan sponsors and annuity providers. The widely recognized Lee-Carter model often assumes linearity in its period effect as an integral part of the model. Nevertheless, deviation from linearity has been observed in historical mortality data. In this paper, we investigate the applicability of four nonlinear time-series models: threshold autoregressive model, Markov switching model, structural change model, and generalized autoregressive conditional heteroskedasticity model for mortality data. By analyzing the mortality data from England and Wales and Italy spanning the years 1900 to 2019, we compare the goodness of fit and forecasting performance of the four nonlinear models. We then demonstrate the implications of nonlinearity in mortality modeling on the pricing of longevity bonds as a practical illustration of our findings.

Li, J. S.-H., Liu, Y., and Chan, W.-S. (2023b). "Hedging longevity risk under non-Gaussian state-space stochastic mortality models: A mean-variance-skewness-kurtosis approach." In: *Insurance: Mathematics and Economics* 113, pp. 96–121.

Longevity risk has recently become a high profile risk among insurers and pension plan sponsors. One way to mitigate longevity risk is to build a hedge using derivatives that are linked to mortality indexes. Longevity hedging methods are often based on the normality assumption, considering only the variance but no other (higher) moments. However, strong empirical evidence suggests that mortality improvement rates are driven by asymmetric and fat-tailed distributions, so that existing longevity hedging methods should be expanded to incorporate higher moments. This paper fills the gap by adopting a mean-variance-skewness-kurtosis approach based on non-Gaussian extensions of commonly-used stochastic mortality models, formulated in a state-space setting. On the basis of a general representation of these models, the authors derive (approximate) analytical expressions for the moments of the present values of the hedging instruments and the liability being hedged. These expressions are then integrated with a polynomial goal programming model, from which the optimal hedge portfolio is identified. Finally, the paper demonstrates the theoretical results with a real mortality data set and a range of hedger preferences.

Li, X. (2023). "Bond Implied Risks around Macroeconomic Announcements." In: *The Journal of Fixed Income* 33(2), pp. 50–97.

Using a large panel of Treasury futures and options, this study constructs model-free measures of bond uncertainty and tail risks. The author mainly studies the behavior of bond risk measures around FOMC announcements and document three novel findings. First, bond uncertainty risk displays a rise and resolution similar to the stock VIX index, while tail risks don't respond to announcements. Second, pre-FOMC announcement drift exists in terms of Treasury yields declining by 1 bps on the day before the announcement. Third, option-implied uncertainty cannot help explain the pre-FOMC announcement drift.

Li, Y. and Forsyth, P. A. (2019). "A data-driven neural network approach to optimal asset allocation for target based defined contribution pension plans." In: *Insurance: Mathematics and Economics* 86, pp. 189–204.

A data-driven Neural Network (NN) optimization framework is proposed to determine optimal asset allocation during the accumulation phase of a defined contribution pension scheme. In contrast to parametric model based solutions computed by a partial differential equation approach, the proposed computational framework can scale to high dimensional multi-asset problems. More importantly, the proposed approach can determine the optimal NN control directly from market returns, without assuming a particular parametric model for the return process. We validate the proposed NN learning solution by comparing the NN control to the optimal control determined by solution of the Hamilton-Jacobi-Bellman (HJB) equation. The HJB equation solution is based on a double exponential jump model calibrated to the historical market data. The NN control achieves nearly optimal performance. An alternative data-driven approach (without the need of a parametric model) is based on using the historic bootstrap resampling data sets. Robustness is checked by training with a blocksize different from the test data. In both two and three asset cases, we compare performance of the NN controls directly learned from the market return sample paths and demonstrate that they always significantly outperform constant proportion strategies.

Liberman, J. and Sosner, N. (2025a). "A Brief Guide to Pricing and Taxation of Variable Prepaid Forwards." In: *The Journal of Wealth Management* 27(4), pp. 99–116.

Variable prepaid forward (VPF) contracts have been developed as a solution to hedging the risk of concentrated, low-basis stock. Using options pricing theory, we develop a VPF pricing model that helps understand the VPF prepayment amount and the cash flows and tax liabilities upon VPF rolls. We also provide a detailed discussion of the application of tax straddle rules to VPF transactions. We show that, depending on the circumstances, a VPF can offer a prepayment exceeding 90% of the value of the stock committed to the contract, with the VPF floor having a major impact on the prepayment amount. While for some price ranges, rolling a VPF may generate a positive cash flow for the investor, we find that rolling a VPF can be very costly when the stock price declines significantly-which is the very scenario from which the investor seeks protection by entering into the VPF contract in the first place.

Liberman, J. and Sosner, N. (2025b). "Combining Variable Prepaid Forwards and Tax-Aware Strategies to Diversify Low-Basis Stock." In: *The Journal of Wealth Management* 27(4), pp. 117–135.

We illustrate how combining variable prepaid forward (VPF) contracts with tax-aware strategies can help diversify low-basis stock and thereby improve after-tax wealth accumulation. We find that direct-indexing strategies have limited ability to offset VPF settlement gains. Thus, in the context of a VPF transaction, direct indexing adds value only when managed at a very low cost. Tax-aware long-short factor strategies offer two advantages over direct indexing. First, they can outperform a passive index before tax. Second, they realize significantly higher net losses than direct-indexing strategies, allowing the investor to offset a larger fraction of the VPF settlement gain. As a result, long-run after-tax wealth outcomes are significantly better when a VPF is combined with tax-aware long-short factor strategies rather than with other alternatives, such as a direct-indexing strategy or a market index fund.

Linnainmaa, J. T., Melzer, B., and Previtero, A. (2021). "The Misguided Beliefs of Financial Advisors." In: *SSRN e-Print*.

A common view of retail finance is that conflicts of interest contribute to the high cost of advice. Within a large sample of Canadian financial advisors and their clients, however, we show that advisors typically invest personally just as they advise their clients. Advisors trade frequently, chase returns, prefer expensive, actively managed funds, and underdiversify. Advisors' net returns of -3% per year are similar to their clients' net returns. Advisors do not strategically hold expensive portfolios only to convince clients to do the same; they continue to do so after they leave the industry.

Liu, D. (2019). "Analytical solutions of optimal portfolio rebalancing." In: *Quantitative Finance* 19(4) (683-697), pp. 1–15.

We study optimal portfolio rebalancing in a mean-variance type framework and present new analytical results for the general case of multiple risky assets. We first derive the equation of the no-trade region, and then provide analytical solutions and conditions for the optimal portfolio under several simplifying yet important models of asset covariance matrix: uncorrelated returns, same non-zero pairwise correlation, and a one-factor model. In some cases, the analytical conditions involve one or two parameters whose values are determined by combinatorial, rather than numerical, algorithms. Our results provide useful and interesting insights on portfolio rebalancing, and sharpen our understanding of the optimal portfolio.

Liu, P. (2023). "A Review on Derivative Hedging Using Reinforcement Learning." In: *The Journal of Financial Data Science* 5(2), pp. 136–145.

Hedging is a common trading activity to manage the risk of engaging in transactions that involve derivatives such as options. Perfect and timely hedging, however, is an impossible task in the real market that characterizes discrete-time transactions with costs. Recent years have witnessed reinforcement learning (RL) in formulating optimal hedging strategies. Specifically, different RL algorithms have been applied to learn the optimal offsetting position based on market conditions, offering an automatic risk management solution that proposes optimal hedging strategies while catering to both market dynamics and restrictions. In this article, the author provides a comprehensive review of the use of RL techniques in hedging derivatives. In addition to highlighting the main streams of research, the author provides potential research directions on this exciting and emerging field.

Liu, X.-Y., Yang, H., Gao, J., and Wang, C. (2021). "FinRL: Deep Reinforcement Learning Framework to Automate Trading in Quantitative Finance." In: *SSRN e-Print*.

Deep reinforcement learning (DRL) has been envisioned to have a competitive edge in quantitative finance. However, there is a steep development curve for quantitative traders to obtain an agent that automatically positions to win in the market, namely to decide where to trade, at what price and what quantity, due to the error-prone programming and arduous debugging. In this paper, we present the first open-source framework FinRL as a full pipeline to help quantitative traders overcome the steep learning curve. FinRL is featured with simplicity, applicability and extensibility under the key principles, full-stack framework, customization, reproducibility and hands-on tutoring. Embodied as a three-layer architecture with modular structures, FinRL implements fine-tuned state-of-the-art DRL algorithms and common reward functions, while alleviating the debugging work- loads. Thus, we help users pipeline the strategy design at a high turnover rate. At multiple levels of time granularity, FinRL simu- lates various markets as training environments using historical data and live trading APIs. Being highly extensible, FinRL reserves a set of user-import interfaces and incorporates trading constraints such as market friction, market liquidity and investor's risk-aversion. Moreover, serving as practitioners' stepping stones, typical trad- ing tasks are provided as step-by-step tutorials, e.g., stock trading, portfolio allocation, cryptocurrency trading, etc.

Liu, X. and Viswanathan, V. (2020). "The term structure of the rebalancing premium." In: *The Journal of Investing* 29(4), pp. 116–125.
The optimal rebalance interval that maximizes the expected geometric return of passive strategies decreases in relation to increasing asset volatility, increasing dispersion in asset volatility, and increasing negative 1-month autocorrelation of asset returns. Conversely, the optimal rebalance interval increases in relation to increasing positive 1-month return autocorrelation and increasing dispersion in expected return of the underlying assets. Much of the literature on the rebalancing premium has focused on the effect of asset volatility. But empirically, the positive autocorrelation of asset returns and the dispersion in asset expected returns have a much larger impact on the rebalancing premium. Depending on the set of assets chosen, the optimal rebalance frequency can vary between 6 months for an equal-weighted portfolio of multiple asset classes and 10 years for an equal-weighted portfolio of regional indices. The large variation in optimal rebalance frequency challenges the notion of a universal rebalancing premium.

Liu, Y. (2024). "Goal Based Dynamic Spending and Investing." In: *SSRN e-Print*.
Motivated by the need of a theoretically sound and practically applicable personalized retirement planning solution, we develop a goal based model for solving the lifetime spending and investing problems. We apply a simple goal value function that measures the percentage of goal targets being fulfilled in each period, and formulate the dynamic programming problem to smooth the consumption over the time by balancing the current spending and expected future spending. The consumption and investment decisions reflect the interaction between the investor's wealth level, goal target and risk attitude. In addition to solving the multi-period financial planning problem, we also connect the goal based value functions with classical (risk-averse) utility function and the value function in prospect theory, and illustrate how the goal based framework can help us understand the decision under risk in a more clear and coherent way.

Lo, A. W. and Ross, J. (2024). "Can ChatGPT Plan Your Retirement?: Generative AI and Financial Advice." In: *SSRN e-Print*.
We identify some of the most pressing issues facing the adoption of large language models (LLMs) in practical settings, and propose a research agenda to reach the next technological inflection point in generative AI. We focus on four challenges facing most LLM applications: domain-specific expertise, an ability to tailor that expertise to a user's unique situation, trustworthiness and adherence to the user's moral and ethical standards, and conformity to regulatory guidelines and oversight. These challenges apply to virtually all industries and endeavors in which LLMs can be applied, such as medicine, law, accounting, education, psychotherapy, marketing, and corporate strategy. For concreteness, we focus on the narrow context of financial advice, which serves as an ideal test bed both for determining the possible shortcomings of current LLMs, and for exploring ways to overcome them. Our goal is not to provide solutions to these challenges-which will likely take years to develop-but to propose a framework and road map for solving them as part of a larger research agenda for improving generative AI in any application.

Lobel, H., Jaconetti, C. M., and Cuff, R. (2019). *The replacement ratio: Making it personal*. Tech. rep. Vanguard.
A replacement ratio is a rule of thumb that estimates what percentage of a person's pre-retirement income will be needed to maintain their lifestyle at retirement. Most studies suggest aiming for a target of between 70 and 85 percent of pre-retirement income. Knowing which end of that range would be more appropriate, however, is an important step in developing a retirement plan. With our approach to calculating the replacement ratio, investors begin with their current annual consumption and then factor in the changes in taxes and health-care costs. The replacement ratio is that total amount expressed as a percentage of the investor's pre-retirement income. Because people-and thus their retirement goals-are unique, otherwise-similar investors can have different replacement ratios. Variables affecting the desired ratio include broad demographic differences (marital status

and income level, for example) as well as more subtle, personalized influences (homeownership, health status, the type of accounts dedicated for retirement, for example).

Lohre, H., Rother, C., and Schafer, K. A. (2020). "Hierarchical Risk Parity: Accounting for Tail Dependencies in Multi-asset Multi-factor Allocations." In: *Machine Learning for Asset Management: New Developments and Financial Applications*. Ed. by E. Jurczenko. Wiley, pp. 329–368.

This chapter examines the use and merits of hierarchical clustering techniques in the context of multi-asset multi-factor investing. In particular, it contrasts these techniques with several competing risk-based allocation paradigms, such as 1/N, minimum-variance, standard risk parity and diversified risk parity. The chapter introduces hierarchical risk parity (HRP) strategies based on the Pearson correlation coefficient and also introduces hierarchical clustering based on the lower tail dependence coefficient. The chapter provides an overview of traditional risk-based allocation strategies and outlines a framework to measure and manage portfolio diversification. It examines the performance of the introduced HRP strategies relative to the traditional alternatives. The chapter discusses Meucci's approach to managing diversification, which serves to construct a diversified risk parity strategy based on economic factors.

Lopez de Prado, M. (2019). "A Data Science Solution to the Multiple-Testing Crisis in Financial Research." In: *The Journal of Financial Data Science* 1(1), pp. 99–110.

Most discoveries in empirical finance are false, as a consequence of selection bias under multiple testing. Although many researchers are aware of this problem, the solutions proposed in the literature tend to be complex and hard to implement. In this article, the author reduces the problem of selection bias in the context of investment strategy development to two sub-problems: determining the number of essentially independent trials and determining the variance across those trials. The author explains what data researchers need to report to allow others to evaluate the effect that multiple testing has had on reported performance. He applies his method to a real case of strategy development and estimates the probability that a discovered strategy is false.

Lopez de Prado, M. and Lewis, M. J. (2019). "Detection of false investment strategies using unsupervised learning methods." In: *Quantitative Finance* 19(9), pp. 1555–1565.

In this paper we address the problem of selection bias under multiple testing in the context of investment strategies. We introduce an unsupervised learning algorithm that determines the number of effectively uncorrelated trials carried out in the context of a discovery. This estimate is critical for computing the familywise false positive probability, and for filtering out false investment strategies.

Lopez de Prado, M., Lipton, A., and Zoonekynd, V. (2025). "Causal Factor Analysis is a Necessary Condition for Investment Efficiency." In: *SSRN e-Print*.

This article reveals the dire consequences of factor model misspecification in the context of portfolio optimization. We show that it is not possible to estimate the efficient frontier without a causal factor model, and that the prevailing (associational) factor modeling paradigm leads to investment inefficiency. Furthermore, we show that the biases introduced by factor model misspecification can be so large as producing portfolios where investors buy what they should sell and vice versa, even when investors have perfect knowledge of the mean and covariances. Our findings challenge the scientific soundness of the portfolio construction paradigms currently used by the asset management industry. To overcome these pitfalls, academics and practitioners should rebuild the financial economics literature on the more scientifically rigorous grounds of causal factor investing.

Love, D. and Phelan, G. (2015). "Hyperbolic discounting and life-cycle portfolio choice." In: *Journal of Pension Economics and Finance* 14, pp. 492–524.

This paper studies how hyperbolic discounting affects stock market participation, asset allocation, and saving decisions over the life cycle in an economy with Epstein Zin preferences. Hyperbolic discounting affects saving and portfolio decisions through at least two channels: (1) it lowers desired saving, which decreases financial wealth relative to future earnings; and (2) it lowers the incentive to pay a fixed cost to enter the stock market. We find that hyperbolic discounters accumulate less wealth relative to their geometric counterparts and that they participate in the stock market at a later age. Because they have lower levels of financial wealth relative to future earnings, hyperbolic discounters who do participate in the stock market tend to hold a higher share of equities, particularly in the retirement years. We find that increasing the elasticity of intertemporal substitution, holding risk aversion constant, greatly magnifies the impact of hyperbolic discounting on all of the model's decision rules and simulated levels of participation, allocation, and wealth. Finally, we introduce endogenous financial knowledge accumulation and find that hyperbolic discounting leads to lower financial literacy and inefficient stock market investment.

Lozada, G. A. (2018a). *Financing Retirement using U.S. Treasury Bonds: Safe Withdrawal Rates, Mean/Standard-Deviation Frontiers,and Endpoint-Dependence of the Safest Maturity*. Tech. rep. University of Utah.
Calculating, for a fixed asset allocation, completely-SafeWithdrawal Rates over various historical periods of one fixed lengthgenerates a "SWR" distribution. Combining its minimum, or meanand standard deviation, with those of other allocations implies amaximin portfolio, or a SWR Mean/Standard-Deviation Frontier.Considering portfolios only of Treasurys of one constant maturity, the lowest-withdrawal-variance asset was sometimes not theshortest-maturity, and sometimes was the longest. Constructingmost subperiods, these anomalies were very rare for real SWRs andoccurred about one-fifth of the time for nominal SWRs, happeningwhen money-market yields changed greatly.

Lozada, G. A. (2018b). *Fixed income for retirement saving: TIAA traditional lessons on quality, duration, risk, and gradual withdrawals*. Tech. rep. University of Utah.
TIAA Traditional annuity has supported retirements for a century. It resembles a stable-value fund. The author investigates the holdings supporting it, constructs readily available alternatives resembling those holdings, compares the returns of those alternatives with Traditional, and constructs a new measure of risk to compare Traditional risk with that of its alternatives in a more appropriate way than by using short-term standard deviation of returns. Under this new measure of risk, which is appropriate for retirement investors, some alternatives exhibited second-degree stochastic dominance over Traditional using 1987-2015 data. However, Traditional may still be a better choice for unsophisticated investors.

Lozada, G. A. (2020). "Fixed income for retirement saving: TIAA traditional lessons on quality, duration, risk, and gradual withdrawals." In: *The Journal of Retirement* 7(4), pp. 39–59.
TIAA Traditional annuity has supported retirements for a century. It resembles a stable-value fund. The author investigates the holdings supporting it, constructs readily available alternatives resembling those holdings, compares the returns of those alternatives with Traditional, and constructs a new measure of risk to compare Traditional risk with that of its alternatives in a more appropriate way than by using short-term standard deviation of returns. Under this new measure of risk, which is appropriate for retirement investors, some alternatives exhibited second-degree stochastic dominance over Traditional using 1987-2015 data. However, Traditional may still be a better choice for unsophisticated investors.

Lu, Y. and Zhu, D. (2023). "Modelling mortality: A bayesian factor-augmented var (favar) approach." In: *ASTIN Bulletin* 53(1), pp. 29–61.
Longevity risk is putting more and more financial pressure on governments and pension plans worldwide due to pensioners' increasing trend of life expectancy and the growing numbers of people reaching retirement age. Lee and Carter (1992, Journal of the American Statistical Association, 87(419), 659-671.) applied a one-factor dynamic factor model to forecast the trend of mortality improvement, and the model has since become the field's workhorse. It is, however, well known that their model is subject to the limitation of overlooking cross-dependence between different age groups. We introduce Factor-Augmented Vector Autoregressive (FAVAR) models to the mortality modelling literature. The model, obtained by adding an unobserved factor process to a Vector Autoregressive (VAR) process, nests VAR and Lee-Carter models as special cases and inherits both frameworks' advantages. A Bayesian estimation approach, adapted from the Minnesota prior, is proposed. The empirical application to the US and French mortality data demonstrates our proposed method's efficacy in both in-sample and out-of-sample performance.

Luger, R. (2024). "Regularizing stock return covariance matrices via multiple testing of correlations." In: *Journal of Econometrics*, p. 105753.
This paper develops a large-scale inference approach for the regularization of stock return covariance matrices. The framework allows for the presence of heavy tails and multivariate GARCH-type effects of unknown form among the stock returns. The approach involves simultaneous testing of all pairwise correlations, followed by setting non-statistically significant elements to zero. This adaptive thresholding is achieved through sign-based Monte Carlo resampling within multiple testing procedures, controlling either the traditional familywise error rate, a generalized familywise error rate, or the false discovery proportion. Subsequent shrinkage ensures that the final covariance matrix estimate is positive definite and well-conditioned while preserving the achieved sparsity. Compared to alternative estimators, this new regularization method demonstrates strong performance in simulation experiments and real portfolio optimization.

Lumby, J. (2017). "Three Essays on Managing Risk in Retirement." PhD thesis. Texas Tech University.
Individuals face a number of risks when entering retirement, such as unknown life expectancy, variable medical expenses, and uncertain financial market returns. Understanding these risks in advance will allow retirees to

pursue appropriate risk management strategies to reduce the likelihood of experiencing negative events that could jeopardize the retirement period. Unfortunately, many retirees either do not understand the risks that they face, or are ill-prepared to handle these risks when they inevitably arise. In this dissertation, I provide new insights into the dynamic relationship between consumer knowledge, risk exposure, and insurance coverage. The first essay provides evidence that many elderly individuals do not understand the risks associated with unknown late-in-life medical expenses, such as nursing care. When educated, many retirees indicate an increased willingness to purchase a suitable insurance policy as a hedge against the risk. The second essay finds that a small percentage of retirees are insured against multiple retirement risks, and that this behavior is associated with increased financial wealth and successful retirement outcomes. Although insurance can provide a hedge against a multitude of financial risks, the third essay suggests that life satisfaction in retirement is about more than material wealth.

Lung, E., Roodt, C., Ryan, L., Warren, G. J., and Wymer, K. (2021). "A Framework for Designing Investment Strategies for Default Retirement Plans." In: *The Journal of Retirement* 8(3), pp. 40–60.
We identify and discuss four key elements to address when designing the investmentstrategy for default retirement plans: whether to cater for member needs or their wants, objectives, the member for which the default is being designed, and risk appetite. Addressing these design elements requires making assumptions about the member, of which the literature provides limited guidance to plan sponsors. We outline the main assumptions and demonstrate the potential impact on retirement experience through illustrative models. We find that mismatches between the member and the way that they are characterized can adversely impact on welfare.

Lurtz, M. R., Archuleta, K., Kothakota, M., and Jorgensen, T. J. (2021). "A deeper dive: A mixed methods approach to risk tolerance." In: *Financial Planning Review* 4(2) (e112).
Most risk tolerance studies are quantitative, even though many factors that may affect the manifestation of risk tolerance are qualitative. This study employed a mixed-methods approach to investigate how individuals consider risk tolerance as it relates to their financial situation. Fuzzy-trace theory, a psycholinguistic theory of risk processing rooted in prospect theory that is becoming increasingly popular in the medical field, guided the study. Quantitative results indicate that stated versus revealed measures of risk tolerance are not consistent for most people. However, higher risk literacy increases the likelihood of consistency. Qualitative results reveal that individuals perceive risk tolerance through various lenses, including knowledge, values, emotions, and personal experience.

Lussier, J. (2019). "Secure retirement: connecting financial theory and human behavior." In: *CFA Institute Research Foundation*.
Investors fear return uncertainty and drawdowns associated with owning relatively risky asset classes, such as equity. The fact that greater risk is associated with greater expected return does not preclude the possibility that realized returns may be far less than a low-risk asset could provide, even with horizons as long as 5 to 10 years. Fear prompts the average investor to sometimes act against his own best interest. Therefore, the average investor portfolio often underperforms a static benchmark, even before fees. The average investor tends to increase allocation to riskier assets after the market has already significantly risen and decrease allocation after a significant decline.Given that financial planning in the context of retirement is a multidecade endeavor that can last 50 years or more, investors face an additional challenge. Financial planners cannot be concerned solely with the behavior of investors facing short-term return uncertainty, which remains an important challenge, but must also be concerned with how to address longer-term uncertainties. For example, there is significant uncertainty about the assumption for long-term expected returns, real as well as nominal. Some lucky investors may accumulate much of their wealth before retirement during an exceptionally strong bull market (e.g., 1982 to 2000), whereas less fortunate investors may have been planning to retire in 2008 or early 2009, just after the most dramatic global liquidity crisis since the Great Depression and a significant decline in interest rates. Moreover, average expected returns represent only part of the story. Some investors may have recorded above-average returns at a time when their accumulated wealth was already significant, whereas others may have recorded above-average returns when their accumulated wealth was low. The implication of the interactions among savings patterns (in accumulation), withdrawal patterns (in decumulation), and timing of above- and below-average portfolio returns is important to understand, particularly with respect to how the interactions should affect the allocation policy.Furthermore, investors face significant risks in the transition period - say, the last 5 to 10 years - from accumulation to decumulation. Considering the uncertainty in long-term expected returns, patterns of returns, and patterns of savings, it is unlikely that a financial plan established when an individual is 30 years old can remain static thereafter. The plan must be revaluated periodically, which requires

an effective feedback mechanism or tool. The investor has many more options available before retirement, however, although some may not necessarily be pleasant. In the event of lower than expected accumulated wealth, the investor could decide to save more, postpone retirement, and/or adjust retirement plans. These options may be unavailable or may be harder to implement once the decision to retire has been made. The investor must implement a transition strategy that reduces the likelihood and/or significance of the unplanned adjustments that may be necessary as the targeted or desired retirement date draws near.In addition, longevity remains uncertain, and retirement plans often are based on expected longevity. Even though the median life expectancy for a 65-year-old individual in the United States is 83.3 years for men and 85.9 years for women, a significant percentage of people will live past age 90. Furthermore, as someone ages, the older she is expected to live. A dynamic issue, life expectancy becomes even more complicated in the context of a couple - one or both individuals could live a very long time. This possibility should also influence the allocation policy over time and the potential need for longevity insurance.Finally, governments in many countries have implemented policies and programs to support the retirement effort. The most important program in the United States is Social Security, but its purpose is to provide only a minimum level of inflation-adjusted income, not to sustain the standard of living that an individual had before retirement. Other programs seek to encourage savings and facilitate wealth accumulation, such as 401(k)s, traditional and Roth IRAs (Registered Retirement Savings Plans and Tax-Free Savings Accounts in Canada), and health savings accounts (HSAs). Most investors, however, underestimate the savings effort required to maintain the standard of living to which they are accustomed, cannot implement a comprehensive and coherent adaptive retirement plan, cannot optimize across all relevant parameters, and lack access to the feedback mechanism needed to make the appropriate adjustments over time.

Lyngdoh, S. W., Das, S., and Das, T. (2024). "A Bibliometric Analysis of Asset Allocation for Retirement." In: *The Journal of Retirement* 12(1), pp. 61–83.

Allocation of investment assets is key in attaining a sustainable retirement portfolio. In this research article, the authors analyzed the most recent research publications in the area related to asset allocation for retirement and identified those which have the highest impact. The authors' research was conducted using the bibliometric analysis technique of research articles collected from the Scopus database. Most of the research articles were published in reputed journals in the United States, United Kingdom, Australia, and Germany. It was also observed that most of the highly cited research articles in the research area of asset allocation for retirement are focused on financial literacy, increase in retirement age, aging, and pension reforms. The authors' findings identified six research themes in asset allocation for retirement such as 1) asset allocation for retirement planning, 2) methods to increase efficiency, 3) investment preferences for retirement savings 4) financial literacy and retirement planning, 5) reforms on retirement savings, and 6) annuities for retirement income. Furthermore, nineteen future research directions are also provided. In conclusion, the authors aim to assist future researchers in identifying highly cited articles, key authors, contributing countries and research themes in asset allocation for retirement. Overall, the analysis provides comprehensive information in addressing research questions in the field of asset allocation for retirement.

MacDonald, B.-J., Jones, B., Morrison, R. J., Brown, R. L., and Hardy, M. (2013). "Research and Reality: A Literature Review on Drawing Down Retirement Financial Savings." In: *North American Actuarial Journal* 17(3), pp. 181–215.

How do, could, and should retirees draw down their financial savings? This article reviews over 100 papers on this topic from the perspective of individuals, families, governments, and financial institutions.
Three significant conceptual/methodological weaknesses in the existing literature are identified:

1) analysts have examined a limited range of self-managed drawdown strategies;
2) nearly all have ignored home ownership, pensions, debt, and government taxes and transfers when quantitatively evaluating alternative drawdown strategies;
3) there is a well-acknowledged gap between the behavior implied by economic models and that of real-life individuals, particularly when it comes to voluntary annuitization.

Expanding the set of drawdown strategies evaluated (e.g., including larger payouts when life expectancy is reduced after the onset of a significant health condition, or using savings as bridge income to delay the take-up of Social Security payments), refining the income concept used, and more exact modeling of the trade-offs underlying individual decision-making will likely increase the appeal of self-managed drawdown strategies and help resolve the "annuity puzzle" that has long dominated this line of research. It may also lead to advice and financial products that will better meet the needs of retirees.

MacDonald, B.-J., Morrison, R. J., Avery, M., and Osberg, L. (2018). "Drawing Down Retirement Savings - Do Pensions, Taxes and Government Transfers Matter Much for Optimal Decisions?" In: *ASTIN Bulletin* 48(3), pp. 1277–1306.
This paper examines the importance of pensions (employment and social security), taxes and government transfers for alternative retirement savings drawdown strategies (DS), compared to the conventional approach in published literature of using a gross income concept obtainable from retirement savings alone. Using a lifetime utility framework, our longitudinal dynamic micro-simulation model incorporates risk aversion, stochastic markets, stochastic mortality and the interactions among sources of retirement income within the complex Canadian tax and social benefit system, enabling us to rank commonly advocated DS and to ask whether incorporating pensions, taxes and transfers alters those rankings. Our findings show the importance of treating the evaluation of alternative DS as a comprehensive and integrated problem by including all sources of income - including pensions, taxes and government transfers. Using restricted income measures can potentially lead to simplistic, and possibly misleading, conclusions.

Machkour, J., Palomar, D. P., and Muma, M. (2024). "FDR-Controlled Portfolio Optimization for Sparse Financial Index Tracking." In: *arXiv e-Print*.
In high-dimensional data analysis, such as financial index tracking or biomedical applications, it is crucial to select the few relevant variables while maintaining control over the false discovery rate (FDR). In these applications, strong dependencies often exist among the variables (e.g., stock returns), which can undermine the FDR control property of existing methods like the model-X knockoff method or the T-Rex selector. To address this issue, we have expanded the T-Rex framework to accommodate overlapping groups of highly correlated variables. This is achieved by integrating a nearest neighbors penalization mechanism into the framework, which provably controls the FDR at the user-defined target level. A real-world example of sparse index tracking demonstrates the proposed method's ability to accurately track the S&P 500 index over the past 20 years based on a small number of stocks. An open-source implementation is provided within the R package TRexSelector on CRAN.

MacLean, L. and Zhao, Y. (2017). "Asset Price Dynamics: Shocks and Regimes." In: *Optimal Financial Decision Making under Uncertainty*. Ed. by G. Consigli, D. Kuhn, and P. Brandimarte. Vol. 245. Springer International Publishing, pp. 35–53.
Security prices changes are known to have a non-normal distribution, with heavy tails. There are modifications to the standard geometric Brownian motion model which accomodate heavy tails, most notably (1) adding point processes to the Brownian motion or (2) classifying time into regimes. With regimes the prices follow Brownian motion dynamics within regime, but the parameters vary by regime. The unconditional distribution of returns is a mixture of normals, with the mixing coefficients being Markov transition probabilities. The contrasting approaches have a common link-risk factors. In the case of the point processes, the intensity of "shocks" depends on a set of factors, eg. bond-stock yield differential, credit spread, implied volatility, exchange rates. The factors drive shocks, which are a component of the returns. With regimes, the economic state is hidden (latent) and is determined by the period by period observations on factors. The characterization of regimes follows from description in terms of the set of risk factors. In this paper the link between the shocks and regimes is explored. The shocks times defined by risk factors are an alternative method of determining regimes and the classifications by shocks and by the Expectation-Maximization algorithm are examined. The connections factors -> regimes -> shocks further justifies a classification of financial markets into homogeneous epochs. The regime structure leads to improved estimates for distribution parameters. The methods are applied to the prediction of returns on Sector Exchange Traded Funds (ETFs).The allocation of investment capital to funds based on predicted returns generates favorable wealth accumulation over a planning horizon.

Maclean, Ziemba, W. T., and Li, Y. (2005). "Time to wealth goals in capital accumulation." In: *Quantitative Finance* 5(4), pp. 343–355.
This paper considers the problem of investment of capital in risky assets in a dynamic capital market in continuous time. The model controls risk, and in particular the risk associated with errors in the estimation of asset returns. The framework for investment risk is a geometric Brownian motion model for asset prices, with random rates of return. The information filtration process and the capital allocation decisions are considered separately. The filtration is based on a Bayesian model for asset prices, and an (empirical) Bayes estimator for current price dynamics is developed from the price history. Given the conditional price dynamics, investors allocate wealth to achieve their financial goals efficiently over time. The price updating and wealth reallocations occur when control limits on the wealth process are attained. A Bayesian fractional Kelly strategy is optimal at each rebalancing, assuming that the risky assets are jointly lognormal distributed. The strategy minimizes

the expected time to the upper wealth limit while maintaining a high probability of reaching that goal before falling to a lower wealth limit. The fractional Kelly strategy is a blend of the log-optimal portfolio and cash and is equivalently represented by a negative power utility function, under the multivariate lognormal distribution assumption. By rebalancing when control limits are reached, the wealth goals approach provides greater control over downside risk and upside growth. The wealth goals approach with random rebalancing times is compared to the expected utility approach with fixed rebalancing times in an asset allocation problem involving stocks, bonds, and cash.

MacMinn, R. D. and Ren, Y. (2023). "The Life Annuity Puzzles?" In: *SSRN e-Print*.
The annuity and life puzzles exist because current economic theory shows that individuals should fully annuitize and not insure. This analysis introduces life insurance as a third asset in a portfolio model. The model is developed with and without a bequest motive. Without the bequest motive, the model reveals not only the annuity puzzle but also the life insurance puzzle. With a bequest motive, the model reveals the existence of an arbitrage direction which shows that the annuity only becomes part of an optimal portfolio in one of three possible cases. The results are consistent with market observations and so do not yield puzzles.

Madan, D. B. and Wang, K. (2022). "Folly and Fantasy in Finance." In: *SSRN e-Print*.
Strategies for selecting hedging measures that both respect certain market values of cash flows and yet maintain control on their distance from physical measures are advocated, proposed and implemented. The hedging criterion is the maximization of a conservative valuation of the hedged position. Such values are modeled as nonlinear expectations based on measure distortions. Measure selections and conservative value maximizing hedges are illustrated for options on SPY and nine sector ETF's.

Maeso, J.-M. and Martellini, L. (2020). "Measuring portfolio rebalancing benefits in equity markets." In: *The Journal of Portfolio Management* 46(4), pp. 94–109.
The potential source of additional performance because of the simple act of resetting portfolio weights back to the original weights is referred as the rebalancing premium. It is also sometimes known as the volatility pumping effect or diversification bonus because volatility and diversification turn out to be key components of the rebalancing premium. The purpose of this article is to provide a thorough empirical analysis of the volatility pumping effect in equity markets and to examine the conditions under which it can be maximized. The authors main contribution to the understanding of the rebalancing premium is an effort to disentangle and separately measure the isolated impact of various components of the total effect. Using the Fama-French-Carhart four-factor model, they find that, after controlling for factor exposures, the average outperformance of the rebalanced strategy with respect to the corresponding buy-and-hold strategy remains substantial at an annualized level above 100 basis points over a 5-year time horizon for stocks in the S&P 500 universe. They also find that size, value, momentum, and volatility are sorting characteristics that have a significant out-of-sample impact on the rebalancing premium. In particular, the selection of small-cap, low book-to-market, past loser, and high-volatility stocks generates a higher out-of-sample rebalancing premium compared to random portfolios for time horizons from 1 year to 5 years. They also show that the initial weighting scheme has a significant impact on the size of the rebalancing premium. Taken together, these results suggest that a substantial rebalancing premium can potentially be harvested in equity markets over reasonably long horizons for suitably selected types of stocks.

Mahaney, J. (2020). *Innovative Strategies to Help Maximize Social Security Benefits*. Tech. rep. Prudential.
Developing a Social Security claiming strategy is an important component to helping enhance retirement security. This updated 2020 edition incorporates changes made to certain claiming strategies as a result of the Bipartisan Budget Act of 2015. In addition, it highlights ways in which withdrawals from retirement savings vehicles, such as 401(k) plans and IRAs, can be integrated with a Social Security claiming strategy to maximize retirement income.

Mahayni, A. and Schneider, J. C. (2012). "Variable annuities and the option to seek risk: Why should you diversify?" In: *Journal of Banking and Finance* 36(9), pp. 2417–2428.
We analyze the impacts of an additional rider incorporated in recent retirement products. The payoff is linked to the performance of a multi asset investment strategy and includes a minimum interest rate guarantee. In addition, the buyer receives the option to decide on the investments dynamically. Prominent examples are so called variable annuities. Due to the embedded guarantee, these products are interesting for risk averse investors who benefit from diversification. However, the price setting of the provider takes into account the most risky strategy. This implies that the investor mitigates optimally between the diversification and the worst case strategy. We analyze the distortion and utility effects caused by the additional rider in the presence of background risk and borrowing

constraints. A simulation analysis sheds light on the question if the additional rider is worth its costs. Analysis of impacts of the rider to decide on investment decisions dynamically. Illustration of investment and utility effects under background risk. Simulation study shedding light on the question if the rider is worth its costs.

Malavasi, M., Lozza, S. O., and Truck, S. (2021). "Second order of stochastic dominance efficiency vs mean variance efficiency." In: *European Journal of Operational Research* 290(3), pp. 1192–1206.

In this paper, we compare two of the main paradigms of portfolio theory: mean variance analysis and expected utility. In particular, we show empirically that mean variance efficient portfolios are typically sub-optimal for non satiable and risk averse investors. We illustrate that the second order stochastic dominance (SSD) efficient set is the solution of a multi-objective optimization problem. We further show that the market portfolio is not necessarily a solution to this optimization problem. We also conduct an empirical analysis, examining the ex ante and ex post performance of SSD and mean variance efficient portfolios, using a bootstrap approach. In an ex ante analysis, we compare empirical moments, the level of diversification and set distances of mean variance and SSD efficient sets. We also show that the global minimum variance (GMV) portfolio and the part of the mean variance efficient frontier (MVEF) composed of highly diversified portfolios is second order stochastically dominated. This result also provides a possible alternative explanation for the diversification puzzle. Conducting an ex post analysis, we construct second order stochastic dominating strategies that outperform the GMV portfolio in terms of wealth and various other performance measures, producing a positive ex post opportunity cost.

Malhotra, D. K. (2024). "Evaluating the Role of Mutual Funds in Retirement Portfolios: A Comprehensive Performance Analysis." In: *The Journal of Retirement* 12(1), pp. 22–39.

Using monthly returns from January 2000 to November 2022, this article compares the performance of mutual funds available for retirement plans with US and global equities. The author found that historically, mutual funds generally outperformed both US and global equities in terms of absolute performance and return per unit of risk. However, the extent of this outperformance varied depending on the specific investment objectives of the mutual funds. Notably, mutual funds with more focused objectives, such as those centered on specialty investments, were able to generate significantly higher absolute returns. On the other hand, funds with an income investment objective achieved the highest risk-adjusted monthly returns. The author also discovered that mutual funds generated a negative and statistically insignificant alpha. The author's analysis also considered different market conditions. The results showed that alphas did not consistently have positive or statistically significant values across all periods and fund categories. While on average, mutual funds outperformed benchmark indexes and displayed exposure to systematic factors, the examination of alpha suggested that mutual fund managers had limited ability to consistently generate excess returns beyond what the benchmark offered.

Malhotra, M. (2012). "A Framework for Finding an Appropriate Retirement Income Strategy." In: *Journal of Financial Planning*.

Even though retirement distribution is the most actively researched area in the world of financial planning, the profession still lacks a comprehensive analysis framework for comparing retirement distribution strategies. Such a framework should do an apples-to-apples comparison among various retirement distribution strategies such as the appropriate time to claim Social Security or the use of systematic portfolio withdrawals in combination with fixed annuities, variable annuities, and bond ladders. The proposed framework in this paper includes two reward and three risk metrics. The widely used metric of probability of success is complemented with two additional risk metrics: a view into what happens when the plan fails and the amount of income generated from fixed sources of cash flow. The proposed reward metrics are retirement income and average legacy. A general outline of the effect of various income strategies on the metrics is provided, which can help planners practice the art of building an income plan mixing a systematic withdrawal portfolio with a fixed source of cash flow such that the plan provides acceptable confidence and reasonable protection in unfavorable markets with a potential to leave a legacy (if so desired) in favorable markets.

Mantilla-Garcia, D., Martinez-Carrasco, M., Garcia Huitron, M. E., and Muralidhar, A. (2020). "From Defined-Contribution Towards Target-Income Retirement Systems." In: *SSRN e-Print*.

The current trend towards Defined Contribution (DC) retirement systems around the world has rendered the risk management of pension funds crucial for the financial health of millions of people. Severe market downturns have put in evidence the need for more effective practices to control losses of retirement income for pension funds' investors. Although the move from the static allocation of policy portfolios toward target-date funds (TDFs) encouraged by regulators features significant improvements, the latter strategies completely ignore the variations in the cost of financing future consumption, which lessens the strategies' ability to control losses in retirement income. While optimal long-term portfolio selection literature emphasizes the importance of hedging against changes in the discount rates that determine the cost of financing future consumption, TDFs strategies, regulatory incentives, and reporting to pension funds' investors disregard the variations in long-term discount rates by focusing in absolute returns. We address these shortcomings by (i) developing performance metrics, such as DC funding-ratios, that foster appropriate incentives for pension fund managers and more sensible and simple reporting to their investors, (ii) introducing a series of asset allocation rules designed to secure a minimum level of target-income in retirement regardless of the returns of the risky assets in the portfolio. The strategies are consistent with insights from long-term portfolio theory and have the critical advantage of being free of any model or parameter estimation risks. We illustrate its advantages relative to a standard TDF strategy in terms of retirement security.

Marekwica, M., Schaefer, A., and Sebastian, S. (2013). "Life cycle asset allocation in the presence of housing and tax-deferred investing." In: *Journal of Economic Dynamics and Control* 37(6), pp. 1110–1125.
We study the dynamic consumption-portfolio problem over the life cycle, with respect to tax-deferred investing for investors who acquire housing services by either renting or owning a home. The joint existence of these two investment vehicles creates potential for tax arbitrage. Specifically, investors can deduct mortgage interest payments from taxable income, while simultaneously earning interest in tax-deferred accounts tax-free. Matching empirical evidence, our model predicts that investors with higher retirement savings choose higher loan-to-value ratios to exploit this tax arbitrage opportunity. However, many households could benefit from more effectively taking advantage of tax arbitrage.

Markwat, T., Molenaar, R., and Rodriguez, J. C. (2016). "Purchasing an Annuity: Now or Later? The Role of Interest Rates." In: *Bankers, Markets and Investors* (142), pp. 4–17.
This paper investigates whether the option to delay annuitization in times of low interest rates has value for retirees. The retiree who chooses to wait bets on a fall in the annuity price brought about by a rise in the interest rates; the cost of such a bet is the loss of the mortality credit during the waiting period. We show that an investor who can only invest in bonds during the waiting period will never find waiting ex-ante profitable if annuities are fairly priced, because waiting is costly and buying a fairly priced annuity is a zero-npv project. In contrast, an investor allowed to invest part of her wealth in the stock market will be able to attain more consumption on average, but at the cost of a sharp increase in risk. A retiree may choose to wait, however, if she believes her views are not priced in the term structure. For such a retiree, waiting is optimal if the expected increase in the interest rate is larger than the square of the hazard rate.

Martel, R. and Sharon, A. (2019a). *An Untimely Retirement: The Dangers of 'Sequence Risk' for Retirees – Introducing "Income to Outcome" retirement framework*. Tech. rep. PIMCO.
A major market downturn at the beginning of one's retirement journey is among the most potentially destructive scenarios investors face. Sharp declines can lead to de-risking and withdrawals from portfolios – both of which tend to crystallize losses and make recovery harder or even impossible.
Fortunately, investors are not completely at the mercy of the markets when it comes to sequence risk and retirement.
By setting up two dedicated portfolios – one, a bond portfolio designed to deliver necessary income in future years ("paycheck replacement"), the other composed of higher-risk long-term growth-oriented assets such as equities – retirees may more easily avoid costly, knee-jerk reactions that could undermine long-term growth.
This is the essence of PIMCO's "Income to Outcome" retirement framework. It's an approach that addresses both the hard issues of asset allocation and the softer, but no less significant, challenges of behavioral finance.

Martel, R. and Sharon, A. (2019b). *Financial Advisors and Retirement: The Decumulation Dilemma*. Tech. rep. PIMCO.
The retirement of the baby boomer generation represents a secular money-in-motion opportunity for financial advisors - one which may transform the asset management industry and individual advisory practices. But the withdrawal of assets over an uncertain remaining lifetime – decumulation – remains both the solution most in demand and the problem the industry has been least able to solve.
Our Income to Outcome decumulation framework incorporates behavioral insights that have become central to how the industry understands investor decisions.
It seeks to help investors set and automate regular withdrawals, protect against sequence-of-returns risk, embed better advice around optimal strategies for claiming Social Security, address longevity risk and pursue long-term growth in an endowment-like portfolio to support unknown expenditures and maximize potential for bequests.

Martel, R. and Sharon, A. (2020). "Don't Drop The Baton: Sequence Risk And The Decumulation Dilemma." In: *Investments and Wealth Monitor* (May/Hune), pp. 33–39.
Key challenges to the retirement handoff include:

- New and sudden headwinds to growth caused by shifting the source of spending from the saver's salary to the retiree's accumulated savings (which is the defining characteristic of a decumulation portfolio).
- The spending portfolio's increased vulnerability to market drawdowns, particularly deep declines, especially during the years just following the transition (i.e., sequence risk).
- The susceptibility of a risk-averse retiree to erratic decisions in the wake of market declines (e.g., a sud- den move to cash, particularly in the early years) that may crystallize port- folio losses and jeopardize recovery, severely limiting the longevity of retirement assets.

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Martellini, L. and Milhau, V. (2020). *Advances in Retirement Investing*. Cambridge University Press. 167 pp.
To supplement replacement income provided by Social Security and employer-sponsored pension plans, individuals need to rely on their own saving and investment choices during accumulation. Once retired, they must also decide at which rate to spend their savings, with the usual dilemma between present and future consumption in mind. This Element explains how financial engineering and risk management techniques can help them in these complex decisions. First, it introduces 'retirement bonds', or retirement bond replicating portfolios, that provide stable and predictable replacement income during the decumulation period. Second, it describes investment strategies that combine the retirement bond with an efficient performance-seeking portfolio so as to reduce uncertainty over the future amount of income while offering upside potential. Finally, strategies using risk insurance techniques are proposed to secure minimum levels of replacement income while giving the possibility of reaching higher levels of income.

Martellini, L., Milhau, V., and Mulvey, J. (2019). "'flexicure' Retirement Solutions: A Part of the Answer to the Pension Crisis?" In: *The Journal of Portfolio Management* 45(5), pp. 136–151.
Individuals preparing for retirement are currently left with an unsatisfactory choice between security with no flexibility with annuity products and flexibility without security with investment products such as balanced funds or target date funds. To get out of this impasse, the authors introduce a range of flexicure retirement goal-based investing strategies that offer both security and flexibility with respect to the objective of generating replacement income in decumulation. Recent advances in financial engineering and digital technologies make it possible to apply goal-based investing principles to a much broader population of investors than the few traditional clients who can afford customized mandates or private banking services, which suggests that these flexicure retirement solutions can be used as part of the solution to the global pension crisis.

Martellini, L., Milhau, V., and Mulvey, J. (2020). "Securing Replacement Income with Goal-Based Retirement Investing Strategies." In: *The Journal of Retirement* 7(4), pp. 8–26.
Individuals preparing for retirement are currently left with an unsatisfactory choice between security with no flexibility with annuity products and flexibility without security with investment products such as balanced funds or target date funds. To get out of this impasse, the authors introduce a range of flexicure retirement goal-based investing strategies that offer both security and flexibility with respect to the objective of generating replacement income in decumulation. Recent advances in financial engineering and digital technologies make it possible to apply goal-based investing principles to a much broader population of investors than the few traditional clients who can afford customized mandates or private banking services, which suggests that these flexicure retirement solutions can be used as part of the solution to the global pension crisis.

Martellini, L., Milhau, V., and Suri, A. (2025). "Efficient Decumulation Investing and Applications to a New Generation of Fully Amortizing Retirement Solutions." In: *The Journal of Portfolio Management*, jpm.2024.1.654.
Individuals in retirement not only face the challenge of efficiently allocating their wealth but also the decision of how much to withdraw from their retirement account over time. Although the asset management industry has so far mostly focused on the investment side of the equation, designing investment products in isolation from the associated withdrawal strategy potentially leads to suboptimal outcomes for retirees. In an effort to improve the adequacy of retirement funds with respect to the needs of investors in decumulation, we build on Waring and Siegel's (2015) insight that a meaningful withdrawal strategy should preserve consumption levels to propose an integrated framework for optimizing the interplay between investment and payout decisions made by asset managers on behalf of retirees. More specifically, we propose that retirement funds include a stream of scheduled retirement payments given by a market-based measure of the purchasing power of outstanding wealth invested

in the fund in terms of retirement income over a fixed horizon. We show that substantial welfare improvement for investors in decumulation would be generated by the joint use of this payout strategy and a retirement bond as the associated risk-free instrument, which can easily be replicated using standard fixed-income products and packaged within a fully amortizing fund structure.

Martin, P. P. and Kintzel, D. (2016). "A Comparison of Free Online Tools for Individuals Deciding When to Claim Social Security Benefits." In: *SSRN e-Print*.

When to claim Social Security retirement benefits is one of the most important financial decisions many people make. The Social Security Administration (SSA) provides a variety of online tools and publications to help individuals decide on their own when to claim benefits, but maintains a neutral stance on when a person should claim. Because SSA remains neutral, other government, nonprofit, academic, and for-profit entities have developed tools to assist the public with their claiming decision. This note analyzes the advantages and limitations of six online benefit calculators. Users of these online tools should consider the source of their information and understand that the benefit estimates they produce are based on different underlying assumptions, which can result in different estimated benefit amounts.

Marwood, D. and Minnen, D. (2020). "Safely Boosting Retirement Income by Harmonizing Drawdown Paths." In: *Journal of Financial Planning* 33(11), pp. 46–60.

Building on Bengen's famous 4 percent rule (Bengen 1994), this paper introduces the DMSWR, a technique for calculating withdrawal rates and retirement income levels that are safe, steady, and adjusted for inflation.

The DMSWR (named after the authors' initials, plus the acronym for safe withdrawal rate) addresses two commonly cited shortcomings of the 4 percent rule. First, it addresses the starting point paradox (Kitces 2008) by ensuring that initial retirement incomes are stable regardless of the precise retirement date and short-term fluctuations in the stock market. Second, the DMSWR significantly reduces the risk of unnecessarily low retirement income that leads to large, unspent portfolios at the end retirement.

The DMSWR provides withdrawal rates that are often substantially higher, and never lower, than the 4 percent rule, which allows financial planners to recommend higher retirement incomes that are still safe.

The core insight underpinning the DMSWR is that a retiree can safely follow the "drawdown path" of an earlier retiree who follows the 4 percent rule and planned for a longer retirement. Since the earlier retiree's withdrawals are safe, the withdrawals of the new retiree must also be safe.

The DMSWR is safe in the same way the 4 percent rule is safe: the withdrawal rates have never failed in the past and, were DMSWR to fail in the future, the 4 percent rule would also fail. Unlike many similar efforts to boost safe withdrawal rates, the DMSWR does not include additional parameters or heuristics that might fail at a time when following the 4 percent rule would succeed.

Historically, the mean DMSWR for 30-year retirements was 5.48 percent, the highest DMSWR was 13.3 percent, and the DMSWR exceeded the baseline withdrawal rate by more than 100 basis points over 55 percent of the time.

"Re-retirement" using the DMSWR often allows incomes to increase during retirement. These higher incomes succeed whenever the 4 percent rule succeeds. The software implementation is at `github.com/minnend/dmswr`.

Massa, M., Moussawi, R., and Simonov, A. (2021). "To Target a Date or Not to Target a Date? That is the Question: The Unintended Consequences of Investing for the Long Run." In: *SSRN e-Print*.

We study how managers of funds created to invest for the long run behave when shielded from liquidity constraints and their investors' short-term needs. Using the universe of US target-date funds (TDFs), we document that asset managers exploit lower investor attention to deliver lower performance. This results in a hypothetical cumulative return loss of 21% for the average investor holding the fund for 50 years. This underperformance is driven by fund families using the TDFs to smoothen the flow shocks of the affiliated open-ended funds. It is also due to higher fees arising from investing in the affiliated expensive share classes. We use the Pension Protection Act of 2006 as an exogenous shock that made TDFs the default investment options within 401(k) retirement plans.

McLean, R. (2021). "Rebalancing Frequency and Safe Withdrawal Rates." In: *Advisor Perspectives*.

Safe-withdrawal rate (SWR) research has traditionally assumed annual portfolio rebalancing, and this is the frequency suggested by many advisors (although some suggest quarterly, monthly, or even daily rebalancing!). But is such frequent fiddling necessary? Rebalancing generates costs in fees and commissions and, depending on the type of account, will trigger tax liabilities. I used the "Big Picture" client education software to examine the impact of rebalancing frequency on historical SWRs and other retirement outcomes. The results argue against the notion to rebalance often.

McNamee, J. L., Paradise, T., and Bruno, M. A. (2019). *Getting back on track: A guide to smart rebalancing*. Tech. rep. Vanguard.
A portfolio's asset allocation reflects an investor's goals and temperament – the need for return and ability to withstand the financial markets' inevitable turbulence. Over time, as the returns of higher- and lower-risk assets diverge, a portfolio can take on exposures that are inconsistent with the investor's risk and return objectives. Rebalancing from one asset class to another can put the portfolio back on track. We review the benefits of rebalancing, analyze the impact of different rebalancing frequencies and thresholds, and highlight strategies to minimize rebalancing costs. Our analysis suggests three best practices when setting expectations for and executing a rebalancing strategy: 1) rebalance to manage risk and emotion 2) set a rebalancing trigger 3) minimize the transaction and tax costs of rebalancing.

McQuarrie, E. F. (2022a). "The Annuity Wager: Can Bonds Outperform a Life Annuity?" In: *SSRN e-Print*.
This paper frames the decision to purchase a fixed immediate annuity as a wager that can be won or lost. The alternative bet is to hold a portfolio of long government bonds, drawing income and liquidating principal as necessary. There is some probability of ruin in either case: the bond portfolio may be exhausted while the bettor is still alive, or the bettor may die before the bond portfolio runs down. The paper applies a historical rather than mathematical lens to calculate the odds of each outcome. I examine 81 start dates from 1918 to 1998 inclusive, with an out-of-sample test from 1878 to 1917. The historical analysis finds that wager outcomes are contingent. Most notably, the annuity is the odds-on favorite when bond yields are very low. However, if the annuity provider cannot harvest a spread over the safe government bond yield and pass on part of that spread in the form of a higher annuity payout, then the analysis shows the bond portfolio to be the odds-on favorite over the stretch of history examined. Mortality credits alone are not enough to favor the annuity. In the middle case – given a yield spread but absent very low yields – the odds do not greatly favor either side of the bet. The historical account is further supported by analysis of the arithmetic of government bond returns and the mechanics of annuity payouts.

McQuarrie, E. F. (2022b). "When and for Whom are Roth Conversions Most Beneficial? A New Set of Guidelines, Cautions and Caveats." In: *SSRN e-Print*.
Much has changed since penalty-free Roth conversions were inaugurated in 2010. Tax rates have gone up and down. The re-characterization provision went away. Heirs can no longer stretch out inherited Roth accounts over a lifetime. Medicare surcharges were expanded and began to adjust for inflation. The age to begin Required Minimum Distributions was pushed out to age 72 and the IRS changed the RMD divisor tables to further slow the pace of distribution.
Given these developments it seemed worthwhile to re-examine the rationale for Roth conversions. That effort exposed multiple flaws in conventional wisdom:

1) Future tax rates need not be higher for a conversion to pay off;
2) Nor is it all that helpful to pay the tax on conversion from outside funds;
3) Nor are Roth conversions especially beneficial for top bracket taxpayers as compared to middle class taxpayers;
4) Rather, the greatest benefit accrues to taxpayers who can make the conversion partly in the zero percent tax bracket, i.e., during a year with no other taxable income.

While the benefits from a Roth conversion are often small and slow to arrive, a Roth conversion will almost always pay off if given enough time, i.e., for life spans that extend past 90 and so long as annual distributions from converted amounts are not taken. Roth conversions work because of compounding, which requires the conversion to be left undisturbed for a long time. The paper elucidates the role played by the mathematics of compounding in underwriting the success of Roth conversions.

McQuarrie, E. F. (2022c). "Will Required Minimum Distributions Exhaust My Savings and Leave Me in Penury?" In: *SSRN e-Print*.
This paper probes the question of how long retirement savings can be sustained under mandated withdrawals whose rate increases with age. It begins with idealized, constant rates of return typical of balanced funds. These laboratory analyses show that funds are unlikely to be exhausted before age 100, and that exhaustion, when it comes, occurs because of Bengen's test, i.e., the eventual need to withdraw more than the minimum to maintain the initial age 72 income in constant dollars. Next, the paper examines sequence-of-returns risk using selected historical data. The worst decade for balanced fund returns in the US was located, but even with this bad start the crudest sort of balanced portfolio could still be made to sustain income to age 101. The paper proceeds to

examine a variety of worst-case scenarios drawn from the two-century historical record, in the US and globally, with the worst cases ex-USA limited to markets not suffering a national disaster. In the US, it was always possible to sustain income to age 100. Globally, this could not always be achieved, with the worst shortfalls explained by either inflation or prolonged periods of depressed asset returns. In the 20th century US, the analyses find a 30/70 allocation to be inferior to a 60/40 allocation when the goal is to sustain inflation-adjusted income for as long as possible; globally and historically, that generalization did not hold. Next, the paper considers whether small changes to the balanced mix, consistent with those seen in Target Date Retirement funds, could improve outcomes; in all the cases examined, small alterations were able to improve the longevity of savings, sometimes dramatically. The paper ends on an optimistic note: two centuries of global market history indicate that exhaustion of tax-sheltered retirement savings before the age of 100 is unlikely to occur for retirees who model their asset mix after those seen in Target Date funds.

McQuarrie, E. F. (2025a). "Benchmarks Differ. Then They Get Revised." In: *The Journal of Portfolio Management* 51(4), pp. 24–39.

For periods beginning 1926, it is conventional to suppose that historical market returns are known and exact. This article challenges that comfortable certainty. Multiple indexes of market return are examined to show that return estimates do not closely agree across indexes and have been unstable within index across revisions. Implications are developed for evaluating portfolio manager performance and back testing investment strategies.

McQuarrie, E. F. (2025b). "Best Asset Location for a TIPS Ladder." In: *SSRN e-Print*.

Tax treatments vary across assets and account types. The literature on asset location developed in response to these challenges. The goal is to find the best location for each asset in the portfolio, defined as the account type where it will receive the most favorable tax treatment. Given multiple assets, such as stocks and bonds, and multiple account types, such as a brokerage account versus a 401(k) plan, a joint solution must be pursued: each asset has to be located such that the overall tax treatment is the most favorable that can be obtained, net, across assets and accounts. This paper develops the best location for a ladder of Treasury Inflation Protected Securities when the portfolio also contains stocks.

McQuarrie, E. F. (2025c). "How the 4% Rule Would Have Failed in the 1960s: Reflections on the Folly of Fixed Rate Withdrawals." In: *SSRN e-Print*.

The 4% withdrawal rule was formulated by William Bengen to satisfy a specific historical test. Bengen sought to determine the initial rate which, applied against a balanced portfolio of stocks and bonds, and adjusted annually for inflation, had always sustained withdrawals for at least 30 years over the span of available data. This paper probes the flaws in Bengen's US historical data which led to the mistaken nomination of 4% as a safe withdrawal rate. After documenting the historical failures of the 4% rate, the paper develops the conditions which determine whether any fixed rate of withdrawal can be sustained and assesses the impact of more versus less conservative asset allocations. Planners will gain a new appreciation for the importance of flexibility in designing withdrawal strategies.

McQuinn, N., Thapar, A., and Villalon, D. (2021). "Portfolio Protection? It's a Long (Term) Story ..." In: *The Journal of Portfolio Management* 47(3), pp. 35–50.

Investors have a natural urge to protect their portfolios from sudden crashes. The authors argue that investors should instead focus on bad outcomes that unfold over longerperiods because those tend to be more detrimental to the long-term goal of wealth accumulation. The authors show that options-based hedging can be effective over shorter periods but tends to weaken over time. Worse still, returns tend to be very punitive during prolonged bull markets. In contrast, risk-mitigating and diversifying strategies such as defensive equities, risk parity, alternative risk premiums, and trend-following have more consistently added value in the longer-lasting market drawdowns that matter most to investors-and, unlike puts, can profit in up as well as down markets. This latter point suggests a crucial advantage for these strategies: that unlike options-based hedging, it is never too late to consider diversifying into them.

Medeiros, M. C., Vasconcelos, G. F. R., Veiga, A., and Zilberman, E. (2021). "Forecasting Inflation in a Data-Rich Environment: The Benefits of Machine Learning Methods." In: *Journal of Business & Economic Statistics* 39, pp. 98–119.

Inflation forecasting is an important but difficult task. Here, we explore advances in machine learning (ML) methods and the availability of new datasets to forecast U.S. inflation. Despite the skepticism in the previous literature, we show that ML models with a large number of covariates are systematically more accurate than the benchmarks. The ML method that deserves more attention is the random forest model, which dominates all other models. Its good performance is due not only to its specific method of variable selection but also the potential nonlinearities between past key macroeconomic variables and inflation. Supplementary materials for this article are available online.

Mendu, H. (2021). "A critical review of 'optimal' annuitization strategies." MA thesis. University of Waterloo.

This paper is an analysis of different self-annuitization strategies advised to a retiree. At the time of retirement, an individual has the choice between annuitizing immediately with their wealth or delaying until a future date while making the most from returns earned through financial markets. This is a similar structure to how a Defined Contribution (DC) pension plan works. Various researchers have assisted retirees on this dilemma of whether or not to annuitize. They designed strategies that work under general market settings and any individual preferences that maximize the income generation. However, these studies assume that individuals are making decisions as if they are maximizing or behaving optimally. In reality, what might be "optimal" in a general sense may not match the optimality defined in a normative sense of giving advice.

Merton, R. C. (2014). "The Crisis in Retirement Planning." In: *Harvard Business Review*.

Corporate America began to really take notice of the looming retirement crisis in the wake of the dot-com crash, when companies in major industries went bankrupt in large part because of their inability to meet their pension obligations. The result was an acceleration of America's shift away from employer-sponsored pension plans toward defined-contribution plans – epitomized by the ubiquitous 401(k) – which transfer the investment risk from the company to the employee.

With that transfer has come a dangerous shift in investment focus, argues Nobel Laureate Robert C. Merton. Traditional pension plans were conceived and managed to provide members with a guaranteed income. And because that objective filtered right through the scheme, members thought of their benefits in those terms. Ask a member what her pension is worth and she'll reply with an income figure: "two-thirds of my final salary," for example. Most DC schemes, however, are designed and managed as investment accounts with the goal of accumulating the largest possible pot of savings. Communication with savers is framed entirely in terms of assets and returns. Ask a saver what his 401(k) is worth and you'll hear a cash amount and perhaps a lament to the value lost in the financial crisis.

The trouble is that investment value and asset volatility are simply the wrong measures if your goal is to secure a particular future income. In this article, Merton explains a liability-driven investment strategy whose aim is to improve the probability of achieving a desired retirement income rather than to maximize the capital value of the savings.

Merton, R. C. and Muralidhar, A. (2020a). "A Six-Component Integrated Approach to Addressing the Retirement Funding Challenge." In: *Journal Of Investment Management* 18(4), pp. 28–54.

This paper offers an integrated approach to addressing the global retirement funding challenge, especially in light of the coronavirus shock that has created an unanticipated and unprecedented impact on lifetime income/consumption. It frames the problem in a six-component approach to the funding challenge with an integrated package presented in a transparent, detailed modular fashion, so that any one module can be replaced with a different version and the rest of the system works. This also means that all six components need not be employed simultaneously, but can be done in a secular fashion. Finally, it develops and proposes in detail a new financial instrument, SeLFIES (Standard-of-Living indexed, Forward-starting, Income-only Securities)-a single financial innovation that provides greatly improved efficiency of implementation to four of the six components. SeLFIES can help complete financial markets and could be a timely innovation given the coronavirus crisis because they are beneficial to governments that seek long-term, local currency debt financing.

Merton, R. C. and Muralidhar, A. (2020b). "Selfies: A new pension bond and currency for retirement." In: *Journal of Financial Transformation* 51.

There is a looming retirement crisis, as individuals are increasingly being asked to take responsibility for their own retirement planning and a majority of these individuals are financially unsophisticated. They cannot perform basic compounding calculations and do not understand the impact of inflation, both critical aspects of retirement planning. Yet, these individuals are being tasked with the responsibility for three complex, interconnected decisions: how much to save, how to invest (with many additional decisions), and how to decumulate one portfolio at retirement. Compounding these challenges, current financial instruments and products (e.g. T-Bills, TIPs, or Target Date Funds) are risky because they focus on the wrong goal - wealth at retirement, as opposed to how much retirement income can be guaranteed to support pre-retirement standard-of-living. Moreover, annuities are complex, costly, and illiquid and seldom used. Without financial innovation and a change in the metric for measuring retirement success, many individuals will retire poor - a financially and socially undesirable outcome for any country. This paper presents an easy, quick and efficient solution for countries to address all these challenges and improve retirement security by creating and issuing an innovative new bond - SeLFIES (Standard-of-Living indexed, Forward-starting, Income-only Securities). The SeLFIES bond is a single, liquid, low-cost, low-risk instrument, easy-to-understand for even the most financially unsophisticated individual, because it embeds accumulation, decumulation, compounding and inflation-adjustments. SeLFIES is good for governments too, as the bond lowers the risk of individuals retiring poor, improves balance sheet management, and funds infrastructure. The paper also discusses key design aspects of SeLFIES to show how they can ensure longevity risk protection and hedge standard-of-living risk, a key unmanaged risk globally today. Additionally, the paper by concludes by demonstrating the universality of the SeLFIES design as well as by showing how it serves a useful purpose by becoming the currency of retirement.

Meyer, W. and Reichenstein, W. (2013). "Adding Longevity through Tax-Efficient Withdrawal Strategies." In: *The Journal of Wealth Management* 16(1), pp. 57–64.

Suppose a client just retired and has funds in a tax-deferred account like a 401(k), a tax-exempt account like a Roth IRA, and a taxable account. She needs to withdraw sufficient funds to finance her spending plans in retirement. This study explains how, just using the tax code, she can tax-efficiently withdraw funds from her financial portfolio to meet her spending goal while allowing her financial portfolio to last several years longer.

Michaud, P.-C. and St-Amour, P. (2023). "Longevity, Health and Housing Risks Management in Retirement." In: *SSRN e-Print*.

Annuities, long-term care insurance and reverse mortgages remain unpopular to manage longevity, medical and housing price risks after retirement. We analyze low demand using a life-cycle model structurally estimated with a unique stated-preference survey experiment of Canadian households. Low risk aversion, substitution between housing and consumption and low marginal utility when in poor health explain most of the reduced demand. Bequests motives are found to be a luxury good and play a limited role. The remaining disinterest is explained by information frictions and behavioural status-quo biases. We find evidence of strong spousal co-insurance motives motivating LTCI and of responsiveness to bundling with a near doubling of demand for annuities when reverse mortgages can be used to annuitize, instead of consuming home equity.

Mikkila, O. and Kanniainen, J. (2023). "Empirical deep hedging." In: *Quantitative Finance* 23(1), pp. 111–122.

Existing hedging strategies are typically based on specific financial models: either the strategies are directly based on a given option pricing model or stock price and volatility models are used indirectly by generating synthetic data on which an agent is trained with reinforcement learning. In this paper, we train an agent in a pure data-driven manner. Particularly, we do not need any specifications on volatility or jump dynamics but use large empirical intra-day data from actual stock and option markets. The agent is trained for the hedging of derivative securities using deep reinforcement learning (DRL) with continuous actions. The training data consists of intra-day option price observations on S&P500 index over 6 years, and top of that, we use other data periods for validation and testing. We have two important empirical results. First, a DRL agent trained using synthetic data generated from a calibrated stochastic volatility model outperforms the classic Black-Scholes delta hedging strategy. Second, and more importantly, we find that a DRL agent, which is empirically trained directly using actual intra-day stock and option prices without the prior specification of the underlying volatility or jump processes, has superior performance compared with the use of synthetic data. This implies that DRL can capture the dynamics of S&P500 from the actual intra-day data and to self-learn how to hedge actual options efficiently.

Milevsky, M. A. (2006). *Calculus of Retirement Income: Financial Models for Pension Annuities and Life Insurance*. Cambridge University Press.

This 2006 book introduces and develops the basic actuarial models and underlying pricing of life-contingent pension annuities and life insurance from a unique financial perspective. The ideas and techniques are then applied to the real-world problem of generating sustainable retirement income towards the end of the human life-cycle. The role of lifetime income, longevity insurance, and systematic withdrawal plans are investigated in a parsimonious framework. The underlying technology and terminology of the book are based on continuous-time financial economics by merging analytic laws of mortality with the dynamics of equity markets and interest rates. Nonetheless, the book requires a minimal background in mathematics and emphasizes applications and examples more than proofs and theorems. It can serve as an ideal textbook for an applied course on wealth management and retirement planning in addition to being a reference for quantitatively-inclined financial planners.

Milevsky, M. A. (2020a). "Calibrating Gompertz in reverse: What is your longevity-risk-adjusted global age?" In: *Insurance: Mathematics and Economics* 92, pp. 147–161.

This paper develops a computational framework for inverting Gompertz-Makeham mortality hazard rates, consistent with compensation laws of mortality for heterogeneous populations, to define a longevity-risk-adjusted global (L-RaG) age. To illustrate its salience and possible applications, the paper calibrates and presents L-RaG values using country data from the Human Mortality Database (HMD). Among other things, the author demonstrates that when properly benchmarked, the longevity-risk-adjusted global age of a 55-year-old Swedish male is 48, whereas a 55-year-old Russian male is closer in age to 67. The paper also discusses the connection between the proposed L-RaG age and the related concept of Biological age, from the medical and gerontology literature. Practically speaking, in a world of growing mortality heterogeneity, the L-RaG age could be used for pension and retirement policy. In the language of behavioral finance and economics, a salient metric that adjusts chronological age for longevity risk might help capture the public attention, educate them about lifetime uncertainty and induce many of them to take action —such as working longer and/or retiring later.

Milevsky, M. A., Huang, H., and Young, V. R. (2015). "A Glide Path for Target Date Fund Annuitization." In: *The Journal of Retirement* 3(1), pp. 27–37.

We describe a recursive algorithm that computes the timing and quantity of purchase of deferred income annuities (DIAs) within target-date funds (TDF) in defined contribution (DC) plans, although the algorithm could also be applied within any retirement account. We map a relatively small number of statistical parameters into a rule that conveys the dollar amount of DIAs to be purchased at any given age and time. Our model is of particular relevance given the recent announcement by the U.S. Treasury Department approving the inclusion of life annuities in 401(k) plans and in TDFs in particular. Note that to qualify as a TDF requires a methodology based on generally accepted investment theories using a consistent investment strategy. This article offers one possible such theory in the context of DIAs.

Milevsky, M. A. and Posner, S. E. (2014). "Can collars reduce retirement sequencing risk? analysis of portfolio longevity extension overlays (leo)." In: *The Journal of Retirement* 1(4), pp. 46–56.

Practitioners are well aware of the pernicious effect of the of investment returns on retirement income sustainability. Poor markets early in the withdrawal phase increase the ruin probability and reduce the longevity of a portfolio. In this article, the authors investigate how and when traded equity options can be used to extend the life of a retiree investments. They label this class of strategies longevity extension overlays (LEOs) and use simulation techniques to analyze the strategy theoretical properties. They also provide evidence on the efficacy of simple LEOs during the 2007-2013 period. Our results are encouraging and offer a justifiable alternative for wealth managers who want to avoid using (more complex and opaque) insurance-product solutions.

Milevsky, M. A., Salisbury, T. S., and Chigodaev, A. (2018). "The implied longevity curve: How long does the market think you are going to live?" In: *arXiv e-Print*.

We use life annuity prices to extract information about human longevity using a framework that links the term structure of mortality and interest rates. We invert the model and perform nonlinear least squares to obtain implied longevity forecasts. Methodologically, we assume a Cox-Ingersoll-Ross (CIR) model for the underlying yield curve, and for mortality, a Gompertz-Makeham (GM) law that varies with the year of annuity purchase. Our main result is that over the last decade markets implied an improvement in longevity of of 6-7 weeks per year for males and 1-3 weeks for females. In the year 2004 market prices implied a 40.1% probability of survival to the age 90 for a 75-year old male (51.2% for a female) annuitant. By the year 2013 the implied survival probability had increased to 46.1% (and 53.1%). The corresponding implied life expectancy has increased (at the age of 75) from 13.09 years for males (15.08 years for females) to 14.28 years (and 15.61 years.) Although these values are implied directly from markets, they are consistent with demographic projections. Similar to implied volatility in option pricing, we believe that our implied survival probabilities (ISP) and implied life expectancy (ILE) are relevant for the financial management of assets post-retirement and very important for the optimal timing and allocation to annuities; procrastinators are swimming against an uncertain but rather strong longevity trend.

Milevsky, M. A. (2018). "Swimming with Wealthy Sharks: Longevity, Volatility and the Value of Risk Pooling." In: *SSRN e-Print*.

Who values life annuities more? Is it the healthy retiree who expects to live long and might become a centenarian, or is the unhealthy retiree with a short life expectancy more likely to appreciate the pooling of longevity risk? What if the unhealthy retiree is pooled with someone who is much healthier and thus forced to pay an implicit loading? To answer these and related questions this paper examines the empirical conditions under which retirees benefit (or may not) from longevity risk pooling by linking the economics of annuity equivalent wealth (AEW) to actuarially models of aging. I focus attention on the Compensation Law of Mortality which implies that individuals with higher relative mortality (e.g. lower income) age more slowly and experience greater longevity uncertainty. Ergo, they place higher utility value on the annuity. The impetus for this research today is the increasing evidence on the growing disparity in longevity expectations between rich and poor.

Milevsky, M. A. (2020b). "Biological (and Other) Ages." In: *Retirement Income Recipes in R*. Springer International Publishing, pp. 259–279.
This chapter examines the implications of mortality heterogeneity, that is the dispersion of longevity prospects within the population. It begins by discussing the extended Gompertz-Makeham model, as well as the compensation law of mortality, linking moments of the remaining lifetime random variable. It then introduces non-chronological measures of age, such as biological age and (especially) longevity risk-adjusted age to illustrate its dispersion. This chapter illustrates how true age can differ around the world and even within countries, based on wealth and income. The main computational implication of this chapter is that the human longevity random variable Tx, depends on (much) more than just chronological age x. Such heterogeneity must be accounted for in any intelligent drawdown methodology or pensionization scheme.

Milevsky, M. A. (2020c). "Intelligent Drawdown Rates." In: *Retirement Income Recipes in R*. Springer International Publishing, pp. 209–232.
This chapter, which could have been called rational decumulation, introduces dynamic risk-adjusted approaches to spending during retirement. The intelligent drawdown philosophy is contrasted with static approaches, such as the 4% rule and its variants, the focus of prior chapters. The material begins with a light-hearted game that develops an intuition for how longevity uncertainty should affect retirement spending as well as a discussion of the benefits from risk pooling. Moving on to the technical content, after a brief crash course on utility theory, the chapter develops algorithms for (1) computing optimal withdrawal rates based on risk aversion preferences and (2) adjusting ongoing spending based on realized financial and longevity variables. One of the surprising aspects of an intelligent approach to retirement spending is that planning to deplete one's liquid wealth at some advanced age is "rationale" if you have ample pension annuity income.

Milevsky, M. A. (2020d). "Life Annuities: From Immediate to Deferred." In: *Retirement Income Recipes in R*. Springer International Publishing, pp. 181–208.
This chapter develops a methodology for valuing simple cash-flow streams that last a lifetime, which are part of most Defined Benefit (DB) pensions. The focus is on the longevity-contingent building blocks of: (1) immediate, (2) temporary, and (3) deferred income annuities. The chapter begins with a discussion of the value of a longevity-contingent claim and how it differs from the market price versus the manufacturing cost of the product. The algorithms and user-defined R functions are mostly based on the Gompertz law of mortality, although a number of alternative continuous and discrete mortality models are discussed as well. The chapter concludes with a mathematical derivation and implementation of a closed-form expression for the Gompertz Annuity Valuation Model.

Milevsky, M. A. (2020e). "Modeling Human Longevity and Life Tables." In: *Retirement Income Recipes in R*. Springer International Publishing, pp. 111–130.
Up to this point in the book, I have assumed a great fiction, namely that human longevity is known and finite. The success or failure of a retirement income plan was monitored and measured until a finite, e.g. 30 year, horizon. This chapter is the first to focus on the uncertainty or randomness in human longevity versus portfolio longevity. It begins with a detailed description and analysis of (historical) cohort life tables from the Human Mortality Database. The cohort life tables are used to extract population survival and death rates, which are then used to reconstruct life tables. The chapter concludes with a high-level discussion of mortality projections and improvements for future birth cohorts. The main emphasis is on gaining familiarity with the basic atomic structure (qx) of actuarial life-science.

Milevsky, M. A. (2020f). "Modeling the Risk of Sequence-of-Returns." In: *Retirement Income Recipes in R*. Springer International Publishing, pp. 85–109.
This chapter is focused on a phenomenon known by professionals in the retirement income business, as the sequence-of-returns effect. Broadly speaking – and using terminology introduced in the previous chapter – this relates to the disproportionate sensitivity of portfolio longevity to realized investment returns in the early stages of retirement withdrawals. More specifically this chapter proposes some formal metrics that measure the extent and magnitude of the risk using statistical correlation and regression methodologies. The chapter concludes by analyzing some derivative-based strategies, using put and call options, that can be used to mitigate the risk of sequence-of-returns.

Milevsky, M. A. (2020g). *Retirement Income Recipes in R: From Ruin Probabilities to Intelligent Drawdowns*. Springer International Publishing. 302 pp.

This book provides computational tools that readers can use to flourish in the retirement income industry. Each chapter describes recipe-like algorithms and explains how to implement them via simple scripts in the freely available R coding language. Students can use those skills to generate quantitative answers to the most common questions in retirement income planning, as well as to develop a deeper understanding of the finance and economics underlying the field itself. The book will be an excellent asset for experienced students who are interested in advanced wealth management, and specifically within courses that focus on holistic modeling of the retirement income process. The material will also be useful to current and future wealth management professionals within the financial services industry. Readers should have a solid understanding of financial principles, as well as a rudimentary background in economics and accounting.

Miller, A. K. (2015). "Improving Withdrawal Rates in a Low Yield World." In: *SSRN e-Print*.

Decreased real bond yields substantially increase failure rates for portfolios with an initial 4% withdrawal rate. One way to increase the safe withdrawal rate of a portfolio is to decrease the allocation to bonds and to increase the allocation to stocks. Unfortunately, increasing the allocation to stocks dramatically changes the risk profile of the portfolio. I propose that a solution to this conundrum is to use a low volatility stock portfolio as the primary asset class in the portfolio. A portfolio constructed primarily of low volatility stocks allows an investor to decrease the allocation to lower yielding bonds while still maintaining a similar risk profile to a traditional 50% stock and 50% bond portfolio.

Millossovich, P., Bacinello, A. R., Olivieri, A., and Pitacco, E. (2011). "Variable Annuities: A Unifying Valuation Approach." In: *SSRN e-Print*.

Life annuities and pension products usually involve a number of guarantees, such as minimum accumulation rates, minimum annual payments or a minimum total payout. Packaging different types of guarantees is the feature of so-called variable annuities. Basically, these products are unit-linked investment policies providing a post-retirement income. The guarantees, commonly referred to as GMxBs (namely, Guaranteed Minimum Benefits of type 'x'), include minimum benefits both in case of death and survival. In this paper we propose a unifying framework for the valuation of variable annuities under quite general model assumptions. We compute and compare contract values and fair fee rates under 'static' and 'mixed' valuation approaches, via ordinary and least squares Monte Carlo methods, respectively.

Mills, E. F. E. A. and Anyomi, S. K. (2023). "Optimal lifetime income annuity without bequest: Single and annual premiums." In: *Finance Research Letters* 53, p. 103613.

This paper studies the optimal annuitization, risk-free investment, and consumption problem in a post-retirement lifecycle model without bequest motives. We characterize annuitization in scenarios where a retiree allocates wealth to a lifetime income annuity in exchange for single or annual premiums. We show that demand for an annuity is influenced by risk aversion in exchange for a single premium and by wealth for annual premiums. The effect of allocating wealth to purchase a lifetime income annuity increases with higher risk aversion for a single premium and decreases with residual wealth for annual premiums. We compare the certainty equivalence of investment yielding the same utility in an actuarially fair pricing framework and show that the product with annual premiums is more attractive. Finally, we show that partial annuitization can be optimal for different risk levels. Our findings thus reflect an explanation for the "annuity puzzle" in the literature on lifecycle consumption.

Minney, A., Zhu, Z., Guo, Y., Li, J., Toscas, P., Koo, B., and Pantelous, A. A. (2022). "Using the pension multiple to measure retirement outcomes." In: *Finance Research Letters* 49, p. 103149.

This paper presents a simple and intuitive measure of retirement income adequacy: the Pension Multiple which captures and quantifies the level of income for each future retirement year as a multiple of the government-provided social security pension. This Pension Multiple at each future retirement year is then mortality-weighted to produce an average Pension Multiple for the entire retirement. An expected shortfall from this average Pension Multiple is introduced to measure the potential shortfall at each future retirement year from the average Pension Multiple. A single measure of retirement income and potential shortfall is then calculated as the averaged Pension Multiple minus the averaged expected shortfall, which is called the adjusted Pension Multiple. Finally, any residual estate is also included as part of the retirement income adequacy measure. The Pension Multiple and estate residue can be calculated by any forecast model for future retirement income. In this paper, we use a robust stochastic forecasting model to demonstrate the effectiveness of using the Pension Multiple to measure and compare different retirement income strategies.

Mislinski, J. (2021). "How to Illustrate Planning Risks to Clients." In: *Advisor Perspectives*.

Clients face three big risks in retirement from the sequence of returns, volatility and asset shortfalls. Michael Hirthler, the founder and chief investment officer at Pennsylvania-based Jacobi Capital Management, explained to me how he uses the Big Picture app to explain those risks to his clients.

Mitchell, J. B. (2009a). "A Mean-Variance Approach to Withdrawal Rate Management: Theory and Simulation." In: *SSRN e-Print*.

Withdrawal rate management of retirement portfolios (decumulation) is explored in a traditional mean-variance context. The paper demonstrates that improvements in the composition of portfolios and in the management of withdrawal rates can both be thought of as increasing the opportunity cost price of annuitizing and reducing the optimal risk of portfolios. Increases in age are shown to have three separate components: decreasing remaining expected lifespan; decreasing variability of remaining lifespan; and increasing variability of returns over remaining holding period. The difficulty of employing a mean-variance approach when a portion of the variation in returns is absorbed by variation in the ending portfolio is noted.

The second portion of the paper demonstrates these principles with simulation analysis. A zero-tolerance (MAXSAFE) approach, as proposed by Bengen, is combined with a multiple-factor withdrawal management strategy, as developed by Stout and Mitchell, to generate a series of risk-return frontiers based on data from 1926-2008.

Improvements in risk-return relationships are found using more restrictive withdrawal rate management approaches than previously reported in the literature. Restricting withdrawal rates when retirees no longer have 1.45 times the present value of their expected withdrawals combined with limitations on withdrawal rate increases allow most retirees to experience average withdrawal rates in excess of 5.5% with average withdrawal rates in excess of 6% and with only a .1% chance of ruin before death.

Mitchell, J. B. (2009b). "Withdrawal Rate Strategies for Retirement Portfolios: Preventive Reductions and Risk Management." In: *SSRN e-Print*.

This paper builds on the work of Stout and Mitchell (2006), Stout (2008), and Blanchett and Frank (2009) by creating a preventive approach to withdrawal management. Proactive strategies, which reduce the withdrawal rate before there are insufficient funds, are shown to significantly reduce the probability of ruin (shortfall) while maintaining the average withdrawal rate. The paper also explores the micro effects of strategy changes by dividing the simulation iterations into groups which have been positively or negatively affected by any particular change, and demonstrates that conventional reporting of the effectiveness of withdrawal rate management techniques can be improved by examining additional moments of the distribution. Data covers 1926-2008 and the mortality table is extended to 108 years.

Mitchell, J. B. (2016). "Migrating with Black Swans: Climate Risk and Retirement Planning." In: *The Journal of Retirement* 4(2), pp. 24–36.

This article combines the thoughts of several authors to describe the migration of retirees through the retirement phase of their lives during a time of climate change. Taleb [2007] introduced the concept of the Black Swan, the (hopefully) infrequent but significant disruption of investment returns. Bernstein [2013] discussed sources of risk in portfolio design and the concept of deep risk, the long-term loss of real capital. Frank, Mitchell, and Blanchett [2012], among others, demonstrated how retirees move through retirement, frequently adjusting to changes in market performance and personal expectations of longevity. Nordhaus [2013] investigated the potential effects of climate change on economic performance and equity returns. The Intergovernmental Panel on Climate Change reports outlined many of the possible underlying climate changes that are likely to drive these risks and necessitate adjustment on the part of retirees. These sources, and others, motivate this article's exploration of climate change, its economic implications, and how retirees might cope as they migrate through their final years. Suggestions are made for how financial planners should prepare themselves and their clients for climate change-related risks.

Mitchell, J. B. and Felton, Z. S. (2015). "Same-Sex Couples and Sustainable Retirement Withdrawal Rates." In: *SSRN e-Print*.

This paper compares sustainable retirement withdrawal rates for same-sex versus heterosexual couples. Due to convergence of expected longevity for males and females at typical retirement ages, systematic differences between same-sex versus heterosexual couples are slight, ranging from .03% at age 50 to less than .4% at age 85. Therefore, same-sex couples can employ conventional withdrawal rate recommendations with, at most, slight modification. Due to a lack of public data regarding mortality differences based on sexual orientation, differences in mortality expectations between individual members of same-sex and heterosexual couples cannot be explored.

Members of same-sex couples, like all retirees, should adjust withdrawal rate expectations periodically to reflect individual conditions.

Mitchell, O. S., Clark, R., and Maurer, R. (2018). "How Persistent Low Returns Will Shape Saving and Retirement." In: *How Persistent Low Returns Will Shape Saving and Retirement*. Oxford University Press.

Financial market developments over the past decade have undermined what was once thought to be conventional wisdom about saving, investment, and retirement spending. How Persistent Low Returns Will Shape Saving and Retirement explores how the weak capital market performance predicted for the next several years will shape pension saving, investment, and decumulation plans. Academics, policymakers, and industry leaders debate alternative strategies to cope with these challenges globally, as economic growth remains slow and low returns become the 'new normal.' This volume includes contributions from plan sponsors, benefit specialists, actuaries, academics, regulators, and others working to design resilient pensions for the next decades. Together, they identify several new tools for retirement savers and pension managers.

Mitchell, O. S. and Utkus, S. P. (2021). "Target-date funds and portfolio choice in 401(k) plans." In: *Journal of Pension Economics and Finance*, pp. 1–18.

Target-date funds in corporate retirement plans grew from $5 billion in 2000 to $734 billion in 2018, partly because federal regulation sanctioned these as default investments in automatic enrollment plans. We show that adopters delegated pension investment decisions to fund managers selected by plan sponsors. Inclusion of these funds in retirement saving menus raised equity shares, boosted bond exposures, curtailed cash/company stock holdings, and reduced idiosyncratic risk. The adoption of low-cost target-date funds may enhance retirement wealth by as much as 50% over a 30-year horizon.

Mladina, P. (2014). "Dynamic Asset Allocation with Horizon Risk: Revisiting Glide Path Construction." In: *The Journal of Wealth Management* 16(4), pp. 18–26.

We compare the empirical distributions of equity and fixed-income returns at different time horizons and find that the risk of equities relative to fixed income is more acute at short time horizons than long time horizons, confirming previous research. This creates the opportunity to develop a dynamic asset allocation process that exploits the reduced horizon risk of equities relative to fixed income. We highlight key data on changing relative risk with time and leverage this information to introduce methods and concepts that inform glide path construction-the building blocks for a dynamic asset allocation process that can support lifecycle, target date retirement, and goals-based investing frameworks.

Mladina, P. (2016). "Optimal Lifetime Asset Allocation with Goals-Based, Lifecycle Glide Paths." In: *The Journal of Wealth Management* 19(1), pp. 10–22.

Assets should serve a purpose: to fund a lifetime of financial goals. If assets serve the purpose of funding lifetime goals, it naturally follows that optimal lifetime asset allocation should be goals-based and multi-period, requiring customization according to goals, human capital, and risk preference. Ideally, risk preference is defined intuitively for private investors. We present a dynamic asset allocation method based on an intertemporal capital asset pricing model that incorporates these features to produce a goals-based lifecycle glide path – an asset allocation roadmap that optimally funds lifetime consumption goals while adapting to evolving conditions.

Mladina, P. (2020). "Refining After-Tax Return and Risk Parameters." In: *The Journal of Wealth Management* 23(2), pp. 8–17.

Taxes introduce certain complexities, requiring proper adjustments to return and risk parameters. The author offers a refined set of after-tax return and risk equations for use in practice and validates them with a stochastic future value cash flow model. The refined after-tax return and risk parameters can be used in portfolio optimization, Monte Carlo simulation, and deterministic present/future value portfolio modeling with internally consistent results. The refinements improve the discovery of the optimal after-tax portfolio and enhance long-term wealth planning in the presence of risk.

Mladina, P. and Grant, C. (2019). "Glide Paths Based on a Retirement Goal and Depleting Human Capital." In: *The Journal of Investing* 28(1), pp. 8–16.

The retirement goals of many Americans are underfunded. The problem is compounded by the complexity of self-managing distribution portfolios, particularly as DC plans replace DB plans. We believe most retirement glide paths are satisfactory but suboptimal solutions. We introduce a glide path of financial assets over the life cycle based on a retirement goal and depleting human capital. The method is anchored to the foundational principles of intertemporal portfolio theory while borrowing heavily from goals-based asset allocation. The result is a dynamic asset allocation over the life cycle that is a function of critical input variables relevant to retirement planning such as retirement savings, retirement consumption and risk aversion. The glide path can be customized to individuals, or semi-customized to discrete subpopulations of DC plan participants.

Mladina, P. and Grant, C. (2024). "An ICAPM for Goals-Based Investing." In: *The Journal of Wealth Management* 27(2), pp. 75–83.

A goals-based methodology built on an intertemporal CAPM (ICAPM) asset allocation framework provides a goal-relative definition of risk for portfolio selection and risk management, efficient capital allocation to fund goals, and efficient goal-aligned subportfolios that accrue to an efficient total portfolio. It offers an elegant goals-based solution for both high-net-worth investors with significant financial assets and retirement accumulators with significant human capital.

Moehle, N., Kochenderfer, M. J., Boyd, S., and Ang, A. (2021). "Tax-Aware Portfolio Construction via Convex Optimization." In: *arXiv e-Print*.

We describe an optimization-based tax-aware portfolio construction method that adds tax liability to standard Markowitz-based portfolio construction. Our method produces a trade list that specifies the number of shares to buy of each asset and the number of shares to sell from each tax lot held. To avoid wash sales (in which some realized capital losses are disallowed), we assume that we trade monthly, and cannot simultaneously buy and sell the same asset. The tax-aware portfolio construction problem is not convex, but it becomes convex when we specify, for each asset, whether we buy or sell it. It can be solved using standard mixed-integer convex optimization methods at the cost of very long solve times for some problem instances. We present a custom convex relaxation of the problem that borrows curvature from the risk model. This relaxation can provide a good approximation of the true tax liability, while greatly enhancing computational tractability. This method requires the solution of only two convex optimization problems: the first determines whether we buy or sell each asset, and the second generates the final trade list. In our numerical experiments, our method almost always solves the nonconvex problem to optimality, and when it does not, it produces a trade list very close to optimal. Backtests show that the performance of our method is indistinguishable from that obtained using a globally optimal solution, but with significantly reduced computational effort.

Moenig, T. (2021). "It's RILA Time: An Introduction to Registered Index-Linked Annuities." In: *SSRN e-Print*.

Registered index-linked annuities (RILAs) are increasingly popular equity-based retirement savings products offered by U.S. life insurance companies. They combine features of fixed-index annuities and traditional variable annuities (TVAs), offering investors equity exposure with downside protection in a tax-deferred setting. This article introduces RILAs to the academic literature by describing the products' key features, developing a general pricing model, and deriving the providers' hedging strategy by decomposing their liabilities into short-term European options.

Numerical illustrations show that RILAs offer investors similar risk profiles (in the long run) as TVAs with maturity guarantees, and that many products currently sold appear to be priced quite favorably for investors. For providers, RILAs may be a preferable alternative or complement to TVAs as they greatly simplify the management of the embedded equity risk and can naturally reduce the TVA capital requirements. These features position RILAs as a viable long-term solution for this product space.

Mohammed, S., Bealer, R., and Cohen, J. (2021). "Embracing advanced AI/ML to help investors achieve success: Vanguard Reinforcement Learning for Financial Goal Planning." In: *arXiv e-Print*.

In the world of advice and financial planning, there is seldom one right answer. While traditional algorithms have been successful in solving linear problems, its success often depends on choosing the right features from a dataset, which can be a challenge for nuanced financial planning scenarios. Reinforcement learning is a machine learning approach that can be employed with complex data sets where picking the right features can be nearly impossible. In this paper, we will explore the use of machine learning for financial forecasting, predicting economic indicators, and creating a savings strategy. Vanguard ML algorithm for goals-based financial planning is based on deep reinforcement learning that identifies optimal savings rates across multiple goals and sources of income to help clients achieve financial success. Vanguard learning algorithms are trained to identify market indicators and behaviors too complex to capture with formulas and rules, instead, it works to model the financial success trajectory of investors and their investment outcomes as a Markov decision process. We believe that reinforcement learning can be used to create value for advisors and end-investors, creating efficiency, more personalized plans, and data to enable customized solutions.

Momen, O., Esfahanipour, A., and Seifi, A. (2020). "A robust behavioral portfolio selection: model with investor attitudes and biases." In: *Operational Research* 20(1), pp. 427–446.

This study develops a behavioral portfolio selection model that uses a robust estimator for expected returns in order to produce portfolios with less need to change over consecutive periods. We also consider investor attitudes toward risk through spectral risk measure as well as investor expectations on future returns by means of the Black-Litterman model, and finally, our model includes a varying risk aversion depending on investor behavioral biases and his latest realized return. In order to evaluate the proposed model and make comparisons possible, we conducted a survey on investor biases and attitudes along with market data of Tehran Stock Exchange. The results reveal that although our model is not mean-variance efficient, it recommends portfolios that are robust, well diversified, and have less utility loss compared to a famous behavioral portfolio model.

Mooney, T., Rapaka, R., and Vera, T. (2020). "Dynamic Regime Strategy for Stress Testing and Optimizing Institutional Investor Portfolios." In: *SSRN e-Print*.

Our work aims to develop a stand-alone trading system to construct portfolios that show the benefits of value and momentum style integration and presents the effectiveness of alternative integration methods for long-only absolute return funds, which seeks absolute returns that are not highly correlated to traditional assets such as stocks and bonds. Our approach uses the CRoss Industry Standard Process for Data Mining (CRISP-DM) model to guide the necessary steps, processes, and workflows for executing our project.

Moriggia, V., Consigli, G., and Iaquinta, G. (2012). "Optimal Stochastic Programming-Based Personal Financial Planning with Intermediate and Long-Term Goals." In: *Stochastic Programming*. WORLD SCIENTIFIC, pp. 43–68.

We present an asset-liability management (ALM) model for individuals facing a long term allocation problem over a diversified investment universe including mutual and pension funds as well as unit linked contracts with corporate, fixed income or equity exposure. Inflation-adjusted living costs and investment or consumption targets over the planning horizon lead to the definition of a comprehensive decision support tool, whose key building blocks are discussed with reference to a case-study. A structured stochastic model for long-term scenario generation is also briefly analyzed.

Mousa, A., Pinheiro, D., and Pinto, A. (2016). "Optimal life-insurance selection and purchase within a market of several life-insurance providers." In: *Insurance: Mathematics and Economics* 67, pp. 133–141.

We consider the problem faced by a wage-earner with an uncertain lifetime having to reach decisions concerning consumption and life-insurance purchase, while investing his savings in a financial market comprised of one risk-free security and an arbitrary number of risky securities whose prices are determined by diffusive linear stochastic differential equations. We assume that life-insurance is continuously available for the wage-earner to buy from a market composed of a fixed number of life-insurance companies offering pairwise distinct life-insurance contracts. We characterize the optimal consumption, investment and life-insurance selection and purchase strategies for the wage-earner with an uncertain lifetime and whose goal is to maximize the expected utility obtained from his family consumption, from the size of the estate in the event of premature death, and from the size of the estate at the time of retirement. We use dynamic programming techniques to obtain an explicit solution in the case of discounted constant relative risk aversion (CRRA) utility functions.

Mueller-Glissmann, C. and Ferrario, A. (2025). "Macro-Based Strategic Tilting of Multi-Asset Portfolios." In: *The Journal of Portfolio Management* 51(5), pp. 84–113.

Even over long investment horizons, equity and bond returns have varied materially, suggesting opportunities for so-called "strategic tilting" of multi-asset portfolios. In this article, the authors develop a long-term return forecasting framework for equity and bonds combining valuations and structural macro conditions. The predictive power of their block approach for 10-year returns is better than predictive regressions on valuations alone, especially out of sample. They run several robustness checks illustrating the benefits of incorporating macro conditions into long-term return forecasting. They then use those macro-based return forecasts to estimate optimal portfolios for the subsequent 10 years. Based on the expected Sharpe ratios they implement two additional strategic tilts: 1) increase allocations to cash or use leverage and 2) increase allocations to real assets or growth stocks. The macro-based strategic tilts materially enhanced risk-adjusted returns compared with a static 60/40 portfolio since 1950 but also relative to tilts based on valuations alone.

Mulvey, J. M. and Hao, H. (2020). "Setting Realistic Goals for Personal Retirement Planning via Micro-Macro Analyses." In: *The Journal of Retirement* 8(2), pp. 23–38.

The vulnerability of individuals planning for retirement has been growing as a result of the conversion from defined benefit plans to defined contribution plans, the steady increase in life longevity, and the uncertainty of asset returns under an ever-changing global environment. A serious problem is the lack of appropriate planning for retirement. How much should an individual in the United States save beyond the Social Security tax to maintain a reasonable lifestyle after retirement? The article designs a framework to facilitate the process of setting realistic goals for retirement planning, featuring the concept of agent-based simulations. Focusing on policy-rule-based investment strategies, the simulation framework includes multiple investable asset categories and explores dynamic allocation based on the investor's age, current salary, and Social Security accumulation situation. Empirical results demonstrate a stylized application of the planning framework.

Mulvey, J. M., Martellini, L., Hao, H., and Li, N. (2019). "A Factor- and Goal-Driven Model for Defined Benefit Pensions: Setting Realistic Benefits." In: *The Journal of Portfolio Management* 45 (3), pp. 165–177.

A factor and goal-driven framework for assessing asset allocation and contribution decisions within defined-benefit pension plans is developed in this article. A critical element is setting future benefits with reference to the ability of the pension sponsors to support liabilities under reasonable investment expectations. The approach suggested by the authors combines a micro study of a representative cohort of individuals with an aggregation across a target population to estimate consistency between the micro and macro environments. A stochastic inflation risk factor affects both contribution and spending cash flows. This agent-based model suggested by the authors provides a more realistic framework than traditional approaches for setting pension benefits.

Munk, C. (2024). "Optimal Retirement Saving and Dissaving." In: *SSRN e-Print*.

Applying a rich model of individuals' life-cycle utility maximization, we conduct a comprehensive evaluation of retirement saving plans for workers determining their contributions to the plan, emulating retirement saving with 401(k)s in the U.S. Across a range of individual characteristics, the access to basic plans with constant expected payouts and either no or full annuitization leads to individual welfare gains of up to 5.06% of initial wealth and lifetime income (around $43,700 in present value terms), and almost all the individuals prefer full annuitization and a target-date fund investment strategy. With flexible plans allowing for partial annuitization, non-constant expected payouts, and unscheduled withdrawals from the pension account, the welfare gains go up to 6.20%, and most individuals prefer a high degree of annuitization and expected payouts being increasing through retirement.

Munnell, A. H., Sanzenbacher, G., Webb, A., and Gillis, C. M. (2016). "Are Early Claimers Making a Mistake?" In: *SSRN e-Print*.

Using Health and Retirement Study (HRS) data and Latent Class Analysis for three cohorts (those born in 1931-1936, 1937-1941, and 1942-1947), this paper explores: 1) who claims Social Security benefits at age 62; 2) what percentage of households claiming at 62 are unprepared for retirement; and 3) whether the unprepared early claimers were pushed into claiming through job shocks and/or poor health or simply decided to take benefits early. Looking across three cohorts makes it possible to see whether these patterns have changed as the average claim age has increased and pension coverage has shifted away from defined benefit (DB) plans. That is, have those who have moved out of age-62 claiming been educated, financially prepared households or unprepared households that have recognized the need to delay claiming?

The paper found that:

- Consistent with previous research, the HRS shows a decline in those claiming at 62.
- Age-62 claimers are less well off than "postponers" in some ways and better off in others.
- Latent class analysis shows that this mixed picture reflects the average of: 1) those with little education and poor job prospects (disadvantaged); and 2) those with at least some college and sufficient resources to claim early (advantaged).
- The percentage of the age-62 claimers in each of these groups has remained virtually constant over the three cohorts.
- Comparing the calculated household replacement rates with target rates from previous research shows that, overall, roughly 65 percent of households claiming at 62 are not prepared; the rate for the disadvantaged group is twice the rate of the advantaged group.
- The percentage unprepared at 62 has increased over time, reflecting an overall trend toward less preparedness.
- A simple probit regression suggests that health and employment shocks and the absence of a DB pension are related to the lack of preparedness for both the disadvantaged and advantaged.

The policy implications of the findings are:

- Given the increasing trend in unpreparedness, further cuts to Social Security benefits would exacerbate this problem.

- Workers claiming at 62 with DB plans were especially likely to be prepared; these plans are not coming back, so the challenge is whether the 401(k) system can be enhanced.

Srtudy whether Early Claimers Making a Mistake.

Munnell, A. H. and Webb, A. (2021). "The Impact of Leakages from 401(k)s and IRAs." In: *The Journal of Retirement* 9(3), pp. 32–54.

This article summarizes what is known about leakages from existing studies and relates these results to detailed data on leakages in 2013 provided by Vanguard's How America Saves. It then uses two data sets - the Survey of Consumer Finances and the Survey of Income and Program Participation - to estimate the impact of leakages on wealth at retirement. The Vanguard data are a critical component because they provide a comprehensive picture of assets and participant flows, whereas the surveys on which earlier studies were based tended to focus on one component, such as loans. A key limitation is that Vanguard's population is probably wealthier than the general population.

Munnell, A. H., Wettstein, G., and Hou, W. (2019). "How Best to Annuitize Defined Contribution Assets?" In: *SSRN e-Print*.

Unlike defined benefit pensions that provide participants with steady benefits for as long as they live, 401(k) plans and Individual Retirement Accounts (IRAs) provide little guidance on how to turn accumulated assets into income. As a result, retirees have to decide how much to withdraw each year and face the risk of either spending too quickly and outliving their resources or spending too conservatively and consuming too little. Surveys of individuals' plans and several recent studies suggest that people will not draw down their accumulations for fear that they will exhaust their money and be unable to cover end-of-life health care costs. They also must consider how to invest their savings after retirement. These are difficult decisions.

Better strategies are possible that will ensure a higher level of lifetime income, reduce the likelihood that people will outlive their resources, and alleviate some of the anxiety associated with post-retirement investing. Workers could use a portion of their 401(k) and IRA assets to purchase an immediate annuity that pays a fixed amount throughout their lives, typically starting at age 65. Or they could purchase an advanced life deferred annuity (ALDA) that requires a smaller share of accumulated assets and begins payments at a later age like 85. Alternatively, they could use their assets to delay claiming Social Security - essentially purchasing an inflation-indexed annuity. Right now, none of these three options is commonly used. Very few workers choose to purchase immediate or deferred annuities (the first two options). And few retirees appear to be deferring claiming in order to receive the maximum annuity income from Social Security - most people simply retire earlier and claim immediately.

Increasing annuitization in a meaningful way would require embedding annuities in 401(k) plans, with annuitization as the default. Recent proposed federal legislation, such as the SECURE Act (Setting Every Community Up for Retirement Enhancement), encourages plan sponsors to offer annuities in their plans by establishing a fiduciary safe harbor when specific statutory conditions are followed in selecting an insurance company. This legislation does not address, however, the question of defaults or the possibility of using 401(k) assets to purchase additional Social Security benefits. Moving forward on these fronts would require some consensus about the appropriate share of 401(k) assets to be annuitized and the best method for annuitizing them.

To address these issues, this paper compares the level of lifetime utility generated by alternative annuitization approaches - immediate annuities, deferred annuities, and additional Social Security through delayed claiming. The analysis also tests different assumptions for the share of initial wealth that participants use to purchase these products.

Muralidhar, A. (2020). "Asset Pricing, Asset Allocation and Risk-Adjusted Performance with Multiple Goals and Agency: The Goals and Risk-based Asset Pricing Model." In: *SSRN e-Print*.

Investment managers require a consistent asset pricing model, asset allocation recommendations and risk-adjusted performance measures (or the "three facets of investing") to be effective in managing portfolios. Incorporating three critical realities of investing into these models (i.e., that investors have many stochastic goals, seek to delegate to skillful agents, and maximize risk-adjusted returns as opposed to expected utility) provides recommendations on the three facets that are markedly different from the foundational papers of Modern Portfolio Theory (MPT). This is important as Goals-based Investing (GBI) and delegation are now the norm, and investors globally are not meeting their goals by adopting traditional MPT. The paper briefly surveys the literature on MPT, GBI, and agency before providing a normative Goals- and Risk-Based Asset Pricing Model (GRAPM) that includes these three realities of investing and articulates the three new facets. GRAPM exploits a simple idea that a relatively risk-free asset for one stochastic goal is a risky asset for another, and vice versa.

These two assets, plus the traditional absolute risk-free rate of MPT, allow us to triangulate to establish returns for all other assets based on the return of any goal-replicating asset and multiple correlations (as opposed to a single relationship with the unobservable "market" portfolio). This approach creates a "pair-wise equilibrium" for all assets - very different from MPT - and also lends itself easily to a new asset pricing model with heterogeneous investors (i.e., each investor has a unique goal). GRAPM incorporates a "risk-aversion" parameter that is also easily observable, unlike MPT, and appears to explain why seemingly similar investors can have markedly different asset allocations or expected returns.

Muralidhar, A. (2023). "Goals-based Arrow-Debreu Securities." In: *SSRN e-Print*.

Arrow-Debreu (AD) securities, while foundational from an academic perspective of asset pricing, do not and will never exist as there is no natural issuer and "states" are hard to specify. Moreover, because of the spanning argument, to price all securities, would require an inordinate number of AD securities. In reality, investors invest to achieve investment goals like retirement, saving for a child's education, health etc. The creation and issuance of unique and innovative retirement and education bonds by Brazil in 2023 offers a chance to view these securities as "Goals-based Arrow Debreu securities"(GADs). We show that the existence of just two of these securities, under certain realistic market conditions, along with the absolute risk-free asset, can be used to determine the price of all assets. More GADs just require more equilibrium conditions, thereby addressing the John Cochrane "free parameter problem" in typical asset pricing models.

Muralidhar, A., Brink, R. van den, Groenendijk, P., and Wouden, R. van der (2023a). "The Importance of Goals-Based (and Values-Based) Liability Indices: Applied to Impact and Green Investing." In: *The Journal of Portfolio Management* 50(2), pp. 134–149.

Global investors are creating portfolios that align with their specific goals and values. The authors posit that goals and values must be translated into goals-based liability indices (GLIdes). GLIdes can be used to achieve goals, benchmark the asset portfolio, and improve governance, as they allow boards to measure progress to their overall (or multiple) goal(s) and also the relative risks of their investment policies and implementation. The authors first introduce the GLIdes concept as a simple extension of the investible liability index developed in the early 2000s. They also show how GLIdes can be applied broadly, and specifically to goals such as "green" investing or impact/sustainable investing. For some goals/values, GLIdes can be created using existing instruments; for others, new instruments will need to be created, much like Retirement Security Bonds were created to be the relative safe asset for retirement investing. The authors also show how investors might set a clear path to achieve dynamic, long-term goals (e.g., net zero compliance), through clear GLIde paths, offering potentially a transparent and auditable process.

Muralidhar, A., Kobor, A., and Vitorino, A. (2023b). "Improving Retirement Security with a Dynamic Asset Allocation Strategy using Retirement Bonds: The Case of Brazil." In: *SSRN e-Print*.

The goal of investing for retirement is to secure a target level of income that maintains the individual's pre-retirement lifestyle. Current "safe harbor" glide-path products shift investments from stocks to bonds on the basis of the individual's age. This approach is unlikely to secure a target retirement income because the glide path is focused on the wrong goal and guarantees neither a wealth nor an income level. We tested a dynamic asset allocation strategy that takes no view of future market performance and is based on a retirement income goal (via a funded status), but leverages a new instrument issued in Brazil - RendA+ (or a retirement bond). We show how this dynamic strategy could dominate standard portfolio choices. The article uses intermediate retirement targets and explores the implications of the dynamic asset allocation strategy for the level of savings required to achieve a retirement goal.

Muralidhar, A., Shin, S. H., and Ohashi, K. (2016). "The Most Basic Missing Instrument in Financial Markets: The Case for Forward Starting Bonds." In: *The Journal of Investment Consulting* 17(2).

There is a looming retirement crisis globally with the three pillars of retirement threatened because of insufficient funding, improper investment decisions, and transferring risk to individuals who are least capable of bearing such risk. This paper argues that the introduction of a unique financial instrument, basically an inflation linked bond which pays coupons when you need it, might help ameliorate this crisis.

The Life Cycle Hypothesis (LCH) demonstrated why people save; namely, they try to set aside resources during their working lives, to be able to tap into them to ensure retirement income when labor income stops. Modern Portfolio Theory (MPT) attempted to help individuals make optimal investment decisions on these savings by investing in stocks, bonds and other assets to ensure sufficient retirement wealth. There are three problems with using MPT approaches for retirement planning: (a) MPT ignores the uses of funds; (b) focuses on wealth as opposed to retirement income maximization; and (b) none of these assets is an ideal hedge for a desired

retirement income. As a result, seemingly safe MPT assets are risky from a retirement income perspective (because the Capital Asset Pricing Model (CAPM) is a specific case of a more general Relative Asset Pricing Model (RAPM)).
The need for our bond is simple: the riskless retirement asset is an inflation-indexed, coupon-only bond that defers payment until retirement and pays till death. All attempts to recreate this profile through traditional stocks and bonds, or purchase such a profile through annuities are sub-optimal, risky, complex or expensive thereby threatening retirement security. The paper goes further to demonstrate that there is a potentially willing supplier of such bonds, and that this bond offers the perfect offsetting cash flow profile to infrastructure projects, thereby completing the market. It also addresses challenges, issues and opportunities surrounding such an instrument and examines issues relating to the creation of a market for FSBs.

Muralidhar, A., van den Brink, R., Groenendijk, P., and Wouden, R. van der (2023c). "The Importance of Goals-based (and Values-based) Liability Indices: Applied to Impact and Green Investing." In: *SSRN e-Print*.
Investors are creating portfolios that align with their specific goals and values. We posit that goals and values should be translated into Goals-based Liability Indices ("GLIdes"). GLIdes can be used to achieve goals, benchmark the asset portfolio, and improve governance.They allow Boards to measure progress to their overall (or multiple) goal(s) and also the relative risks of their investment policies and implementation. We first introduce the GLIdes concept as an extension of the Investible Liability Index developed in the early 2000s. The paper also shows how GLIdes can be applied broadly, and specifically to goals such as "green" investing or impact/sustainable investing. For some goals/values, GLIdes can be created using existing instruments. For others, new instruments will need to be created much like Retirement Security Bonds were created to be the relative safe asset for retirement investing. The paper shows how investors might set a clear path to achieve dynamic, long-term goals (e.g., Net Zero compliance), through clear GLIde paths, offering a transparent and auditable process.

Murguia, A. and Pfau, W. D. (2021a). "A Model Approach to Selecting a Personalized Retirement Income Strategy." In: *SSRN e-Print*.
This study identifies and validates a set of scorable retirement income factors to define preferences for an overall retirement income style. This investigation further attempts to create a workable model for retirement income planning by showing how the factors connect to four main retirement income strategies: systematic withdrawals with total return investing, risk wrap with deferred annuities, protected income with immediate annuities, and time segmentation or bucketing. Approaching retirement income agnostically and matching retirement income strategies based on an individual's personal retirement income style may lead to improved outcomes that achieve greater "buy in" and comfort.

Murguia, A. and Pfau, W. D. (2021b). "Retirement Income Beliefs and Financial Advice Seeking Behaviors." In: *SSRN e-Print*.
This investigation identifies and validates a series of salient behavioral finance and psychological constructs that influence retirement income planning. We show how these scales are related to each other as well as retirement income concerns and investment behaviors. We also describe how four investment personas can be linked with the Advisor Usefulness and Retirement Income Self-Efficacy scales to successfully identify preferred financial implementation methods. This can assist individuals in more readily recognizing their relative strengths and weaknesses when implementing a retirement income strategy, and financial professionals can present advice in a manner that addresses a client's concerns and preferred implementation.

Murphy, R. O., Lamas, S., and Sin, R. (2020). "Identifying What Investors Value in a Financial Adviser: Uncovering Opportunities and Pitfalls." In: *Journal of Financial Planning* 33(7), pp. 44–52.
Understanding what investors value and look for in a financial adviser is critical to both the client's and adviser's success. This research identified the alignment and discordance between what non-retired investors value, and what financial advisers think their clients value. These insights show opportunities for advisers to better articulate their value and, in doing so, improve attraction, retention, and promote client satisfaction. Using an online platform, investors were asked to rank a set of 15 adviser attributes from most important to least important. Advisers were asked to rank the same attributes, predicting what they thought investors valued. Results show that both groups recognized that reaching financial goals is a top priority, with notable disagreements on how to achieve this. Investors underestimated the importance of behavioral coaching in helping them stay on course. Advisers underestimated the importance of tax-efficient strategies to their clients. A follow-up study tested different ways of describing behavioral coaching to potentially find better wording that more naturally resonates

with investors, and results suggest that simple changes in phrasing and communication can improve how investors perceive the value behaviorally attuned advisers bring to the relationship.

Neo, P. L. and Tee, C. W. (2023). "Tail Risk Hedging: The Search for Cheap Options." In: *The Journal of Portfolio Management* 50(1), pp. 106–119.

The authors discover that a simple heuristic of sorting liquid equity options by dollar price to construct a portfolio of cheap put options leads to a surprisingly robust tail risk hedge – the superior performance holds even when compared against advanced empirical option strategies. Further investigation reveals the asymmetry in market correlation under different market conditions as the mechanism of this robust hedging performance. The put options selected by the heuristic comprises stocks with diverse firm characteristics. The correlation spike accompanying tail risk events leads to most of these options moving into the money, compensating for the losses incurred on a broad-base equity index holding. During normal market conditions, these options benefited from the diversification effect because of a lower market correlation, thus mitigating the portfolio drag effect.

Nersesian, J. (2022). "Net Unrealized Appreciation: A Retirement Distribution Strategy to Reduce Taxes." In: *Investments and Wealth Monitor* 2022(September/October).

Net unrealized appreciation NUA is a retirement distribution strategy that may provide timely benefits because of (1) the significant spread between ordinary income and long-term capital gains tax rates, (2) the increased availability of company stock as an investment option in defined contribution plans, and (3) the significant gains produced in a positive equity market during the past few decades. A Under the NUA strategy, employees take a lump-sum, in-kind distribution of company stock from their qualified retirement plan (employee stock option plan, 401(k) plan, etc.) upon separation from service, recognizing ordinary income taxes only on the stock's cost basis. The difference between the basis and the fair market value (FMV) – the NUA – is taxed at long-term capital gains rates only when the stock is sold, regardless of the holding period. This option may produce better results than rolling the stock into an individual retirement account (IRA) where the entire value eventually will be taxed as ordinary income upon distribution. A Internal Revenue Code 402(e)(4) technically refers to "employer securities," including common shares and units of "stock funds" of employer stock that may be eligible if the distribution can be made in-kind. A If a 10-percent early withdrawal penalty is triggered, it applies only to the stock's original cost basis, not the full distribution amount. A NUA is treated as income in respect of a decedent if the employee dies before the NUA is recognized and taxed. The NUA portion of the position does not receive a step-up in basis as is typically provided for appreciated assets on death. A The NUA strategy also is available to beneficiaries for stock not distributed from a qualified plan before death.

Neville, H., Draaisma, T., Funnell, B., Harvey, C. R., and Hemert, O. van (2021a). "The Best Strategies for Inflationary Times." In: *SSRN e-Print*.

Over the past three decades, a sustained surge in inflation has been absent in developed markets. As a result, investors face the challenge of having limited experience and no recent data to guide the repositioning of their portfolios in the face of heighted inflation risk. We provide some insight by analyzing both passive and active strategies across a variety of asset classes for the U.S., U.K., and Japan over the past 95 years. Unexpected inflation is bad news for traditional assets, such as bonds and equities, with local inflation having the greatest effect. Commodities have positive returns during inflation surges but there is considerable variation within the commodity complex. Among the dynamic strategies, we find that trend-following provides the most reliable protection during important inflation shocks. Active equity factor strategies also provide some degree of hedging ability. We also provide analysis of alternative asset classes such as fine art and discuss the economic rationale for including cryptocurrencies as part of a strategy to protect against inflation.

Neville, H., Draaisma, T., Funnell, B., Harvey, C. R., and Hemert, O. V. (2021b). "The Best Strategies for Inflationary Times." In: *The Journal of Portfolio Management* 47(8), pp. 8–37.

Over the past three decades, a sustained surge in inflation has been absent in developed markets. As a result, investors face the challenge of having limited experience and no recent data to guide the repositioning of their portfolios in the face of heightened inflation risk. In this article, the authors provide some insight by analyzing both passive and active strategies across a variety of asset classes for the United States, the United Kingdom, and Japan over the past 95 years. Unexpected inflation is bad news for traditional assets, such as bonds and equities, with local inflation having the greatest effect. Commodities have positive returns during inflation surges, but there is considerable variation within the commodity complex. Among the active strategies, the authors find that trend-following provides the most reliable protection during important inflation shocks. Active equity factor strategies also provide some degree of hedging ability. The authors also provide an analysis of alternative asset classes such as fine art and discuss the economic rationale for including cryptocurrencies as part of a strategy to protect against inflation.

Ngai, A. and Sherris, M. (2011). "Longevity risk management for life and variable annuities: The effectiveness of static hedging using longevity bonds and derivatives." In: *Insurance: Mathematics and Economics* 49(1), pp. 100–114.

For many years, the longevity risk of individuals has been underestimated, as survival probabilities have improved across the developed world. The uncertainty and volatility of future longevity has posed significant risk issues for both individuals and product providers of annuities and pensions. This paper investigates the effectiveness of static hedging strategies for longevity risk management using longevity bonds and derivatives (qq-forwards) for the retail products: life annuity, deferred life annuity, indexed life annuity, and variable annuity with guaranteed lifetime benefits. Improved market and mortality models are developed for the underlying risks in annuities. The market model is a regime-switching vector error correction model for GDP, inflation, interest rates, and share prices. The mortality model is a discrete-time logit model for mortality rates with age dependence. Models were estimated using Australian data. The basis risk between annuitant portfolios and population mortality was based on UK experience. Results show that static hedging using qq-forwards or longevity bonds reduces the longevity risk substantially for life annuities, but significantly less for deferred annuities. For inflation-indexed annuities, static hedging of longevity is less effective because of the inflation risk. Variable annuities provide limited longevity protection compared to life annuities and indexed annuities, and as a result longevity risk hedging adds little value for these products.

Nguyen, H., Sherris, M., Villegas, A. M., and Ziveyi, J. (2024). "Scenario selection with LASSO regression for the valuation of variable annuity portfolios." In: *Insurance: Mathematics and Economics* 116, pp. 27–43.

Variable annuities (VAs) are increasingly becoming popular insurance products in many developed countries which provide guaranteed forms of income depending on the performance of the equity market. Insurance companies often hold large VA portfolios and the associated valuation of such portfolios for hedging purposes is a very time-consuming task. There have been several studies focusing on inventing techniques aimed at reducing the computational time including the selection of representative VA contracts and the use of a metamodel to estimate the values of all contracts in the portfolio. In addition to the selection of representative contracts, this paper proposes using LASSO regression to select a set of representative scenarios, which in turn allows for the set of representative contracts to expand without significant increase in computational load. The proposed approach leads to a remarkable improvement in the computational efficiency and accuracy of the metamodel.

Nicolas, F. (2019). *Investment Strategies for Retirement*. World Scientific. 416 pp.

The issue of pension financing is evolving everywhere, becoming more of a corporate or individual matter rather than a state one. Demographic changes are making sharing mechanisms hard to control, and social deficits often lead governments to pull back from their obligations. This raises many questions for the individual: 1) At what age should I start saving? When should I increase or reduce my savings? 2) How do I secure my income? 3) What products should I choose to supplement my pension and savings? 4) What degree of risk should I take on? Despite the burden for securing one's retirement increasingly placed on individuals, many are often badly prepared to tackle this very long savings process, which is often complicated by the specific characteristics of a pension plan. This publication, intended for investment professionals, customer advisors, and individuals interested in personal finance and asset management, looks at some of the fundamental elements of investment strategies and techniques for retirement.

Ning, Y., Yang, S., and Zheng, W. (2022). "Dynamic rebalancing of a risk parity investment portfolio." In: *Journal of Investment Strategies* 11(4).

This paper examines a popular risk parity investment portfolio, the so-called All Weather portfolio, from January 2005 to May 2020. We find that the All Weather portfolio outperforms several other portfolios in the long term based on the riskadjusted Sharpe ratio, Calmar ratio and maximum drawdown. We further examine the impact of various static and dynamic portfolio-rebalancing strategies (eg, periodic rebalancing, dynamic range rebalancing and moving-average-distance-based trend-following rebalancing) on the All Weather portfolio. We find that the riskadjusted performance of the portfolio can be improved by implementing the optimal range-rebalancing strategy. Further, we find that the moving-average-distancebased trend-following rebalancing strategy can improve the All Weather portfolio's performance under some circumstances. It seems that the two dynamic rebalancing strategies can be used interchangeably.

Novotny, L. E. and Mansson, A. (2020). "Finding the Optimal Withdrawal Rate on a Retirement Portfolio." MA thesis. Copenhagen Business School.

This thesis seeks to illuminate how investors may withdraw optimally from their retirement portfolio, considering personal attributes, to maximize their lifetime utility. It thereby challenges the popular existing research conducted by Cooley et al. (1999)which investigates a sustainable withdrawal rate, mainly by using the overlapping period method. The main finding of the thesis is, that the personal attributes of the investors heavily impacts their recommended payment plans, suggesting that the generic sustainable withdrawal rate presented by Cooley et al. (1999) hardly can be used as a basis for planning retirement. To do this, this study specifies a utility framework, which is being used as a tool for assessing investor utility by given payout plans, considering a series of personal traits, mainly including subjective discount rate, constant relative risk aversion, time horizon, habit consumption and the loss aversion of the investor. The thesis then assesses optimal investor payout plans with an exogenously given rate of return, thus excluding risk. Subsequently the study introduces uncertainty on the investment assets, making the optimization problem more complex. To assess risk, the thesis considers two assets with their simulated returns being provided by Monte Carlo simulation and bootstrapping simulation, over a 30-year time horizon. These are then indexed and used for providing optimal payment plans considering lower returns given by the utilization of tail risk measures such as Value at Risk. The thesis finally attempts to show the diversity in optimal withdrawals by applying the utility framework to the fictious case examples, and by using the data from the simulations. Despite only being a fictious case, the stereotypical investors each represent qualities which are relatable to real life investors.

O'Donovan, J. and Yu, G. Y. (2024). "Transaction Costs and Cost Mitigation in Option Investment Strategies." In: *SSRN e-Print*.

We examine the impact of transaction costs on the profitability of long-short portfolios of delta-hedged option returns. Of the 24 variables studied, 17 generate positive and significant gross returns, but none remain profitable after accounting for trading costs. We explore cost-mitigation strategies and propose a novel approach that outperforms existing methods, restoring profitability to 7 long-short portfolios. Our findings emphasize the crucial role of implementation costs in assessing the investment opportunity set in equity-option markets and underscore the importance of incorporating transaction costs when evaluating option-based strategies.

Oehler, A. and Horn, M. (2024). "Does ChatGPT provide better advice than robo-advisors?" In: *Finance Research Letters* 60, p. 104898.

We develop three investor profiles with different risk attitudes and consult ChatGPT and 17 robo-advisors to recommend an investment portfolio. We compare the recommendations with a benchmark derived from academic literature. ChatGPT's recommendations align with the three investor profiles and the benchmark. In contrast, only three out of the 17 robo-advisors come close to meeting the benchmark for all three investor profiles. Three robo-advisors fail to meet the benchmark for all investor profiles. Our findings reveal that ChatGPT provides better financial advice for one-time investments than robo-advisors. A policy implication is to clearly disclose that independent sophisticated chatbots can be recommended as a trustworthy source of information for retail investors.

Ohana, J.-J., Benhamou, E., Guez, B., Lefort, B., Saltiel, D., and Jacquot, T. (2024). "Crafting Portfolios Tailored to Investor Preferences with Generative AI." In: *SSRN e-Print*.

Investors often face challenges aligning portfolios with specific preferences or regulatory constraints, such as avoiding commodities. Can machine learning offer a solution? Using Graphical Models, we frame this as an inference problem to replicate a desired fund within set constraints. Our approach achieves a high correlation with the target portfolio, allowing cost-effective replication. An example to decoding global macro hedge funds is provided. It shows very stable correlation emphasizing that the method is able to replicate the strategy even under some strong investment constraints.

Olkhov, V. (2024). "Market-Based 'Actual' Returns of Investors." In: *SSRN e-Print*.

We describe how the market-based average and volatility of the 'actual' return, which the investors gain within their market sales, depend on the statistical moments, volatilities, and correlations of the current and past market trade values. We describe three successive approximations. First, we derive the dependence of the market-based average and volatility of a single sale return on market trade statistical moments determined by multiple purchases in the past. Then, we describe the dependence of average and volatility of return that a single investor gains during the 'trading day." Finally, we derive the market-based average and volatility of return of different investors during the 'trading day" as a function of volatilities and correlations of market trade values. That highlights the distribution of the 'actual' return of market trade and can serve as a benchmark for 'purchasing' investors.

Ouazad, A. (2022). "Do Investors Hedge Against Green Swans? Option-Implied Risk Aversion to Wildfires." In: *SSRN e-Print*.
Measuring beliefs about natural disasters is challenging. Deep out-of-the-money options allow investors to hedge at a range of strikes and time horizons, thus the 3-dimensional surface of firm-level option prices provides information on (i) skewed and fat-tailed beliefs about the impact of natural disaster risk across space and time dimensions at daily frequency; and (ii) information on the covariance of wildfire-exposed stocks with investors' marginal utility of wealth. Each publicly-traded company's daily surface of option prices is matched with its network of establishments and wildfire perimeters over two decades. First, wildfires affect investors' risk neutral probabilities at short and long maturities; investors price asymmetric downward tail risk and a probability of upward jumps. The volatility smile is more pronounced. Second, comparing risk-neutral and physical distributions reveals the option-implied risk aversion with respect to wildfire-exposed stock prices. Investors' marginal utility of wealth is correlated with wildfire shocks. Option-implied risk aversion identifies the wildfire-exposed share of portfolios. For risk aversions consistent with Barro (2012), equity options suggest (i) investors hold larger shares of wildfire-exposed stocks than the market portfolio; or (ii) investors may have more pessimistic beliefs about wildfires' impacts than what observed returns suggest, such as pricing low-probability unrealized downward tail risk. We calibrate options with models featuring both upward and downward risk. Results are consistent a significant pricing of downward jumps.

Owadally, I., Jang, C., and Clare, A. (2021a). "Optimal investment for a retirement plan with deferred annuities." In: *Insurance: Mathematics and Economics* 98, pp. 51–62.
We construct an optimal investment portfolio model with deferred annuities for an individual investor saving in a retirement plan. The objective function consists of power utility in terms of consumption of all secured retirement income from the deferred annuity purchases, as well as bequest from remaining wealth invested in equity, bond, and cash funds. The asset universe is governed by a vector autoregressive model incorporating the Nelson-Siegel term structure and equity returns. We use multi-stage stochastic programming to solve the optimization problem numerically. Deferred annuity purchases are made continuously over the working lifetime of the investor, increasing particularly in the years before retirement. The investment strategy hedges price changes in deferred annuities, and bond holding and deferred annuity purchases increase when interest rates are high. Optimal investment and deferred annuity choices depend on realized and expected values of state variables. The optimal strategy is also compared with typical retirement plan strategies such as glide paths. Our results provide support for deferred annuities as a major source of retirement income.

Owadally, I., Mwizere, J.-R., Kalidas, N., Murugesu, K., and Kashif, M. (2021b). "Long-Term Sustainable Investment for Retirement." In: *Sustainability* 13(9), p. 5000.
We consider whether sustainable investment can deliver performance comparable to conventional investment in investors' long-term retirement plans. On the capital markets, sustainable investment can be achieved through various instruments and strategies, one of them being investment in mutual funds that subscribe to ESG (environmental, social, and governance) principles. First, we compare the investment performance of ESG funds with matched conventional funds over the period 1994-2020, in Europe and the U.S. We find no significant evidence of differing performance (at 5% level) despite using a number of investment performance metrics. Second, we perform a historical backtest to model a UK personal retirement plan from 2000 till 2020, taking full account of investment management fees and transaction costs. We find that investing in an index-tracker fund overlaid with ESG screening delivers a pension which is 10.4% larger than is achieved if the index-tracker fund is used without screening. This is also 20.2% larger than is achieved by investing in a collection of actively managed funds with a sustainable purpose. We conclude that an ESG-screened long-term passive investment approach for retirement plans is likely to be successful in satisfying the twin objectives of a secure retirement income and of sustainability.

Pae, Y. and Atra, R. (2020). "Rules of Thumb versus Industry Glide Paths; Some Bootstrapping Evidence." In: *The Journal of Investing* 29(3), pp. 23–27.
We compare the performance of retirement portfolios using the average glide path of five popular target date funds to general rules of thumb for asset allocation. Surprisingly, the industry average target date fund has similar return and risk as the "120 minus your age rule". In addition, a simple "140 minus your age rule" produces greater expected savings at retirement and a lower failure rate for average US investors retiring in their early 60s. A naive approach such as the "120 minus your age" rule or the "140 minus your age" can benefit average US employees by reducing transaction costs, improving retirement balances and increasing the probability of a comfortable retirement through an easy-to-understand investing rule.

Pakula, D. (2020). *Guiding your clients through stormy weather: Sustainable withdrawal rates in times of crisis*. Tech. rep. Vanguard.

Many investors are concerned that a severe market crisis can dramatically affect their portfolio wealth and spending needs.

This research note uses simulated market scenarios to explore the interaction among market crises, expected returns, and sustainable withdrawal rates.

This analysis yields two insights. First, the inverse relationship between market downturns and expected returns may limit, but not eliminate, a downturn's impact on sustainable spending. And second, modest spending adjustments in response to a downturn can preserve much of a portfolio's long-term spending power.

Paradise, T. and Kahler, J. R. (2020). *What to do with your next dollar: A quantitative framework*. Tech. rep. Vanguard.

Our financial lives can quickly become complex. Various tools and account types are available to help us save for an uncertain future, including employer plans, IRAs, 529s, and HSAs. But in addition to managing their assets, most households are also juggling various liabilities, such as a mortgage or student loans. How should investors prioritize cash flows on the household balance sheet? Should they invest more in a 401(k) or focus on paying off their mortgage early? The goal is to maximize long-term after-tax net worth-assets minus debts-while also meeting shorter-term goals and spending needs. This paper introduces a framework for evaluating the best use of your next discretionary dollar. Key to success is smart management of both sides of the balance sheet-your assets as well as your liabilities. Our framework has three components 1) Understanding your investment options: It is important to understand the costs, tax benefits, and rules of the investment opportunities available to you. We quantify the relative return potential of several types of accounts, showing the benefits of prioritizing tax-advantaged vehicles. 2) Debt: the other side of the balance sheet What are the relative benefits and constraints of prioritizing debt prepayment over further investment? We show that prioritizing the prepayment of high-interest debt can lead to better outcomes and explore the behavioral, liquidity, and risk tolerance considerations involved in the trade-off. 3) Balancing wealth maximization with other financial objectives Finally, we analyze the cash allocation decision and suggest the following steps. First, prioritize matched savings and high-cost debt. Next, make sure you have adequate emergency savings. Last, select the highest-returning opportunities appropriate for your goals.

Park, H. (2022). "Influence of risk tolerance on long-term investments: a Malliavin calculus approach." In: *Stochastics*, pp. 1–32.

This study investigates the influence of risk tolerance on the expected utility in the long run. We estimate the extent to which the expected utility of optimal portfolios is affected by small changes in the risk tolerance. For this purpose, we adopt the Malliavin calculus method and the Hansen-Scheinkman decomposition, through which the expected utility is expressed in terms of the eigenvalues and eigenfunctions of an operator. We conclude that the influence of risk aversion on the expected utility is determined by these eigenvalues and eigenfunctions in the long run.

Park, K., Wong, H. Y., and Yan, T. (2023). "Robust retirement and life insurance with inflation risk and model ambiguity." In: *Insurance: Mathematics and Economics* 110, pp. 1–30.

We study a robust consumption-investment problem with retirement and life insurance decisions for an agent who is concerned about inflation risk and model ambiguity. Assuming that an inflation-linked index bond and a stock are available in the market, this paper considers a comprehensive setup of ambiguity in the return, volatility, and correlation parameters in the joint dynamics of their market prices. With a finite planning horizon, the agent has a general utility function with different marginal utilities of consumption before and after retirement. Combining the classical dual approach and the G-stopping time theory, we derive the novel robust strategies using integral equation representations. We numerically and extensively investigate the effects of ambiguity from different sources on the robust decisions. While model ambiguity generally leads the ambiguity- and risk-averse agent to decrease the consumption rate, life insurance purchase, and investment demands, it also generates contrasting effects on robust retirement time and wealth level. Specifically, model ambiguity lowers the target wealth level to immediate retirement of a young agent but increases the retirement time of an older agent compared to the case of known parameters. A rich agent takes ambiguity more seriously than a poor agent in the sense of adjusting the strategies on a more significant scale. Our simulation and comparison study demonstrate the significance of addressing the ambiguity in volatility and correlation in addition to the ambiguity in return.

Park, S. (2023). "Optimal Annuitization with Markov Regime Switching Model." In: *SSRN e-Print*.

We develop a new dynamic model of annuitization in which a constant relative risk aversion utility maximizing individual receives non-traded labor income and has to decide on her allocation between a stock and a risk-free asset, as well as decide on the time when she starts her retirement annuity at one distinct point in time i.e., at around optimal retirement age. The distinct feature of the paper results from its consideration for a two-state regime-switching market environment. We analytically derive regime-dependent thresholds of wealth for annuitization and quantitatively identify the target annuitization wealth levels under the carefully chosen baseline parameter values. In particular, we find that the target annuitization wealth level is smaller in the Bull market than in the bear market, thus accounting for the reality that U.S. individuals opted for early annuitization in the Bull market as observed in the late 1990's.

Parker, F. J. (2016a). "Goal-Based Portfolio Optimization." In: *The Journal of Wealth Management* 19(3), pp. 22–30.

The article presents a goal-based portfolio optimization approach that is truly native to the goal-based environment. It begins by redefining risk as the probability of failing to attain a specified goal and redefining reward as the excess wealth over and above what is required by the goal. It then presents an optimization procedure that seeks to minimize goal failure and maximize excess return. In preliminary tests, it finds that this goal-based procedure lowers the probability of failing to achieve a specified goal while delivering higher excess wealth than the procedures currently available.

Parker, F. J. (2016b). "Portfolio Selection in a Goal-Based Setting." In: *The Journal of Wealth Management* 19(2), pp. 41–46.

Using different portfolio ratios, we illustrate the deficiencies of using only modern portfolio theory (MPT) metrics and assumptions when selecting portfolios in a goal-based setting. Through the lenses of the ratios, we show how MPT can choose the wrong, albeit efficient, portfolio to accomplish a goal. These facts illustrate the need for goal-based practitioners to factor in other variables, such as time to a goal and maximum loss thresholds, when managing a portfolio to a goal-oriented mandate.

Parker, F. J. (2020). "Allocation of wealth both within and across goals: a practitioner guide." In: *The Journal of Wealth Management* 23(1), pp. 8–21.

Although the goals-based investment literature has grown, there remain two unsolved problems. First, there is no cohesive theory for the allocation of wealth across goals. If, for example, a client wants to retire in thirty years, send a child to university in eight years, and buy a vacation home in four, how she should divvy her wealth across those goals has been an open question. Restating the same problem: The vesting of shorter-dated goals carries a loss of achievement probability for longer-dated ones. How much probability loss is acceptable? Second, mean-variance portfolios yield lower probabilities of goal achievement than goals-based portfolios. I demonstrate use of the goals-based method. Parker (2020) introduced a cohesive goals-based allocation model that solves these problems. The approach, however, carries some practical challenges that are addressed in this discussion. Finally, I discuss some possible implications of the approach on the structure of firms, the regulatory environment, and the industry as a whole.

Parker, F. J. (2021a). "A Goals-Based Theory of Utility." In: *Journal of Behavioral Finance* 22(1), pp. 10–25.

Theoretical frameworks to date have prescribed how an investor should allocate wealth within mental accounts. However, there is no fully cohesive solution to prescribe how an investor should rationally allocate resources across mental accounts. It is the aim of this discussion to fill that theoretical gap. We present a framework which can be used to rationally allocate resources both within and across mental accounts or goals. We then compare and contrast this method with mean-variance optimization and behavioral portfolio theory, showing that both are stochastically dominated by the goals-based utility framework. In further analysis of empirical validity, we discuss the Samuelson Paradox, Friedman-Savage puzzle, and probability weighting functions.

Parker, F. J. (2021b). "Achieving Goals While Making an Impact: Balancing Financial Goals with Impact Investing." In: *The Journal of Impact and ESG Investing* 1(3), pp. 27–38.

For investors who wish to engage in impact investing and who have specific goals to achieve, there exists the potential for a trade-off. When impact investments yield lower returns than nonimpact portfolios, how much return should an investor be willing to give up to incorporate it? Using recent advances in goals-based utility theory, this article explores an answer to that question and offers practical and concrete advice for advisors to individual investors and fiduciaries of trusts. Using the goals-based framework, the author shows how an investor's willingness to sacrifice return for an impact investing mandate changes in response to market and portfolio conditions.

Parker, F. J. (2023). "Goals-Based Investing and JWM: Where They Have Been, Where They Are Headed." In: *The Journal of Wealth Management* 26(1), pp. 29–38.
The Journal of Wealth Management has played a key role in supporting research that benefits individual investors. What began as concern over taxes, has grown into its own branch of portfolio theory, now known as goals-based investing. Though originally a behavioral adaptation to mean-variance portfolio theory, goals-based investing now boasts its own normative and axiomatic support, as well as making way for behavioral considerations. I review the story of these ideas over the last quarter-century and highlight the role JWM has played in supporting them.

Parker, F. J. (2021c). "Multi-Period Goals-Based Portfolio Optimization." In: *SSRN e-Print*.
Portfolio managers regularly have views on capital markets that span multiple periods. A portfolio manager, for example, may expect a recession with high probability over the coming period, followed by a recovery in the subsequent period. Currently, there are few methods for optimal portfolio allocation across these multiple periods that match how practitioners operate in the real world. Herein, we present a multi-period optimization method using a goals-based framework that allows practitioners to develop multi-period capital market expectations and optimize their portfolios across those periods.

Pashchenko, S. and Porapakkarm, P. (2019). "Accounting for Social Security Claiming Behavior." In: *SSRN e-Print*.
Why do most individuals claim Social Security benefits before the full retirement age? Claiming benefits early results in a substantial reduction in pension income, yet many people claim as early as possible (age 62) or soon thereafter. We argue that by answering this question, we can make two additional contributions to the literature. First, early claiming is equivalent to low demand for Social Security annuity, thus it offers a unique context for studying the well-known annuity puzzle. Since participation in Social Security is nearly universal, the low demand for this annuity cannot be explained away by market failures. Second, we show that claiming decisions are closely linked to the subjective rate of time preferences and thus can provide a new angle for the identification of this parameter. We provide a quantitative analysis of claiming decisions using a rich structural life-cycle model that matches many important features of the data.
We find that the claiming puzzle can be attributed to a combination of three factors:

- the discrepancy between individuals' subjective valuation of Social Security annuity and its implicit price,
- strong bequest motives,
- pre-annuitized wealth.

We show that if individuals were rewarded for delaying claiming not with additional annuity income but with equivalent (in present value terms) lump-sum payments, the fraction of early claimers would be significantly reduced. effective in inducing individuals to delay claiming.

Patel, S. and diBartolomeo, D. (2024). "Additive Determination of an Investor's Risk Tolerance in Asset Allocation." In: *The Journal of Investing* 33(5), pp. 34–54.
This study aims to improve asset allocation by combining traditional finance theory and behavioral finance. The authors construct efficient portfolios using modern portfolio theory (MPT) and identify an appropriate portfolio for a particular investor through a risk assessment score (RAS). The identified portfolio is not only optimal but also suitable based on the investor's risk profile. An RAS questionnaire is designed to produce an additive score in the range of 0 to 100 (i.e., a percentage) that maps linearly to the range of the tangency slope. Their framework is highly flexible and cost effective in terms of both capital market assumptions and the delineation of investor preferences. To emphasize the flexibility of the process, they provide an example questionnaire for retail investors in India, as well as an illustration of a three-asset-class efficient frontier. They also briefly discuss the ability of mapping responses to qualitative inquiries to the tangency slope in the allocation decisions of institutional investors (e.g., defined-benefit pension funds).

Peijnenburg, K., Nijman, T., and Werker, B. J. M. (2017). "Health Cost Risk: A Potential Solution To the Annuity Puzzle." In: *The Economic Journal* 127(603), pp. 1598–1625.
We find that health cost risk lowers optimal annuity demand at retirement. If medical expenses can be sizeable early in retirement, full annuitisation at retirement is no longer optimal because agents do not have enough time to build a liquid wealth buffer. Furthermore, large deviations from optimal annuitisation levels lead to small utility differences. Our results suggest that health cost risk can explain a large proportion of empirically observed annuity choices. Finally, allowing additional annuitisation after retirement results in welfare gains of at most 2.5% when facing health cost risk, and negligible gains without this risk.

Pellerin, M. (2021). "Investing for Retirement Income: A Comparison of Asset Allocations and Spending Strategies." In: *SSRN e-Print*.
We study the performance of different investment and spending strategies for retirement. Investment strategies include wealth-focused glide paths that combine equities with short-term, high-quality fixed income. We also consider an income-focused glide path that combines a moderate equity allocation at retirement and an inflation-protected bond portfolio that uses liability-driven investing. Spending rules include fixed spending (similar to the 4% rule), flexible spending, as well as nominal and real annuitization. We examine simulated lifetimes with either stochastic longevity or fixed longevity of 30 years in retirement.
We find that, for all spending strategies, an income-focused asset allocation delivers similar retirement income to the wealth-focused allocations we study while offering better protection against market, interest rate, and inflation risk. We also find that a glide path with an LDI portfolio offers a better tradeoff between income growth and income risk management. Finally, our results suggest that high equity exposure in retirement is an inadequate tool to manage longevity risk.
Peng, X. and Li, B. (2023). "Optimal investment, consumption and life insurance purchase with learning about return predictability." In: *Insurance: Mathematics and Economics* 113, pp. 70–95.
This paper studies the optimal investment, consumption and life insurance purchase problem for a wage earner under the condition that the return on the risky asset is predictable. We assume that the market price of risk is an affine function consisting of an observable and an unobservable factor that follow the O-U processes, while the evolution of the interest rate is described by the Vasicek model. The optimal investment, consumption and life insurance strategies and the corresponding value function are derived by adopting the filtering technique and the dynamical programming principle approach. In addition, for comparative analysis, the suboptimal strategies and the utility losses are presented when the wage earner ignores learning or the randomness of the interest rate. Finally, some numerical examples are presented to illustrate the results.
Pepperell, R., Greenwood, D., and Alsharman, M. (2020). "Death or Bust? The Risk with Post-Retirement Income Models." In: *SSRN e-Print*.
This paper reviews the key models used to forecast and evaluate sustainable withdrawal rates from pension portfolios and enumerates their key risks so that financial advisors are better able to judge model limitations and thus the risks associated with the recommendations they are making.
We focus on the model risk in isolation to determine the single model class or group of models best suited to retirement income planning from the wider set of candidates. We also look at the implementation issues surrounding these choices. We conclude by discussing the industry and policy making implications with reference to the UK.
Perry, J. and Randazzo, A. (2019). "Building Guaranteed Retirement Income Into Defined Contribution Plans: Life Cycle Retirement Planning with Income Insurance Products." In: *SSRN e-Print*.
Retirement outcomes would be improved if retirees had access to insurance products that linked guaranteed, inflation-adjusted income to life cycle investment portfolio strategies. Suitable retirement planning for individual, defined contribution plans must concurrently account for trade-offs during the pre-retirement wealth accumulation phase of life as well as income preferences for the decumulation phase (retirement income drawdown). This combined focus would take into account an individual's unique characteristics while suitably quantifying real asset return risk and longevity risk. To accomplish this complex goal, we develop a realistic simulation framework that illustrates how reasonably priced insurance products can be linked with defined contribution life cycle tools such that individuals can smooth their standard of living over time.
Pesenti, S. M. and Jaimungal, S. (2023). "Portfolio Optimization within a Wasserstein Ball." In: *SIAM Journal on Financial Mathematics* 14(4), pp. 1175–1214.
We study the problem of active portfolio management where an investor aims to outperform a benchmark strategy's risk profile while not deviating too far from it. Specifically, an investor considers alternative strategies whose terminal wealth lies within a Wasserstein ball surrounding a benchmark's terminal wealth-being distributionally close-and that have a specified dependence/copula tying state-by-state outcomes to it. The investor then chooses the alternative strategy that minimizes a distortion risk measure of terminal wealth. In a general complete market model, we prove that an optimal dynamic strategy exists and provide its characterization through the notion of isotonic projections. We further propose a simulation approach to calculate the optimal strategy's terminal wealth, making our approach applicable to a wide range of market models. Finally, we illustrate how investors with different copula and risk preferences invest and improve upon the benchmark using the Tail Value-at-Risk, inverse S-shaped, and lower- and upper-tail distortion risk measures as examples. We find

that investors' optimal terminal wealth distribution has larger probability masses in regions that reduce their risk measure relative to the benchmark while preserving the benchmark's structure.

Pfau, W. (2021). *Retirement Planning Guidebook: Navigating the Important Decisions for Retirement Success*. Retirement Researcher Media. 476 pp.

From the author: Writing this book has been a nine-year long passion project. This is the book I always intended to write, and the other three books in the Retirement Researcher Guide Series were spun-off along the way as my writing on some of these topics became so long that they constituted separate books on their own. Three of the chapters in this book summarize other books in this series, and ten of the chapters present new and original content.

Let me show with Retirement Planning Guidebook's Table of Contents how the other books in this series fit in:

1) Chapter 1: Retirement Income Styles and Decisions
2) Chapter 2: Retirement Risks
3) Chapter 3: Quantifying Goals and Assessing Preparedness
4) Chapter 4: Sustainable Spending from Investments [Summarizes How Much Can I Spend in Retirement?]
5) Chapter 5: Annuities and Risk Pooling [Summarizes Safety-First Retirement Planning]
6) Chapter 6: Social Security
7) Chapter 7: Medicare and Health Insurance
8) Chapter 8: Long-Term Care Planning
9) Chapter 9: Housing Decisions in Retirement [Summarizes Reverse Mortgages and adds new content]
10) Chapter 10: Tax Planning for Efficient Retirement Distributions
11) Chapter 11: Legacy and Incapacity Planning
12) Chapter 12: The Non-Financial Aspects of Retirement Success
13) Chapter 13: Putting it All Together

Retirement Planning Guidebook: Navigating the Important Decisions for Retirement Success.

Pfau, W. D. (2015a). "Choosing a Retirement Income Strategy: A New Evaluation Framework." In: *SSRN e-Print*. This article presents the initial stages of a new evaluation framework for choosing among retirement income strategies. The investigation includes eight retirement income strategies: constant inflation-adjusted withdrawal amounts, a constant withdrawal percentage of remaining assets, a withdrawal percentage based on remaining life expectancy, a more aggressive hybrid withdrawal percentage, inflation-adjusted and fixed single premium immediate annuities, a variable annuity with a guaranteed living withdrawal benefit rider, and a strategy which annuitizes the flooring level to meet basic needs and uses the hybrid withdrawal percentage for remaining assets. These eight strategies will be analyzed with six retirement outcome measures over a 30-year retirement period: the average amount whereby spending falls below the minimally acceptable level, the average spending amount, the remaining bequest at the end of the retirement period, the minimum spending amount for any year in the retirement period, a measure of whether spending increases or decreases over time defined as spending in the first year divided by spending in the 30th year, and the value of total spending after accounting for diminishing returns from increased spending for a client with somewhat inflexible spending needs. The model is applied to three client scenarios representing a cross-section of RIIA's client segmentation matrix. It is built using Monte Carlo simulations which reflect current market conditions, so that systematic withdrawals and guaranteed products share compatible underlying assumptions.

Pfau, W. D. (2015b). "How to Link Retirement Strategies to Sustainable-Spending Rates." In: *Advisor Perspectives*. -Sustainable-spending rates for retirement depend on many factors: asset- and product-allocation, market valuations at the start of retirement (particularly, current interest rates), the desired spending pattern over retirement, the degree of flexibility to adjust spending in response to market performance and the length of the client's planning horizon. Last week, my article introduced the Retirement Accumulation and Retirement Affordability indices, which help clients determine if they are retiring at a good time. In this article, I will present my new Retirement Dashboard. More specifically, I will explain the section of the dashboard on "The Cost of Retiring Today - Sustainable Spending Rates for Retirement," with these factors in mind. This analysis is conducted for a couple who both turn 65 on January 1, 2015. I estimated the sustainable-spending rates for two very different types of retirement-income strategies: those based on dedicated-income sources that can match client spending needs without exposing the client to market volatility, and those based on investment portfolios in which market volatility will play a much larger role in determining retirement sustainability.

Pfau, W. D. (2015c). "Making Sense Out of Variable Spending Strategies for Retirees." In: *SSRN e-Print*.
Variable spending strategies can be situated on a continuum between two extremes: spending a constant amount from the portfolio each year without regard for the remaining portfolio balance, and spending a fixed percentage of the remaining portfolio balance. Variable spending strategies seek compromise between these extremes by avoiding too many spending cuts while also protecting against the risk that spending must subsequently fall to uncomfortably low levels. Two basic categories for variable spending rules explored include decision rule methods and actuarial methods. Ten strategies will be compared using a consistent set of portfolio return and fee assumptions, and using an XYZ formula to calibrate initial spending: the client willingly accepts an X percent probability that spending falls below a threshold of Y dollars (in inflation-adjusted terms) by year Z of retirement. Presenting the distribution of spending and wealth outcomes for different strategies in which the initial spending rate is calibrated with the XYZ formula will allow for a more meaningful comparison of strategies. The article provides a framework for identifying appropriate spending strategies based on client preferences.

Pfau, W. D. (2018). "An Overview of Retirement Income Planning." In: *Journal of Financial Counseling and Planning* 29(1), pp. 114–120.
Retirement income planning has emerged as a distinct field within financial planning with the realization that risks change dramatically in retirement. The combination of longevity risk, increasing market risk triggered by taking distributions from assets, and spending shocks create new challenges. Wealth management has traditionally focused on accumulating assets without applying further thought to these differences happening after retirement. Retirees experience reduced capacity to bear financial market risk once they have retired. This calls for different thought processes from those typically included in traditional investment management. Risk pooling becomes an important retirement income tool combined with a traditional investment portfolio. Retirement income challenges and a framework for helping individuals develop more efficient and successful retirement income plans are discussed.

Pfau, W. D. (2019). "The four approaches to managing retirement income risk." In: *SSRN e-Print*.
Sequence-of-returns risk amplifies the impacts of investment volatility when taking distributions from a volatile investment portfolio in retirement. There are four techniques for managing this sequence risk: reduce the spending rate, adjust spending to portfolio performance, reduce portfolio volatility in the early retirement years, and draw from a buffer asset outside the portfolio to support spending when the portfolio is underperforming. We examine the peculiarities of sequence risk using an example with U.S. market returns which leads to wealth depletion after a thirty-year retirement. We show how minor spending adjustments can lead to dramatically different outcomes for remaining investment wealth at the end of the thirty-year retirement period. We provide examples for how each of these risk management techniques can be implemented in practice by integrating a wider range of tools to create the potential for greater retirement efficiency in terms of supporting the retirement financial goals with the available asset base. Examples include delaying Social Security, using annuities with lifetime spending protections, employing a rising equity glidepath in retirement, using a time segmentation or bucketing strategy, and creating a asset with a reverse mortgage or whole life insurance.

Pfau, W. D. (2022). "The Role and Inner Workings of Variable Annuities with Guaranteed Lifetime Withdrawal Benefits in Retirement." In: *The Journal of Retirement* 10(1), pp. 48–62.
A deferred variable annuity with a guaranteed lifetime withdrawal benefit is a retirement tool that can support protected lifetime income without having to annuitize assets until the contract value of the annuity depletes. This tool is controversial, with critics labeling it as too high cost and too complex for practical use. However, these annuities can have uses in a retirement income plan. For retirees expressing preferences for market growth, strategy commitment, asset liquidity, and longevity risk aversion, the variable annuity with a living benefit may serve as a suitable source for a protected lifetime income floor. Thus, this article aims to explain more about these potential uses and provide an understanding about how the guarantees for variable annuities work. Details such as the benefit base, contract value, rollup rate, step-up opportunities, and guaranteed withdrawal rates are explored to provide readers with a better sense of how to assess available variable annuity options and when they may be appropriate for a client.

Pfau, W. D. and Dokken, W. (2015). "Rethinking Retirement: Sustainable Withdrawal Rates for New Retirees in 2015." In: *Financial Advisor Magazine*.
Recent research that Dr. Wade Pfau and I have conducted finds that investors retiring as of January 1, 2015, who pursue a traditional 4% withdrawal rate from their savings to fund retirement have a more than 50% chance of outliving their savings. Last month in Financial Advisor, we detailed our research on safe withdrawal rates for retirees and the findings that could disrupt the current idea about what savings can safely be generated

during a person's retirement. Taking into account the person's investment decisions, the real safe withdrawal rate is much closer to 2% and could be lower. But we also believe that understanding investor behavior is key to understanding the withdrawal figure.

Pfau, W. D. and Parrish, S. (2023). "Which Social Security Claiming Strategy Generates the Highest Legacy Value?" In: *Journal of Financial Planning*.

Delaying Social Security can be framed as longevity insurance that helps to support the increasing costs associated with living a long life. It provides inflation-adjusted lifetime benefits for a retiree and a surviving spouse, and the monthly benefits will be 77 percent larger in inflation-adjusted terms for those who claim at 70 instead of at 62. But sometimes this insurance value is questioned, with the idea being that an early claiming decision will provide an opportunity for investments to grow and support better retirement outcomes. Simple extrapolations about suggesting that claiming Social Security early and investing the benefits to earn historical stock market returns misses several important points about retirement income: retirees may not invest this aggressively, and retirees must fund spending from assets and do not experience simple time-weighted returns. To generate the returns needed to beat the benefit of delaying Social Security, there would need to be a high tolerance for risk and an aggressive asset allocation. We find evidence using the historical data that it is uncommon for investment returns to beat the implied benefit of delaying Social Security for long-lived retirees even with relatively aggressive asset allocation strategies.

Pfau, W. D., Tomlinson, J., and Vernon, S. (2016a). "A Portfolio Approach to Retirement Income Security." In: *The Journal of Retirement* 4(2), pp. 11–22.

Workers with defined contribution retirement plans face a daunting array of choices as they enter retirement. They need to decide how to allocate their savings among asset classes and how much to withdraw from savings to support retirement. They also need to make choices about when to claim Social Security and whether or not to purchase annuities (and what type to purchase) to provide guaranteed lifetime income in addition to Social Security. In this article, we report on a recent collaboration between the Stanford Center on Longevity (SCL) and the Society of Actuaries (SOA) aimed at analyzing the various alternatives and finding optimal combinations of asset and product allocations, and strategies for withdrawing funds from savings. An important objective in this work has been finding solutions that are sufficiently straightforward to be implemented by retiring workers in employer-sponsored retirement plans or by utilizing low-cost advice options. Solutions that require substantial adviser involvement may be too expensive for typical middle-income workers. While the report focuses on solutions for defined contribution retirement plans, its framework can also be applied to out-of-plan solutions.

Pfau, W. D., Tomlinson, J., and Vernon, S. (2016b). *Optimizing Retirement Income Solutions in Defined Contribution Retirement Plans: A Framework for Building Retirement Income Portfolios*. Tech. rep. Stanford Center on Longevity.

This report helps plan sponsors, advisors, and retirees achieve these goals by demonstrating an analytical framework and criteria for helping them evaluate and compare a variety of possible retirement income solutions. Our goal is to further understanding of how to use various retirement income generators (RIGs) to meet specific retirement planning goals.

Pfau, W. D., Tomlinson, J., and Vernon, S. (2017). *Optimizing Retirement Income by Integrating Retirement Plans, IRAs, and Home Equity: A Framework for Evaluating Retirement Income Decisions*. Tech. rep. Stanford Center on Longevity.

In this report, the authors present a framework of analyses and methods that financial advisers, financial institutions, plan sponsors, and retirees can use to compare and assess strategies for developing lifetime retirement income. We recommend that financial advisers, plan sponsors, and financial institutions use disciplined analyses to demonstrate they are acting in the best interests of their clients who are approaching and entering retirement.

Phoa, W. (2024). "Further Applications of Mean-Variance Optimization." In: *The Journal of Portfolio Management* 50(8), pp. 250–259.

In 1952, Harry Markowitz published 'Portfolio Selection' and introduced mean-variance optimization. The importance of that article goes beyond mean-variance optimization as a practical tool for portfolio choice. From a technical point of view, mean-variance optimization continues to provide the foundation for building additional tools, used across a range of application areas not foreseen by Markowitz. We present two simple illustrations, addressing the problems of saving for retirement and the theoretical return premium on illiquid assets. In each case, an approach via mean-variance optimization provides useful insights and practical guidance alongside the existing literature, despite being technically much simpler.

Pittman, S. (2014). "Is a Portfolio Built to Produce Yield a Sensible Retirement Income Portfolio?" In: *Journal of Financial Planning* 27(8), pp. 52–60.
This paper focuses on the efficiency of allowing yield payments to dictate retirement spending, as well as the efficiency of tilting a portfolio away from an "efficient" portfolio for the sake of producing more yield.
Much of the literature suggests that a total return objective is a better choice than focusing on portfolio yield for retirees. This analysis furthers this literature by revisiting the cash flow and estate objectives of a retiree.
Results suggest that (1) using a yield portfolio forces a tradeoff between retirement spending and estate goals; (2) current yields may not be high enough to meet retirement income needs; (3) yields change over time, leading to inconsistent spending; (4) drawing down an efficient portfolio with the same risk as a portfolio emphasizing yield has higher expected principal accumulation; and (5) drawing down a portfolio is more tax-efficient than using yield to generate retirement income.

Pittman, S. (2015). "Use Your Client's Funded Ratio to Simplify and Improve Retirement Planning Decisions." In: *The Journal of Retirement* 3(2), pp. 93–104.
This article shows investment advisors how to compute a funded ratio and, subsequently, how to use this metric to assess retirement readiness for individual clients. The funded ratio allows the advisor to measure the relative size of assets and liabilities of an individual's retirement plan, providing an alternate perspective to that obtained from Monte Carlo simulation. Upon discussing how to calculate and use a funded ratio, a simple graphic is presented and described to show advisors how to quickly determine if their clients are on track to retire given their current retirement plan.

Platanakis, E., Sutcliffe, C. M., and Ye, X. (2021). "Horses for Courses: Mean-Variance for Asset Allocation and 1/N for Stock Selection." In: *European Journal of Operational Research* 288(1), pp. 302–317.
For various organizational reasons, large investors typically split their portfolio decision into two stages - asset allocation and stock selection. We hypothesise that mean-variance models are superior to equal weighting for asset allocation, while the reverse applies for stock selection, as estimation errors are less of a problem for mean-variance models when used for asset allocation than for stock selection. We confirm this hypothesis in separate analyses of US and international equities using four different types of mean-variance model (Bayes-Stein, Black-Litterman, Bayesian diffuse prior and Markowitz), a range of parameter settings, and a simulation analysis calibrated to US data.

Podkaminer, E., Tollette, W., and Siegel, L. (2022). "Protecting Portfolios Against Inflation." In: *The Journal of Investing* 31(2), pp. 23–44.
Inflation is a perennial threat to the real value of portfolios, even though current inflation rates are low. To protect portfolios against inflation, cash, inflation-indexed bonds, equities, real estate, and commodities are the usual candidates. We examine each, plus other assets and, importantly, various kinds of liabilities, to examine their historical and prospective responses to expected and unexpected inflation. Our article is integrative, bringing together ideas and data from many different sources in one place.

Pomante, U., Farina, V., and Cucurachi, P. A. (2024). "A "Plug-and-Play," Goal-Based, Dynamic Asset Allocation Model with Genetic Algorithms." In: *The Journal of Wealth Management* 26(4), pp. 33–48.
This work aims to reconcile theory and practice by proposing a dynamic model of asset allocation that allows, with the help of genetic algorithms, the creation of reasonable portfolios for investors chasing a variety of objectives over multiple time horizons. Our results are interesting and confirm the efficiency of the model, due to its good convergence towards the space of best solutions in a limited computation time. The methodology may be applied in the fields of: 1) wealth management, to define approaches that are useful for investors whose wealth must be invested to offset future liabilities; and 2) pension funds, to define-based on an individual's pension needs-personalized investment dynamics that are useful for meeting retirement needs.

Prendergast, J. R. (2021). "Replicating Maximum-Yield Annuities with US Treasury Funds." In: *The Journal of Fixed Income* 30(4), pp. 81–99.
Many individuals rely on annuity purchases to provide a steady stream of income during their post-retirement years. Life insurance companies provide a variety of annuity contracts, but they often come with sizable fees that are extremely detrimental to the purchaser in a low-interest-rate environment. This article develops a methodology that enables retail investors to structure annuities using commonly available US Treasury exchange-traded funds (ETFs) or mutual funds. Only historical price or NAV data and Treasury yield data are required to implement the strategy. The funds compose a dynamically managed portfolio requiring only an initial investment. Each period, the portfolio is rebalanced to match the modified duration of a reference annuity. This minimizes interest rate risk from parallel shifts in future yield curves. The weights are optimized to provide the highest yield

possible. The objective is for the portfolio to provide the required periodic annuity payments and have a zero balance at annuity maturity under a variety of interest rate scenarios, while providing the maximum possible yield. Scenario tests show that the strategy is effective under parallel yield curve shifts, but may have shortfalls for curve steepening and gains for curve flattening. Investors may choose to add to their initial investment to reduce the risk of shortfalls if the curve steepens. The article concludes with an implementation of the strategy using actual Treasury ETFs.

Prendergast, M. (2022). "Econometric Models for Computing Safe Withdrawal Rates." In: *OSF*.
This paper describes a methodology for estimating safe withdrawal rates during retirement that is based on a retiree's age, risk tolerance and investment strategy, and then provides results obtained from using that methodology. The estimates are generated by a three-step process. In the first step, Monte Carlo simulations of future inflation rates, 10-year treasury rates, corporate bond rates (AAA and BAA), the S&P 500 index values and S&P 500 dividend yields are performed. In the second step, portfolio composition and withdrawal rate combinations are evaluated against each of the Monte Carlo simulations in order to calculate portfolio longevity likelihood tables, which are tables that show the likelihood that a portfolio will survive a certain number of years for a given withdrawal rate. In the third and final step, portfolio longevity tables are compared with standard mortality tables in order to estimate the likelihood that the portfolio outlasts the retiree. This three-step approach was then applied using both a Monte Carlo random walk model and an ARIMA/GARCH model based upon over 100 years of monthly historical data. The end result was estimates of the likelihood of portfolio survival to mortality for over 500,000 retiree age/sex/portfolio/withdrawal rate combinations, each combination supported by at least 10,000 Monte Carlo economic simulation points per model. Both models are supported by 100 years of historical data. The first model is a random walk with step sizes determined by bootstrapping, and the second is an ARIMA/GARCH regression model. This data is analyzed to predict safe withdrawal rates and portfolio composition strategies appropriate for the late 2021 economic environment.

Pruser, J. (2021). "Forecasting US inflation using Markov dimension switching." In: *Journal of Forecasting* 40(3), pp. 481–499.
This study considers Bayesian variable selection in the Phillips curve context by using the Bernoulli approach of Korobilis (Journal of Applied Econometrics, 2013, 28(2), 204-230). The Bernoulli model, however, is unable to account for model change over time, which is important if the set of relevant predictors changes. To tackle this problem, this paper extends the Bernoulli model by introducing a novel modeling approach called Markov dimension switching (MDS). MDS allows the set of predictors to change over time. It turns out that only a small set of predictors is relevant and that the relevant predictors exhibit a sizable degree of time variation for which the Bernoulli approach is not able to account, stressing the importance and benefit of the MDS approach. In addition, this paper provides empirical evidence that allowing for changing predictors over time is crucial for forecasting inflation.

Qian, E. E. (2018). *Portfolio Rebalancing*. Chapman and Hall/CRC. 262 pp.
The goal of Portfolio Rebalancing is to provide mathematical and empirical analysis of the effects of portfolio rebalancing on portfolio returns and risks. The mathematical analysis answers the question of when and why fixed-weight portfolios might outperform buy-and-hold portfolios based on volatilities and returns. The empirical analysis, aided by mathematical insights, will examine the effects of portfolio rebalancing in capital markets for asset allocation portfolios and portfolios of stocks, bonds, and commodities.

Quaedvlieg, R. and Schotman, P. (2022). "Hedging Long-Term Liabilities." In: *Journal of Financial Econometrics*.
Pension funds and life insurers face interest rate risk arising from the duration mismatch of their assets and liabilities. With the aim of hedging long-term liabilities, we estimate variations of a Nelson-Siegel model using swap returns with maturities up to 50 years. We consider versions with three and five factors, as well as constant and time-varying factor loadings. We find that we need either five factors or time-varying factor loadings in the three-factor model to accommodate the long end of the yield curve. The resulting factor hedge portfolios perform poorly due to strong multicollinearity of the factor loadings in the long end, and are easily beaten by a robust, near Mean-Squared-Error- optimal, hedging strategy that concentrates its weight on the longest available liquid bond.

Quinn, J. F. and Cahill, K. E. (2018). "Challenges and Opportunities of Living and Working Longer." In: *How Persistent Low Returns Will Shape Saving and Retirement*. Oxford University Press.
This chapter describes the challenges and opportunities that older Americans face, with a focus on retirement income security and the role of continued work later in life. We first overview the new world of retirement income security including a discussion of how a low return environment (e.g. low interest rates) exacerbates existing

retirement income security challenges. We then document how older people have responded to the evolving retirement income landscape, especially when and how they exit the labor force, and we explore how continued work later in life can help mitigate some of the anticipated retirement security challenges. We then pose some important outstanding questions. The implications of societal aging depend in large part on how we harness or squander the labor resources of older individuals.

Rabitti, G. and Borgonovo, E. (2020). "Is mortality or interest rate the most important risk in annuity models? A comparison of sensitivity analysis methods." In: *Insurance: Mathematics and Economics* 95, pp. 48–58.
Demographic and financial factors are key risk-drivers for insurance companies and pension funds. This paper proposes a systematic investigation for deepening our understanding how these risk drivers affect the annuity cost. We employ local and global sensitivity methods. For local sensitivity, we derive closed form expressions for the differential importance measures of perturbed annuities and connect them to the entropy of the annuity cost. For global sensitivity, we compare variance-based, moment-independent sensitivity measures and Shapley effects. In particular, moment-independent sensitivity measures and Shapley effects are compared for the first time in the case of dependent risk factors. Our framework encompasses and extends several previous results on the sensitivity analysis of annuity models. From a methodological viewpoint, the techniques compared in this paper can support analysts in building annuity models and in verifying the impact of risk drivers in their models. Numerical results using the U.S. 1990 and the U.K. 1990-1994 mortality tables show that the demographic factor is the most important risk source in low-interest rate contexts. However, when uncertainty on the two risk sources is taken into account, the financial factor becomes the global key-driver of risk. Also, interactions among the two factors appear quantitatively significant.

Radovanov, B. and Marcikic, A. (2014). "Testing The Performance Of The Investment Portfolio Using Block Bootstrap Method." In: *Economic Themes* 52(2).
The aim of this paper is to create a stable model of investment portfolio optimization through a high degree of diversification and reduction of sudden changes in the allocation with monitoring of the dynamics of the impact factor. In this sense, there is bootstrap application procedure, which, without an excessive number of constraints involved in the optimization process provides solutions based on uncertain information. Thus defined, the optimization method has been patented by Michaud (1999) entitled re-sampled efficiency. Accordingly, this paper offers a comparison of the performance block bootstrap optimization models and traditional Markowitz's model inside and outside of the sample by applying the most frequently traded stocks on the BSE. The results show a better performance out of the sample and the presence of a larger number of shares forming the portfolio through bootstrap methodology. However, only through the traditional optimization process could be attained optimum according to the required limits. Such effects can be observed by comparing the limits of efficiency obtained through these optimization models. However, optimization-based methods bootstrap finds its place in reducing errors of assessment resulting from the limited sample size.

Ramaswamy, R. and Peng, J. (2024). "Tax Optimization of Active Strategies: Moving beyond Direct Indexing." In: *The Journal of Beta Investment Strategies* 15(3), pp. 20–33.
In this article, the authors discuss how to evaluate the efficacy of a tax-loss harvesting overlay for active strategies while describing key factors impacting loss harvesting. To do this, the authors introduce a concept called the tax efficiency ratio, which is somewhat analogous to an information ratio. To better understand the tax efficiency ratio, the authors examine strategies from different asset classes and characteristics. The tax efficiency ratio may help investors better understand the efficacy of a tax-loss harvesting overlay.

Raskie, S. (2017). "Navigating Uncertainties in Accumulation and Decumulation of Retirement Portfolios." In: *Journal of Economics and Public Finance* 3(3), p. 470.
Individuals face many challenges when developing a retirement plan. Hurdles arise at different stages of the retirement planning lifecycle. In the pre-retirement period, a significant obstacle arises when individuals must save for retirement to maximize their utility in retirement. The question of how much to save along with where to save impacts the amount the individual has in retirement. Post-retirement individuals must overcome the obstacle of how to optimally withdraw from their retirement savings to mitigate sequence risk and longevity risk to reduce the chance of portfolio failure. Individuals in post-retirement must develop strategies that not only mitigate these risks but also allow them to enjoy the retirement they have envisioned.

Rattray, S., Granger, N., Harvey, C. R., and Hemert, O. V. (2020). "Strategic Rebalancing." In: *The Journal of Portfolio Management* 46(6), pp. 10–31.
A mechanical rebalancing strategy, such as a monthly or quarterly reallocation toward fixed portfolio weights, is an active strategy. Winning asset classes are sold, and losers are bought. During crises, when markets are

often trending, this can lead to substantially larger drawdowns than a buy-and-hold strategy. This article shows that the negative convexity induced by rebalancing can be substantially mitigated, taking the popular 60-40 stock-bond portfolio as the use case. One alternative is an allocation to a trend-following strategy. The positive convexity of this overlay tends to counter the impact on drawdowns of the mechanical rebalancing strategy. The second alternative is called strategic rebalancing, which uses smart rebalancing timing based on trend-following signals, without a direct allocation to a trend-following strategy. For example, if the trend-following model suggests that stock markets are in a negative trend, rebalancing is delayed.

Rebonato, R. (2019). "A financially justifiable and practically implementable approach to coherent stress testing." In: *Quantitative Finance* 19(5), pp. 827–842.

We present an approach to stress testing that is both practically implementable and solidly rooted in well-established financial theory. We present our results in a Bayesian-net context, but the approach can be extended to different settings. We show (i) how the consistency and continuity conditions are satisfied; (ii) how the result of a scenario can be consistently cascaded from a small number of macrofinancial variables to the constituents of a granular portfolio; and (iii) how an approximate but robust estimate of the likelihood of a given scenario can be estimated. This is particularly important for regulatory and capital-adequacy applications.

Reichenstein, W. (2006). "After-Tax Asset Allocation." In: *Financial Analysts Journal* 62(4), pp. 14–19.

Several studies have found fundamental flaws in the traditional approach to managing individual investors' portfolios, including a failure to distinguish between 1USD of pretax funds in a 401(k) and 1USD of after-tax funds in either a taxable account or Roth IRA. This study recommends that an individual's asset values be converted to after-tax values and the asset allocation be based on the after-tax values. In general, within the target asset allocation, individuals should hold bonds and other assets subject to ordinary income tax rates in retirement accounts and hold stocks, especially passively managed stocks, in taxable accounts.

Reichenstein, W. (2020). "Saving in Roth Accounts and Making Roth Conversions before Retirement in Today's Low Tax Rates." In: *Journal of Financial Planning*.

There is a strong possibility that tax rates will rise in the next few years by more than the tax rate increases that are already scheduled to occur in 2026 according to the Tax Cuts and Jobs Act (TCJA). And, due to the taxation of Social Security benefits and income-based Medicare premiums, many retirees will have substantially higher marginal tax rates in retirement than their then-current tax brackets. As a result, many investors can expect to have higher marginal tax rates in their retirement years than in their pre-retirement years. This column recommends advice financial advisers can provide this year and perhaps the next few years to help clients-especially pre-retirement-age clients-take advantage of what my colleagues and I believe are today's temporarily relatively low tax rates. In the next few years, while tax rates are relatively low, pre-retirement-age investors should consider: (1) saving in tax-exempt accounts like a Roth 401(k) instead of tax-deferred accounts like a 401(k); and (2) making Roth conversions to fill today's relatively low tax brackets.

Reichenstein, W. (2021). "Minimizing the damage of tax torpedo." In: *Journal of Financial Planning* 34(9), pp. 62–65.

By coordinating a Social Security (SS) claiming strategy with a tax-efficient withdrawal strategy, financial planners can add substantial value to many clients' accounts by helping them reduce their lifetime income taxes. The case below illustrates how a planner can add substantial value to a mass-affluent client with $1 million of savings by helping them minimize the adverse effects of the "tax torpedo." The tax torpedo refers to the income range wherein each dollar of additional income causes another $0.50 or $0.85 of SS benefits to be taxable. Thus, taxable income increases by $1.50 or $1.85. So, the marginal tax rate (MTR) is 150 percent or 185 percent of the tax bracket, where MTR denotes the additional taxes paid on the next dollar of income. This column explains how financial planners can help many clients avoid, or at least minimize, the adverse effects of this tax torpedo.

Reichenstein, W. (2022a). "Optimal Social Security Claiming and Withdrawal Strategies in a Low-Return Environment." In: *Journal of Financial Planning* 35(6), pp. 58–62.

Today's projected real returns on bonds are historically very low. Assuming the historically average equity risk premium will continue going forward, projected real returns on stocks are also historically low. This study is designed to examine what impact these projected low real returns have (1) on retirees' optimal Social Security claiming strategy, and (2) on their most tax-efficient withdrawal strategy from their financial portfolio. To examine these issues, I utilized the www.incomesolver.com software. I consider a 65-year-old moderate-income retiree with $1 million in financial assets with a life expectancy of 90 years. As we shall see, today's low-return environment encourages retirees to delay the start of their Social Security benefits, even more so than if we had an average-return environment. Separately, in both low-return and average-return environments, the same general withdrawal strategy proves optimal, and it can add substantial value to this moderately wealthy retiree in both return environments.

Reichenstein, W. (2022b). "Social Security Redo Strategies for 2022." In: *Journal of Financial Planning* 35(3), pp. 58–63.

In Reichenstein and Meyer (2016), I coauthored an article that described then-available Social Security redo strategies. However, some of those strategies are no longer legal. Thus, in this column, I update that discussion by describing four redo strategies that are currently available for Social Security claiming decisions.

In the analysis of competing Social Security claiming strategies in this column, I compare the expected real (i.e., inflation-adjusted) lifetime benefits of competing claiming strategies. Suppose a single individual's monthly benefits are $2,000 the first year. Since COLAs are designed to keep the (pretax) purchasing power of Social Security benefits constant, the analyses assume a constant real benefit of $2,000 per month for this individual. We find that most clients appreciate this framework.

Reichenstein, W. (2023). "Why Multiphase Withdrawal Strategies Beat Single-Phase Withdrawal Strategies." In: *Journal of Financial Planning* 36(6).

In this column, I illustrate the advantage of adopting a multiphase withdrawal strategy in retirement that can exploit the varying patterns of marginal tax rates (MTRs) across years and the varying pattern of MTRs within many retirement years, where MTR denotes the additional taxes paid on the last dollar of income. As explained in Reichenstein (2019, 2020, 2021a, 2021b) and Reichenstein and Meyer (2020a, 2020b, 2021a, 2021b, 2022a, 2022b), the MTR within a year when Social Security benefits (SSB) are received is usually 150 percent or 185 percent of the tax bracket for a wide range of income. In addition, if the retired household will be on Medicare two years hence then, due to IRMAAs, there can be up to five huge spikes in MTRs. A tax-efficient withdrawal strategy in retirement requires that the household navigate this varying pattern of MTRs within these years, in which the MTR will rise and fall sharply as income rises. It also requires households to navigate the very different pattern of MTRs across years, such as in years before or after Social Security benefits begin. Some factors that may cause a new phase in the withdrawal strategy include the following: (1) SSB begin, (2) the Tax Cuts and Jobs Act (TCJA) expires and higher tax rates and MTRs return, and (3) required minimum distributions (RMDs) begin, which may affect both the taxable amount of SSB and the level of Medicare premiums two years hence. In this column, I present a case for a higher-income retired married couple that needs to consider how their withdrawal strategy in retirement will affect (1) the taxation of their SSB, (2) their lifetime Medicare premiums, and (3) their lifetime income taxes. However, the same lessons apply to single retirees.

Reichenstein, W. and Meyer, W. (2016a). "Redo Strategies: When Can You Redo a Prior Social Security Claiming Decision?" In: *Journal of Financial Planning*.

Many financial planners are aware that a client has the one-time right to withdraw his or her application for retirement benefits if made within 12 months after benefits began. This article explains two other redo strategies that planners may not be aware of. One strategy is the ability to suspend retirement benefits at full retirement age (FRA) or later in order to earn delayed retirement credits, and then to reinstate suspended benefits at a later date. The second strategy applies to disabled individuals where the disabled person begins disability benefits before attaining FRA then suspends disability benefits at FRA in order to earn delayed retirement credits. Prior to the passage of the Bipartisan Budget Act of 2015, other redo strategies were available, however changes embedded in this Act eliminated these other strategies as of April 29, 2016. This article discusses these recent rule changes, explaining that there are now three sets of rules, noting which set of rules apply to individuals, depending upon when they were born.

Reichenstein, W. and Meyer, W. (2016b). "Social security claiming strategies for widows and widowers." In: *The Journal of Retirement* 3(4), pp. 77–86.

This study presents rules governing survivor benefits and provides several cases to illustrate these rules. It then presents guidelines to help financial advisors quickly determine the best strategy for a widow or widower who is eligible for survivor benefits. The authors present separate advice for four groups of clients who differ in terms of the age of the surviving spouse. In addition, they present examples that illustrate the logic of the recommended strategies for each group of survivors. As seen in several examples, there are many times when a widow or widower should not simply take the larger of his or her own retirement benefit or his or her survivor benefits.

Reichenstein, W. and Meyer, W. (2017). "Valuing Roth Conversion and Recharacterization Options." In: *Journal of Financial Planning* 30(11), pp. 48–56.

This paper explains the options provided by the tax code for Roth conversions and recharacterizations.

Models are presented of the after-tax value of Roth conversion strategies to the strategy of retaining funds in a

TDA and withdrawing these funds in a later year.
Four reasonable cases illustrate how the Roth conversion and recharacterization options in the tax code could allow a taxpayer to increase the after-tax value of the funds converted to a Roth account by 5.56 percent, compared to the strategy of retaining these same funds in a tax-deferred account until later in retirement. One particular case, in which the Roth conversion increased the after-tax value by 62.79 percent, reflects the situation many taxpayers face. An example explains what a financial adviser should do to take advantage of the Roth conversion/recharacterization options available in the tax code.

Reichenstein, W. and Meyer, W. (2018). "Understanding the Tax Torpedo and Its Implications for Various Retirees." In: *Journal of Financial Planning* 31(7), pp. 38–45.
The "tax torpedo" refers to the sharp rise and then sharp fall in marginal tax rates caused by the taxation of Social Security benefits. For many middle-income households, this tax torpedo causes their marginal tax rate within a wide range of income to be 150 percent or 185 percent of their tax bracket. Many middle-income retirees can have a federal-alone marginal tax rate of 40.7 percent in 2018. If we return to the 2017 tax brackets in 2026, as called for in the Tax Cuts and Jobs Act, then many middle-income retirees will face a federal marginal tax rate of 46.25 percent. This paper provides income ranges in 2018 within which the tax torpedo likely will apply for singles, qualifying widow(er)s, married couples filing separately that lived apart for the entire year, and married couples filing jointly. Examples illustrate how singles and couples younger than age 70 may be able to reduce the adverse effects of the tax torpedo and thus extend the longevity of their financial portfolio by delaying Social Security benefits until 70. This paper also explains how singles and couples already subject to required minimum distributions may be able to reduce the adverse effects of the tax torpedo and extend the longevity of their financial portfolio using Roth conversions.

Reichenstein, W. and Meyer, W. (2019a). "Medicare and Tax Planning for Higher-Income Households." In: *The Journal of Wealth Management* 22(3), pp. 28–40.
This article explains how Medicare Part B and D premiums vary with a household Modified Adjusted Gross Income. Each time MAGI increases by 1 above several income threshold levels, a couple Medicare premiums two years hence can rise by more than 2,400. These premium hikes represent spikes in the marginal tax rate exceeding 240,000 %. It then explains how certain life-changing events can affect a household level of Medicare premiums. Next, it presents two cases that illustrate the value that financial advisors can add to clients portfolios by helping them coordinate a smart Social Security claiming strategy with a tax-efficient withdrawal strategy. These cases demonstrate that higher-income households should consider making Roth conversions from 2019 through 2025, when tax rates are scheduled to be temporarily lower, and before required minimum distributions begin. These Roth conversions may allow these households to greatly reduce the size of both their lifetime income taxes and their lifetime Medicare premiums. Finally, it explains the advantages for households with someone at least age 70.5 of making charitable contributions through Qualified Charitable Distributions from Individual Retirement Accounts.

Reichenstein, W. and Meyer, W. (2019b). "Optimizing Social Security Benefits Is Still Complicated." In: *The Journal of Retirement* 6(3), pp. 69–79.
The Bipartisan Budget Act of 2015 made significant changes to the rules affecting Social Security benefits. Some people think that these rule changes made it relatively easy to determine when someone should claim their Social Security benefits. In this article, the authors explain the rule changes and provide 10 reasons why making an optimal Social Security claiming decision is still complicated, even if we knew how long the single claimant or each partner in a married couple will live. Deciding when to claim Social Security benefits remains a complex decision.

Reichenstein, W. and Meyer, W. (2020). "Investment implications of the rising and falling pattern of marginal tax rates for retirees." In: *The Journal of Retirement* 8(1), pp. 53–64.
This article begins by explaining how income-based increases in Medicare premiums produce dramatic increases in marginal tax rates for retirees with relatively high levels of income. It then explains how the taxation of Social Security benefits causes most lower- and middle-income households to have marginal tax rates for a wide range of income after Social Security benefits begin that are either 150% or 185% of their tax bracket. By combining these two factors, the authors show that most retirees will have marginal tax rates in retirement that rise and fall frequently and sharply as their income increases. This article then explains how this rising and falling pattern of marginal tax rates should affect retired households withdrawal strategies from their financial portfolio, where withdrawal strategy is defined broadly to include Roth conversions.

Reichenstein, W. and Meyer, W. (2021a). "Advice for Married Couples When One Spouse Will Die Year(s) Before the Other Spouse." In: *Journal of Financial Planning* 34(1).

In a recent article, we explained how the taxation of Social Security benefits causes many retirees' marginal tax rate to be 150 percent or 185 percent of their tax bracket. We further explained that a $0.01 increase in 2018 income above any of three income-threshold levels can cause a married couple's 2020 Medicare annual premiums to increase by more than $2,540. Each of these increases in annual premiums represents a spike in marginal tax rates exceeding 254,000 percent. In this column, we discuss the implications of the taxation of Social Security benefits and income-based Medicare premiums for retired married couples filing jointly, when one spouse dies at least one calendar year before the other spouse. Thus, this column applies to most retired married couples.

Reichenstein, W. and Meyer, W. (2021b). "How Social Security Coordination Can Add Value to a Tax-Efficient Withdrawal Strategy." In: *The Journal of Retirement* 9(2), pp. 37–57.

This study describes a tax-efficient withdrawal strategy that can add substantial value to many clients of financial advisors with financial portfolios worth up to $2 million. In early retirement years, these households can delay the start of their Social Security benefits and make Roth conversions to fill relatively low tax brackets, which are generally also their marginal tax rates. Once Social Security benefits begin, they can make tax-free Roth withdrawals to minimize the amount of tax-deferred account (e.g., 401(k)) withdrawals that are taxed at 185% of their tax bracket, due to the taxation of Social Security benefits. With a series of cases, we show that a financial advisor can add substantial value to many of their clients' financial portfolios by recommending such a withdrawal strategy.

Reichenstein, W. and Meyer, W. (2021c). "Social Security Claiming Strategies for Singles and Their Implications for Couples." In: *Journal of Financial Planning* 34(5).

This report presents important insights related to the optimal Social Security claiming age for a single individual. The first section demonstrates that in order to maximize expected real (inflation-adjusted) lifetime Social Security benefits, it is seldom optimal for a single retiree to claim benefits in any month near either of two dates. The first date is 36 months prior to the full retirement age (FRA), while the second date is the FRA. The second section discusses four other factors that the single retiree should consider when selecting their claiming age. Each of these factors should encourage single individuals to delay claiming Social Security beyond this lifetime maximizing age. The report then explains when the lessons from the first section related to maximizing a single retiree's expected real lifetime benefits also apply to the claiming strategy for each partner of a married couple. As the reader will see, the lessons for single retirees apply to the claiming strategy for some married couples, but not for others.

Reichenstein, W. and Meyer, W. (2022). "Tax efficient withdrawal strategies for Five groups." In: *Journal of Financial Planning* 35(5).

We present tax-efficient withdrawal strategies for five groups, where "withdrawal" is interpreted broadly to include Roth conversions. The key to developing a tax-efficient withdrawal strategy is to manage a household's marginal tax rates, where marginal tax rate reflects the additional taxes paid on the next dollar of ordinary income. In preretirement years, the marginal tax rate (MTR) for a single individual or married couple filing jointly is usually the same as their tax bracket. However, as explained in the next two sections, their MTR in retirement years is often substantially higher than their tax bracket, due to the taxation of Social Security benefits and income-based Medicare premiums. As illustrated with an example, as a single retiree's income rises, her MTR rises from her current tax bracket to 150 percent of her tax bracket, then to 185 percent, before falling back to her initial tax bracket when her income exceeds the level where 85 percent of Social Security benefits are taxable, which is the maximum. Furthermore, as her income continues to rise, there are up to five huge spikes in her MTR due to income-based ncreases in annual Medicare premiums. Tax-efficient withdrawal strategies require navigating this rollercoaster pattern of MTRs. The final section presents tax-efficient withdrawal strategies for five groups: four groups of retirees and the fifth group consisting of taxpayers who are younger than their retirement years.

Reichenstein, W. and Meyer, W. (2023). "Why Multi-Phase Withdrawal Strategies Beat Proportional Withdrawal Strategies." In: *The Journal of Wealth Management* 26(2), pp. 35–51.

Fidelity and Schwab recommend proportional withdrawal strategies in retirement. Proportional withdrawal strategies ignore the rising and falling pattern of marginal tax rates due to the taxation of Social Security benefits and income-based Medicare premiums. We present several cases that demonstrate that multi-phase withdrawal strategies that navigate this marginal-tax-rates pattern can add substantial additional value to retirees' portfolios compared to proportional withdrawal strategies.

Reichenstein, W. R. (2007). "Note on Applying After-Tax Asset Allocation." In: *The Journal of Wealth Management* 10(2), pp. 94–97.
The author discusses an area within the field of after-tax asset allocation in which he and his co-author William Jennings disagree with the point proposed in the preceding article by Stephen Horan. But, more importantly, the author expresses the importance of those areas in which Horan, Jennings and he, as well as other scholars, agree. In particular, there appears to be wide agreement among scholars that, when calculating an individual's asset allocation, one should distinguish between pretax funds and after-tax funds. In short, taxes matter! He concludes with a crucial message: if it is indeed right to argue that funds in tax-deferred or tax-exempt portfolios should be accounted differently than those held through fully taxable structures, then the profession is currently miscalculating individuals' asset allocations, and the measurement errors can be substantial.

Reilly, C. and Byrne, A. (2018). "Investing for Retirement in a Low Returns Environment." In: *How Persistent Low Returns Will Shape Saving and Retirement*. Oxford University Press.
Low returns on financial assets and increasing longevity mean saving for retirement is becoming more challenging than it has been in the past. Generations retiring in the near term face increased longevity but have lived through periods with strong market returns boosting their assets, and many also have defined benefit plan entitlements. Younger generations, who also face increasing longevity, are unlikely to earn historical investment returns on their retirement portfolios, and few have traditional pensions. We model the likely outcomes for different cohorts under scenarios for savings behavior, investment returns, and longevity. While younger generations do face substantial challenges, we show that plausible courses of action involve increased contributions and delayed or partial retirement, which can provide reasonable income replacement rates in retirement. We map out the steps that the retirement industry (government, employers, and financial services providers) must take to support people in following these courses of action, such as providing more flexibility over social security.

Renshaw, A. (2025). "Sorting vs. Optimizing: Holding the Largest N Names Is Less Efficient than Minimizing Tracking Error Using Only N Names." In: *The Journal of Beta Investment Strategies* 16(1), pp. 50–67.
We document the superior risk-adjusted performance (IR) of a hedged portfolio constructed by minimizing tracking error to a parent benchmark while holding at most N names (the Optimal N portfolio) over a portfolio that cap-weights the largest N names in a parent benchmark (the Top N portfolio). Both portfolios have large size (market-cap) biases, so the absolute performance is consistent with the historical performance of the size factor: poor performance prior to 2015, excellent performance since 2015. Nevertheless, across different N (50, 100, 200), universes (large/mid cap, small cap; developed, emerging), and time periods (both when size is and is not working), the IRs for the Optimal N portfolio generally are better than those of the Top N portfolio. We show that the outperformance is not driven by other well-known, non-size factors such as value, quality, industries, or countries, as those active exposures are small. We show that the outperformance can be attributed to a Selection Effect for the non-Top N names using performance attribution. We suggest that the outperformance may be driven by market participants preferring the non-Top N names when they hedge their own portfolios, but we have no direct evidence to support this. The superiority of the Optimal N approach over the Top N approach is a useful insight that can be taken advantage of by portfolio managers.

Reus, L. and Prado, R. (2022). "Need to Meet Investment Goals? Track Synthetic Indexes with the SDDP Method." In: *Computational Economics* 60(1), pp. 47–69.
This work presents a novel application of the Stochastic Dual Dynamic Problem (SDDP) to large-scale asset allocation. We construct a model that delivers allocation policies based on how the portfolio performs with respect to user-defined (synthetic) indexes, and implement it in a SDDP open-source package. Based on US economic cycles and ETF data, we generate Markovian regime-dependent returns to solve an instance of multiple assets and 28 time periods. Results show our solution outperforms its benchmark, in both profitability and tracking error.

Reynolds, M. and Grisier, K. (2024). "Don't Let Inflation Spike Your Financial Plan: A Goals-Based Analysis of Purchasing Power Erosion." In: *The Journal of Wealth Management* 27(2), pp. 84–98.
Inflation rates and expectations have far-reaching effects, influencing asset returns, cash requirements to meet spending needs, and various other aspects relevant to achieving financial goals. To understand the comprehensive impact of inflation shocks, we conducted sensitivity analyses using Monte Carlo simulations for various investor archetypes in a goals-based framework. Results show that larger inflation shocks reduce the likelihood of meeting financial goals. However, several factors influence the degree of sensitivity to inflation shocks, including, but not limited to, asset allocation, time horizon, and net saver status.

Reznik, G., Alleva, B., Song, J., Sarney, M., and Olsen, A. (2020). *Analysis of Benefit Estimates Shown in the Social Security Statement*. Tech. rep. Social Security Administration.
The Social Security Statement is one of the Social Security Administration's (SSA's) most important ways to communicate with the public. Because a worker's complete lifetime earnings are unknown at the time his or her Statement is prepared, SSA estimates his or her future benefits by using the worker's historical earnings to project future earnings until retirement. Prior literature has examined the accuracy of benefit estimates shown in the Statement; however, there have not been efforts to test the accuracy of alternative methods of estimating retirement benefits. This paper documents a study by SSA's Office of Retirement and Disability Policy of the accuracy of the current Statement estimation method, the current method's assumption that 2 years of zero earnings predict no future earnings, and the accuracy of potential alternative methods of projecting earnings and benefits. Using administrative data from the Continuous Work History Sample (CWHS), the paper finds that the Statement's current estimation method performs as well as or better than any of the other methods tested. The paper concludes with recommendations for the Social Security Statement.

Rietz, R., Blumenschein, T., Crough, S., and Cohen, A. (2018). "Analyzing Retirement Withdrawal Strategies." In: *Preprints*.
An optimal withdrawal strategy beginning at age 65 provides a lifetime income from a portfolio, adjusted annually for inflation, while reducing the probability of living in financial ruin to an ac-ceptable level. This paper analyzes the probability of living in financial ruin, potentially for multiple years, rather than just the event of ruin. A stochastic Excel model was developed to simulate the effect of varying investment returns on a portfolio with two asset classes; large company stocks and long-term government bonds. A stochastic model is also applied to retiree mortality. The following variables were analyzed to determine their relative impact on withdrawal strategies: Withdrawing a constant percentage of the portfolio, Gender, Initial asset allocation, Asset allocation rebalancing methods, and Low investment return environments. For both genders and most withdrawal rates, an approximately equal initial asset allocation of stocks and bonds, combined with a level rebalancing function, provided the lowest probability of living in financial ruin. Because each investment return followed its own probability distribution function, some retirees experienced financial ruin even in the most conservative simulations.

Rietz, R. J., Blumenschein, T., Crough, S., Cohen, A., and Coleman, J. (2020). "A Simulation for Minimizing both the Probability and the Length of Financial Ruin in Retirement." In: *SSRN e-Print*.
Retirees worry about depleting their portfolio, but a greater concern could be how long they might live without income from that portfolio. A retiree may accept a 4% probability of portfolio depletion, but object to the possibility of living five or six years afterwards without that income. Thus, a retirement portfolio's exhaustion is not a terminal event, but rather is only the beginning of a retiree's living in poverty. This analysis simulates not only the event of financial ruin, but also its duration during the retiree's remaining lifetime. This paper analyzes withdrawing a constant percentage of the portfolio, gender, initial asset allocation, asset allocation rebalancing methods, and low investment return environments to determine their relative impact on withdrawal strategies.

Rob, B. (2023). "Time to Retire: The 4% Withdrawal Rule." In: *The Journal of Investing* 32(4), pp. 91–111.
The so-called "4% rule" is perhaps the single most commonly referenced retirement withdrawal approach. Most discussion of this rule centers on the appropriate percentage to use, 4%, 3%, or 5%. My objective is to suggest a superior retirement rule. Superior in the sense that it will deliver a higher standard of living during one's retirement years. I start with the observation that past retirement withdrawal analysis suffers from two key deficiencies, that is, a reliance on past US stock and bond market returns (ex-post cherry picking the best performing market) and an assumption that asset class returns can be modeled with independent and identically distributed random variables, serving to mask their important time series properties. This article repairs these two deficiencies. A new retirement distribution rule is proposed. One that distributes/liquidates an ever-increasing proportion of the retiree's then-current portfolio as they age, but subject to a minimum monthly distribution expressed in dollar terms. This approach provides simplicity, ease of execution, and transparency. It also results in a standard of living for the retiree as expressed by the median and average monthly distributions that are 101.3% and 174.2%, respectively, above those delivered by the best possible constant-dollar withdrawal rule. This pleasing result occurs for two primary reasons. First, this new rule realizes significant benefit from selling more/less shares when markets are high/low (which is the exact opposite of what transpires with all constant-dollar rules). Or to use other words, it delivers a measurably higher IRR despite utilizing an identical asset mix. Second, the new rule effectively consumes the retiree's entire portfolio over the retirement life, leaving no unused residual balance (again, the exact opposite of all constant-dollar rules).

Rogalla, R. (2021). "Optimal Portfolio Choice in Retirement With Participating Life Annuities." In: *North American Actuarial Journal* 25(sup1), S182–S195.
This article derives optimal consumption, investment, and annuitization patterns for retired households that have access to German-style participating payout life annuities (PLAs), allowing for capital market risks as well as idiosyncratic and systematic longevity risks. PLAs provide guaranteed minimum benefits in combination with participation in insurers' surpluses. Minimum benefits are calculated based on conservative assumptions regarding capital market and mortality developments, while surpluses distributed to annuitants bridge the gap between the insurers' actual investment and mortality experiences and the projections used in pricing. Through the participation scheme, systematic longevity risk is shared between insurers and annuitants, as unanticipated longevity shocks result in benefit adjustments via the surplus mechanism. We show that the retiree draws substantial utility from access to PLAs, equivalent to 20% of initial wealth in the presence of systematic longevity risk. We also find that stochasticity in mortality rates only has minor impact on the appeal of PLAs to the retiree. Even if the interest rate guarantee is reduced to zero in adverse capital market environments, PLAs prove to provide substantial utility for retirees. Overall, the participating life annuity design produces substantial welfare gains over a no-annuity world, while being an effective setup that helps providers manage long-term risks that are difficult to hedge otherwise, such as systematic longevity risks.
Roncalli, T. (2019). "How Machine Learning Can Improve Portfolio Allocation of Robo-Advisors." In: *SwissQuant Conference*.
solving portfolio optimization with machine learning algorithms.
Roncalli, T. (2021). "Advanced Course in Asset Management." In: *SSRN e-Print*.
These presentation slides have been written for the Advanced Course in Asset Management (theory and applications) given at the University of Paris-Saclay. They contain 15 tutorial exercises and 5 main lectures:

1) Portfolio Optimization
2) Risk Budgeting
3) Smart Beta, Factor Investing and Alternative Risk Premia
4) Green and Sustainable Finance, ESG Investing and Climate Risk
5) Machine Learning in Asset Management

The Table of contents is the following:
Part 1. Portfolio Optimization 1. Theory of portfolio optimization 1.a. The Markowitz framework 1.b. Capital asset pricing model (CAPM) 1.c. Portfolio optimization in the presence of a benchmark 1.d. Black-Litterman model 2. Practice of portfolio optimization 2.a. Covariance matrix 2.b. Expected returns 2.c. Regularization of optimized portfolios 2.d. Adding constraints 3. Tutorial exercises 3.a. Variations on the efficient frontier 3.b. Beta coefficient 3.c. Black-Litterman model
Part 2. Risk Budgeting 1. The ERC portfolio 1.a. Definition 1.b. Special cases 1.c. Properties 1.d. Numerical solution 2. Extensions to risk budgeting portfolios 2.a. Definition of RB portfolios 2.b. Properties of RB portfolios 2.c. Diversification measures 2.d. Using risk factors instead of assets 3. Risk budgeting, risk premia and the risk parity strategy 3.a. Diversified funds 3.b. Risk premium 3.c. Risk parity strategies 3.d. Performance budgeting portfolios 4. Tutorial exercises 4.a. Variation on the ERC portfolio 4.b. Weight concentration of a portfolio 4.c. The optimization problem of the ERC portfolio 4.d. Risk parity funds
Part 3. Smart Beta, Factor Investing and Alternative Risk Premia 1. Risk-based indexation 1.a. Capitalization-weighted indexation 1.b. Risk-based portfolios 1.c. Comparison of the four risk-based portfolios 1.d. The case of bonds 2. Factor investing 2.a. Factor investing in equities 2.b. How many risk factors? 2.c. Construction of risk factors 2.d. Risk factors in other asset classes 3. Alternative risk premia 3.a. Definition 3.b. Carry, value, momentum and liquidity 3.c. Portfolio allocation with ARP 4. Tutorial exercises 4.a. Equally-weighted portfolio 4.b. Most diversified portfolio 4.c. Computation of risk-based portfolios 4.d. Building a carry trade exposure
Part 4. Green and Sustainable Finance, ESG Investing and Climate Risk 1. ESG investing 1.a. Introduction to sustainable finance 1.b. ESG scoring 1.c. Performance in the stock market 1.d. Performance in the corporate bond market 2. Climate risk 2.a. Introduction to climate risk 2.b. Climate risk modeling 2.c. Regulation of climate risk 2.d. Portfolio management with climate risk 3. Sustainable financing products 3.a. SRI Investment funds 3.b. Green bonds 3.c. Social bonds 3.d. Other sustainability-linked strategies 4. Impact investing 4.a. Definition 4.b. Sustainable development goals (SDG) 4.c. Voting policy, shareholder activism and engagement 4.d. The challenge of reporting 5. Tutorial exercises 5.a. Probability distribution of an ESG score 5.b. Enhanced ESG score and tracking error control

Part 5. Machine Learning in Asset Management 1. Portfolio optimization 1.a. Standard optimization algorithms 1.b. Machine learning optimization algorithms 1.c. Application to portfolio allocation 2. Pattern learning and self-automated strategies 3. Market generators 4. Tutorial exercises 4.a. Portfolio optimization with CCD and ADMM algorithms 4.b. Regularized portfolio optimization.

Rook, C. J. (2015a). "Minimizing the Probability of Ruin in Retirement." In: *arXiv e-Print*.

Retirees who exhaust their savings while still alive are said to experience financial ruin. These savings are typically grown during the accumulation phase then spent during the retirement decumulation phase. Extensive research into invest-and-harvest decumulation strategies has been conducted, but recommendations differ markedly. This has likely been a source of concern and confusion for the retiree. Our goal is to find what has heretofore been elusive, namely an optimal decumulation strategy. Optimality implies that no alternate strategy exists or can be constructed that delivers a lower probability of ruin, given a fixed inflation-adjusted withdrawal rate.

Rook, C. J. (2015b). "Optimal Equity Glidepaths in Retirement." In: *SSRN e-Print*.

Dynamic retirement glidepaths evolve over time based on some measure such as the retiree's funded status or current market valuations. Conversely, static glidepaths are fixed at a starting point and selected under the assumption that they will not change. In practice, new static glidepaths may be derived periodically making them more flexible. The optimal static retirement glidepath would be the one that performs better than all others with respect to some metric. When systematic withdrawals are made from a retirement portfolio, glidepaths are often assessed via the probability of ruin (or success). Our goal here is to derive the optimal static glidepath with respect to this metric. It is a result new to the literature and the shape will be of special interest to retirees, financial advisors, retirement researchers, and target-date fund providers.

Rook, C. J. (2017a). "Multivariate Density Modeling for Retirement Finance." In: *SSRN e-Print*.

Prior to the financial crisis mortgage securitization models increased in sophistication as did products built to insure against losses. Layers of complexity formed upon a foundation that could not support it and as the foundation crumbled the housing market followed. That foundation was the Gaussian copula which failed to correctly model failure-time correlations of derivative securities in duress. In retirement, surveys suggest the greatest fear is running out of money and as retirement decumulation models become increasingly sophisticated, large financial firms and robo-advisors may guarantee their success. Similar to an investment bank failure the event of retirement ruin is driven by outliers and correlations in times of stress. It would be desirable to have a foundation able to support the increased complexity before it forms however the industry currently relies upon similar Gaussian (or lognormal) dependence structures. We propose a multivariate density model having fixed marginals that is tractable and fits data which are skewed, heavy-tailed, multimodal, i.e., of arbitrary complexity allowing for a rich correlation structure. It is also ideal for stress-testing a retirement plan by fitting historical data seeded with black swan events. A preliminary section reviews all concepts before they are used and fully documented C/C source code is attached making the research self-contained. Lastly, we take the opportunity to challenge existing retirement finance dogma and also review some recent criticisms of retirement ruin probabilities and their suggested replacement metrics.

Rook, C. J. (2017b). "Multivariate Density Modeling for Retirement Finance." In: *arXiv e-Print*.

Prior to the financial crisis mortgage securitization models increased in sophistication as did products built to insure against losses. Layers of complexity formed upon a foundation that could not support it and as the foundation crumbled the housing market followed. That foundation was the Gaussian copula which failed to correctly model failure-time correlations of derivative securities in duress. In retirement, surveys suggest the greatest fear is running out of money and as retirement decumulation models become increasingly sophisticated, large financial firms and robo-advisors may guarantee their success. Similar to an investment bank failure the event of retirement ruin is driven by outliers and correlations in times of stress. It would be desirable to have a foundation able to support the increased complexity before it forms however the industry currently relies upon similar Gaussian (or lognormal) dependence structures. We propose a multivariate density model having fixed marginals that is tractable and fits data which are skewed, heavy-tailed, multimodal, i.e., of arbitrary complexity allowing for a rich correlation structure. It is also ideal for stress-testing a retirement plan by fitting historical data seeded with black swan events. A preliminary section reviews all concepts before they are used and fully documented C/C++ source code is attached making the research self-contained. Lastly, we take the opportunity to challenge existing retirement finance dogma and also review some recent criticisms of retirement ruin probabilities and their suggested replacement metrics.

Root, E., Mallett, C., Zwerling, N., and Toale, T. (2013). "Strategic Liability Management: Building a Glide Path

Strategy to Manage Interest Rate and Longevity Risks." In: *Institutional Investor Guides: Special Issues*, pp. 56–60.
Glide Path Strategy (GPS) may be thought of as a term for a transition from a traditional market-based asset allocation and returns model to one in which a defined-benefit (DB) plan's benefit liabilities are the primary reference point for asset allocations and returns assessment as a DB plan's funded status improves. This article discusses how the GPS provides a durable framework to support holistic and strategic DB plan management as the sponsor's understanding of plan liabilities deepens, how a GPS can serve to facilitate managing risks that are important today, such as interest rate risk, and those that are expected to become more apparent in the future, such as longevity risk, and the ways in which major risk reduction methods work in building and sustaining a GPS over time.

Rosenthal, D. (2018). "Joint effect of random years of longevity and mean reversion in equity returns on the safe withdrawal rate in retirement." In: *SSRN e-Print*.
Using historical data on inflation-adjusted total equity returns (price change plus dividend) from the S&P 500 from 1926 to 2017, this paper develops a simulation-based model to determine the safe, inflation-adjusted withdrawal rate from a portfolio of assets. The model, named the Realistic Retirement Simulator (RRS), improves upon other simulation models by directly addressing two factors that significantly affect the safe withdrawal rate: (1) uncertainty about the number of years of retirement, i.e., at what age will the retiree die; and (2) mean reversion in equity returns. RRS models the number of years in retirement as a random factor based on the Social Security Administration 2015 Actuarial Life Table. The mean-reverting stock return model within RRS is statically calibrated to the 1926 to 2017 S&P 500 data. With these two key factors addressed and assuming future equity returns follow the historical record, RRS shows that a 65-year-old male retiree can withdraw 6% of the starting portfolio balance each year from a 100% stock portfolio with a 90% success rate; a 4% withdrawal rate is 99% successful. A simulation model that does not address these two key factors the author is not aware of single model that addresses both factors that a 4% withdrawal rate results in a 90% success rate for a retirement lasting 30 years. At the 90% success level, about half of the increase from 4% to 6% comes from treating the length of the retirement as a random factor and other half comes from the mean-reverting model. Many scenarios are run to show how the success rate changes when RRS input assumptions are changed, e.g., age of retiree or stock/bond mix of retiree portfolio. Of particular importance is the assumption that future equity returns will repeat the historical record. If the future, long-run trend for equity returns is a 4% compound annual growth rate (CAGR) instead of the 6.93% observed in the historical data, RRS shows that a withdrawal rate of 4% has a success rate of 95%. Regardless of the assumption about future equity returns, directly modeling uncertainty in the length of retirement and mean reversion in equity returns results in more accurate and higher estimates of the safe withdrawal rates compared to models that do not directly address these factors.

Rossi, A. G. and Utkus, S. P. (2021). "The Needs and Wants in Financial Advice: Human versus Robo-advising." In: *SSRN e-Print*.
We use a broad survey to elicit investor needs and their satisfaction in the context of financial advice. We provide evidence that traditionally-advised individuals do not hire financial advisors mainly to maximize portfolio returns. They instead hire financial advisors to satisfy a broader set of needs. These needs include acquiring "peace of mind," having access to the opinions of an expert and delegating financial decisions. We also show the majority of investors do not know how much they pay for financial advice, but the miscalculation of the costs of financial advice are unlikely to drive the decision to hire financial advisors. Robo-advised investors are more interested in the financial performance of their portfolio. They also view robo-advising as an empowering tool that can contribute to their own self-improvement. Even robo-advised investors, however, value greatly the possibility to reach out and interact with humans. Our results provide novel evidence on why individual investors choose to hire financial advisors. They also inform on the optimal design of robo-advisers.

Roux, E.-M. and de Villiers, J. (2020). "A simplified approach to estimate the sustainable lifestyle level for retirement planning." In: *Investment Analysts Journal* 49, pp. 232–242.
In this article we offer a simplified version of the alternative retirement planning model we originally proposed (De Villiers and Roux, 2019). Our method focuses on determining the sustainable lifestyle level (SLL) that an individual can currently afford while still saving enough towards retirement to sustain this lifestyle level up to retirement and beyond. The model is simplified by assuming that the real rate of return on retirement savings before retirement will be the same as the withdrawal rate of income from the accumulated savings during retirement. This method yields a much simpler SLL relationship in that it is more generally applicable albeit

possibly less accurate. This approach should improve communication of the extent of the retirement savings challenge, possibly leading to better savings outcomes.

Roy, K. and Kim-Steiner, Y. (2019). *Three retirement spending surprises*. Tech. rep. J.P. Morgan Asset Management.
We analyzed the spending patterns of more than 5 million households to evaluate real-world retirement behaviors. Our research uncovered three surprising patterns: a lifetime spending curve, a retirement spending surge and a high degree of retirement spending volatility.
These new insights indicate helping investors make the most of their retirement assets may require a more dynamic approach to withdrawals than the static rules of the past.

Rudys, V. (2023). "How does retirement affect optimal life cycle portfolio allocation between stocks and bonds?" In: *Journal of Asset Management*.
Portfolio allocation decisions over the life cycle depend, among many factors, on the retirement income as well as risks and choices faced in retirement. However, due to computational complexity, many retirement factors are typically assumed away. By building on the standard life cycle investment-consumption model that includes a more realistic progressive retirement income program, I discuss how each aspect of retirement income affects optimal portfolio allocation over the life cycle. I find that all investors across all scenarios maintain high levels of stocks in their portfolios at a young age. However, investors who face low net replacement rates, risk of forced retirement, or retirement income uncertainty hedge these risks by accumulating higher private savings and reducing risky portfolio shares at an earlier age. In a realistic setting with early forced retirement risk and endogenous retirement timing, optimal equity portfolio share temporarily increases once the investor becomes eligible for retirement. Retirement income's dependency on workers' lifetime labor earnings is not, however, an important factor in portfolio allocation over the life cycle.

Rundle, J. (2018). "A Social Security Purchase: Is Delaying Social Security More Effective than Purchasing a Deferred Income Annuity?" In: *The Journal of Retirement* 6(1), pp. 8–21.
This article analyzes a retirement strategy of delaying Social Security commencement past Social Security retirement age to effectively increase annuity income.
Using Monte Carlo simulations for returns on a pre-tax retirement portfolio and simplified tax, marriage, and Social Security benefit assumptions, I compare three retirement strategies:

1) following the 4% rule with retirement at Social Security normal commencement age (66),
2) delayed commencement to age 70 with forgone Social Security income replaced by portfolio withdrawals
3) following the 4% rule with normal retirement age commencement and purchase of deferred life annuity of equivalent expected actuarial value to the increase in Social Security that would occur if commencement were delayed to age 70.

Results for the scenarios analyzed generally support delaying Social Security as a preferred method to annuitize assets over the direct purchase of a deferred annuity, particularly under the married scenario. Compared with Strategy 1, delaying Social Security increases portfolio fail rates by year 30 (age 96) by 10 percentage points (to about 20%) and results in reduced portfolio values across outcomes.
The trade-off is increased inflation-protected income under portfolio failure scenarios by 32% of the primary benefit.

Ruthbah, U. (2020). "The Retirement Puzzle." In: *SSRN e-Print*.
The present-day retirees may not be as well off as they expect to be during their retired life. Given the current state of the world - higher life expectancy, close to zero real interest rate and the economic turmoil caused by the Covid-19 pandemic, a superannuation fund as large as 545,000 USD may not be enough to support a comfortable lifestyle. The rules of means-tested age pension provide incentives for a broad spectrum of people to deplete their superannuation in the early years of retirement, with significant negative consequences for government finances. Spending at retirement does not increase proportionally with assets. For some range of superannuation balances, the net present value of spending could increase more than the increase in assets, as the net present value of the age pension does not decline at par with increases in assets. In other asset levels, the opposite happens. The distributional impact of the age pension is unequal across different asset levels and is regressive to some extent.

Ryack, K. (2016). "Incorporating Financial Risk Tolerance Research into the Financial Planning Process." In: *Journal of Financial Planning* 29(10), pp. 54–61.
An individual's financial risk tolerance ultimately drives his or her financial decisions and is therefore of critical importance in personal financial planning. Two psychometrically valid measures of an individual's financial risk tolerance are highlighted, and academic research findings on the nine determinants of an individual's financial

risk tolerance are discussed. For example, financial risk tolerance tended to increase with the level of the client's education, income and wealth, and financial knowledge and experience. Financial risk tolerance also appeared to vary with gender, age, and market conditions. A framework was developed to organize the nine discrete factors that influence a client's financial risk tolerance on the basis of foreseeability and manageability to make it easier for financial planners to integrate financial risk tolerance into the financial planning process.

Rzepczynski, M. S., Brunner, A., and Wild, P. (2024). ""I Have Never Seen a Bad Backtest": Modeling Reality in Quantitative Investing." In: *The Journal of Investing* 33(1), pp. 142–155.
Backtests often are discounted based on the expected failure between live and simulated (theoretical) trading. These failures are associated with the modeling assumptions that address market conditions. Deviations between live trading and strategy model performance will be biased downward by poor backtesting methodologies that do not account for model uncertainty and trading cost assumptions. However, model reality can be improved through explicitly accounting for all trading costs, endogenizing costs as part of the overall backtesting methodology, and forming systematic backtesting methodologies that account for sources of randomness. This article presents a framework for assessing backtested performance that can be employed by both investors and managers as a checklist for improving model reality, reducing trading cost biases, and accounting for uncertainty.

Sahm, C. (2017). "How Much Does Risk Tolerance Change?" In: *SSRN e-Print*.
Stability of preferences is central to how economists study behavior. This paper uses panel data on hypothetical gambles over lifetime income in the Health and Retirement Study to quantify changes in risk tolerance over time and differences across individuals. The maximum-likelihood estimation of a correlated random effects model utilizes information from 12,000 respondents in the 1992-2002 HRS. The results are consistent with constant relative risk aversion and career selection based on preferences. While risk tolerance changes with age and macroeconomic conditions, persistent differences across individuals account for 73% of the systematic variation.

Sakurai, Y., Wakabayashi, D., and Ishizaki, F. (2024). "Formulations to select assets for constructing sparse index tracking portfolios." In: *The Journal of Investment Strategies* 13(2), pp. 1–15.
In this paper, we study asset selection methods to construct a sparse index tracking portfolio. For its advantage over full replication portfolio, the concept of sparse index tracking portfolio has significant attention in the field of finance and investment management. We propose useful formulations to select assets for sparse index tracking portfolio. Our formulations are described as combinatorial optimization problems, and they can yield various asset selection methods, including some existing methods, by adjusting the values of parameters. As a result, the proposed formulations can provide a well-balanced asset selection to create successful sparse index tracking portfolios. We also provide numerical examples to compare the tracking performance of resulting sparse index tracking portfolios.

Salampasis, D., Mention, A.-L., and Kaiser, A. O. (2019). "Wealth Management in Times of Robo: Towards Hybrid Human-Machine Interactions." In: *SSRN e-Print*.
Breakthrough technological developments are grounded to disruptive business models that are leading foundational changes within the financial services industry. In this frame of reference, wealth management is experiencing transformational forces reaching the fundamental core function of the business, redefining existing narratives and practices. Advisory services in management wealth are deeply embedded within a human-to-human interaction and robo-advising is going to augment the existing value chain leading to the creation of a hybrid advisory model for complex investment portfolios. Therefore, the fundamental question is not about 'either or' but 'together for'.

Sammon, M. and Shim, J. J. (2025). "Index Rebalancing and Stock Market Composition: Do Index Funds Incur Adverse Selection Costs?" In: *SSRN e-Print*.
We find that index funds incur adverse selection costs from changes in the composition of the stock market. This is because indices rebalance in response to composition changes, both on the extensive margin (IPOs/delistings or additions/deletions) and intensive margin (issuance/buybacks). This rebalancing approach successfully captures the market as it evolves, but effectively buys at high prices and sells at low prices. A long-short portfolio capturing the intensive margin rebalancing trades of index funds has an average alpha of -3% to -4% per year. Despite representing a size of less than 10% of AUM, this rebalancing portfolio does poorly enough to drag down overall index fund returns. We estimate that a "sleepy" strategy that rebalances annually improves fund returns by 40 bps per year relative to rebalancing quarterly. We argue this is because sleepy rebalancing avoids the short-and medium-term adverse selection associated with relatively quickly taking the other side of firms' primary and secondary market activity.

Sandidge, J. (2022a). "The Vital Signs of Retirement Income." In: *Retirement Management Journal* 11(1), pp. 46–63.

Creative problem-solvers manage complexity and simplify problems by focusing on the essence of the problem and ignoring the distraction of superfluous information, and fields as diverse as aviation, medicine, and high-rise construction use checklists to manage complexity. This paper explains why managing the negative momentum created by market losses and withdrawals is the essence of successful retirement-income management. It describes the vital signs of a sustainable distribution portfolio as well as a checklist for detecting the level of negative momentum embedded in a sustainable distribution portfolio. It closes with examples of how the checklist can change advisors' current annual review process from an abstract Monte Carlo discussion-one that likely fails to resonate with retirees-to a checklist-driven discussion focused on concrete items that investors can understand as the basis for cash-flow recommendations in retirement.

Sandidge, J. B. (2022b). "Why So Serious About Retirement Income?" In: *Investments and Wealth Monitor* 2022(September/October), pp. 46–50.

This article will discuss the seer-sucker theory of forecasting, focus on three illusory patterns that cause experts to paint an overly pessimistic narrative around retirement income, and present the genuine pattern advisors and retirees should focus on.

Sapra, S., Klein, S., and Martel, R. (2023). "Asset Allocation for Retirement Income: A Framework for Income-Oriented Investors." In: *The Journal of Portfolio Management* 49(4), pp. 127–141.

Certain investor types, particularly those approaching or in retirement, care about the income properties of investment portfolios. This article addresses the income characteristics of equities and bonds and provides a framework for building multi-asset portfolios with varying degrees of income orientation, both ex post, investigating the historical income behavior of equities and bonds, as well as through a forward-looking lens, based on today's much lower dividend and bond yields. Because bond yields and prices are inversely related, bonds imbed a natural time-diversification property across the two dimensions that income-oriented investors care most about: portfolio income and expected wealth. Despite the long-term growth potential of equity income and the prospect for higher bond yields over the next several years, we still find that a meaningful allocation to bonds is likely optimal for investors with an orientation toward organic portfolio income.

Sapra, S. and Pedersen, N. (2017). "The Role of Long-Maturity TIPS in Retirement Portfolios." In: *The Journal of Retirement* 4(4), pp. 95–106.

This article argues that long-duration Treasury Inflation-Protected Securities (TIPS) should play an important role in the portfolios of workers who are within 10 to 20 years of retirement. Long TIPS effectively help hedge retirees against fluctuations in the real wealth required to sustain a given level of consumption in their retirement years. As such, long-duration TIPS act as the "risk-free asset"in the context of retirement savings, and, for investors on a reasonable savings path, an allocation to this asset class may increase the likelihood of a successful retirement outcome.

Sarantsev, A. (2023). "A New Stock Market Valuation Measure with Applications to Retirement Planning." In: *arXiv e-Print*.

We generalize the classic Shiller cyclically adjusted price-earnings ratio (CAPE) used for prediction of future total returns of the stock market. We treat earnings growth as exogenous. The difference between log wealth and log earnings is modeled as an autoregression of order 1 with linear trend 4.5% and Gaussian innovations. Detrending gives us a new valuation measure. This autoregression is significantly different from the random walk. Therefore, our results disprove the Efficient Market Hypothesis. Therefore, long-run total returns equal long-run earnings growth plus 4.5%. We apply results to retirement planning. A withdrawal process governs how a retired capital owner withdraws a certain fraction of wealth annually. The fraction can vary from year to year. We study the long-term behavior of such processes.

Sarkar, S. (2023). "Managing Portfolio for Maximizing Alpha and Minimizing Beta." In: *arXiv e-Print*.

Portfolio management is an essential component of investment strategy that aims to maximize returns while minimizing risk. This paper explores several portfolio management strategies, including asset allocation, diversification, active management, and risk management, and their importance in optimizing portfolio performance. These strategies are examined individually and in combination to demonstrate how they can help investors maximize alpha and minimize beta. Asset allocation is the process of dividing a portfolio among different asset classes to achieve the desired level of risk and return. Diversification involves spreading investments across different securities and sectors to minimize the impact of individual security or sector-specific risks. Active management involves security selection and risk management techniques to generate excess returns while minimizing losses. Risk management strategies, such as stop-loss orders and options strategies, aim to minimize losses in adverse market conditions. The importance of combining these strategies for optimizing portfolio performance is emphasized in this paper. The proper implementation of these strategies can help investors achieve their investment goals over the long-term, while minimizing exposure to risks. A call to action for investors to utilize portfolio management strategies to maximize alpha and minimize beta is also provided.

Sato, M. (2016). "Optimal Withdrawal Rate Under Longevity Risks: A Criterion for Life Planning after Retirement." In: *SSRN e-Print*.
We discuss the optimal withdrawal rate in the existence of longevity risks. The longevity is realized as the factor of conditional survival rate in the model. We solve a optimal consumption problem under longevity risks, while managing the assets. The objective of the current work is giving a rational criterion that can be practically used in planning the life after retirement.

Schabort, K. P. (2022). "Sequence of return risk in South African post-retirement portfolios: the effectiveness of volatility-focused asset allocation strategies to address sequence and associated risks." MA thesis. University of Cape Town.
equence of return risk (which is the risk of unfavourable investment outcomes at the most unfavourable time) is an important consideration for efficiently funding retirement portfolio spending goals. This study examines the sensitivity of retirement decumulation portfolios to sequence of return risk ("SOR risk", also referred to as "sequence risk") in a South African context and evaluates the effectiveness of five volatility-focused asset allocation strategies in addressing the risk. Using monthly asset index returns separated into bull and bear market regimes for the period 1991 to 2020, correlated non-normal returns were simulated and combined with independently simulated inflation to generate 10 000 independent 30-year simulation trials. The performance of each of the five strategies was measured against a benchmark set of portfolios using simulated data and was validated using partial out-of-sample historical data. Performance was assessed using the sustainable withdrawal rate (SWR) and actuarial coverage ratio (ACR) metrics. The sensitivity results showed that SOR is greatest at the retirement date, declining asymptotically (halving within the first ten years of retirement). The geographic diversification strategy showed clear benefit to reducing SOR whereas the results for the risk parity strategy were not conclusive. The analysis and comparison of the lowrisk, rising equity glidepath, and dynamic cash buffer strategies formed the focus of the study. All three showed considerable SOR potential with the dynamic cash buffer strategy outperforming the others and substantially reducing SOR relative to the benchmark.

Scherer, B. and Lehner, S. (2021). "What Drives Robo-Advice?" In: *SSRN e-Print*.
The promise of robo-advisory firms is to provide low cost access to diversified portfolios built in accordance with the academic literature on normative portfolio choice. We investigate the latter claim. How much normative advice does robo-advice contain? For this purpose we web-scrap portfolio recommendations for 151200 investor types (input combinations from an online questionnaire) for one of the largest US robo-advisors. Our results show that the type of investment goal and the length of time horizon are dominating inputs with significant influence on recommended equity allocations. Normative advice in the form of Merton type hedging demands plays no role at all.

Schlanger, T., O'Connor, B., and Ahluwalia, H. (2022). "A Holistic Approach to Creating High Income Portfolios That Are Risk–Return Efficient and Tax Aware." In: *The Journal of Portfolio Management* 49(1), pp. 198–212.
Retirees and other investors may prefer to use dividends from equities and interest from fixed income to fund their spending needs. This mental accounting phenomenon of spending the portfolio's income without drawing down the invested principal makes them seek high income-producing asset classes. These investors, either directly or via advisors, adopt static high income portfolios as an investment strategy, often ignoring risk-return efficiency and tax efficiency principles. In this article, the authors introduce a methodology to systematically construct high income portfolios through an expected utility of wealth maximization approach while allowing for the incorporation of an investor's income preferences. However, because asset return expectations are conditional on asset valuations, the authors find that popular static high income portfolios that are created on an ad hoc basis-by substituting broad market exposure with high-yielding assets-by design are not optimal. More importantly, the authors find that high income portfolios are tax inefficient when it comes to high tax bracket investors, making their use only appropriate when taxes are not a primary concern-for example, for low- to moderate-income investors or in a tax-deferred account.

Schroeder, M. and Bonn, T. (2024). "Why Direct Indexing Is a Tectonic Shift for Private Markets: A Guide to Direct Start-Up Investments." In: *The Journal of Beta Investment Strategies* 15(3), pp. 66–80.

To the best of our knowledge, this article is the first to present a conceptual approach for a private market-based direct index. Based on the growing data availability of capital table management vendors, the authors propose a new way of determining a start-up's fair value. Using hedonic methods as an enabler, the authors consider changing value-determining components, thus creating a dynamic valuation model. Finally, the authors outline several implications for the index constituents, such as increased information efficiency and benefits from decreasing capital costs. In this way, the authors introduce a new niche for direct indexing (DI) applications that pressures indexing vendors and creates extended investment diversification opportunities for preliminary institutional investors. This allows the creation of major benchmarks that act as private equity (PE) proxies.

Schroeder, M. and Kronseder, C. (2024). "AI-Powered Direct Indexing: Exploring Thematic Universes for Enhanced Risk-Adjusted Returns." In: *SSRN e-Print*.

This paper explores direct indexing (DI) in the stock market using FINDALL. FINDALL is our self-engineered Transformers-based search engine which detects thematically relevant stock tickers in websites and PDFs. We demonstrate creating thematic indices in minutes, compiling strategies from passive to leveraged momentum positions with external trading signals. We study thematic indices such as Bionic, Defense, Energy, and Luxury. We benchmark our returns against flagship thematic exchange traded funds (ETFs) from major issuers. Using OLS regression, we test for index outperformance over its thematic ETF counterpart. First, our key finding is that the rule-based FINDALL approach is more accurate in selecting thematically relevant stocks compared to portfolio managercurated ETF stock selection. Additionally, our expense ratios are in all cases at least 66% lower than the industry average for thematic ETFs. Secondly, the Sharpe ratio indicators for all indices are higher than the ETF benchmarks showing enhanced risk-adjusted returns.

Schumann, E. (2019). "Backtesting." In: *SSRN e-Print*.

We discuss the backtesting of investment and trading strategies. We start with the challenges and pitfalls: overfitting, data preparation, and the effects of randomness. Then, we introduce and describe R software for backtesting. We demonstrate how to use the software for univariate and multivariate strategies (i.e. portfolio strategies) for two equity data sets. Specifically, we discuss the implementation and testing of momentum and portfolio optimization models. Throughout, we stress the analysis of sensitivity and robustness checks. Since such analyses require to run many backtests, we also discuss how backtests can be run in parallel.

Scruggs, J. T. (2019). "Asset Allocation and Withdrawal Strategies: Three Levers for Managing Retirement Outcomes." In: *Journal of Financial Planning*.

This paper demonstrates how moving three choice levers (asset allocation, initial withdrawal rate, and withdrawal flexibility) affects the joint distribution of retirement outcomes in terms of future consumption and terminal wealth. Using Monte Carlo simulation, the joint effects of the three choices on retirement outcomes in terms of income and terminal wealth were investigated. Results indicate that withdrawal flexibility effectively transferred some portfolio risk from a household's wealth to its income, and allowing for even modest withdrawal flexibility dramatically reduced the probability of running out of money. Because retiree goals in terms of lifestyle and legacy can be related to income and terminal wealth, a case may be made for featuring regular Monte Carlo simulation in the retiree decision-making process.

Sebastian, M. (2024). "A Goals-Based Approach to ESG Investing." In: *The Journal of Impact and ESG Investing* 4(4), pp. 30–41.

Discussions of environmental, social, and governance (ESG) investing tend to be complex and vague at the same time, in an attempt to serve the potentially conflicting interests of activists, investment managers and consultants, stakeholders and asset owners. Impact ("doing good") is difficult to achieve with most investment choices, therefore complicating the most common rationale for ESG. Investment service providers, activists, and groups like the United Nations Principles for Responsible Investment know this but usually don't say it loudly. Asset owners need to be realistic about what they can achieve and line that up with what, if anything, they want to achieve with ESG, such as the following. Alignment of portfolios with values, without expecting impact, is achievable for those who want it. Active investment managers should be expected to use (integrate) ESG data in their investment processes-with a risk-return goal, separate from impact or values-based investing. Large investors, or like-minded groups, may be able to influence others with their investment choices (like divestment), especially by using them to stigmatize what they view as bad actors. Asset owners can vote proxies in line with their interests and values. Investors can leverage their consumer power, such as by working with more emerging/diverse managers-which some argue have an alpha advantage. Direct impact can be achieved, largely with targeted private market investments.

Serbin, V., Borkovec, M., and Chigirinskiy, M. (2011). "Robust Portfolio Rebalancing with Transaction Cost Penalty an Empirical Analysis." In: *Journal of Investment Management*.

The goal of this paper is to study and compare two popular techniques used by practitioners to reduce the sensitivity of optimal portfolios to uncertainty in expected return for a typical portfolio optimization problem. Specifically, we investigate whether including transaction costs into the optimization problems objective function addresses the robustness issue.We weight this approach against the robust optimization method described in Goldfarb and Iyengar (2003). The latter directly incorporates the distribution of estimation errors in the optimization problem and determines the optimal portfolio allocation by selecting the least favorable realization of the expected returns in the investors uncertainty region.Our analysis focuses on the return maximization problem with constraints on total risk or tracking error and a transaction cost penalty in the objective function. We demonstrate that not only are the effects of incorporating a transaction cost penalty into the optimization problem similar to those of modeling uncertainty in expected returns, but that there are also some interesting differences. We offer some insights into the observed interplay between modeling transaction costs and modeling return uncertainty.

Šestanović, T. and Arnerić, J. (2021). "Neural network structure identification in inflation forecasting." In: *Journal of Forecasting* 40(1), pp. 62–79.

Neural networks (NNs) are appropriate to use in time series analysis under conditions of unfulfilled assumptions, i.e., non-normality and nonlinearity. The aim of this paper is to propose means of addressing identified shortcomings with the objective of identifying the NN structure for inflation forecasting. The research is based on a theoretical model that includes the characteristics of demand-pull and cost-push inflation; i.e., it uses the labor market, financial and external factors, and lagged inflation variables. It is conducted at the aggregate level of euro area countries from January 1999 to January 2017. Based on the estimated 90 feedforward NNs (FNNs) and 450 Jordan NNs (JNNs), which differ in variable parameters (number of iterations, learning rate, initial weight value intervals, number of hidden neurons, and weight value of the context unit), the mean square error (MSE), and the Akaike Infor- mation Criterion (AIC) are calculated for two periods: in-the-sample and out- of-sample. Ranking NNs simultaneously on both periods according to either MSE or AIC does not lead to the selection of the 'best' NN because the optimal NN in-the-sample, based on MSE and/or AIC criteria, often has high out-of- sample values of both indicators. To achieve the best compromise solution, i.e., to select an optimal NN, the preference ranking organization method for enrichment of evaluations (PROMETHEE) is used. Comparing the optimal FNN and JNN, i.e., FNN(4,5,1) and JNN(4,3,1), it is concluded that under approximately equal conditions, fewer hidden layer neurons are required in JNN than in FNN, confirming that JNN is parsimonious compared to FNN. Moreover, JNN has a better forecasting performance than FNN.

Sestok, B. (2021). "Implementing Advice based on "A Comparison of the Tax Efficiency of Decumulation Strategies"." In: *Journal of Financial Planning* 34(11).

This paper revisits research published in March 2021 to provide guidance on effective decumulation strategies that go contrary to the conventional wisdom of which accounts to pull from first in retirement.

Sexauer, S. C., Peskin, M. W., and Cassidy, D. (2015). "Making Retirement Income Last a Lifetime." In: *Financial Analysts Journal* 71(1), pp. 79–89.

To enable investors to spend down the assets in their defined contribution accounts more easily, the authors propose a decumulation benchmark comprising a laddered portfolio of TIPS for the first 20 years (consuming 88 percent of available capital) and a deferred life annuity purchased with the remaining 12 percent. This portfolio can be used directly by the investor (akin to indexing) or as a benchmark for evaluating the performance of a more aggressive strategy.

Seymour, A., Flint, E. J., and Chikurunhe, f. (2018). "Dynamic portfolio management strategies: A framework for historical analysis." In: *SSRN e-Print*.

The performance of dynamic trading and investment strategies can be difficult to predict. Although not without its problems, analysis of the historical performance of a strategy can provide valuable insight into its general risk and return properties. Furthermore, historical analysis allows one to compare variations of a strategy and examine the impact of various parameter choices and implementation rules. Dynamic strategy applications in three areas are considered, namely derivatives, asset allocation and equity factor portfolios. Firstly, the analysis of a strategy involving single-stock derivatives is examined in which call options on certain constituents of an index portfolio are sold as an alternative method of under-weighting the underlying. Secondly, the historical performance of an optimization-based asset allocation strategy is considered. The assumed aim of the strategy is to outperform a benchmark of CPI 5 via dynamic trading in a portfolio of domestic equities, bonds, property and cash, as well as international equities and bonds. Finally, the effects of portfolio construction on factor performance are studied via an historical analysis in which portfolios corresponding to a selection of fundamental factors are managed according to a range of weighting schemes, rebalance frequencies and portfolio sizes.

Shalett, L. (2015). *An Outcomes-Oriented Approach to Alternatives*. Tech. rep. Morgan Stanley Wealth Management.
Transformational forces are colliding in a way that necessitates a fresh approach to asset allocation guidance for alternative asset classes and strategies: the proliferation of lower-cost alternative investment formats; the normalization of interest rates; and the need to reintroduce alternatives to Financial Advisors and clients, many of whom in the past have been disillusioned by unfulfilled expectations, high fees, tax complexity and liquidity. In this new outcomes-based approach, we have a navigation framework that is intuitive and tests for suitability through alignment with basic portfolio goals. We also posit performance parameters that allow us to compare the trade-offs between alternative mutual funds/ ETFs and private offerings and suggest benchmarks that provide clients with a way to measure success.

Shang, H. L. and Haberman, S. (2020). "Forecasting age distribution of death counts: an application to annuity pricing." In: *Annals of Actuarial Science* 14(1), pp. 150–169.
We consider a compositional data analysis approach to forecasting the age distribution of death counts. Using the age-specific period life-table death counts in Australia obtained from the Human Mortality Database, the compositional data analysis approach produces more accurate 1- to 20-step-ahead point and interval forecasts than Lee-Carter method, Hyndman-Ullah method and two naive random walk methods. The improved forecast accuracy of period life-table death counts is of great interest to demographers for estimating survival probabilities and life expectancy, and to actuaries for determining temporary annuity prices for various ages and maturities. Although we focus on temporary annuity prices, we consider long-term contracts that make the annuity almost lifetime, in particular when the age at entry is sufficiently high.

Sharpe, W. (2019). *Retirement Income Analysis with scenario matrices*. Stanford University. 200 pp.
financial economics textbook covering the various subject areas related to retirement planning. The textbook is titled RISMAT, which stands for Retirement Income Analysis (with scenario matrices). It covers subjects within the scope of financial economics such as utility theory and the valuation of assets, and others from outside such as demographics and building mortality tables. Incorporated in the text is a comprehensive system for retirement planning developed by Sharpe using Matlab software. In modeling investments, Sharpe uses an approach based on the capital asset pricing model (CAPM), which posits that the optimum relationship between return and volatility can be achieved with portfolios that combine a risk-free asset and a highly diversified market portfolio that includes proportionate shares of all publicly-traded investible assets. Sharpe develops a proxy for the market portfolio based on a mix of four Vanguard funds: 1) Total stock market index 2) Total bond market index 3) Total international stock market index 4) Total international bond market index He uses the Vanguard inflation-protected securities fund to represent the risk-free asset.

Shemendyuk, A. and Wagner, J. (2024). "On the factors determining the health profiles and care needs of institutionalized elders." In: *Insurance: Mathematics and Economics* 114, pp. 223–241.
In many developed countries, population aging raises a number of issues related to the organization and financing of long-term care. While the determinants of the overall burden and cost of care are well understood, the organization of institutionalized long-term care must meet the needs of the elderly. One way to optimize management is to use information on health problems to assess the infrastructure needed, the qualifications of staff, and the allocation of new entrants. In this research, we determine the typical health profiles of institutionalized elderly using novel longitudinal data from nursing homes in the canton of Geneva, Switzerland. Our data contain comprehensive information on health factors such as impairments of psychological and sensory functions, levels of limitations, and pathologies for 21 549 individuals covering the period from 1996 to 2018. First, we perform a spectral clustering algorithm and determine the profiles of the institutionalized individuals. Then, we use multinomial logistic regression to study the effects of the factors that determine these health profiles. Our main findings include eight typical health profiles: the largest group consists of the most "healthy" individuals, who, on average, require the least amount of help with their daily needs and who stay in the institution the longest. We show that, in contrast to age at admission and gender, the limitations and the set of pathologies are relevant factors in determining the profile. Our study sheds light on the typical structures of elderly' health profiles, which can be used by institutions to organize their resources and by insurance companies to derive profile-based products that provide additional insurance coverage in case of special needs.

Shen, Y., Sherris, M., Wang, Y., and Ziveyi, J. (2023). "The Valuation and Assessment of Retirement Income Products: a Unified Markov Chain Monte Carlo Framework." In: *SSRN e-Print*.

This paper devises a flexible assessment framework of a catalogue of existing retirement income products which include, account-based pensions, group self-annuities and variable annuity contracts. It utilises the Hamiltonian Monte Carlo approach, a proven computational technique for simulating conditional distributions in higher dimensions. Graphical illustrations for the risk-return trade-offs for each product are presented which can readily be adapted by advisors, and all stakeholders as a tool for enhancing the decision-making process for retirees. The key features of the retirement income products which include variable annuities, account-based pension, and group self-annuities, are presented in an easily understandable way for a retiree with common knowledge. Sensitivity analysis on the investment options of the underlying fund provides insights for retirees to maximise income according to their risk preferences. We also devise a lifetime utility analysis framework for the comparison of lifetime utility from purchasing the products.

Sherrill, D. E. (2019). "Different paths to the same target: variation in target date funds." In: *The Journal of Investing*.

Target date funds (TDFs) provide a simple way to obtain an asset allocation that matches the current and future needs of an average investor with a specific retirement date. This article examines the differences between TDFs with the same target date but from different fund providers and finds considerable differences in asset allocations, risk, performance, and other fund characteristics. These differences are partially predictable based on the information in fund prospectuses; however, TDFs often deviate from the planned asset allocations. Investors and advisors should be aware that TDFs with the same target date can vary dramatically, warranting additional research when selecting between TDFs and then monitoring chosen TDFs asset allocations in the future. Potential solutions to these concerns include increased guidance from regulators, greater discernment from employers in employer retirement plan investment options, and more descriptive fund names.

Sherwood, M. W. (2021). "Risk Capacity Portfolio Construction." In: *The Journal of Investing* 30(2), pp. 31–52.

This paper introduces a portfolio construction framework for risk-averse investors that aim to meet or exceed a client's capital accumulation needs for a future event, such as retirement. Risk Capacity Portfolio Construction (RCPC) presents a risk-optimized alternative to Target Date Funds (TDFs). Risk Capacity Portfolio Construction is an economic sciences innovation that is validated by Skew-Risk Modeling, which was first presented in Targeted Return Portfolio Construction (Sherwood, 2019) and is detailed within this paper in its application to retirement-focused investors. Risk Capacity Portfolio Construction factors skewness and kurtosis into the risk management and asset allocation of an individual's life cycle and can optimize risk beyond simply accounting for the equity risk premium and human capital.

Shi, Y. (2021). "Forecasting Mortality Rates with the Adaptive Spatial Temporal Autoregressive Model." In: *Journal of Forecasting* 40(3), pp. 528–546.

Accurate forecasts of mortality rates are essential to various demographic research like population projection, and the pricing of insurance products such as pensions and annuities. Recent studies have considered a spatial-temporal autoregressive (STAR) model for the mortality surface, where mortality rates of each age depend on the historical values of itself (temporarily) and the neighboring cohorts ages (spatiality). This model has sound statistical properties including co-integrated dependent variables and existence of closed-form solutions. Despite its improved forecasting performance over the famous Lee-Carter (LC) model, the constraint that only the effects of the same and neighboring cohorts are significant can be too restrictive. In this study, we adopt a data-driven adaptive weighted structure and propose the adaptive STAR (ASTAR) model. Retaining all desirable features of the STAR, our model uniformly outperforms the LC and STAR counterparties for forecasting accuracy, when mortality data aged 0–100 of the United Kingdom, France, Italy, Spain and Japan over 1950–2016 are considered. Two sensitivity tests and additional simulation results also lead to robust conclusions. The proposed ASTAR model is therefore a widely useful tool in modelling and forecasting mortality rates in other contents, and may be extensible to multi-population modelling.

Shoven, J. and Slavov, S. (2012). "The Decision to Delay Social Security Benefits: Theory and Evidence." In: *SSRN e-Print*, pp. 121–144.

Social Security benefits may be commenced at any time between age 62 and age 70. As individuals who claim later can, on average, expect to receive benefits for a shorter period, an actuarial adjustment is made to the monthly benefit amount to reflect the age at which benefits are claimed. We investigate the actuarial fairness of this adjustment. Our simulations suggest that delaying is actuarially advantageous for a large subset of people, particularly for real interest rates of 3.5 percent or below. The gains from delaying are greater at lower interest rates, for married couples relative to singles, for single women relative to single men, and for two-earner couples relative to one-earner couples. In a two-earner couple, the gains from deferring the primary earner's benefit are greater than the gains from deferring the secondary earner's benefit. We then use panel data from the Health and Retirement Study to investigate whether individuals' actual claiming behavior appears to be influenced by the degree of actuarial advantage to delaying. We find no evidence of a consistent relationship between claiming behavior and factors that influence the actuarial advantage of delay, including gender and marital status, interest rates, subjective discount rates, or subjective assessments of life expectancy.

Shoven, J. and Slavov, S. (2013). "When Does it Pay to Delay Social Security? The Impact of Mortality, Interest Rates, and Program Rules." In: *SSRN e-Print*.
Social Security benefits may be commenced at any time between ages 62 and 70. As individuals who claim later can, on average, expect to receive benefits for a shorter period, an actuarial adjustment is made to the monthly benefit to reflect the age at which benefits are claimed. In earlier work (Shoven and Slavov, 2012), we investigated the actuarial fairness of this adjustment for individuals with average life expectancy for their cohort. We found that for current real interest rates, delaying is actuarially advantageous for a large subset of people, particularly for primary earners in married couples. In this paper, we quantify the degree of actuarial advantage or disadvantage for individuals whose mortality differs from the average. We find that at real interest rates close to zero, most households - even those with mortality rates that are twice the average - benefit from some delay, at least for the primary earner. At real interest rates closer to their historical average, however, singles with mortality that is substantially greater than average do not benefit from delay; however, primary earners with high mortality can still improve the present value of the household's benefits through delay. We also investigate the extent to which the actuarial advantage of delay has grown since the early 1960s, when the choice of when to claim first became available, and we decompose this growth into three effects: (1) the effect of changes in Social Security's rules, (2) the effect of changes in the real interest rate, and (3) the effect of changes in life expectancy.

Shoven, J. B. and Sialm, C. (2004). "Asset location in tax-deferred and conventional savings accounts." In: *Journal of Public Economics* 88(1-2), pp. 23–38.
This paper derives optimal asset allocations (which assets to hold) and asset locations (in which accounts to hold them) for a risk-averse investor saving for retirement. The investor can hold taxable corporate bonds, tax-exempt municipal bonds, and stocks either in a tax-deferred or a conventional taxable savings account. Taxable bonds have a preferred location in the tax-deferred account and tax-exempt bonds have a preferred location in the taxable account for investors in sufficiently high tax brackets. Tax-efficient stock portfolios (e.g. passively-managed mutual funds) should be held in the taxable account and tax-inefficient stock portfolios (e.g. actively-managed mutual funds) should be held in the tax-deferred account. We show that locating assets optimally can significantly improve the risk-adjusted performance of retirement saving.

Shoven, J. B. and Walton, D. (2023). "Target Retirement Fund: A Variant on Target Date Funds that Uses Deferred Life Annuities Rather than Bonds to Reduce Risk as Retirement Approaches." In: *SSRN e-Print*.
This paper evaluates a new variant of the popular target date funds used in employer-based retirement savings plans. We call this new variant a "target retirement plan." Instead of increasing the allocation to bond funds as retirement approaches, a target retirement fund gradually purchases deferred life annuities beginning at age 50. In the particular straw model target retirement fund examined in the paper, the defined contribution participant makes deferred life annuity purchases at ages 50, 52, 54, 56, 58, 60 and 62. We compare how a target retirement fund participant would fare compared with someone who stays with a traditional TDF until retirement and then buys an immediate life annuity. We examine 1,000 possible 30-year futures for stock returns, bond fund returns and Treasury interest rates. The main result from this paper is that buying a retirement annuity in advance (by accumulating deferred life annuities) is superior to sticking with a Target Date Fund until retirement and then buying an immediate annuity in most scenarios of future stock returns, interest rates and bond returns.

Shoven, J. B. and Walton, D. B. (2021). "An Analysis of the Performance of Target Date Funds." In: *The Journal of Retirement* 8(4), pp. 43–65.
This article presents a thorough evaluation of target date funds (TDFs) for the period 2010-2020. TDFs have grown enormously in assets, reaching USD1.4 trillion at the end of 2019, and account for approximately 24% of all assets in 401(k) accounts. We report on the results of a style analysis evaluation of TDFs that determines their effective asset allocation. It examines both the constant in the style analysis regressions and resulting Sharpe ratios, which reflect the over- or under-performance of the funds relative to a passive benchmark with the same asset allocation. Lower cost TDFs tend to match the benchmark returns, while higher cost TDFs deviate from them considerably. We examine how TDFs performed in the stock market crash between February 19 and March 23, 2020, during which five-week period broad market averages fell by about one-third. We find that the value

of long-dated TDFs (those with a target date of 2045 and beyond) fell by between 30% and 35%, while the 2025 funds, designed for people roughly 60 years old, lost between 20% and 25% of their value. We find that past performance only weakly influences future expected performance. As with equity funds in general in this period, TDFs with actively managed ingredient funds, on average, trailed the performance of their cheaper passively managed counterparts.

Simonato, J.-G. (2023). "Maximizing the Probability to Reach the Goal: An Exploration Exercise in Goal-Based Wealth Management." In: *The Journal of Portfolio Management* 49(5), pp. 189–207.

Goal-based wealth management (GBWM) is a portfolio approach in which the investor associates risk with the probability of not attaining a financial goal. Using several datasets, the author examines the performance of a multiperiod GBWM strategy that maximizes the probability of achieving a financial goal. With varying restrictions about leverage and short sales, he compares the goal-based wealth investor with a standard and a goal-attentive mean-variance investor. Without transaction costs, the results suggest that, in terms of goal achievement, a goal-based wealth investor focusing on the probability of reaching a goal does better than a standard mean-variance investor. Compared to a goal-attentive mean-variance investor, the results still favor the goal-based wealth investor but to a lesser extent. With transaction costs, goal-based wealth and goal-attentive mean-variance investors yield similar results in many cases.

Simsek, K. D., Jeong Kim, M., Chang Kim, W., and Mulvey, J. M. (2018). "Optimal Longevity Risk Management in the Retirement Stage of the Life Cycle." In: *The Journal of Retirement* 5(3), pp. 73–92.

The uncertainty in life expectancy plays a critical role in individual financial planning. Its impact is magnified during the retirement years (the wealth distribution stage of the life cycle), as new sources of income typically are not available to retired persons. Utilizing a multi-stage stochastic program, the authors model and solve the optimal asset allocation problem of a retired couple with uncertain life expectancy in the presence of a term life insurance policy. In the base case, they find optimal policies assuming no longevity risk (i.e., lifetime scenarios are uncertain although life expectancy is fixed on the retirement date). Next, they introduce longevity risk in the scenario generation stage through either a shift in the expected lifetimes or an unexpected cut in periodic retirement income. The authors find that the optimal asset allocation policy depends on the presence and the type of these risks as well as the relative price of insurance and the size of any cut in pension benefits.

Sin, R., Murphy, R., and Lamas, S. (2019). "Goals-Based Financial Planning: How Simple Lists Can Overcome Cognitive Blind Spots." In: *Journal of Financial Planning*.

Prior research has shown that behavioral biases may inhibit investors from identifying and prioritizing investing goals that are important to them. A nationally representative study was conducted to understand if a simple behavioral technique would nudge investors away from using unreliable, top-of-mind notions when it comes to creating investing goals; and if a more sophisticated technique is better at prioritizing investment goals than a simple one. Asking people to self-report their investing goals is insufficient. About 26 percent of the participants in the study changed their top goal when prompted with reminders about other goals. On average, using a more sophisticated ranking technique did not lead to any appreciable difference in how investment goals were ranked, suggesting that when it comes to prioritizing multiple goals, a straightforward rank ordering suffices.

Siskos, C. (2021). "What's Your Strategy for Maximizing Your Social Security Benefits?" In: *Kiplinger*.

Deciding when to start taking your Social Security benefits can have significant financial consequences. Here are some of the factors you should think through.

Sivarajan, S. and De Bruijn, O. (2021). "Risk Tolerance, Return Expectations and Other Factors Impacting Investment Decisions." In: *The Journal of Wealth Management* 23(4).

Do risk tolerance questionnaires predict risk-taking? Traditionally, investor preferences have been identified using risk tolerance questionnaires. However, there has been little focus as to whether the questionnaires are actually useful in predicting risk-taking decisions of investors and advisers. The purpose of this research was to review the existing literature and, using an online analytical survey, conduct a statistical analysis to determine the relationship between risk tolerance and risk-taking. The resulting quantitative analysis was complemented with qualitative interviews. The main findings are that return expectations and demographic variables are important predictors of risk-taking decisions, whereas risk tolerance questionnaires are not. A framework of two models was developed to better understand investment decisions. It forms the basis for a general discussion of the implications of these findings for practitioners and other stakeholders.

Skallsjo, S. R. (2019). "Simple formulas for portfolio rebalancing rules." In: *SSRN e-Print*.

Portfolio rebalancing rules are used by large scale asset managers to manage a trade-off between frequent rebalancing on one hand, and a cost of deviating from the portfolio benchmark on the other. This note contributes to the design of such policies in two ways. First, it clarifies the analytical framework, by showing how the rebalancing problem can be formulated in a standard model for asset management. Although this step is straightforward, it is important as it makes a clear distinction between a risk factor and a risk factor exposure. Keeping this distinction facilitates the design of the optimal policy.Second, for a given policy the note provides closed form expressions for certain statistical quantities relevant for the policy design. These include average rebalancing frequency, average annual transaction amount, mean square deviation from benchmark, and mean absolute deviation from benchmark. The resulting formulas are simple arithmetic expressions, easy to interpret and straightforward to implement.

Slavov, S. (2024). "Retirement, Social Security deferral, and life annuity demand." In: *SSRN e-Print*.
Delaying Social Security is equivalent to purchasing an inflation-indexed life annuity; it involves forgoing current benefits in exchange for higher real benefits in the future. Although the terms for deferring retired worker benefits between ages 62 and 70 are generous, most people claim benefits well before age 70. This behavior may be related to the 'annuity puzzle," i.e., the observation that individuals don't annuitize their retirement assets to the extent that economic models based on consumption-smoothing predict they should.

Smith, G. and Gould, D. P. (2007). "Measuring and controlling shortfall risk in retirement." In: *The Journal of Investing* 16(1), pp. 82–95.
A key challenge for retired investors is determining the stock-bond asset allocation that, for a given spending rate, provides an acceptable probability of shortfall – having real wealth drop below a specified floor during the investor's lifetime. Standard portfolio analysis yields the well known tradeoff between risk and return described by the Markowitz frontier. For retirement planning, we reconceptualize this as a tradeoff between shortfall probability (risk) and the median value of terminal wealth (return). For specified assumptions, there is a stock-bond asset allocation that minimizes shortfall risk. Portfolios with more stocks increase the median values of terminal wealth, but at the expense of higher shortfall risk. Portfolios with fewer stocks are inferior in that they decrease the median value of terminal wealth and increase shortfall risk. We find that for a variety of plausible assumptions about asset returns, investment strategies, and what constitutes a shortfall, the minimum-risk portfolio generally has between 50 and 70% stocks.

Smith, L. (2021). *The Best Age for Social Security Retirement Benefits*. Tech. rep. SmartAsset.
Let's take a look at how Social Security works, and what you need to know when deciding the best age for your retirement.

Smith, L. and Mukherjee, M. (2024). "Long-Run Expectations for S&P 500 Direct Indexing with Tax Loss Harvesting." In: *The Journal of Beta Investment Strategies* 15(3), pp. 81–107.
The authors estimate the tax alpha of an S&P 500 direct index strategy with tax loss harvesting over a 60-year horizon including the effects of dividend reinvestment, unqualified dividends, corporate actions, and management fees. The authors estimate the impact of relaxing the key assumptions of unlimited short-term gains, reinvestment of tax savings, and maximum tax rates. The authors find that realized gains from cash acquisitions offset realizable losses at long horizons and that management fees can render tax loss harvesting inferior to an index ETF even under the most favorable assumptions.

Sneddon, T., Bao, C., and Zhu, Z. (2015). "Optimal Asset Liability Management for Post-retirement stage with Income Protection Product." In: *21st International Congress on Modelling and Simulation*.
Australia has a compulsory defined contribution retirement provision system, whereby employers must contribute a proportion of the pre-tax salary of their employees towards an individual account which cannot be accessed until retirement except in extraordinary circumstances. These funds are generally invested in a portfolio of financial assets from which the retiree may draw throughout retirement. Retirees under this system face two key problems when making investment and withdrawal decisions regarding this portfolio. Firstly, retirees must manage their superannuation investment portfolio to maximise their riskadjusted returns and thereby best financially provide for their own retirement. Secondly, retirees must optimise their withdrawal pattern from the superannuation account throughout retirement so as to maximise their post-retirement lifetime utility given the need to minimise the risk of portfolio ruin prior to death. We model this issue as a dynamic stochastic optimisation problem with constraints. The market value of the portfolio is a function of the annual contributions invested by the individual throughout their career and the returns derived from their investment in uncertain financial markets. The post-retirement lifetime utility function is a function of discounted annual income throughout retirement and is therefore subject to market and inflation risk. We model the financial market uncertainties as correlated stochastic processes as projected by a variant on the Wilkie stochastic investment model developed within CSIRO, the SUPA (Simulation of Uncertainty for Pension Analysis) model. We also

define an income protection asset (an inflation index linked annuity) which is available to the individual as a tool to hedge inflation risk. We then solve the dynamic superannuation/pension portfolio optimisation problem using a numerical approach that is based on the stochastic control algorithm to calculate the conditional value functions of investors for a sequence of discrete decision dates. The algorithm provides optimal decisions for portfolio asset allocation in financial markets and the optimal amount to shift towards the annuity product on an annual basis to achieve maximum post-retirement lifetime utility whilst minimising the risk of portfolio ruin prior to death.

Sneddon, T., Zhu, Z., and O'Hare, C. (2016). "Modelling defined contribution retirement outcomes: A stochastic approach using Australia as a case study." In: *Australian Journal of Actuarial Practice* 4, pp. 5–19.
In this paper we present a stochastic forecast model in a defined contribution pension system for projecting the accumulation and decumulation phases from an individual fund perspective. We use the Australian superannuation system as the context to demonstrate this "SUPA"(Simulation of Uncertainty for Pension Analysis) model. The SUPA model can be used to simulate the evolution of superannuation fund balances across time during the accumulation and decumulation phases. The model comprises four elements: (1) a stochastic projection of investment returns; (2) a stochastic projection of income levels (upon which contributions to the fund are based); (3) a projection of levels of withdrawal in retirement; and (4) a stochastic projection of increasing longevity (life table). The combination of these four elements within the SUPA model is described in detail in this paper. One application of the model is demonstrated through a case study involving recent Australian legislative amendments. In this example, we show how the model can be used to forecast likely outcomes (i.e. whether individuals will have sufficient funds in retirement), under the current superannuation structure and a previous structure. This will demonstrate how the SUPA model can be used to model the potential impact of any changes to a superannuation system.

Society of Actuaries Research (2017). *Deciding When to Claim Social Security*. Tech. rep. Society of Actuaries.
The decision to claim Social Security benefits is one of the most important retirement decisions a person will make. Many people decide to claim their Social Security benefits early without considering other options. Perhaps they do not know that they have options or they may think that one option isn't much different from another. Options do exist, however, and some of them can significantly improve retirement security. This Decision Brief discusses many of those options.

Soni, A. (2023). "The Role of Annuities in Managing Sequence of Returns Risk Approaching and in Retirement." In: *The Journal of Retirement* 11(4), pp. 26–39.
Research and product development efforts aimed at addressing retirement income needs have generally been focused on the accumulation side of financial planning. However, innovation and product development on the decumulation side of the equation has been limited. In this article, we utilize an asset allocation model for retirement that can include non-traditional investments to demonstrate how annuities are an integral component of a multi-asset portfolio, aiding in the management of risks unique to retirees and near-retirees. For optimal decumulation allocation, we propose a methodology that maximizes a measure of household financial well-being that considers the two most important, yet often conflicting, objectives in the decumulation phase of the retirement: a) minimizing shortfall risk, and b) maximizing potential bequest. We use this model to demonstrate how annuities could be a prudent addition to a portfolio of traditional stocks and bond investments to better manage sequence of returns or sequence of cash flows risk.

Sorensen, E., Alonso, N., and Belanger, D. (2023). "The Use and Misuse of Tracking Error." In: *The Journal of Portfolio Management* 49(8), pp. 12–23.
Equity portfolio tracking error to a benchmark is a most ubiquitous restriction for active portfolios as prescribed by fiduciaries. The restriction is typically a tight range with minimum and maximum ex-ante extremes. The typical capitalization-weighted benchmark (i.e., the S&P 500 Index) has significant changes in diversification character overtime. This brings into question the sensibility of holding tracking error (TE) constant for a skilled active manager. The authors demonstrate that as the concentration of names in the S&P 500 increases, its diversification fades. When this happens, constant tracking error is the enemy of the skilled diversified manager, other things equal. Using multivariate classification and regression trees (CART), they show that high tracking dominates low tracking when index concentration is low, trending lower, and return dispersion is high. Low tracking dominates when the opposite index conditions exist. The authors conclude that the power of a flexible TE process in wealth creation is dominant over inflexibility in the benchmark.

Sornwanee, M. (2020). "Multi-Regime, Goal-Based Retirement Solutions: Sensitivity Analysis and Post-Retirement Performance Comparison of Dynamic Strategies." MA thesis. Princeton University.

The shift from defined-benefit to defined-contribution pension plans worldwide has left many individuals who lack access to private banking services without proper retirement financial planning. As individuals have more responsibilities for their retirement planning, there is a growing concern of choosing asset allocation strategies that are not suitable for their goals. Besides, in the light of the current situation regarding the COVID-19 pandemic, individuals must be able to prepare for a possible prolonged "crash" economic regime. Understanding the impact of such a crash regime is important for individuals to make optimal financial decisions. These aspects will be considered in the paper. The objective of this study is to model and comprehend a fully automated retirement planning system for individual investors. By understanding the characteristics of each strategy through various risk measures and analyzing its sensitivity to prolonged crash economic regimes, health risks, savings rate, and replacement income ratio (spending after retirement, divided by the last salary before retirement), we can use the results to recommend the most appropriate asset allocation and savings rate strategy for an individual investor. In this study, we implement multi-regime Monte Carlo simulations to simulate asset returns throughout the accumulation phase (pre-retirement) and decumulation phase (post-retirement). We consider different asset allocation strategies: standard glide path (SGP: linearly reduce equity weight over time), reversed glide path (RGP: linearly increase equity weight over time), "Catch-Up" strategy (CGP: follow SGP and increase equity weight when the wealth falls behind a benchmark portfolio, called "goal portfolio"), "Play-Safe" strategy (PGP: follow SGP and decrease equity weight when the wealth exceeds the goal portfolio), and "Dynamic Zone" glide path (DGP: combine CGP and PGP). We analyze the performances concerning non-goal-based and goal-based risk measures while incorporating the longevity risk and its corresponding risk measures, such as the probability of outliving retirement savings, to further comprehend the characteristics of each policy. We evaluate a goal-based, dynamic strategy that alters the savings rate based on its performance relative to the goal portfolio. That is, it increases the savings rate during a crash regime and decreases the savings rate during a normal regime. Moreover, we conduct sensitivity analyses on 4 parameters: prolonged crash economic regime, health risks, savings rate, and replacement income ratio. The results are analyzed to fully comprehend the strengths and weaknesses of each strategy. First, we recommend PGP for individual investors, especially when there is a prolonged crash regime. Generally, it outperforms other strategies in average maximum drawdown ratio, Goal-at-Risk (GaR), Conditional Goal-at-Risk (CGaR), and the probability of outliving retirement savings. If a 10-year crash regime starts after the middle of the pre-retirement phase, PGP best protects the portfolio from a huge loss during the crisis as it has the lowest probability of outliving retirement savings, which is 18 percentage points lower than that of SGP. With PGP, individuals need to increase their savings rate by 8-31 percentage points, depending on the timing of a 10-year crash regime, to narrow the probability of outliving savings to 10%. However, it is less effective to increase the savings rate only during the crisis or to decrease the replacement income ratio. Besides, for any allocation strategy, the worst time to have a prolonged crash regime is in the middle of the pre-retirement phase, leading to an increase of 69 percentage points in the probability of outliving savings. In terms of savings rate, shifting more savings to an earlier part in the pre-retirement phase can result in a reduction in both the average maximum drawdown and the probability of outliving retirement savings for CGP, PGP, and DGP. Also, the result of the goal-based, dynamic savings rate strategy, in terms of the probability of outliving savings, is equivalent to that of increasing the savings rate for the entire pre-retirement period by 12%.

Sosner, N., Liberman, J., and Liu, S. (2021). "Integration of Income and Estate Tax Planning." In: *The Journal of Wealth Management* 24(1), pp. 78–104.

The preservation and transfer of wealth to future generations are among the central financial goals for most high-net-worth families. Using a stylized theoretical model and Monte Carlo simulations, this article quantifies the benefits of income and estate tax planning for growing wealth over generations. The article shows that a family that invests with income and estate tax efficiency in mind can achieve substantially higher wealth levels than a family oblivious to taxes. More important, the article demonstrates that a significant value accrues from integrating income tax efficiency and estate tax planning: Becoming efficient with respect to one tax should make the family even more eager to become efficient with respect to the other.

Staden, P. van, Forsyth, P., and Li, Y. (2024). "Smart leverage? Rethinking the role of Leveraged Exchange Traded Funds in constructing portfolios to beat a benchmark." In: *arXiv e-Print*.

Leveraged Exchange Traded Funds (LETFs), while extremely controversial in the literature, remain stubbornly popular with both institutional and retail investors in practice. While the criticisms of LETFs are certainly valid, we argue that their potential has been underestimated in the literature due to the use of very simple investment strategies involving LETFs. In this paper, we systematically investigate the potential of including a broad stock market index LETF in long-term, dynamically-optimal investment strategies designed to maximize the outperformance over standard investment benchmarks in the sense of the information ratio (IR). Our results exploit the observation that positions in a LETF deliver call-like payoffs, so that the addition of a LETF to a portfolio can be a convenient way to add inexpensive leverage while providing downside protection. Under stylized assumptions, we present and analyze closed-form IR-optimal investment strategies using either a LETF or standard/vanilla ETF (VETF) on the same equity index, which provides the necessary intuition for the potential and benefits of LETFs. In more realistic settings, we use a neural network-based approach to determine the IR-optimal strategies, trained on bootstrapped historical data. We find that IR-optimal strategies with a broad stock market LETF are not only more likely to outperform the benchmark than IR-optimal strategies derived using the corresponding VETF, but are able to achieve partial stochastic dominance over the benchmark and VETF-based strategies in terms of terminal wealth.

Steinorth, P. and Mitchell, O. S. (2015). "Valuing variable annuities with guaranteed minimum lifetime withdrawal benefits." In: *Insurance: Mathematics and Economics* 64, pp. 246–258.
Variable annuities with guaranteed minimum lifetime withdrawal benefits (VA/GLWB) offer retirees longevity protection, exposure to equity markets, and access to flexible withdrawals in emergencies. We model how risk-averse retirees optimally withdraw from the products, balancing returns and the embedded longevity protection. Assuming reasonable individual preferences, the resulting cash flow generates a Money's Worth Ratio of around 0.9 for our stylized VA/GLWB in the post-crisis product, considerably lower than what was offered prior to 2008. Sensitivity analyses with respect to portfolio choice, mortality, fees, and guaranteed withdrawal rates show that MWRs range from 0.80 to 1.0, with the portfolio choice making the biggest difference. For most parameter choices, the utility value of the VA/GLWB exceeds that of a similar mutual fund, but it is less than for a fixed annuity. Interestingly, VA/GLWB withdrawals in early retirement can optimally exceed contract maximum withdrawals, despite the fact that this reduces future withdrawal guarantees.

Stocker, A. (2022a). "A 'Semi-Historical' Time Series of UK Annuity Rates for Retirement Income Backtesting." In: *SSRN e-Print*.
In this paper, a time series of UK annuity income rates (AR) based on historical interest rates but modern mortality rates (hence the term "semi-historical") has been derived for both nominal and inflation-linked annuities. The errors associated with the calculated rates for nominal annuities are approximately 10% of AR, while those for inflation-linked annuities are likely to be higher at 15% of AR. The results of a historical backtest for a relatively straightforward case where a retiree purchased a single life level annuity at retirement age with half of their available assets indicate that this provides a boost in income for 15 years or more in virtually all historical retirements at the expense of a reduced legacy until more than 25 years after retirement.

Stocker, A. (2022b). "Supporting Retirement Expenditure Using a Combination of State Pension, Investment Portfolio, Bond Ladders, and Annuities." In: *SSRN e-Print*.
In this paper, the effect of using retirement income from a variety of sources on initial income rate and the survivability of the income is investigated. The sources of income include pre-existing ('baseline') inflation linked guaranteed income (e.g., state pension and uncapped direct benefit pensions), immediate annuities (including level, and those with escalation and inflation linked options), bond ladders, and investment portfolio. Assuming an annual decline in expenditure of 1 to 2% over the course of the retirement period, the results indicate that including an inflation linked annuity provides the best performance when the retiree has a baseline annual income of less than about 10% of their initial portfolio value, while a bond ladder provides the best solution when the baseline income is greater than about 20% of the initial portfolio value. The robustness of portfolio approaches were improved by using percentage of portfolio with a ceiling at required income level.

Stocker, A. (2023). "Using Lifetime Annuities to Increase Retirement Portfolio Survivability." In: *SSRN e-Print*.
In this paper, the number of years, Y it took for retirement income derived from the combination of an investment portfolio and a lifetime annuity to fall below a constant, inflation-linked, target income was investigated for retirees in the UK and US. The results indicate that there was a threshold value, RT in the annuity payout rate, AR above which the purchase of the annuity increased Y compared to the case where no annuity was purchased, and vice versa. The value of RT was different for the two countries and depended on which annuity options were used and whether there was a delay in the purchase. For example, for US retirees with a target income of 4% of the initial portfolio value, RT was approximately 7.3%, 5.6%, and 4.1%, for level, 3% escalation, and CPI/RPI annuities purchased at the start of retirement, respectively. RT was much higher when the purchase of the annuity was delayed for D years after the start of retirement since, in the worst retirements, the reduction in real portfolio value was greater than the increase in annuity payout rate with age. However, RT for a delayed purchase was decreased when inflation linked bonds (ILB) with a maturity of D purchased at the start of

retirement were then used to purchase the annuity or when a ladder of ILB was constructed to provide the target income for the delay period D. For example, for US retirees purchasing a level annuity after a delay of 10 years, and assuming the ILB yield to maturity was 1%, RT was 13.3%, 9.6%, and 9.0% for the portfolio, ILB held to maturity, and ILB ladder approaches, respectively.

Stocker, A. (2024a). "Building an Inflation Adjusted Income floor in Retirement Using a Combination of a Tips Ladder and Delayed or Deferred Annuities." In: *SSRN e-Print*.
Retirees whose inflation adjusted guaranteed income floor (e.g., social security, state pension, or defined benefit pension) does not cover their essential expenses could consider using part of their investment portfolio to construct additional flooring. In this paper, a combination of an inflation protected bond (e.g., TIPS) ladder with a deferred or delayed nominal annuity purchase are compared with those from a TIPS ladder (without annuity) and a conventional stock and bond portfolio for a 35 year retirement starting at 65 years old. The approach that was the most robust was strongly dependent on TIPS yields. For yields above approximately 2.1%, the TIPS ladder provided the most robust solution (although there was an element of reinvestment risk), for TIPS yields of about 1.5% to 2.1%, a combination of a TIPS ladder and a delayed annuity was most robust, while for TIPS yields below about 1.5%, a TIPS ladder and deferred annuity provided the best, although relatively poor, solution.

Stocker, A. (2024b). "The Effect of the Maturity of Gilts on 'Safe' Withdrawal Rates in Retirement." In: *SSRN e-Print*.
Using a new database of historical total returns for 3 month bills and gilt sectors (i.e., indices) with various ranges of maturity (e.g., 0 to 5 years, 5 to 10 years, etc.), the effect of maturity on so-called 'safe' withdrawal rates, SWR (i.e., the amount of inflation adjusted income, expressed as a fraction of the initial portfolio value, that can be drawn from a portfolio without running out of money within a defined period) for UK retirees has been investigated. For a portfolio containing 50% equities and 50% fixed income, a mean difference in the SWR between the best and worst performing fixed income sectors of about 60 basis points was found. The mean difference was higher for lower equity allocations and vice versa. By ranking the performance of the sectors for each historical retirement, it was found that intermediate maturities (in particular, the 5 to 10 years, under 10 years, and 5 to 15 years sectors) generally had the best overall performance, while long (over 15 years) and short (3 month bills and under 5 year gilts) maturities tended to have the worst.

Stollman, J. (2024). "Obtaining Leveraged Returns in a Retirement Account." In: *The Journal of Investing* 33(6), pp. 96–106.
A leveraged-index fund structure is presented that does not require reindexing and does not suffer from path dependence. As such, it is appropriate for long-term investment, including being suitable for retirement accounts because investors will never lose more than their initial investment. By applying this leveraged fund structure to an index, such as the S&P 500 or Dow Jones Industrial Average, that is designed to rise over the long term, investors can obtain outsized returns that beat market averages.

Stone, C. A. and Zissu, A. (2024). "Distribution Strategies for Non-Eligible Beneficiaries of Individual Retirement Accounts under the 2019 SECURE Act." In: *The Journal of Retirement*, jor.2024.1.166.
This paper develops a model to guide financial planners and beneficiaries of inherited individual retirement accounts (IRAs) in evaluating distribution strategies under the SECURE Act of 2019. This act changed the previous rule that allowed IRA beneficiaries to stretch out required minimum distributions based on life expectancy, known as a 'stretch IRA." Before 2020, this allowed younger beneficiaries to shield asset returns from income tax for extended periods. The SECURE Act eliminated this option for most beneficiaries, except those categorized as 'eligible designated beneficiaries." We focus on traditional IRA beneficiaries constrained by the 10-year distribution rule, modeling their choice between annuitizing distributions over 10 years or making a single distribution at the end of 10 years. Our model illustrates that, under certain realistic assumptions, these choices offer definitive rankings in terms of wealth maximization over the 10-year horizon after inheritance.

Suarez, E. D. (2020). "The perfect withdrawal amount over the historical record." In: *Financial Services Review* 28(2), pp. 96–132.
What has been the perfect withdrawal amount (PWA) from retirement savings accounts in longterm historical data? The PWA is that which, if taken out in the first year of retirement and used again every year adjusted by inflation, leaves exactly the desired final balance on the account. We present the formula for obtaining this measure and evaluate the values it has taken in the past under varying combinations of the relevant parameters. We find that safety-minded investors should enter retirement with a higher stock allocation than what is currently used in most investment funds designed to provide income during retirement.

Suarez, E. D., Suarez, A., and Walz, D. T. (2015). "The Perfect Withdrawal Amount: A Methodology for Creating Retirement Account Distribution Strategies." In: *Financial Services Review* 24(4).
We present a new way to develop withdrawal strategies from retirement portfolios. It is derived analytically, instead of from empirical testing, and it iterates always in the same manner. It is based on a new measure we develop, the Perfect Withdrawal Amount, for which we discuss how to construct a probability distribution and how to apply it sequentially. We also derive a new measure of sequencing risk. We present new strategies built with this framework.

Suhonen, A., Lennkh, M., and Perez, F. (2017). "Quantifying Backtest Overfitting in Alternative Beta Strategies." In: *The Journal of Portfolio Management* 43 (2), pp. 90–104.
The authors investigate the biases in the backtested performance of "alternative beta"strategies using a unique sample of 215 trading strategies developed and promoted by global investment banks. Their results lend support to the cautions in the recent literature regarding backtest overfitting and lack of robustness in trading strategy performance during the "live"period (out of sample). The authors report a median 73 percent deterioration in Sharpe ratios between backtested and live performance periods for the strategies, and they establish a link between performance deterioration and strategy complexity, with the realized reduction in live versus backtested Sharpe ratios of the most complex strategies exceeding those of the simplest ones by over 30 percentage points. The robustness of strategy exposure to risk factors varies between asset classes and strategies; it appears reasonable in equity volatility and FX carry strategies but quite weak in the equity value strategy in particular.

Sun, J. and Lan, H. (2022). "Solving the Decumulation Puzzle with Dynamic Asset Allocation and Annuities." In: *The Journal of Wealth Management* 25(1), pp. 135–151.
This article presents a framework that solves the post-retirement investment and consumption problem by maximizing the utility of consumption over time. The model produces optimal solutions on asset allocation, consumption, and annuitization in retirement with consideration for longevity/mortality risks. The optimized investment solution is a function of wealth and age that can accommodate for different risk aversions and life-expectancies. We find that annuities offer the most utility for investors with higher wealth relative to their guaranteed income and are more effective when purchased within certain age bands. Compared to traditional glide-path type approaches, the optimized solution provides life-time consumption profiles that are superior in both the average and downside cases.

Sun, Q. (2019). "Dynamic Retirement Financial Planning Model." PhD thesis. University of Connecticut.
We developed a retirement financial planning strategy based on Markov chain modeling of retirement health conditions and Geometric Brownian Motion modeling of asset values. The annual living expenses of a retiree are modeled as basic expenses plus discretionary expenses. Our goal is to solve for the maximum discretionary expenses while healthy, which was obtained using a closed-form solution and quantile optimization technique. The highlight of this model is the use of Kalman Filter for annual recalibration. It allows the model to automatically adjust the suggested amount of discretionary expenses by looking at daily fund values from previous year. After running a lot of simulations and testings, we showed that our dynamic model beats other static models and a naive recalibration model in the sense that it virtually eliminates ruin and is able to let a retiree withdraw the largest possible amount.

Sun, W., Triest, R. K., and Webb, A. (2008). "Optimal Retirement Asset Decumulation Strategies: The Impact of Housing Wealth." In: *Asia-Pacific Journal of Risk and Insurance* 3(1).
A considerable literature examines the optimal decumulation of financial wealth in retirement. We extend this research to incorporate housing, which comprises the majority of most households' non-pension wealth.We estimate the relationship between the returns on housing, stocks, and bonds, and simulate a variety of decumulation strategies incorporating reverse mortgages. We show that homeowner's reversionary interest, the amount that can be borrowed through a reverse mortgage, is a surprisingly risky asset. Under our baseline assumptions, we find that the average household would be as much as 24 percent better off taking a reverse mortgage as a lifetime income relative to what appears to be the most common strategy: delaying tapping housing wealth until financial wealth is exhausted and then taking a line of credit. In addition, we show that housing wealth displaces bonds in optimal portfolios, making the low rate of participation in the stock market even more of a puzzle.

Sutcliffe, C. (2015). "Trading death: The implications of annuity replication for the annuity puzzle, arbitrage, speculation and portfolios." In: *International Review of Financial Analysis* 38, pp. 163–174.
Annuities are perceived as illiquid financial securities. Long positions in annuities can be offset using life insurance and debt. The ability to offset annuities may help to reduce the annuity puzzle. Offsetting annuities may also improve the efficiency of the annuities market. Annuities are perceived as being illiquid financial instruments,

and this has limited their attractiveness to consumers and their inclusion in financial models. However, short positions in annuities can be replicated using life insurance and debt, permitting long positions in annuities to be offset, or short annuity positions to be created. The implications of this result for the annuity puzzle, arbitrage between the annuity and life insurance markets, and speculation on expected longevity are investigated. It is argued that annuity replication could help reduce the annuity puzzle, improve the price efficiency of annuity markets and promote the inclusion of annuities in household portfolios.

Swedroe, L. (2021). "The Role of Financial Risk Tolerance in Investment Policy." In: *Advisor Perspectives*.
Evaluating an investor's ability, willingness and need to take risk, and then designing his or her portfolio accordingly, are the most crucial functions of investment/financial planning and the "fintech" software that supports the profession. Thus, the ability to estimate an investor's financial risk tolerance (FRT) is key to the success of a financial plan. Financial advisors know that psychological factors, such as financial anxiety, financial satisfaction (the knowledge that one has "enough"), obsession with money, personality type, self-esteem and sensation-seeking behavior, are all important contributors in determining an investor's FRT - the willingness to accept financial risk.

Szado, E. (2019). "The portfolio diversification potential of long VIX futures and options strategies." In: *SSRN e-Print*.
It is well established that the VIX Index tends to be negatively correlated with equity markets. This suggests that VIX futures and options may have the potential to provide significant diversification benefits for traditional portfolios. However, since the term structure of VIX futures is generally upward sloping, long VIX futures positions can place a significant drag on portfolio performance.In this paper we consider the performance of strategies that buy VIX futures or VIX call options in a portfolio context in 2008 and 2016, as well as over a 10+ year period from 2006 to 2017. In addition, we consider alternative strategies including long S&P 500 protective put strategies and the dynamic S&P 500 plus VIX call buying strategy of the VXTH index.Meaningful portfolio diversification benefits for risk-averse investors are possible over particular time periods with small allocations to long VIX futures or call options, but there can be substantial portfolio drag if large allocations are made over long time periods during which there are flat to rising stock markets.

Taljaard, B. H. and Maré, E. (2021). "Why has the equal weight portfolio underperformed and what can we do about it?" In: *Quantitative Finance* 21(11), pp. 1855–1868.
It is widely noted that market capitalisation weighted portfolios are inefficient and underperform an equal weighted portfolio over the long-term. However, at least since 2016, an equal weighted portfolio of stocks in the S&P500 has significantly underperformed the market capitalisation weighted portfolio. In this paper, we analyse this underperformance using stochastic portfolio theory. We show that the equal weighted portfolio does appear to outperform the market capitalisation weighted portfolio over the long-term but with periods of significant short-term underperformance. In addition, we find that concentration in the market capitalisation weighted portfolio has increased in recent years and has contributed to the recent underperformance together with a significantly lower level of diversification benefits. Furthermore, we highlight an approach to improve the performance of a portfolio by dynamically selecting a market cap or an equal weighting using a rudimentary linear regression model.

Taljaard, B. and Mare, E. (2019). "Too Much Rebalancing Is Not a Good Thing." In: *SSRN e-Print*.
There is now an abundance of literature showing that the equal weighted portfolio outperforms the value weighted portfolio. However, this has led to claims that the act of rebalancing itself within an equal weight strategy is what leads to outperformance, whether or not individual security returns are mean-reverting. In this paper we show that you can achieve the same, if not better, returns than that of the equal weighted portfolio rebalanced monthly by rebalancing less than half of the time. This is achieved by considering only whether the diversification benefit is increasing or decreasing over some period of time.

Tayali, S. T. (2020). "A novel backtesting methodology for clustering in mean–variance portfolio optimization." In: *Knowledge-Based Systems* 209, p. 106454.
The decisions of asset selection and allocation lie at the heart of financial portfolio management. For these challenging tasks, the mathematical programming model of the mean-variance optimization problem proposes to use the concept of diversification. The novel methodology in this article is a representation of the accumulated knowledge of this model from the modern portfolio theory. It is a practical application for portfolio managers to help synthesize the available historical data and to infer rational decisions. The state-of-the-art backtesting methodology integrates the unsupervised machine learning method of clustering analysis into the mean-variance portfolio optimization model. The test results from the proposed novel methodology show that clustering with

Euclidean distance measures outperform the results of the benchmark and other specified clustering methods for different datasets, backtesting periods, and temporal scales of major stock indices.

Templin, N. (2021). "When to Tap Social Security: The Sometimes Surprising Answer From Online Calculators." In: *Wall Street Journal*.
The conventional wisdom is to wait as long as possible. But that isn't always the best advice.

Tergersen, A. (2018). "Forget the 4% Rule: Rethinking Common Retirement Beliefs." In: *Wall Street Journal*.
Three retirement-savings rules come into question as stocks and bonds get more expensive and retirements last longer.

Tertilt, M. and Scholz, P. (2018). "To Advise, or Not to Advise Robo-Advisors Evaluate the Risk Preferences of Private Investors." In: *The Journal of Wealth Management* 21(2), pp. 70–84.
Robo-advisors promise efficient, rational, and transparent investment advisory. The authors analyze how robo-advisors determine their users risk tolerance and which equity exposure is derived from the individual risk profile. Findings indicate significant differences in the quality of offered investment advice. Robo-advisors usually ask relatively few questions in the assessment of their users risk profile, and it is particularly surprising that some of the questions do not seem to have any impact on the risk categorization. Moreover, the recommended equity exposure is relatively conservative.

Thanki, H., Karani, A., and Goyal, A. K. (2020). "Psychological antecedents of financial risk tolerance." In: *The Journal of Wealth Management* 23(2), pp. 36–51.
Research indicates that financial risk tolerance (FRT) is a subjective and complex phenomenon and may diverge from individual to individual based on their demographics, genetic makeup, socioeconomic profiles, personality types, and psychological constructs. Past researchers have mainly focused on relating the socioeconomic profile, demographics, and personality type of investors to FRT. This article uses a novel approach that relates other critical behavioral and psychological factors, such as financial anxiety, financial satisfaction, obsession with money, personality type, self-esteem, and sensation-seeking behavior, as equally important contributors for determining the FRT. Confirmatory factor analysis and structural equation modeling techniques were used to test the variables and hypotheses. The results show that financial satisfaction is negatively correlated with FRT, whereas financial anxiety, obsession with money, personality type, self-esteem, and sensation-seeking behavior are positively correlated with FRT. All these variables act as antecedents of FRT.

Tharp, D. T. and Kitces, M. E. (2018). "Life-Cycle Earnings Curves and Safe Savings Rates." In: *The Journal of Retirement* 5(3), pp. 109–120.
Analyses of savings rates needed for succcessful retirement rates (SSRs) typically assume constant real earnings growth throughout one's career. However, data on the life-cycle earnings patterns of millions of U.S. workers suggest that earnings growth does not occur at a constant rate that matches inflation. Instead, earnings tend to increase at a decreasing rate during the early years of one's career and decrease at an increasing rate in the later years. Utilizing simulations of saving and dissaving throughout the life cycle based on both historical market returns and forecasted returns, the authors examine the impact of assuming more realistic earnings growth relative to constant inflation-adjusted growth. Results indicate that failing to account for more realistic earnings curves throughout the life cycle may overstate SSRs for lower-income households while understating SSRs for higher-income households, and understate SSRs for younger households while overstating SSRs for older households. Furthermore, historical SSRs of 10% or less are found for all but the highest-income households after accounting for more realistic earnings curves and Social Security benefits-though variations in specific effects may exist based on the simulation methods utilized.

Thompson, J. R., Feng, L., Reesor, R. M., Grace, C., and Metzler, A. (2021). "Measuring Financial Advice: Aligning Client Elicited and Revealed Risk." In: *SSRN e-Print*.
Financial advisors use questionnaires and discussions with clients to determine a suitable portfolio of assets that will allow clients to reach their investment objectives. Financial institutions assign risk ratings to each security they offer, and those ratings are used to guide clients and advisors to choose an investment portfolio risk that suits their stated risk tolerance. This paper compares client Know Your Client (KYC) profile risk allocations to their investment portfolio risk selections using a value-at-risk discrepancy methodology. Value-at-risk is used to measure elicited and revealed risk to show whether clients are over-risked or under-risked, changes in KYC risk lead to changes in portfolio configuration, and cash flow affects a client's portfolio risk. We demonstrate the effectiveness of value-at-risk at measuring clients' elicited and revealed risk on a dataset provided by a private Canadian financial dealership of over 50,000 accounts for over 27,000 clients and 300 advisors. By measuring both elicited and revealed risk using the same measure, we can determine how well a client's portfolio aligns

with their stated goals. We believe that using value-at-risk to measure client risk provides valuable insight to advisors to ensure that their practice is KYC compliant, to better tailor their client portfolios to stated goals, communicate advice to clients to either align their portfolios to stated goals or refresh their goals, and to monitor changes to the clients' risk positions across their practice.

Thorp, S., Bateman, H., Dobrescu, L., Newell, B., and Ortmann, A. (2020). "flicking the switch: Simplifying disclosure to improve retirement plan choices." In: *Journal of Banking & Finance* 121, p. 105955.

Standardized information disclosures aim to help people compare complex financial products and make better choices. We investigate the extent to which information shown in a regulator-mandated dashboard helps retirement savers choose between alternative pension plans. We conduct incentivized experiments that collect participants' repeated choices between two pension plans using the mandatory dashboard, and subsequently test whether an even simpler dashboard improves choices, and by how much. Participants switch quickly from a high- to a low-fee pension plan when they see explicit nominal fees but are significantly more confused by percentage fees and adjust slower. When differences between plan performance arise from gross returns, not fees, we find that complex information formats can seriously hinder participants' recognition and reactions. We present a Bayesian updating model which estimates the relative noisiness of the signals from fees and gross returns across different treatments and use this model to show how better information presentation raises retirement savings.

Tian, W. and Zhu, Z. (2020). "Optimal Investing after Retirement Under Time-Varying Risk Capacity Constraint." In: *arXiv e-Print*.

This paper explores an optimal investing problem for a retiree facing longevity risk and living standard risk. We formulate the optimal investing problem as an optimal portfolio choice problem under a time-varying risk capacity constraint. Under the specific condition on model parameters, we show that the value function is a C*C solution of the HJB equation and derive the optimal investment strategy in terms of second-order ordinary differential equations. The optimal portfolio is nearly neutral to the stock market movement if the portfolio's value is at a sufficiently high level; but, if the portfolio is not worth enough to sustain the retirement spending, the retiree actively invests in the stock market for the higher expected return. In addition, we solve an optimal portfolio choice problem under a leverage constraint and show that the optimal portfolio would lose significantly in stressed markets. This paper shows that the time-varying risk capacity constraint has important implications for asset allocation in retirement.

Todd, T. M., Heckman, S., and MacDonald, M. (2023). "Examining the Conventional Wisdom of Municipal Bond Investments and Use in Financial Planning." In: *Journal of Financial Planning* 36(10), pp. 66–82.

Conventional financial planning wisdom surrounding municipal bonds suggests that as income increases, so should the likelihood of municipal bond investment (to offer legal tax avoidance and minimization). Municipal bonds are generally seen as a "safer" investment due to traditionally lower default rates. Thus, conventional wisdom would also suggest that as willingness to take financial risk decreases, the likelihood of municipal bond investment should increase. Because municipal bonds are a more specialized investment option (i.e., for tax minimization), their use should be associated with higher levels of financial knowledge. Relatedly, because municipal bonds are a more niche investment (compared to corporate stock, for example), their use should increase with use of a financial or other adviser. Using the 2019 Survey of Consumer Finances, this study examines that conventional wisdom empirically and other aspects of municipal bond investment, including the associations between such ownership and professional investment advice. This study finds that municipal bond investment is positively associated with interest and dividend income, net worth, objective financial knowledge, and the use of a financial planner. It also finds that municipal bond investment is negatively associated with willingness to take financial risk.

Tomlinson, J. (2017). "Coping with Sequence Risk: How Variable Withdrawal and Annuitization Improve Retirement Outcomes." In: *Advisor Perspectives*.

Both the level and the sequence of investment returns will have a big impact on retirement outcomes. Poor returns during the early years of retirement are bad news. However, the particular withdrawal strategy used affects sequence risk, and an approach where withdrawals are variable and respond to portfolio performance can improve retirement outcomes. I'll examine the evidence and then use my own modeling to show how a strategy that combines variable withdrawals with partial annuitization using a single-premium immediate annuity (SPIA) maximizes the cash available for consumption.

Tomlinson, J. (2018). "Retirement Strategies in Pictures." In: *Advisor Perspectives*.

Advisors providing retirement recommendations need to evaluate strategies and present options to clients. Communication is key and it can be a challenge to come up with the best way to compare alternatives. One way involves presenting metrics such as expected bequests and plan failure probabilities. However, it may be more helpful to use a graphical approach to show the year-by-year progression of funds available during retirement. I'll demonstrate a graphical approach by comparing a variety of strategies.

Tomlinson, J. (2020a). "New Estimates of the Need for Long-Term Care." In: *Advisor Perspectives*.
When the subject of long-term care (LTC) comes up, advisors need to be able to discuss with clients the likelihood of needing such care and its potential duration. These discussions feed into decisions about whether to purchase LTC insurance. In this February 2020 article (and APViewpoint conversation), Allan Roth got the discussion going in terms of both the probabilities of needing care and an evaluation of insurance options. Recently I have analyzed additional data that improves our understanding of LTC needs and informs the decision-making about insurance.

Tomlinson, J. (2020b). "The Unimportance of Asset Allocation in Retirement Planning." In: *Advisor Perspectives*.
Earlier this year, Jeremy Siegel said that, "75/25 is the new 60/40," a recommendation to raise stock allocations to make up for lower bond yields. However, what matters for investors saving for retirement is not the asset class performance, but how those returns translate into retirement consumption. To examine the impact of increasing stock allocations, I'll focus on retirement planning and show impacts on both the average and variability of retirement withdrawals. Spoiler alert - as the title hints, 75/25 versus 60/40 doesn't matter that much.

Torri, G., Giacometti, R., and Paterlini, S. (2023). "Penalized enhanced portfolio replication with asymmetric deviation measures." In: *Annals of Operations Research* 332(1-3), pp. 481–531.
Passive investment strategies, such as those implemented by Exchange Traded Funds (ETFs), have gained increasing popularity among investors. In this context, smart beta products promise to deliver improved performance or lower risk through the implementation of systematic investing strategies, and they are also typically more cost-effective than traditional active management. The majority of research on index replication focuses on minimizing tracking error relative to a benchmark index, implementing constraints to improve performance, or restricting the number of assets included in portfolios. Our focus is on enhancing the benchmark through a limited number of deviations from the benchmark. We propose a range of innovative investment strategies aimed at minimizing asymmetric deviation measures related to expectiles and quantiles, while also controlling for the deviation of portfolio weights from the benchmark composition through penalization. This approach, as compared to traditional minimum tracking error volatility strategies, places a greater emphasis on the overall risk of the portfolio, rather than just the risk relative to the benchmark. The use of penalization also helps to mitigate estimation risk and minimize turnover, as compared to strategies without penalization. Through empirical analysis using simulated and real-world data, we critically examine the benefits and drawbacks of the proposed strategies in comparison to state-of-the-art tracking models.

Totten, T. L. and Siegel, L. B. (2019). "Combining Conventional Investing with a Lifetime Income Guarantee:A Blueprint for Retirement Security." In: *The Journal of Retirement*.
Both retirees and their employers can benefit from combining conventional DC-type savings plans, designed to be spent down in the early part (say, the first 20 years) of retirement, with a DB-like or deferred life annuity component guaranteeing income for the rest of the retiree life in order to hedge the risk of living too long. Earlier articles proposed to do this using a riskless investment strategy, and the authors build on the previous work by asserting that the DC-type investment will (and should) include equities. Through simulation, the authors show that either a high-risk (equity-heavy) or low-risk (fixed income-heavy) DC component plus a deferred annuity component dominates a conventional DC spend-down, from the standpoint of generating sustainable retirement income, in most states of the world.

Traccucci, P., Dumontier, L., Garchery, G., and Jacot, B. (2019). "A Triptych Approach for Reverse Stress Testing of Complex Portfolios." In: *Risk (Cutting Edge)*.
Pascal Traccucci, Luc Dumontier, Guillaume Garchery and Benjamin Jacot present an extended reverse stress test (ERST) triptych approach with three variables: level of plausibility, level of loss and scenario. Any two of these variables can be derived, provided the third is given as input. A new version of the Levenberg-Marquardt optimisation algorithm is introduced to derive the ERST in certain complex cases.

Trainor, W. J., Chhachhi, I., and Brown, C. L. (2019). "Leaping Black Swans." In: *The Journal of Investing*.
This article examines an option based portfolio insurance strategy where a fixed percentage of the portfolio is used to purchase in the money long-term call options with the remainder invested in a standard investment grade bond fund. Our results show a 90/10 portfolio, where 10% is invested in long term call options, has returns commensurate with the S&P 5000 while mitigating losses. These call option based portfolios don require rebalancing during the term of coverage and are flexible enough to suit investors with very different risk tolerances

and portfolio sizes. As constructed, these portfolios outperform put option based portfolio insurance strategies and perform as well as a CPPI, with the added advantage of not needing to be actively managed.

Trainor, W. J. and Wampler, E. B. (2022). "Leveraged ETF Option Barbells." In: *The Journal of Beta Investment Strategies* 13(3), pp. 66–75.
This article compares how Standard & Poor's (S&P) 500 and Nasdaq-100 barbell strategies invested primarily in fixed income assets and in-the-money call options on the underlying index (or their 2x and 3x leveraged counterparts) achieve a significant percentage of upside appreciation while reducing downside risk. From 2010 to 2022, a barbell strategy composed of 88% 7 to 10-year Treasury bonds and 12% 0.7 delta 6-month call options on the underlying index achieves more than 81% of the geometric annual return of the index while reducing major losses by 21% to 52%, as exemplified by the market declines in the last quarter of 2018 and the 2020 COVID-19 crash. Using call options on the 2x or 3x LETF performs marginally worse than simply using options on the index and only gives a 32% to 47% participation rate of the underlying LETF returns. However, the outlined 88/12 barbell strategy is exposed to interest rate risk, and in the 10 months ending June 30, 2022, the 88/12 barbells lost -16% or more as the options lost more than 90% of their value while treasuries declined -10.4%.

Traut, J., De Nard, G., Cirulli, A., and Walker, P. S. (2025). "Low Risk, High Variability: Practical Guide for Portfolio Construction." In: *SSRN e-Print*.
The low-risk anomaly challenges traditional financial theory by stating that less volatile stocks generate higher risk-adjusted returns. This paper explores how various portfolio construction choices influence the performance of low-risk portfolios. We show that methodological decisions critically influence portfolio outcomes, causing substantial dispersion in performance metrics across weighting schemes and risk estimators. This can only be marginally mitigated by incorporating constraints such as short-sale restrictions and size or price filters. Our analysis reveals that volatility-based estimators yield the most favorable performance distribution, outperforming beta-based approaches. Transaction costs are found to significantly affect performance and are vitally important in identifying the most attractive portfolios, highlighting the importance of realistic implementation constraints. Through rigorous empirical analysis, this study bridges the gap between theoretical insights and practical applications, offering actionable guidance to investors. The findings advocate for a cautious approach to nonstandard errors in portfolio modeling and emphasize the necessity of robust strategies in low-risk investing.

Tretiakova, I. and Yamada, M. S. (2017). "Autonomous portfolio:a decumulation investment strategy that will get you there." In: *The Journal of Retirement* 5(2), pp. 83–95.
The article primary contribution is applying a dynamic decision theory approach to investment management and adding spending flexibility to improve incomes for retirees for whom saving more and working longer are no longer options. We investigate whether a self-driving portfolio, engineered to protect against market risk, can deliver more income than other investment approaches while minimizing the risk of ruin. We consider whether a dynamic strategy, based on a stochastically dominant decision theory algorithm and tested with fixed and variable spending policies, can minimize the chance of running out of money before dying. Outcomes are compared with a number of static approaches. We find that the dynamic strategy provides an average of 51% more income than a target date fund through retirement when combined with a hybrid spending rule over a relevant probability of ruin range of 1% through 25%. Furthermore, when rebalanced to constant risk rather than to fixed asset mixes, sequence of return risk can be reduced. Variable spending policies add effectiveness and should be considered regardless of investment strategy. Importantly, dynamic decision theory establishes a framework for goal-seeking solutions that may have application in other phases of the investor life cycle.

Tsai, C. C.-L. and Cheng, E. S. (2021). "Incorporating statistical clustering methods into mortality models to improve forecasting performances." In: *Insurance: Mathematics and Economics* 99, pp. 42–62.
Statistical clustering is a procedure of classifying a set of objects such that objects in the same class (called cluster) are more homogeneous, with respect to some features or characteristics of objects, to each other than to those in any other classes. In this paper, we apply four clustering approaches to improving forecasting performances of the Lee-Carter and CBD models. First, each of four clustering methods (Ward's hierarchical clustering, divisive hierarchical clustering, K-means clustering, and Gaussian mixture model clustering) is adopted to determine, based on some characteristics of mortality rates, the number and partition of age clusters from the whole study ages 25-84. Next, we forecast 10-year and 20-year mortality rates for each of the age clusters using the Lee-Carter and CBD models, respectively. Finally, numerical illustrations are given with two R packages "NbClust" and "mclust" for clustering. Mortality data for both genders of the US and the UK are obtained from the Human Mortality Database, and the MAPE (mean absolute percentage error) measure is adopted to evaluate forecasting

performance. Comparisons of MAPE values are made with and without clustering, which demonstrate that all the proposed clustering methods can improve forecasting performances of the Lee-Carter and CBD models.

Turvey, P. A., Basu, A. K., and Verhoeven, P. (2013). "Embedded Tax Liabilities and Portfolio Choice." In: *The Journal of Portfolio Management* 39(3), pp. 93–101.

Taxes play an important role in determining after-tax investment risk and returns, but many practitioners still make investment decisions based on pre-tax values. The apparent complexity of dealing with deferred capital gains and the question of how these implied future tax liabilities should be valued are central to this problem. The authors use a simple arbitrage argument to show that a risk-free discount rate is appropriate for calculating the present value of future tax liabilities. This lets analysts adjust risk and returns for effective tax rates and present a more accurate picture to the investor. The results show a taxation-induced preference for holding equities over bonds and a location preference for holding equities in a taxable account and bonds in retirement accounts. These important findings contrast with traditional investment advice that suggests a greater capacity for risk in retirement accounts.

Valentine, K. D., Buchanan, E. M., Scofield, J. E., and Beauchamp, M. T. (2019). "Beyond p values: utilizing multiple methods to evaluate evidence." In: *Behaviormetrika* 46(1), pp. 121–144.

Null hypothesis significance testing is cited as a threat to validity and reproducibility. While many individuals suggest that we focus on altering the p value at which we deem an effect significant, we believe this suggestion is short-sighted. Alternative procedures (i.e., Bayesian analyses and observation-oriented modeling: OOM) can be more powerful and meaningful to our discipline. However, these methodologies are less frequently utilized and are rarely discussed in combination with NHST. Herein, we discuss three methodologies (NHST, Bayesian Model comparison, and OOM), then compare the possible interpretations of three analyses (ANOVA, Bayes Factor, and an Ordinal Pattern Analysis) in various data environments using a frequentist simulation study. We found that changing significance thresholds had little effect on conclusions. Furthermore, we suggest that evaluating multiple estimates as evidence of an effect allows for more robust and nuanced interpretations of results and implies the need to redefine evidentiary value and reporting practices. Recent events in psychological science have prompted concerns within the discipline regarding research practices and ultimately, the validity and reproducibility of published reports (Etz and Vandekerckhove 2016; Lindsay 2015, Open Science Collaboration 2015; van Elk et al. 2015). One often discussed matter is over-reliance, abuse, and potential hacking of p values produced by frequentist null hypothesis significance testing (NHST), as well as misinterpretations of NHST results (Gigerenzer 2004; Ioannidis 2005; Simmons et al. 2011). We agree with these concerns and believe that many before us have voiced sound, generally accepted opinions on potential remedies, such as an increased focus on effect sizes (Cumming 2008; Lakens 2013; Maxwell et al. 2015; Nosek et al. 2012). However, other suggestions have been met with less enthusiasm, including an article by Benjamin et al. (2018) advocating that researchers should begin thinking only of p values less than .005 as "statistically significant", thus changing alpha levels to control Type I error rates. Alternatively, Pericchi and Pereira (2016) promote the use of fluctuating alpha levels as a function of sample size to assist with these errors. Trafimow et al. (2018) critiques this suggestion to broadly lower the alpha level to .005 and suggested that findings should be weighted on the basis of evidence accumulation from multiple studies. We argue that alpha should not be the sole focus of our attention, but rather, we should wonder if a p value should be utilized at all, and, if so, what that p value can tell us in relation with other indicators. While NHST and p values may have merit, researchers have a wealth of other statistical tools available to them. We believe that improvements may be made to the sciences as a whole when individuals become aware of these tools and how these methods may be used, either alone or in combination, to strengthen understanding of data and conclusions. These sentiments have been shared by the American Statistical Association who recently held a conference focusing on going beyond NHST, expanding their previous stance on p values (Wasserstein and Lazar 2016). Therefore, the main goal of this project was to show researchers how two alternative paradigms compare to NHST in terms of their methodological design, statistical interpretations, and comparative robustness. Herein, we will discuss the following methodologies: NHST, Bayes factor comparisons, and observation-oriented modeling. To compare their methodological designs, we first provide historical backgrounds, procedural steps, and limitations for each paradigm. We then simulated data using a three timepoint repeated measures design with a Likert-type scale as the outcome variable to be able to compare the statistical interpretations and comparative robustness. By simulating possible data sets and analyzing them with each of the three paradigms, we will be able to discuss the conclusions these three methods reach given the same data and to compare how often these methodologies agree within different data environments (i.e., given varying sample sizes and effect sizes). Beyond simply comparing methodologies, we also sought to identify how changing the alpha criteria within the NHST framework may alter conclusions. Although previous work has already compared Frequentist NHST to Bayesian approaches (Goodman 1999; Rouder et al. 2012; Wetzels et al. 2011), this manuscript adds a novel contribution: observation-oriented modeling. By introducing social scientists to observation-oriented modeling (OOM), a relatively new paradigm that is readily interpretable, we will show both how useful this paradigm can be in these contexts, and how it compares to two well-known methods. We hope that by discussing these methodologies in terms of a simple statistical analysis researchers will be able to easily compare and contrast methodologies.

Vallarino, D. (2024). "Dynamic Portfolio Rebalancing: A Hybrid new Model Using GNNs and Pathfinding for Cost Efficiency." In: *arXiv e-Print*.

This paper introduces a novel approach to optimizing portfolio rebalancing by integrating Graph Neural Networks (GNNs) for predicting transaction costs and Dijkstra's algorithm for identifying cost-efficient rebalancing paths. Using historical stock data from prominent technology firms, the GNN is trained to forecast future transaction costs, which are then applied as edge weights in a financial asset graph. Dijkstra's algorithm is used to find the least costly path for reallocating capital between assets. Empirical results show that this hybrid approach significantly reduces transaction costs, offering a powerful tool for portfolio managers, especially in high-frequency trading environments. This methodology demonstrates the potential of combining advanced machine learning techniques with classical optimization algorithms to improve financial decision-making processes. Future research will explore expanding the asset universe and incorporating reinforcement learning for continuous portfolio optimization.

van Bilsen, S. and Bovenberg, A. L. (2020). "The decumulation period of a personal pension with risk sharing: investment approach versus consumption approach." In: *Journal of Pension Economics and Finance* 19(2), pp. 262–291.

This paper models the decumulation period of a Personal Pension with Risk sharing (PPR). We derive several relationships between the contract parameters. Individuals can adopt two approaches to the decumulation period of a PPR: the investment approach and the consumption approach. In the investment approach, individuals specify how to invest wealth and how much wealth to withdraw. Retirement consumption follows endogenously. In the consumption approach, in contrast, individuals specify retirement consumption exogenously. Investment and withdrawal policies follow endogenously. We explore these two approaches in detail. Consistent with habit formation, we allow for excess smoothness and excess sensitivity in retirement consumption.

van Bilsen, S. and Linders, D. (2019). "Affordable and adequate annuities with stable payouts: Fantasy or reality?" In: *Insurance: Mathematics and Economics* 86, pp. 19–42.

This paper introduces a class of unit-linked annuities that extends existing annuities by allowing portfolio shocks to be gradually absorbed into the annuity payouts. Consequently, our new class enables insurers to offer an affordable and adequate annuity with a stable payout stream. We show how to price and adequately hedge the annuity payouts in a general financial environment. In particular, our model accounts for various stylized facts of stock returns such as asymmetry and heavy-tailedness. Furthermore, the generality of our framework makes it possible to explore the impact of a parameter misspecification on the annuity price and the hedging performance.

Van Harlow, H. V. and Brown, P. (2016a). "Market Risk, Mortality Risk, And Sustainable Retirement Asset Allocation: A Downside Risk Perspective." In: *Journal of Investment Management* 14(2), pp. 5–32.

Despite its clear importance, there is no consensus on the optimal asset allocation strategy for retirement investors of varying age, gender, and risk tolerance. This study analyzes the allocation question by focusing on the downside risks that result from the joint uncertainty over investment returns and life expectancy. Using a new analytical approach, we show that concentrating on the severity of retirement funding shortfalls, rather than just the probability of ruin, markedly increases the sustainability of a retirement portfolio. We demonstrate that for retirement investors attempting to minimize downside risk while sustaining future withdrawals, appropriate equity allocations range between five and 25 percent, levels that are strikingly low compared to those typically found in life-cycle funds. Further, these optimal portfolio constructions appear to vary little with alternative capital market assumptions.We also show that more aggressive investors having substantial bequest motives should still be relatively conservative in their stock allocations. We conclude that the higher equity allocations commonly employed in practice significantly underestimate the risks that these higher-volatility portfolios pose to the sustainability of retirement savings and incomes.

Van Harlow, W. (2014). *Optimal Asset Allocation in Retirement: A Downside Risk Perspective*. Tech. rep. Putnam Institute.

Once an individual has retired, asset allocation becomes a critical investment decision. Unfortunately, there is no consensus on what the optimal allocation should be for retirees of varying age, gender and risk tolerance. This study analyzes the allocation question through a focus on the downside risks created by uncertainty over investment returns and life expectancy. We find that the range of appropriate equity asset allocations in retirement is strikingly low compared with those of typical lifecycle and retirement funds now in the marketplace. In fact, for retirement portfolios whose primary goal is to minimize the risk of depletion and sustain withdrawals, optimal equity allocations range between 5 percent and 25 percent. This quite conservative level of equity holdings changes little even when we significantly change our assumptions on capital market returns. We even find that more aggressive equity allocations, those that still retain some focus on depletion risk but also seek to provide substantial bequests to heirs, are also relatively conservative. The study suggests, in short, that the higher equity allocations used in many popular retirement investment products today significantly underestimate the risks that these higher-volatility portfolios pose to the sustainability of retirees' savings and to the incomes on which retirees depend.

Van Harlow, W. and Brown, K. C. (2016b). "Improving the Outlook for a Successful Retirement: A Case for Using Downside Hedging." In: *The Journal of Retirement* 3(3), pp. 35–50.

One of the biggest risks to a successful retirement is the exposure of savings to poor investment returns in the early stages of the retirement. Mitigating this "sequence-of-returns"risk is in consequence an important investment question. In this study, we conduct extensive simulation analysis to show that for sustainable withdrawal rates, hedging with costless collars or with put options can eliminate or significantly reduce funding shortfall risk for a retirement portfolio. In addition, we demonstrate with a few examples that, for a given level of shortfall risk, hedging can increase the income generated by retirement savings by almost 40 percent. Thus, downside hedging strategies within retirement portfolios appear to offer attractive benefits to retirees worried about outliving their income resources.

Van Harlow, W. and Brown, K. C. (2017). "Health State and the Savings Required for a Sustainable Retirement." In: *The Journal of Retirement* 4(4), pp. 25–38.

Life expectancy varies greatly with health conditions, but very rarely is this relationship incorporated into retirement investment planning. This is surprising given that the vast majority of older households have one or more adverse health conditions. The authors show that the savings required to fund a successful retirement for someone with one of several diseases whose impact on life expectancy has been estimated can be reduced by as much as 26 percent for females and 33 percent for males relative to the savings required for a healthy individual. Similarly, the savings required to fund healthcare expenses in retirement can be reduced by 29 percent to 39 percent. The interaction of projected health costs and the effect of disease on life expectancy have counterintuitive effects on retirement planning. For example, the higher retiree healthcare expenses associated with conditions such as diabetes and tobacco use are offset by reduced life expectancies. The net effect is that, in certain health states, less savings are required for healthcare than would be necessary for a healthy individual.

Van Harlow, W., Brown, K. C., and Jenks, S. E. (2020). "The Use and Value of Financial Advice for Retirement Planning." In: *The Journal of Retirement* 7(3), pp. 46–79.

Offering professional advice around the retirement planning process represents an important component of the financial services industry. The authors examine the demographic, investment, and behavioral characteristics of individuals who obtain this advice as well as the economic value that it ultimately adds. Using a survey of more than 4,000 working households, they find that wealth and income levels are positively correlated with the decision to engage a professional advisor, as are factors such as marital status, age, and education level. To assess the value added by this advice, the authors develop a unique metric of retirement income replacement that incorporates health-based life expectancy and household-specific financial circumstances. The approach estimates the percentage of annual pre-retirement income that a household will be able to spend each year in retirement. The authors establish the unconditional finding that advised households generate significantly larger proportions of post-employment spending (both gross and net of Social Security benefits) than do nonadvised households. Controlling for additional explanatory factors, we find that an advisor adds more than 15 percentage points of income replacement in retirement. These findings support the conclusion that obtaining and implementing financial advice in the retirement planning process leads to a demonstrable increase in the level of sustainable retirement spending.

van Loon, R. J. M. (2022). "Balancing Long-Term Goals versus Short-Term Risks." In: *The Journal of Investing* 32(1), pp. 132–149.

In investing, there can sometimes be a tension between long-term goals and short-term risks. The investor might have a specific end goal in mind when structuring an investment portfolio, but the realization of a short-term risk in the interim can force a stop out before the end goal is achieved. One can think of margin calls, solvency triggers, or even behavioral effects. These are situations where it is not only the end goal that matters, but also the journey toward it. In this article, the author derives the probability of the investor reaching a target hurdle value at the end of the investment horizon without breaching a lower barrier value during the investment horizon. The addition of an intrahorizon loss constraint can lead to meaningfully different investment behavior. The article describes three stylized examples to demonstrate the practical consequences and provides some tools on how to manage the trade-off.

Vandenbroucke, J. and Fortuna, G. (2019). "Loss Aversion Implied by a Risk-Based Questionnaire." In: *The Journal of Wealth Management* 22(1), pp. 39–48.

This article uses data from an existing classical risk-based questionnaire to define subgroups of investors that are expected to exhibit a different attitude towards loss. Field research confirms that differences in revealed loss attitude match the model prediction even when selecting investors with the same classical risk profile. The study should motivate to define investor profiles based on two coordinates rather than just one, meaning a combination of risk and latitude vis-a-vis losses. Such behavioral investor profiles improve customer centricity, contribute to a long-term relationship and simply increase the likelihood of right-selling individual products.

Vanguard Research (2020). *Dynamic spending: A better way to budget in retirement*. Tech. rep. Vanguard.

Financial markets are unpredictable – they can bring tremendous gains one day and post painful declines the next. If you're close to retirement or already retired, this can pose a significant concern – but it doesn't have to. Investors use a number of methods to figure out how much they can spend in retirement, but one, in particular, stands out: a dynamic spending approach. Dynamic means it's flexible – by making minor changes to your spending in any given year in response to the markets' performance, you can help sustain your portfolio during market downturns (while sacrificing little) and have more to spend following high-performing years.

Vaughan, C. (2017). "RMD Arbitrage: Strategies for Legally Delaying and Reducing RMDs." In: *Journal of Financial Planning* 30(6), pp. 49–57.

Most American households have an age gap of greater than one year between the spouses. For couples with an age gap, required minimum distributions (RMDs) will start in different years. This disparity creates an opportunity to maximize late-life portfolio values by maximizing the assets subject to the younger spouse's RMD schedule. This paper identifies numerous strategies to maximize the assets in the younger spouse's account. The primary benefit of these strategies is that they allow the couple to maximize the tax deferral of the assets by delaying and reducing RMDs. Although a surviving younger spouse would typically want to roll the assets into their own IRA, a surviving older spouse may benefit from keeping the assets in an inherited spousal IRA in certain situations.

Ventura-Marco, M., Vidal-MeliA, C., and Perez-Salamero GonzAlez, J. M. (2023). "Joint life care annuities to help retired couples to finance the cost of long-term care." In: *Insurance: Mathematics and Economics* 113, pp. 122–139.

This paper examines the possibility of including cash-for-care benefits in life care annuities (LCAs) to help retired couples to cope with the cost of long-term care (LTC). The paper develops an actuarial model to assess how much it would cost to add an extra stream of payments to annuities for couples should either or both require LTC. We also analyse how willing couples would be to choose this type of LCA. Using Australian LTC transition probability data for a realistic calibration and assuming independence of the risks involved, we numerically illustrate the model and the theoretical findings implied. The paper highlights the importance of reporting the joint life expectancy and the survivor life expectancy broken down by health state, given that this information makes the computation of the actuarial factors transparent and provides highly useful information to help the couple understand the need to be protected against the cost of LTC services. Our proposal could be welfare-improving, especially for "dual-earner" couples, given that under the assumption that coordinated retirement behaviour exists within the couple, the availability of LCAs for couples may complete the market over and above the use of products for individuals alone.

Veres, B. (2020). "A Comparison of Risk Tolerance Products." In: *Advisor Perspectives*.

Bob Veres compares Risk Tolerance Products from Andes Wealth Tech, Finametrica, Riskalyze and Totum Risk. Bear markets beget portfolio losses, unhappy clients and, sadly, lawsuits against advisors. If you relied on any of the popular risk tolerance products to construct that portfolio, here's how you are likely to fare under the careful scrutiny of an arbitrator.

Vincent, K., Hsu, Y.-C., and Lin, H.-W. (2018). "Analyzing the Performance of Multifactor Investment Strategies under a Multiple Testing Framework." In: *The Journal of Portfolio Management* 44(4), pp. 113–126.
Evaluating portfolios based on numerous combinations of factors using the individual backtesting method could suffer from serious data mining bias and lead to spurious significant findings. Accordingly, the authors employ a multiple hypothesis testing method to examine the multifactor portfolio performance. Their empirical results show that even after they adjust for the multiple comparisons bias, stock-picking strategies with certain combined firm characteristics could generate significantly better liquidity risk-adjusted returns. In addition, the outperforming multifactor strategies that the authors report are robust to alternative definitions of factors. However, they observe that the number of significantly profitable multifactor portfolios has decreased substantially in the era of increased liquidity and trading activity in the U.S. stock market.

Vovk, V. and Wang, R. (2020). "True and false discoveries with e-values." In: *arXiv e-Print*.
The topic of this paper is multiple hypothesis testing based on e-values, which are Bayes factors stripped of their Bayesian content. Using e-values instead of p-values, which are standard in this area, leads to simple and efficient procedures that control the number of false discoveries under arbitrary dependence of the base e-values. We prove an optimality result for our main procedure and demonstrate advantages of our methods over standard methods using simulated and real-world datasets.

Vovk, V. and Wang, R. (2021). "E-values: Calibration, combination, and applications." In: *Annals of Statistics* 49(3), pp. 1736–1753.
Multiple testing of a single hypothesis and testing multiple hypotheses are usually done in terms of p-values. In this paper we replace p-values with their natural competitor, e-values, which are closely related to betting, Bayes factors, and likelihood ratios. We demonstrate that e-values are often mathematically more tractable; in particular, in multiple testing of a single hypothesis, e-values can be merged simply by averaging them. This allows us to develop efficient procedures using e-values for testing multiple hypotheses.

Vrdoljak, N., Laster, D., and Suri, A. (2014). "The Role of Variable Annuities in Addressing Retirement Risks." In: *The Journal of Retirement* 2(2), pp. 55–66.
Investors commonly fund their retirement through portfolios of stocks and bonds or mutual funds, which offer scant protection against the risk of a market downturn or of outliving their wealth. This article examines how adding a variable annuity with a guarantee lifetime withdrawal benefit (VA + GLWB) to a balanced portfolio can help reduce the risk of running out of money in retirement. It quantifies, through simulation analysis, the degree to which an allocation to a VA + GLWB can help mitigate two key retirement risks: longevity and sequence of returns risk. This risk reduction comes at the cost of a decrease in the expected future bequest. The article also explores how the timing of withdrawals and the asset allocation within a VA + GLWB impacts retirement outcomes.

Waggle, D. and Agrrawal, P. (2024). "Guaranteed Income and Optimal Retirement Glide Paths." In: *SSRN e-Print*.
Deciding on asset allocations over time and portfolio withdrawal rates remain key considerations for retirees. These decisions are impacted by factors such as the investor's wealth level, degree of risk aversion, the desire to leave a bequest, and the presence of guaranteed income, such as Social Security. This paper considers these issues as they relate to utility levels for retirees. Initial equity allocations from 0 to 100 percent were examined. For investors with moderate and high levels of risk aversion, higher proportions of Social Security relative to overall wealth led to higher initial optimal allocations to stock. This is somewhat intuitive because the higher level of guaranteed income allows for more risk with the remainder of the portfolio. However, retirees who rely more on Social Security are less wealthy, and lower levels of wealth are generally equated with lower allocations to stock. Five different glide paths, which provide plans for changes in allocations to stocks over time, were considered: increasing the weight of stock slowly, increasing fast, decreasing slowly, decreasing fast, and constant. The typical guidance for retirees is to decrease their allocation to stock over time. This paper finds, however, that increasing glide paths, where the level of stock increases over time, are often optimal.

Waggle, D., Lee, H., and Moon, G. (2024). "Efficient Portfolios with Social Security as an Asset." In: *Journal of Financial Planning* 37(12), pp. 86–101.
Social Security provides retirees with a quasi-guaranteed lifetime income stream that increases with inflation and can be considered a government-backed "annuity." This paper examines the impact of including the value of expected future Social Security payments as an asset, along with stocks and bonds, in efficient portfolios. The value of Social Security to a retiree depends on the individual's life expectancy, the level of benefit payments, and market interest rates, all of which change over time. The mean and standard deviation of the returns on Social Security were estimated based on its expected annual payouts and the changes in its valuation over time.

Based on this paper's estimations, Social Security provides a higher return than government bonds with a lower level of risk. The paper's results show that including Social Security in mean-variance efficient portfolios with stocks and bonds resulted in considerably higher allocations to stocks (relative to the stock-bond total) at the beginning of retirement compared to the guidance when Social Security was not included. The value of Social Security declines over time as retirees age and face higher probabilities of death. As this happens, the optimal allocation to stocks as a percentage of financial assets gradually declines and is offset by an increase in bonds.

Walker, P., Sacks, B. H., and Sacks, S. R. (2021). "To Reduce the Risk of Retirement Portfolio Exhaustion, Include Home Equity as a Non-Correlated Asset in the Portfolio." In: *Journal of Financial Planning* 34(12), pp. 82–97.
The conventional wisdom dictates that you should take distributions from your portfolio first and only from home equity as a backup. But including home equity as part of the portfolio can greatly increase the probability of maintaining constant purchasing power throughout retirement.

Wallick, D. W., Berkowitz, D. B., Clarke, A. S., DiCiurcio, K. J., and Stockton, K. A. (2018). "Getting More from Less in Defined Benefit Plans." In: *How Persistent Low Returns Will Shape Saving and Retirement*. Oxford University Press.
As global interest rates hover near historic lows, defined benefit pension plan sponsors must grapple with the prospect of lower investment returns. We examine three levers that can enhance portfolio outcomes in a low-return world: increased contributions; reduced investment costs; and increased portfolio risk. We use portfolio simulations based on a stochastic asset class forecasting model to evaluate each lever according to two criteria: the magnitude of impact and the certainty that this impact will be realized. We show that increased contributions have the greatest and most certain impact. Reduced costs have a more modest, but equally certain impact. Increased risk can deliver a significant impact, but with the least certainty.

Wang, H., Suri, A., Laster, D., and Almadi, H. (2011). "Portfolio Selection in Goals-Based Wealth Management." In: *The Journal of Wealth Management* 14(1), pp. 55–65.
The authors propose an incremental step toward combining the insights of modern portfolio theory with some of the propensities documented in the literature on behavioral finance. They develop a goals-based wealth management approach that finds a specific subportfolio to address each of an investor's goals and then derive the least-cost solution. T hey relate the closed-form solution for the one-period, two-asset problem to the mean-variance efficient frontier. Consistent with the - lockbox separation- concept proposed by Sharpe, they demonstrate that a multiperiod goal, such as a retirement plan, can be viewed as a collection of single-period problems. Next, they extend their result to a market with many assets, where portfolios are exogenously given. Finally, they illustrate the approach with a case study with multiple asset classes and multiperiod goals.

Wang, Z., Crawford, A., Lee, K. L., and Jaganathan, S. (2023). "reslife: Residual Lifetime Analysis Tool in R." In: *arXiv e-Print*.
Mean residual lifetime is an important measure utilized in various fields, including pharmaceutical companies, manufacturing companies, and insurance companies for survival analysis. However, the computation of mean residual lifetime can be laborious and challenging. To address this issue, the R package reslife has been developed, which enables efficient calculation of mean residual lifetime based on closed-form solution in a user-friendly manner. reslife offers the capability to utilize either the results of a flexsurv regression or user-provided parameters to compute mean residual lifetime. Furthermore, there are options to return median and percentile residual lifetime. If the user chooses to use the outputs of a flexsurv regression, there is an option to input a data frame with unobserved data. In this article, we present reslife, explain its underlying mathematical principles, illustrate its functioning, and provide examples on how to utilize the package. The aim is to facilitate the use of mean residual lifetime, making it more accessible and efficient for practitioners in various disciplines, particularly those involved in survival analysis within the pharmaceutical industry. This package has been approved and available on CRAN: https://cran.r-project.org/web/packages/reslife/index.html.

Waring, M. B. and Siegel, L. B. (2018). "What Investment Risk Means to You, Illustrated Strategic Asset Allocation, the Budget Constraint, and the Volatility of Spending During Retirement." In: *The Journal of Retirement* 6(2), pp. 7–26.
In our experience, many investment professionals don articulate risk well to clients. This research uniquely and graphically reveals the nature of strategic asset allocation investment risk not only for single-period investors, but also for multi-period investors such as those whose savings are held for retirement. Using Monte Carlo simulations, we evaluate and picture the nature of that multi-period consumption risk with spending rules that are subject to a budget constraint and those that aren (such as the 4% rule). Risk in a multi-period context means that realized spending may increase with greater SAA risk, but it may instead be worse.

Warren, G. J. (2019). "Design of Investment Options using Utility Functions: A Demonstration for Products." In: *SSRN e-Print*.
Utility functions can assist in designing a menu of investment options by encoding the objectives and preferences for various investor types with the aim of developing suitably tailored products. This paper demonstrates the process through the construction of an illustrative menu of Australian MyRetirement products. The demonstration shows how utility-based analysis can be combined with more traditional metrics to compare strategies and convey their implications; and provides direction on how product options might be presented to a retiring member. Central to the process is characterizing investor types by selected attributes. This both facilitates specifying utility functions for use in product development, and supports communication based around the type of investor for which a product is designed along with its key features and the outcomes it may deliver. Utility functions thus provide the mechanics that drive the analysis, without becoming an explicit component of investor engagement.

Warshawsky, M. (2018). "Reforming Retirement Income: Annuitization, Combination Strategies, and Required Minimum Distributions." In: *SSRN e-Print*.
Laddered immediate life annuity purchase and asset withdrawal combination strategies represent an excellent way for retirees to manage their retirement assets in order to get lifetime income in a flexible manner while still maintaining growth, liquidity, and bequest potentials. Simulations show that even by age 95, a retiree using this strategy will get higher income, in inflation-adjusted terms, on average and across most scenarios than by using full and complete annuitization or just using the common 4 percent withdrawal rule. Moreover, a significant fund balance remains to the retiree throughout life, on average, but with no risk of running out of assets. The minimum distribution requirements that govern tax-qualified retirement accounts for older retirees should be reformed to encourage the use of these partial annuitization combination strategies. This broad reform of the required minimum distribution rules needs to be done for the same reason the Obama administration made a narrow exception to the required minimum distribution rules for longevity insurance to achieve the reasonable public policy goal of encouraging the use of partial annuitization by retirees.

Weber, E. U. and Klement, J. (2018). "Risk tolerance and circumstances." In: *CFA Institute Research Foundation Briefs* 4(2).
An investor risk attitude is a stable characteristic, like a personality trait, but risk-taking behavior can change based on the investor age, recent market events, and life experiences. These factors change investors perceptions of the risks. Differences in risk tolerance between men and women or in different circumstances trace back to emotional as much as rational considerations. Financial advisers should consider all of these factors when advising clients and can use four simple steps to incorporate best practices: be aware, educate, nudge, and hand hold. The term risk tolerance is defined and used in different ways. Whether risk tolerance is a stable characteristic of a given investor or also takes into account external circumstances (e.g., economic shocks or the domain of the decision) depends on how it is defined and measured. This brief focuses on a definition of risk tolerance prevalent in the practitioner community, an investor willingness to take perceived risk or the trade-off an investor is willing to make between the perceived risk and expected return of different investment choices. This definition derives from a psychological interpretation of the risk-return framework of classical portfolio theory. It treats risk tolerance as an attitude toward risk and decouples this pure attitudinal variable from the perceptions of risks and returns variables in their own right and distinct from the expected value and variance of the distribution of possible outcomes. Defined in this way, risk tolerance may differ among investors as a function of socioeconomic and biological differences but (with the exception of a brief boost during adolescence) shows stability across an investor lifespan, financial shocks, and other circumstances. Risk tolerance, in this sense, is the mediator that translates perceptions of risk and situational needs and constraints into decision and action. The variables that change with market conditions and other circumstances are investors perceptions of investment risks and expectations of return. In contrast to risk tolerance, which attaches to an individual and her biological makeup and personality, these variables change over time in response to changing external conditions. Therefore, an investor risk-taking behavior (as revealed by her investment decisions) can look like it has changed, despite the stability in her risk tolerance. Perceived risks and expected returns are influenced by hopes and fears as much as by past returns and rational expectations and thus need to be assessed in their own right and possibly corrected. Managing investor emotions through the ups and downs of financial markets is arguably a financial adviser most important task. Calm times provide an opportunity to discuss and formulate an investment policy for each client that can be consulted when emotions are running high. Managing risk perceptions requires the financial adviser to act more like a therapist than a mechanic. It is above all about managing expectations and emotions and helping clients to better deal with emotions when it comes to financial decisions. The end result

of this process might be a portfolio that is not in the sense of modern portfolio theory, with its assumption of econs, but rather a portfolio that the human need for investments that can be handled in the presence of changing emotions and changing risk perceptions.

Wesslen, R., Karduni, A., Markant, D., and Dou, W. (2022). "Effect of uncertainty visualizations on myopic loss aversion and the equity premium puzzle in retirement investment decisions." In: *IEEE Transactions on Visualization and Computer Graphics* 28(1), pp. 454–464.

For many households, investing for retirement is one of the most significant decisions and is fraught with uncertainty. In a classic study in behavioral economics, Benartzi and Thaler (1999) found evidence using bar charts that investors exhibit myopic loss aversion in retirement decisions: Investors overly focus on the potential for short-term losses, leading them to invest less in riskier assets and miss out on higher long-term returns. Recently, advances in uncertainty visualizations have shown improvements in decision-making under uncertainty in a variety of tasks. In this paper, we conduct a controlled and incentivized crowdsourced experiment replicating Benartzi and Thaler (1999) and extending it to measure the effect of different uncertainty representations on myopic loss aversion. Consistent with the original study, we find evidence of myopic loss aversion with bar charts and find that participants make better investment decisions with longer evaluation periods. We also find that common uncertainty representations such as interval plots and bar charts achieve the highest mean expected returns while other uncertainty visualizations lead to poorer long-term performance and strong effects on the equity premium. Qualitative feedback further suggests that different uncertainty representations lead to visual reasoning heuristics that can either mitigate or encourage a focus on potential short-term losses. We discuss implications of our results on using uncertainty visualizations for retirement decisions in practice and possible extensions for future work.

Wiafe, O. (2015). "Investment strategies in retirees' decumulation phase." PhD thesis. Queensland University of Technology.

This thesis consists of three studies on investment strategies for Australian retirees. Specifically, it investigates retirees' preference between alternative drawdown strategies in the presence of government pensions, appropriate management of longevity risk through the use of deferred annuities and asset allocation in retirement. It finds drawdown strategies linked to life expectancy to be the best performers. Deferred annuities are found to improve retirement incomes for risk averse retirees. For retirees who want to meet certain wealth thresholds in retirement, equity dominated portfolios provide superior outcomes for higher threshold levels.

Wiafe, O. K., Basu, A. K., and Chen, E. T. (2020). "Portfolio choice after retirement: Should self-annuitisation strategies hold more equities?" In: *Economic Analysis and Policy* 65, pp. 241–255.

We compare the performance of alternative investment strategies in the decumulation phase for retirees who self-annuitise. We find that portfolios with constant high exposure to equities, as well as portfolios that increase exposures to equities over time, consistently outperform conservative portfolios which avoid investment in equities or the conventional lifecycle portfolios which reduce allocation to equities over time. While an increasing equity glidepath improves the performance of an investment strategy, the allocations to equities at the start of retirement is critical. Using a "utility of terminal wealth" approach that allows for loss aversion as in prospect theory of Kahneman and Tversky (1979), we find a growth portfolio with very high (but not total) exposure to equities to dominate the alternative strategies at low and moderate thresholds. With further increases in wealth threshold levels, a strategy with an all equity allocation becomes dominant. The lifecycle portfolio is dominated by the "reverse lifecycle" portfolio at all threshold levels. Finally, for retirees who defer annuitisation, we find strategies with a higher equity component have greater likelihood of outperforming an immediate annuitisation strategy in terms of generating a guaranteed income late in retirement.

Wiafe, O. K., Basu, A. K., and Chen, E.-T. ((2016). "Asset Allocation in Retirement: Does Glide Path Matter?" In: *SSRN e-Print*.

We compare the performance of the commonly nominated default retirement investment option, the lifecycle fund, to alternative investment strategies during retirees' decumulation phase. Under different shortfall risk measures, we find balanced portfolios with constant exposure to equities, equity dominated portfolios as well as 'reverse lifecycle' portfolios that increase exposures to equities over time to consistently outperform the conventional lifecycle portfolio. While an increasing equity glidepath improves the performance of an investment strategy, the starting asset allocations are equally important. Using a utility-of-terminal wealth approach which allows for loss aversion as discussed in prospect theory by Kahneman and Tversky (1979), we find the Growth portfolio to dominate the alternative strategies at low and moderate thresholds. With increasing wealth threshold levels, a strategy with all equity allocations becomes dominant. The lifecycle portfolio is dominated by the 'reverse lifecycle' portfolio at all threshold levels.

Wiafe, O. K., Basu, A. K., and Chen, E.-T. J. (2014). "Portfolio Strategies in Decumulation Phase: Does Lifecycling Fail?" In: *SSRN e-Print*.

We compare the performance of the commonly nominated default retirement investment option, the lifecycle fund, to alternative investment strategies during retirees' decumulation phase. Under different shortfall risk measures, we find balanced portfolios with constant exposure to equities, equity dominated portfolios as well as 'reverse lifecycle' portfolios that increase exposures to equities over time to consistently outperform the conventional lifecycle portfolio. While an increasing equity glidepath improves the performance of an investment strategy, the starting asset allocations are equally important. Using a utility-of-terminal wealth approach which allows for loss aversion as discussed in prospect theory by Kahneman and Tversky (1979), we find the Growth portfolio to dominate the alternative strategies at low and moderate thresholds. With increasing wealth threshold levels, a strategy with all equity allocations become dominant. The lifecycle portfolio is dominated by the 'reverse lifecycle' portfolio at all threshold levels.

Wiecki, T., Campbell, A., Lent, J., and Stauth, J. (2016). "All That Glitters Is Not Gold: Comparing Backtest and Out-of-Sample Performance on a Large Cohort of Trading Algorithms." In: *The Journal of Investing* 25(3), pp. 69–80.

When automated trading strategies are developed and evaluated using backtests on historical pricing data, there exists a tendency to overfit to the past. Using a unique dataset of 888 algorithmic trading strategies developed and backtested on the Quantopian platform, with at least six months of out-of-sample performance, this article studies the prevalence and impact of backtest overfitting. Specifically, the authors find that commonly reported backtest evaluation metrics, such as the Sharpe ratio, offer little value in predicting out-of-sample performance (R2 $<$ 0.025). In contrast, higher-order moments, such as volatility and maximum drawdown, as well as portfolio construction features (e.g., hedging), show significant predictive value of relevance to quantitative finance practitioners. Moreover, in line with prior theoretical considerations, the authors find empirical evidence of overfitting-the more backtesting a quant has done for a strategy, the larger the discrepancy between backtest and out-of-sample performance. Finally, they show that by training nonlinear, machine-learning classifiers on a variety of features that describe backtest behavior, out-of-sample performance can be predicted with much greater accuracy (R2 = 0.17) on hold-out data than when using linear, univariate features. A portfolio constructed by using predictions on hold-out data performed significantly better out-of-sample than one constructed from algorithms with the highest backtest Sharpe ratios.

Wiegrebe, S., Kopper, P., Sonabend, R., Bischl, B., and Bender, A. (2024). "Deep learning for survival analysis: a review." In: *Artificial Intelligence Review* 57(3).

The influx of deep learning (DL) techniques into the field of survival analysis in recent years has led to substantial methodological progress; for instance, learning from unstructured or high-dimensional data such as images, text or omics data. In this work, we conduct a comprehensive systematic review of DL-based methods for time-to-event analysis, characterizing them according to both survival- and DL-related attributes. In summary, the reviewed methods often address only a small subset of tasks relevant to time-to-event data-e.g., single-risk right-censored data-and neglect to incorporate more complex settings. Our findings are summarized in an editable, open-source, interactive table: `https://survival-org.github.io/DL4Survival`. As this research area is advancing rapidly, we encourage community contribution in order to keep this database up to date.

Wilcox, J., Bodie, Z., and di Bartolomeo, D. (2023). "Lean Advice for New Investors." In: *Journal of Investment Consulting* 22(1), pp. 19–30.

As defined benefit pensions have been replaced by investor-directed defined contribution plans, implementing sound investment policies for retail investors of modest means has become a problem of increasing urgency. This group comprises most of the investor population across all countries. In the U.S. it is characterized by inadequate saving, failure to take advantage of materially higher payout for delayed Social Security benefits, extensive credit card debt, and pursuit of naive investment policies. While sophisticated financial advice is available to large investors, the associated high costs prohibit access by many others. This paper provides an affordable prescriptive blueprint for them, particularly for new investors. While somewhat US-centric, our framework can be put to practical use in any country. We envision implementation as a low-cost tier service using simple technology that permits mass customization with limited in-person contact.

Williams, P. D. (2021). "Inflation Expectations in the U.S.: Linking Markets, Households, and Businesses." In: *SSRN e-Print*.

Inflation has been below the Federal Reserve's target for much of the past 20 years, creating worries that inflation may be deanchoring from the FOMC's target. This paper uses a factor model that incorporates information from professional forecasters, household and business surveys, and the market for Treasury inflation protected securities (TIPS) to estimate long-run inflation expectations. These have fallen notably in the past few years (to roughly 1.9 percent for CPI inflation, well below the FOMC's target). It appears that, even before the covid recession, the private sector viewed the economy as likely to suffer from persistent headwinds to inflation.

Winter, P. and Planchet, F. (2022). "Modern tontines as a pension solution: a practical overview." In: *European Actuarial Journal* 11.

In the context of global aging population, improved longevity and ultra-low interest rates, the question of pension plan under-funding and adequate elderly financial planning is gaining awareness worldwide, both among experts, regulatory bodies, and popular media. Additional emergence of societal changes-Peer to Peer business model and Financial Disintermediation-have contributed to the resurgence of the concept of "Tontines" in various papers and the proposal of further models. These generalizations can offer efficient decumulation schemes with high longevity protection which is particularly well adapted for retirement needs-both for its members and carriers. In this paper, we revisit the mechanism proposed by Fullmer and Sabin (Journal of Accounting and Finance, 2019.)-which allows the pooling of Modern Tontines through a self-insured community. This "Tontine" generalization retains the flexibility of an individual design: open contribution for a heterogeneous population, individualized asset allocation and predesigned annuitization plan. The actuarial fairness is achieved by allocating the deceased proceedings to survivors using a specific individual pool share which is a function of the prospective expected payouts for the period considered. After a brief introduction, this article provides a formalization of the mathematical framework with prospective analysis, characterizes the inherent bias, generalizes the mechanism to joint lives, and analyses simulated outcomes based on various assumptions. A reverse moral hazard limit is exposed and discussed (the "Term Dilemma"). Some solutions are then proposed to overcome scheme shortcomings and some requirements for practical implementation are discussed.

Woerheide, W. J. and Nanigian, D. (2011). "Sustainable Withdrawal Rates from Retirement Portfolios: The Historical Evidence on Buffer Zone Strategies." In: *SSRN e-Print*.

This paper evaluates the performance of portfolios that dynamically adjust one's mix of risky and risk-free assets during retirement based on the performance of risky assets. We analyze these "buffer zone" portfolio decumulation strategies over various time periods from 1926 to 2009. We find that they are generally inferior to a simpler static asset allocation at minimizing longevity risk.

Wolfe, B. and Brazier, R. (2018). *Spending retirement assets ... or not?* Tech. rep. BlackRock.

Something unexpected has been the shared experience for our most recent generation of retirees. The vast majority haven't been spending their retirement savings - leaving nest eggs mostly untouched and living on ready sources of income instead. However, future retirees may be less fortunate.

While on the surface this is indeed good news - and appears to support the argument that fears of a future retirement crisis are overstated, the conditions that supported this spending and savings behavior are unlikely to persist. Future retirees will face a much different retirement landscape and will need to adopt new sets of skills – behavioral and financial – that will help them systematically tap into retirement savings to support future spending.

Financial industry norms and academic theories have always assumed assets accumulated for retirement would be systematically withdrawn - following the "4% rule" or some other rule of thumb or system - by retirees in order to maintain a consistent standard of living. Technically, this is referred to as "consumption smoothing" whereby individuals seek to have consistent spending on par with pre-retirement levels. With concerns that retirement savings for individuals may be dangerously low,1 the fear has been that withdrawals for such smoothing could leave retirees running out of funds well short of their passing away.

This research conducted by the BlackRock Retirement Institute (BRI) in conjunction with the Employee Benefit Research Institute (EBRI) found that on average across all wealth levels, most current retirees still have 80% of their pre-retirement savings after almost two decades in retirement.

This is significant because:

- These findings begin to challenge industry norms and academic theories about lifecycle consumption especially during the retirement phase
- Across all wealth levels measured, more than one third of current retirees grew their assets – leaving considerable potential consumption on the table

- Late in life out-of-pocket medical expenses – a major reason to retain assets – do not appear to be warranted except for a very small portion of the population
- The financial landscape for future retirees will most likely be more challenging, requiring different savings and spending behaviors

This paper sets to lay the foundation for how retirees have managed their sources of cash– assets and income– against their spending behaviors. The resulting "husbanding" of assets over the past two decades may be due to a host of favorable environmental factors current retirees benefited from during their working, accumulation years. These included beneficial changes to Social Security and Medicare, a relatively high percentage of jobs that offered defined benefit pensions, strong real estate appreciation and an investment market that generally delivered strong returns and high interest rates.
Has the confluence of these factors created a situation whereby retirees may not have felt the pressure to draw down principal from retirement savings in order to maintain a reasonable standard of living? Perhaps retirees had other plans for their assets beyond themselves – bequests or charitable donations come to mind. Possibly they would have preferred to spend more freely but lacked the financial confidence or tools to efficiently decumulate their assets or were worried about end-of-life healthcare expenses? Looking further, perhaps there were strong emotional biases at play – with fear of outliving retirement assets at the top of the list.
Methodology
The objective of the study was to analyze the "how" and some possible "whys" spending and liquid assets change during retirement, taking into account (non-housing) assets, income, spending, out-of-pocket medical expenses and bequests. Data was collected from the bi-annual Health and Retirement Study (HRS, 1992-2014) and the Consumption and Activities Mail Survey (CAMS, 2005-2015). A sample of 7,148 retiree households provided self-reported asset data and out-of-pocket medical expenditure and a subsample of 1,660 households provided the household expenditure data. Retirees were segmented into three groups based on pre-retirement non-housing retirement assets – $0 to less than $200,000 (lowest wealth), $200,000 to less than $500,000 (medium wealth) and $500,000 and above (highest wealth).

Wolfe, B. and Ferraro, M. (2022). *Decumulation challenges and potential solutions – Helping build a pathway towards retirement spending and income confidence*. Tech. rep. BlackRock.
Something unexpected has been the shared experience for many of our most recent generation of retirees that will reverberate across the investment and insurance industries. The vast majority haven't been spending their retirement savin – leaving nest eggs mostly untouched and living on ready sources of income instead. However, future retirees may be less fortunate, and may need to spend down principal in order to fund their spending needs. In this paper, we summarize the findings of The BlackRock Retirement Institute's research and white paper, Spending retirement assets ... or not? and provide a framework for advisors to engage with their clients around the challenges in store as they transition from asset accumulation to retirement income.
Key findings:

1) On average across all wealth levels, most current retirees still have 80% of their pre-retirement savings after almost two decades in retirement
2) More than one third of current retirees actually grew their assets – leaving considerable potential consumption on the table
3) The financial landscape for future retirees will most likely be more challenging, requiring different savings and spending behaviors
4) Future retirees are going to need retirement income solutions that can provide spending confidence – for both essential spending needs and more discretionary "wants" – and insurance solutions can play an important role within an integrated retirement income and investment strategy
5) The potential role of insurance can be better understood with the use of illustrative tools that show investors how well insurance products can be integrated into investment plans to fully optimize hard-earned savings in retirement and spend with greater certainty

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Wu, S. (2021). "Assessing the relationship between health and household portfolio allocation." In: *Financial Planning Review* 4(4).

This paper surveys the literature on the relationship between health and household portfolio allocation and provides updated empirical analysis based on recent data. Prior research finds robust evidence for cross-sectional correlations between measures of health status and portfolio decisions, but establishing the causal pathways and underlying mechanisms has proven more difficult and complex. Analysis from the most recently available 2016 and 2018 waves of the Health and Retirement Study yields results that are consistent with existing literature. Households with worse self-reported health have a lower probability of holding various types of financial assets and invest a higher share of their portfolios in safe assets, relative to other asset categories. However, there is only weak evidence that new health shocks to a household change portfolio holdings. The paper concludes with a discussion of the implications of this research and directions for future work.

Wysocki, M. and Slepaczuk, R. (2024). "Construction and Hedging of Equity Index Options Portfolios." In: *SSRN e-Print*.

This research presents a comprehensive evaluation of systematic index option-writing strategies, focusing on S&P500 index options. We compare the performance of hedging strategies using the Black-Scholes-Merton (BSM) model and the Variance-Gamma (VG) model, emphasizing varying moneyness levels and different sizing methods based on delta and the VIX Index. The study employs 1-minute data of S&P500 index options and index quotes spanning from 2018 to 2023. The analysis benchmarks hedged strategies against buy-and-hold and naked option-writing strategies, with a focus on risk-adjusted performance metrics including transaction costs. Portfolio delta approximations are derived using implied volatility for the BSM model and market-calibrated parameters for the VG model. Key findings reveal that systematic option-writing strategies can potentially yield superior returns compared to buy-and-hold benchmarks. The BSM model generally provided better hedging outcomes than the VG model, although the VG model showed profitability in certain naked strategies as a tool for position sizing. In terms of rehedging frequency, we found that intraday heding in 130-minute intervals provided both rel.

Xu, G. (2015). "The risk profiles of 401(k) accounts." In: *The Journal of Retirement* 2(3), pp. 67–77.

This article examines the relationship between the riskiness of the portfolios of 401(k) plan participants, as gauged by the total risk, global beta, total equity, and U.S. and foreign equity exposures of their portfolios, and the participant age, gender, approach to investment, and whether he or she participates in a pension plan. The author finds evidence not only of behavioral biases affecting the investment decisions of investors with brokerage accounts, but also of two new apparent behavioral biases. First, participants with a defined-benefit pension plan do not take more risk, other things being equal, than participants without such a plan. Second, young active participants (participants who have made an active choice of their investments) take less risk compared with young passive participants, while older active participants take more risk than older passive participants.

Xu, G. (2018). "Static and Dynamic Tax Diversification of Withdrawals from Multiple Individual Retirement Accounts." In: *The Journal of Retirement* 6(2), pp. 75–87.

This article shows why diversified simultaneous retirement funding drawn from accounts with differing tax treatments will save more on taxes than sequential withdrawal. We utilize dynamic programming to quantify the optimal funding under the assumption that the goal is to maximize the total discounted after-tax consumption and bequest amounts. The dynamic programming setting uses the actual tax schedule, considers the required minimum distribution and life expectancy, and constrains consumption with upper and lower bounds. We also find the optimal funding through static diversification, a simple optimization schema. We compare the static and full dynamic programming solutions in two cases, one for a single filer and another for a married joint filer. The static and the full dynamic programming solution differ more in the single filer case than in the married joint filer case. The static optimal withdrawal is close to sequential withdrawal. The full dynamic programming optimal withdrawal is close the sequential withdrawal when both types of accounts have a substantial balance.

Xu, G. and Anichini, T. (2016). "Mean-Variance Analysis in Post-Retirement Planning." In: *The Journal of Retirement* 3(3), pp. 62–76.

In this study we analyze the variability and average level ofspending in retirement under two strategies: the popular 4 percent rule and what we will term the self-funded variable annuity strategy (SVA strategy). The 4 percent rule stipulates that each year a retiree should spend 4 percent of her initial wealthat retirement; the value of spending grows with the rateof inflation. Under the SVA strategy, each year the retiree determines her spending level as equivalent to the income she could theoretically obtain by purchasing an annuity(but without actually purchasing one). The 4 percent rule entailslower average spending and little variability, while theSVA strategy entails higher average spending and highervariability. The SVA strategy is based on a theoretical foundation of expected utility maximization. We demonstrate how a retiree can make an informed decision about allocating among stocks, bonds, and annuities, and whether to combine the 4 percent spending rule with the SVA strategy to achieve higher average spending. In particular, if future returns turnout to be lower than historical returns, as many experts predict,combining the 4 percent rule and SVA strategy may achieve spending of more than 4 percent of initial wealth on average while reducing the failure rate to less than 10 percent. We also conside rhow mortality might affect a mixed strategy that combinesa purchased annuity with the SVA strategy.

Xu, G., Markowitz, H., and Guerard, J. B. (2019). "Shortfall risk and shortfall duration for portfolio choice in decumulation." In: *The Journal of Retirement* 7(1), pp. 24–34.
The article studies the portfolio selection problem in the retirement phase by using the habit formation utility function in the context of traditional utility maximization. The habit formation utility can be further simplified to a linear combination of shortfall risk and shortfall duration. A retiree who can easily adapt to a new spending level should emphasize shortfall duration whereas a retiree who is rigid in spending should emphasize shortfall risk. The article provides the conditions in which current practitioners favorite choices of shortfall risk, as a criterion to choose retirement portfolios, are consistent with utility maximization.

Xu, J. and Hoesch, A. (2018). "Predicting longevity: an analysis of potential alternatives to life expectancy reports." In: *The Journal of Retirement* 5(4), pp. 9–24.
Retirees, pension funds, and the insurance industry have all been negatively affected by the wrongful estimation of longevity. The inaccuracies in current life expectancy (LE) reports primarily result from misinterpretations of the influence of resilience factors on longevity. This study examines different and more accurate measurement metrics to minimize the risks related to biased LE calculations. By using both qualitative and quantitative research approaches, this research develops a new conceptual model: a two-factor-LE-analysis model with a telomere test as a medical basis (physiological factors) and a big data approach to filter the psychological factors to longevity. The authors suggest that the new model, together with the insights of the existing LE-projection methodologies, has considerable potential to improve LE predictions.

Xu, M., Alonso-Garcia, J., Sherris, M., and Shao, A. W. (2023). "Insuring longevity risk and long-term care: Bequest, housing and liquidity." In: *Insurance: Mathematics and Economics* 111, pp. 121–141.
We study the impact of housing wealth and individual preferences on demand for annuities and long-term care insurance (LTCI). We build a multi-state lifecycle model that includes longevity risk and health shocks. The preference is represented by a recursive utility function that separates risk aversion and elasticity of intertemporal substitution (EIS). When health shocks are considered, a higher level of risk aversion lowers the annuity demand, while a lower level of the EIS has the opposite effect. The impact diminishes with a weaker bequest motive, more liquid wealth, or access to LTCI, all of which increase the demand for annuities. The presence of home equity can enhance annuity demand, but the enhancement is marginal when LTCI is available. The presence of home equity has a crowding-out effect on LTCI demand, and the effect is strengthened by a lack of bequest motives or a lower degree of risk aversion. The cash poor but asset rich may demand more LTCI coverage than their renter counterparts to preserve bequests. When both life annuities and LTCI are available, we find that the product demand is robust to changes in risk aversion and the EIS, providing insights into product designs that bundle annuities and LTCI.

Yang, B. (2017). "Longevity and statistical modelling." PhD thesis. Nanyang Technological University Singapore.
This dissertation consists of two studies on the modelling aspects of mortality (or longevity). In the first paper, we examine cohort extensions of the Poisson common factor model for modelling mortality of both genders jointly. Several alternatives are specified and applied to datasets from five developed regions. We find that direct parameterisation of cohort effect could improve model fitting, reduce the need for additional period factors, and produce consistent mortality forecasts for females and males. Furthermore, we find that the cohort effect appears to be gender indifferent for the populations examined and has an interaction effect with age in certain cases. The second paper explores the prediction error in mortality projection. This is important given the increasing longevity risk and the rising demand for longevity-linked products. Insofar, only parameter error and process error have been considered jointly while the issue of model error has been little studied. Here, we propose a method to account for process error, parameter error and model error in an integrated manner by modifying the semi-parametric bootstrapping technique. We apply the method to two datasets from the Continuous Mortality Investigation (CMI) and use the simulated scenarios to price the q-forward contracts with a risk-neutral approach. We find that model selection has a significant impact on the valuation results and thus it is crucial to incorporate model error in mortality projection. The third part of the dissertation surveys the current landscape of the longevity market and discusses some open issues related to the pricing of longevity products in the context of the broader literature.

Yang, Y., Chen, S., Cui, Z., and Zhang, Z. (2024). "Valuation of Guaranteed Lifelong Withdrawal Benefit with the Long-term Care Option." In: *Insurance: Mathematics and Economics*.

In this paper, under the stochastic interest rate framework, we consider the valuation of a Guaranteed Lifelong Withdrawal Benefit (GLWB) annuity product by explicitly incorporating the health state of the policyholder through the long-term care (LTC) option. The product provides policyholders with protection against longevity risk and market downturns, as well as financial support when facing LTC needs. Within the context of dynamic withdrawals, the valuation of the GLWB annuity with the LTC option is characterized as a stochastic optimal control problem. We introduce a novel bang-bang analysis approach without the usual convexity assumption in literature and prove that the optimal withdrawal strategies for the policyholder are constrained to a finite set. Furthermore, we perform a sensitivity analysis on the price determinants of GLWB annuities with and without the LTC option, and provide economic interpretations. Lastly, we investigate the impact of gender on the optimal withdrawal strategy and the fair fee of the annuity with the LTC option.

Yap, Z. J., Pathmanathan, D., and Dabo-Niang, S. (2025). "Forecasting mortality rates with functional signatures." In: *ASTIN Bulletin*, pp. 1–24.

This study introduces an innovative methodology for mortality forecasting, which integrates signature-based methods within the functional data framework of the Hyndman-Ullah (HU) model. This new approach, termed the Hyndman-Ullah with truncated signatures (HUts) model, aims to enhance the accuracy and robustness of mortality predictions. By utilizing signature regression, the HUts model is able to capture complex, nonlinear dependencies in mortality data which enhances forecasting accuracy across various demographic conditions. The model is applied to mortality data from 12 countries, comparing its forecasting performance against variants of the HU models across multiple forecast horizons. Our findings indicate that overall the HUts model not only provides more precise point forecasts but also shows robustness against data irregularities, such as those observed in countries with historical outliers. The integration of signature-based methods enables the HUts model to capture complex patterns in mortality data, making it a powerful tool for actuaries and demographers. Prediction intervals are also constructed with bootstrapping methods.

Yoon, Y. (2010). "Glide path and dynamic asset allocation of target date funds." In: *Journal of Asset Management* 11(5), pp. 346–360.

The glide path of typical target date funds is based on the relatively simple assumption of risk. If an explicit term structure of risk is present or risk is time-varying, the conventional glide path may not be adequate to fulfil the purpose of target date funds. We introduce a new approach to define the glide path of target date funds. Our starting point is to determine the level of risk budget for each target date. According to the pre-defined risk budget, we derive the asset allocation of target date funds by explicitly incorporating the current term structure of risk. As risk does change through different market phases, we implement a dynamic asset allocation strategy for target date funds that considers simultaneously both the pre-defined risk budget and the prevailing market risks. The main difference is that at any given time our risk-controlled dynamically rebalanced target date funds would not exceed the pre-defined risk budget regardless of market movements.

Youmbi, D. and Pagnani, R. (2023). "Variable Annuity Modelling and Pricing." In: *SSRN e-Print*.

In this paper we explain how to model and price a Variable Annuity. We show that selling a variable annuity contract involves being short a put option on the sub-account. We explain how to analytically estimate the forward value of the sub-account, by solving an differential equation. We provide some PnL simulations, of a typical Variable Annuity contract.

Young, R. (2020a). *Tax-Efficient Withdrawal Strategies*. Tech. rep. T. Rowe Price.

There are alternatives to the conventional strategy of drawing on a taxable account first, followed by tax-deferred and then Roth accounts. Many people can take advantage of income in a low tax bracket or tax-free capital gains. If planning to leave an estate to heirs, consider which assets will ultimately maximize the after-tax value.

Young, R. (2020b). "The Roth/Pretax Decision in Late Career Years: The Increasing Importance of Accumulated Assets in Light of the SECURE Act." In: *Journal of Financial Planning* 33(12), pp. 59–68.

A late-career worker with a choice between pretax and Roth retirement contributions should consider the level of tax-deferred assets already accumulated. That asset level affects the worker's tax rates over the course of retirement, due to required minimum distributions (which may not be needed to support spending), and also affects taxes on beneficiaries. This study analyzes the decision for a variety of scenarios by calculating breakeven asset levels (as a multiple of income) where Roth and pretax contribution strategies yield the same after-tax value for beneficiaries. Contrary to the conventional view that people in peak earning years should save using pretax contributions, the analysis shows that many people who are well on track for retirement would help their beneficiaries-without sacrificing retirement spending-by choosing the Roth strategy. The SECURE Act, passed in late 2019, requires most non-spouse beneficiaries of retirement accounts to withdraw all of the funds within 10 calendar years. That requirement can increase beneficiaries' tax rates, compared with a "stretch" strategy allowable under prior law. This study shows that breakeven asset levels are significantly lower after passage of the SECURE Act, indicating that more people would benefit from a Roth strategy now. Among the scenarios evaluated were different beneficiary characteristics, which showed that beneficiaries' future incomes and tax situations can significantly affect the breakeven asset levels.

Yu, L. (2021). "Comparing Classical Portfolio Optimization and Robust Portfolio Optimization on Black Swan Events." MA thesis. University of Waterloo.
Black swan events, such as natural catastrophes and manmade market crashes, historically have a drastic negative influence on investments; and there is a discrepancy on losses caused by these two types of disasters. In general, there is a recovery and it is of interest to understand what type of investment strategies lead to better performance for investors. In this thesis we study classical portfolio optimization, robust portfolio optimization and some historical black swan events. We compare two main strategies: mean variance optimization vs robust portfolio optimization on two types of black swan events: natural vs anthropogenic. The comparison illustrates that robust portfolio optimization is much more conservative, and has a shorter recovery time than classical portfolio optimization. Moreover, the losses in the stock investment resulted from a natural disaster are very minor compared to the losses resulted from an anthropogenic market crash.

Yu, S., Chen, Y., and Dong, C. (2021a). "Learning Time Varying Risk Preferences from Investment Portfolios using Inverse Optimization with Applications on Mutual Funds." In: *arXiv e-Print*.
The fundamental principle in Modern Portfolio Theory (MPT) is based on the quantification of the portfolio's risk related to performance. Although MPT has made huge impacts on the investment world and prompted the success and prevalence of passive investing, it still has shortcomings in real-world applications. One of the main challenges is that the level of risk an investor can endure, known as *risk-preference*, is a subjective choice that is tightly related to psychology and behavioral science in decision making. This paper presents a novel approach of measuring risk preference from existing portfolios using inverse optimization on the mean-variance portfolio allocation framework. Our approach allows the learner to continuously estimate real-time risk preferences using concurrent observed portfolios and market price data. We demonstrate our methods on real market data that consists of 20 years of asset pricing and 10 years of mutual fund portfolio holdings. Moreover, the quantified risk preference parameters are validated with two well-known risk measurements currently applied in the field. The proposed methods could lead to practical and fruitful innovations in automated/personalized portfolio management, such as Robo-advising, to augment financial advisors' decision intelligence in a long-term investment horizon.

Yu, S., Wang, H., and Dong, C. (2021b). "Learning Risk Preferences from Investment Portfolios Using Inverse Optimization." In: *arXiv e-Print*.
The fundamental principle in Modern Portfolio Theory (MPT) is based on the quantification of the portfolio's risk related to performance. Although MPT has made huge impacts on the investment world and prompted the success and prevalence of passive investing, it still has shortcomings in real-world applications. One of the main challenges is that the level of risk an investor can endure, known as *risk-preference*, is a subjective choice that is tightly related to psychology and behavioral science in decision making. This paper presents a novel approach of measuring risk preference from existing portfolios using inverse optimization on the mean-variance portfolio allocation framework. Our approach allows the learner to continuously estimate real-time risk preferences using concurrent observed portfolios and market price data. We demonstrate our methods on real market data that consists of 20 years of asset pricing and 10 years of mutual fund portfolio holdings. Moreover, the quantified risk preference parameters are validated with two well-known risk measurements currently applied in the field. The proposed methods could lead to practical and fruitful innovations in automated/personalized portfolio management, such as Robo-advising, to augment financial advisors' decision intelligence in a long-term investment horizon.

Yuan, M. and Zhou, G. (2022). "Why Naive 1/N Diversification Is Not So Naive, and How to Beat It?" In: *SSRN e-Print*.
In this paper, we study portfolio choice problem under estimation risk and show why the 1/N rule is very difficult to beat in applications and studies. First, as long as the dimensionality is high relative to sample size, we show that the usual estimated investment strategies are biased even asymptotically. Second, we show that the 1/N rule is optimal in a one-factor model with diversifiable risks as dimensionality increases, irrespectively of the

sample size, making investment theory-based rules inadequate as they suffer from estimation errors. Third, we provide strategies that can outperform the 1/N under suitable conditions.

Zhang, C., Li, Y., Chen, X., Jin, Y., Tang, P., and Li, J. (2020a). "DoubleEnsemble: A New Ensemble Method Based on Sample Reweighting and Feature Selection for Financial Data Analysis." In: *IEEE International Conference on Data Mining (ICDM)*. IEEE.

Modern machine learning models (such as deep neural networks and boosting decision tree models) have become increasingly popular in financial market prediction, due to their superior capacity to extract complex non-linear patterns. However, since financial datasets have very low signal-to-noise ratio and are non-stationary, complex models are often very prone to overfitting and suffer from instability issues. Moreover, as various machine learning and data mining tools become more widely used in quantitative trading, many trading firms have been producing an increasing number of features (aka factors). Therefore, how to automatically select effective features becomes an imminent problem. To address these issues, we propose DoubleEnsemble, an ensemble framework leveraging learning trajectory based sample reweighting and shuffling based feature selection. Specifically, we identify the key samples based on the training dynamics on each sample and elicit key features based on the ablation impact of each feature via shuffling. Our model is applicable to a wide range of base models, capable of extracting complex patterns, while mitigating the overfitting and instability issues for financial market prediction. We conduct extensive experiments, including price prediction for cryptocurrencies and stock trading, using both DNN and gradient boosting decision tree as base models. Our experiment results demonstrate that DoubleEnsemble achieves a superior performance compared with several baseline methods.

Zhang, F., Guo, R., and Cao, H. (2020b). "Information Coefficient as a Performance Measure of Stock Selection Models." In: *arXiv e-Print*.

Information coefficient (IC) is a widely used metric for measuring investment managers' skills in selecting stocks. However, its adequacy and effectiveness for evaluating stock selection models has not been clearly understood, as IC from a realistic stock selection model can hardly be materially different from zero and is often accompanies with high volatility. In this paper, we investigate the behavior of IC as a performance measure of stick selection models. Through simulation and simple statistical modeling, we examine the IC behavior both statically and dynamically. The examination helps us propose two practical procedures that one may use for IC-based ongoing performance monitoring of stock selection models.

Zhang, J. and Ma, J. (2024). "Contribution to portfolio tracking error volatility with geometric illustration." In: *Applied Economics Letters* 31(20), pp. 2176–2181.

In this study, we formulate Tracking Error Volatility (TEV) of a portfolio with matrices based on pre-defined risk factors and betting sizes and then we derive contributions to TEV from each factor and each security using the corresponding second derivatives as marginal contributions. This breakdown helps portfolio managers identify where TEV mostly comes from. Geometric illustrations are used to visualize and prove that contributions to TEV are orthogonal projections of each standalone TEV onto portfolio total TEV.

Zhang, S. (2018). "Optimal Retirement Planning: Scenario Generation, Preferences, and Objectives." PhD thesis. University of Waterloo.

The global trend of shifting from defined benefit (DB) to defined contribution (DC) workplace pension plans is putting growing pressure on individuals to take more ownership in retirement planning and financial decision-making. The essence of the DB is the life-long income guarantee, which requires limited financial planning decisions to be made, either in the accumulation or decumulation phase. The DC on the other hand, is significantly more complex. The lump sum payment at retirement burdens individuals with the task of income generation, in the presence of challenges stemming from an uncertain future lifetime, economic conditions, and evolving consumption needs. The average retiree has limited competency to navigate these challenges, due to low financial literacy, lack of willpower, or deteriorating cognitive abilities with older ages. The high stake of these challenges calls for a normative solution to be proposed - a solution that considers the intricacy of risks, preferences, and normative objective formulations. The objective of this thesis is to explore such a solution. This thesis comprises three inter-related research directions: long-term economic scenario generators (ESGs), recursive preferences in life-cycle portfolio selection, and retirement objective formulation. A brief description of the subsequent chapters will now follow. The first chapter conducts a review of Wilkie's ESG, with analysis restricted to series pertinent to retirement planning. Our main findings indicate that there exist challenges in modelling long-term economic series due to the presence of multiple structural shifts in the historical time series. Consequently, certain assumptions of stationarity are violated, and parameters are sensitive to the calibration period. A backtest based on 30-year out-of-sample data indicated that over that period the model had tended to overestimate inflation, underestimate total return on stocks, and performed relatively well for long-term interest rates. Additionally, Wilkie's ESG can be under-representative of the risk in long-term stock investment, particularly in the tails. The second chapter provides an introductory discussion of Epstein-Zin preferences, which are adopted in the succeeding chapter as a normative preference model. The purpose is to first investigate the implied optimal behaviour and its plausibility. We pay particular attention to whether the output leads to plausible behaviour given the context of retirement planning. Specifically, analytical solutions for a simple consumption problem are derived, isolating the impact of relative risk aversion (RRA), elasticity of intertemporal substitution (EIS), time discounting, and risks stemming from mortality, investment, and inflation. We investigate three Epstein-Zin models employed in the literature, which differ in their treatment of mortality risk, and find that some lead to normatively implausible solutions. Importantly, we find that the EIS is not always monotone in its effect on consumption volatility over time, meaning that its interpretation can be ambiguous when considering an uncertain future lifetime. This has been misinterpreted in the literature to date. We also show that one particular Epstein-Zin specification is not necessarily a generalization of expected utility maximization under constant relative risk aversion, as many works wrongly claim. The third chapter investigates the normative validity of the optimal consumption and investment strategies of a discrete-time Epstein-Zin utility maximizing DC retiree who wishes to benefit from stock investment, longevity insurance, and inflation protection. A comparison of three Epstein-Zin specifications is conducted. We use a combination of qualitative and quantitative criteria to evaluate the adequacy of the optimal consumption profile, with special attention paid to the downside risk at extreme old ages. We find that it remains optimal to fully annuitize, but agents with high relative risk aversion hold precautionary savings, the level of which is impacted by the EIS and the preference specification. As discussed in the preceding chapter, the interpretation of EIS on consumption volatility is found to be ambiguous. Investigations of the optimal consumption profile reveal that agents are exposed to relatively high levels of downside risk in the long run. This is partially attributed to a time discounting factor less than 1, which implicitly (and contradictorily) assumes myopia in normative decision-making. An investigation of zero time discounting is conducted, with downside risk found be to significantly reduced in the long run. The fourth chapter focuses on retirement objective formulation. This chapter is motivated by the unsatisfactory normative solutions found in the preceding chapter under mathematically convenient objective functions. In order to develop more actionable prescriptive solutions, we seek to holistically explore actual retirement decision-making. To this end, we conduct a survey study of 1,000 Canadian (pre-)retirees age 50 to 80, on topics of retirement consumption, wealth, income, risk perception, decision making, and planning objectives. Additionally, we investigate the descriptive validity of the expected lifetime discounted utility maximization framework in predicting optimal planning behaviours. Overall, there is overwhelming evidence of heterogeneity in wealth, income, concerns, and objectives. We find a prevalence of low retirement assets, a severe underestimation of survival probabilities to an extreme old age of 95, and a strong aversion toward life annuities. Pre-retirees appear to have reasonable expectations regarding income and assets in retirement, with the median retiree relying heavily on public pension sources. (Pre-)retirees are primarily concerned with liquidity needs, consumption smoothing, inflation, and longevity in retirement, and are least concerned with bequests. We elicited risk and time preferences, and found an average RRA parameter between 1.74 to 2.48 for pre-retirees and 2.48 to 3.74 for retirees, and a median subjective time discount factor of 0.997. A study of decision-making under risky scenarios reveals dramatic differences between the actual and implied choices under the expected utility maximization framework. Particularly, in the presence of inflation risk, agents lack the understanding of the long-term cumulative impact of inflation on the cost of living. In the presence of investment risk, the upside gain drives decision-making, and the presence of minimum income protection effectively provided by public pension income induces more risk-taking behaviour. The last chapter concludes the thesis, and proposes general directions for future work in retirement planning research.

Zhang, Y. and De Smedt, J. (2024). "Index tracking using shapley additive explanations and one-dimensional pointwise convolutional autoencoders." In: *International Review of Financial Analysis* 95, p. 103487.

The aim of index tracking is to mimic the performance of a benchmark index via minimizing the tracking error between the returns of the market index and the tracking portfolio. Lately, various deep learning solutions have been proposed to perform stock prediction or active investment. However, there remains a gap in literature to explore the application of deep learning to index tracking. In this paper, the one-dimensional Pointwise Convolutional Autoencoder is proposed to capture the main market characteristics and the Shapley Additive Explanations feature importance ranking is applied to select stocks to implement the partial replication index tracking with and without Covid-19 data. Moreover, portfolios with different holding periods and with different rebalancing frequency are created on different financial markets to check the effectiveness of the proposed strategy. Compared with different benchmark stock selection strategies, including Pearson correlation, mutual information, and Euclidean distance, the proposed strategy achieves state-of-the-art performance on different financial markets.

Zhang, Y. and Ahluwalia, H. (2024). "A Rational Multi-Asset Portfolio Rebalancing Decision-Making Framework." In: *The Journal of Portfolio Management* 50(5), pp. 11–24.

The authors propose a comprehensive, rigorous, and quantitative decision-making framework that considers the important elements for selecting an optimal multi-asset portfolio rebalancing strategy. These aspects are typically ignored in traditional rebalancing research, which either leverages one historical sample path for asset returns in the analysis or ignores transaction costs (or assumes static transaction costs) or does not propose an optimal rebalancing strategy. In particular, the proposed framework incorporates a simulation engine to model the inherent return uncertainty of assets as well as a dynamic transaction cost simulations model that links the bid-ask spread with the market condition, and a utility-based optimization engine that determines the optimal rebalancing strategy while simultaneously quantifying its benefit. Additionally, the framework allows for attributing the source of benefit of one rebalancing approach relative to another and in which market environments those benefits typically play out.

Zhang, Z., Zohren, S., and Roberts, S. (2020c). "Deep Learning for Portfolio Optimization." In: *The Journal of Financial Data Science* 22(4), pp. 8–20.

In this article, the authors adopt deep learning models to directly optimize the portfolio Sharpe ratio. The framework they present circumvents the requirements for forecasting expected returns and allows them to directly optimize portfolio weights by updating model parameters. Instead of selecting individual assets, they trade exchange-traded funds of market indexes to form a portfolio. Indexes of different asset classes show robust correlations, and trading them substantially reduces the spectrum of available assets from which to choose. The authors compare their method with a wide range of algorithms, with results showing that the model obtains the best performance over the testing period of 2011 to the end of April 2020, including the financial instabilities of the first quarter of 2020. A sensitivity analysis is included to clarify the relevance of input features, and the authors further study the performance of their approach under different cost rates and different risk levels via volatility scaling.

Zhao, S., Dong, Z., Cao, Z., and Douady, R. (2024). "Hedge Fund Portfolio Construction Using PolyModel Theory and iTransformer." In: *arXiv e-Print*.

When constructing portfolios, a key problem is that a lot of financial time series data are sparse, making it challenging to apply machine learning methods. Polymodel theory can solve this issue and demonstrate superiority in portfolio construction from various aspects. To implement the PolyModel theory for constructing a hedge fund portfolio, we begin by identifying an asset pool, utilizing over 10,000 hedge funds for the past 29 years' data. PolyModel theory also involves choosing a wide-ranging set of risk factors, which includes various financial indices, currencies, and commodity prices. This comprehensive selection mirrors the complexities of the real-world environment. Leveraging on the PolyModel theory, we create quantitative measures such as Long-term Alpha, Long-term Ratio, and SVaR. We also use more classical measures like the Sharpe ratio or Morningstar's MRAR. To enhance the performance of the constructed portfolio, we also employ the latest deep learning techniques (iTransformer) to capture the upward trend, while efficiently controlling the downside, using all the features. The iTransformer model is specifically designed to address the challenges in high-dimensional time series forecasting and could largely improve our strategies. More precisely, our strategies achieve better Sharpe ratio and annualized return. The above process enables us to create multiple portfolio strategies aiming for high returns and low risks when compared to various benchmarks.

Zhao, Z., Xu, F., Du, D., and Meihua, W. (2021). "Robust portfolio rebalancing with cardinality and diversification constraints." In: *Quantitative Finance* 21(10), pp. 1707–1721.

In this paper, we develop a robust conditional value at risk (CVaR) optimal portfolio rebalancing model under various financial constraints to construct sparse and diversified rebalancing portfolios. Our model includes transaction costs and double cardinality constraints in order to capture the trade-off between the limit of investment scale and the diversified industry coverage requirement. We first derive a closed-form solution for the robust CVaR portfolio rebalancing model with only transaction costs. This allows us to conduct an industry risk analysis for sparse portfolio rebalancing in the absence of diversification constraints. Then, we attempt to remedy the hidden industry risk by establishing a new robust portfolio rebalancing model with both sparse and diversified constraints. This is followed by the development of a distributed-version of the Alternating Direction Method of Multipliers (ADMM) algorithm, where each subproblem admits a closed-form solution. Finally, we conduct empirical tests to compare our proposed strategy with the standard sparse rebalancing and no-rebalancing strategies. The computational results demonstrate that our rebalancing approach produces sparse and diversified portfolios with better industry coverage. Additionally, to measure out-of-sample performance, two superiority indices are created based on worst-case CVaR and annualized return, respectively. Our ADMM strategy outperforms the sparse rebalancing and no-rebalancing strategies in terms of these indices.

Zhu, S., Zhu, W., Pei, X., and Cui, X. (2020). "Hedging crash risk in optimal portfolio selection." In: *Journal of Banking & Finance* 119, p. 105905.
When almost all underlying assets suddenly lose a certain part of their nominal value in a market crash, the diversification effect of portfolios in a normal market condition no longer works. We integrate the crash risk into portfolio management and investigate performance measures, hedging and optimization of portfolio selection involving derivatives. A suitable convex conic programming framework based on parametric approximation method is proposed to make the problem a tractable one. Simulation analysis and empirical study are performed to test the proposed approach.

Zhu, X., Zhou, K. Q., and Wang, Z. (2024). "A new paradigm of mortality modeling via individual vitality dynamics." In: *arXiv e-Print*.
The significance of mortality modeling extends across multiple research areas, ranging from life insurance valuation to optimal lifetime decision-making. Existing approaches, such as mortality laws and factor-based models, often fall short in capturing the complexity of individual mortality, hindering their ability to address specific research needs. To overcome these limitations, this paper introduces a novel approach to mortality modeling centered on the dynamics of individual vitality. A four-component framework is developed to account for initial conditions, natural aging processes, stochastic fluctuations, and accidental events over an individual's lifetime. We demonstrate the framework's analytical capabilities across various settings and explore its practical implications in solving life insurance problems and deriving optimal lifetime decisions. Our results show that the proposed framework not only encompasses existing mortality models but also provides individualized mortality outcomes and offers an intuitive explanation for survival biases.

Ziemba, W. T. (2016). "An Approach to Financial Planning of Retirement Pensions with Scenario-Dependent Correlation Matrixes and Convex Risk Measures." In: *The Journal of Retirement* 4(1), pp. 99–111.
The article describes an approach to asset-liability modeling using discrete time stochastic linear programming. The model uses future scenarios and optimizes the asset-liability mix subject to various constraints and is applicable to insurance companies, bank trading departments, overall bank asset-liability management, and other financial institutions. An application to the Siemens Austria pension fund, where the model has been in use since 2000, is described. The model has had considerable success and has been used by regulators to determine the effect of various possible pension fund policy changes. It has also been used by pension fund advisors who deal with uncertain assets and liabilities subject to various legal and policy constraints.

Zilbering, Y., Jaconetti, C. M., and Kinniry Jr., F. M. (2015). *Best practices for portfolio rebalancing*. Tech. rep. Vanguard.
The primary goal of a rebalancing strategy is to minimize risk relative to a target asset allocation, rather than to maximize returns. Over time, asset classes produce different returns that can change the portfolio's asset allocation. To recapture the portfolio's original risk-and-return characteristics, the portfolio should therefore be rebalanced. In theory, investors select a rebalancing strategy that weighs their willingness to assume risk against expected returns net of the cost of rebalancing. Vanguard research has found that there is no optimal frequency or threshold for rebalancing, since risk-adjusted returns do not differ meaningfully from one rebalancing strategy to another. As a result, we conclude that for most broadly diversified stock and bond fund portfolios (assuming reasonable expectations regarding return patterns, average returns, and risk), annual or semiannual monitoring, with rebalancing at 5 percent thresholds, is likely to produce a reasonable balance between risk control and cost minimization for most investors. Annual rebalancing is likely to be preferred when taxes or substantial time/costs are involved.

Zilbering, Y., Zhu, V., Banis, G., and Ahluwalia, H. (2025). "Tax-Aware Portfolio Construction: A Multi-Asset Approach." In: *The Journal of Portfolio Management* 51(5), pp. 64–83.
For taxable accounts, the industry norm when constructing tax-efficient portfolios is to substitute tax-inefficient bonds with tax-free, municipal bonds. This simplified approach only considers one dimension: the tax inefficiency of taxable bonds versus the tax-free treatment of municipal bonds. This ad-hoc substitution ignores the additional credit risk associated with municipal bonds, however, resulting in a portfolio that is less risk-return efficient or diversified. In this article, the authors discuss their methodology of systematically constructing tax-aware multi-

asset portfolios that target the best of both worlds-diversification and tax efficiency. Using the new methodology, the authors show that the industry approach is not ideal, and an optimal allocation for generating after-tax wealth is highly dependent on the investor profile, including: 1) their tax bracket, 2) market conditions and expected returns, 3) whether they draw all, some, or none of their income from the portfolio, and 4) whether they plan on liquidating their portfolio at the end of the horizon or not. The authors conclude that in most cases, the optimal tax-aware portfolio should include some exposure to taxable bonds.

Zuo, W., Damle, A., and Tuljapurkar, S. (2024). "Sensitivity and uncertainty in the Lee-Carter mortality model." In: *International Journal of Forecasting*.
The Lee-Carter model (LC) is widely used for forecasting age-specific mortality, and typically performs well regardless of the uncertainty and often limited quality of mortality data. Why? We analyze the robustness of LC using sensitivity analyses based on matrix perturbation theory, coupled with simulations that examine the effect of unavoidable randomness in mortality data. The combined effects of sensitivity and uncertainty determine the robustness of LC. We find that the sensitivity of LC and the uncertainty of death rates both have non-uniform patterns across ages and years. The sensitivities are small in general, with the largest sensitivities at both ends of the period. The uncertainties of death rates are high in young ages (5-19 years) and old ages (90+ years), rising in young ages but dropping in old ages. Our results reveal that LC is robust against random perturbation and sudden short-term changes.

Zuss, N. (2022). *Participants Will Need Support to Understand Lifetime Income Projections*. Tech. rep. PlanSponsor.
Recordkeepers are bolstering education, projection modeling tools and tailored advice capabilities to support plan participants and encourage them to remain on track for retirement income planning.
Plan sponsors and advisers are preparing for defined contribution (DC) plan participants' reactions to the lifetime income estimates that will be coming on their plan statements this year with custom tools and planning advice, as well as investment menu evaluations.
In 2019's Setting Every Community Up for Retirement Enhancement (SECURE) Act, Congress required that DC plans provide retirement plan participants with projections illustrating monthly retirement income amounts generated from their accumulated retirement savings. The Department of Labor (DOL)'s interim final rule on the matter, published in September, requires that participants' projected income be illustrated as an account balance conversion to a lifetime income annuity.